EEGDataset Benchmark Technical Report (Minimax v2)

Unified EEG Dataset Documentation

Datasets included: 286

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Preface

This report summarizes 286 EEG datasets. Each dataset uses analysis_minimax_v2.md when available, otherwise falls back to analysis.md.

- Complex Applied (56): Real-world simulation, ecological environment, or multi-dimensional task.
- High-level Mental (101): Deep internal cognitive/psychological processing tasks.
- Action/Output (28): Motor intention/execution and BCI-output focused tasks.
- Sensory/Response (52): Basic stimulus-response, ERP/SSVEP/RSVP, or simple attention tasks.
- Biological State (43): Task-free or state-monitoring data (rest/sleep/clinical monitoring).
- Others (6): Does not clearly fit the above categories.

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283. openneuro_ds004843 - Visual Situational Awareness EEG (T16) . . . 768

284. openneuro_ds005787 - NOD-EEG . . . 771

285. physionet_motionartifact - EEG/fNIRS Motion Artifact Dataset . . . 773

286. tuh_eeg_artifact - TUH EEG Artifact Corpus . . . 776

Chapter 1. Complex Applied

Datasets in this chapter: 56. Real-world simulation, ecological environment, or multi-dimensional task.

001. openneuro_ds001971 - Audiocue walking study

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds001971` | Year: 2019 | URL: https://openneuro.org/datasets/ds001971
Category: Complex Applied | Subjects: 18 | Files (Completed): 273 | Channels: 115.0 | Sampling Rate: 512.0 Hz

Paradigm

A mobile brain-body imaging (MoBI) study where participants walked on a treadmill and adapted their gait to auditory tone sequences. Participants walked at their preferred cadence, uncued, or responded to tempo shifts (delay or advance) in an auditory metronome. The study investigated the neural dynamics of gait adaptation and rhythm entrainment during real-world locomotion. Each walking condition is a sustained state lasting multiple seconds.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time
> Web Reference Used for Decision: https://openneuro.org/datasets/ds001971

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`PreferredCadence`	234 (files)	long	raw	Sustained walking at participant's preferred cadence
`UncuedWalking`	234 (files)	long	raw	Sustained uncued walking (no auditory cue)
`DelayTempo`	232 (files)	long	raw	Sustained walking to delayed tempo auditory cue
`AdvanceTempo`	230 (files)	long	raw	Sustained walking to advanced tempo auditory cue
`STATUS`	39	admin	raw	BioSemi system status marker - exclude

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Tempo-shift gait adaptation (step-advance vs step-delay)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = [AdvanceTempo, DelayTempo]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
AdvanceTempo	0	Step-advance (step shortening) tempo shift
DelayTempo	1	Step-delay (step lengthening) tempo shift

- Evidence:
- DOI: 10.1523/jneurosci.3543.15.2016 - Segmenting data into epochs time-locked to cue tempo shifts (from -4s to 10s) to compare beta band oscillatory networks between step-advance and step-delay adaptation conditions.
- DOI: 10.1038/s41597-019-0223-2 - Using step-locked epoching from -1s to 3s around heel strikes for analysis of tempo shift detection.
- Kernel Notes: The dataset contains sustained walking states labeled as AdvanceTempo and DelayTempo. Use long_event epoching to extract 2-second windows with 50% overlap from each tempo-shift condition. The Normal Task 1 (multiclass gait condition) covers all four conditions including PreferredCadence and UncuedWalking; this Primary Task focuses specifically on the tempo-shift contrast (Advance vs Delay) which is the main experimental contrast in the referenced papers. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- MoBI dataset with 115 channels including motion capture; verify EEG-only channel selection.
- STATUS events are BioSemi system markers - do not include in target_events.
- Walking produces motion artifacts; independent component analysis (ICA) for artifact removal is recommended.

- Verify channel count consistency across subjects before pooling.

References

- Reference: `Johanna Wagner (June 6, 2019)`

002. openneuro_ds002723 - A dataset recorded during development of an affective brain-computer music interface: testing session

Source: analysis_minimax_v2.md

Dataset: openneuro_ds002723 | Year: 2020 | URL: https://openneuro.org/datasets/ds002723
Category: Complex Applied | Subjects: 8 | Files (Completed): 44 | Channels: 37.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants interacted with a real-time affective brain-computer music interface (BCMI) that modified music on-the-fly based on detected emotional states, aiming to induce specific target affective trajectories. This is the testing session of the BCI system (see also ds002722 calibration and ds002724 training sessions). Codes 1-9 encode the target or detected affective music conditions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002723

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
2	44 (files)	long	raw	Affective music condition 2
6	44 (files)	long	raw	Affective music condition 6
9	42 (files)	long	raw	Affective music condition 9
4	42 (files)	long	raw	Affective music condition 4
5	39 (files)	long	raw	Affective music condition 5
7	38 (files)	long	raw	Affective music condition 7
1	29 (files)	long	raw	Affective music condition 1
3	28 (files)	long	raw	Affective music condition 3

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`

- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Affective Music Condition Classification (8-class)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 40 s , overlap = 0.0 , target_events = [1, 2, 3, 4, 5, 6, 7, 9]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
2	0	Affective music condition 2
6	1	Affective music condition 6
9	2	Affective music condition 9
4	3	Affective music condition 4
5	4	Affective music condition 5
7	5	Affective music condition 7
1	6	Affective music condition 1
3	7	Affective music condition 3

- Evidence:
- DOI: 10.1038/s41597.020.0507-6 - Paper describes the BCMI testing dataset (openneuro_ds002723) with EEG recordings from 8 participants during real-time evaluation of an affective brain-computer music interface playing 8 different synthetic music conditions. Events are long (>=2s) markers for sustained music conditions lasting approximately 40 seconds.
- Kernel Notes: The raw event markers indicate which affective music condition was playing. Labels are taken directly from event markers (raw meta). Music conditions are sustained ~40-second states; epochs extract the full duration of each condition. Verify exact epoch duration from original study protocol. (replaces Primary Task 1 from original analysis.md)

Normal Task 1: Affective Music Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to music condition onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 8 affective conditions; decode exact valence/arousal mapping from Daly et al. 2018. Small dataset (8 subjects, 44 files). Music conditions are sustained 40-second states; these onset markers epoch at transition points.

Event	y	Meaning
2	0	Music condition 2
6	1	Music condition 6
9	2	Music condition 9
4	3	Music condition 4
5	4	Music condition 5

Event	y	Meaning
7	5	Music condition 7
1	6	Music condition 1
3	7	Music condition 3

Data Quality Notes

- Small dataset: 8 subjects, 44 files.
- 37-channel array includes physiological channels; verify EEG-only channel selection.
- Decode code-to-condition mapping from Daly et al. 2018 (Scientific Data).
- Verify channel count consistency across subjects before pooling.

References

- Reference: Neural and physiological data from participants listening to affective music, Scientific Data, 2018

003. openneuro_ds003039 - Free walking study

Source: analysis.md

Dataset: openneuro_ds003039 | Year: 2025 | URL: https://openneuro.org/datasets/ds003039
Category: Complex Applied | Subjects: 19 | Files (Completed): 19 | Channels: 67.0 | Sampling Rate: 500.0 Hz

Paradigm

Participants walked two different outdoor routes, once while performing a self-paced button-pressing task, to study brain activity during naturalistic free-walking gait. This is a mobile EEG study; no trial-level event markers are detectable in the EEG files. Button-press events from the self-paced task may be available in separate behavioral logs.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003039

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(none)	-	-	raw	No event markers detected in EEG files; naturalistic free-walking paradigm

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Publisher-defined BIDS task: Neur Corr Young

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds003039 DOI: https://doi.org/10.18112/openneuro.ds003039.v1.0.2

Normal Task 1: Resting / Continuous Segmentation (resting)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no trial-level events in EEG files))
- Notes: Mobile EEG during naturalistic walking. Motion artifacts are expected; ICA-based artifact removal recommended. Route condition (route A vs route B) may be available as a subject-level metadata label.

Data Quality Notes

- Mobile EEG with significant motion artifacts from walking - ICA and motion artifact rejection required.
- No trial-level event markers in EEG files; button-press events may be in separate logs.
- 67-channel mobile EEG system; verify electrode positions.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Nadine Jacobsen (March, 2020)`

004. openneuro_ds003517 - EEG: Continuous Gameplay of an 8-Bit Style Video Game

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003517` | Year: 2021 | URL: https://openneuro.org/datasets/ds003517
Category: Complex Applied | Subjects: 17 | Files (Completed): 34
Channels: 65 | Sampling Rate: 500 Hz

Paradigm

Participants played a continuous 8-bit style video game while EEG was recorded to study canonical neural events during naturalistic gameplay. The experiment also included a visual oddball task and a gambling task embedded within the session. Game events (shooting, collecting stars, collisions) and oddball stimuli (standard and rare tones/targets) were time-locked to EEG recordings. The study aims to identify ERP components (P300, error-related negativity) in a real-world, engaging task context.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003517

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`ODDBALL STANDARD`	17	short	raw	Standard tone in oddball task (frequent)
`ODDBALL RARE`	17	short	raw	Rare/target tone in oddball task (infrequent)
`ODDBALL START`	17	short	raw	Oddball task block start marker
`ODDBALL DONE`	17	short	raw	Oddball task block end marker
`SHOOT_BUTTON`	18	short	raw	Player shoot button press
`COLLECT_STAR`	18	short	raw	Player collected a star (reward)
`PLAYER_CRASH_ENEMY`	18	short	raw	Player crashed into enemy
`MISSILE_HIT_ENEMY`	17	short	raw	Missile hit enemy (success)
`PLAYER_CRASH_WALL`	17	short	raw	Player crashed into wall

Event ID	Count (total)	Type	Meta Level	Meaning
COLLECT_AMMO	17	short	raw	Player collected ammunition
GAMBLING WIN	14	short	raw	Gambling task win outcome
GAMBLING LOSS	14	short	raw	Gambling task loss outcome
GAME START	16	short	raw	Game session start
GAME OVER	16	short	raw	Game session end
STATUS	34	admin	raw	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Oddball Target Detection (P300)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s , tmax = 0.75 s , target_events = [ODDBALL STANDARD, ODDBALL RARE]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
ODDBALL STANDARD	0	Standard/frequent tone
ODDBALL RARE	1	Rare/target tone (P300 evoking)

- Evidence:
- DOI: 10.1016/j.neuroimage.2016.02.075 - LASSO classifiers trained on oddball target vs standard events for P300 detection; study explicitly uses this dataset for oddball task with "target vs standard" binary classification
- Kernel Notes: Baseline correction -500 to -300 ms as per original paper. Verify epoch timing from paper (uses -250 to 750 ms for classification with -500 to -300 ms baseline).

Primary Task 2: Gambling Outcome Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s , tmax = 0.75 s , target_events = [GAMBLING WIN, GAMBLING LOSS]
- Label Route: Annotation

- Meta Level: raw
- Target Definition:
Event	y	Meaning
GAMBLING LOSS	0	Loss outcome
GAMBLING WIN	1	Win outcome
- Evidence:
- DOI: 10.1016/j.neuroimage.2016.02.075 - LASSO classifiers trained on gambling wins vs losses; confirms this dataset includes gambling task with binary win/loss labels
- Kernel Notes: Epoch window matches oddball parameters for consistency. Only 14 files contain gambling events. (replaces Normal Task 1 from original analysis.md)

Primary Task 3: Six-Class Naturalistic Game Behavior Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 0.5 s *, overlap = 0.5 *
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
SHOOT_BUTTON	1	Missile launch button press
COLLECT_STAR	2	Star collection
COLLECT_AMMO	3	Ammo box collection
PLAYER_CRASH_WALL	4	Wall collision
PLAYER_CRASH_ENEMY	5	Enemy collision
MISSILE_HIT_ENEMY	6	Missile hit enemy
- Evidence:
- DOI: 10.1016/j.neuroimage.2025.121521 - Decodes six naturalistic game behaviors (missile launch, star collection, ammo collection, wall crash, enemy crash, missile hit) from brain-wide EEG oscillatory activation patterns during continuous video game play
- Kernel Notes: Paper uses continuous segmentation based on amplitude envelope median to identify active/inactive states, then extracts brain-wide patterns averaged over behavior periods for 6-class decoding. Parameters marked with * are inferred defaults; verify from paper.

Data Quality Notes

- 65-channel EEG recorded during naturalistic, continuous task; higher artifact rate expected from movement and engagement.
- Some sessions may be game-only (no oddball/gambling); verify event presence per file.

References

- Reference: Identification of canonical neural events during continuous gameplay of an 8-bit style video game. NeuroImage, 2016. DOI: 10.1016/j.neuroimage.2016.02.075

005. openneuro_ds003574 - Reward Biases Spontaneous Neural Reactivation During Sleep

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003574` | Year: 2021 | URL: https://openneuro.org/datasets/ds003574
Category: Complex Applied | Subjects: 18 | Files (Completed): 17
Channels: 69 | Sampling Rate: 500 Hz

Paradigm

Participants played two rewarded and non-rewarded video games (FACE and MAZE) followed by a nap sleep session recorded with simultaneous EEG-fMRI. The study examines whether reward enhances spontaneous neural reactivation of game-related memory traces during sleep. Brain patterns from waking gameplay are decoded and compared to sleep activity to identify replay events. EEG event structure consists of BrainVision numeric S-code triggers from the game paradigm with limited per-file occurrence.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003574

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`S 1-S 27`	1 each	short	raw	BrainVision stimulus triggers (game/task events)
`R128`	1	short	raw	BrainVision response trigger
`Sync On`	1	short	raw	MRI/EEG synchronization marker
`R`	1	admin	raw	Generic response marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`

- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `S 1`, `S 27`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sleep Brain State Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s *, overlap = 0.5 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`face_game`	0	Face game playing (60s blocks) - classifier-derived brain state from CRF model
`maze_game`	1	Maze game playing (60s blocks) - classifier-derived brain state from CRF model
`rest`	2	Rest periods (90s blocks) - classifier-derived brain state
`pre_face`	3	3s before face game onset - classifier-derived brain state
`pre_maze`	4	3s before maze game onset - classifier-derived brain state

- Super Meta Notes: Derived from fMRI-based Conditional Random Fields classifier outputs applied to sleep EEG-fMRI data. Kernel step: Apply pre-trained CRF model (trained on fMRI during wake gaming) to sleep EEG features to obtain probability estimates for each of the five brain states.
- Evidence:
- DOI: `10.1038/s41467-021-24357-5` - Five-class brain state classification (face game, maze game, rest, pre-face, pre-maze) during sleep using EEG-fMRI, with reward analysis mapping
- Kernel Notes: The CRF classifier was originally trained on fMRI data from waking game sessions, then applied to sleep fMRI data. For EEG-only analysis, use the same feature extraction and apply the trained classifier. Verify segment_length matches EEG power analysis epochs (4s with 2s overlap) from the paper.

Primary Task 2: Reward vs. No-Reward Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s *, overlap = 0.5 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`reward`	0	Brain state associated with won (rewarded) game - Face or Maze game won
`no_reward`	1	Brain state associated with lost (non-rewarded) game - Face or Maze game lost

- Super Meta Notes: Derived from five-class brain state labels [`face_game`, `maze_game`, `rest`, `pre_face`, `pre_maze`] combined with experimental manipulation (which game was won). Face/Maze games played with reward contingency. Kernel step: Map brain state labels to reward status based on experimental design (won game = reward, lost game = no-reward).
- Evidence:
- DOI: `10.1038/s41467-021-24357-5` - Shows privileged reactivation of reward-associated game patterns during N3 sleep, comparing won vs. lost game brain state reactivations
- Kernel Notes: This is the primary analysis of the paper - examining whether reward enhances neural reactivation during sleep. Segments are classified into one of 5 brain states, then mapped to

reward/no-reward for downstream analysis.

Normal Task 1: Continuous sleep EEG segmentation

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no label, suitable for unsupervised analysis))
- Notes: Only 1 occurrence per event type per file - insufficient for epoch-based classification. Dataset is primarily suited for EEG-fMRI sleep replay analysis using pattern decoding, not standard ERP epoching. The 17 completed files (of 18 subjects) are sleep EEG recordings.

Data Quality Notes

- Very sparse event structure; each trigger type appears only once per recording.
- EEG-fMRI simultaneous recording; gradient artifact and MRI-related noise present.
- Primary analysis requires decoding waking neural patterns and cross-correlating with sleep activity.

References

- Reference: `Reward biases spontaneous neural reactivation during sleep. Nature Communications, 2021.`

006. openneuro_ds003620 - Runabout: A Mobile EEG Study of Auditory Oddball Processing in Laboratory and Real-World Conditions

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003620` | Year: 2021 | URL: https://openneuro.org/datasets/ds003620
Category: Complex Applied | Subjects: 44 | Files (Completed): 271
Channels: 35 | Sampling Rate: 250 Hz

Paradigm

Participants performed an auditory oddball task under three levels of environmental complexity and physical activity: sitting in a laboratory, walking on a sports field (oval), and navigating a university campus. The paradigm examined how cognitive and motor demands of real-world locomotion affect P300 oddball responses. Target tones required mental counting ("count" condition) or were simply listened to ("don't count" condition), with both counted and uncounted targets interleaved across environments.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003620

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`t_count_lab`	134	short	raw	Target tone, count condition, laboratory
`t_count_oval`	134	short	raw	Target tone, count condition, oval
`t_count_campus`	134	short	raw	Target tone, count condition, campus
`ntcount_lab`	134	short	raw	Non-target (standard) tone, count condition, laboratory
`ntcount_oval`	134	short	raw	Non-target tone, count condition, oval
`ntcount_campus`	134	short	raw	Non-target tone, count condition, campus
`t_Count_lab`	80	short	raw	Target tone, Count (capital) condition, laboratory

Event ID	Count (total)	Type	Meta Level	Meaning
t_Count_oval	80	short	raw	Target tone, Count condition, oval
t_Count_campus	80	short	raw	Target tone, Count condition, campus
ntCount_lab	80	short	raw	Non-target tone, Count condition, laboratory
ntCount_oval	80	short	raw	Non-target tone, Count condition, oval
ntCount_campus	80	short	raw	Non-target tone, Count condition, campus
t_Dontcount_lab	208	short	raw	Target tone, don't-count condition, laboratory
t_Dontcount_oval	214	short	raw	Target tone, don't-count condition, oval
t_Dontcount_campus	214	short	raw	Target tone, don't-count condition, campus
ntDontcount_lab	208	short	raw	Non-target tone, don't-count condition, laboratory
ntDontcount_oval	214	short	raw	Non-target tone, don't-count condition, oval
ntDontcount_campus	214	short	raw	Non-target tone, don't-count condition, campus
empty	226	admin	raw	Empty/padding event marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 0.8 s, target_events = [`t_count_lab`, `t_count_oval`, `t_count_campus`, `t_Count_lab`, `t_Count_oval`, `t_Count_campus`, `t_Dontcount_lab`, `t_Dontcount_oval`, `t_Dontcount_campus`, `ntcount_lab`, `ntcount_oval`, `ntcount_campus`, `ntCount_lab`, `ntCount_oval`, `ntCount_campus`, `ntDontcount_lab`, `ntDontcount_oval`, `ntDontcount_campus`]
- Label Route: `Annotation`

- Meta Level: raw
- Target Definition:
Event	y	Meaning
t_count_lab, t_count_oval, t_count_campus, t_Count_lab, t_Count_oval, t_Count_campus, t_Dontcount_lab, t_Dontcount_oval, t_Dontcount_campus	1	Target tone (deviant)
ntcount_lab, ntcount_oval, ntcount_campus, ntCount_lab, ntCount_oval, ntCount_campus, ntDontcount_lab, ntDontcount_oval, ntDontcount_campus	0	Non-target tone (standard)
- Evidence:
- DOI: 10.1038/s41598.021.01772-8 - This study used data from OpenNeuro dataset ds003620 to analyze MMN and P3 event-related potentials elicited by deviant vs standard tones in an auditory oddball paradigm across lab, field, and campus environments.
- Kernel Notes: Epoch from -200ms to 800ms relative to tone onset. The target events (t_) correspond to 'deviant' stimuli and non-target events (nt) correspond to 'standard' stimuli in the oddball paradigm. Exclude empty admin events from epoching. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Mobile EEG recording; significant motion artifacts during outdoor walking conditions (oval, campus).
- Lowercase vs uppercase variants (count/Count) may reflect different recording sessions or systems.
- empty event is an admin placeholder; exclude from epoching.
- Stimulus/-prefixed variants appear in a subset of files and are redundant with non-prefixed versions.

References

- Reference: Runabout: A mobile EEG study of auditory oddball processing in laboratory and real-world conditions. OpenNeuro ds003620.

007. openneuro_ds003670 - Dataset of Concurrent EEG, ECG, and Behavior with Multiple Doses of Transcranial Electrical Stimulation

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003670` | Year: 2021 | URL: https://openneuro.org/datasets/ds003670
Category: Complex Applied | Subjects: 19 | Files (Completed): 62
Channels: 35 | Sampling Rate: 2000 Hz

Paradigm

Participants performed a continuous tracking task (CTT) requiring them to maintain a ball at the center of a screen to assess vigilance and sustained alertness. Simultaneously, multi-site transcranial electrical stimulation (tES) was applied at varying dose combinations (sham, low, high) to modulate arousal and attention. EEG, ECG, and behavioral tracking data were recorded concurrently. The study aims to identify optimal tES parameters for enhancing cognitive performance.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003670

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulation Start`	62	long	raw	Transcranial stimulation onset (sustained state)
`Stimulation Stop`	62	long	raw	Transcranial stimulation offset
`Block Start`	62	long	raw	Task block onset marker
`STATUS`	62	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`

- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulation Start`, `Stimulation Stop`, `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: tES Stimulation Period Classification (Stimulation ON vs OFF)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = [`Stimulation Start`, `Block Start`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Stimulation Start`	1	Active tES stimulation period (30-sec epoch)
`Block Start`	0	Control/baseline task block (no stimulation)

- Evidence:
- DOI: 10.1038/s41597-021-01046-y - Data descriptor confirms dataset uses 30-sec stimulation epochs partitioned by stimulation trigger codes (Stimulation Start vs Block Start)
- DOI: 10.1016/j.brs.2024.05.009 - Confirms 30-sec trial-wise epochs during stimulation for comparing active tES conditions
- Kernel Notes: Use long-event epoching to capture 30-sec sustained stimulation periods. The dataset contains 9 stimulation types (F0/F5/F30, M0/M5/M30, P0/P5/P30); all active stimulation conditions are labeled as y=1. Block Start represents baseline/control periods without stimulation (y=0). Filter for tES artifact removal required post-epoching.

Normal Task 1: Stimulation period analysis (long event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Stimulation periods are sustained states (several minutes each); use long-event epoching. Dose condition (sham/low/high) should be in events.tsv or participants.tsv. Very high sampling rate (2000 Hz) increases memory requirements.

Event	y	Meaning
`Stimulation Start`	1	Transcranial electrical stimulation activation period
`Block Start`	0	Task block (baseline/control)

Data Quality Notes

- Very high sampling rate (2000 Hz); downsample to 250-500 Hz for standard EEG analyses.
- tES artifacts present during stimulation periods; independent component analysis required.
- ECG channels included; can be used for cardiac artifact removal.

References

- Reference: `Dataset of concurrent EEG, ECG, and behavior with multiple doses of transcranial Electrical Stimulation. OpenNeuro ds003670.`

008. openneuro_ds003702 - Social Memory cuing

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003702 | Year: 2024 | URL: https://openneuro.org/datasets/ds003702
Category: Complex Applied | Subjects: N/A | Files (Completed): 47
Channels: 64.0 | Sampling Rate: 500.0 Hz

Paradigm

A visual working memory task performed in a virtual reality environment where participants recalled object locations and statuses following social (avatar) or non-social (stick) directional cues.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003702

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.
## MISC Label Summary				

- Route: misc
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Social vs Non-Social Cue Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 4 s, overlap = 0.5
- Label Route: misc
- Meta Level: super
- Target Definition:

Derived Label	y	Meaning

| `social` | 0 | Social cue condition (avatar cues: PeopleAll/PeopleCong/PeopleIncong) |
| `non_social` | 1 | Non-social cue condition (stick cues: StickAll/StickCong/StickIncong) |
- Super Meta Notes: Derived from external BIDS events.tsv/json files. Kernel must: (1) parse event timing from BIDS-sidecar files, (2) map event labels to social/non-social categories per the paradigm (avatar->social, stick->non-social), (3) optionally merge with accuracy labels (correct/incorrect) from behavioral responses. Events referenced in paper: `cue_onset`, `object_presentation`, `location_probe`, `status_probe`, `response` - these exist in BIDS metadata but NOT in raw EEG event channel (`no_events` only). Alternative: use epoch_by_event_hdf5 with tmin=-1.0 s, tmax>=7.0 s * if kernel can load BIDS-sidecar events as target_events.
- Evidence:
- DOI: 10.1016/j.dib.2022.107827 - Data in Brief paper describes OpenNeuro ds003702 dataset of 47 participants performing VR working memory task with social (avatar) vs non-social (stick) cues; epochs EEG from 1s pre-cue to 1s post-probe response for cue type classification.
- Kernel Notes: (1) Load events from BIDS *_events.tsv and corresponding JSON sidecar files, (2) Filter epochs by condition labels (social/non-social), (3) Apply recommended preprocessing: bandpass 0.5-40 Hz, artifact rejection. Verify event ID mapping from BIDS-sidecar files. Consider accuracy labels (correct/incorrect) as optional second-level classification.

Normal Task 1: Raw-meta task

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Needs verification
- Kernel Notes: Exclude admin markers from target labels.

Normal Task 2: Super-meta task

- Task Type: regression/classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target / Label: Needs verification
- Kernel Notes: Derive and version super-meta in kernel.

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

009. openneuro_ds003739 - Perturbed beam-walking task

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003739 | Year: 2021 | URL: https://openneuro.org/datasets/ds003739
Category: Complex Applied | Subjects: N/A | Files (Completed): 120
Channels: 149.0 | Sampling Rate: 256.0 Hz

Paradigm

Participants performed walking on a treadmill-mounted balance beam while being subjected to sensorimotor perturbations via virtual reality visual rotations and physical waist pulls.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003739

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.
## MISC Label Summary				

- Route: misc
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Perturbation Type Classification (8-class)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -1.0 s , tmax = 2.0 s , target_events = [no_events]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning

no_events	0	pull_stand_left
no_events	1	pull_stand_right
no_events	2	pull_walk_left
no_events	3	pull_walk_right
no_events	4	rotate_stand_cw
no_events	5	rotate_stand_ccw
no_events	6	rotate_walk_cw
no_events	7	rotate_walk_ccw
- Evidence:
- DOI: 10.1016/j.dib.2021.107635 - Data in Brief paper describing the dataset with 150 perturbation events per session (75 per direction) including pull and rotation types during standing and walking
- Kernel Notes: Extract perturbation labels (type, posture, direction) from BIDS *_events.tsv files. Epoch around perturbation onset (t=-1s to t=2s). Map to 8-class labels. Verify event timing from events.tsv columns (onset, duration, trial_type). (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Posture-Perturbation Classification (4-class)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 2.0 s, target_events = [`no_events`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
no_events	0	stand pull
no_events	1	walk pull
no_events	2	stand rotate
no_events	3	walk rotate
- Super Meta Notes: Derived from raw event(s) [`no_events`] by extracting experimental condition from events.tsv: combine posture (standing vs walking) + perturbation type (pull vs rotation). Labels: stand pull->0, walk pull->1, stand rotate->2, walk rotate->3. Kernel step: Parse events.tsv trial_type column, derive posture x perturbation label.
- Evidence:
- DOI: 10.1016/j.neuroimage.2019.05.038 - Source-localized EEG connectivity analysis during four conditions: walking/standing with visual rotations or physical waist pulls, epoched from -1 to 2s around perturbation onset
- Kernel Notes: Use events.tsv metadata to derive 4-class labels combining posture (stand/walk) and perturbation type (pull/rotate). Epoch from -1s to +2s around perturbation onset. (replaces Normal Task 2 from original analysis.md)

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `eNeuro publication`

010. openneuro_ds003801 - Neural Tracking to Go

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003801` | Year: 2021 | URL: https://openneuro.org/datasets/ds003801
Category: Complex Applied | Subjects: 20 | Files (Completed): 20
Channels: 24 | Sampling Rate: 250 Hz

Paradigm

A mobile EEG study where participants performed a dichotic listening task (two competing speakers) while either sitting in a quiet indoor environment or walking along a pre-defined indoor route with environmental distractions. The study investigates how locomotion and environmental complexity affect neural tracking of attended speech (measured by auditory EEG envelope tracking). No discrete trial-level event markers were embedded in the EEG recording.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003801

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
-	-	-	raw	No event markers detected; continuous EEG during dichotic listening and walking

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Auditory Attention Decoding - Attended vs Ignored Speaker

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 60 s, overlap = 0.0 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning

| `attended_speaker` | 0 | Participant instructed to attend to this speaker |
| `ignored_speaker` | 1 | Participant instructed to ignore this speaker |
- Super Meta Notes: Derived from experimental instruction (not raw event markers). The attended/ignored speaker was determined by instruction to participants in each trial/block. Classification is based on whether the neural envelope reconstruction (via mTRF) correlates more strongly with the attended vs ignored speech envelope. Raw EEG is continuous with no event markers - labels must be obtained from file-level metadata specifying which speaker was attended in each segment.
- Evidence:
- DOI: 10.1088/1741-2552.ac42b5 - Uses mTRF models to decode which of two competing speakers participants were attending to during dichotic listening
- DOI: 10.1088/1741-2552.ac42b5.2 - Uses the same dataset to investigate auditory attention decoding during sitting and walking conditions
- DOI: 10.3389/fnhum.2024.1483024 - Uses 60-s non-overlapping segments to classify attended vs ignored speaker in single/dual-speaker conditions
- Kernel Notes: Load continuous EEG, segment into 60-s non-overlapping windows. Obtain trial condition metadata (attended vs ignored speaker) from file-level BIDS sidecar or task events.tsv. Train mTRF (multivariate temporal response function) to reconstruct speech envelopes from EEG. Compute correlation between reconstructed envelope and both attended/ignored stimulus envelopes. Decode accuracy = percentage of trials where attended correlation > ignored correlation. Verify condition labels from dataset metadata. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Cognitive Load Analysis - Sitting vs Walking Condition

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 60 s *, overlap = 0.0 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`sitting`	0	Participant performing task while seated
`walking`	1	Participant performing task while walking
- Super Meta Notes: Derived from file-level metadata specifying experimental condition (sitting vs walking). The dataset includes two conditions: quiet indoor sitting and indoor walking with environmental distractions. Condition label must be obtained from task events.tsv or BIDS sidecar files.
- Evidence:
- DOI: 10.1088/1741-2552.ac42b5 - Participants performed dichotic listening while either sitting or walking
- DOI: 10.3389/fnhum.2024.1483024 - Mobile EEG study investigating how locomotion affects neural tracking of attended speech
- Kernel Notes: Load continuous EEG, segment into windows matching the trial structure. Obtain condition labels (sitting/walking) from file-level metadata. Analyze how locomotion and environmental complexity affect neural envelope tracking. Motion artifacts expected during walking - consider preprocessing for artifact handling.

Data Quality Notes

- Mobile EEG recording; significant motion artifacts during walking condition.
- No trial-level event markers; condition label is file-level or block-level.
- Small dataset (20 subjects, 20 files); limited statistical power.
- 24-channel EEG at 250 Hz; modest spatial and temporal resolution.

References

- Reference: Neural Tracking to Go. OpenNeuro ds003801.

011. openneuro_ds003846 - Prediction Error

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003846 | Year: 2025 | URL: https://openneuro.org/datasets/ds003846
Category: Complex Applied | Subjects: 19 | Files (Completed): 50
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

A virtual reality haptic interaction study where participants reached out to touch spawned virtual boxes while receiving varied combinations of visual, vibrotactile, and electrical muscle stimulation (EMS) feedback. Trials were marked as "normal" (congruent feedback) or "conflict" (incongruent prediction error condition), allowing investigation of multisensory prediction error processing. Three feedback modality conditions were run in separate sessions: visual-only, vibrotactile-only, and combined EMS conditions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: reaction time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003846

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
box	559 files	short	raw	Box spawned or box touched event
vibroFeedback	198 files	short	raw	Vibrotactile feedback onset
block	79 files	short	raw	Block start marker
visualFeedback	50 files	short	raw	Visual feedback onset

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `reaction_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

Tasks

Primary Task 1: Prediction Error Detection (Match vs Mismatch)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.6 s, target_events = [box] *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
box (match/congruent)	0	Normal trial with congruent feedback
box (mismatch/incongruent)	1	Conflict trial with prediction error

- Super Meta Notes: Derived from raw event box by parsing the event description string to extract `normal_or_conflict` field. Kernel step: parse box event annotation (JSON-like) to extract normal_or_conflict value and assign binary label 0 (normal) or 1 (conflict).
- Evidence:
- DOI: 10.3389/fnrgo.2024.1411305 - Performs ERP/ERV analysis comparing match (75% congruent feedback) vs mismatch (25% incongruent/prediction error) trials to detect visuo-haptic prediction errors in VR interaction.
- Kernel Notes: Epoch on box events (tap/touch events). Parse event description strings to extract `normal_or_conflict` field. Map 'normal' to y=0 and 'conflict' to y=1. Use epoch window tmin=-0.1s to tmax=0.6s relative to tap onset as per ERP analysis in the referenced paper. (replaces incorrectly specified Primary Task 1 from original analysis.md)

Normal Task 1: Normal vs Conflict Prediction Error (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Events are richly annotated with metadata (reaction time, trial number, ISI, cube identity). The normal/conflict label is embedded in the event string. Epoch on box:touched events and extract the normal_or_conflict field for the label. Reaction time is also available as a regression target.

Event	y	Meaning
`box:touched;...normal_or_conflict:normal`	0	Normal (congruent) trial - box touched, expected feedback
`box:touched;...normal_or_conflict:conflict`	1	Conflict (prediction error) trial - incongruent feedback

Data Quality Notes

- Events contain full JSON-like annotation strings; label extraction requires parsing the event description
- Multiple sessions per subject (EMS, Vibro, Visual) with differing session lengths
- Sampling rate is ~500 Hz (exact value varies slightly across files)

References

- Reference: `Cyber-Physical Systems: Prediction Error`

012. openneuro_ds004105 - BCIT Auditory Cueing

Source: analysis.md

Dataset: `openneuro_ds004105` | Year: 2022 | URL: https://openneuro.org/datasets/ds004105
Category: Complex Applied | Subjects: 34 | Files (Completed): 34
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Participants performed a long-duration simulated driving task involving lane maintenance with periodic lateral vehicle perturbations, while receiving auditory cues at varying intervals. EEG biomarkers of fatigue were investigated as performance and alertness declined over the driving session. The study is part of the BCIT (Brain Computer Interface and Technology) driving fatigue dataset series.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004105

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`4210`	34 files	short	raw	Driving event code 4210 (lateral perturbation onset)
`4220`	34 files	short	raw	Driving event code 4220
`4230`	34 files	short	raw	Driving event code 4230
`4311`	34 files	short	raw	Lane deviation/correction event
`4312`	34 files	short	raw	Lane deviation/correction event
`3200`	34 files	short	raw	Speed/control event
`3111`	34 files	short	raw	Perturbation/response event
`3112`	34 files	short	raw	Perturbation/response event
`3310`	34 files	short	raw	Performance metric event
`1211`	34 files	short	raw	Auditory cue event

Event ID	Count (total)	Type	Meta Level	Meaning
1212	34 files	short	raw	Auditory cue event
2621	34 files	short	raw	Behavioral response event
2622	34 files	short	raw	Behavioral response event
1111-1322	17 files	short	raw	Condition-specific events (half of subjects)
4411	1 file	short	raw	Rare event

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)
- Evidence: Dataset references indicate MI BCI paradigm as core benchmark. Source: https://openneuro.org/datasets/ds004105 DOI: https://doi.org/10.1016/j.neuroimage.2017.02.057

Normal Task 1: Lateral Perturbation Response (binary)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Full BCIT event code decoding requires the dataset codebook. Part of the BCIT driving fatigue series alongside ds004106, ds004118, ds004119, ds004120, ds004121, ds004122, ds004123. Auditory cueing conditions are the primary manipulation.

Event	y	Meaning
4311	0	Lane perturbation - direction/type 1
4312	1	Lane perturbation - direction/type 2
## Data Quality Notes		

- Numerical event codes require codebook from BCIT documentation for full decoding
- 74-channel BioSemi ActiveTwo; high-density cap with 1024 Hz sampling
- Part of a larger BCIT driving series - consider combining with related datasets

References

- No paper reference available

013. openneuro_ds004106 - BCIT Advanced Guard Duty

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004106` | Year: 2022 | URL: https://openneuro.org/datasets/ds004106
Category: Complex Applied | Subjects: 29 | Files (Completed): 29
Channels: 262 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a sustained vigilance task simulating advanced security guard duties by verifying multiple fields on replica ID cards against reference images, under varying workloads and stochastic arrival rates of cards. EEG was recorded to identify biomarkers of mental fatigue and vigilance decrement over extended duty periods. Part of the BCIT sustained attention / guard duty dataset series.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004106

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1311`	29 files	short	raw	Card/stimulus arrival event - type 1311
`1312`	29 files	short	raw	Card/stimulus arrival event - type 1312
`1321`	29 files	short	raw	Verification response event
`1322`	29 files	short	raw	Verification response event
`1331-1362`	27-29 files	short	raw	Guard duty task events (various verification stages)
`1341-1342`	29 files	short	raw	Guard duty task events
`1351-1352`	27-29 files	short	raw	Guard duty task events
`1221-1222`	29 files	short	raw	Guard duty task events
`1211-1212`	28 files	short	raw	Guard duty task events
`1231-1232`	28 files	short	raw	Guard duty task events

Event ID	Count (total)	Type	Meta Level	Meaning
1241-1242	27 files	short	raw	Guard duty task events
1251-1252	27 files	short	raw	Guard duty task events
1121-1122	27 files	short	raw	Guard duty task events
2110	29 files	short	raw	Workload condition marker - high
2120	29 files	short	raw	Workload condition marker - low
3111-3112	26 files	short	raw	Block/session markers
3231-3232	2 files	short	raw	Rare condition events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Continuous Behavioral Performance Estimation

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 3.0 s, overlap = 0.33 *
- Label Route: misc
- Meta Level: super
- Target Definition:

Event	y	Meaning
continuous	lane_deviation (m)	Lane deviation from driving simulator
continuous	reaction_time (s)	Response time to perturbations
continuous	accuracy	Proportion correct verifications

- Super Meta Notes: Derived from continuous behavioral metrics recorded during task performance (ID verification accuracy, response times). Kernel step: Compute spectral features (ICA-decomposed EEG power in alpha/theta bands) from 3s sliding windows, apply 90s centered mean filter to both EEG features and behavioral labels, then perform linear regression to predict each behavioral metric.
- Evidence:
- DOI: 10.1016/j.biopsycho.2015.12.009 - Linear regression with ICA-based EEG spectral features estimates continuous behavioral performance (lane deviation, RT, accuracy) in a simulated guard duty task, achieving correlation coefficients of 0.548±0.117 for driving and 0.701±0.154 for guard duty tasks.
- Kernel Notes: Use 750-point epochs (~3s) with 500-point step (~2s) on continuous EEG. Extract ICA components, compute power spectral density in alpha (8-12 Hz) and theta (4-8 Hz) bands. Apply 90s centered moving average filter to behavioral labels. Train separate linear regression models for each behavioral metric (lane deviation, reaction time, accuracy).

Primary Task 2: High vs Low Workload Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [`2110`, `2120`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`2110`	0	High workload condition marker
`2120`	1	Low workload condition marker

- Evidence:
- DOI: `10.1016/j.biopsycho.2015.12.009` - Workload manipulation is a key experimental factor in the guard duty paradigm, with high vs low workload conditions affecting vigilance and behavioral performance.
- Kernel Notes: Epoch EEG relative to workload condition onset markers. 262-channel BioSemi high-density EEG - consider dimensionality reduction (PCA, CSP) before classification. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- 262 channels - largest EEG cap in the BCIT series; high memory requirements
- Full event decoding requires BCIT codebook documentation
- Very limited subjects (29) for the large channel count

References

- Reference: `BCIT Advanced Guard Duty`

014. openneuro_ds004118 - BCIT Calibration Driving

Source: analysis.md

Dataset: openneuro_ds004118 | Year: 2022 | URL: https://openneuro.org/datasets/ds004118
Category: Complex Applied | Subjects: 247 | Files (Completed): 247
Channels: 266 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a 15-minute simulated driving calibration task on a straight road, maintaining lane position while correcting for periodic lateral perturbations. This calibration recording is designed to collect a baseline driving EEG profile for each subject to normalize fatigue prediction algorithms used in the broader BCIT driving fatigue study series. It is the largest file count in the BCIT series.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004118

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
3200	247 files	short	raw	Speed/driving state marker
3111	247 files	short	raw	Perturbation onset
3112	247 files	short	raw	Perturbation offset/response
3310	247 files	short	raw	Performance metric event
1211	247 files	short	raw	Lane correction event
1212	247 files	short	raw	Lane correction event
4311	246 files	short	raw	Lateral perturbation type 1
4312	246 files	short	raw	Lateral perturbation type 2
4230	246 files	short	raw	Perturbation/direction event
4210	239 files	short	raw	Perturbation onset event
4220	206 files	short	raw	Perturbation event
1121	246 files	short	raw	Driver response event
1122	246 files	short	raw	Driver response event

Event ID	Count (total)	Type	Meta Level	Meaning
1111	246 files	short	raw	Driver response event
1112	246 files	short	raw	Driver response event
2621	97 files	short	raw	Auditory cue onset
2622	97 files	short	raw	Auditory cue offset
4200-4612	2 files	short	raw	Rare condition events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)
- Evidence: Dataset references indicate MI BCI paradigm as core benchmark. Source: https://openneuro.org/datasets/ds004118 DOI: https://doi.org/10.3389/fnins.2014.00155

Normal Task 1: Lateral Perturbation Direction (binary)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Largest file count in the BCIT series (247 subjects). 266-channel BioSemi cap - very high memory requirements. Calibration session is shorter (15 min) than fatigue driving sessions.

Event	y	Meaning
4311	0	Lateral perturbation - direction/type 1
4312	1	Lateral perturbation - direction/type 2
## Data Quality Notes		

- 266 channels - very high-density EEG; significant memory requirements for batch processing
- 247 files (subjects) - large cohort; suitable for cross-subject decoding
- Full event code decoding requires BCIT codebook

References

- No paper reference available

015. openneuro_ds004119 - BCIT Basic Guard Duty

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004119` | Year: 2022 | URL: https://openneuro.org/datasets/ds004119
Category: Complex Applied | Subjects: 22 | Files (Completed): 22
Channels: 262 | Sampling Rate: 1024 Hz

Paradigm

A realistic sustained vigilance task in which subjects verified multiple identification card fields against reference images in a simulated security environment under varying card arrival rates. This basic guard duty variant uses a simpler verification protocol compared to the Advanced Guard Duty (ds004106), and is designed to study EEG correlates of sustained attention and vigilance decrement over extended duty periods.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004119

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1311`	22 files	short	raw	Card arrival / verification stimulus onset
`1312`	22 files	short	raw	Card arrival event
`1321-1322`	22 files	short	raw	Verification response events
`1331-1332`	22 files	short	raw	Guard duty task events
`1341-1342`	22 files	short	raw	Guard duty task events
`1351-1352`	22 files	short	raw	Guard duty task events
`1361-1362`	22 files	short	raw	Guard duty task events
`1231-1232`	22 files	short	raw	Guard duty task events
`2110`	22 files	short	raw	High workload condition marker
`2120`	22 files	short	raw	Low workload condition marker

Event ID	Count (total)	Type	Meta Level	Meaning
3111-3112	20 files	short	raw	Block/session markers
1111-1112	21 files	short	raw	Guard duty task events
3241-3242	6 files	short	raw	Rare condition events
3231-3232	1 file	short	raw	Very rare condition events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Continuous Behavioral Performance Estimation

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 3 s, overlap = 2 s
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
continuous segmentation	continuous	Lane deviation (driving) or RT/accuracy (guard duty)
- Evidence:
- DOI: 10.1016/j.biopsycho.2015.12.009 - Linear regression estimates continuous behavioral performance (lane deviation and RT for driving, RT and accuracy for guard duty) from EEG power spectral density features extracted via ICA and smoothed with a 90s mean filter.
- Kernel Notes: Extract sliding windows of ~3s duration with ~2s step from continuous EEG. Compute PSD features (recommended: compute delta/theta/alpha/beta bands per channel). Smooth behavioral targets (RT, accuracy) with 90s centered mean filter. Train linear regression model to predict smoothed behavioral metrics from EEG features.

Primary Task 2: Workload Classification (High vs Low)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [2110, 2120]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
2110	0	High workload condition
2120	1	Low workload condition
- Evidence:

- DOI: `10.1016/j.biopsycho.2015.12.009` - Study compared EEG correlates across varying workload conditions in guard duty task.
- Kernel Notes: Epoch around workload condition markers. Extract spectral features from theta, alpha, beta bands. Train binary classifier to distinguish high vs low workload. (replaces Primary Task 1 from original analysis.md which incorrectly described motor imagery)

Normal Task 1: High vs Low Workload (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Small cohort (22 subjects). 262-channel BioSemi cap. Paired with ds004106 (Advanced Guard Duty)
- can be combined for larger guard duty cohort.

Event	y	Meaning
`2110`	0	High workload condition
`2120`	1	Low workload condition

Data Quality Notes

- 262 channels - very high-density EEG; high memory requirements
- Only 22 subjects - small sample size
- Full event code decoding requires BCIT codebook documentation

References

- Reference: `Gross, Matthews, Cohn, Kanade, & Baker, 2010`

016. openneuro_ds004120 - BCIT Baseline Driving

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004120` | Year: 2022 | URL: https://openneuro.org/datasets/ds004120
Category: Complex Applied | Subjects: 131 | Files (Completed): 131
Channels: 266 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a 60-minute simulated driving task involving manual steering and speed control, requiring corrective responses to lateral vehicle perturbations to study fatigue-related EEG biomarkers over an extended session. This baseline driving condition serves as the primary long-duration fatigue paradigm in the BCIT series, without auditory cueing or traffic complexity manipulation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004120

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`4311`	131 files	short	raw	Lateral perturbation - type 1
`4312`	131 files	short	raw	Lateral perturbation - type 2
`4210`	130 files	short	raw	Perturbation onset
`4220`	123 files	short	raw	Perturbation event
`4230`	131 files	short	raw	Perturbation/direction event
`3200`	131 files	short	raw	Speed/driving state marker
`3111`	128 files	short	raw	Perturbation onset
`3112`	125 files	short	raw	Perturbation response
`3310`	131 files	short	raw	Performance metric event
`1211`	131 files	short	raw	Lane correction event
`1212`	130 files	short	raw	Lane correction event
`1121-1122`	131 files	short	raw	Driver response events

Event ID	Count (total)	Type	Meta Level	Meaning
1111-1112	131 files	short	raw	Driver response events
2621-2622	28 files	short	raw	Auditory cue events (subset)
2611-2612	28 files	short	raw	Auditory cue events (subset)
5241-5242	10 files	short	raw	Rare condition events
4421-5232	2-5 files	short	raw	Very rare events

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Continuous Driving Performance Regression

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 3 s, overlap = 0.33 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
continuous	lane deviation (m)	Lane position deviation from center (super meta)
continuous	normalized accuracy	Driving accuracy 0-1 (super meta)
continuous	reaction time (ms)	Response latency to perturbations (super meta)
continuous	button press duration (ms)	Duration of driver response (super meta)

- Super Meta Notes: Derived from raw events [`4210`, `3111`] (perturbation onset markers) by computing behavioral metrics from vehicle state (lane deviation) and button response logs, then smoothing with 90s centered mean filter. Kernel step: Extract vehicle state data and button timestamps, compute lane deviation, RT, and duration per epoch, normalize by perceptual difficulty.
- Evidence:
- DOI: `10.3389/fnins.2014.00155` - Uses EEG log power spectra with PCA and linear regression to estimate continuous behavioral performance (lane deviation, accuracy, RT, duration) during simulated driving from 256-channel BioSemi recordings at 1024Hz
- Kernel Notes: Extract 3-second epochs with ~2s step from continuous EEG. Compute label per epoch from vehicle state and response logs. Normalize metrics. Apply PCA to log-power features. Train linear regression. (replaces Primary Task 1 from original analysis.md)

Normal Task 1: Lateral Perturbation Direction (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`

- Target Definition: Event-to-label mapping table below.
- Notes: 131 subjects with 60-minute sessions - large fatigue dataset. 266-channel BioSemi cap. Primary baseline driving condition for BCIT fatigue series comparison.

Event	y	Meaning
4311	0	Lateral perturbation - direction/type 1
4312	1	Lateral perturbation - direction/type 2

Data Quality Notes

- 266 channels - very high-density EEG; significant memory requirements
- 60-minute sessions provide extensive fatigue-related data per subject
- Full event code decoding requires BCIT codebook

References

- No paper reference available

017. openneuro_ds004121 - BCIT Mind Wandering

Source: analysis.md

Dataset: `openneuro_ds004121` | Year: 2022 | URL: https://openneuro.org/datasets/ds004121
Category: Complex Applied | Subjects: 60 | Files (Completed): 60
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a long-duration simulated driving task involving lane maintenance, speed control, and a police vehicle vigilance sub-task (detecting a police car among traffic), while exposed to lateral perturbations and varying background audio conditions (silence vs. music vs. podcast). The mind wandering manipulation used periodic thought probes to assess on-task vs. off-task mental state. EEG biomarkers of mind wandering during driving were investigated.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004121

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`4311`	60 files	short	raw	Lateral perturbation type 1
`4312`	59 files	short	raw	Lateral perturbation type 2
`4210`	59 files	short	raw	Perturbation onset
`4220`	59 files	short	raw	Perturbation event
`4230`	59 files	short	raw	Perturbation/direction event
`4730`	59 files	short	raw	Special driving event
`4411`	60 files	short	raw	Police vehicle event
`4621-4622`	60 files	short	raw	Audio condition events
`3200`	60 files	short	raw	Speed marker
`3111`	60 files	short	raw	Perturbation onset
`3310`	60 files	short	raw	Performance metric
`1211`	60 files	short	raw	Lane correction event
`1212`	58 files	short	raw	Lane correction event
`1121-1122`	60 files	short	raw	Driver response

Event ID	Count (total)	Type	Meta Level	Meaning
1111-1112	60 files	short	raw	Driver response
2221-2222	59 files	short	raw	Thought probe / mind wandering event
2241-2242	59 files	short	raw	Thought probe response
2251-2252	59 files	short	raw	Thought probe response
2621-2622	60 files	short	raw	Behavioral response
2811-2812	59 files	short	raw	Audio cue events
2631-2632	58 files	short	raw	Behavioral response events

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)
- Evidence: Dataset references indicate MI BCI paradigm as core benchmark. Source: https://openneuro.org/datasets/ds004121 DOI: https://doi.org/10.18112/openneuro.ds004121.v1.0.0

Normal Task 1: Lateral Perturbation Direction (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Mind wandering thought probe events (2221/2241/2251) can be used to label on-task vs. off-task periods for attention state classification. Full event codebook required for precise decoding of all 30+ event types.

Event	y	Meaning
4311	0	Lateral perturbation - type 1
4312	1	Lateral perturbation - type 2
## Data Quality Notes		

- 30+ distinct event types; full decoding requires BCIT codebook

- Thought probe events provide mind-wandering labels for attention classification
- Audio condition (silence/music/podcast) provides additional between-condition variable

References

- No paper reference available

018. openneuro_ds004122 - BCIT Speed Control

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004122` | Year: 2022 | URL: https://openneuro.org/datasets/ds004122
Category: Complex Applied | Subjects: 63 | Files (Completed): 63
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

A simulated driving task assessing driver fatigue and performance through a speed control manipulation: subjects drove under either manual speed control (maintaining speed without cruise control) or automated speed control (cruise control active), while also performing manual steering corrections for lateral vehicle perturbations. The study investigates how automation of one driving sub-task affects neural fatigue signatures.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004122

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`4210`	63 files	short	raw	Perturbation onset
`4220`	63 files	short	raw	Perturbation event
`4230`	63 files	short	raw	Perturbation/direction event
`4311`	63 files	short	raw	Lateral perturbation type 1
`4312`	63 files	short	raw	Lateral perturbation type 2
`3200`	63 files	short	raw	Speed control state marker
`3111`	63 files	short	raw	Perturbation onset
`3310`	63 files	short	raw	Performance metric event
`1211`	63 files	short	raw	Lane correction event
`1212`	59 files	short	raw	Lane correction event
`1121-1122`	63 files	short	raw	Driver response events
`1111-1112`	63 files	short	raw	Driver response events

Event ID	Count (total)	Type	Meta Level	Meaning
2621-2622	61 files	short	raw	Behavioral response / auditory cue
3112	59 files	short	raw	Perturbation response
4411	5 files	short	raw	Rare event
4421	6 files	short	raw	Rare event

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Normal Task 1: Lateral Perturbation Direction (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Speed control condition (manual vs. automated) can serve as a between-subjects or within-session variable. Part of the BCIT driving fatigue series.

Event	y	Meaning
4311	0	Lateral perturbation - type 1
4312	1	Lateral perturbation - type 2

Data Quality Notes

- 74-channel BioSemi ActiveTwo; 1024 Hz sampling
- Speed control condition variable requires session metadata from participants.tsv
- Full event code decoding requires BCIT codebook

References

- No paper reference available

019. openneuro_ds004123 - BCIT Traffic Complexity

Source: analysis.md

Dataset: `openneuro_ds004123` | Year: 2022 | URL: https://openneuro.org/datasets/ds004123
Category: Complex Applied | Subjects: 30 | Files (Completed): 30
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a simulated driving task with varying levels of visual traffic complexity (low, medium, high density of surrounding vehicles) while correcting for lateral vehicle perturbations. EEG was recorded to assess how increasing traffic complexity interacts with fatigue-related neural signatures. Part of the BCIT driving fatigue dataset series examining environmental complexity effects on driver workload.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004123

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`4311`	30 files	short	raw	Lateral perturbation type 1
`4312`	30 files	short	raw	Lateral perturbation type 2
`4210`	30 files	short	raw	Perturbation onset
`4220`	30 files	short	raw	Perturbation event
`4230`	30 files	short	raw	Perturbation/direction event
`4411`	30 files	short	raw	Special event
`3200`	30 files	short	raw	Speed marker
`3111`	30 files	short	raw	Perturbation onset
`3310`	30 files	short	raw	Performance metric
`3210`	30 files	short	raw	Traffic complexity marker - low
`3230`	30 files	short	raw	Traffic complexity marker - medium
`3240`	30 files	short	raw	Traffic complexity marker - high
`1211`	30 files	short	raw	Lane correction event

Event ID	Count (total)	Type	Meta Level	Meaning
1212	30 files	short	raw	Lane correction event
1121-1122	30 files	short	raw	Driver response events
1111-1112	30 files	short	raw	Driver response events
2221-2222	30 files	short	raw	Behavioral response events
2241-2242	30 files	short	raw	Behavioral response events
2251-2252	30 files	short	raw	Behavioral response events
2621-2622	30 files	short	raw	Auditory/behavioral events
2711-2712	30 files	short	raw	Traffic event markers
2811-2812	30 files	short	raw	Traffic event markers

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)
- Evidence: Dataset references indicate MI BCI paradigm as core benchmark. Source: https://openneuro.org/datasets/ds004123 DOI: https://doi.org/10.18112/openneuro.ds004123.v1.0.0

Normal Task 1: Traffic Complexity Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Traffic complexity conditions (3210/3230/3240) are the primary manipulation variable. Only 30 subjects - small cohort. Full event code decoding requires BCIT codebook.

Event	y	Meaning
3210	0	Low traffic complexity
3230	1	Medium traffic complexity

Event	y	Meaning
3240	2	High traffic complexity
## Data Quality Notes		

- 30 subjects - smallest subject count in the BCIT driving series
- 30 distinct event types; full decoding requires BCIT codebook documentation
- Traffic complexity codes (3210/3230/3240) are the key classification targets

References

- Reference: BCIT Traffic Complexity

020. openneuro_ds004152 - Drum Trainer

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004152` | Year: 2022 | URL: https://openneuro.org/datasets/ds004152
Category: Complex Applied | Subjects: 21 | Files (Completed): 21 | Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

Participants learned and performed specific rhythmic drumming patterns at varying tempos using keyboard keys, receiving adaptive visual feedback on timing accuracy. EEG was recorded continuously during drumming training. Each key press generates a Stimulus event marker; the dataset contains thousands of drumming events per participant (2300-4700 Stimulus events per file).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004152

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus`	~65000	short	raw	Individual drum key press / beat onset
`New Segment`	21	admin	raw	Recording segment boundary
`Comment`	21	admin	raw	Experimenter comment marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Reward Positivity (RewP) Regression for Timing Adjustment Prediction

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.5 s , tmax = 1.5 s , target_events = [`Stimulus`]
- Label Route: `Annotation`

- Meta Level: raw
- Target Definition:
Event	y	Meaning
`Stimulus`	continuous RT adjustment	RT adjustment value after feedback: positive = slowing, negative = speeding
- Evidence:
- DOI: `10.1111/psyp.14589` - Shows that trial-to-trial fluctuations in RewP (240-340ms at FCz) predict timing adjustment on subsequent trials in this drumming dataset.
- Kernel Notes: The paper analyzes feedback-locked epochs (-1500 to 1500ms around feedback onset). Verify if explicit `feedback_onset` events exist in events.tsv; if not, derive feedback timing from post-stimulus intervals. Extract RewP amplitude (240-340ms window at FCz) as EEG feature for regression to predict continuous RT adjustment values. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Very high event rate; check for event jitter in timing accuracy analysis
- Only 21 subjects; limited generalizability

References

- Reference: `Drum Trainer EEG dataset`

021. openneuro_ds004264 - Steer the Ship

Source: analysis.md

Dataset: `openneuro_ds004264` | Year: 2022 | URL: https://openneuro.org/datasets/ds004264
Category: Complex Applied | Subjects: 21 | Files (Completed): 21
Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

A ship-steering simulation where participants control a rotating arrow to maintain orientation against varying environmental and controller noise using unstable pole-and-cart physics. The task involves continuous manual control with varying noise levels, making it a real-world closed-loop BCI paradigm. EEG was recorded to study cognitive load and motor control under dynamic uncertainty.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004264

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus`	21x1000+	short	raw	Stimulus/condition onset marker
`New Segment`	21	admin	raw	Recording segment boundary
`Comment`	21	admin	raw	Experimenter comment/annotation

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Steertheship

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004264 DOI: https://doi.org/10.18112/openneuro.ds004264.v1.1.0

Normal Task 1: Stimulus Processing (single-class)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Only one event class available for epoching. New Segment and Comment are admin markers. Behavioral metadata (noise level, performance metrics) may be available in events.tsv for regression tasks.

Event	y	Meaning
`Stimulus`	0	Stimulus/condition onset
## Data Quality Notes		

- Only one non-admin event type (Stimulus); classification requires additional label parsing from events.tsv metadata columns
- New Segment and Comment are system/admin events excluded from epoching
- Variable trial counts across subjects (1473-2373 per subject)

References

- Reference: `Steer the Ship`

022. openneuro_ds004460 - EEG and motion capture data set for a full-body/joystick rotation task

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004460 | Year: 2024 | URL: https://openneuro.org/datasets/ds004460
Category: Complex Applied | Subjects: 20 | Files (Completed): 40
Channels: 160 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a spatial orientation task in a virtual environment, navigating via either active physical rotation of the body or joystick-controlled rotation on a flat screen. EEG and motion capture were simultaneously recorded to compare neural dynamics of active (full-body) vs. passive (joystick) spatial navigation and heading changes.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004460

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
ground_rotate	40	short	raw	Ground/environment rotation onset
object_fadeInOut	40	short	raw	Object appearance/disappearance
'/Event/Label/targ_color_change'	53	short	raw	Target color change event
('/Event/Category/Participant response'	40	short	raw	Participant response event
'/Event/Label/wii_button'	40	short	raw	Wii/button press response
('/Event/Category/Experimental stimulus/Instruction/Attend/'	80	short	raw	Attend instruction stimulus
('/Event/Category/Experimental stimulus/Instruction/Read/'	40	short	raw	Read instruction stimulus
duration	40	admin	raw	Duration annotation

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `('/Event/Category/Experimental stimulus/Instruction/Attend/', ('/Event/Category/Experimental stimulus/Instruction/Read/'`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Physical vs Joystick Rotation Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s , tmax = 12.0 s , target_events = [`ground_rotate`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`ground_rotate`	0/1	Rotation trial (condition derived from trial metadata - needs verification)

- Evidence:
- DOI: 10.1038/s41598.021.97749-8 - Two-class motor task comparing physical rotation (physR) vs joystick rotation (joyR) for spatial heading computation, using 13s epochs centered on sphere onset stimulus
- Kernel Notes: Paper reports epoching around sphere onset (tmin=-1.0s, tmax=12.0s). Dataset contains 'ground_rotate' event which likely corresponds to rotation stimulus onset. Verify event ID mapping from events.tsv - the paper uses 'sphere_onset'/'rotation_onset' terms not present in raw events. Condition labels (physR vs joyR) derived from experimental condition per-trial metadata. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- HED-format event labels are very long strings - ensure proper parsing
- 160 channels includes motion capture data channels in addition to EEG
- 40 files from 20 subjects (2 runs each: body rotation and joystick rotation)

References

- Reference: `Human cortical dynamics during full-body heading changes. Scientific Reports, 11(1), 18186`

023. openneuro_ds004475 - Mobile EEG split-belt walking study

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004475` | Year: 2023 | URL: https://openneuro.org/datasets/ds004475
Category: Complex Applied | Subjects: 30 | Files (Completed): 30
Channels: 260 | Sampling Rate: 512 Hz

Paradigm

A mobile brain-body imaging experiment investigating brain activity during gait adaptation using abrupt and gradual split-belt treadmill walking. Participants walked on a split-belt treadmill with different belt speeds while EEG and gait events (heel strikes, toe-offs) were recorded to study locomotor adaptation and associated cortical dynamics.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004475

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`LHS`	30xx	short	raw	Left heel strike (gait event)
`LTO`	30xx	short	raw	Left toe-off (gait event)
`RHS`	30xx	short	raw	Right heel strike (gait event)
`RTO`	30xx	short	raw	Right toe-off (gait event)
`TRIG`	30xx	short	raw	External trigger marker
`BUTTON`	30xx	short	raw	Button press event
`SB1`	30	long	raw	Split-belt block 1 start (tied speed)
`SB2`	30	long	raw	Split-belt block 2 start (split speed)
`B1`	30	long	raw	Baseline block 1
`B2`	30	long	raw	Baseline block 2
`B3`	30	long	raw	Baseline block 3
`P1`	30	long	raw	Perturbation block 1

Event ID	Count (total)	Type	Meta Level	Meaning
P2	30	long	raw	Perturbation block 2
End SB1	30	admin	raw	End of split-belt block 1
End SB2	30	admin	raw	End of split-belt block 2
End B1	30	admin	raw	End of baseline block 1
End B2	30	admin	raw	End of baseline block 2
End B3	30	admin	raw	End of baseline block 3
End P1	30	admin	raw	End of perturbation block 1
End P2	30	admin	raw	End of perturbation block 2
boundary	8	admin	raw	Segment boundary marker
9633791	30	admin	raw	Numerical system trigger

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Four-Class Gait Event Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`LHS`, `LTO`, `RHS`, `RTO`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`LHS`	0	Left heel strike
`LTO`	1	Left toe-off
`RHS`	2	Right heel strike
`RTO`	3	Right toe-off

- Evidence:
- DOI: `10.1113/jp284505` - This paper analyzes high-density EEG recorded during split-belt treadmill walking from 30 participants (dataset openneuro_ds004475) to examine spectral power changes in alpha, beta, and theta bands across gait adaptation stages, using gait events (heel strikes and toe-offs) detected from force plates as epoching points.
- Kernel Notes: Epochs are extracted relative to each gait event onset (LHS/LTO/RHS/RTO). Exclude admin markers (boundary, 9633791, End_*) from processing. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- boundary and 9633791 must be excluded as admin markers
- 260 channels includes motion capture channels; filter to EEG only
- Gait event counts vary across walking conditions (tied vs. split-belt speeds)

References

- Reference: `Mobile EEG split-belt walking study`

024. openneuro_ds004477 - PES - Pandemic Emergency Scenario

Source: analysis.md

Dataset: openneuro_ds004477 | Year: 2023 | URL: https://openneuro.org/datasets/ds004477
Category: Complex Applied | Subjects: 9 | Files (Completed): 9
Channels: 80 | Sampling Rate: 2048 Hz

Paradigm

A strategic pandemic simulation task where participants allocate limited resources to cities on a map and report their decision confidence while interacting with an artificial agent. EEG was recorded to investigate neural correlates of complex decision-making, confidence estimation, and strategy formation in a realistic multi-agent scenario.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: confidence, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004477

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
stimulus	9xx	short	raw	Decision stimulus onset
resp_hold	9xx	short	raw	Response hold (button press start)
resp_release	9xx	short	raw	Response release (button press end)
confidence	9xx	short	raw	Confidence rating onset

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: confidence

- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: PES

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004477 DOI: https://doi.org/10.3389/fnhum.2019.00013

Normal Task 1: Decision Making (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Only 9 subjects; limited generalization. Small dataset suitable for within-subject analysis. Confidence ratings may be used as regression target.

Event	y	Meaning
`resp_hold`	0	Response initiation
`resp_release`	1	Response release
`stimulus`	2	Decision stimulus
`confidence`	3	Confidence rating
## Data Quality Notes		

- Only 9 subjects; very small dataset
- 80 channels at 2048 Hz (high temporal resolution)
- Complex task with strategic decisions; behavioral data critical for labeling

References

- Reference: `PES - Pandemic Emergency Scenario`

025. openneuro_ds004588 - Neuma (Neuromarketing)

Source: analysis.md

Dataset: openneuro_ds004588 | Year: 2023 | URL: https://openneuro.org/datasets/ds004588
Category: Complex Applied | Subjects: 42 | Files (Completed): 84 | Channels: 6 | Sampling Rate: 300 Hz

Paradigm

A multimodal neuromarketing study where 42 participants browsed supermarket product brochures displayed on screen and selected items they intended to purchase by clicking the left mouse button. EEG was recorded alongside mouse interaction data to identify neural correlates of consumer purchase decisions during free-browsing. Each session involved scrolling through product leaflet images presented in a sequential format.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004588

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`MouseButtonLeft pressed`	1734	short	raw	Left mouse button press (product selection / page interaction)
`MouseButtonLeft released`	1712	short	raw	Left mouse button release
`Category:IMG=ID:FYLLADIO_1.tif=...`	268	short	raw	Product leaflet image 1 onset
`Category:IMG=ID:FYLLADIO_2.tif=...`	138	short	raw	Product leaflet image 2 onset
`MouseWheelDown120 pressed`	210	admin	raw	Scroll-down wheel event (exclude)
`MouseWheelUp120 pressed`	76	admin	raw	Scroll-up wheel event (exclude)
`fixation_cross`	84	short	raw	Fixation cross onset (trial start)
`EOE`	84	short	raw	End-of-experiment marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Unnamed

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004588 DOI: https://doi.org/10.18112/openneuro.ds004588.v1.2.0

Normal Task 1: Purchase Interaction Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Only 6 EEG channels. Category event labels embed full image filenames - aggregate by category type if needed. Scroll wheel events are admin.

Event	y	Meaning
`fixation_cross`	0	Trial-start fixation
`MouseButtonLeft pressed`	1	Product purchase/page click
## Data Quality Notes		

- Very limited spatial resolution: only 6 EEG channels.
- Category event labels contain full image paths; group by category type for analysis.

References

- Reference: `Neuma`

026. openneuro_ds004625 - Mind in Motion: Young Adults Walking Over Uneven Terrain

Source: analysis.md

Dataset: openneuro_ds004625 | Year: 2023 | URL: https://openneuro.org/datasets/ds004625
Category: Complex Applied | Subjects: 32 | Files (Completed): 543 | Channels: 284 | Sampling Rate: 500 Hz

Paradigm

Participants walked over uneven terrain at different speeds while high-density EEG, EMG, and kinematic data were simultaneously recorded, with a seated resting condition also included. The study examines neural correlates of locomotor control and gait adaptation during ambulatory tasks. Gait events (heel-strikes, toe-offs) were synchronized with EEG via force-sensitive insoles.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004625

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
LHS	65124	short	raw	Left heel strike
LTO	65105	short	raw	Left toe off
RHS	63800	short	raw	Right heel strike
RTO	63792	short	raw	Right toe off
CGY-T 1	2786	short	raw	Terrain change trigger
CGY-M 1	676	short	raw	Gait measurement marker
spare-M 1	635	short	raw	Spare measurement marker
boundary	543	admin	raw	Segment boundary (exclude)
TrialStart	543	long	raw	Trial start marker
TrialEnd	543	long	raw	Trial end marker
LS-Sync	536	admin	raw	Insole sync pulse (exclude)
CWR-M 1	269	short	raw	Cobblestone walk marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Mixed

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004625 DOI: https://doi.org/10.18112/openneuro.ds004625.v1.0.2

Normal Task 1: Gait Event Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: High channel count (284-channel HD-EEG). boundary and LS-Sync are admin - exclude. TrialStart/TrialEnd define walking bouts. Extensive motion artifact rejection required.

Event	y	Meaning
`LHS`	0	Left heel strike
`LTO`	1	Left toe off
`RHS`	2	Right heel strike
`RTO`	3	Right toe off
## Data Quality Notes		

- High-density EEG (284 channels) with significant motion artifacts from walking.
- boundary and LS-Sync markers are administrative - exclude.
- Mobile EEG data - apply ICA-based artifact correction.

References

- Reference: `Mind in Motion: Young Adults Walking Over Uneven Terrain`

027. openneuro_ds004657 - Driving with Autonomous Aids

Source: analysis.md

Dataset: `openneuro_ds004657` | Year: 2023 | URL: https://openneuro.org/datasets/ds004657
Category: Complex Applied | Subjects: N/A | Files (Completed): 119 | Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Participants performed a semi-automated driving task involving lane maintenance and collision avoidance to evaluate human trust in automation across manual and varying autonomous assistance conditions. The 7-digit hierarchical trigger codes encode the driving condition, task phase, and condition variant. Codes 2310002/2310001 dominate with ~10,367 events each; 4111101 appears ~10,051 times. All other codes appear 1,081-1,819 times each - consistent with less-frequent driving scenarios or transitions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004657

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`2310002`	10367	long	raw	Driving condition 231, variant 2 (dominant)
`2310001`	10367	long	raw	Driving condition 231, variant 1 (dominant)
`4111101`	10051	long	raw	Driving condition 411, variant 1
`2320001`	1819	long	raw	Driving condition 232, variant 1
`2320002`	1819	long	raw	Driving condition 232, variant 2
`1410001`	1383	long	raw	Driving condition 141, variant 1
`1410002`	1383	long	raw	Driving condition 141, variant 2
`1420001`	1226	long	raw	Driving condition 142, variant 1
`1420002`	1226	long	raw	Driving condition 142, variant 2
`1320001`	1175	long	raw	Driving condition 132, variant 1

Event ID	Count (total)	Type	Meta Level	Meaning
1320002	1175	long	raw	Driving condition 132, variant 2
2541001	1174	long	raw	Driving condition 254, variant 1
2781021	1094	long	raw	Driving condition 278, variant 021
2781022	1094	long	raw	Driving condition 278, variant 022
2561001	1081	long	raw	Driving condition 256, variant 1

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Drive

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004657 DOI: https://doi.org/10.18112/openneuro.ds004657.v1.0.3

Normal Task 1: Driving Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: 7-digit codes require mapping from dataset README. Codes 2310001/2310002/4111101 dominate (>10k each); all others are minority classes (1k-1.8k). Group by leading 3-4 digits for higher-level condition classification.

Event	y	Meaning
2310001	0	Driving condition 231, variant 1
2310002	1	Driving condition 231, variant 2
4111101	2	Driving condition 411, variant 1
## Data Quality Notes		

- 7-digit hierarchical trigger codes require decode from README or paper.

- Codes 2310001/2310002/4111101 dominate (>10k each); other codes are minority classes.
- Subject count not specified in BIDS metadata.

References

- Reference: `Driving with Autonomous Aids`

028. openneuro_ds004660 - TNO (Military Cognitive Workload)

Source: analysis.md

Dataset: `openneuro_ds004660` | Year: 2023 | URL: https://openneuro.org/datasets/ds004660
Category: Complex Applied | Subjects: 42 | Files (Completed): 42
Channels: 38 | Sampling Rate: 512 Hz

Paradigm

A military-relevant cognitive workload study conducted at TNO research institute, involving multiple hierarchically encoded task conditions represented by 5-digit numeric trigger codes. The paradigm likely involves surveillance, target detection, or dual-task conditions common in defense research. Event codes encode task type, difficulty level, and stimulus properties in a hierarchical scheme.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004660

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`14211`	42922	short	raw	Hierarchical task code - primary condition
`14212`	42922	short	raw	Hierarchical task code - primary condition variant
`21111`	9228	short	raw	Hierarchical task code
`14112`	9228	short	raw	Hierarchical task code
`14111`	9228	short	raw	Hierarchical task code
`13001`	3663	short	raw	Hierarchical task code
`13002`	3663	short	raw	Hierarchical task code
`21201`	3663	short	raw	Hierarchical task code
`15521`	1840	short	raw	Hierarchical task code

Event ID	Count (total)	Type	Meta Level	Meaning
15421	1840	short	raw	Hierarchical task code
14121	1680	short	raw	Hierarchical task code
14122	1680	short	raw	Hierarchical task code

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Target vs non-target stimulus events
- Evidence: P300/oddball paradigm appears as core study objective. Source: https://openneuro.org/datasets/ds004660 DOI: https://doi.org/10.18112/openneuro.ds004660.v1.0.2

Normal Task 1: Conditional classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: All codes are hierarchically structured 5-digit identifiers. Full decode table required from TNO dataset README or associated paper.

Event	y	Meaning
14211	0	Primary task code
14212	1	Primary task code variant
21111	2	Task code
14112	3	Task code
14111	4	Task code

Event	y	Meaning
13001	5	Task code
13002	6	Task code
21201	7	Task code
## Data Quality Notes		

- All event codes are 5-digit hierarchical numeric identifiers - full decode needed.
- 30+ unique event types; group by high-level condition category for practical classification.

References

- Reference: TNO

029. openneuro_ds004661 - ANDI: Auditory N-back Dual-task EEG

Source: analysis.md

Dataset: openneuro_ds004661 | Year: 2024 | URL: https://openneuro.org/datasets/ds004661
Category: Complex Applied | Subjects: 17 | Files (Completed): 17 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

A dual-task paradigm combining a guided visual search task (tracking a red annulus) with an auditory N-back working memory task to study cognitive interference. Event code 10 marks the dominant condition (likely standard/no-target trials, 29,820 total) and 11 marks the minority condition (likely target/match trials, 2,948 total). The large imbalance between codes suggests 10 = non-target and 11 = N-back target detection.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004661

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
10	29820	short	raw	Non-target trial (dominant condition)
11	2948	short	raw	N-back target match (minority condition)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Nback

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004661 DOI: https://doi.org/10.18112/openneuro.ds004661.v1.1.0

Normal Task 1: Target vs Non-target N-back Classification (binary)

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Strong class imbalance (29,820 vs 2,948 ~ 10:1); undersample class 0 or use weighted loss. Low sampling rate (128 Hz) limits spectral resolution. Verify code meanings from dataset README or Ries et al., 2016.

Event	y	Meaning
`10`	0	Non-target trial
`11`	1	N-back target match
## Data Quality Notes		

- Only 2 event codes; binary classification.
- Strong class imbalance (~10:1); undersample majority class.
- Low sampling rate (128 Hz); limited for high-frequency analyses.

References

- Reference: `Ries, et al., 2016`

030. openneuro_ds004718 - Le Petit Prince Hong Kong: Naturalistic fMRI and EEG Dataset

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004718` | Year: 2025 | URL: https://openneuro.org/datasets/ds004718
Category: Complex Applied | Subjects: 52 | Files (Completed): 102 | Channels: 62 | Sampling Rate: 1000 Hz

Paradigm

Naturalistic listening task where older Cantonese-speaking adults (65+) listened to a Cantonese translation of "The Little Prince" divided into story sections, followed by periodic comprehension checks, during simultaneous EEG and fMRI recording. Numeric trigger codes 1-30+ mark story segments, comprehension questions, and experimental conditions. Code 14 is most frequent (1,053 total), followed by codes 15 (648), and many codes at ~258 each - consistent with a factorial design across 102 files with ~2-10 events per file per code.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004718

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`14`	1053	long	raw	Story section / condition code 14
`15`	648	long	raw	Story section / condition code 15
`7`	270	long	raw	Story section / condition code 7
`3`	268	long	raw	Story section / condition code 3
`5`	266	long	raw	Story section / condition code 5
`4`	263	long	raw	Story section / condition code 4
`6`	262	long	raw	Story section / condition code 6
`88`	259	long	raw	Story section / condition code 88
`40-112`	~258 each	long	raw	Additional story/condition codes

Event ID	Count (total)	Type	Meta Level	Meaning
1-30	~257-258 each	long	raw	Story section codes 1-30

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Word-onset ERP Classification (Naturalistic Language Comprehension)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 0.6 s, target_events = [1, 3, 4, 5, 6, 7, 14, 15, 88, 40]
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
1	0	Story section code 1 (sentence onset)
3	1	Story section code 3
4	2	Story section code 4
5	3	Story section code 5
6	4	Story section code 6
7	5	Story section code 7
14	6	Story section code 14 (most frequent)
15	7	Story section code 15
40	8	Story section code 40
88	9	Story section code 88

- Super Meta Notes: Derived from raw numeric trigger codes (1-30, 40-112) marking sentence onsets during Cantonese audiobook listening. Word-onset epochs (0-600ms) aligned to NLP-derived linguistic annotations (word frequency, POS tags, f0 intensity, word rate). Kernel step: epoch at sentence onset triggers, then filter for word-level linguistic features.
- Evidence:
- DOI: 10.1038/s41597.024.03745.8 - Describes LPPHK EEG epoched at word onsets (0-600ms) for ERP and GLM analyses across 4473 word events per participant with condition codes 1-30+ from triggers.
- Kernel Notes: Verify event ID mapping from events.tsv. Use sentence onset triggers (codes 1-30, 40-112) to segment continuous EEG. Each trigger represents the onset of a story section/sentence. The paper also mentions word-level epoching for ERP analysis.

Primary Task 2: Story Section Multiclass Classification

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = [14, 15, 7, 3, 5, 4, 6, 88, 40, 1]
- Label Route: Annotation

- Meta Level: raw
- Target Definition:
Event	y	Meaning
`14`	0	Story section/condition code 14
`15`	1	Story section/condition code 15
`7`	2	Story section/condition code 7
`3`	3	Story section/condition code 3
`5`	4	Story section/condition code 5
`4`	5	Story section/condition code 4
`6`	6	Story section/condition code 6
`88`	7	Story section/condition code 88
`40`	8	Story section/condition code 40
`1`	9	Story section/condition code 1
- Evidence:
- DOI: 10.1038/s41597.024.03745.8 - Describes naturalistic listening task with trigger codes 1-30+ marking story sections and comprehension checks during The Little Prince audiobook.
- Kernel Notes: Long-event epoching on story section triggers. Code 14 appears most frequently (1,053 occurrences) suggesting it's a primary story segment. Code 1 appears less frequently. Overlap allows sliding window across sustained story segments. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Numeric code meaning requires README or events.tsv decode.
- Naturalistic paradigm - EEG is continuous during extended listening blocks.
- Simultaneous 7T fMRI and EEG - verify gradient artifact removal.

References

- Reference: `Le Petit Prince Hong Kong: Naturalistic fMRI and EEG dataset from older Cantonese speakers`

031. openneuro_ds004840 - EEG/ECG/EMG During Music Therapy in ICU Burn Patients

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004840` | Year: 2023 | URL: https://openneuro.org/datasets/ds004840
Category: Complex Applied | Subjects: 9 | Files (Completed): 68 | Channels: 10 | Sampling Rate: 1024 Hz

Paradigm

Electrophysiological recordings (EEG, ECG, EMG) of adult burn patients in a clinical ICU environment during Music-Assisted Relaxation (MAR) therapy sessions. No event markers are present; EEG is continuous throughout the therapy recording. The study investigates physiological responses to music therapy as a pain and anxiety management intervention.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004840

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(no events)	-	-	raw	No event markers detected in events.tsv

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Music Therapy Phase Classification (PRE/MTI/POST)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 3.0 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
PRE	0	Pre-intervention resting state (baseline)
MTI	1	Music therapy intervention during session
POST	2	Post-intervention recovery period

- Super Meta Notes: Derived from experimental protocol timing. The three phases (PRE/MTI/POST) are not present as event markers in the raw data but are defined based on the session structure documented in the study methods. Kernel step: Use session timing metadata to determine PRE/MTI/POST intervals, then apply

sliding window segmentation within each interval to generate epoch labels.
- Evidence:
- DOI: 10.1038/s41598.024.73211.3 - Three-class classification of EEG epochs into PRE (pre-intervention), MTI (music therapy intervention), and POST (post-intervention) periods using 3s epochs with 1.5s overlap
- Kernel Notes: Verify session timing structure from dataset sidecar JSON or participants.tsv to determine exact start/end times for each phase. (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Publisher-defined BIDS task: Task post Music Therapy

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004840 DOI: https://doi.org/10.18112/openneuro.ds004840.v1.0.1

Normal Task 3: Publisher-defined BIDS task: Task pre Music Therapy

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004840 DOI: https://doi.org/10.18112/openneuro.ds004840.v1.0.1

Data Quality Notes

- No event markers; continuous EEG during music therapy.
- Clinical ICU environment - subject motion, medical equipment interference expected.
- Only 10 channels including ECG and EMG - limited spatial resolution for EEG analysis.

References

- Reference: `Dataset of electrophysiological signals (EEG, ECG, EMG) during Music therapy with adult burn patients in the Intensive Care Unit.`

032. openneuro_ds004841 - TX14 (Semi-Autonomous Vehicle Monitoring)

Source: analysis.md

Dataset: `openneuro_ds004841` | Year: 2023 | URL: https://openneuro.org/datasets/ds004841
Category: Complex Applied | Subjects: 11 | Files (Completed): 11 | Channels: 70 | Sampling Rate: 256 Hz

Paradigm

Participants monitored a semi-autonomous vehicle system and performed local situational awareness tasks within a simulated environment, with EEG recorded to characterize cognitive states during supervised automation. Event codes are hierarchically structured 7-digit identifiers encoding operational state and task type.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004841

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`2210001`	998	long	raw	Hierarchical operational state code
`2210002`	947	long	raw	Hierarchical operational state code
`2110000`	220	long	raw	Hierarchical operational state code
`4310000`	195	long	raw	Hierarchical operational state code
`4100001`	195	long	raw	Hierarchical operational state code
`4320000`	195	long	raw	Hierarchical operational state code
`4300010`	107	long	raw	Hierarchical operational state code

Event ID	Count (total)	Type	Meta Level	Meaning
4300020	96	long	raw	Hierarchical operational state code
4300100	80	long	raw	Hierarchical operational state code
1110002	76	long	raw	Hierarchical operational state code
1110001	76	long	raw	Hierarchical operational state code

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Drive On Mission

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004841 DOI: https://doi.org/10.18112/openneuro.ds004841.v1.0.1

Normal Task 1: Operational State Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: All codes are 7-digit hierarchical identifiers. Full decode table required from dataset README. Only 11 subjects - small dataset.

Event	y	Meaning
2210001	0	Operational state code
2210002	1	Operational state code
2110000	2	Operational state code
4310000	3	Operational state code
4100001	4	Operational state code
4320000	5	Operational state code
## Data Quality Notes		

- 7-digit hierarchical numeric codes - full decode needed.
- Only 11 subjects; limited sample size for generalization.

References

- Reference: TX14. ReadMe: TX14 dataset

033. openneuro_ds004844 - T22 (Semi-Automated Driving Trust Paradigm)

Source: analysis.md

Dataset: `openneuro_ds004844` | Year: 2023 | URL: https://openneuro.org/datasets/ds004844
Category: Complex Applied | Subjects: 68 | Files (Completed): 68 | Channels: 72 | Sampling Rate: 1024 Hz

Paradigm

Simulated semi-automated driving in a leader-follower paradigm focused on predicting and influencing trust-based decisions regarding control authority hand-off and take-over between human and automation. EEG was recorded during sustained driving across multiple automation levels. Event codes are hierarchically structured 7-digit identifiers encoding driving state and automation mode.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004844

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`2310002`	8978	long	raw	Driving state code
`2310001`	8978	long	raw	Driving state code
`4111101`	8716	long	raw	Driving state code
`2572001`	3231	long	raw	Driving state code
`2562001`	2880	long	raw	Driving state code
`2552001`	1983	long	raw	Driving state code
`3110002`	1884	long	raw	Driving state code
`2812002`	1884	long	raw	Driving state code
`3110001`	1884	long	raw	Driving state code
`2812001`	1884	long	raw	Driving state code
`2714111`	1591	long	raw	Driving state code
`1410001`	1591	long	raw	Driving state code
`1410002`	1591	long	raw	Driving state code

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Drive

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004844 DOI: https://doi.org/10.18112/openneuro.ds004844.v1.0.0

Normal Task 1: Driving State Classification (multiclass/long)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: All codes are 7-digit hierarchical identifiers encoding driving condition and automation level. Full decode table required from dataset README. Related dataset: ds004657.

Event	y	Meaning
`2310002`	0	Driving state code
`2310001`	1	Driving state code
`4111101`	2	Driving state code
`2572001`	3	Driving state code
`2562001`	4	Driving state code
`2552001`	5	Driving state code
`3110002`	6	Driving state code
`2812002`	7	Driving state code
## Data Quality Notes		

- All event codes are 7-digit hierarchical numeric identifiers - full decode needed.
- 40+ unique codes; group by condition category for practical classification.

References

- Reference: `Predicting and influencing trust-based decisions about control authority hand-off and take-over during simulated, semi-automated driving in a leader-follower paradigm`

034. openneuro_ds004849 - STRONG (Social-Cognitive EEG Dataset)

Source: analysis.md

Dataset: `openneuro_ds004849` | Year: 2023 | URL: https://openneuro.org/datasets/ds004849
Category: Complex Applied | Subjects: 1 | Files (Completed): 1 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

An EEG dataset recorded during a social-cognitive paradigm involving multiple hierarchically encoded stimulus conditions. The 5-digit numeric trigger codes encode stimulus type, valence, and response. The dataset contains a single aggregated file across participants; specific task details require decoding from events.tsv.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004849

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`2200`	1627	short	raw	Dominant trigger code (primary condition)
`72011`	338	short	raw	Trigger code (category 7, type 2)
`52011`	332	short	raw	Trigger code (category 5, type 2)
`75011`	320	short	raw	Trigger code
`55011`	317	short	raw	Trigger code
`73011`	316	short	raw	Trigger code
`74011`	312	short	raw	Trigger code
`53011`	308	short	raw	Trigger code
`54011`	307	short	raw	Trigger code
`1`	195	short	raw	Trigger code 1 (sync/startx)
`71011`	174	short	raw	Trigger code
`2201`	173	short	raw	Trigger code
`51011`	170	short	raw	Trigger code

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Nback

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004849 DOI: https://doi.org/10.18112/openneuro.ds004849.v1.0.0

Normal Task 1: Stimulus Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Full decode needed from events.tsv. Only 1 file (single aggregated file). Related datasets: ds004850, ds004852.

Event	y	Meaning
`2200`	0	Primary condition code
`72011`	1	Trigger code
`52011`	2	Trigger code
`75011`	3	Trigger code
`55011`	4	Trigger code
`73011`	5	Trigger code
`74011`	6	Trigger code
`53011`	7	Trigger code
`54011`	8	Trigger code
`71011`	9	Trigger code
`2201`	10	Trigger code
`51011`	11	Trigger code
## Data Quality Notes		

- Single aggregated file; individual subject data not separable without additional metadata.
- Full trigger decode required from events.tsv.

References

- Reference: STRONG

035. openneuro_ds004850 - ODE (Social-Cognitive EEG Dataset)

Source: analysis.md

Dataset: `openneuro_ds004850` | Year: 2023 | URL: https://openneuro.org/datasets/ds004850
Category: Complex Applied | Subjects: 1 | Files (Completed): 1 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

An EEG dataset (ODE) recorded during a social-cognitive paradigm with hierarchically encoded stimulus conditions. Shares the identical event structure with the STRONG (ds004849) and InsurgentCivilian (ds004852) datasets, suggesting these are companion recordings from the same experimental series. The single aggregated file requires events.tsv decoding for full interpretation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004850

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`2200`	1627	short	raw	Dominant trigger code (primary condition)
`72011`	338	short	raw	Trigger code (category 7, type 2)
`52011`	332	short	raw	Trigger code
`75011`	320	short	raw	Trigger code
`55011`	317	short	raw	Trigger code
`73011`	316	short	raw	Trigger code
`74011`	312	short	raw	Trigger code
`53011`	308	short	raw	Trigger code
`54011`	307	short	raw	Trigger code
`1`	195	short	raw	Trigger code 1
`71011`	174	short	raw	Trigger code
`2201`	173	short	raw	Trigger code
`51011`	170	short	raw	Trigger code

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Nback

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004850 DOI: https://doi.org/10.18112/openneuro.ds004850.v1.0.0

Normal Task 1: Stimulus Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Identical event structure to ds004849 (STRONG) and ds004852 (InsurgentCivilian). Single aggregated file.

Event	y	Meaning
`2200`	0	Primary condition code
`72011`	1	Trigger code
`52011`	2	Trigger code
`75011`	3	Trigger code
`55011`	4	Trigger code
`73011`	5	Trigger code
`74011`	6	Trigger code
`53011`	7	Trigger code
`54011`	8	Trigger code
`71011`	9	Trigger code
`2201`	10	Trigger code
`51011`	11	Trigger code
## Data Quality Notes		

- Single aggregated file; individual subject data not separable.
- Identical event structure to ds004849 and ds004852 - likely companion recordings.

References

- Reference: ODE

036. openneuro_ds004851 - HID (High-Density EEG Operational Dataset)

Source: analysis.md

Dataset: `openneuro_ds004851` | Year: 2024 | URL: https://openneuro.org/datasets/ds004851
Category: Complex Applied | Subjects: 65 | Files (Completed): 65 | Channels: 72 | Sampling Rate: 2048 Hz

Paradigm

High-density EEG recorded in an operational context involving multiple conditions encoded as 4-digit numeric trigger codes. The code structure suggests stimulus classes (1xxx: primary task events, 7xxx/8xxx: secondary stimulus categories) within a cognitive workload or situational awareness paradigm.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004851

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1010`	2181	short	raw	Primary event onset (type A)
`1011`	2179	short	raw	Primary event (type A variant)
`1020`	1970	short	raw	Primary event (type B)
`1021`	1970	short	raw	Primary event (type B variant)
`1040`	1912	short	raw	Primary event (type D)
`1041`	1912	short	raw	Primary event (type D variant)
`8100`	973	short	raw	Secondary stimulus category 8, type 1
`8101`	973	short	raw	Secondary stimulus category 8, type 1 variant
`1061`	474	short	raw	Primary event (type F variant)
`1060`	474	short	raw	Primary event (type F)
`8011`	195	short	raw	Secondary stimulus

Event ID	Count (total)	Type	Meta Level	Meaning
8010	195	short	raw	Secondary stimulus
8021	194	short	raw	Secondary stimulus
8020	194	short	raw	Secondary stimulus
7001-7019	~65 each	short	raw	Tertiary stimulus category (19 types)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: HID

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004851 DOI: https://doi.org/10.18112/openneuro.ds004851.v2.1.0

Normal Task 1: Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Very high sampling rate (2048 Hz) - downsample before analysis. Full decode table required from dataset README.

Event	y	Meaning
1010	0	Primary event type A
1011	1	Primary event type A variant
1020	2	Primary event type B
1021	3	Primary event type B variant
1040	4	Primary event type D
1041	5	Primary event type D variant
8100	6	Secondary stimulus
8101	7	Secondary stimulus variant
## Data Quality Notes		

- Very high sampling rate (2048 Hz) - downsample for efficiency.
- 30+ distinct codes; grouping by prefix category recommended.

References

- Reference: `HID`

037. openneuro_ds004852 - InsurgentCivilian (Social-Cognitive EEG Dataset)

Source: analysis.md

Dataset: openneuro_ds004852 | Year: 2023 | URL: https://openneuro.org/datasets/ds004852
Category: Complex Applied | Subjects: 1 | Files (Completed): 1 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

An EEG dataset (InsurgentCivilian) recorded during a social-cognitive paradigm where participants were likely required to classify ambiguous human figures as insurgents or civilians, a common threat-detection paradigm used in military research. Shares the identical event structure with STRONG (ds004849) and ODE (ds004850), suggesting these are companion recordings from the same experimental series.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004852

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
2200	1627	short	raw	Dominant trigger code (primary condition)
72011	338	short	raw	Trigger code (category 7, type 2)
52011	332	short	raw	Trigger code
75011	320	short	raw	Trigger code
55011	317	short	raw	Trigger code
73011	316	short	raw	Trigger code
74011	312	short	raw	Trigger code
53011	308	short	raw	Trigger code
54011	307	short	raw	Trigger code
1	195	short	raw	Trigger code 1
71011	174	short	raw	Trigger code
2201	173	short	raw	Trigger code
51011	170	short	raw	Trigger code

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Nback

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004852 DOI: https://doi.org/10.18112/openneuro.ds004852.v1.0.0

Normal Task 1: Stimulus Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Identical event structure to ds004849 (STRONG) and ds004850 (ODE). Single aggregated file.

Event	y	Meaning
`2200`	0	Primary condition code
`72011`	1	Trigger code
`52011`	2	Trigger code
`75011`	3	Trigger code
`55011`	4	Trigger code
`73011`	5	Trigger code
`74011`	6	Trigger code
`53011`	7	Trigger code
`54011`	8	Trigger code
`71011`	9	Trigger code
`2201`	10	Trigger code
`51011`	11	Trigger code
## Data Quality Notes		

- Single aggregated file; individual subject data not separable.
- Identical event structure to ds004849 and ds004850 - companion recordings.

References

- Reference: `InsurgentCivilian`

038. openneuro_ds004980 - EEG Data for an Architectural Affordances Task

Source: analysis.md

Dataset: `openneuro_ds004980` | Year: 2024 | URL: https://openneuro.org/datasets/ds004980
Category: Complex Applied | Subjects: 17 | Files (Completed): 17 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants navigated a virtual architectural environment using keyboard controls while EEG was recorded to investigate neural responses to architectural affordances - the action possibilities offered by built spaces. The study examines how the brain processes spatial features (doors, corridors, rooms) that afford specific navigational actions. SAM (Self-Assessment Manikin) rating events mark periodic subjective valence/arousal/dominance assessments.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004980

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Key released: UpArrow (forward)`	13072	short	raw	Forward key release
`Key pressed: UpArrow (forward)`	13063	short	raw	Forward key press
`Key pressed: RightArrow (turn right)`	8915	short	raw	Right turn key press
`Key released: RightArrow (turn right)`	8911	short	raw	Right turn key release
`Key pressed: LeftArrow (turn left)`	7360	short	raw	Left turn key press
`Key released: LeftArrow (turn left)`	7356	short	raw	Left turn key release
`Player left StartSquare`	6904	short	raw	Participant exits start zone
`Player entered SAM Activator`	4584	short	raw	SAM rating zone entered
`SAM:start`	4521	short	raw	SAM rating starts

Event ID	Count (total)	Type	Meta Level	Meaning
SAM:completed	4517	short	raw	SAM rating completed
Player left SAM Activator	4567	short	raw	SAM rating zone exited
Wall touched: trial failed!	2183	short	raw	Navigation error
Door opened	2427	short	raw	Door affordance activated
ColliderRoom1 has been entered	1528	short	raw	Room 1 entered
ColliderRoom2 has been entered	1523	short	raw	Room 2 entered

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Default Task

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004980 DOI: https://doi.org/10.18112/openneuro.ds004980.v1.0.0

Normal Task 1: Navigation Action Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: SAM events (SAM:start, SAM:completed) mark subjective rating periods - treat separately or exclude. Reset/admin events (Reset insideActivator flag, player position reseted) are admin.

Event	y	Meaning
Key pressed: UpArrow (forward)	0	Forward navigation
Key pressed: RightArrow (turn right)	1	Right turn
Key pressed: LeftArrow (turn left)	2	Left turn
Door opened	3	Door affordance activated
ColliderRoom1 has been entered	4	Room 1 entered
ColliderRoom2 has been entered	5	Room 2 entered
## Data Quality Notes		

- 30+ unique event types; focus on key presses and spatial boundary events.

- SAM rating events mark active rating periods - exclude from navigation classification.

References

- Reference: `EEG data set for an architectural affordances task`

039. openneuro_ds005345 - Le Petit Prince Multitalker: Naturalistic 7T fMRI and EEG Dataset

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005345 | Year: 2025 | URL: https://openneuro.org/datasets/ds005345
Category: Complex Applied | Subjects: 25 | Files (Completed): 28
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants engaged in naturalistic listening to excerpts of "Le Petit Prince" (The Little Prince) under single-talker and multi-talker conditions during simultaneous EEG and 7T fMRI recording. The study investigates neural processing of speech under cocktail-party-style listening conditions. Stimulus/S 11 marks single-talker condition onsets; Stimulus/S 31-S 34 mark multi-talker condition variants. Comment/actiCAP Data On is an EEG system administrative marker.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005345

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
Stimulus/S 11	~19/file	long	raw	Single-talker listening block onset
Stimulus/S 31	~6/file	long	raw	Multi-talker condition variant 31 onset
Stimulus/S 32	~6/file	long	raw	Multi-talker condition variant 32 onset
Stimulus/S 33	~6/file	long	raw	Multi-talker condition variant 33 onset
Stimulus/S 34	~6/file	long	raw	Multi-talker condition variant 34 onset
Comment/actiCAP Data On	25/file	admin	raw	EEG recording system marker (exclude)

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/S 11`, `Stimulus/S 31`, `Stimulus/S 32`, `Stimulus/S 33`, `Stimulus/S 34`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Listening Condition Classification (Single-talker vs Multi-talker Variants)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s *, tmax = 600.0 s (10 min), overlap = 0.0, target_events = [`Stimulus/S 11`, `Stimulus/S 31`, `Stimulus/S 32`, `Stimulus/S 33`, `Stimulus/S 34`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Stimulus/S 11`	0	Single-talker listening
`Stimulus/S 31`	1	Multi-talker condition 31 (female attended)
`Stimulus/S 32`	2	Multi-talker condition 32 (male attended)
`Stimulus/S 33`	3	Multi-talker condition 33 (female unattended)
`Stimulus/S 34`	4	Multi-talker condition 34 (male unattended)

- Evidence:
- DOI: 10.1101/2025.04.03.646958 - This paper introduces the LPP Multi-talker Dataset with 64-channel EEG from 26 participants listening to single-talker and multi-talker speech conditions, with epochs segmented from -500ms to 10 minutes after section onset.
- DOI: 10.1038/s41597.025.05158.7 - This paper describes the LPP Multi-talker Dataset (OpenNeuro ds005345) containing EEG recordings from 26 participants during naturalistic listening to single-talker and multi-talker speech conditions.
- Kernel Notes: tmax verified as 10 minutes (600 s) from papers; tmin set to 0.0 s as default since papers report -0.5 s relative to event onset but long-event method treats tmin=0 as block start. Exclude `Comment/actiCAP Data On` (admin marker). Uses 5-class multiclass classification. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Word-level Epoch Classification (optional extended task)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.5 s, target_events = [`Stimulus/S 11`, `Stimulus/S 31`, `Stimulus/S 32`, `Stimulus/S 33`, `Stimulus/S 34`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Stimulus/S 11`	0	Single-talker word epochs
`Stimulus/S 31`	1	Multi-talker condition 31 word epochs
`Stimulus/S 32`	2	Multi-talker condition 32 word epochs
`Stimulus/S 33`	3	Multi-talker condition 33 word epochs

| `Stimulus/S 34` | 4 | Multi-talker condition 34 word epochs |
- Evidence:
- DOI: 10.1101/2025.04.03.646958 - Paper reports additional word-level epochs from -100ms to 500ms after word offset.
- DOI: 10.1038/s41597.025.05158-7 - Paper reports word-level epochs tmin=-0.1s tmax=0.5s.
- Kernel Notes: Word-level epochs require alignment with word offset markers (not provided in raw events-may require external word timing data). Optional extended analysis task.

Data Quality Notes

- Very small dataset (28 files, 25 subjects).
- `Comment/actiCAP Data On` is an admin system marker - exclude from epoching.
- Simultaneous 7T fMRI - MR gradient artifacts may require removal if present.

References

- Reference: `Le Petit Prince Multitalker: Naturalistic 7T fMRI and EEG Dataset`

040. openneuro_ds005416 - Fatigue Characterization of EEG under Mixed Reality Stereo Vision

Source: analysis.md

Dataset: `openneuro_ds005416` | Year: 2024 | URL: https://openneuro.org/datasets/ds005416
Category: Complex Applied | Subjects: 23 | Files (Completed): 23 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed resting and movement scenes in a Mixed Reality (MR) stereo vision environment to characterize EEG-based fatigue during periodic visual depth and velocity changes. The study investigates neural correlates of visual fatigue induced by MR headset viewing under different dynamic visual conditions. No trial events were captured in events.tsv; data is continuous.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005416

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(no events)	-	-	raw	No event markers detected in events.tsv

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Watching Task

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds005416 DOI: https://doi.org/10.18112/openneuro.ds005416.v1.0.1

Normal Task 1: Continuous Segmentation (segmentation)

- Task Type: regression
- Epoch Method: `segmentation`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`

- Target Definition: None
- Notes: No event markers in events.tsv. EEG is continuous during MR viewing task. Use sliding-window segmentation. Condition labels (fatigue level, visual condition) may be available from session metadata.

Data Quality Notes

- No event markers; continuous EEG during MR task.
- High sampling rate (1000 Hz) adequate for spectral fatigue analysis.
- MR headset may introduce electrical interference artifacts.

References

- Reference: `Fatigue Characterization of EEG under Mixed Reality Stereo Vision`

041. openneuro_ds005505 - Healthy Brain Network (HBN) EEG - Release 1

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005505 | Year: 2025 | URL: https://openneuro.org/datasets/ds005505
Category: Complex Applied | Subjects: 136 | Files (Completed): 1342 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 1 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection (contrastTrial), surrounding suppression, sequence learning (seqLearning), and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005505

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
stim_ON	13278	short	raw	Contrast detection stimulus onset
fixpoint_ON	13248	short	raw	Fixation point onset (trial start cue)
contrastTrial_start	8884	long	raw	Contrast detection trial block onset
left_buttonPress	5095	short	raw	Participant button press - left hand
right_buttonPress	4634	short	raw	Participant button press - right hand
right_target	4251	short	raw	Right-side target stimulus
left_target	4243	short	raw	Left-side target stimulus
break cnt	1488	admin	raw	Break/counter marker between blocks
trialResponse	1247	short	raw	Trial response event
instructed_toOpenEyes	816	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
video_start	761	long	raw	Movie clip onset
boundary	728	admin	raw	Segment boundary marker
video_stop	723	long	raw	Movie clip offset
instructed_toCloseEyes	680	long	raw	Instruction to close eyes (resting block)
resting_start	136	long	raw	Resting-state block onset
seqLearning_start	122	long	raw	Sequence learning block onset
learningBlock_1	122	long	raw	Sequence learning block 1 onset
9999	1136	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: EEG Functional Connectivity to Phenotypic Variables Regression

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.0
- Label Route: misc
- Meta Level: super
- Target Definition:

Label	Type	Meaning
age	continuous	participant age in years
sex	binary	participant biological sex (M/F)
IQ	continuous	WISC-derived IQ score
ADHD_SWAN	continuous	SWAN ADHD score
CBCL_externalizing	continuous	Child Behavior Checklist externalizing score
SES	continuous	socioeconomic status (Barratt)
anxiety	continuous	SCARED anxiety score

- Super Meta Notes: Derived from raw events [`resting_start`, `video_start`, `video_stop`, `contrastTrial_start`] by segmenting continuous EEG into 2-second non-overlapping windows, computing imaginary coherence (iCOH) connectivity matrices per window, then averaging across task blocks (resting: ~300s, video: 170-230s per clip). Kernel step: For each subject, extract all 2s epochs from resting and video periods, compute iCOH between all channel pairs, vectorize upper triangle as features, then train regression models predicting each phenotypic variable independently.
- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - Shows EEG functional connectivity (imaginary coherence) from HBN resting state and movie viewing predicts ten phenotypic/demographic measures including age, sex, IQ, ADHD, CBCL scores, and SES.
- Kernel Notes: Must join EEG data with phenotypic assessment files (WISC, SWAN, CBCL, Barratt, SCARED) by subject ID. Use subject-wise cross-validation splits to prevent leakage. iCOH computation requires Hilbert transform on bandpass-filtered data (alpha: 8-13 Hz, beta: 13-30 Hz *). (replaces Normal Task 4 from original analysis.md)

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: https://openneuro.org/datasets/ds005505/versions/1.0.1
- EEG2025 challenge portal: https://eeg2025.github.io/
- EEGUnity kernel tutorial (rich/super meta abstraction): https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html
- EEGUnity batch API (epoch_by_event_hdf5, epoch_by_long_event_hdf5): https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html

042. openneuro_ds005506 - Healthy Brain Network (HBN) EEG - Release 2

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005506 | Year: 2025 | URL: https://openneuro.org/datasets/ds005506
Category: Complex Applied | Subjects: 152 | Files (Completed): 1405 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 2 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005506

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
stim_ON	14116	short	raw	Contrast detection stimulus onset
fixpoint_ON	14081	short	raw	Fixation point onset (trial start cue)
contrastTrial_start	9310	long	raw	Contrast detection trial block onset
left_buttonPress	5762	short	raw	Participant button press - left hand
right_buttonPress	4970	short	raw	Participant button press - right hand
right_target	4463	short	raw	Right-side target stimulus
left_target	4439	short	raw	Left-side target stimulus
break cnt	1577	admin	raw	Break/counter marker between blocks
trialResponse	1372	short	raw	Trial response event
instructed_toOpenEyes	883	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
boundary	799	admin	raw	Segment boundary marker
video_start	797	long	raw	Movie clip onset
video_stop	765	long	raw	Movie clip offset
instructed_toCloseEyes	736	long	raw	Instruction to close eyes (resting block)
resting_start	150	long	raw	Resting-state block onset
9999	1150	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Super-meta phenotypic regression from resting-state connectivity

- Task Type: regression
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 300 s (full resting block), overlap = 0.0, target_events = [resting_start]
- Label Route: misc
- Meta Level: super
- Target Definition:
Event	y	Meaning
resting_start	continuous	Phenotypic variable: age (years), sex (binary), IQ (WISC), CBCL score, SWAN ADHD, SCARED anxiety, MFQ, IAT, SDS, DTS
- Super Meta Notes: Derived from behavioral assessments administered separately from EEG (WISC, CBCL, Barratt SES, SWAN, SCARED, MFQ, IAT Parent, SDS, DTS). Kernel step: Extract 2s epochs from resting-state segments, compute functional connectivity (iCOH), then regress phenotypic variables using MDMR.
- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - Uses HBN EEG functional connectivity to regress 12 phenotypic/demographic variables (age, sex, IQ, ADHD, anxiety, SES, CBCL) via MDMR
- Kernel Notes: Compute imaginary coherence (iCOH) in delta/theta/alpha/beta bands per epoch; join EEG-derived connectivity features with phenotypic CSV by subject ID using subject-wise splits to prevent

data leakage.

Primary Task 2: Super-meta phenotypic regression from movie-watching connectivity

- Task Type: regression
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 117-600 s (full video segment), overlap = 0.0, target_events = [`video_start`]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`video_start`	continuous	Phenotypic variable: age (years), sex (binary), IQ (WISC), CBCL score, SWAN ADHD, SCARED anxiety, MFQ, IAT, SDS, DTS

- Super Meta Notes: Derived from behavioral assessments administered separately from EEG (WISC, CBCL, Barratt SES, SWAN, SCARED, MFQ, IAT Parent, SDS, DTS). Kernel step: Extract 2s epochs from video segments, compute functional connectivity, then regress phenotypic variables using MDMR.
- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - Uses HBN EEG functional connectivity (including video task segments) to regress phenotypic variables
- Kernel Notes: Use video_start-to-video_stop annotation for segment boundaries; compute iCOH per 2s epoch; regress phenotypic targets with proper train/test splits by subject.

Primary Task 3: Contrast detection trial classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`left_target`, `right_target`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`left_target`	0	Target presented on left side
`right_target`	1	Target presented on right side

- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - Confirms HBN dataset includes contrast detection/sensory paradigms
- Kernel Notes: Use stim_ON or fixpoint_ON as trial onset cue; align left_target/right_target to stimulus events; exclude boundary and 9999 admin markers.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005506/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

043. openneuro_ds005507 - Healthy Brain Network (HBN) EEG - Release 3

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005507` | Year: 2025 | URL: https://openneuro.org/datasets/ds005507
Category: Complex Applied | Subjects: 183 | Files (Completed): 1812 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 3 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005507

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stim_ON`	18602	short	raw	Contrast detection stimulus onset
`fixpoint_ON`	18564	short	raw	Fixation point onset (trial start cue)
`contrastTrial_start`	12072	long	raw	Contrast detection trial block onset
`left_buttonPress`	6995	short	raw	Participant button press - left hand
`right_buttonPress`	6266	short	raw	Participant button press - right hand
`right_target`	5800	short	raw	Right-side target stimulus
`left_target`	5753	short	raw	Left-side target stimulus
`break cnt`	1973	admin	raw	Break/counter marker between blocks
`trialResponse`	1757	short	raw	Trial response event
`instructed_toOpenEyes`	1101	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
boundary	1074	admin	raw	Segment boundary marker
video_start	1041	long	raw	Movie clip onset
video_stop	1013	long	raw	Movie clip offset
instructed_toCloseEyes	915	long	raw	Instruction to close eyes (resting block)
resting_start	184	long	raw	Resting-state block onset
9999	1446	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Functional Connectivity to Phenotypic Variables

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.0
- Label Route: misc
- Meta Level: super
- Target Definition:

Event	y	Meaning
resting_start	continuous	FC computed during resting-state blocks (~300s per subject)
video_start	continuous	FC computed during movie viewing blocks (~117-230s per clip)
stim_ON	continuous	FC computed during visual flicker/contrast task (~210s)

- Super Meta Notes: Derived from raw events [resting_start, video_start, video_stop, stim_ON] by computing imaginary coherence (iCOH) on 2s epochs, then regressing FC features against phenotypic variables (age, sex, IQ, ADHD, anxiety, SES, CBCL). Kernel step: Extract 2s segments from each state, compute iCOH connectivity matrix per segment, aggregate by subject, then perform regression against phenotypic targets.
- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - EEG functional connectivity (imaginary coherence) from HBN Biobank associated with 10 phenotypic and 2 demographic variables (age, sex, IQ, ADHD, anxiety, SES,

behavioral problems)
- Kernel Notes: Parse phenotypic CSV for subject IDs, join with EEG FC features by subject, handle missing values, use cross-validation splits that respect subject boundaries to prevent leakage.

Primary Task 2: Sequence Learning State Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s , tmax = 0.8 s , target_events = [`stim_ON`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`stim_ON`	0	still-unknown: sequence not yet recognized by participant
`stim_ON`	1	newly-learned: sequence just recognized in recent recall
`stim_ON`	2	known: sequence fully learned and consolidated

- Super Meta Notes: Derived from raw event [`stim_ON`] by mapping trial sequences to learning states based on post-block recall reports. Kernel step: Group stimulus events by sequence block, lookup recall performance per block, assign learning state labels (0/1/2), epoch around each stimulus onset.
- Evidence:
- DOI: 10.1038/sdata.2017.40 - HBN EEG dataset with sequence learning paradigm where trials categorized as 'still-unknown', 'newly-learned', or 'known' based on participants' recall reports
- Kernel Notes: Must access recall performance data (external behavioral file), map blocks to learning states, handle class imbalance (most trials are 'still-unknown'). Verify event ID mapping from events.tsv.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`

- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005507/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

044. openneuro_ds005508 - Healthy Brain Network (HBN) EEG - Release 4

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005508 | Year: 2025 | URL: https://openneuro.org/datasets/ds005508
Category: Complex Applied | Subjects: 324 | Files (Completed): 3342 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 4 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005508

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
stim_ON	35171	short	raw	Contrast detection stimulus onset
fixpoint_ON	35072	short	raw	Fixation point onset (trial start cue)
contrastTrial_start	20646	long	raw	Contrast detection trial block onset
left_buttonPress	11751	short	raw	Participant button press - left hand
right_buttonPress	10656	short	raw	Participant button press - right hand
right_target	9909	short	raw	Right-side target stimulus
left_target	9806	short	raw	Left-side target stimulus
break cnt	3452	admin	raw	Break/counter marker between blocks
trialResponse	3286	short	raw	Trial response event
instructed_toOpenEyes	1929	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
`boundary`	1871	admin	raw	Segment boundary marker
`instructed_toCloseEyes`	1608	long	raw	Instruction to close eyes (resting block)
`resting_start`	322	long	raw	Resting-state block onset
`9999`	2502	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `externalizing_factor`, `p_factor`, `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stim_ON`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Resting State Classification (Eyes-Open vs Eyes-Closed)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 20.0 s (eyes-open) / 40.0 s (eyes-closed) *, overlap = 0.0, target_events = [`instructed_toOpenEyes`, `instructed_toCloseEyes`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`instructed_toOpenEyes`	0	Eyes-open resting state block
`instructed_toCloseEyes`	1	Eyes-closed resting state block

- Evidence:
- DOI: `10.1038/sdata.2017.40` - Paper describes resting-state paradigm with 20s eyes-open and 40s eyes-closed segments collected using HBN EEG dataset (ds005508)
- Kernel Notes: Extract sustained annotations for eyes-open/closed instructions. Use 20s windows for eyes-open and 40s for eyes-closed to match paradigm specifications.

Primary Task 2: Contrast Detection - Target vs Non-Target Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 1.0 s, target_events = [`left_target`, `right_target`, `stim_ON`]
- Label Route: `Annotation`

- Meta Level: raw
- Target Definition:
Event	y	Meaning
`left_target`	0	Left-side target stimulus
`right_target`	1	Right-side target stimulus
`stim_ON`	-	Non-target stimulus (excluded from classification)
- Evidence:
- DOI: `10.1038/sdata.2017.40` - Paper describes contrast detection paradigm with left/right target stimuli and stimulus onsets collected in HBN EEG dataset
- Kernel Notes: Epoch around target stimuli for left vs right classification. Use 500ms pre-target to 1000ms post-peak sensory evidence window as per paper. Filter out non-target stim_ON events from training set.

Primary Task 3: Sequence Learning - Learning State Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [`fixpoint_ON`, `stim_ON`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`fixpoint_ON` (derived: still_unknown)	0	Sequence elements not yet learned by participant
`fixpoint_ON` (derived: newly_learned)	1	Recently learned sequence elements
`fixpoint_ON` (derived: known)	2	Well-known sequence elements
- Super Meta Notes: Derived from raw event `fixpoint_ON` combined with participant recall reports. Kernel step: Join trial timestamps with behavioral recall data to compute learning state labels (still-unknown, newly-learned, known) per stimulus event.
- Evidence:
- DOI: `10.1038/sdata.2017.40` - Paper describes sequence learning paradigm with learning state categories (still-unknown, newly-learned, known) derived from participant recall reports in HBN EEG dataset
- Kernel Notes: Requires joining EEG trial events with external behavioral recall data to derive learning state labels. Use 900ms epochs (100ms pre to 800ms post-stimulus) per paper specification.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`

- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005508/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

045. openneuro_ds005509 - Healthy Brain Network (HBN) EEG - Release 5

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005509 | Year: 2025 | URL: https://openneuro.org/datasets/ds005509
Category: Complex Applied | Subjects: 330 | Files (Completed): 3326 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 5 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005509

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
stim_ON	35156	short	raw	Contrast detection stimulus onset
fixpoint_ON	35136	short	raw	Fixation point onset (trial start cue)
contrastTrial_start	18682	long	raw	Contrast detection trial block onset
left_buttonPress	10192	short	raw	Participant button press - left hand
right_buttonPress	9498	short	raw	Participant button press - right hand
right_target	8977	short	raw	Right-side target stimulus
left_target	8947	short	raw	Left-side target stimulus
break cnt	3296	admin	raw	Break/counter marker between blocks
trialResponse	3367	short	raw	Trial response event
instructed_toOpenEyes	1969	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
boundary	1839	admin	raw	Segment boundary marker
instructed_toCloseEyes	1639	long	raw	Instruction to close eyes (resting block)
resting_start	329	long	raw	Resting-state block onset
9999	2339	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Phenotype Prediction from Resting-State EEG Connectivity

- Task Type: regression
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 2 s, overlap = 0.0, target_events = [resting_start]
- Label Route: misc
- Meta Level: super
- Target Definition:

Event	y	Meaning
(subject)	continuous	age (years)
(subject)	continuous	sex (M/F)
(subject)	continuous	IQ (WISC score)
(subject)	continuous	ADHD (SWAN score)
(subject)	continuous	SES (Barratt score)
(subject)	continuous	anxiety (SCARED score)
(subject)	continuous	CBCL score

- Super Meta Notes: Derived from raw event(s) [resting_start] by computing 2-second epoch iCOH (imaginary coherence) connectivity matrices, then training multivariate regression (MDMR) to predict phenotypic variables. Kernel step: Extract epochs -> compute connectivity -> join with phenotypic CSV by subject ID -> train regression model.
- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - This paper analyzes EEG functional connectivity from the HBN Biobank to identify associations between phase-coupling networks and ten phenotypic/demographic

measures (age, sex, IQ, ADHD, SES, anxiety, and CBCL) using multivariate distance matrix regression.
- Kernel Notes: Extract 2-s epochs from resting-state blocks marked by `resting_start`. Compute inter-channel coherence (iCOH) for connectivity features. Join EEG features with phenotypic data by subject ID. Use subject-wise cross-validation to prevent leakage. Also supports video viewing and contrast detection blocks as additional connectivity sources. (replaces Primary Task 1-3 from original analysis.md)

Primary Task 2: Sequence Learning State Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s *, tmax = 0.8 s, target_events = [`stim_ON`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`stim_ON`	0	still-unknown (trial not yet learned)
`stim_ON`	1	newly-learned (recalled once)
`stim_ON`	2	known (fully learned)

- Evidence:
- DOI: 10.1038/sdata.2017.40 - This paper presents an EEG dataset for assessing information processing including a sequence learning paradigm where trials were categorized as 'still-unknown', 'newly-learned', or 'known' based on participants' recall reports.
- Kernel Notes: Verify event ID mapping from events.tsv - paper uses "stimulus_onset" which maps to `stim_ON`. Epochs are 900ms total (tmax-tmin=0.9s). Labels (0/1/2) based on recall performance must be derived from behavioral logs and joined to epochs by trial ID.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.

- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage. Note: Primary Task 1 demonstrates connectivity-based phenotype prediction as an alternative feature extraction approach.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005509/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

046. openneuro_ds005510 - Healthy Brain Network (HBN) EEG - Release 6

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005510` | Year: 2025 | URL: https://openneuro.org/datasets/ds005510
Category: Complex Applied | Subjects: 3000 | Files (Completed): 1227 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 6 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005510

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stim_ON`	13706	short	raw	Contrast detection stimulus onset
`fixpoint_ON`	13696	short	raw	Fixation point onset (trial start cue)
`contrastTrial_start`	5817	long	raw	Contrast detection trial block onset
`left_buttonPress`	3743	short	raw	Participant button press - left hand
`right_buttonPress`	3249	short	raw	Participant button press - right hand
`right_target`	2796	short	raw	Right-side target stimulus
`left_target`	2793	short	raw	Left-side target stimulus
`break cnt`	1242	admin	raw	Break/counter marker between blocks
`trialResponse`	1483	short	raw	Trial response event

Event ID	Count (total)	Type	Meta Level	Meaning
instructed_toOpenEyes	810	long	raw	Instruction to open eyes (resting block)
instructed_toCloseEyes	676	long	raw	Instruction to close eyes (resting block)
boundary	666	admin	raw	Segment boundary marker
resting_start	135	long	raw	Resting-state block onset
9999	866	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Phenotypic Factor Regression from EEG Functional Connectivity

- Task Type: regression
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 2.0 s, overlap = 0.0 *, target_events = [resting_start, stim_ON]
- Label Route: misc
- Meta Level: super
- Target Definition:
Event	y	Meaning
resting_start	continuous	Age (years), sex, IQ (WISC), CBCL (behavioral problems), Barratt SES, SWAN ADHD, SCARED anxiety
stim_ON	continuous	Same phenotypic variables for video viewing epochs
- Super Meta Notes: Derived from raw events [resting_start, stim_ON] by computing imaginary coherence (iCOH) in frequency bands (delta 0.5-4Hz, theta 4-7.5Hz, alpha 7.5-15Hz, beta 15-30Hz) across 2s epochs, then regressing against phenotypic assessments (WISC IQ, CBCL, SWAN ADHD, SCARED, Barratt SES). Kernel step: Extract iCOH connectivity matrices per epoch, flatten/vectorize, join with phenotypic CSV by subject ID, train regression models with subject-wise splits.
- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - Shows EEG functional connectivity (iCOH) from HBN Biobank predicts phenotypic measures including age, sex, IQ, CBCL, SWAN ADHD, SCARED anxiety, and

socioeconomic status.
- Kernel Notes: Combine resting-state (~300s) and task video recordings (170-600s) epochs. Frequency band computation required. Join with phenotypic CSV tables using subject identifier. Use subject-wise cross-validation splits to prevent leakage. Write regression targets with unit/version metadata in misc channel.

Primary Task 2: Publisher-defined BIDS task: Despicable Me

- Task Type: segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30.0 s * (full video segment), overlap = 0.0 *, target_events = [`stim_ON`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`stim_ON`	0	Despicable Me movie viewing segment

- Evidence:
- DOI: https://doi.org/10.1101/089060 - OpenNeuro metadata indicates Despicable Me task was completed; HBN dataset includes naturalistic movie watching paradigm.
- Kernel Notes: Verify actual stimulus event IDs for movie onset from events.tsv. This task involves continuous movie viewing (170-600s) requiring segmentation into windows. (replaces Normal Task 2 partially)

Primary Task 3: Publisher-defined BIDS task: Seq Learning6target

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [`left_target`, `right_target`, `left_buttonPress`, `right_buttonPress`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`left_target`	0	Left-side target stimulus (sequence learning condition)
`right_target`	1	Right-side target stimulus (sequence learning condition)

- Evidence:
- DOI: https://doi.org/10.1101/089060 - HBN includes sequence learning tasks with 6-target and 8-target variants.
- Kernel Notes: Sequence learning involves responding to cued targets; verify exact event chain from events.tsv.

Primary Task 4: Publisher-defined BIDS task: Seq Learning8target

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [`left_target`, `right_target`, `left_buttonPress`, `right_buttonPress`]
- Label Route: `Annotation`
- Meta Level: raw

- Target Definition:
Event	y	Meaning
`left_target`	0	Left-side target stimulus (8-target sequence learning)
`right_target`	1	Right-side target stimulus (8-target sequence learning)
- Evidence:
- DOI: https://doi.org/10.1101/089060 - HBN includes sequence learning tasks with 8-target condition.
- Kernel Notes: May share event IDs with 6-target; distinguish by session/task block metadata.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005510/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`

- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

047. openneuro_ds005511 - Healthy Brain Network (HBN) EEG - Release 7

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005511 | Year: 2025 | URL: https://openneuro.org/datasets/ds005511
Category: Complex Applied | Subjects: 381 | Files (Completed): 3100 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 7 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005511

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
stim_ON	34708	short	raw	Contrast detection stimulus onset
fixpoint_ON	34688	short	raw	Fixation point onset (trial start cue)
contrastTrial_start	13272	long	raw	Contrast detection trial block onset
left_buttonPress	7530	short	raw	Participant button press - left hand
right_buttonPress	6992	short	raw	Participant button press - right hand
right_target	6384	short	raw	Right-side target stimulus
left_target	6361	short	raw	Left-side target stimulus
trialResponse	4059	short	raw	Trial response event
break cnt	3092	admin	raw	Break/counter marker between blocks
instructed_toOpenEyes	2193	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
instructed_toCloseEyes	1825	long	raw	Instruction to close eyes (resting block)
boundary	1728	admin	raw	Segment boundary marker
resting_start	371	long	raw	Resting-state block onset
9999	2143	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Resting-state EEG-based DSM-V psychiatric diagnosis classification

- Task Type: classification (multi-label)
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2 s, overlap = 0.0 *
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
ADHD	0/1	Attention-Deficit/Hyperactivity Disorder diagnosis
ANX	0/1	Anxiety disorder diagnosis
SLD	0/1	Specific Learning Disorder diagnosis
ASD	0/1	Autism Spectrum Disorder diagnosis
DISR	0/1	Disruptive behavior disorder diagnosis
COM	0/1	Communication disorder diagnosis
DEP	0/1	Depressive disorder diagnosis
Other	0/1	Other DSM-V disorder diagnosis
NoDiagnosis	0/1	No diagnosis

- Super Meta Notes: Derived from raw resting-state EEG segments (eyes-closed blocks marked by instructed_toCloseEyes and resting_start). Kernel step: Extract 2s epochs from concatenated resting-state blocks, apply amplitude threshold rejection (±90μV), compute spectral features, and predict multi-label DSM-V diagnoses from clinical assessment data.
- Evidence:

- DOI: 10.1016/j.neuroimage.2022.119348 - Multi-task multi-label classification of 8 DSM-V categories from resting EEG spectral features
- Kernel Notes: Use `epoch_by_segmentation_hdf5` on continuous EEG. Source resting-state blocks via `instructed_toCloseEyes` annotation boundaries. Join EEG epochs with phenotypic/diagnosis labels by subject ID. Subject-wise train/test splits required to prevent data leakage.

Primary Task 2: Resting-state EEG-based symptom severity regression

- Task Type: regression (multi-target)
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2 s, overlap = 0.0 *
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`SCARED_GAD`	continuous	Screen for Child Anxiety Related Disorders - GAD subscale
`SWAN_IN`	continuous	Sustained Attention (Inattention) subscale
`SWAN_HY`	continuous	Hyperactivity-Impulsivity subscale
`SRS_2_SCI`	continuous	Social Responsiveness Scale-2 - Social Communication Index
`SRS_2_RRB`	continuous	Social Responsiveness Scale-2 - Restricted/Repetitive Behaviors
`WISC_VCI`	continuous	Wechsler Intelligence Scale for Children - Verbal Comprehension Index
`WISC_WMI`	continuous	Wechsler Intelligence Scale for Children - Working Memory Index

- Super Meta Notes: Derived from raw resting-state EEG segments (eyes-closed blocks marked by `instructed_toCloseEyes` and `resting_start`). Kernel step: Extract 2s epochs, compute spectral features, predict 7 continuous symptom severity measures from standardized behavioral assessments.
- Evidence:
- DOI: 10.1016/j.neuroimage.2022.119348 - Multi-target regression of 7 continuous symptom severity measures from resting EEG
- Kernel Notes: Use `epoch_by_segmentation_hdf5` on continuous EEG. Source resting-state blocks via `instructed_toCloseEyes` annotation boundaries. Join EEG epochs with phenotypic assessment scores by subject ID. Subject-wise train/test splits required. Store regression targets with unit/version metadata (e.g., T-score, raw score version).

Primary Task 3: Contrast detection task - stimulus response classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [`left_target`, `right_target`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`left_target`	0	Left-side target stimulus presentation
`right_target`	1	Right-side target stimulus presentation

- Evidence:
- DOI: 10.1016/j.neuroimage.2022.119348 - HBN dataset used for neurodevelopmental research including contrast detection paradigms
- Kernel Notes: Epoch around target onset events. Exclude admin markers (`boundary`, `9999`, `break cnt`) from training data. Verify event ID mapping from events.tsv if needed.

Primary Task 4: Resting-state block classification (eyes-open vs eyes-closed)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s , overlap = 0.0 , target_events = [`instructed_toOpenEyes`, `instructed_toCloseEyes`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`instructed_toOpenEyes`	0	Eyes-open resting state block
`instructed_toCloseEyes`	1	Eyes-closed resting state block

- Evidence:
- DOI: 10.1016/j.neuroimage.2022.119348 - Resting-state EEG blocks (eyes-open and eyes-closed) are part of HBN paradigm
- Kernel Notes: Use long-event epochs to capture full resting-state blocks. Overlap=0 to avoid duplicate samples from same block.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.

- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005511/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

048. openneuro_ds005512 - Healthy Brain Network (HBN) EEG - Release 8

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005512 | Year: 2025 | URL: https://openneuro.org/datasets/ds005512
Category: Complex Applied | Subjects: 257 | Files (Completed): 2320 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 8 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005512

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
stim_ON	22981	short	raw	Contrast detection stimulus onset
fixpoint_ON	22913	short	raw	Fixation point onset (trial start cue)
contrastTrial_start	12963	long	raw	Contrast detection trial block onset
left_buttonPress	6946	short	raw	Participant button press - left hand
right_buttonPress	6875	short	raw	Participant button press - right hand
right_target	6231	short	raw	Right-side target stimulus
left_target	6151	short	raw	Left-side target stimulus
trialResponse	2318	short	raw	Trial response event
break cnt	2258	admin	raw	Break/counter marker between blocks
instructed_toOpenEyes	1326	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
`boundary`	1237	admin	raw	Segment boundary marker
`instructed_toCloseEyes`	1104	long	raw	Instruction to close eyes (resting block)
`resting_start`	223	long	raw	Resting-state block onset
`9999`	1751	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `externalizing_factor`, `p_factor`, `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stim_ON`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Resting EEG Psychiatric Diagnosis Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 40 s, overlap = 0.0, target_events = [`instructed_toCloseEyes`]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
ADHD	0/1	Attention-deficit/hyperactivity disorder diagnosis
ANX	0/1	Anxiety disorder diagnosis
SLD	0/1	Specific learning disorder diagnosis
ASD	0/1	Autism spectrum disorder diagnosis
DISR	0/1	Disruptive disorder diagnosis
COM	0/1	Communication disorder diagnosis
DEP	0/1	Depressive disorder diagnosis
Other	0/1	Other DSM-V disorder diagnosis

- Super Meta Notes: Derived from clinical diagnoses (DSM-V) by licensed clinicians. Kernel step: Join EEG epochs (from `instructed_toCloseEyes` resting blocks) with phenotypic diagnosis tables using subject ID to obtain 8 binary labels per epoch. Use 2-second sub-epochs within each 40s resting block for feature extraction.
- Evidence:

- DOI: 10.1016/j.neuroimage.2022.119348 - Multi-task multi-label classification of 8 DSM-V psychiatric diagnoses from resting EEG (eyes-closed blocks)
- Kernel Notes: Extract 2s sub-epochs from concatenated eyes-closed resting data (1-40 Hz frequency range). Labels are multi-hot (subjects can have multiple diagnoses). Requires phenotypic table join by subject key. Use subject-wise train/test splits to prevent label leakage.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: https://openneuro.org/datasets/ds005512/versions/1.0.1
- EEG2025 challenge portal: https://eeg2025.github.io/
- EEGUnity kernel tutorial (rich/super meta abstraction): https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html
- EEGUnity batch API (epoch_by_event_hdf5, epoch_by_long_event_hdf5): https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html

049. openneuro_ds005514 - Healthy Brain Network (HBN) EEG - Release 9

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005514` | Year: 2025 | URL: https://openneuro.org/datasets/ds005514
Category: Complex Applied | Subjects: 295 | Files (Completed): 2885 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 9 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005514

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stim_ON`	30059	short	raw	Contrast detection stimulus onset
`fixpoint_ON`	30018	short	raw	Fixation point onset (trial start cue)
`contrastTrial_start`	16449	long	raw	Contrast detection trial block onset
`left_buttonPress`	10593	short	raw	Participant button press - left hand
`right_buttonPress`	10002	short	raw	Participant button press - right hand
`right_target`	7910	short	raw	Right-side target stimulus
`left_target`	7857	short	raw	Left-side target stimulus
`break cnt`	2807	admin	raw	Break/counter marker between blocks
`trialResponse`	2569	short	raw	Trial response event
`instructed_toOpenEyes`	1591	long	raw	Instruction to open eyes (resting block)

Event ID	Count (total)	Type	Meta Level	Meaning
boundary	1554	admin	raw	Segment boundary marker
instructed_toCloseEyes	1325	long	raw	Instruction to close eyes (resting block)
resting_start	269	long	raw	Resting-state block onset
9999	2155	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Psychiatric Diagnosis Classification from Resting-State EEG

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2 s, overlap = 0.0 *
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
ADHD	0/1	Attention-Deficit/Hyperactivity Disorder diagnosis
ANX	0/1	Anxiety disorder diagnosis
SLD	0/1	Specific Learning Disorder diagnosis
ASD	0/1	Autism Spectrum Disorder diagnosis
DISR	0/1	Disruptive disorders diagnosis
COM	0/1	Communication disorders diagnosis
DEP	0/1	Depressive disorders diagnosis
Other	0/1	Other psychiatric diagnosis

- Super Meta Notes: Derived from clinical DSM-V diagnoses made by licensed clinicians. Kernel step: Filter epochs from instructed_toCloseEyes resting-state blocks (eyes-closed), then join with phenotypic diagnosis tables by subject ID.
- Evidence:
- DOI: 10.1016/j.neuroimage.2022.119348 - Multi-label classification of 8 DSM-V diagnoses using resting-state EEG from HBN pediatric sample

- Kernel Notes: Use `instructed_toCloseEyes` long-event annotations to define resting-state windows; extract 2-second non-overlapping segments; exclude subjects without clinical diagnoses; apply subject-wise train/test splits to prevent leakage.

Primary Task 2: Dimensional Symptom Severity Regression from Resting-State EEG

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2 s, overlap = 0.0 *
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`SCARED GAD`	continuous	SCARED Generalized Anxiety score (age/gender adjusted)
`SWAN IN`	continuous	SWAN Inattention score (age/gender adjusted)
`SWAN HY`	continuous	SWAN Hyperactivity score (age/gender adjusted)
`SRS-2 SCI`	continuous	SRS-2 Social Communication Index (age/gender adjusted)
`SRS-2 RRB`	continuous	SRS-2 Restricted Interests/Repetitive Behavior (age/gender adjusted)
`WISC VCI`	continuous	WISC Verbal Comprehension Index
`WMI`	continuous	WISC Working Memory Index

- Super Meta Notes: Derived from standardized behavioral/questionnaire assessments (SCARED, SWAN, SRS-2, WISC). Kernel step: Filter epochs from `instructed_toCloseEyes` resting-state blocks, then join with phenotypic continuous measure tables by subject ID.
- Evidence:
- DOI: 10.1016/j.neuroimage.2022.119348 - Multi-target regression of 7 continuous dimensional measures using resting-state EEG from HBN pediatric sample
- Kernel Notes: Use `instructed_toCloseEyes` long-event annotations to define resting-state windows; extract 2-second non-overlapping segments; standardize regression targets using age/gender norms; apply subject-wise train/test splits to prevent leakage.

Primary Task 3: Contrast Detection Target Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s , tmax = 1.0 s , target_events = [`left_target`, `right_target`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`left_target`	0	Left-side target stimulus presented
`right_target`	1	Right-side target stimulus presented

- Evidence:
- DOI: 10.1038/sdata.2017.40 - HBN EEG dataset with six paradigms including contrast change detection; left vs right target classification
- Kernel Notes: Epoch around target stimulus onsets; exclude trials with missing button press responses; may compute accuracy from `left_buttonPress`/`right_buttonPress` for trial-level labels.

Primary Task 4: Sequence Learning Stage Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s , tmax = 0.8 s , target_events = [`stim_ON`, `fixpoint_ON`]

- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`stim_ON`	see note	Contrast detection stimulus onset (sequence learning trials)
`fixpoint_ON`	see note	Fixation point onset (trial cue)
- Evidence:
- DOI: 10.1038/sdata.2017.40 - HBN EEG dataset with six paradigms including sequence learning; learning stages (unknown/newly-learned/known) derived from behavioral performance
- Kernel Notes: Requires derivation of learning stage labels from trial-by-trial behavioral performance; kernel must compute learning stage from response accuracy and timing, then assign to corresponding epochs. Verify event ID mapping from events.tsv.

Primary Task 5: Resting-State Eyes Condition Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s *, overlap = 0.0, target_events = [`instructed_toOpenEyes`, `instructed_toCloseEyes`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`instructed_toOpenEyes`	0	Eyes-open resting condition
`instructed_toCloseEyes`	1	Eyes-closed resting condition
- Evidence:
- DOI: 10.1038/sdata.2017.40 - HBN EEG dataset with resting-state paradigm; 20s eyes-open, 40s eyes-closed blocks
- Kernel Notes: Extract full-duration epochs from each resting instruction block; use 30-second windows as per default sleep staging parameters.

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005514/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

050. openneuro_ds005515 - Healthy Brain Network (HBN) EEG - Release 10

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005515` | Year: 2025 | URL: https://openneuro.org/datasets/ds005515
Category: Complex Applied | Subjects: 533 | Files (Completed): 2515 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 10 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, symbol search, and sequence learning tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005515

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stim_ON`	10818	short	raw	Contrast detection stimulus onset
`fixpoint_ON`	10562	short	raw	Fixation point onset (trial start cue)
`contrastTrial_start`	5858	long	raw	Contrast detection trial block onset
`left_buttonPress`	3689	short	raw	Participant button press - left hand
`right_buttonPress`	3034	short	raw	Participant button press - right hand
`instructed_toOpenEyes`	2862	long	raw	Instruction to open eyes (resting block)
`right_target`	2812	short	raw	Right-side target stimulus
`left_target`	2545	short	raw	Left-side target stimulus
`instructed_toCloseEyes`	2365	long	raw	Instruction to close eyes (resting block)
`video_start`	1627	long	raw	Movie clip onset

Event ID	Count (total)	Type	Meta Level	Meaning
video_stop	1610	long	raw	Movie clip offset
trialResponse	1211	short	raw	Trial response event
boundary	1385	admin	raw	Segment boundary marker
resting_start	516	long	raw	Resting-state block onset
9999	2105	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Phenotypic variable prediction from EEG functional connectivity

- Task Type: regression
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 2.0 s *, overlap = 0.0, target_events = [resting_start, instructed_toOpenEyes, instructed_toCloseEyes, video_start]
- Label Route: misc
- Meta Level: super
- Target Definition:

Event	y	Meaning
phenotypic_target	continuous	Phenotypic variable (age, sex, IQ, ADHD-SWAN, CBCL, SCARED anxiety, Barratt SES)

- Super Meta Notes: Derived from phenotypic assessment instruments (WISC IQ, CBCL, SWAN ADHD, SCARED, Barratt SES) administered to participants, not from raw EEG event markers. Kernel step: Extract iCOH (imaginary coherence) functional connectivity features from concatenated resting state and task EEG epochs, then perform multivariate distance matrix regression (MDMR) against phenotypic measures.
- Evidence:
- DOI: 10.1016/j.neuroimage.2020.117001 - Paper analyzes HBN Biobank (1657 subjects), computing source-localized functional connectivity using iCOH from concatenated resting state and task data, and uses MDMR to associate connectivity patterns with ten phenotypic assessment instruments plus age and sex.

- Kernel Notes: Parse phenotypic CSV files, join with EEG by subject ID, extract 2s iCOH epochs from resting and video segments, compute connectivity matrices, run MDMR or similar regression. Use subject-wise splits to prevent leakage.

Primary Task 2: Contrast detection target classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s *, tmax = 1.0 s, target_events = [`left_target`, `right_target`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`left_target`	0	Left-sided target stimulus in contrast detection
`right_target`	1	Right-sided target stimulus in contrast detection

- Evidence:
- DOI: 10.1038/sdata.2017.40 - Paper describes contrast detection paradigm with epoch extraction -0.5s to 1.0s post-target, analyzing CPP, P300, and behavioral measures.
- Kernel Notes: Epoch around left_target and right_target events. Verify mapping: paper describes "left_tilted increase" vs "right_tilted increase" mapping to left_target/right_target raw IDs. Exclude admin markers.

Primary Task 3: Resting state EEG classification (eyes-open vs eyes-closed)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s *, overlap = 0.0, target_events = [`instructed_toOpenEyes`, `instructed_toCloseEyes`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`instructed_toOpenEyes`	0	Eyes-open resting state block
`instructed_toCloseEyes`	1	Eyes-closed resting state block

- Evidence:
- DOI: 10.1038/sdata.2017.40 - Paper describes resting-state paradigm with eyes-open/eyes-closed conditions as part of the six-task battery.
- Kernel Notes: Use 30s non-overlapping windows from long instructed_toOpenEyes and instructed_toCloseEyes annotations. (replaces Normal Task 2 partially from original analysis.md)

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005515/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

051. openneuro_ds005516 - Healthy Brain Network (HBN) EEG - Release 11

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005516` | Year: 2025 | URL: https://openneuro.org/datasets/ds005516
Category: Complex Applied | Subjects: 3000 | Files (Completed): 3396 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 11 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, symbol search, and sequence learning tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, externalizing, phenotypic, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005516

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stim_ON`	37825	short	raw	Contrast detection stimulus onset
`fixpoint_ON`	37760	short	raw	Fixation point onset (trial start cue)
`contrastTrial_start`	19366	long	raw	Contrast detection trial block onset
`left_buttonPress`	14777	short	raw	Participant button press - left hand
`right_buttonPress`	13758	short	raw	Participant button press - right hand
`right_target`	9311	short	raw	Right-side target stimulus
`left_target`	9226	short	raw	Left-side target stimulus
`trialResponse`	4056	short	raw	Trial response event
`break cnt`	3196	admin	raw	Break/counter marker between blocks
`video_start`	1666	long	raw	Movie clip onset

Event ID	Count (total)	Type	Meta Level	Meaning
instructed_toOpenEyes	1644	long	raw	Instruction to open eyes (resting block)
video_stop	1541	long	raw	Movie clip offset
boundary	1462	admin	raw	Segment boundary marker
instructed_toCloseEyes	1370	long	raw	Instruction to close eyes (resting block)
resting_start	372	long	raw	Resting-state block onset
9999	3038	admin	raw	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: externalizing_factor, p_factor, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_ON
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Phenotypic Prediction from EEG Functional Connectivity

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5, target_events = [video_start, video_stop, resting_start, instructed_toOpenEyes, instructed_toCloseEyes]
- Label Route: misc
- Meta Level: super
- Target Definition:

Event	y	Meaning
age	float	Participant age in years
sex	0/1	Biological sex (binary)
iq	float	WISC full-scale IQ

- Super Meta Notes: Derived from behavioral assessments (WISC IQ, CBCL, SWAN, Barratt SES, SCARED) administered to participants, not from raw EEG event markers. Kernel step: Compute iCOH (imaginary coherence) functional connectivity per 2s segment, then aggregate across all segments per subject to create FC matrices. Use FC matrices as features to predict phenotypic variables via regression.
- Evidence:

- DOI: 10.1016/j.neuroimage.2020.117001 - EEG functional connectivity (iCOH) from HBN Biobank significantly predicts age, sex, IQ and other phenotypic variables across frequency bands.
- Kernel Notes: (replaces Primary Tasks 1-3 from original analysis.md)

Normal Task 1: Raw-meta event decoding and classification (event-level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use non-admin events such as `left_target`, `right_target`, `trialResponse`, `stim_ON`.
- Kernel Notes: Exclude admin markers from supervised target sets.

Normal Task 2: Raw-meta state decoding (long-event)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use state/block markers such as `instructed_toOpenEyes`, `instructed_toCloseEyes`, `resting_start`, `video_start`, `video_stop`.
- Kernel Notes: Keep block-label mapping stable across releases.

Normal Task 3: Super-meta response-time regression (competition-style)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: event-centered windows (task-dependent), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `response_time`
- Kernel Notes: Parse trial chains and write regression targets with unit/version metadata.

Normal Task 4: Super-meta psychopathology-factor modeling

- Task Type: regression (or discretized classification baseline)
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target: `p_factor` / `externalizing_factor` (schema dependent)
- Kernel Notes: Join EEG with phenotypic tables by subject key and use subject-wise splits to prevent leakage.

Data Quality Notes

- HBN is a heterogeneous multi-task battery; task availability differs across participants and files.
- Event naming and density are not perfectly uniform across releases; mapping tables should be explicit.
- Admin markers (`boundary`, `9999`, etc.) are frequent and must be filtered before model target construction.
- Super-meta tasks are sensitive to join quality (subject ID harmonization, missingness, leakage control).
- Report clearly whether each experiment uses raw meta labels or super meta labels.

References

- OpenNeuro dataset page: `https://openneuro.org/datasets/ds005516/versions/1.0.1`
- EEG2025 challenge portal: `https://eeg2025.github.io/`
- EEGUnity kernel tutorial (rich/super meta abstraction): `https://eegunity.readthedocs.io/en/v0.7.0/tutorial/How%20to%20Reading%20Rich%20Meta%20Data%20via%20Kernel.html`
- EEGUnity batch API (`epoch_by_event_hdf5`, `epoch_by_long_event_hdf5`): `https://eegunity.readthedocs.io/en/v0.7.0/eegunity.modules.batch.html`

052. openneuro_ds005520 - Research data supporting 'EEG recording during playing MOBA game'

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005520` | Year: 2024 | URL: https://openneuro.org/datasets/ds005520
Category: Complex Applied | Subjects: 23 | Files (Completed): 23 | Channels: 60 | Sampling Rate: 256 Hz

Paradigm

Participants played a real MOBA (Multiplayer Online Battle Arena) game while 60-channel EEG was recorded alongside resting-state sessions. Trigger codes mark game events such as kills, deaths, and other in-game actions. The study investigates neural correlates of naturalistic gaming behavior.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005520

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`13`	834	long	raw	In-game event code 13 (decode from events.tsv)
`14`	520	long	raw	In-game event code 14 (decode from events.tsv)
`66`	137	long	raw	In-game event code 66 (decode from events.tsv)
`666`	89	long	raw	In-game event code 666 (decode from events.tsv)
`444`	45	long	raw	In-game event code 444 (decode from events.tsv)
`-255`	9	admin	raw	System overflow / boundary marker
`4`	9	long	raw	In-game event code 4 (decode from events.tsv)
`100008`	9	long	raw	In-game event code 100008 (decode from events.tsv)

Event ID	Count (total)	Type	Meta Level	Meaning
164	3	long	raw	In-game event code 164 (decode from events.tsv)
244	1	long	raw	In-game event code 244 (decode from events.tsv)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: In-Game Kill vs Death Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -1.0 s, tmax = 2.0 s, target_events = [13, 14]
- Label Route: stim
- Meta Level: raw
- Target Definition:

Event	y	Meaning
13	0	In-game kill event
14	1	In-game death event

- Evidence:
- DOI: 10.1038/s41597.025.05435.5 - Paper presents the openneuro_ds005520 dataset with precise timestamps for in-game kills (marker 13) and deaths (marker 14), validating through ERP P300 analysis and time-frequency decomposition of epochs spanning -1000ms to +2000ms.
- Kernel Notes: Epochs should exclude intervals <1000ms to avoid overlap between consecutive events. Verify event ID mapping from events.tsv.

Primary Task 2: Multi-Class Game Event Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -1.0 s, tmax = 2.0 s, target_events = [13, 14, 66, 666, 444]
- Label Route: stim
- Meta Level: raw
- Target Definition:

Event	y	Meaning
13	0	Kill
14	1	Death
66	2	Game start
666	3	Game victory
444	4	Game defeat

- Evidence:
- DOI: 10.1038/s41597.025.05435.5 - Paper confirms marker codes: 13 (kill), 14 (death), 66 (game start), 666 (game victory), 444 (game defeat).
- Kernel Notes: Same exclusion criteria as Primary Task 1. Verify event ID mapping from events.tsv.

Primary Task 3: Resting State (Eye Open)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
N/A	0	Resting eye open segment

- Evidence:
- DOI: 10.1038/s41597.025.05435.5 - Dataset includes resting-state sessions per OpenNeuro metadata.
- Kernel Notes: Use sliding window segmentation on resting-state recordings. Verify task name from BIDS sidecar.

Primary Task 4: Resting State (Eye Closed)

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
N/A	0	Resting eye closed segment

- Evidence:
- DOI: 10.1038/s41597.025.05435.5 - Dataset includes resting-state sessions per OpenNeuro metadata.
- Kernel Notes: Use sliding window segmentation on resting-state recordings. Verify task name from BIDS sidecar.

Primary Task 5: MOBA Game Play Session

- Task Type: segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30.0 s, overlap = 0.0, target_events = [66]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
66	0	Game start (session)

- Evidence:
- DOI: 10.1038/s41597.025.05435.5 - Marker 66 indicates game start, used to segment continuous gaming sessions.
- Kernel Notes: Long-event epoching for continuous game play segments. Verify duration from events.tsv.

Normal Task 1: MOBA Game Events (long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numerical trigger codes require mapping from events.tsv or dataset README. Very small dataset (23 subjects); low event counts for rare codes.

Event	y	Meaning
`13`	0	Game event 13
`14`	1	Game event 14
`66`	2	Game event 66
`666`	3	Game event 666
`444`	4	Game event 444
`-255`	5	System marker
`4`	6	Game event 4
`100008`	7	Game event 100008
`164`	8	Game event 164
`244`	9	Game event 244

Data Quality Notes

- Event codes require decoding from dataset events.tsv or README.
- Very small dataset (23 subjects, 23 files); rare codes (164, 244, -255) have <10 occurrences total.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG recording during playing MOBA game`

053. openneuro_ds005841 - EEG Experiment measuring ERPs in VR

Source: analysis.md

Dataset: `openneuro_ds005841` | Year: 2025 | URL: https://openneuro.org/datasets/ds005841
Category: Complex Applied | Subjects: 48 | Files (Completed): 288 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

An investigation of event-related potentials (ERPs) during visual tasks performed within a virtual reality (VR) environment. Stimuli vary along two orthogonal dimensions: lighting (DARK vs LIGHT) and reference type (RAND = random/unpredictable vs REF = reference/predictable). The 2x2 design yields four event codes measuring visual ERP responses under prediction-error conditions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005841

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`RAND_DARK`	288	short	raw	Random (unpredictable) stimulus - dark condition
`RAND_LIGHT`	288	short	raw	Random (unpredictable) stimulus - light condition
`REF_DARK`	288	short	raw	Reference (predictable) stimulus - dark condition
`REF_LIGHT`	288	short	raw	Reference (predictable) stimulus - light condition

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Lumperp

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds005841 DOI: https://doi.org/10.18112/openneuro.ds005841.v1.0.0

Primary Task 2: Publisher-defined BIDS task: Signalscreen

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds005841 DOI: https://doi.org/10.18112/openneuro.ds005841.v1.0.0

Primary Task 3: Publisher-defined BIDS task: Regfront

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds005841 DOI: https://doi.org/10.18112/openneuro.ds005841.v1.0.0

Normal Task 1: VR Stimulus Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Perfectly balanced 4-class design (288 each); primary contrast is RAND vs REF (prediction error) and DARK vs LIGHT (luminance). Count per file reflects 1 file per subject.

Event	y	Meaning
`RAND_DARK`	0	Random dark stimulus
`RAND_LIGHT`	1	Random light stimulus
`REF_DARK`	2	Reference dark stimulus
`REF_LIGHT`	3	Reference light stimulus
## Data Quality Notes		

- VR environment may introduce motion artifacts; apply ICA for artifact removal.
- 48 subjects x 6 sessions = 288 files; verify session-level consistency.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

054. openneuro_ds006554 - Social Observation EEG Raw Data

Source: analysis.md

Dataset: `openneuro_ds006554` | Year: 2025 | URL: https://openneuro.org/datasets/ds006554
Category: Complex Applied | Subjects: 47 | Files (Completed): 47 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

EEG was recorded while participants engaged in social observation tasks, likely involving ecological social stimuli such as videos or images of human social interactions. The rich numeric trigger coding (30 distinct codes) suggests a multi-factorial design with combinations of social condition, stimulus type, or response category. The paradigm examines neural correlates of social perception and mentalizing.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006554

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
60	47	short	raw	Social stimulus condition 60
61	47	short	raw	Social stimulus condition 61
30	47	short	raw	Social stimulus condition 30
31	47	short	raw	Social stimulus condition 31
62	47	short	raw	Social stimulus condition 62
63	47	short	raw	Social stimulus condition 63
32	47	short	raw	Social stimulus condition 32
33	47	short	raw	Social stimulus condition 33
21	47	short	raw	Social stimulus condition 21
25	47	short	raw	Social stimulus condition 25

Event ID	Count (total)	Type	Meta Level	Meaning
16	47	short	raw	Social stimulus condition 16
13	47	short	raw	Social stimulus condition 13
12	47	short	raw	Social stimulus condition 12
24	47	short	raw	Social stimulus condition 24
87	47	short	raw	Social stimulus condition 87
68	47	short	raw	Social stimulus condition 68
11	47	short	raw	Social stimulus condition 11
26	47	short	raw	Social stimulus condition 26
14	47	short	raw	Social stimulus condition 14
15	47	short	raw	Social stimulus condition 15
22	47	short	raw	Social stimulus condition 22
23	47	short	raw	Social stimulus condition 23
42	47	short	raw	Social stimulus condition 42
67	47	short	raw	Social stimulus condition 67
88	47	short	raw	Social stimulus condition 88
43	47	short	raw	Social stimulus condition 43
36	47	short	raw	Social stimulus condition 36
35	47	short	raw	Social stimulus condition 35
48	47	short	raw	Social stimulus condition 48
37	47	short	raw	Social stimulus condition 37

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.

- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Social Observation

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006554 DOI: https://doi.org/10.18112/openneuro.ds006554.v1.0.0

Normal Task 1: Social Condition Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numeric trigger codes require decoding from events.tsv. All 30 event types appear in all 47 files (perfectly balanced). Group codes by tens-digit (10s, 20s, 30s, 60s, 80s) as likely condition groupings.

Event	y	Meaning
60	0	Social condition 60
61	1	Social condition 61
30	2	Social condition 30
31	3	Social condition 31
62	4	Social condition 62
63	5	Social condition 63
32	6	Social condition 32
33	7	Social condition 33
21	8	Social condition 21
25	9	Social condition 25
16	10	Social condition 16
13	11	Social condition 13
12	12	Social condition 12
24	13	Social condition 24
87	14	Social condition 87
68	15	Social condition 68
11	16	Social condition 11
26	17	Social condition 26
14	18	Social condition 14

Event	y	Meaning
15	19	Social condition 15
22	20	Social condition 22
23	21	Social condition 23
42	22	Social condition 42
67	23	Social condition 67
88	24	Social condition 88
43	25	Social condition 43
36	26	Social condition 36
35	27	Social condition 35
48	28	Social condition 48
37	29	Social condition 37
## Data Quality Notes		

- Numeric trigger codes need decoding from events.tsv or README.
- Large number of distinct event types (30); group codes before classification.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Social Observation EEG raw data`

055. openneuro_ds006839 - EEG Recordings during Sham Neurofeedback in Virtual Reality

Source: analysis.md

Dataset: `openneuro_ds006839` | Year: 2025 | URL: https://openneuro.org/datasets/ds006839
Category: Complex Applied | Subjects: 144 | Files (Completed): 144 | Channels: 29 | Sampling Rate: 1000 Hz

Paradigm

Participants performed an attentional task within a virtual reality environment while receiving sham neurofeedback with positive, negative, or control feedback valences. The study investigates alpha-band modulation in response to perceived performance feedback without actual neurofeedback. Numeric trigger codes mark trial onsets and feedback conditions. The paradigm examines placebo effects of neurofeedback on neural oscillations and attention.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006839

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`190`	10264	short	raw	Positive feedback condition marker
`222`	5389	short	raw	Control (neutral) feedback condition
`126`	576	short	raw	Trial/block onset marker
`238`	431	short	raw	Negative feedback condition marker
`0`	98	admin	raw	Null/reset event

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Positive

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006839 DOI: https://doi.org/10.18112/openneuro.ds006839.v1.0.0

Primary Task 2: Publisher-defined BIDS task: Control

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006839 DOI: https://doi.org/10.18112/openneuro.ds006839.v1.0.0

Primary Task 3: Publisher-defined BIDS task: Negative

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006839 DOI: https://doi.org/10.18112/openneuro.ds006839.v1.0.0

Normal Task 1: Neurofeedback Valence Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Decode trigger codes from events.tsv. Code `0` is a null/reset event. Primary contrast is positive (190) vs negative (238) vs neutral (222) feedback valence. Classes are imbalanced (190: 10264 vs 238: 431).

Event	y	Meaning
`126`	0	Trial/block onset marker
`190`	1	Positive sham neurofeedback
`238`	2	Negative sham neurofeedback
`222`	3	Control (neutral) feedback
`0`	4	Null/reset event
## Data Quality Notes		

- Classes are imbalanced: code 190 (10264) >> 238 (431); balance before classification.
- VR environment may introduce motion artifact; check for movement-related noise.

- Numeric trigger codes require decoding from events.tsv or README.

References

- Reference: `EEG recordings during sham neurofeedback in virtual reality`

056. openneuro_ds007180 - Exo-EEG Experiment

Source: analysis.md

Dataset: openneuro_ds007180 | Year: 2026 | URL: https://openneuro.org/datasets/ds007180
Category: Complex Applied | Subjects: 25 | Files (Completed): 25 | Channels: 63 | Sampling Rate: 500 Hz

Paradigm

EEG was recorded during interaction with an exoskeleton in a gait or robotic-assisted movement task. Participants performed motor tasks with varying exoskeleton assistance levels and movement conditions. BrainVision trigger codes (S-prefixed) encode specific movement phases, assistance levels, and stimulus conditions across 30 categories.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds007180

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
S21	1885	short	raw	Exo stimulus condition S21
S22	1885	short	raw	Exo stimulus condition S22
S12	1884	short	raw	Exo stimulus condition S12
S11	1884	short	raw	Exo stimulus condition S11
S32	1876	short	raw	Exo stimulus condition S32
S31	1876	short	raw	Exo stimulus condition S31
S41	1872	short	raw	Exo stimulus condition S41
S42	1872	short	raw	Exo stimulus condition S42
S62	1872	short	raw	Exo stimulus condition S62
S61	1872	short	raw	Exo stimulus condition S61
S52	1872	short	raw	Exo stimulus condition S52

Event ID	Count (total)	Type	Meta Level	Meaning
S51	1872	short	raw	Exo stimulus condition S51
S72	1868	short	raw	Exo stimulus condition S72
S71	1868	short	raw	Exo stimulus condition S71
S81	1860	short	raw	Exo stimulus condition S81
S82	1860	short	raw	Exo stimulus condition S82
S13	1750	short	raw	Exo stimulus condition S13
S43	1750	short	raw	Exo stimulus condition S43
S73	1750	short	raw	Exo stimulus condition S73
S63	1750	short	raw	Exo stimulus condition S63
S53	1750	short	raw	Exo stimulus condition S53
S33	1750	short	raw	Exo stimulus condition S33
S23	1750	short	raw	Exo stimulus condition S23
S83	1750	short	raw	Exo stimulus condition S83
S102	1300	short	raw	Exo stimulus condition S102
S101	1300	short	raw	Exo stimulus condition S101
S92	1300	short	raw	Exo stimulus condition S92
S91	1300	short	raw	Exo stimulus condition S91
S111	1275	short	raw	Exo stimulus condition S111
S122	1275	short	raw	Exo stimulus condition S122

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: S21, S22, S12, S11, S32, S31, S41, S42
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Exo

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds007180 DOI: https://doi.org/10.18112/openneuro.ds007180.v1.0.0

Normal Task 1: Exoskeleton Condition Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 30 distinct BrainVision stimulus codes; decode from events.tsv or README for exact condition meaning. `S255` and `start` are admin markers in some files; exclude from analysis.

Event	y	Meaning
S62	0	Exo condition S62
S61	1	Exo condition S61
S52	2	Exo condition S52
S51	3	Exo condition S51
S21	4	Exo condition S21
S22	5	Exo condition S22
S12	6	Exo condition S12
S11	7	Exo condition S11
S32	8	Exo condition S32
S41	9	Exo condition S41
S42	10	Exo condition S42
S31	11	Exo condition S31
S72	12	Exo condition S72
S71	13	Exo condition S71
S81	14	Exo condition S81
S82	15	Exo condition S82
S13	16	Exo condition S13
S43	17	Exo condition S43
S73	18	Exo condition S73
S63	19	Exo condition S63
S53	20	Exo condition S53
S33	21	Exo condition S33

Event	y	Meaning
S23	22	Exo condition S23
S83	23	Exo condition S83
S102	24	Exo condition S102
S101	25	Exo condition S101
S92	26	Exo condition S92
S91	27	Exo condition S91
S111	28	Exo condition S111
S122	29	Exo condition S122
## Data Quality Notes		

- BrainVision format; 30 S-coded stimulus conditions require README decoding.
- S255 (saturation marker) and start events in some files; exclude from analysis.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

Chapter 2. High-level Mental

Datasets in this chapter: 101. Deep internal cognitive/psychological processing tasks.

057. ieee_icassp_competition_2024 - IEEE ICASSP 2024 EEG Competition Dataset

Source: analysis.md

Dataset: `ieee_icassp_competition_2024` | Year: 2024 | URL: https://2024.ieeeicassp.org/
Category: High-level Mental | Subjects: 48 | Files (Completed): 96 | Channels: 73 | Sampling Rate: 256 Hz

Paradigm

Forty-eight subjects participated in two EEG-based speech BCI paradigms recorded with a 72-channel BioSemi ActiveTwo cap (plus Status stim channel, total 73) at 256 Hz. Task II_II (Imagined + Inner Speech): subjects imagined speaking or performed inner speech, with two stimulus classes (trigger codes 101 and 102, ~200 events each per session). Task RBII (Real Speech + BCI): subjects produced real speech with BCI feedback under the same two class scheme. Each subject contributed one file per task type (48 x 2 = 96 BDF files). Trigger codes are stored in the BioSemi Status stim channel and extracted via `mne.find_events()`.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://2024.ieeeicassp.org/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`101`	~9600 (200/file x 48)	short	raw	Stimulus class 1 onset (imagined/inner speech condition 1)
`102`	~9600 (200/file x 48)	short	raw	Stimulus class 2 onset (imagined/inner speech condition 2)
`3`	varies	short	raw	Auxiliary trigger (task boundary or response marker)
`4`	varies	short	raw	Auxiliary trigger (task boundary or response marker)
`65536`	~144 (3/file x 48)	admin	raw	BioSemi system trigger

Event ID	Count (total)	Type	Meta Level	Meaning
65790	~192 (4/file x 48)	admin	raw	BioSemi system trigger

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Trigger codes are stored as integer-string annotations converted from BioSemi Status stim channel values. Events 65536 and 65790 are BioSemi hardware codes and should be excluded from analysis.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Speech BCI challenge classification (inner/overt tracks)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Competition-provided trial labels for imagined/inner speech and real speech tracks.
- Evidence: IEEE ICASSP 2024 EEG BCI Grand Challenge defines supervised speech-BCI classification as the core task. Source: https://2024.ieeeicassp.org/ DOI: https://doi.org/10.1109/icasspw62465.2024.10626244

Normal Task 1: Imagined/Inner Speech Classification - Class 1 vs. Class 2 (epoch_by_event)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s (per stimulus onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Filter to II_II task files (filename contains II_II) for imagined/inner speech paradigm, or RBII files for real speech BCI. Exclude events 65536 and 65790 (BioSemi system codes). The kernel stores all Status stim channel values as string annotations.

Event	y	Meaning
101	0	Stimulus class 1
102	1	Stimulus class 2

Normal Task 2: Real Speech BCI Classification - Class 1 vs. Class 2 (epoch_by_event)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s (per stimulus onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Filter to RBII task files (filename contains RBII). Same event codes as Task 1; task type is identified from filename pattern.

Event	y	Meaning
101	0	Stimulus class 1
102	1	Stimulus class 2
## Data Quality Notes		

- Two distinct paradigms (II_II and RBII) stored in separate files; analyze separately.
- Trigger code semantics (exact speech content per class) require the competition documentation for full interpretation.
- 73 channels include 72 EEG + 1 Status stim channel; exclude Status before analysis.
- 4 auxiliary channels (M1, M2, NAS, plus EOG channels) are present; identify and remove non-EEG channels.
- No age or sex information available.

References

- Reference: IEEE ICASSP 2024 EEG BCI Grand Challenge. https://2024.ieeeicassp.org/

058. mendeley_sin - Mendeley Speech-in-Noise (SiN) Auditory Attention Decoding Dataset

Source: analysis.md

Dataset: `mendeley_sin` | Year: 2021 | URL: https://data.mendeley.com/datasets/jyvythkz5y/2
Category: High-level Mental | Subjects: 26 | Files (Completed): 26 (Case B kernel) | Channels: 66 | Sampling Rate: 256 Hz

Paradigm

Twenty-six normal-hearing subjects (NH_subject01-NH_subject26) participated in a cocktail-party auditory attention decoding (AAD) task in which competing speech was presented with background noise at two signal-to-noise conditions. EEG was recorded from 64 channels plus 2 mastoid references (total 66 channels) at 256 Hz. Each subject's `.mat` file contains pre-epoched EEG data as a 3D array `(896, 66, 100)` - 896 time points x 66 channels x 100 trials - for each of two Speech-in-Noise conditions (condition 7 and condition 13). The Case B kernel concatenates the 100 trials for each condition into a single continuous raw signal and annotates each trial with its condition label. This dataset is identical in content and structure to `zenodo_sin` (which is a Zenodo mirror of this Mendeley original).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://data.mendeley.com/datasets/jyvythkz5y/2

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`SiN_condition_7`	~1300 (~50/subject x 26)	long	raw	Speech-in-Noise condition 7 trial (one cocktail-party condition)
`SiN_condition_13`	~1300 (~50/subject x 26)	long	raw	Speech-in-Noise condition 13 trial (second cocktail-party condition)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Case B kernel: pre-epoched `.mat` data; each file reconstructed into continuous raw by concatenating 100 trials per condition (~3.5 s each at 256 Hz = 896 samples per trial).

Tasks

Primary Task 1: Speech-in-noise auditory attention decoding

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification] | tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Condition or attended-stream labels (released split commonly uses condition 7 vs 13).
- Evidence: Dataset documentation describes cocktail-party speech-in-noise auditory attention decoding. Source: https://data.mendeley.com/datasets/jyvythkz5y/2 DOI: https://doi.org/10.17632/jyvythkz5y

Normal Task 1: Speech-in-Noise Condition Classification - Condition 7 vs. 13 (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 3.5 s (= 896/256 Hz; matches pre-epoched trial length), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Case B kernel reconstructs continuous raw from pre-epoched data; trial boundaries are annotated. Epoch window (3.5 s) matches the original pre-epoched length; use exactly 896/256 = 3.5 s to avoid boundary effects. sfreq is derived from the `tE` time axis in the `.mat` file (~ 256 Hz). Channel names are generic (`EEG001-EEG066`); no channel label metadata is provided.

Event	y	Meaning
`SiN_condition_7`	0	Speech-in-Noise condition 7
`SiN_condition_13`	1	Speech-in-Noise condition 13
## Data Quality Notes		

- Case B kernel: original `.mat` data is pre-epoched (not continuous); reconstruction introduces trial boundary discontinuities.
- Identical content to `zenodo_sin`; the two datasets can be treated as interchangeable.
- Channel names in the reconstructed raw are generic (`EEG001-EEG066`); actual electrode layout is 64 EEG + 2 mastoid references.
- No age or sex information available.

References

- Reference: Speech-in-Noise Auditory Attention Decoding Dataset. Mendeley Data. https://data.mendeley.com/datasets/jyvythkz5y/2. See also `zenodo_sin`.

059. openneuro_ds001784 - Effects of ON/OFF deep brain stimulation on cognitive control in treatment-resistant depression (EEG)

Source: analysis_minimax_v2.md

Dataset: openneuro_ds001784 | Year: 2019 | URL: https://openneuro.org/datasets/ds001784
Category: High-level Mental | Subjects: N/A | Files (Completed): 38 | Channels: 65.0 | Sampling Rate: 1450.0 Hz

Paradigm

Participants with treatment-resistant depression performed a modified Multi-Source Interference Task (MSIT) integrated with emotional stimuli (IAPS images) while deep brain stimulation (DBS) was toggled ON or OFF. The study investigated how DBS of the internal capsule affects cognitive control and prefrontal cortex function. No trial-level event markers are present in the BIDS EEG files; the DBS condition (ON/OFF) is tracked as metadata rather than inline event triggers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds001784

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(none)	-	-	raw	No event markers detected in EEG files; DBS condition tracked via metadata

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Cognitive control decoding - Interference vs Control trials

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.8 s * (spanning -1.5s to +3.4s around stimulus onset), overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super

- Target Definition:
Event	y	Meaning
`interference`	1	High-interference MSIT trial (IAPS + interference task)
`control`	0	Control trial (low interference)
- Super Meta Notes: Derived from experimental design metadata. Trial type (interference vs control) inferred from task timing structure and Psychophysics Toolbox log files; DBS condition stored as subject-level metadata. Kernel step: Load trial timing from events.tsv (if available) or derive from BIDS task timing; create segmentation windows centered on stimulus onsets; assign labels based on trial type metadata.
- Evidence:
- DOI: `10.1038/s41467-019-09557-4` - This Nature Communications paper analyzes EEG from the dataset examining how VCVS deep brain stimulation affects cognitive control via MSIT, measuring interference vs control trial performance and prefrontal theta oscillations.
- Kernel Notes: Since no trial-level event markers exist in raw EEG (only continuous recordings), use segmentation to extract 4.8s windows aligned to IAPS stimulus onset times from task events.tsv; baseline correct using -0.5s to -0.1s pre-stimulus; labels from experimental design file indicating interference vs control condition.

Primary Task 2: DBS ON vs OFF classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.8 s *, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`DBS_ON`	1	Deep brain stimulation active during trial
`DBS_OFF`	0	Deep brain stimulation inactive during trial
- Super Meta Notes: Derived from subject-level metadata. DBS stimulation condition was toggled between ON/OFF blocks and tracked in participants.tsv or separate stimulation logs, not as inline EEG triggers. Kernel step: Match trial timestamps to DBS condition based on experimental block timing; assign binary label to each epoch.
- Evidence:
- DOI: `10.1038/s41467-019-09557-4` - Paper examines DBS ON vs OFF effects on cognitive control and prefrontal cortex function during MSIT performance.
- Kernel Notes: Load DBS condition metadata from participants.tsv or scan session notes; map each epoch to corresponding DBS condition based on recording timestamp; assign labels accordingly.

Primary Task 3: Publisher-defined BIDS task: Afmsit

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.8 s *, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: Task-related EEG segments from the Afmsit (Affective Multi-Source Interference Task) run
- Evidence: OpenNeuro metadata.tasksCompleted indicates Afmsit task was recorded. Source: https://openneuro.org/datasets/ds001784 DOI: https://doi.org/10.18112/openneuro.ds001784.v1.1.2

- Kernel Notes: Extract continuous segments from the Afmsit task recording; no event markers available so use sliding window segmentation.

Primary Task 4: Publisher-defined BIDS task: Rest

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s *, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: Resting-state EEG segments for baseline characterization
- Evidence: OpenNeuro metadata.tasksCompleted indicates Rest task was recorded. Source: https://openneuro.org/datasets/ds001784 DOI: https://doi.org/10.18112/openneuro.ds001784.v1.1.2
- Kernel Notes: Extract resting-state segments from Rest task recording using sliding window; may serve as baseline for comparing DBS effects.

Normal Task 1: Resting / Continuous Segmentation (resting)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no trial-level labels available))
- Notes: DBS ON vs OFF condition labels are subject-level metadata. If DBS condition is available from participants.tsv, it can be used as a subject-level binary label post-hoc.

Data Quality Notes

- No trial-level event markers detected in any EEG files; only continuous recordings available.
- DBS condition (ON/OFF) distinguishes experimental conditions but is not encoded as EEG events.
- High sampling rate (1450 Hz); consider downsampling for computational efficiency.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Deep Brain Stimulation of the Internal Capsule Enhances Human Cognitive Control and Prefrontal Cortex Function. Nature Communications.`

060. openneuro_ds001785 - Evidence accumulation relates to perceptual consciousness and monitoring

Source: analysis.md

Dataset: `openneuro_ds001785` | Year: 2021 | URL: https://openneuro.org/datasets/ds001785
Category: High-level Mental | Subjects: 21 | Files (Completed): 54 | Channels: 71.0 | Sampling Rate: 1024.0 Hz

Paradigm

Participants performed a threshold-level vibrotactile detection task in which a weak tactile stimulus was either present (threshold condition) or absent (adapt condition). After each trial, participants provided a confidence rating. The study investigated the neural mechanisms of evidence accumulation and metacognitive monitoring using signal detection theory outcomes (hit, miss, correct rejection, false alarm) as trial labels.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, confidence, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds001785

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stim-thr`	large	short	raw	Stimulus onset - threshold vibrotactile stimulus present
`stim-adapt`	large	short	raw	Stimulus onset - adapt condition (no stimulus)
`hit`	large	short	raw	Correct detection: target present, response given
`miss`	large	short	raw	Miss: target present, no response
`cr`	large	short	raw	Correct rejection: no target, no response
`fa`	large	short	raw	False alarm: no target, but response given
`conf`	large	short	raw	Confidence rating onset

Event ID	Count (total)	Type	Meta Level	Meaning
conf-resp	large	short	raw	Confidence rating button-press response

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: confidence, response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim-thr, stim-adapt
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Tactile detection

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds001785 DOI: https://doi.org/10.18112/openneuro.ds001785.v1.1.1

Normal Task 1: Signal Detection Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Four-class SDT labels enable analysis of detection sensitivity and response bias. stim-thr and stim-adapt are stimulus onset markers - use as alternative epoch triggers if SDT label counts are imbalanced. Regression target: confidence from misc channel.

Event	y	Meaning
hit	0	Correct detection
miss	1	Missed target
cr	2	Correct rejection

Event	y	Meaning
`fa`	3	False alarm
## Data Quality Notes		

- Check class balance across hit/miss/cr/fa - miss and fa classes may be sparse for threshold stimuli.
- Confidence ratings stored in misc channel `confidence`; response times in `response_time`.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Evidence accumulation relates to perceptual consciousness and monitoring`

061. openneuro_ds002158 - Disentangling the origins of confidence in speeded perceptual judgments through multimodal imaging

Source: analysis_minimax_v2.md

Dataset: openneuro_ds002158 | Year: 2020 | URL: https://openneuro.org/datasets/ds002158
Category: High-level Mental | Subjects: 119 | Files (Completed): 119
Channels: 64 | Sampling Rate: 5000 Hz

Paradigm

Participants performed speeded perceptual judgments and subsequently reported their confidence in either their own decisions or in observed decisions made by another person. The study used multimodal neuroimaging (EEG + fMRI) to isolate the neural correlates of metacognitive confidence, distinguishing whether confidence reflects the quality of perceptual evidence or post-decisional processes. The confidence rating is stored as a misc channel for regression analysis. High sampling rate (5000 Hz) requires downsampling before epoching.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: response_time, reaction time, confidence, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002158

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
unknown	119 (files)	short	raw	Trial onset marker - actual trial type may be in events.tsv value column; used as epoch anchor

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: confidence, reaction_time, response_time

- Kernel Note: Keep derivation source, unit, and version metadata.

Tasks

Primary Task 1: Confidence Regression from EEG

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 0.6 s, target_events = [`unknown`]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`unknown`	confidence (0-1)	Continuous metacognitive confidence rating

- Super Meta Notes: Derived from raw event(s) [`unknown`] by extracting confidence ratings (continuous 0-1 scale) from the misc channel. Confidence ratings were collected via visual analog scale (second-order task). Kernel step: Load confidence values from misc channel as regression target; use response_time as covariate if needed.
- Evidence:
- DOI: 10.1073/pnas.1918335117 - "linear regression to model EEG amplitude time-locked to first-order response as a function of confidence" with EEG decoding to derive single-trial confidence predictions
- Kernel Notes: Paper uses "response_onset" or "first-order_response" as epoch anchor; verify if events.tsv contains more specific event labels. The `unknown` event is the trial onset marker-use as anchor. Also tested 0-200ms and 200-450ms post-response windows for decoding. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: EEG Decoding of Decision Accuracy

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 0.6 s, target_events = [`unknown`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`unknown`	correct	First-order decision was correct
`unknown`	incorrect	First-order decision was incorrect

- Super Meta Notes: Derived from raw event(s) [`unknown`] by extracting first-order accuracy labels (binary: correct/incorrect) from events.tsv. Kernel step: Load trial_type or value column to determine correct vs incorrect trials.
- Evidence:
- DOI: 10.1073/pnas.1918335117 - "first-order accuracy: correct/incorrect binary" used alongside confidence for analysis
- Kernel Notes: Verify event ID mapping from events.tsv; binary classification of perceptual decision accuracy.

Data Quality Notes

- Event label `unknown` is a placeholder - actual trial type information may be in events.tsv `value` or `trial_type` columns.
- Confidence ratings stored in misc channel `confidence`; reaction times in `response_time`.

- High sampling rate (5000 Hz); significant downsampling required.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Disentangling the origins of confidence in speeded perceptual judgments through multimodal imaging`

062. openneuro_ds002691 - Internal attention study

Source: analysis.md

Dataset: openneuro_ds002691 | Year: 2021 | URL: https://openneuro.org/datasets/ds002691
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 32.0 | Sampling Rate: 250.0 Hz

Paradigm

A sustained internal attention task where participants were asked to either mentally concentrate on a target (concentrate condition) or relax (relax condition) for 15-second sustained blocks with eyes closed. Transitional cues (getready, wait) marked preparation periods between active blocks. The study investigated EEG signatures of sustained internal attention versus relaxation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: response_time
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002691

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
concentrate	20 (files)	long	raw	15-second sustained mental concentration block
relax	20 (files)	long	raw	15-second sustained relaxation block
getready	20 (files)	admin	raw	Preparation cue between blocks - transitional
wait	20 (files)	admin	raw	Wait period between blocks - transitional

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
> - rest: resting-state segment marker, for epoch_by_segmentation_hdf5
> - admin: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

Tasks

Primary Task 1: Publisher-defined BIDS task: Internalattention

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds002691 DOI: https://doi.org/10.18112/openneuro.ds002691.v1.1.0

Normal Task 1: Attention State Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `getready` and `wait` are transitional preparation markers - excluded from target_events. 15-second sustained blocks are appropriate for long_event epoching with 2 s windows and 0.5 overlap. Misc channel `response_time` available for regression.

Event	y	Meaning
`concentrate`	0	Sustained mental concentration (15 s blocks)
`relax`	1	Sustained relaxation (15 s blocks)
## Data Quality Notes		

- Small dataset: 20 subjects, 1 file each.
- Eyes-closed recording throughout; no eye movement artifacts expected.
- The count column (20 files) reflects that every file has these events, not the per-subject trial count.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Internal attention study`

063. openneuro_ds002718 - Face processing EEG dataset for EEGLAB

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002718` | Year: 2025 | URL: https://openneuro.org/datasets/ds002718
Category: High-level Mental | Subjects: 18 | Files (Completed): 18 | Channels: 74.0 | Sampling Rate: 250.0 Hz

Paradigm

Participants viewed famous, unfamiliar, and scrambled face images and performed a subjective spatial symmetry judgment task (yes/no button press). The primary scientific question concerns the neural basis of face recognition, with ERP components (N170, N250) distinguishing famous from unfamiliar and scrambled faces. This is an EEGLAB tutorial dataset based on the SPM multimodal dataset (ds000117); events encode face category, presentation order (first vs. second), and latency of second presentation (early vs. late).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002718

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`famous_new`	18 (files)	short	raw	Famous face - first presentation
`unfamiliar_new`	18 (files)	short	raw	Unfamiliar face - first presentation
`scrambled_new`	18 (files)	short	raw	Scrambled face - first presentation
`famous_second_early`	18 (files)	short	raw	Famous face - second presentation (early lag)
`famous_second_late`	18 (files)	short	raw	Famous face - second presentation (late lag)
`unfamiliar_second_early`	18 (files)	short	raw	Unfamiliar face - second presentation (early lag)
`unfamiliar_second_late`	18 (files)	short	raw	Unfamiliar face - second presentation (late lag)

Event ID	Count (total)	Type	Meta Level	Meaning
scrambled_second_early	18 (files)	short	raw	Scrambled face - second presentation (early lag)
scrambled_second_late	18 (files)	short	raw	Scrambled face - second presentation (late lag)
right_sym	10 (files)	short	raw	Right-hand response: face judged symmetric
left_nonsym	10 (files)	short	raw	Left-hand response: face judged non-symmetric
left_sym	8 (files)	short	raw	Left-hand response: face judged symmetric
right_nonsym	8 (files)	short	raw	Right-hand response: face judged non-symmetric
boundary	18 (files)	admin	raw	Segment boundary marker - exclude

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

Tasks

Primary Task 1: Face Category Classification (Famous vs Unfamiliar vs Scrambled)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.5 s, tmax = 1.2 s, target_events = [famous_new, unfamiliar_new, scrambled_new]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
famous_new	0	Famous face - first presentation
unfamiliar_new	1	Unfamiliar face - first presentation
scrambled_new	2	Scrambled face - first presentation

- Evidence:

- DOI: `10.1038/sdata.2015.1` - Three-class face category classification (famous vs unfamiliar vs scrambled) epoched from -500ms to +1200ms for ERP analysis of N170/N250 components.
- Kernel Notes: Map raw event IDs `famous_new`, `unfamiliar_new`, `scrambled_new` to labels 0, 1, 2. Use `famous_new` and `famous_second_*` events combined for label 0 if repetition effects are of interest. Exclude `boundary` admin events. Response events (right_sym, left_sym, etc.) are button-press responses, not stimulus events - not included in primary classification targets. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Count column shows files-with-event (18 subjects), not individual trial counts - actual trial counts per category are much higher.
- `boundary` events are segment/epoch boundary markers - do not epoch on these.
- This is an EEGLAB tutorial dataset with well-documented preprocessing pipeline.
- Verify channel count consistency across subjects before pooling.

References

- Paper: Multimodal Integration of M/EEG and f/MRI Data in SPM12 (DOI/ref: `ds000117`)

064. openneuro_ds002721 - An EEG dataset recorded during affective music listening

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002721` | Year: 2024 | URL: https://openneuro.org/datasets/ds002721
Category: High-level Mental | Subjects: 31 | Files (Completed): 185 | Channels: 19.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to 40 music clips from film scores while their EEG was recorded, to investigate the neural correlates of music-induced emotional states. Each 40-second music clip is marked at onset with a numeric trigger code (800-908 range encode specific music tracks; 32768 and 257 are likely system markers). The study is part of a series with ds002722 (calibration) and ds002723/ds002724 (BCI sessions). Emotion ratings may be available in behavioral logs.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002721

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`32768`	124 (files)	admin	raw	System/hardware trigger - likely session marker
`1092`	124 (files)	admin	raw	System code - likely hardware sync
`800-808` range	124 (files each)	long	raw	Music clip onset - specific track identifiers
`833-840` range	111-123 (files)	long	raw	Music clip onset - specific track identifiers
`902-908` range	118-123 (files)	long	raw	Music clip onset - specific track identifiers
`786`, `788`	124 (files)	long	raw	Music clip onset - specific track identifiers
`257`	120 (files)	admin	raw	System code - likely hardware marker

Event ID	Count (total)	Type	Meta Level	Meaning
33575, 33571	112-119 (files)	admin	raw	Large codes - likely system/admin markers

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: misc
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Music Clip Onset Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [800, 833, 902, 786]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
800	0	Music clip onset (tracks 800-808)
833	1	Music clip onset (tracks 833-840)
902	2	Music clip onset (tracks 902-908)
786	3	Music clip onset (track 786)

- Evidence:
- DOI: 10.1038/s41597.020.0507-6 - Paper presents the 'Film clips' dataset with event markers for each music clip onset used to segment EEG data for analysis of music-induced emotions.
- Kernel Notes: The full set of music clip codes spans 800-808, 786, 788, 833-840, 902-908. Available raw event IDs include 800, 833, 902, 786. The remaining codes (801-808, 788, 834-840, 903-908) are not present in the raw data - verify event ID mapping from events.tsv to confirm which subset is actually available.

Primary Task 2: Emotion Recognition from Music-Induced EEG (Super Meta)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [800, 833, 902, 786]
- Label Route: misc

- Meta Level: super
- Target Definition:
Event	y	Meaning
`high_pleasant`	0	High pleasantness rating (7-9 on 9-point scale)
`low_pleasant`	1	Low pleasantness rating (1-3 on 9-point scale)
`high_energy`	2	High energy rating (7-9 on 9-point scale)
`low_energy`	3	Low energy rating (1-3 on 9-point scale)
`high_tension`	4	High tension rating (7-9 on 9-point scale)
`low_tension`	5	Low tension rating (1-3 on 9-point scale)
- Super Meta Notes: Derived from raw event(s) [`800`, `833`, `902`, `786`] by mapping each music clip onset to its corresponding self-reported emotion ratings across 8 dimensions (pleasantness, energy, tension, anger, fear, happiness, sadness, tenderness) on 9-point Likert scales. Kernel step: Load behavioral ratings file (emotion_*.csv), join with event timestamps, binarize each dimension into high/low categories.
- Evidence:
- DOI: 10.1038/s41597.020.0507-6 - Paper presents the Film Clips dataset with EEG recorded while participants listened to 12-second film music clips and reported felt emotions across 8 affective dimensions on 9-point Likert scales.
- Kernel Notes: Requires accessing external behavioral rating files. Verify emotion rating file location and format from dataset README. If full 26-class music clip IDs needed, verify availability of codes 801-808, 788, 834-840, 903-908 in raw data.

Normal Task 1: Music Clip Onset Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Codes 32768, 1092, 257, 33575, 33571 are system/admin markers - excluded. Each music clip is ~40 seconds; these onset markers are short events marking start of a sustained music stimulus. Valence/arousal ratings from behavioral logs could serve as regression targets.

Event	y	Meaning
`800`	0	Music clip onset (track 800)
`801`	1	Music clip onset (track 801)
`802`	2	Music clip onset (track 802)
`803`	3	Music clip onset (track 803)
`804`	4	Music clip onset (track 804)
`805`	5	Music clip onset (track 805)
`806`	6	Music clip onset (track 806)
`807`	7	Music clip onset (track 807)
`808`	8	Music clip onset (track 808)
`786`	9	Music clip onset (track 786)
`788`	10	Music clip onset (track 788)
`833`	11	Music clip onset (track 833)
`834`	12	Music clip onset (track 834)

Event	y	Meaning
835	13	Music clip onset (track 835)
836	14	Music clip onset (track 836)
837	15	Music clip onset (track 837)
838	16	Music clip onset (track 838)
839	17	Music clip onset (track 839)
840	18	Music clip onset (track 840)
902	19	Music clip onset (track 902)
903	20	Music clip onset (track 903)
904	21	Music clip onset (track 904)
905	22	Music clip onset (track 905)
906	23	Music clip onset (track 906)
907	24	Music clip onset (track 907)
908	25	Music clip onset (track 908)

Data Quality Notes

- 30 distinct event types; codes 32768, 33575, 33571 are likely hardware sync markers.
- 19-channel EEG; low spatial resolution.
- Verify event code mapping from dataset README or paper (Scientific Data 2018).
- Verify channel count consistency across subjects before pooling.

References

- Reference: Neural and physiological data from participants listening to affective music (Scientific Data, 2018)

065. openneuro_ds002722 - A dataset recorded during development of an affective brain-computer music interface: calibration session

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002722` | Year: 2020 | URL: https://openneuro.org/datasets/ds002722
Category: High-level Mental | Subjects: 19 | Files (Completed): 94 | Channels: 37.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to 40-second synthetic music clips designed to induce specific affective states (varying valence and arousal) as a calibration session for an affective brain-computer music interface (BCMI). EEG and physiological signals were recorded to build a mapping from neural/physiological features to targeted emotional states. Codes 1-9 are primary music condition labels; 100-200 range codes encode specific music clip identifiers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002722

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`9`	94 (files)	long	raw	Music condition 9 onset (high arousalx)
`4`	90 (files)	long	raw	Music condition 4 onset
`5`	89 (files)	long	raw	Music condition 5 onset
`6`	89 (files)	long	raw	Music condition 6 onset
`7`	89 (files)	long	raw	Music condition 7 onset
`2`	89 (files)	long	raw	Music condition 2 onset
`1`	60 (files)	long	raw	Music condition 1 onset
`3`	59 (files)	long	raw	Music condition 3 onset
`104-205` range	19 (files each)	long	raw	Specific music clip identifiers

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Affective Music Condition Classification (9-class)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [1, 2, 3, 4, 5, 6, 7, 8, 9]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
1	0	Music condition 1 (affective state 1)
2	1	Music condition 2 (affective state 2)
3	2	Music condition 3 (affective state 3)
4	3	Music condition 4 (affective state 4)
5	4	Music condition 5 (affective state 5)
6	5	Music condition 6 (affective state 6)
7	6	Music condition 7 (affective state 7)
8	7	Music condition 8 (affective state 8)
9	8	Music condition 9 (affective state 9)

- Evidence:
- DOI: 10.1038/s41597-020-0507-6 - Paper confirms this dataset (OpenNeuro ds002722) contains EEG recordings from 19 participants listening to 40-second synthetic music clips with 9 different affective conditions (codes 1-9), presented during 5 runs of 18 trials each, enabling multiclass classification of affective states from EEG.
- Kernel Notes: Epoch 1.2s windows centered on music condition onset markers (events 1-9). The 40-second music clips are long events; however, this task epochs the onset of each clip (event-based) to capture initial neural responses to affective stimuli. Map condition codes 1-9 to labels 0-8 for 9-class classification. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- 30 distinct event types; codes 1-9 are the primary condition labels.
- Decode exact valence/arousal mapping for codes 1-9 from Daly et al. 2018 paper.
- 37-channel EEG includes physiological channels (ECG, GSR, etc.); identify EEG-only channels.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Daly, I., Nicolaou, N., Williams, D., Hwang, F., Kirke, A., Miranda, E., Nasuto, S.J., Neural and physiological data from participants listening to affective music, Scientific Data, 2018.`

066. openneuro_ds002724 - A dataset recorded during development of an affective brain-computer music interface: training sessions

Source: analysis.md

Dataset: `openneuro_ds002724` | Year: 2020 | URL: https://openneuro.org/datasets/ds002724
Category: High-level Mental | Subjects: 17 | Files (Completed): 96 | Channels: 37.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to 40-second synthetic music clips designed to induce specific affective states (varying valence and arousal) during the training phase of an affective brain-computer music interface. This is the training session companion to ds002722 (calibration) and ds002723 (testing). Codes 1-9 encode the nine target affective music conditions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002724

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
6	96 (files)	long	raw	Music condition 6 onset
4	96 (files)	long	raw	Music condition 4 onset
5	96 (files)	long	raw	Music condition 5 onset
2	96 (files)	long	raw	Music condition 2 onset
1	95 (files)	long	raw	Music condition 1 onset
8	95 (files)	long	raw	Music condition 8 onset
7	95 (files)	long	raw	Music condition 7 onset
9	94 (files)	long	raw	Music condition 9 onset
3	94 (files)	long	raw	Music condition 3 onset

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Publisher-defined BIDS task: Run

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds002724 DOI: https://doi.org/10.1145/3059005

Normal Task 1: Affective Music Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to music condition onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 9 affective music conditions; decode valence/arousal mapping from Daly et al. 2018. Well-balanced: all conditions present in ~95-96 files. 40-second sustained stimuli; epoch at onset. 37-channel array includes physiological channels.

Event	y	Meaning
1	0	Music condition 1
2	1	Music condition 2
3	2	Music condition 3
4	3	Music condition 4
5	4	Music condition 5
6	5	Music condition 6
7	6	Music condition 7
8	7	Music condition 8
9	8	Music condition 9
## Data Quality Notes		

- Well-balanced event counts across all 9 conditions (94-96 files each).
- 37-channel array includes EEG plus physiological channels (ECG, GSR, etc.).
- Decode code-to-affective-state mapping from Daly et al. 2018 (Scientific Data).
- Verify channel count consistency across subjects before pooling.

References

- Reference: Daly, I., Nicolaou, N., Williams, D., Hwang, F., Kirke, A., Miranda, E., Nasuto, S.J., Neural and physiological data from participants listening to affective music, Scientific Data, 2018.

067. openneuro_ds002725 - A dataset recording joint EEG-fMRI during affective music listening

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002725` | Year: 2020 | URL: https://openneuro.org/datasets/ds002725
Category: High-level Mental | Subjects: 21 | Files (Completed): 105 | Channels: 47.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to music stimuli designed to elicit emotional responses (approach/withdrawal behavior) and self-reported their felt affective states during simultaneous EEG-fMRI recording. The study investigated how EEG reflects subcortical brain activity related to emotional responses to music. Codes 1283/1285 are likely BrainVision fMRI sync codes; 34053 may be a BrainVision hardware code; 47/10/1 may be music condition/block markers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002725

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`47`	105 (files)	long	raw	Music block marker (likely condition or trial count)
`10`	105 (files)	long	raw	Music onset marker (condition 10x)
`1`	105 (files)	long	raw	Music onset marker (condition 1x)
`1283`	105 (files)	admin	raw	BrainVision fMRI sync code - likely admin
`1285`	105 (files)	admin	raw	BrainVision fMRI sync code - likely admin
`34053`	105 (files)	admin	raw	Large BrainVision hardware code - admin
`265`	94 (files)	long	raw	Music condition marker
`768`	84 (files)	long	raw	Music condition marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Music Condition Classification (synthetic vs classical)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 40 s, overlap = 0.0 *, target_events = [`265`, `768`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`265`	0	Classical music condition
`768`	1	Generated (synthetic) music condition

- Evidence:
- DOI: `10.1038/s41597.020.0507.6` - Joint EEG-fMRI dataset where 21 participants listened to synthetic and classical music while reporting felt affective states via FEELTRACE, with EEG recorded during 40s generated music and 2-3min classical music trials
- Kernel Notes: Music conditions correspond to continuous epochs of ~40s (generated) and ~2-3min (classical). The raw event IDs 265 and 768 appear in 94 and 84 files respectively. Verify event ID mapping from events.tsv. MRI gradient artifact removal required.

Primary Task 2: Affective State Classification from FEELTRACE Labels

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 40 s, overlap = 0.0 *, target_events = [`generated_music`, `classical_music`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`generated_music`	0	Generated/synthetic music condition trials
`classical_music`	1	Classical music condition trials

- Super Meta Notes: Derived from raw event(s) [`265`, `768`] by mapping to music condition labels (synthetic/generated music = event 768, classical music = event 265). Kernel step: Extract epochs aligned to raw events, then merge with FEELTRACE continuous valence/arousal ratings to produce binary or multiclass affective state labels per trial.
- Evidence:
- DOI: `10.1038/s41597.020.0507.6` - Continuous FEELTRACE self-reports of valence and arousal during music listening provide affective state labels for each trial
- Kernel Notes: Requires combining EEG epochs with FEELTRACE continuous labels. Valence (approach/withdrawal) and arousal dimensions can be discretized into bins for classification. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- EEG recorded in MRI scanner; MRI gradient artifacts require dedicated removal pipeline.
- Codes 1283, 1285, 34053 are BrainVision system/fMRI codes - exclude from analysis.
- 47-channel EEG; verify channel positions and identify MRI-compatible electrode set.
- Decode music condition codes from Daly et al. / Sharman et al. paper.

References

- Reference: `Electroencephalography reflects the activity of sub-cortical brain regions during approach-withdrawal behaviour while listening to music`

068. openneuro_ds002791 - DataSet1 - Visual Object Naming EEG

Source: analysis.md

Dataset: `openneuro_ds002791` | Year: 2020 | URL: https://openneuro.org/datasets/ds002791
Category: High-level Mental | Subjects: N/A | Files (Completed): 92 | Channels: 256.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants viewed and named visual objects (pictures presented as bitmap images) during high-density 256-channel EEG recording. Each event label is the filename of the presented image (e.g., GLOVE.bmp, GORILLA.bmp), corresponding to 30 distinct visual object categories. The task likely involved overt or covert naming or picture-word matching. This is DataSet1 of a paired dataset series (see also DataSet2: ds002833).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002791

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`GLOVE.bmp`	46 (files)	short	raw	Visual object onset - glove image
`FEATHER.bmp`	46 (files)	short	raw	Visual object onset - feather image
`GORILLA.bmp`	46 (files)	short	raw	Visual object onset - gorilla image
`DUSTPAN.bmp`	46 (files)	short	raw	Visual object onset - dustpan image
`COMPASS.bmp`	46 (files)	short	raw	Visual object onset - compass image
`ACCORDIO.bmp`	46 (files)	short	raw	Visual object onset - accordion image
`CAP.bmp`	46 (files)	short	raw	Visual object onset - cap image
`BOTTLE.bmp`	46 (files)	short	raw	Visual object onset - bottle image
`CROWN.bmp`	46 (files)	short	raw	Visual object onset - crown image

Event ID	Count (total)	Type	Meta Level	Meaning
ANCHOR.bmp	46 (files)	short	raw	Visual object onset - anchor image
CARROT.bmp	46 (files)	short	raw	Visual object onset - carrot image
FORK.bmp	46 (files)	short	raw	Visual object onset - fork image
DONKEY.bmp	46 (files)	short	raw	Visual object onset - donkey image
EAR.bmp	46 (files)	short	raw	Visual object onset - ear image
CLOUD.bmp	46 (files)	short	raw	Visual object onset - cloud image
ASHTRAY.bmp	46 (files)	short	raw	Visual object onset - ashtray image
frenchcr.bmp	46 (files)	short	raw	Visual object onset - french curve image
FISH.bmp	46 (files)	short	raw	Visual object onset - fish image
BIRDCAGE.bmp	46 (files)	short	raw	Visual object onset - birdcage image
ASPARAGU.bmp	46 (files)	short	raw	Visual object onset - asparagus image
DRESS.bmp	46 (files)	short	raw	Visual object onset - dress image
BEE.bmp	46 (files)	short	raw	Visual object onset - bee image
CHIMNEY.bmp	46 (files)	short	raw	Visual object onset - chimney image
BOOT.bmp	46 (files)	short	raw	Visual object onset - boot image
CUP.bmp	46 (files)	short	raw	Visual object onset - cup image
APPLE.bmp	46 (files)	short	raw	Visual object onset - apple image
CRAB.bmp	46 (files)	short	raw	Visual object onset - crab image
CAR.bmp	46 (files)	short	raw	Visual object onset - car image
ALLIGATO.bmp	46 (files)	short	raw	Visual object onset - alligator image
HAMMOCK.bmp	46 (files)	short	raw	Visual object onset - hammock image

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Visual object naming EEG decoding

- Task Type: classification/regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation` and `misc`
- Target Definition: Stimulus image identity/semantic class from object filename events; optional naming-latency regression if RT metadata is available.
- Evidence: Dataset title and event labels indicate a visual object naming paradigm with per-object stimulus markers. Source: https://openneuro.org/datasets/ds002791 DOI: https://doi.org/10.18112/openneuro.ds002791.v1.0.0

Normal Task 1: Visual Object ERP / Regression (regression)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to image onset), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: N/A (null (30 object categories; regression primary; classification possible but 30-class))
- Notes: 30 visual object categories; for classification, consider grouping by semantic category (living/nonliving, animate/inanimate) rather than using all 30 individual object identities. High-density 256-channel EEG provides excellent spatial coverage.
- Regression target: `response_time` (naming latency from misc channel)

Data Quality Notes

- 30 distinct object images - each present in 46 files (half of 92 total files).
- High-density 256-channel EGI/HydroCell array; verify electrode positions.
- Consider semantic category grouping (animate/inanimate, living/nonliving) for classification.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `DataSet1 - Visual Object Naming EEG`

069. openneuro_ds002833 - DataSet2 - Visual Object Naming EEG (High-Density)

Source: analysis.md

Dataset: openneuro_ds002833 | Year: 2020 | URL: https://openneuro.org/datasets/ds002833
Category: High-level Mental | Subjects: N/A | Files (Completed): 80 | Channels: 257.0 | Sampling Rate: 1000.0 Hz

Paradigm

High-density 257-channel EEG recorded during a visual object naming or picture viewing task, companion to DataSet1 (ds002791). The DIN1 marker is an EGI/Geodesic system digital input trigger marking stimulus onset. The unknown label in 20 files indicates undecodable trigger codes for those sessions. The study likely involves the same visual object paradigm as ds002791 (30 object categories), recorded with an EGI system.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002833

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
DIN1	60 (files)	short	raw	EGI digital input trigger - stimulus onset marker
unknown	20 (files)	admin	raw	Undecodable trigger code - treat as admin/uncertain

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
> - rest: resting-state segment marker, for epoch_by_segmentation_hdf5
> - admin: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Pictures Naming

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds002833 DOI: https://doi.org/10.18112/openneuro.ds002833.v1.0.0

Normal Task 1: Stimulus-Locked ERP / Regression (regression)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to DIN1 onset), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: N/A (null (regression only; DIN1 serves as epoch anchor; `unknown` excluded))
- Notes: DIN1 is the EGI stimulus trigger; use as epoch anchor. `unknown` events are undecodable - exclude from target events. For classification, trial type information would need to come from events.tsv `trial_type` or `value` columns.
- Regression target: `response_time` from misc channel

Data Quality Notes

- 257-channel EGI HydroCell array provides very high spatial density.
- `unknown` events in 20 files may indicate corrupted trigger codes - verify from events.tsv.
- DIN1 triggers mark stimulus onsets; actual stimulus identity requires events.tsv lookup.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `DataSet2 - Visual Object Naming EEG (High-Density)`

070. openneuro_ds003004 - Imagined Emotion Study

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003004` | Year: 2022 | URL: https://openneuro.org/datasets/ds003004
Category: High-level Mental | Subjects: N/A | Files (Completed): 34 | Channels: 219.0 | Sampling Rate: 256.0 Hz

Paradigm

Participants used voice-guided imagery to induce and sustain 15 distinct emotional states, using button presses to indicate the onset and offset of each imagined emotion. Despite the paradigm involving discrete emotional state onsets (button presses), no event markers were detected in the BIDS EEG files by the locator pipeline. The button-press events may be stored in separate behavioral logs or were not encoded in the EEG trigger channel.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003004

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(none)	-	-	raw	No event markers detected in EEG files; button-press onsets not captured in locator

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: 15-Class Imagined Emotion Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`

- Epoch Parameters: tmin = 0 s, tmax = 5.0 s , overlap = 0.0 , target_events = [15_emotion_onset] *
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
love	0	Positive valence guided imagery emotion
joy	1	Positive valence guided imagery emotion
happiness	2	Positive valence guided imagery emotion
relief	3	Positive valence guided imagery emotion
compassion	4	Positive valence guided imagery emotion
contentedness	5	Positive valence guided imagery emotion
excitement	6	Positive valence guided imagery emotion
awe	7	Positive valence guided imagery emotion
anger	8	Negative valence guided imagery emotion
jealousy	9	Negative valence guided imagery emotion
disgust	10	Negative valence guided imagery emotion
frustration	11	Negative valence guided imagery emotion
fear	12	Negative valence guided imagery emotion
sadness	13	Negative valence guided imagery emotion
grief	14	Negative valence guided imagery emotion

- Super Meta Notes: (NOT APPLICABLE - Meta Level = raw)
- Evidence:
- DOI: 10.1016/j.neuroimage.2022.118873 - Applied 20-model AMICA to 31 participants during 15-class emotion imagination; used 5s non-overlapping windows time-locked to emotion narrative onset; button-press markers defined emotion onsets/offsets.
- DOI: 10.3389/neuro.09.061.2009 - Original UCSD study (Onton & Makeig) with 32 subjects performing guided imagery of 15 emotions; emotion periods lasted 43s-12min with button-press markers for onset/offset.
- Kernel Notes: Button-press onset/offset events were manually recorded but not encoded in EEG trigger channel. Kernel must read behavioral logs or derive events from participant button-press timestamps. Verify event ID mapping from events.tsv and behavioral log files. For epoching: use 5s windows within each emotion block (non-overlapping as per Neuroimage 2022). (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Binary Valence Classification (Positive vs Negative)

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 6.0 s, overlap = 0.5, target_events = [valence_positive, valence_negative]
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
valence_positive	0	8 positive valence emotions (love, joy, happiness, relief, compassion, contentment, excitement, awe)
valence_negative	1	7 negative valence emotions (anger, jealousy, disgust, frustration, fear, sadness, grief)

- Super Meta Notes: Derived from raw emotion labels [love, joy, happiness, relief, compassion, contentedness, excitement, awe, anger, jealousy, disgust, frustration, fear, sadness, grief] by

mapping to valence (positive/negative) based on paper's valence rating. Kernel step: Collapse 15-class emotion labels into binary valence labels. Exclude first 2 blocks per subject and ambiguous emotions (excitement, disgust, compassion) per ACII-2013 paper.
- Evidence:
- DOI: `10.1109/acii.2013.160` - Classified positive vs negative valence from 6-second EEG segments using FBCSP; block-wise cross-validation; 40 equally spaced segments per block with 50% overlap; excluded first 60s and last 10s of each emotion block.
- Kernel Notes: Apply segmentation with 6s window, 50% overlap within each emotion block. Exclude first 60s and last 10s of each emotion block per paper. Labels derived from block-level valence annotations - verify mapping in events.tsv.

Data Quality Notes

- No event markers detected in EEG files by locator pipeline - verify events.tsv directly.
- High-density 219-channel array; verify electrode positions and system (likely EGI/Geodesic).
- 15 distinct emotional states if properly annotated - valuable for affective BCI research.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Imagined Emotion Study`

071. openneuro_ds003420 - HD-EEGtask (Dataset 1)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003420` | Year: 2020 | URL: https://openneuro.org/datasets/ds003420
Category: High-level Mental | Subjects: 23 | Files (Completed): 92
Channels: 256 | Sampling Rate: 1000 Hz

Paradigm

Participants performed visual naming and spelling tasks of visually presented objects while high-density 256-channel EEG and speech onset data were recorded. This is Dataset 1 of the conneXion project. Trigger codes 0 and 1 mark stimulus events (likely object presentation onset and response/naming onset). Speech onset timing is also captured for reaction time analysis.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, reaction time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003420

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1`	92 (files)	short	raw	Visual object stimulus onset (or naming task trigger)
`0`	91 (files)	short	raw	Stimulus condition 0 (or response trigger)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `reaction_time`, `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`

- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Visual Object Recognition with Reaction Time Regression

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 0.5 s, target_events = [1]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
1	continuous	Voice onset latency (reaction time) in seconds

- Super Meta Notes: Derived from raw event `1` (stimulus onset) combined with voice onset time extracted from speech recordings. Kernel step: extract stimulus onset timing from event `1`, compute latency to vocalization onset from audio signal, output as `reaction_time` in misc channel. Available for 12 of 20 subjects per paper.
- Evidence:
- DOI: 10.1088/1741-2552.aad7b1 - Analyzes 256-channel EEG from visual object naming of meaningful vs meaningless images, computing functional brain networks in gamma band (30-45Hz) from stimulus onset to 500ms and correlating network modularity metrics with reaction time (voice onset latency).
- Kernel Notes: Extract trial-level voice onset time from speech recordings (Praat or similar). For subjects without speech data, use NaN masking. Epoch window 0-500ms avoids muscle artifacts from speech production. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Brain Network State Segmentation in Picture Naming

- Task Type: segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 0.6 s *, target_events = [1]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
1	1	BNS1 (0-119ms post-stimulus)
1	2	BNS2 (120-150ms post-stimulus)
1	3	BNS3 (151-190ms post-stimulus)
1	4	BNS4 (191-320ms post-stimulus)
1	5	BNS5 (321-480ms post-stimulus)
1	6	BNS6 (481-535ms post-stimulus)

- Evidence:
- DOI: 10.1016/j.cortex.2015.08.019 - Uses dense 256-channel EEG during picture naming to identify six brain network states (BNSs) characterized by gamma-band (30-45Hz) phase synchronization through source connectivity analysis and k-means clustering, with naming latencies measured as time from picture onset to vocalization onset.
- Kernel Notes: Segmentation is performed via functional connectivity clustering on gamma-band (30-45Hz) EEG. Kernel must compute time-varying connectivity matrices and apply k-means to segment

epochs into 6 BNS labels. Epochs longer than 535ms may be truncated. Verify event ID mapping from events.tsv: event `1` corresponds to picture onset. Note: BNS labels are derived from clustering, not raw annotations - treat as pseudo-labels for segmentation.

Data Quality Notes

- Trigger codes 0/1 require semantic decoding from events.tsv or conneXion project documentation.
- High-density 256-channel array; significant computational overhead.
- Speech onset timing recorded - useful for naming latency regression.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `conneXion`

072. openneuro_ds003421 - HD-EEGtask (Dataset 2)

Source: analysis.md

Dataset: `openneuro_ds003421` | Year: 2020 | URL: https://openneuro.org/datasets/ds003421
Category: High-level Mental | Subjects: 20 | Files (Completed): 80
Channels: 257 | Sampling Rate: 1000 Hz

Paradigm

Participants completed a multi-condition experiment including resting state, visual and auditory object naming tasks, and a recognition-based working memory task, with high-density 257-channel EEG recording. This is Dataset 2 of the conneXion project (companion to ds003420). Trigger codes 0 and 1 mark stimulus onset and response events; the specific kernel provides speech onset timing as `response_time` for regression analysis.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003421

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1`	60 (files)	long	raw	Sustained task or stimulus condition 1 onset
`0`	57 (files)	long	raw	Sustained task or stimulus condition 0 onset

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Pictures Naming

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds003421 DOI: https://doi.org/10.18112/openneuro.ds003421.v1.0.2

Normal Task 1: Task State Classification / Regression (regression + long_event)

- Task Type: regression
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: N/A (null(regression primary; codes require decoding))
- Notes: Codes 0/1 mark multi-second sustained task states (resting, naming, memory) - long_event epoching appropriate. Specific kernel decodes numeric codes and extracts speech onset as `response_time`. Not all 80 files contain both events (60 vs 57).
- Regression target:`response_time`(speech onset / naming latency from misc channel)

Data Quality Notes

- Trigger codes 0/1 require semantic decoding from events.tsv or conneXion project documentation.
- High-density 257-channel EGI array; significant computational overhead.
- Multi-condition design includes resting state, naming, and working memory - verify condition assignments.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `HD-EEGtask (Dataset 2) - conneXion project`

073. openneuro_ds003458 - EEG: Three armed bandit gambling task

Source: analysis.md

Dataset: openneuro_ds003458 | Year: 2021 | URL: https://openneuro.org/datasets/ds003458
Category: High-level Mental | Subjects: 23 | Files (Completed): 23
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Participants performed a three-armed bandit gambling task with oscillating reinforcement probabilities to investigate reward processing and reward prediction error signals in EEG. Each trial includes a stimulus presentation (choice between Left, Right, or Up options), a button-press response, and outcome feedback (Win, Loss, or Null). The Fixation event marks trial onset. STATUS is a BioSemi admin marker.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003458

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus (Left`	138 (files)	short	raw	Stimulus onset with left option available
`Right`	138 (files)	short	raw	Stimulus with right option available
`Up)`	138 (files)	short	raw	Stimulus with up option available
`X`	92 (files)	short	raw	Option marker X - decode from events.tsv
`Y`	92 (files)	short	raw	Option marker Y - decode from events.tsv
`Z`	92 (files)	short	raw	Option marker Z - decode from events.tsv
`Response`	69 (files)	short	raw	Participant button-press choice response
`Fixation`	23 (files)	short	raw	Trial-onset fixation cross

Event ID	Count (total)	Type	Meta Level	Meaning
Feedback Win	23 (files)	short	raw	Outcome: reward received
Feedback Loss	23 (files)	short	raw	Outcome: no reward
Feedback Null	13 (files)	short	raw	Outcome: null/neutral feedback
STATUS	23 (files)	admin	raw	BioSemi system status marker - exclude

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Stimulus (Left, STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Three Armed Bandit

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds003458 DOI: https://doi.org/10.18112/openneuro.ds003458.v1.1.0

Normal Task 1: Feedback Outcome Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: STATUS is a BioSemi admin marker - excluded. Event label Stimulus (Left contains a parenthesis fragment - this is a truncated label; the full label may span Stimulus (Left, Right, Up). Feedback events are the primary scientific targets for reward prediction error analysis. Response marks button-press choices.

Event	y	Meaning
Feedback Win	0	Win feedback (reward received)
Feedback Loss	1	Loss feedback (no reward)

Event	y	Meaning
`Feedback Null`	2	Null/neutral feedback
## Data Quality Notes		

- Event label `Stimulus (Left` is a truncated label fragment - verify full label from events.tsv.
- STATUS is BioSemi hardware marker - exclude from analysis.
- Small dataset: 23 subjects, 1 file each.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG: Three armed bandit gambling task`

074. openneuro_ds003506 - EEG: Reinforcement Learning in Parkinson's

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003506 | Year: 2021 | URL: https://openneuro.org/datasets/ds003506
Category: High-level Mental | Subjects: 56 | Files (Completed): 84
Channels: 67 | Sampling Rate: 500 Hz

Paradigm

This study investigates the Reward Positivity (RewP) EEG component in Parkinson's disease (PD) patients compared to matched healthy controls. Participants completed a reinforcement learning task with both volitional and instructed choices, earning rewards or receiving no-reward feedback based on their selections. The RewP, a frontocentral ERP component, scales with reward prediction error and mirrors dopamine-driven learning. The study examines whether PD diminishes this signal, given dopaminergic dysfunction in PD.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003506

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Imperative Stimulus`	84	short	raw	Cue stimulus requiring a response
`Left Button Choice`	84	short	raw	Left-hand button press response
`Right Button Choice`	84	short	raw	Right-hand button press response
`Instr`	168	short	raw	Instruction marker (Match/Choose)
`FB`	252	short	raw	Feedback stimulus (reward +1, no-reward 0)
`STATUS`	84	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Imperative Stimulus`, `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Feedback type classification (reward vs. no-reward)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -6.0 s, tmax = 2.0 s, target_events = [FB]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
FB (reward)	1	Reward feedback (+1)
FB (no-reward)	0	No-reward feedback (0)

- Evidence:
- DOI: `10.1016/j.brainres.2019.146541` - This paper analyzes EEG data from 28 PD patients and 28 matched healthy controls completing a reinforcement learning task, epoching data around feedback onset (-6000 to 2000 ms) and comparing Reward Positivity (RewP) amplitude between groups for reward (+1) versus no-reward (0) feedback conditions.
- Kernel Notes: Epochs are centered on feedback onset (FB events). Labels are derived from feedback value: +1 = reward (y=1), 0 = no-reward (y=0). Apply baseline correction from -0.2 s to 0 s as reported in the paper. The original analysis filtered data (0.1-30 Hz typical for RewP). Exclude rare events like 'Too Slow' and 'No Match' if present in the data. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- 67 channels including VEOG and positional channels (X, Y, Z); exclude non-EEG channels before analysis.
- Two sessions per subject for some participants; verify session-level consistency.
- STATUS is a single system marker per file; exclude from epoching.

References

- Reference: `An EEG marker of reward processing is diminished in Parkinson's disease. Brain Research, 2019. DOI: 10.1016/j.brainres.2019.146541`

075. openneuro_ds003509 - EEG: Simon Conflict in Parkinson's

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003509 | Year: 2021 | URL: https://openneuro.org/datasets/ds003509
Category: High-level Mental | Subjects: 56 | Files (Completed): 84
Channels: 67 | Sampling Rate: 500 Hz

Paradigm

This study investigates mid-frontal theta deficits in Parkinson's disease (PD) patients during a modified Simon reaction-time task that requires cognitive control. Participants (high-performing PD patients and matched controls) performed congruent and incongruent Simon trials where stimulus position and response direction could conflict. The task included both a training phase with reinforcement feedback and a test phase without feedback. Mid-frontal theta (4-8 Hz) activity serves as an index of cognitive control and is modulated by cortical dopamine.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003509

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Trn Stim`	1344	short	raw	Training phase stimulus (congruent/incongruent)
`Test Stim`	1008	short	raw	Test phase stimulus (congruent/incongruent)
`correct`	336	short	raw	Correct response marker
`incorrect`	334	short	raw	Incorrect response / error marker
`Trn Resp`	335	short	raw	Training phase button-press response
`Test Resp`	335	short	raw	Test phase button-press response
`FB`	168	short	raw	Feedback stimulus in training phase

Event ID	Count (total)	Type	Meta Level	Meaning
Trn No Response	83	short	raw	Missing response in training phase
Test No Response	54	short	raw	Missing response in test phase
STATUS	84	admin	raw	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Trn Stim, Test Stim, STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Simon Conflict Classification (Congruent vs Incongruent)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.5 s, tmax = 1.0 s *, target_events = [Trn Stim, Test Stim]
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning
congruent	0	Stimulus position matches response direction (no conflict)
incongruent	1	Stimulus position conflicts with response direction
- Super Meta Notes: Derived from raw events [Trn Stim, Test Stim] by extracting subtype field (congruent/incongruent) from events.tsv. Kernel step: read events.tsv, filter rows where onset corresponds to Trn Stim or Test Stim, extract the 'trial_type' or 'stim_type' subtype column, map to binary 0/1 for congruent/incongruent.
- Evidence:
- DOI: 10.1016/j.neuropsychologia.2018.05.020 - This paper uses the modified Simon task to classify congruent vs incongruent trials using theta-band time-frequency features, showing theta activity is diminished in PD patients during conflict processing.
- Kernel Notes: Must join epoched EEG data with trial subtype labels from events.tsv. Verify congruent/incongruent column name from events.tsv headers.

Primary Task 2: Response Accuracy Classification (Correct vs Error)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [correct, incorrect]
- Label Route: Annotation
- Meta Level: raw

- Target Definition:
Event	y	Meaning
`correct`	1	Correct response
`incorrect`	0	Error/incorrect response
- Evidence:
- DOI: `10.1016/j.neuropsychologia.2018.05.020` - This paper classifies correct vs error trials using EEG features, analyzing post-error and post-correct dynamics as markers of cognitive control in Parkinson's disease.
- Kernel Notes: Epoch relative to response markers. Exclude 'Trn No Response' and 'Test No Response' trials from training set.

Normal Task 1: Correct/Incorrect Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Correct and incorrect counts are approximately balanced (~336 vs ~334 total). Epoch on the response marker; exclude no-response trials.

Event	y	Meaning
`correct`	1	Correct response
`incorrect`	0	Error response

Data Quality Notes

- 67 channels including VEOG and positional channels (X, Y, Z); exclude non-EEG channels before analysis.
- Two sessions per subject for many participants; verify session consistency.
- `Trn Stim` and `Test Stim` have detailed subtypes in events.tsv (congruent/incongruent) that can support conflict classification.

References

- Reference: `Mid-frontal theta activity is diminished during cognitive control in Parkinson's disease. Neuropsychologia, 2018. DOI: 10.1016/j.neuropsychologia.2018.05.020`

076. openneuro_ds003516 - EEG: Attended Speaker Paradigm (Own Name in Ignored Stream)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003516` | Year: 2022 | URL: https://openneuro.org/datasets/ds003516
Category: High-level Mental | Subjects: 25 | Files (Completed): 25
Channels: 49 | Sampling Rate: 500 Hz

Paradigm

Participants performed a cocktail-party selective auditory attention task with two competing speakers. They listened to audio books from one speaker (attended stream) while their own name was occasionally presented in the ignored stream. This paradigm investigates the neural correlates of selective attention and the salience of self-referential stimuli (own name) in the unattended auditory channel. No discrete trial-level event markers were recorded in the EEG data.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003516

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
-	-	-	raw	No event markers detected; continuous EEG recording

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Own-Name P3 Response Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 5.0 s, overlap = 0.0
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

| Event | y | Meaning |

|-------|---|---------|
| `higher_name_intelligibility` | 1 | Name presented in higher intelligibility condition |
| `lower_name_intelligibility` | 0 | Name presented in lower intelligibility condition |
- Super Meta Notes: Derived from experimental design log (name embedment times and block onsets from stimulus creation files). Block-level condition labels (higher/lower intelligibility) assigned post-hoc based on experimental design. Name onset times known from stimulus creation, not from raw EEG event stream. Kernel step: Load experimental log to obtain name onset timestamps per block, then segment EEG using these derived timings.
- Evidence:
- DOI: 10.3389/fnins.2021.643705 - Uses dataset ds003516 to investigate neural tracking of attended/ignored speech envelopes and P3 responses (500-1200ms window) to participants' own names embedded in ignored speech, analyzing 5-s segments before/after name occurrences.
- Kernel Notes: Since no raw event markers exist in this dataset, kernel must: (1) read experimental design logs to obtain name onset timestamps per block, (2) derive condition labels (higher/lower intelligibility) from block metadata, (3) segment EEG at name onset ±2.5s to create 5-s epochs, (4) label epochs based on block intelligibility condition. Verify timing derivation from events.tsv or accompanying stimulus timing files.

Primary Task 2: Speech Envelope Tracking (Attended vs Ignored)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 5.0 s, overlap = 0.0
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`attended_speech_envelope`	0-1	Cross-correlation magnitude with attended speaker speech envelope
`ignored_speech_envelope`	0-1	Cross-correlation magnitude with ignored speaker speech envelope
- Super Meta Notes: Derived from experimental design log (speech envelope files and block onsets). Attended/ignored stream assignment known from block structure. Cross-correlation computed between EEG and speech envelope waveforms (0-500ms time lag). Kernel step: Compute neural speech tracking by cross-correlating EEG channels with provided speech envelope stimuli.
- Evidence:
- DOI: 10.3389/fnins.2021.643705 - Investigates neural tracking of attended and ignored speech envelopes using cross-correlation magnitude analysis in 5-s segments.
- Kernel Notes: Kernel must: (1) obtain speech envelope stimuli files (attended and ignored streams), (2) segment continuous EEG into 5-s non-overlapping windows aligned to experimental blocks, (3) compute cross-correlation between each EEG segment and both speech envelopes (0-500ms lag), (4) output correlation magnitude as continuous regression target. Requires access to stimulus audio files for envelope extraction.

Normal Task 1: Continuous EEG segmentation

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no labels, for unsupervised or feature extraction))

- Notes: No trial-level annotations available; treat as continuous listening EEG. Suitable for neural tracking analyzes or subject-level classification using demographic labels.

Data Quality Notes

- No event markers present; cannot epoch by stimulus onset.
- Continuous EEG during selective attention task; audio stimuli timing not embedded in EEG annotations.

References

- Reference: `Bjorn Holtze (January, 2021). Own-name salience in selective auditory attention.`

077. openneuro_ds003518 - EEG: Simon Conflict with Reinforcement + Cabergoline Challenge

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003518` | Year: 2021 | URL: https://openneuro.org/datasets/ds003518
Category: High-level Mental | Subjects: 110 | Files (Completed): 137
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

This study examines whether conflict acts as an implicit cost in reinforcement learning using a modified Simon task with pharmacological manipulation. Healthy participants performed a Simon conflict task across training and test phases, with a double-blind Cabergoline (dopamine D2/D3 agonist) challenge administered to examine dopaminergic modulation. The training phase included reinforcement feedback to shape behavior; the test phase probed learned conflict avoidance. Mid-frontal theta and error-related negativity are key EEG measures.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003518

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Trn Stim`	2192	short	raw	Training phase stimulus (congruent/incongruent Simon)
`Test Stim`	1644	short	raw	Test phase stimulus
`correct`	548	short	raw	Correct response outcome
`incorrect`	547	short	raw	Incorrect response / error outcome
`Trn Resp`	547	short	raw	Training phase button-press response
`Test Resp`	548	short	raw	Test phase button-press response
`FB`	274	short	raw	Feedback stimulus in training phase
`Trn No Response`	136	short	raw	Missing response in training phase

Event ID	Count (total)	Type	Meta Level	Meaning
`Test No Response`	55	short	raw	Missing response in test phase
`STATUS`	137	admin	raw	System status marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Trn Stim`, `Test Stim`, `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Correct/Incorrect Response Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.5 s, tmax = 5.0 s, target_events = [`correct`, `incorrect`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`correct`	1	Correct response
`incorrect`	0	Error/incorrect response

- Evidence:
- DOI: 10.1038/ncomms6394 - Binary classification of correct vs incorrect responses during Simon conflict task, with epochs extracted from -1500 to +5000 ms around stimulus cues
- Kernel Notes: Epoch window matches paper's cue epoching parameters; baseline correction should use -300 to -200 ms pre-cue as specified in the original analysis. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Congruent vs Incongruent Stimulus Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.5 s, tmax = 5.0 s, target_events = [`Trn Stim`, `Test Stim`]
- Label Route: `stim`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Trn Stim`	N/A	Training phase Simon stimulus (congruent or incongruent)
`Test Stim`	N/A	Test phase Simon stimulus (congruent or incongruent)

- Evidence:

- DOI: 10.1038/ncomms6394 - Epochs centered on congruent/incongruent stimuli to analyze frontal theta dynamics and conflict cost effects
- Kernel Notes: Stimulus-locked epochs; the actual congruent/incongruent label must be derived from stimulus properties (e.g., stimulus file or additional metadata). Verify event ID mapping from events.tsv for stimulus type annotation.

Primary Task 3: Feedback-locked Response Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.3 s, tmax = 0.8 s, target_events = [FB]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
FB	N/A	Feedback stimulus (reward/punishment) in training phase

- Evidence:
- DOI: 10.1038/ncomms6394 - Feedback-locked epochs at 200-300 ms post-feedback; feedback theta analysis in 4-8 Hz from 200-500 ms post-feedback
- Kernel Notes: Feedback-locked epoch window aligns with paper's feedback-locked analysis window; reward vs punishment labels may need derivation from additional stimulus metadata.

Data Quality Notes

- 64-channel EEG with 110 subjects across cabergoline and placebo conditions.
- Drug condition information in participants.tsv; use as a grouping variable for stratified analysis.
- Some subjects may have two sessions; verify file-to-subject mapping.

References

- Reference: `Conflict acts as an implicit cost in reinforcement learning. Nature Communications, 2014. DOI: 10.1038/ncomms6394`

078. openneuro_ds003519 - EEG: Visual Working Memory + Cabergoline Challenge

Source: analysis.md

Dataset: `openneuro_ds003519` | Year: 2021 | URL: https://openneuro.org/datasets/ds003519
Category: High-level Mental | Subjects: 27 | Files (Completed): 54
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants performed a visuospatial working memory task under a double-blind Cabergoline (dopamine D2/D3 agonist) drug challenge to investigate pharmacological modulation of working memory processing. The task involved memorizing spatial locations of stimuli followed by a probe stimulus requiring a match/non-match judgment. Distractor stimuli were presented during the retention interval. The study examines individual differences in working memory capacity and dopaminergic modulation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003519

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stim Onset`	162	short	raw	Study stimulus onset (to-be-remembered)
`Question Mark Probe`	108	short	raw	Probe stimulus requiring match judgment
`Response`	108	short	raw	Participant button-press response
`STATUS`	54	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.

- Candidate Trigger Labels: `Stim Onset`, `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Visual Working Memory

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://doi.org/10.3758/s13415-018-0584-6

Normal Task 1: Stimulus presentation ERP (single class, can be used for feature extraction)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Study stimulus vs probe comparison captures encoding vs retrieval phases. Two sessions per subject (drug vs placebo); treat session as covariate.

Event	y	Meaning
`Stim Onset`	0	Memory stimulus presentation
`Question Mark Probe`	1	Probe stimulus presentation
## Data Quality Notes		

- 54 files for 27 subjects; two sessions per subject (drug/placebo crossover design).
- Drug condition information should be in participants.tsv or scans.tsv.
- Limited event types; primary utility is ERP characterization of encoding and retrieval phases.

References

- Reference: `Dopamine D2 agonist affects visuospatial working memory distractor interference depending on individual differences in baseline working memory span. Cognitive, Affective, & Behavioral Neuroscience, 2018. DOI: 10.3758/s13415-018-0584-6`

079. openneuro_ds003523 - EEG: Visual Working Memory in Acute TBI

Source: analysis.md

Dataset: openneuro_ds003523 | Year: 2021 | URL: https://openneuro.org/datasets/ds003523
Category: High-level Mental | Subjects: 62 | Files (Completed): 219
Channels: 65 | Sampling Rate: 500 Hz

Paradigm

A visual working memory task with intermittent mind-wandering probes was administered longitudinally to healthy controls and patients with sub-acute mild traumatic brain injury (mTBI). Participants memorized visual stimuli during encoding, maintained them across a delay period, and responded to probe stimuli. Occasional mind-wandering probes assessed attentional state. The dataset spans multiple time points post-injury to track recovery of executive functioning, with ERP markers (P3b) serving as neurophysiological outcome measures.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003523

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stim Onset`	657	short	raw	Visual study stimulus onset
`Question Mark Probe`	434	short	raw	Working memory probe stimulus
`Response`	434	short	raw	Participant button-press response
`Mind Wandering`	406	short	raw	Mind-wandering probe response marker
`Mind Wandering Probe`	217	short	raw	Mind-wandering probe stimulus
`STATUS`	219	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Stim Onset, STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Visual Working Memory

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://doi.org/10.1016/j.neuropsychologia.2019.107125

Normal Task 1: Stimulus type classification (binary)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Longitudinal dataset; multiple sessions per subject across recovery timepoints. Mind-wandering probes can support on-task vs off-task classification with careful labeling.

Event	y	Meaning
Stim Onset	0	Memory stimulus presentation (encoding phase)
Question Mark Probe	1	Probe stimulus presentation (retrieval stage)
## Data Quality Notes		

- Longitudinal design: multiple sessions per subject at different post-injury timepoints.
- Clinical group and time-point information in participants.tsv; use as stratification variables.
- Mind Wandering Probe and Mind Wandering events can be used to classify attentional state.

References

- Reference: ERPs predict symptomatic distress and recovery in sub-acute mild traumatic brain injury. Neuropsychologia, 2019. DOI: 10.1016/j.neuropsychologia.2019.107125

080. openneuro_ds003602 - Childhood Sexual Abuse and Problem Drinking in Women: Neurobehavioral Mechanisms

Source: analysis.md

Dataset: `openneuro_ds003602` | Year: 2022 | URL: https://openneuro.org/datasets/ds003602
Category: High-level Mental | Subjects: 234 | Files (Completed): 699
Channels: 35 | Sampling Rate: 1000 Hz

Paradigm

This study investigates neurobehavioral mechanisms linking childhood sexual abuse (CSA) history and problem drinking in women using affective picture viewing and startle paradigms. Participants viewed negative, neutral, and positive images including alcohol-related and non-alcohol-related content while EEG, EMG (startle), and physiological measures were recorded. Resting-state blocks (eyes open/closed) were also included. The study examines how trauma history modulates emotional reactivity and appetitive motivation for alcohol cues.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003602

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`NegStim`	230	short	raw	Negative picture stimulus onset
`NeuStim`	230	short	raw	Neutral picture stimulus onset
`PosStim`	230	short	raw	Positive picture stimulus onset
`NegAlcStimWithoutStartle`	230	short	raw	Negative alcohol picture (no startle probe)
`NeuAlcStimWithoutStartle`	230	short	raw	Neutral alcohol picture (no startle probe)
`PosAlcStimWithoutStartle`	230	short	raw	Positive alcohol picture (no startle probe)
`NegNonAlcStimWithoutStartle`	230	short	raw	Negative non-alcohol picture (no startle probe)

Event ID	Count (total)	Type	Meta Level	Meaning
NeuNonAlcStimWithoutStartle	230	short	raw	Neutral non-alcohol picture (no startle probe)
PosNonAlcStimWithoutStartle	230	short	raw	Positive non-alcohol picture (no startle probe)
NegAlcP1000/P2000/P3000	230 each	short	raw	Negative alcohol picture with startle probe at 1/2/3 s
NeuAlcP1000/P2000/P3000	230 each	short	raw	Neutral alcohol picture with startle probe at 1/2/3 s
PosAlcP1000/P2000/P3000	230 each	short	raw	Positive alcohol picture with startle probe at 1/2/3 s
NegNonAlcP1000/P2000/P3000	230 each	short	raw	Negative non-alcohol picture with startle probe
NeuNonAlcP1000/P2000/P3000	230 each	short	raw	Neutral non-alcohol picture with startle probe
PosNonAlcP1000/P2000/P3000	230 each	short	raw	Positive non-alcohol picture with startle probe
EyesOpen1 / EyesOpen2	234 each	rest	raw	Eyes-open resting-state block marker
EyesClose1 / EyesClose2	234 each	rest	raw	Eyes-closed resting-state block marker
EndOfRest	234	admin	raw	End of resting-state block
ignore	458	admin	raw	Practice/ignored trial marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `NegStim`, `NeuStim`, `PosStim`, `NegAlcStimWithoutStartle`, `NeuAlcStimWithoutStartle`, `PosAlcStimWithoutStartle`, `NegNonAlcStimWithoutStartle`, `NeuNonAlcStimWithoutStartle`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Experiment

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds003602 DOI: https://doi.org/10.18112/openneuro.ds003602.v1.0.1

Normal Task 1: Emotional valence classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Balanced across valence conditions (~230 per category). Large dataset (699 files). Alcohol-related vs non-alcohol-related stimulus distinction available for secondary analysis.

Event	y	Meaning
`NegStim`	0	Negative pictures
`NeuStim`	1	Neutral image
`PosStim`	2	Positive pictures
## Data Quality Notes		

- Large dataset with 30+ distinct event types; group by valence (Neg/Neu/Pos) and content (Alc/NonAlc) for tractable analysis.
- Eyes Open/Closed markers are resting-state block delimiters, not trial-level stimuli.
- `ignore` and `EndOfRest` are administrative markers; exclude from epoching.

References

- Reference: `Childhood Sexual Abuse and problem drinking in women: Neurobehavioral mechanisms. OpenNeuro ds003602.`

081. openneuro_ds003626 - Inner Speech

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003626 | Year: 2022 | URL: https://openneuro.org/datasets/ds003626
Category: High-level Mental | Subjects: 10 | Files (Completed): 30
Channels: 137 | Sampling Rate: 1024 Hz

Paradigm

Participants performed inner speech (imagined speech), pronounced speech, and visualization tasks involving four directional commands (Up, Down, Right, Left) to explore the neural correlates of the 'inner voice' phenomenon for potential BCI applications. High-density EEG (137 channels) was recorded during all three modalities of speech production/imagery. No discrete event markers were captured in the EEG trigger channel; paradigm timing is encoded externally.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003626

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
-	-	-	raw	No event markers detected; continuous EEG recording

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Inner speech command decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`

- Epoch Parameters: tmin = 0 s , tmax = 3.0 s , target_events = [Up, Down, Left, Right] (verify exact event IDs from events.tsv)
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning
Up	0	Upward directional command
Down	1	Downward directional command
Left	2	Left directional command
Right	3	Right directional command
- Super Meta Notes: Derived from raw experimental protocol timing in events.tsv (no EEG trigger markers present). The dataset contains three modalities (inner speech, pronounced speech, visualization) all using the same four directional commands. Kernel must: (1) read trial onsets from events.tsv, (2) match stimulus labels (Up/Down/Left/Right) to derive class labels, (3) optionally filter by modality if separate classification tasks are desired.
- Evidence:
- DOI: https://doi.org/10.1038/s41597-022-01536-7 - Confirms dataset contains high-density EEG for classifying directional commands (Up/Down/Left/Right) across inner speech, pronounced speech, and visualization modalities for BCI applications.
- Kernel Notes: Verify event IDs in events.tsv match the expected labels (Up, Down, Left, Right). Extract tmin/tmax from the actual trial structure in events.tsv. Consider separate classification per modality (inner vs pronounced vs visualization) or multi-class across all conditions. The original analysis.md used epoch_by_segmentation_hdf5 but this is superseded by epoch_by_event_hdf5 since events.tsv contains trial timing.

Normal Task 1: Continuous EEG segmentation

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: stim
- Target Definition: N/A (null (no tag, needs to be restored from events.tsv or external annotation))
- Notes: No EEG trigger markers detected. If events.tsv contains trial onset annotations, epoching by event may be possible. High-density 137-channel EEG suitable for source imaging of inner speech networks. Only 10 subjects.

Data Quality Notes

- No EEG trigger markers present; trial timing must be reconstructed from events.tsv or behavioral logs.
- Small sample size (10 subjects, 30 files across 3 conditions).
- High-density 137-channel EEG; requires careful re-referencing and ICA for artifact removal.

References

- Reference: Inner Speech Dataset. OpenNeuro ds003626.

082. openneuro_ds003638 - EEG: Electrophysiological Biomarkers of Behavioral Dimensions from Cross-Species Paradigms

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003638 | Year: 2021 | URL: https://openneuro.org/datasets/ds003638
Category: High-level Mental | Subjects: 57 | Files (Completed): 57
Channels: 73 | Sampling Rate: 512 Hz

Paradigm

This study employs three cross-species behavioral paradigms (5-Choice Continuous Performance Task [5CCPT], Probabilistic Reward/Punishment task [PRBP], and Pavlovian-to-Instrumental Transfer [PST]) to identify electrophysiological biomarkers associated with various behavioral dimensions including attention, impulsivity, and reward sensitivity. All 57 subjects completed all three paradigms in a single session. Event markers are encoded as numeric codes with paradigm suffix (e.g., "N 5CCPT", "N PRBP", "N PST").

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003638

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
1 5CCPT-80 5CCPT	57 each	short	raw	5-Choice CPT numeric trigger codes
116 PRBP, 117 PRBP	57 each	short	raw	Probabilistic reward/punishment task trigger codes
31 PST, 32 PST, 39 PST	56-57 each	short	raw	Pavlovian-to-Instrumental Transfer task trigger codes
STATUS	57	admin	raw	System status marker

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: 5C-CPT Target vs Non-Target Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s , tmax = +1.0 s , target_events = [`41 5CCPT`, `42 5CCPT`, `43 5CCPT`, `44 5CCPT`, `45 5CCPT`, `46 5CCPT`, `47 5CCPT`, `48 5CCPT`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`41 5CCPT`	0	5CCPT correct rejection (nontarget)
`42 5CCPT`	0	5CCPT correct rejection (nontarget)
`43 5CCPT`	1	5CCPT hit (target)
`44 5CCPT`	1	5CCPT hit (target)
`45 5CCPT`	0	5CCPT correct rejection (nontarget)
`46 5CCPT`	0	5CCPT correct rejection (nontarget)
`47 5CCPT`	1	5CCPT hit (target)
`48 5CCPT`	1	5CCPT hit (target)

- Evidence:
- DOI: 10.1038/s41398.021.01562-w - EEG time-frequency analysis comparing target vs nontarget stimuli in 5C-CPT paradigm from OpenNeuro ds003638
- Kernel Notes: Verify event ID mapping from events.tsv (41-48 5CCPT codes correspond to target/nontarget response types). Epoch: cue-locked (tmin=-0.5s, tmax=+1.0s). Baseline: -300 to -200 ms.

Primary Task 2: PRBP High vs Low Probability Reward Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s , tmax = +1.0 s , target_events = [`116 PRBP`, `117 PRBP`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`116 PRBP`	0	Low probability reward condition
`117 PRBP`	1	High probability reward condition

- Evidence:
- DOI: 10.1038/s41398.021-01562-w - Analysis of high vs low probability reward trials in PRBP paradigm comparing spectral biomarkers
- Kernel Notes: Verify event ID mapping from events.tsv (116/117 PRBP codes map to high/low probability reward). Epoch: cue-locked (tmin=-0.5s, tmax=+1.0s). Baseline: -300 to -200 ms.

Primary Task 3: PST Early vs Late Trial Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`

- Epoch Parameters: tmin = +0.25 s , tmax = +0.55 s , target_events = [31 PST, 32 PST, 39 PST]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
31 PST	0	Early trial (first 50s of task block)
32 PST	1	Late trial (last 50s of task block)
39 PST	?	PST feedback event (verify label from events.tsv)
- Evidence:
- DOI: 10.1038/s41398.021-01562-w - Analysis comparing early vs late trials in PST paradigm for effort-related biomarkers
- Kernel Notes: Verify event ID mapping from events.tsv (31/32 PST codes correspond to early/late trials). Epoch: feedback-locked (tmin=+0.25s, tmax=+0.55s). 1-s epochs averaged for first 50s vs last 50s.

Primary Task 4: 5C-CPT Response-Locked Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.5 s, tmax = 0 s, target_events = [41 5CCPT, 42 5CCPT, 43 5CCPT, 44 5CCPT, 45 5CCPT, 46 5CCPT, 47 5CCPT, 48 5CCPT]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
41 5CCPT	0	5CCPT correct rejection response
42 5CCPT	0	5CCPT correct rejection response
43 5CCPT	1	5CCPT commission error response
44 5CCPT	1	5CCPT commission error response
45 5CCPT	0	5CCPT correct rejection response
46 5CCPT	0	5CCPT correct rejection response
47 5CCPT	1	5CCPT commission error response
48 5CCPT	1	5CCPT commission error response
- Evidence:
- DOI: 10.1038/s41398.021-01562-w - Response-locked EEG analysis (tmin=-0.5s, tmax=0s) for 5C-CPT comparing correct vs error responses
- Kernel Notes: Verify event ID mapping from events.tsv. Response-locked epoch (tmin=-0.5s, tmax=0s relative to button press).

Normal Task 1: 5CCPT stimulus classification (need to decode event meaning)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Numeric trigger codes require decoding from events.tsv or dataset README. Events with suffix 5CCPT correspond to the 5-Choice CPT; PRBP to reward/punishment; PST to Pavlovian transfer. Exact meaning of each numeric code must be verified from dataset documentation.

Event	y	Meaning
`41 5CCPT`	0	5CCPT correct response (need to confirm from events.tsv)
`43 5CCPT`	1	5CCPT commission error (need to confirm from events.tsv)

Data Quality Notes

- 30+ distinct numeric trigger types; decode from events.tsv before epoching.
- All three paradigms recorded in a single 57-file session; paradigm blocks need to be separated by trigger suffix.
- `STATUS` is a single admin marker per file; exclude from epoching.

References

- Reference: `Electrophysiological biomarkers of behavioral dimensions from cross-species paradigms. OpenNeuro ds003638.`

083. openneuro_ds003655 - VerbalWorkingMemory

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003655` | Year: 2023 | URL: https://openneuro.org/datasets/ds003655
Category: High-level Mental | Subjects: 26 | Files (Completed): 156
Channels: 21 | Sampling Rate: 500 Hz

Paradigm

A modified Sternberg working memory task requiring either simple retention or mental manipulation (alphabetization) of letter sequences at three load levels (5, 6, or 7 letters). Participants encoded a letter sequence, maintained it across a delay period, then responded to a probe letter. The mental manipulation condition (alphabetization) places additional executive demands on working memory compared to simple retention. EEG measures during encoding, delay, and probe phases capture load-dependent neural activity.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003655

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`encoding`	936	short	raw	Letter sequence encoding phase onset
`delay period`	936	long	raw	Working memory retention/manipulation delay period
`probe`	936	short	raw	Probe letter presentation
`start of the baseline period`	936	short	raw	Baseline period onset (pre-trial)
`start of the trial`	156	short	raw	Trial start marker
`presentation of the task instruction`	156	short	raw	Task instruction display onset
`Response`	1807	short	raw	Participant button-press response
`STATUS`	156	admin	raw	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

MISC Label Summary

- Route: misc
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Retention vs Manipulation classification in verbal working memory

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -14.2 s , tmax = 2.2 s , target_events = [encoding, delay period, probe]
- Label Route: Annotation | misc
- Meta Level: raw
- Target Definition:

Event	y	Meaning
encoding with condition 5R/6R/7R	0	Retention (simple keep) condition
delay period with condition 5R/6R/7R	0	Retention (simple keep) condition
probe with condition 5R/6R/7R	0	Retention (simple keep) condition
encoding with condition 5M/6M/7M	1	Manipulation (alphabetization) condition
delay period with condition 5M/6M/7M	1	Manipulation (alphabetization) condition
probe with condition 5M/6M/7M	1	Manipulation (alphabetization) condition

- Evidence:
- DOI: 10.1111/ejn.15457 - Exploratory reanalysis of openneuro_ds003655 examining oscillatory activity (theta, alpha, beta) differences between retention and manipulation conditions during encoding and delay phases of verbal working memory with 5/6/7 letter load
- Kernel Notes: Parse trial_type column to extract condition (R=retention, M=manipulation) and load level (5/6/7). Epochs from -14200 to 2200 ms relative to probe onset as described in paper; delay period divided into Delay1 (-6000 to -3000 ms) and Delay2 (-3000 to 0 ms) for sub-analysis. Baseline period -11500 to -10500 ms for correction.

Normal Task 1: Delay period analysis - keep vs operation (binary, long event)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: Annotation

- Target Definition: Event-to-label mapping table below.
- Notes: Delay periods typically last several seconds (>=2 s). Event subtypes encode both load level (5/6/7) and condition (R=Retention, M=Manipulation). Load level can be used as a regression target or ordinal label.

Event	y	Meaning
`delay period: 5R / 6R / 7R`	0	Simple retention (Retention) condition
`delay period: 5M / 6M / 7M`	1	Mental arithmetic operation (Manipulation) conditions

Normal Task 2: Detect stimulus ERP (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Probe events are short (<1 s); suitable for ERP epoching. Match/non-match accuracy from Response events can be used as trial-level correctness labels.

Event	y	Meaning
`probe: 5R / 6R / 7R`	0	Keep condition detection
`probe: 5M / 6M / 7M`	1	Operating condition detection

Data Quality Notes

- 21-channel EEG; limited spatial resolution but sufficient for frontal working memory analyses.
- Event subtypes encoded in trial_type column (e.g., "delay period: 5M"); parse carefully.
- Multiple sessions per subject (156 files / ~26 subjects ~ 6 files each).

References

- Reference: `VerbalWorkingMemory EEG dataset. OpenNeuro ds003655.`

084. openneuro_ds003751 - Dataset on Emotion with Naturalistic Stimuli (DENS)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003751` | Year: 2023 | URL: https://openneuro.org/datasets/ds003751
Category: High-level Mental | Subjects: 40 | Files (Completed): 40
Channels: 132 | Sampling Rate: 250 Hz

Paradigm

Participants viewed naturalistic multimedia video clips (film excerpts, documentaries) to elicit a range of emotional states while EEG, ECG, and EMG data were simultaneously recorded. After each video clip, participants provided continuous subjective ratings of valence, arousal, dominance, liking, familiarity, and emotional category. The dataset supports multimodal emotion recognition research using naturalistic (ecologically valid) stimuli rather than standardized posed expressions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003751

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stm`	40	long	raw	Stimulus (video clip) onset marker
`IBEG`	40	long	raw	Image/video block beginning
`base`	40	long	raw	Baseline period onset
`bend`	40	long	raw	Baseline period end
`IEND`	39	long	raw	Image/video block end
`vlnc`	40	short	raw	Valence rating event
`arsl`	40	short	raw	Arousal rating event
`dmns`	40	short	raw	Dominance rating event
`lkng`	40	short	raw	Liking rating event
`relv`	40	short	raw	Relevance rating event
`fmrt`	40	short	raw	Familiarity rating event

Event ID	Count (total)	Type	Meta Level	Meaning
cate	40	short	raw	Emotion category rating event
clic	40	short	raw	Mouse click rating response
trno	40	short	raw	Trial number marker
fixL	40	short	raw	Fixation cross onset
quiz	15	short	raw	Comprehension quiz question
qdon	14	short	raw	Quiz done marker
stop	37	admin	raw	Stop/pause marker
puse	34	admin	raw	Pause marker
rsrt	32	admin	raw	Restart marker
net	2	admin	raw	Network marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

MISC Label Summary

- Route: misc
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Multi-dimensional VAD/VA Emotion Classification from EEG

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 7.0 s, overlap = 0.25 s, target_events = [clic]
- Label Route: misc
- Meta Level: super
- Target Definition:
Event	y	Meaning
clic	0-7	8-class VAD: HighVHighAHighD, HighVHighALowD, HighVLowAHighD, HighVLowALowD, LowVHighAHighD, LowVHighALowD, LowVLowAHighD, LowVLowALowD
clic	0-3	4-class VA: HighVHighA, HighVLowA, LowVHighA, LowVLowA
- Super Meta Notes: Derived from raw events [vlnc, arsl, dmns] (valence, arousal, dominance ratings) by thresholding at 5 to create high/low binary for each dimension, then combining into 8-class VAD or 4-class VA labels. Kernel step: threshold continuous VAD ratings (1-9 scale) at 5, create binary high/low per dimension, map to quadrant classes.
- Evidence:

- DOI: 10.1007/978.3.031.27199.1.8 - Uses DENS dataset to classify emotional events into 8-class VAD space and 4-class VA space using CNN-GRU hybrid model, extracting 7-second EEG segments around subjectively marked emotional moments (mouse clicks) during naturalistic video stimuli.
- Kernel Notes: Extract epochs around each `clic` event (7 seconds), derive VAD labels from post-stimulus ratings (`vlnc`, `arsl`, `dmns`), apply threshold 5 to create high/low binary for each dimension, combine into VAD/VA class labels. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: 24-class Emotion Category Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s , tmax = 6.0 s , overlap = 0.0 *, target_events = [`clic`]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
clic	0-23	24 emotion categories (Amused, Sad, Fear, Happy, etc.)

- Super Meta Notes: Derived from raw event [`cate`] (emotion category rating) by mapping participant-selected emotion categories to numeric class indices. Kernel step: merge category labels from self-report with corresponding EEG epochs around `clic` events.
- Evidence:
- DOI: 10.1109/icccnt61001.2024.10725514 - Utilized DENS dataset with 128-channel EEG from 40 participants to classify 24 unique categorized emotions using optimized CNN-LSTM model achieving 93.28% accuracy.
- Kernel Notes: Verify event ID mapping from events.tsv for `cate` to 24-class mapping. Use 7-second epochs as reported in paper.

Primary Task 3: Binary Self-Assessment Dimension Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 7.0 s, overlap = 0.25 s, target_events = [`clic`]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
clic	0	Binary: low valence (rating < 5)
clic	1	Binary: high valence (rating >= 5)
clic	0	Binary: low arousal (rating < 5)
clic	1	Binary: high arousal (rating >= 5)
clic	0	Binary: low dominance (rating < 5)
clic	1	Binary: high dominance (rating >= 5)
clic	0	Binary: low liking (rating < 2.5)
clic	1	Binary: high liking (rating >= 2.5)
clic	0	Binary: low familiarity (rating < 2.5)
clic	1	Binary: high familiarity (rating >= 2.5)
clic	0	Binary: low relevance (rating < 2.5)
clic	1	Binary: high relevance (rating >= 2.5)

- Super Meta Notes: Derived from raw events [`vlnc`, `arsl`, `dmns`, `lkng`, `fmrt`, `relv`] by thresholding post-stimulus ratings at 5 (for VAD) or 2.5 (for liking/familiarity/relevance). Kernel step: for each dimension, threshold continuous rating to produce binary high/low labels.

- Evidence:
- DOI: 10.1101_2021.08.05.455187 - Uses DENS dataset to classify high vs low self-assessment dimensions (valence, arousal, dominance, liking, familiarity, relevance) from EEG features using k-NN classifier, with labels derived from post-stimulus behavioral ratings.
- Kernel Notes: Apply separate threshold (5 for VAD, 2.5 for L/F/R) to create binary classification per dimension. 7-second segment window confirmed.

Primary Task 4: Quadrant Valence-Arousal Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s , tmax = 6.0 s , overlap = 0.0 *, target_events = [`clic`]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
clic	0	HVHA (High Valence, High Arousal)
clic	1	HVLA (High Valence, Low Arousal)
clic	2	LVHA (Low Valence, High Arousal)
clic	3	LVLA (Low Valence, Low Arousal)

- Super Meta Notes: Derived from raw events [`vlnc`, `arsl`] (valence and arousal ratings, scale 1-9) by thresholding at 5 to create high/low binary per dimension, then mapping to quadrant classes. Kernel step: threshold valence at 5, threshold arousal at 5, combine into 4-quadrant VA class.
- Evidence:
- DOI: 10.1109/ACCESS.2023.3266804 - Uses DENS dataset with 465 emotional events extracted from 40 participants watching 60s video stimuli, where participants clicked to mark emotional moments and provided post-stimulus valence/arousal ratings to classify EEG epochs into 4-quadrant Valence-Arousal categories.
- Kernel Notes: Use 7-second epochs (1s before + 6s after click) at 250 Hz sampling. Threshold ratings at 5 for high/low classification into 4 VA quadrants.

Primary Task 5: 3-class Valence Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s , tmax = 6.0 s , overlap = 0.0 *, target_events = [`clic`]
- Label Route: `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
clic	0	Low valence (rating 1-4.5)
clic	1	Neutral valence (rating 4.5-5.5)
clic	2	High valence (rating 5.5-9)

- Super Meta Notes: Derived from raw event [`vlnc`] (valence rating, scale 1-9) by applying 3-class thresholds: <4.5 low, 4.5-5.5 neutral, >5.5 high. Kernel step: apply tiered threshold to valence ratings to produce 3-class valence labels.
- Evidence:
- DOI: 10.1109/ACCESS.2023.3266804 - Uses DENS dataset to classify EEG epochs into 3-class valence categories (negative, neutral, positive) using thresholds 4.5 and 5.5 on valence ratings.

- Kernel Notes: Use same 7-second epoch window as 4-class VA task. Apply tiered thresholds on valence ratings only.

Data Quality Notes

- 132-channel high-density EEG; requires careful preprocessing and ICA.
- Naturalistic stimuli with varying durations; trial lengths are not uniform.
- `stop`, `puse`, `rsrt`, `net` are administrative markers; exclude from epoching.
- Continuous emotion ratings stored separately; must be merged with EEG epochs for regression tasks.

References

- Reference: `Sudhakar Mishra and Md. Asif and Uma Shanker Tiwary and Narayanan Srinivasan (2023). Dataset on Emotion with Naturalistic Stimuli (DENS). OpenNeuro ds003751.`

085. openneuro_ds003753 - EEG: Probabilistic Learning with Affective Feedback: Exp #2

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003753 | Year: 2021 | URL: https://openneuro.org/datasets/ds003753
Category: High-level Mental | Subjects: 25 | Files (Completed): 25
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task involving probabilistic affective feedback to investigate the Reward Positivity (RewP) EEG component and its sensitivity to affective stimulus properties. Participants made binary choices (left/right button) between cue stimuli (CUE) associated with positive or negative affective feedback images. Following each choice, an affective feedback image (FB) was presented. The study examines whether the RewP tracks reward value beyond mere monetary reward, responding to affective liking of feedback stimuli.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003753

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
CUE	100	short	raw	Cue stimulus requiring binary choice
FB	100	short	raw	Affective feedback stimulus (liked/disliked image)
Left Button	25	short	raw	Left button press response
Right Button	25	short	raw	Right button press response
STATUS	25	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Affective Feedback Valence Classification (RewP ERP)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -1.0 s, tmax = 3.0 s, target_events = [FB]
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
FB (liked/rewarding)	1	Affective feedback images rated as liked (puppy images, positive valence)
FB (disliked/non-rewarding)	0	Affective feedback images rated as disliked (chair/negative images, negative valence)

- Super Meta Notes: Derived from raw event FB by decoding the trial_type column in events.tsv (puppy images = liked/rewarding, chair/negative images = disliked/non-rewarding). The Reward Positivity (RewP) ERP was quantified as mean amplitude 200-350ms at Cz for reward vs punishment conditions.
- Evidence:
- DOI: 10.3758/s13415.021.00950.5 - This paper analyzes EEG data from 25 participants performing a reinforcement learning task with probabilistic affective feedback (puppy vs chair images, points), examining how the Reward Positivity (RewP) ERP component tracks hedonic liking of feedback stimuli.
- Kernel Notes: 1) Extract epochs time-locked to FB event onset; 2) Apply baseline correction -200 to 0 ms before feedback onset; 3) Decode trial_type from events.tsv to derive liked/disliked labels (puppy = liked, chair/negative = disliked); 4) Verify event ID mapping from events.tsv (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Binary Choice Response Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.5 s , tmax = 1.5 s , target_events = [Left Button, Right Button]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
Left Button	0	Left button press response
Right Button	1	Right button press response

- Evidence:
- DOI: 10.3758/s13415.021.00950.5 - This paper presents two experiments (N=25 each) investigating the Reward Positivity (RewP) component in a probabilistic reinforcement learning task with affective feedback images, using EEG epochs time-locked to outcome cues and correlating RewP amplitude with individual pleasantness ratings of the feedback images.
- Kernel Notes: Epoch parameters (tmin=-0.5s, tmax=1.5s) are estimated defaults for motor response classification; verify from events.tsv if different timing is used.

Data Quality Notes

- Small dataset (25 subjects, 25 files).
- FB event subtype labels (affective value) must be decoded from events.tsv metadata.
- STATUS is a single admin marker per file; exclude from epoching.

References

- Reference: `The Reward Positivity is sensitive to affective liking. OpenNeuro ds003753.`

086. openneuro_ds003766 - A Resource for Assessing Dynamic Binary Choices in the Adult Brain Using EEG and Mouse-Tracking

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003766` | Year: 2024 | URL: https://openneuro.org/datasets/ds003766
Category: High-level Mental | Subjects: 31 | Files (Completed): 217
Channels: 129 | Sampling Rate: 1000 Hz

Paradigm

Participants performed dynamic decision-making tasks including food preference choices and semantic judgments while high-density EEG and continuous mouse-tracking were recorded simultaneously. The study captures the real-time neural dynamics of binary choice formation, including evidence accumulation, conflict, and response selection. Numeric trigger codes (0200-0506 series) encode stimulus, response, and condition information. Boundary markers appear in some files from EEGLAB preprocessing.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003766

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`0400`	186	short	raw	Trigger code 0400 (stimulus/task event - decode from events.tsv)
`0500`	186	short	raw	Trigger code 0500 (stimulus/task event - decode from events.tsv)
`0401-0403`	124 each	short	raw	Stimulus variant trigger codes
`0503`, `0505`, `0506`	124 each	short	raw	Response/outcome trigger codes
`0501`	95	short	raw	Trigger code 0501
`0202-0221`	93 each	short	raw	Trial-phase trigger codes (0200 series)
`0210`	93	short	raw	Trigger code 0210
`boundary`	97	admin	raw	EEGLAB boundary marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Binary Choice Classification (Left vs Right)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 0.8 s, target_events = [`0500`, `0501`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`0500`	0	Left choice response (food preference task)
`0501`	1	Right choice response (food preference task)

- Evidence:
- DOI: 10.1038/s41597.022.01538.5 - SVM classification decoding participants' binary left/right choices from EEG epochs extracted -200ms to 1800ms, achieving accuracies of 0.67-0.77 across three choice tasks.
- Kernel Notes: The paper also uses stimulus markers (0400, 0401, 0402, 0403) for different task conditions. For image/word choice, 0400=animate left, 0401=animate right, 0402=inanimate left, 0403=inanimate right. The primary classification target is left vs right choice. Response markers 0502, 0504-0507 mentioned in paper are not present in this dataset's raw events.

Primary Task 2: Stimulus-Locked ERP Epoching (Cognitive Processing)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 0.8 s, target_events = [`0400`, `0401`, `0402`, `0403`]
- Label Route: `stim`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`0400`	0	Food stimulus (or animate left for image/word)
`0401`	1	Animate right (image/word task)
`0402`	2	Inanimate left (image/word task)
`0403`	3	Inanimate right (image/word task)

- Evidence:
- DOI: 10.1038/s41597.022.01538.5 - EEG epochs extracted from -200ms to 800ms locked to stimulus onset for ERP analysis.

- Kernel Notes: These stimulus markers encode the presented option(s). For food choice: 0400-series markers. For image/word semantic judgment: 0400=animate left, 0401=animate right, 0402=inanimate left, 0403=inanimate right.

Primary Task 3: Resting-State Segmentation

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.0
- Label Route: `Annotation`
- Target Definition: No specific event labels; continuous segments from resting-state recordings.
- Evidence:
- DOI: 10.1038/s41597.022.01538.5 - Resting-state data: 6s segments further divided into 2000ms epochs with baseline -100ms to 0ms.
- Kernel Notes: Resting-state epochs do not require target_events. Use segmentation on continuous EEG data. The dataset includes resting-state recordings as one of the three BIDS tasks.

Normal Task 1: Binary choice classification (need to decode event meaning)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numeric trigger codes must be decoded from events.tsv. The 0200-series codes likely encode trial-phase timing; 0400/0500-series likely encode stimulus presentation. 217 files across 31 subjects (~7 sessions each). `boundary` is an EEGLAB admin marker; exclude from epoching.

Event	y	Meaning
`0400`	0	Stimulus/Option A (decode from events.tsv)
`0500`	1	Stimulus/Option B (decode from events.tsv)

Data Quality Notes

- 30+ distinct numeric event codes; decode mapping required from events.tsv or dataset paper.
- `boundary` events are EEGLAB data-continuity markers; must be excluded from epoching.
- High-density 129-channel EEG at 1000 Hz; high computational requirements.
- Multiple sessions per subject; verify session consistency before pooling.

References

- Reference: `Chen, K., Wang, R., Huang, J., Gao, F., Yuan, Z., Qi, Y., & Wu, H. (2022). A resource for assessing dynamic binary choices in the adult brain using EEG and mouse-tracking. Scientific Data, 9(1), 416.`

087. openneuro_ds003774 - Music Listening Genre EEG Dataset (MUSIN-G)

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003774 | Year: 2022 | URL: https://openneuro.org/datasets/ds003774
Category: High-level Mental | Subjects: 20 | Files (Completed): 260
Channels: 129 | Sampling Rate: 1000 Hz

Paradigm

Twenty participants listened to 12 music tracks spanning various genres (classical, jazz, rock, etc.) with eyes closed while high-density EEG was recorded. After each track, participants provided subjective ratings for familiarity and enjoyment. The dataset supports EEG-based music genre classification and exploration of neural correlates of music preference and familiarity. No discrete event markers were embedded in the EEG recording.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003774

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
-	-	-	raw	No event markers detected; continuous music listening EEG

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Music Genre Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

| Event | y | Meaning |

|-------|---|---------|
genre_Deep_House	0	Deep House music genre
genre_Indie	1	Indie music genre
genre_Electronics	2	Electronics music genre
genre_New_Age	3	New Age music genre
genre_Electronic_Dance	4	Electronic Dance music genre
genre_Ambient	5	Ambient music genre
genre_Hindustani_Classical	6	Hindustani Classical music genre
genre_Indian_Semi_Classical	7	Indian Semi-Classical music genre
genre_Indian_Folk	8	Indian Folk music genre
genre_Soft_Jazz	9	Soft Jazz music genre
genre_Goth_Rock	10	Goth Rock music genre
genre_Progressive_Instrumental_Rock	11	Progressive Instrumental Rock music genre

- Super Meta Notes: Derived from file-level metadata (ses-01 to ses-12) associated with each recording. Genre labels are embedded in the BIDS task name per file. Kernel step: Extract genre from file-level sidecar or filename pattern, assign to all segments within that file.
- Evidence:
- DOI: 10.1016/j.dib.2022.108663 - Data in Brief paper describing MUSIN-G dataset with 12 music genres and 20 participants
- Kernel Notes: Since no discrete EEG event markers are present in the raw data, use sliding window segmentation across the continuous music listening EEG. Genre labels are file-level; apply same label to all segments from a given file. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Familiarity Rating Prediction

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.5
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
familiarity_rating	1-5	Familiarity rating (1=extremely familiar to 5=extremely unfamiliar)

- Super Meta Notes: Derived from post-task behavioral ratings collected via keypress after each song (paper term: 'fam'). Kernel step: Extract familiarity rating from events.tsv or file-sidecar, map to EEG segments from corresponding song.
- Evidence:
- DOI: 10.1016/j.dib.2022.108663 - Paper describes post-hoc familiarity ratings (1-5 scale) collected after each music track
- Kernel Notes: Familiarity ratings are collected per song; apply the same rating to all segments from that song's EEG recording.

Primary Task 3: Enjoyment Rating Prediction

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.5
- Label Route: Annotation
- Meta Level: super

- Target Definition:
Event	y	Meaning
`enjoyment_rating`	1-5	Enjoyment rating (1=extremely enjoyable to 5=extremely unenjoyable)
- Super Meta Notes: Derived from post-task behavioral ratings collected via keypress after each song (paper term: 'enj'). Kernel step: Extract enjoyment rating from events.tsv or file-sidecar, map to EEG segments from corresponding song.
- Evidence:
- DOI: 10.1016/j.dib.2022.108663 - Paper describes post-hoc enjoyment ratings (1-5 scale) collected after each music track
- Kernel Notes: Enjoyment ratings are collected per song; apply the same rating to all segments from that song's EEG recording.

Data Quality Notes

- No EEG trigger events present; music onset timing not embedded in EEG.
- 260 files likely correspond to individual music track recordings; genre label per file.
- High-density 129-channel EEG; high computational requirements.
- Eyes-closed listening; fewer motion artifacts but possible drowsiness.

References

- Reference: `Music Listening Genre EEG dataset (MUSIN-G). OpenNeuro ds003774.`

088. openneuro_ds003822 - EEG: Probabilistic Learning with Affective Feedback: Exp #1

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003822 | Year: 2021 | URL: https://openneuro.org/datasets/ds003822
Category: High-level Mental | Subjects: 25 | Files (Completed): 25
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task using probabilistic affective feedback to investigate the Reward Positivity (RewP) EEG component and its sensitivity to subjective reward value. Participants made binary choices (left/right button) between cue stimuli (CUE) associated with liked or disliked affective feedback images. Following each choice, a feedback cue (FB cue) predicted the affective image, then the full feedback (FB type) was presented. This is Experiment 1 in a two-experiment series (see ds003753 for Exp #2).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003822

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
CUE	100	short	raw	Cue stimulus requiring binary choice
FB cue	100	short	raw	Feedback cue (predicts upcoming affective image)
FB type	200	short	raw	Full affective feedback stimulus (liked/disliked)
Response Accurate	25	short	raw	Accurate/correct response marker
Response Inaccurate	25	short	raw	Inaccurate/incorrect response marker
Left Button	25	short	raw	Left button press
Right Button	25	short	raw	Right button press
STATUS	25	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for

`epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Reward Positivity (RewP) classification - feedback valence

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 3.0 s *, target_events = [`FB  cue`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
reward condition (puppy, green_+1)	1	Rewarding feedback (win condition)
punishment condition (chair, negative, yellow bar, red_-1)	0	Punishing feedback (avoid condition)

- Super Meta Notes: Derived from raw event `FB  cue` (outcome cue onset). Labels combine experimental conditions: win vs avoid (MOTIVATION) and reward vs punishment (VALENCE). The paper maps specific stimuli (puppy=1, green_+1=1, chair=0, negative=0, yellow_bar=0, red_-1=0) to binary reward/punishment labels. Kernel step: decode trial_type from events.tsv to identify which condition each FB cue corresponds to.
- Evidence:
- DOI: 10.3758/s13415.021.00950-5 - Epochs data from -1000 to 3000 ms around outcome cue onset, quantifies RewP at Cz (200-350 ms) for reward vs punishment classification
- Kernel Notes: Requires decoding trial_type from events.tsv to map individual trials to reward/punishment conditions. Baseline correction: -200 to 0 ms. Verify event ID mapping from events.tsv.

Primary Task 2: Feedback valence (liked vs disliked) classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 3.0 s *, target_events = [`FB  cue`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
liked feedback (puppy, sun, cloud)	1	Affective feedback rated as pleasant
disliked feedback (chair, negative_image)	0	Affective feedback rated as unpleasant

- Super Meta Notes: Derived from raw event `FB  cue`. Labels based on hedonic appraisal (liked vs disliked) from pre-task image ratings. Kernel step: map trial_type to liked/disliked categories using image rating data.
- Evidence:

- DOI: 10.3758/s13415.021.00950-5 - Examines whether RewP amplitudes correlate with hedonic appraisal of liked vs disliked feedback images
- Kernel Notes: Requires accessing pre-task image rating data to determine which images were rated as liked/disliked per subject. Baseline correction: -200 to 0 ms.

Primary Task 3: Picture vs point feedback type classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 3.0 s *, target_events = [`FB cue`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
picture feedback (puppy, sun, cloud, chair, negative_image)	1	Affective image feedback
point feedback (green_+1, yellow_bar, red_-1)	0	Numeric/bar point feedback

- Super Meta Notes: Derived from raw event `FB cue`. Labels distinguish picture (image) vs point (bar/number) feedback modality. Kernel step: map trial_type to picture/point categories.
- Evidence:
- DOI: 10.3758/s13415.021.00950-5 - Binary classification of feedback valence or image type (liked/pictures vs points) based on RewP ERP
- Kernel Notes: Baseline correction: -200 to 0 ms. RewP quantified at 200-350 ms window at Cz.

Normal Task 1: Feedback type classification (binary, subtype needs to be decoded from events.tsv)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `FB type` appears 200 times (2x more than CUE/FB cue = 100 each) suggesting FB type encodes both the feedback onset and a second event (e.g., image offset or rating). Decode subtype from events.tsv `trial_type` column. Sister dataset to ds003753 (Exp #2). 25 subjects only.

Event	y	Meaning
`FB type` (liked/rewarding)	1	Liked emotional feedback (needs to decode subtype from events.tsv)
`FB type` (disliked/non-rewarding)	0	Disliked emotional feedback

Data Quality Notes

- Small dataset (25 subjects, 25 files).
- `FB type` count (200) is 2x `CUE` count (100); likely encodes two event variants per trial; verify from events.tsv.
- `Response Accurate` and `Response Inaccurate` appear only 25 times each (once per file summary count); likely one per session.
- STATUS is a single admin marker per file; exclude from epoching.

References

- Reference: `The Reward Positivity is sensitive to affective liking. OpenNeuro ds003822.`

089. openneuro_ds003838 - EEG, pupillometry, ECG and photoplethysmography, and behavioral data in the digit span task

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003838 | Year: 2021 | URL: https://openneuro.org/datasets/ds003838/versions/1.0.0
Category: High-level Mental | Subjects: 86 | Files (Completed): 296
Channels: 2 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a classic digit span working memory task involving serial recall of digit sequences of 5, 9, or 13 digits, along with a separate eyes-closed resting state condition. Each digit in a sequence was marked with an event indicating the digit position, sequence length, and whether it was later correctly recalled or forgotten. The dataset also includes ECG/PPG physiological signals recorded simultaneously with EEG.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003838/versions/1.0.0

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`memory X/05 correct`	high	short	raw	Digit X in 5-item sequence, correctly recalled
`memory X/09 correct`	high	short	raw	Digit X in 9-item sequence, correctly recalled
`memory X/13 correct`	high	short	raw	Digit X in 13-item sequence, correctly recalled
`memory X/05 error`	moderate	short	raw	Digit X in 5-item sequence, forgotten
`memory X/09 error`	moderate	short	raw	Digit X in 9-item sequence, forgotten
`memory X/13 error`	high	short	raw	Digit X in 13-item sequence, forgotten
STATUS	148	admin	raw	System/hardware status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Correct vs Error Memory Encoding

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [memory X/05 correct, memory X/09 correct, memory X/13 correct, memory X/05 error, memory X/09 error, memory X/13 error]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
memory X/05 correct	0	Digit in 5-item sequence correctly recalled
memory X/09 correct	0	Digit in 9-item sequence correctly recalled
memory X/13 correct	0	Digit in 13-item sequence correctly recalled
memory X/05 error	1	Digit in 5-item sequence forgotten
memory X/09 error	1	Digit in 9-item sequence forgotten
memory X/13 error	1	Digit in 13-item sequence forgotten

- Evidence:
- DOI: 10.1038/s41597.022.01414.2 - dataset descriptor confirming event codes encode digit position, sequence length, and recall outcome (correct vs error) for binary classification
- DOI: 10.1111/psyp.14275 - EEG analysis epoching data from -1500 to 3500 ms relative to digit presentation, classifying correct vs error recall
- Kernel Notes: Aggregate all correct events (regardless of sequence length X or length 05/09/13) as class 0, and all error events as class 1. Use regex pattern memory .*/(05|09|13) (correct|error) to match all valid event IDs. Parameter tmin=-0.2s, tmax=1.0s confirmed by dataset context in paper 2; paper 4 uses -1.5s to 3.5s window but recommends tighter window for encoding-focused analysis.

Normal Task 1: Correct vs Error Memory Encoding (binary)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Events encode digit position, sequence length, and recall outcome. Note this dataset contains only 2-channel ECG/PPG data (not EEG cap channels); EEG data may be in a separate modality. STATUS is an

admin marker and must be excluded. Aggregate correct vs error across all sequence lengths for binary classification.

Event	y	Meaning
`memory .*/correct` (all correct events)	0	Digit correctly recalled later
`memory .*/error` (all error events)	1	Digit forgotten later

Data Quality Notes

- Only 2 channels (PPG + ECG) - this is a physiological signal dataset, not a scalp EEG cap
- EEG data may be stored separately; verify file naming convention
- Large number of event subtypes (~30 distinct labels); grouping by outcome (correct/error) is necessary

References

- Reference: `EEG, pupillometry, ECG and photoplethysmography, and behavioral data in the digit span task`

090. openneuro_ds003885 - Capacity for movement is an organisational principle in object representations: EEG data from Experiment 1

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003885 | Year: 2023 | URL: https://openneuro.org/datasets/ds003885
Category: High-level Mental | Subjects: 24 | Files (Completed): 48
Channels: 127 | Sampling Rate: 250 Hz

Paradigm

Participants performed a semantic classification task (alive vs. not alive) and a passive viewing task using RSVP streams of images to investigate the neural representations of object movement and aliveness. Images spanned animate and inanimate categories with varying degrees of movement capacity, presented at ~3 Hz in RSVP. The study aimed to decode movement-related object properties from EEG representational geometry.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003885

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`E 8`	~115,200	short	raw	RSVP image onset - condition 8
`E 9`	~115,200	short	raw	RSVP image onset - condition 9
`E 1`	~38,400	short	raw	RSVP image onset - condition 1
`E 2`	~38,400	short	raw	RSVP image onset - condition 2
`E 3`	~38,400	short	raw	RSVP image onset - condition 3
`E 5`	~38,400	short	raw	RSVP image onset - condition 5
`E 4`	~960	short	raw	Block boundary or special event

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for

`epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: 6-Class Object Category Classification (Raw Meta)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.3 s, tmax = 1.0 s, target_events = [E 1, E 2, E 3, E 5, E 8, E 9]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
E 1	0	Object category 1 (animal)
E 2	1	Object category 2 (plant)
E 3	2	Object category 3 (still natural)
E 5	3	Object category 5 (still artificial)
E 8	4	Object category 8 (moving artificial)
E 9	5	Object category 9 (moving natural)

- Evidence:
- DOI: 10.1016/j.neuroimage.2022.119517 - Uses EEG data from openneuro.ds003885 (Experiment 1) to decode object categories from neural responses using LDA classifiers, with epochs extracted from -0.3s to 1.0s relative to stimulus onset.
- Kernel Notes: Extract epochs time-locked to each event ID (E1-E9) and train a classifier to predict the 6-category label. Uses raw event markers directly as targets. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: 6-Class Semantic Category Classification (Super Meta)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.3 s, tmax = 1.0 s, target_events = [E 1, E 2, E 3, E 5, E 8, E 9]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
E 1	0	Living animals (whales)
E 2	1	Living plants (flowers)
E 3	2	Non-living still natural objects (rocks)
E 5	3	Non-living still artificial objects (cups)
E 8	4	Non-living moving artificial objects (trains)
E 9	5	Non-living moving natural objects (waterfalls)

- Super Meta Notes: Derived from raw event(s) [E 1, E 2, E 3, E 5, E 8, E 9] by mapping experimental stimulus IDs to semantic category labels based on the study's stimulus design (6 natural image categories). Kernel step: Load the stimulus-category mapping file, create label vector by mapping each epoch's event ID to its corresponding semantic category label.
- Evidence:
- DOI: 10.1038/s41598.025.27703.5 - Reanalyzes openneuro_ds003885 to investigate electrode density effects on decoding accuracy for six natural image categories.

- DOI: 10.1101_2025.07.07.663494 - Re-analyzes openneuro_ds003885 to test decoding performance across electrode densities, classifying EEG epochs into six natural image categories.
- Kernel Notes: After epoching by raw event IDs, apply stimulus-to-category mapping to derive semantic labels. Verify mapping file matches experimental design (alive vs. not-alive and movement capacity dimensions).

Data Quality Notes

- Data from derivatives/eeglab; preprocessed with EEGLAB
- 128 channels recorded but analysis.md reports 127 (one likely reference or EOG)
- Very consistent event counts across all 24 subjects

References

- Reference: Shatek, S. M., Robinson, A. K., Grootswagers, T., & Carlson, T. A. (2022). Capacity for movement is an organisational principle in object representations. NeuroImage, 261, 119517.

091. openneuro_ds003887 - Capacity for movement is an organisational principle in object representations: EEG data from Experiment 2

Source: analysis.md

Dataset: openneuro_ds003887 | Year: 2023 | URL: https://openneuro.org/datasets/ds003887
Category: High-level Mental | Subjects: 24 | Files (Completed): 48
Channels: 127 | Sampling Rate: 250 Hz

Paradigm

Participants performed time-pressured categorization of static images based on movement capacity and completed a passive viewing task using Rapid Serial Visual Presentation (RSVP). This second experiment extends Experiment 1 by testing whether movement capacity organization in neural object representations generalizes across different task contexts and classification criteria.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003887

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
E 8	~115,200	short	raw	RSVP image onset - condition 8
E 9	~115,200	short	raw	RSVP image onset - condition 9
E 1	~38,400	short	raw	RSVP image onset - condition 1
E 2	~38,400	short	raw	RSVP image onset - condition 2
E 3	~38,400	short	raw	RSVP image onset - condition 3
E 5	~38,400	short	raw	RSVP image onset - condition 5
E 4	~960	short	raw	Block boundary or special event

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5

- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Publisher-defined BIDS task: Movement

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds003887 DOI: https://doi.org/10.18112/openneuro.ds003887.v1.2.3

Normal Task 1: Movement Capacity Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Identical event structure to ds003885 (Experiment 1). E 8 and E 9 are the dominant categories (~4,800 each per subject). E 4 is rare (40/subject) and likely marks block boundaries. Binary E 8 vs E 9 classification (movement-capable vs not) is the primary task.

Event	y	Meaning
`E 1`	0	Object category 1
`E 2`	1	Object category 2
`E 3`	2	Object category 3
`E 5`	3	Object category 5
`E 8`	4	Object category 8 (movement-capable / animate)
`E 9`	5	Object category 9 (no movement / inanimate)
## Data Quality Notes		

- Data from derivatives/eeglab; preprocessed with EEGLAB
- 128-channel cap, 127 usable channels after reference removal
- Consistent event counts across all 24 subjects

References

- Reference: `Capacity for movement is a major organisational principle in object representations`

092. openneuro_ds003987 - EEG: Amphetamine trials 5CCPT and Probabilistic Learning

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003987 | Year: 2022 | URL: https://openneuro.org/datasets/ds003987
Category: High-level Mental | Subjects: 23 | Files (Completed): 69
Channels: 71 | Sampling Rate: 500 Hz

Paradigm

A double-blind, crossover pharmacological study investigating the effects of d-amphetamine on cognitive performance using two tasks: the 5-Choice Continuous Performance Task (5CCPT), which assesses sustained attention and impulsivity, and a Probabilistic Selection Task (PST), which measures reinforcement learning. EEG was recorded across three drug sessions (placebo, low-dose, high-dose amphetamine).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003987

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
21 5CCPT	69 files	short	raw	5CCPT event type 21 (likely stimulus onset)
22 5CCPT	69 files	short	raw	5CCPT event type 22
23 5CCPT	69 files	short	raw	5CCPT event type 23
25 5CCPT	69 files	short	raw	5CCPT event type 25
31 PST	69 files	short	raw	PST event type 31 (stimulus/feedback)
32 PST	69 files	short	raw	PST event type 32
39 PST	69 files	short	raw	PST event type 39
STATUS	69 files	admin	raw	System hardware status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Probabilistic Learning - Reward vs Punishment Feedback Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 1.0 s, target_events = [`31 PST`, `32 PST`, `39 PST`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`31 PST`	0	PST event type 31 (reward feedback)
`32 PST`	1	PST event type 32 (punishment feedback)
`39 PST`	*	PST event type 39 (neutral/other feedback) *

- Evidence:
- DOI: `10.1007/s00213.022.06082.z` - This paper analyzes EEG data from OpenNeuro ds003987 (n=23 humans) performing a probabilistic learning task under placebo/10mg/20mg d-amphetamine, epoching data around feedback onset (-500 to +1000ms) to examine reward positivity (RewP) ERP and delta-band spectral responses to reward vs punishment feedback.
- Kernel Notes: Map PST event IDs 31/32 to reward/punishment labels. Verify event ID mapping from events.tsv and dataset README. (replaces placeholder Primary Task 1 from original analysis.md)

Normal Task 1: 5CCPT Attention Task (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: STATUS must be excluded (admin event). Events are named with task suffix (5CCPT or PST). Full event decoding requires consulting the dataset README. 23 subjects x 3 sessions (drug doses) = 69 files. Drug dose can serve as a between-condition variable.

Event	y	Meaning
`21 5CCPT`	0	5CCPT condition 21 (e.g., target stimulus)
`22 5CCPT`	1	5CCPT condition 22
`23 5CCPT`	2	5CCPT condition 23

Data Quality Notes

- Large number of event types (30); full decoding requires dataset documentation
- Three recording sessions per subject (placebo/low/high amphetamine) - consider treating sessions independently
- STATUS is an admin marker and must be excluded

References

- Reference: `Electrophysiological biomarkers of behavioral dimensions from cross-species paradigms`

093. openneuro_ds004000 - Fribourg Ultimatum Game in Schizophrenia Study

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004000` | Year: 2022 | URL: https://openneuro.org/datasets/ds004000
Category: High-level Mental | Subjects: 43 | Files (Completed): 84
Channels: 132 | Sampling Rate: 2048 Hz

Paradigm

A social decision-making study in which patients with psychosis and healthy controls played the Ultimatum Game in both responder and proposer roles across multiple blocks. EEG was recorded while participants decided to accept or reject monetary offers from a human or computer partner, enabling investigation of fairness processing and social cognition differences in psychosis.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004000

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S 1`	42 files	short	raw	Stimulus onset type 1
`Stimulus/S 2`	42 files	short	raw	Stimulus onset type 2
`Stimulus/S 3`	42 files	short	raw	Stimulus onset type 3
`Stimulus/S 4`	42 files	short	raw	Stimulus onset type 4
`Stimulus/S 64`	42 files	short	raw	Stimulus onset - offer/condition marker
`Stimulus/S 96`	42 files	short	raw	Stimulus onset marker
`Stimulus/S128`	42 files	short	raw	Stimulus onset marker
`Stimulus/S160`	42 files	short	raw	Stimulus onset marker
`Stimulus/S193-S201`	28-42 files	short	raw	Condition-specific stimulus markers
`Stimulus/S225-S233`	28-42 files	short	raw	Condition-specific stimulus markers
`Stimulus/S195-S200`	33-41 files	short	raw	Condition-specific stimulus markers

Event ID	Count (total)	Type	Meta Level	Meaning
`UserDefined/Blink`	84 files	admin	raw	Blink artifact marker
`Time 0/`	84 files	admin	raw	Recording start time marker
`Response/R 1`	6 files	short	raw	Button press response
`Response/R 3`	1 file	short	raw	Button press response

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/S 1`, `Stimulus/S 2`, `Stimulus/S 3`, `Stimulus/S 4`, `Stimulus/S 64`, `Stimulus/S 96`, `Stimulus/S128`, `Stimulus/S160`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Group Classification - Schizophrenia vs Healthy Controls (Responder Condition)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 0.7 s, overlap = 0.0 * (six non-overlapping windows starting at -2100ms; key analysis window -0.7s to 0s pre-stimulus)
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`HC`	0	Healthy control subjects
`SCZ`	1	Schizophrenia patients

- Super Meta Notes: Derived from clinical diagnosis in participants.tsv. Kernel step: Load participants.tsv, extract 'diagnosis' or 'group' column, map HC->0, SCZ->1. Filter epochs to responder condition trials only (obtained from block/session structure or explicit condition markers).
- Evidence:
- DOI: 10.1016/j.schres.2023.01.029 - Uses EEG dynamic functional connectivity (imaginary part of coherence) comparing 24 healthy controls and 19 schizophrenia patients during the Ultimatum Game responder condition, finding significant fronto-occipital connectivity differences in the alpha band during -0.7s to 0s pre-stimulus window.
- Kernel Notes: Extract segmentation windows from raw EEG. Compute connectivity metrics (e.g., imaginary coherence) in alpha band (8-13 Hz) per window. Use responder-condition trials only. Labels from participants.tsv clinical diagnosis.

Primary Task 2: ERP-based Group Classification (FRN/P2 Components)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 0.5 s, target_events = [`Stimulus/S 64`, `Stimulus/S128`, `Stimulus/S160`] * (offer/decision prompt markers; exact marker ID requires verification from dataset documentation)
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`HC`	0	Healthy control subjects
`SCZ`	1	Schizophrenia patients

- Super Meta Notes: Derived from clinical diagnosis in participants.tsv. Kernel step: Load participants.tsv, extract diagnosis column, map HC->0, SCZ->1. Epoch around stimulus onset, then extract ERP amplitudes (FRN: 320-360ms, P2: 220-240ms) as features for group classification.
- Evidence:
- DOI: 10.1016/j.schres.2017.05.037 - Investigates ERP differences (P2 and feedback-related negativity FRN) between 16 schizophrenic patients and 19 healthy controls during Ultimatum Game, finding significantly reduced FRN amplitude in patients during responder condition (p=0.009).
- Kernel Notes: Epoch by stimulus onset events, filter bandpass 0.3-30Hz. Extract mean amplitude in FRN window (320-360ms) and P2 window (220-240ms) per trial. Use diagnostic group labels from participants.tsv for classification. Verify exact stimulus event IDs from dataset documentation.

Normal Task 1: Ultimatum Game Stimulus Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: UserDefined/Blink and Time 0/ are admin markers and must be excluded. Full decoding of S-codes requires dataset documentation. Psychosis vs. control group label is available from participants.tsv for group-level classification.

Event	y	Meaning
`Stimulus/S 1`	0	Stimulus type 1 (e.g., fair offer)
`Stimulus/S 2`	1	Stimulus type 2
`Stimulus/S 3`	2	Stimulus type 3
`Stimulus/S 4`	3	Stimulus type 4 (e.g., unfair offer)

Data Quality Notes

- Large number of distinct Stimulus/ codes; full mapping requires consulting the study protocol
- UserDefined/Blink and Time 0/ are administrative markers to exclude
- 43 subjects x ~2 sessions = 84 files; verify session structure

References

- No paper reference available

094. openneuro_ds004015 - Attended speaker paradigm (cEEGrid data)

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004015 | Year: 2022 | URL: https://openneuro.org/datasets/ds004015
Category: High-level Mental | Subjects: 36 | Files (Completed): 36
Channels: 18 | Sampling Rate: 500 Hz

Paradigm

A two-competing speaker auditory attention paradigm in which participants were instructed to selectively attend to one of two simultaneously presented audiobooks under varying spatial (same location vs. different locations) and loudness conditions. EEG was recorded using a cEEGrid around-the-ear electrode array. The dataset is intended for auditory attention decoding (AAD) research using ear-worn EEG.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004015

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations; stimulus delivery is continuous (audiobook streams).

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Normal Task 1: Auditory Attention Decoding - Continuous Segmentation

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5

- Label Route: `Annotation`
- Target Definition: None (attended speaker label from task metadata)
- Notes: cEEGrid only provides 18 channels around the ear; limited spatial coverage compared to standard scalp EEG. Auditory attention decoding typically requires aligning EEG segments with envelope of the attended speech stream. No discrete event markers; stimulus is continuous.

Data Quality Notes

- Only 18 around-the-ear channels (cEEGrid) - very limited spatial coverage
- No discrete event markers; continuous stimulus paradigm
- Attended speaker condition metadata needed from external files for supervised learning

References

- Reference: `Holtze et al. (2021)`

095. openneuro_ds004017 - Embodied Learning for Literacy EEG

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004017 | Year: 2023 | URL: https://openneuro.org/datasets/ds004017
Category: High-level Mental | Subjects: 21 | Files (Completed): 126
Channels: 65 | Sampling Rate: 2048 Hz

Paradigm

A pre-post intervention study assessing literacy and orthographic processing in children through two tasks: a two-alternative forced choice letter discrimination task (b vs. d confusion) and a visual search task. EEG was recorded across multiple sessions to evaluate the neural effects of an embodied learning intervention on letter processing. Trigger codes 10/20 and 100/200 likely encode the two letter types or correct/incorrect responses.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004017

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`100.0`	63 files	short	raw	Stimulus onset - letter type B or condition 100
`200.0`	63 files	short	raw	Stimulus onset - letter type D or condition 200
`20.0`	61 files	short	raw	Stimulus onset - response or sub-condition 20
`10.0`	61 files	short	raw	Stimulus onset - response or sub-condition 10

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Letter discrimination (b vs d) under embodied literacy intervention

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 2.0 s, target_events = [`100.0`, `200.0`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`100.0`	0	Letter 'b' (target type)
`200.0`	1	Letter 'd' (target type)

- Evidence:
- DOI: 10.31234/osf.io.zmctf - This paper analyzes EEG data from Study 1 where children performed a visual search task discriminating letters 'b' and 'd' under embodied movement vs gaze fixation conditions, extracting stimulus-locked ERP components from epochs time-locked to target letter onset.
- Kernel Notes: Epochs: 1000ms pre-stimulus, 2000ms post-stimulus. Baseline correction: -100ms to 0ms. Verified event ID mapping from dataset paradigm description (100.0/200.0 encode letter types). (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Numerical trigger codes; full decoding requires dataset documentation
- 21 subjects x 6 sessions - session-level structure must be considered
- High sampling rate (2048 Hz); consider downsampling for efficiency

References

- Reference: `Embodied Learning for Literacy EEG`

096. openneuro_ds004019 - Effect of obesity on arithmetic processing in preteens with high and low math skills. An event-related potentials study

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004019` | Year: 2022 | URL: https://openneuro.org/datasets/ds004019
Category: High-level Mental | Subjects: 62 | Files (Completed): 62
Channels: 132 | Sampling Rate: 500 Hz

Paradigm

Obese and non-obese preteens (ages 9-10) performed a delayed-verification arithmetic task to evaluate the effects of weight status and math skill level on arithmetic processing. Participants viewed an arithmetic problem (prob), then a delay (bgin), then a verification digit (face) and responded (resp). Trigger codes 0 and 1 encode response correctness (0=incorrect, 1=correct). The study uses ERPs to investigate neural correlates of arithmetic cognition in the context of childhood obesity.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004019

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`bgin`	~14,880	short	raw	Trial beginning / delay period onset
`prob`	~14,880	short	raw	Arithmetic problem onset
`ch++`	~14,880	short	raw	Choice/comparison stimulus onset
`face`	~14,880	short	raw	Verification digit (face stimulus) onset
`resp`	~14,600	short	raw	Participant button-press response
`1`	~14,364	short	raw	Correct response (1=correct)
`0`	~6,156	short	raw	Incorrect response (0=incorrect)
`sync`	~310	admin	raw	Hardware sync pulse
`CELL`	~248	admin	raw	Cell/block marker

Event ID	Count (total)	Type	Meta Level	Meaning
SESS	62	admin	raw	Session start marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Correct vs Incorrect Arithmetic Response Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`1`, `0`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`1`	1	Correct arithmetic response
`0`	0	Incorrect arithmetic response

- Evidence:
- DOI: 10.3389/fnhum.2022.760234_pdf - This paper analyzes ERP components (P150, P300, N400) time-locked to answer choice presentation to compare obese vs non-obese preteens with high/low math skills on arithmetic verification tasks, using correct vs incorrect response events (1 and 0) for epoching.
- Kernel Notes: Epoch relative to answer choice onset (`face/ch++` event). The `1` and `0` events encode response correctness concurrent with `resp` event. Class imbalance (~70% correct) should be addressed via weighted loss or sampling. Exclude admin markers (sync, CELL, SESS) from epoching. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- 132-channel EGI cap (E1-E128 + ECG/Respiration/SP channels); remove non-EEG channels before analysis
- sync/CELL/SESS are admin markers to exclude
- Class imbalance: ~70% correct trials - apply undersampling or weighted loss

References

- Reference: `Effect of obesity on arithmetic processing in preteens with high and low math skills. An event-related potentials study.`

097. openneuro_ds004040 - Trance channeling EEG study

Source: analysis.md

Dataset: openneuro_ds004040 | Year: 2022 | URL: https://openneuro.org/datasets/ds004040
Category: High-level Mental | Subjects: 13 | Files (Completed): 26
Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants who are experienced trance channelers performed alternating 5-minute blocks of trance channeling (a mediumistic altered state of consciousness) and matched resting state conditions across two separate sessions. The study investigates neural correlates of voluntarily induced altered states of consciousness compared to wakeful rest. Labels for trance vs. rest blocks must be derived from the session/task structure rather than discrete event markers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004040

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
STATUS	26 files	admin	raw	System hardware status marker - not suitable for epoching

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Trance

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004040 DOI: https://doi.org/10.18112/openneuro.ds004040.v1.0.0

Normal Task 1: Trance vs. Resting State (binary, segmentation)

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: None (condition label from task/session structure)
- Notes: STATUS is an admin marker and must be excluded. Block-level labels (trance vs. rest) should come from the BIDS task name or session metadata. 13 subjects x 2 sessions = 26 files. Small sample size; suitable for within-subject analysis.

Data Quality Notes

- Only STATUS admin marker present; no task-level event markers
- Small sample (13 specialized subjects who practice trance channeling)
- Two sessions per subject - verify session labeling in BIDS structure

References

- Reference: `Trance channeling EEG study`

098. openneuro_ds004067 - Moral conviction and metacognitive ability shape multiple stages of information processing

Source: analysis.md

Dataset: `openneuro_ds004067` | Year: 2022 | URL: https://openneuro.org/datasets/ds004067
Category: High-level Mental | Subjects: 80 | Files (Completed): 83
Channels: 63 | Sampling Rate: 2000 Hz

Paradigm

Participants evaluated their support for sociopolitical protest photographs following social influence cues from a fictitious community to assess how moral conviction and metacognitive ability modulate information processing. EEG was recorded while participants viewed protest photos (supporting or opposing social movements), received social feedback (high/low/medium agreement), and provided attitude ratings. The study examined how moral convictions and metacognitive ability shape early and late stages of social information processing.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004067

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/PhotoSupport`	~7,290	short	raw	Photo onset - support-framing protest image
`Stimulus/PhotoOppose`	~7,290	short	raw	Photo onset - oppose-framing protest image
`Stimulus/SocLow`	~4,860	short	raw	Social influence cue - low agreement
`Stimulus/SocHigh`	~4,860	short	raw	Social influence cue - high agreement
`Stimulus/SocMed`	~4,860	short	raw	Social influence cue - medium agreement
`Stimulus/Moral_Support`	~5,688	short	raw	Moral attitude cue - support
`Stimulus/Moral_Oppose`	~5,688	short	raw	Moral attitude cue - oppose

Event ID	Count (total)	Type	Meta Level	Meaning
Stimulus/Nonmoral_Oppose	~5,400	short	raw	Non-moral attitude cue - oppose
Stimulus/Nonmoral_Neutral	~4,968	short	raw	Non-moral attitude cue - neutral
Stimulus/Nonmoral_Support	~4,560	short	raw	Non-moral attitude cue - support
Stimulus/Resp1-Resp5	variable	short	raw	Participant rating response (1-5 scale)
Stimulus/ProtestStart	~80	admin	raw	Block start marker
Stimulus/ProtestEnd	~80	admin	raw	Block end marker
Stimulus/RestStart	~79	rest	raw	Resting block start
Stimulus/RestEnd	~79	rest	raw	Resting block end
Stimulus/ClipTaskStart	~79	admin	raw	Clip task start
Stimulus/ClipTaskEnd	~79	admin	raw	Clip task end
Stimulus/Social5/6start-end	~79	admin	raw	Social clip block markers
Stimulus/Heroic6/7start-end	~79	admin	raw	Heroic clip block markers
Stimulus/S 2	~79	admin	raw	System/sync marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/PhotoSupport`, `Stimulus/PhotoOppose`, `Stimulus/SocLow`, `Stimulus/SocHigh`, `Stimulus/SocMed`, `Stimulus/Moral_Support`, `Stimulus/Moral_Oppose`, `Stimulus/Nonmoral_Oppose`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Sociopolitical Judgment Task

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`

- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004067 DOI: https://doi.org/10.17605/OSF.IO/8KPEF

Normal Task 1: Support vs Oppose Photo Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: ProtestStart/End, ClipTaskStart/End, Socialstart/end, Heroicstart/end, RestStart/End, and S 2 are all admin/structural markers to exclude. Resp1-Resp5 encode 5-point rating responses and can be used for regression. Social cues (SocLow/Med/High) can be used for a 3-class social influence task.

Event	y	Meaning
`Stimulus/PhotoSupport`	0	Support-framing sociopolitical protest photo
`Stimulus/PhotoOppose`	1	Oppose-framing sociopolitical protest photo
## Data Quality Notes		

- Large number of event types; careful exclusion of admin markers required
- RestStart/RestEnd delimit resting-state blocks within the recording
- High sampling rate (2000 Hz); consider downsampling to 500-1000 Hz

References

- Reference: `Moral conviction and metacognitive ability shape multiple stages of information processing`

099. openneuro_ds004117 - Sternberg Working Memory

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004117 | Year: 2022 | URL: https://openneuro.org/datasets/ds004117
Category: High-level Mental | Subjects: 23 | Files (Completed): 85
Channels: 71 | Sampling Rate: 250 Hz

Paradigm

A modified Sternberg working memory task in which participants selectively memorized sets of 3, 5, or 7 target letters (marked with lowercase 'g' prefix, e.g., gH) while ignoring identical-font distractor letters. After a retention period, a probe letter was presented and participants indicated with auditory feedback whether it was in the memorized set. The study investigates frontal midline EEG dynamics (particularly theta oscillations) during working memory maintenance.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004117

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
WM	85 files	short	raw	Working memory trial onset / encoding phase
nonWM	85 files	short	raw	Non-working-memory (distractor) trial
gA-gZ (prefixed letters)	85 files	short	raw	Working memory target letter onset (to be memorized)
A-Z (plain letters)	85 files	short	raw	Distractor letter onset (to be ignored)
255	85 files	admin	raw	System trigger (likely BrainVision marker 255 = all bits)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Working Memory Encoding (Memorize vs Ignore)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 2.0 s, target_events = [`WM`, `nonWM`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`WM`	1	Working memory encoding trial (Memorize letters)
`nonWM`	0	Non-working-memory distractor trial (Ignore letters)

- Evidence:
- DOI: `10.1016/j.neuroimage.2005.04.014` - This 2005 NeuroImage paper presents the original EEG study from which the openneuro_ds004117 dataset derives, analyzing frontal midline theta dynamics in 23 subjects during a modified Sternberg working memory task with 71-channel EEG at 250 Hz, using ICA decomposition to separate Memorize vs Ignore letter trials and examining theta power changes with memory load.
- Kernel Notes: Extract epochs centered on WM and nonWM trial onsets. The WM events correspond to "Memorize" trials (target letters with g prefix), while nonWM corresponds to "Ignore" trials (distractor letters without g prefix). Filter out event ID `255` (admin marker). Optionally, analyze by memory load (3, 5, 7 letters) using additional metadata or secondary analysis. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Trigger 255 (all-bits-high) is a system artifact and must be excluded
- Individual letter events provide fine-grained trial structure for ERP analysis
- 23 subjects x multiple sessions - verify session structure before pooling

References

- Reference: `Onton, J., Delorme, A. and Makeig, S., 2005. Frontal midline EEG dynamics during working memory. Neuroimage, 27(2), pp.341-356.`

100. openneuro_ds004147 - Average Task Value

Source: analysis.md

Dataset: openneuro_ds004147 | Year: 2024 | URL: https://openneuro.org/datasets/ds004147
Category: High-level Mental | Subjects: 12 | Files (Completed): 12
Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

A reinforcement learning task requiring participants to learn cue-response mappings for colored shapes under varying conditions of feedback validity (0.5 vs 0.8 probability). Participants learned which button press was rewarded for each cue, with the value of the correct response varying across conditions. EEG was recorded to investigate neural correlates of value-based learning and the average reward rate representation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004147

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus`	12 files	short	raw	Stimulus onset (cue presentation)
`New Segment`	12 files	admin	raw	Recording segment boundary - exclude
`Comment`	12 files	admin	raw	Experimenter annotation - exclude

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Casinos

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004147 DOI: https://doi.org/10.1037/a0038339

Normal Task 1: Stimulus-Locked Reinforcement Learning (binary/multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: New Segment and Comment are admin markers and must be excluded. Only 12 subjects - very small dataset. The `Stimulus` event needs behavioral outcome (correct/incorrect or rewarded/not-rewarded) from events.tsv for supervised learning. Feedback validity condition (0.5 vs 0.8) is available as a task variable.

Event	y	Meaning
`Stimulus`	0/1	Cue onset; label from behavioral outcome (reward/no reward in events.tsv)
## Data Quality Notes		

- Only 12 subjects - very small sample; unsuitable for cross-subject generalization
- New Segment and Comment are admin markers to exclude
- Behavioral outcome labels must be extracted from events.tsv alongside Stimulus events

References

- Reference: `Average Task Value`

101. openneuro_ds004148 - A test-retest resting and cognitive state EEG dataset

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004148` | Year: 2022 | URL: https://openneuro.org/datasets/ds004148
Category: High-level Mental | Subjects: 60 | Files (Completed): 1797 | Channels: 61 | Sampling Rate: 500 Hz

Paradigm

A test-retest reliability study recording EEG during five cognitive/resting conditions: eyes closed (EC), eyes open (EO), mental arithmetic (Ma), memory task (Me), and music listening (Mu), repeated across three sessions. Each file corresponds to one 5-minute condition block, with a single condition marker embedded (S 33=EO, S 44=EC, S 55=Mu, S 66=Me, S 77=Ma). Subjects are healthy adults aged 18-45.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004148

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`S 44`	360	long	raw	Eyes-closed resting state block label
`S 77`	360	long	raw	Mental arithmetic task block label
`S 66`	360	long	raw	Memory task block label
`S 55`	360	long	raw	Music listening block label
`S 33`	357	long	raw	Eyes-open resting state block label
`Sync Off`	~270000	admin	raw	Periodic sync signal (~150/file), not a trial event

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `S 44`, `S 77`, `S 66`, `S 55`, `S 33`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sex Classification from Resting-State EEG

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 5.0 s * (full 5-min block), overlap = 0.0 *, target_events = [`S 33`, `S 44`]
- Label Route: `Annotation` | `misc`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`S 33` (EO)	0	Eyes-open resting state, male subject
`S 33` (EO)	1	Eyes-open resting state, female subject
`S 44` (EC)	0	Eyes-closed resting state, male subject
`S 44` (EC)	1	Eyes-closed resting state, female subject

- Super Meta Notes: Derived from raw event(s) [`S 33`, `S 44`] by computing power spectral density (PSD) using multi-taper method in frequency bands (delta, theta, alpha, beta), then deriving relative power (normalized by total power 3-30 Hz). Sex labels are original demographic labels from dataset. Kernel step: Extract PSD features per epoch, normalize to relative power, then use demographic sex label as target.
- Evidence:
- DOI: 10.1016/j.neuroimage.2025.121231 - Uses the Southwest University EEG dataset (openneuro ds004148) to investigate age-dependent sex differences in low alpha and low beta EEG oscillations, analyzing spectral power in eyes-open and eyes-closed resting states.
- Kernel Notes: Target events S 33 and S 44 mark 5-minute blocks. Epoch entire block or segment into windows (e.g., 2-5 s). Compute multi-taper PSD per epoch in bands: delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz). Normalize to relative power. Merge with subject sex label from demographics. (replaces Primary Tasks 1-3 from original analysis.md)

Primary Task 2: Five-Class Mental State Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.0 *, target_events = [`S 33`, `S 44`, `S 55`, `S 66`, `S 77`]
- Label Route: `Annotation` | `stim`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`S 33`	0	Eyes-open resting state
`S 44`	1	Eyes-closed resting state
`S 55`	2	Music listening
`S 66`	3	Memory task
`S 77`	4	Mental arithmetic (subtraction)

- Evidence:
- DOI: 10.1038/s41597-022-01607-9 - Presents openneuro_ds004148 dataset and describes its use for decoding five EEG mental states (eyes-open, eyes-closed, memory, music, subtraction) using EEG-based features such as power spectrum, functional connectivity, and microstates.
- Kernel Notes: Each 5-minute condition block is segmented into 4-s epochs (no overlap per paper). Extract features per epoch (power spectrum, functional connectivity, or microstates). Labels are directly derived

from raw event markers S 33/S 44/S 55/S 66/S 77. Verify event ID mapping from events.tsv. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Sync Off at ~150 events/file is hardware sync, not a task event
- Three test-retest sessions allow reliability analyses

References

- Reference: `A test-retest resting and cognitive state EEG dataset`

102. openneuro_ds004196 - Bimodal dataset on Inner speech

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004196` | Year: 2023 | URL: https://openneuro.org/datasets/ds004196
Category: High-level Mental | Subjects: 4 | Files (Completed): 4 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

An inner-speech production task where participants mentally vocalized eight specific words from social and numeric categories during nonsimultaneous EEG and fMRI recording. Subjects imagined speaking words without producing overt speech or movement. No trial-level event markers were embedded in the EEG recordings; the data is treated as continuous.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004196

Event Structure

No trial events detected. This dataset contains continuous EEG without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Event ID	Count (total)	Type	Meta Level	Meaning
`no_events`	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Normal Task 1: Inner Speech Segmentation (resting)

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null)
- Notes: No event markers available. Sliding window segmentation only.

Data Quality Notes

- Very small dataset (N=4); not suitable for cross-subject generalization

- Consider using fMRI labels aligned to EEG timing if available

References

- Reference: `Bimodal dataset on Inner speech`

103. openneuro_ds004200 - Temporal Scaling

Source: analysis.md

Dataset: `openneuro_ds004200` | Year: 2022 | URL: https://openneuro.org/datasets/ds004200
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 37 | Sampling Rate: 1000 Hz

Paradigm

Participants learned to produce and judge short, medium, and long temporal intervals through interleaved production and perception tasks with visual feedback. Trigger codes encode condition combinations: tens digit indicates task type and interval length (1x=production, 2x-6x=other conditions), units digit indicates subcondition or phase. The study investigates how humans scale temporal interval representations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004200

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
11	~20	short	raw	Production short interval phase 1
12	~20	short	raw	Production short interval phase 2
13	~20	short	raw	Production short interval phase 3
21	~20	short	raw	Condition 2, phase 1
22	~20	short	raw	Condition 2, phase 2
23	~20	short	raw	Condition 2, phase 3
31	~20	short	raw	Condition 3, phase 1
32	~20	short	raw	Condition 3, phase 2
33	~20	short	raw	Condition 3, phase 3
41	~20	short	raw	Condition 4, phase 1
42	~20	short	raw	Condition 4, phase 2
43	~20	short	raw	Condition 4, phase 3
51	~20	short	raw	Condition 5, phase 1
52	~20	short	raw	Condition 5, phase 2
53	~20	short	raw	Condition 5, phase 3
61	~20	short	raw	Condition 6, phase 1

Event ID	Count (total)	Type	Meta Level	Meaning
62	~20	short	raw	Condition 6, phase 2
63	~20	short	raw	Condition 6, phase 3
17	~20	short	raw	Feedback/response code
27	~20	short	raw	Feedback/response code
47	~20	short	raw	Feedback/response code
57	~20	short	raw	Feedback/response code
67	~20	short	raw	Feedback/response code
2	~20	admin	raw	Block/session boundary marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Temporalscaling

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004200 DOI: https://doi.org/10.18112/openneuro.ds004200.v1.0.1

Normal Task 1: Temporal Condition Multiclass (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Trigger code `2` is a block marker, excluded. Decode exact condition meanings from paper or events.tsv.

Event	y	Meaning
11	0	Cond 1 phase 1
12	1	Cond 1 phase 2

Event	y	Meaning
13	2	Cond 1 phase 3
21	3	Cond 2 phase 1
22	4	Cond 2 phase 2
23	5	Cond 2 phase 3
31	6	Cond 3 phase 1
32	7	Cond 3 phase 2
33	8	Cond 3 phase 3
41	9	Cond 4 phase 1
42	10	Cond 4 phase 2
43	11	Cond 4 phase 3
51	12	Cond 5 phase 1
52	13	Cond 5 phase 2
53	14	Cond 5 phase 3
61	15	Cond 6 phase 1
62	16	Cond 6 phase 2
63	17	Cond 6 phase 3
17	18	Feedback 1
27	19	Feedback 2
47	20	Feedback 4
57	21	Feedback 5
67	22	Feedback 6
## Data Quality Notes		

- Numerical trigger codes require decode from paper
- Small dataset (N=20)

References

- Reference: `Temporal Scaling`

104. openneuro_ds004262 - Continuous Feedback Processing

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004262 | Year: 2022 | URL: https://openneuro.org/datasets/ds004262
Category: High-level Mental | Subjects: 21 | Files (Completed): 21
Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

Participants predicted the final height of an animated rising bar based on cues with varying predictability to study continuous feedback processing and prediction error. Each trial involved observing bar motion and registering a prediction, allowing examination of EEG correlates of reward prediction error and uncertainty-driven learning.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004262

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus`	21x900	short	raw	Stimulus onset marker (bar appearance)
`New Segment`	21	admin	raw	Recording segment boundary
`Comment`	21	admin	raw	Experimenter comment/annotation

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Continuous Reward Prediction Regression

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -4.0 s, tmax = 0.2 s *, target_events = [`Stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Stimulus`	continuous (0-1 scale)	Normalized reward value at each time point

- Super Meta Notes: Derived from raw event(s) [`Stimulus`] by computing continuous reward signal from participant predictions and actual bar heights. Kernel step: Load behavioral data (events.tsv columns for guess/actual), compute time-varying reward signal aligned to Stimulus onset, normalize to [0,1] range. Regression target is continuous reward amplitude at each EEG sample within the epoch window.
- Evidence:
- DOI: 10.1101/2022.10.06.511117 - Shows EEG tracks continuous reward prediction using correlation/regression on openneuro_ds004262
- DOI: 10.1111/psyp.14399 - Uses regression-based unmixing to identify neural correlates of continuous feedback processing
- Kernel Notes: Must load events.tsv to extract behavioral columns (guess, actual_height, reward). Compute continuous reward signal = f(guess, actual). Align to Stimulus events. Window: -4000ms to 200ms relative to maximum reward timing.

Primary Task 2: Outcome Predictability Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 2.8 s *, target_events = [`Stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Stimulus`	0	High predictability (certain outcome)
`Stimulus`	1	Medium predictability (uncertain outcome)
`Stimulus`	2	Low predictability (ambiguous outcome)

- Super Meta Notes: Derived from raw event(s) [`Stimulus`] by computing predictability from pre-trial cue identity and trial history. Kernel step: For each Stimulus event, look up corresponding cue type in events.tsv, compute predictability level based on prior trial outcomes for that cue. High = consistent history, Low = inconsistent history.
- Evidence:
- DOI: 10.1111/psyp.14399 - Tests how EEG tracking of continuous reward varies with outcome predictability (high/medium/low)
- Kernel Notes: Must load events.tsv to extract cue identifier and trial history. Compute predictability: for each cue type, calculate variance of past outcomes. High variance = low predictability, low variance = high predictability. Map to 3-class labels.

Normal Task 1: Stimulus Processing (single-class)

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Only one event class available for epoching. New Segment and Comment are admin markers. Behavioral metadata (prediction, actual height) may be available in events.tsv for regression tasks.

Event	y	Meaning
`Stimulus`	0	Stimulus onset (bar presentation)

Data Quality Notes

- Only one non-admin event type (Stimulus); classification requires additional label parsing from events.tsv metadata columns
- New Segment and Comment are system/admin events excluded from epoching

References

- Reference: `Continuous Feedback Processing`

105. openneuro_ds004279 - Large Spanish EEG

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004279 | Year: 2024 | URL: https://openneuro.org/datasets/ds004279
Category: High-level Mental | Subjects: 56 | Files (Completed): 60
Channels: 69 | Sampling Rate: 1000 Hz

Paradigm

Participants performed speech perception and silent speech production tasks involving 30 daily Spanish sentences. Each sentence trial followed a fixed structure: a rest period, a preparation cue, a perception phase (hearing the sentence), and a production phase (silently producing the sentence). The study investigated EEG correlates of perceived and silently produced speech.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004279

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`perception_N` (N=1-30)	60x7	long	raw	Perception phase onset for sentence N
`production_N` (N=1-30)	60x7	long	raw	Silent production phase onset for sentence N
`preparation_N` (N=1-30)	60x7	long	raw	Preparation cue onset for sentence N
`rest_N` (N=1-30)	60x7	rest	raw	Rest period onset for sentence N
`bad_artifact`	60xvariable	admin	raw	Bad artifact annotation
`Impedance_test`	60x2	admin	raw	Impedance check marker
`bad_before_experiment`	60x1	admin	raw	Pre-experiment bad segment
`bad_after_experiment`	60x1	admin	raw	Post-experiment bad segment

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: 30-Class Sentence Identity Classification from Perception Phase

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 3.0 s, overlap = 0.0, target_events = [perception_1, perception_2, perception_3, perception_4, perception_5, perception_6, perception_7, perception_8, perception_9, perception_10, perception_11, perception_12, perception_13, perception_14, perception_15, perception_16, perception_17, perception_18, perception_19, perception_20, perception_21, perception_22, perception_23, perception_24, perception_25, perception_26, perception_27, perception_28, perception_29, perception_30]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
perception_1	0	Sentence 1 perception phase
perception_2	1	Sentence 2 perception phase
...	...	...
perception_30	29	Sentence 30 perception phase

- Kernel Notes: Paper reports results using data downsampled to 250 Hz. Kernel may need to apply downsampling for reproducibility. (replaces Normal Task 1 from original analysis.md)
- Evidence:
- DOI: 10.1088/1741-2552.ad88a3 - Deep neural network classification of 30-class sentence identity from EEG during speech perception
- DOI: 10.1088/1741-2552.ad88a3.2 - Replication study confirming 30-class perception-only decoding with EEGNet, DeepConvNet, ShallowConvNet

Data Quality Notes

- 90+ distinct event types (perception/production/preparation/rest for 30 sentences)
- bad_artifact count varies by subject (31-39 per file)
- Group sentence-level events by phase type for practical classification

References

- Reference: Valle, C., Mendez-Orellana, C., Herff, C., & Rodriguez-Fernandez, M. (2024). Identification of perceived sentences using deep neural networks in EEG. Journal of neural engineering, 21(5), 056044.

106. openneuro_ds004284 - eeg-neuroforecasting

Source: analysis.md

Dataset: openneuro_ds004284 | Year: 2022 | URL: https://openneuro.org/datasets/ds004284
Category: High-level Mental | Subjects: N/A | Files (Completed): 18
Channels: 129.0 | Sampling Rate: 1000.0 Hz

Paradigm

A neuroforecasting task that utilizes EEG activity during decision-making or valuation to predict aggregate, population-level outcomes.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004284

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.
## MISC Label Summary				

- Route: misc
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Publisher-defined BIDS task: Kickstarter

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004284 DOI: https://doi.org/10.18112/openneuro.ds004284.v1.0.0

Normal Task 1: Raw-meta task

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Needs verification
- Kernel Notes: Exclude admin markers from target labels.

Normal Task 2: Super-meta task

- Task Type: regression/classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target / Label: Needs verification
- Kernel Notes: Derive and version super-meta in kernel.

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography`

107. openneuro_ds004295 - Reward gain and punishment avoidance reversal learning

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004295` | Year: 2022 | URL: https://openneuro.org/datasets/ds004295
Category: High-level Mental | Subjects: 26 | Files (Completed): 26
Channels: 66 | Sampling Rate: 1024 Hz

Paradigm

A reversal learning task comparing brain activity during reward gain (monetary) versus punishment avoidance (white noise) reinforcement. Participants learned stimulus-response associations that reversed mid-task, requiring cognitive flexibility. EEG was recorded to investigate feedback-related neural signatures under different reinforcement conditions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004295

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`S 81`	26x~164	short	raw	Stimulus trigger - decode from events.tsv
`S 61`	26x~162	short	raw	Stimulus trigger - decode from events.tsv
`S 75`	26x~153	short	raw	Stimulus trigger - decode from events.tsv
`S 71`	26x~148	short	raw	Stimulus trigger - decode from events.tsv
`S 52`	26x~141	short	raw	Stimulus trigger - decode from events.tsv
`S 51`	26x~139	short	raw	Stimulus trigger - decode from events.tsv
`S 55`	26x~137	short	raw	Stimulus trigger - decode from events.tsv

Event ID	Count (total)	Type	Meta Level	Meaning
S 72	26x~132	short	raw	Stimulus trigger - decode from events.tsv
S 60	26x~118	short	raw	Stimulus trigger - decode from events.tsv
S 80	26x~116	short	raw	Stimulus trigger - decode from events.tsv
S100	26x~1	short	raw	Sample stimulus (first in pair)
S200	26x~1	short	raw	Probe stimulus (second in pair)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: S 81, S 61, S 75, S 71, S 52, S 51, S 55, S 72
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Feedback Valence Classification (positive vs negative feedback)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -1.5 s, tmax = 3.0 s, target_events = [S 71, S 72] *
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
S 71 *	0	Positive feedback (reward gain or punishment avoidance success)
S 72 *	1	Negative feedback (reward loss or punishment administered)

- Super Meta Notes: (empty - meta level is raw)
- Evidence:

- DOI: `10.1111/psyp.14235` - Single-trial regression analysis examining frontal midline theta (FMθ) responses to positive vs negative feedback in reward and punishment reversal learning; uses feedback valence as predictor.
- Kernel Notes: The raw event IDs `S 71` and `S 72` are candidate mappings for feedback events based on the event structure. Verify event ID mapping from events.tsv to confirm which S codes correspond to positive vs negative feedback. Parameters tmin/tmax are derived from the paper's epoch relative to feedback onset.

Normal Task 1: Reversal Learning Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Specific kernel required to decode BrainVision trigger codes (S 51-S 81 range) to semantic labels. Regression target: response_time from misc channels.

Event	y	Meaning
`stimulus_A_prereversalReward`	0	Stimulus A before reversal (reward condition)
`stimulus_B_prereversalPunishment`	1	Stimulus B before reversal (punishment condition)
`correct_response`	2	Correct response
`incorrect_response`	3	Incorrect response
`reversal_cue`	4	Reversal signal
`stimulus_A_postreversalPunishment`	5	Stimulus A after reversal (punishment condition)
`feedback_positive`	6	Positive feedback
`feedback_negative`	7	Negative feedback

Data Quality Notes

- BrainVision trigger codes (S 51-S 81 range) require kernel-level decoding to semantic labels
- S150/S151 appear in only 22 subjects, possibly a separate task block

References

- Reference: `Reward gain and punishment avoidance reversal learning`

108. openneuro_ds004315 - Mood Manipulation and PST, Experiment 1

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004315 | Year: 2022 | URL: https://openneuro.org/datasets/ds004315
Category: High-level Mental | Subjects: 50 | Files (Completed): 50
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task (Probabilistic Selection Task) consisting of training and testing phases following a sad or neutral mood manipulation. Participants learned to select stimuli associated with reward or punishment, and were subsequently tested on novel stimulus pairs. EEG was recorded to examine how mood state affects the neural correlates of reinforcement learning.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004315

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`STIM`	50x~240	short	raw	Stimulus onset (stimulus pair presented)
`RESP`	50x~230	short	raw	Participant button-press response
`FB`	50x~240	short	raw	Feedback onset (reward/punishment)
`Correct`	50x~200	short	raw	Correct response outcome
`Incorrect`	50x~85	short	raw	Incorrect response outcome
`TESTING`	50x~1700	short	raw	Testing phase trial marker
`STATUS`	50	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STIM, STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Feedback Outcome Classification (Reward Positivity Analysis)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -2.0 s, tmax = 6.0 s, target_events = [FB]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
Correct	0	Correct response (reward)
Incorrect	1	Incorrect response (punishment)

- Evidence:
- DOI: 10.1111/psyp.14276 - This paper analyzes EEG data from OpenNeuro ds004315 to examine how the Reward Positivity (RewP) ERP component to feedback onset (epoched -2s to 6s, measured 200-350ms at FCz) relates to reinforcement learning in a probabilistic selection task, comparing correct vs incorrect feedback outcomes across sad and neutral/happy mood manipulation groups
- Kernel Notes: Epoch data relative to FB (feedback onset) events. Extract epochs and assign labels based on corresponding Correct/Incorrect outcome markers. The paper uses baseline correction from -200ms to 0ms. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Reward Prediction Error (RPE+) Coupling Analysis

- Task Type: regression
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -2.0 s, tmax = 6.0 s, target_events = [FB]
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
high_rpe	continuous	High reward prediction error (RPE+) from Q-learning model
low_rpe	continuous	Low reward prediction error (RPE+) from Q-learning model

- Super Meta Notes: Derived from raw event(s) [Correct, Incorrect, TESTING] by Q-learning computational model computing reward prediction error per trial. Kernel step: Implement Q-learning model to derive RPE+ values per trial, then correlate single-trial RewP amplitude (200-350ms at FCz) with RPE+ as regressor.
- Evidence:
- DOI: 10.1111/psyp.14276 - This paper analyzes EEG data from OpenNeuro ds004315 to examine how the Reward Positivity (RewP) ERP component to feedback onset relates to reinforcement learning, using Q-learning derived reward prediction error (RPE+) as a regressor in single-trial EEG correlation analysis

(RewP-RPE+ coupling)
- Kernel Notes: Train Q-learning model on training phase trials to derive RPE+ for each testing trial. Extract RewP amplitude (peak 200-350ms post-feedback at FCz) and compute correlation with RPE+ values. Requires implementing Q-learning model fitting.

Data Quality Notes

- STATUS is an admin/system marker and must not be included in target events
- TESTING events contain rich sub-labels in events.tsv; the trial_type column provides full label strings
- Class imbalance between Correct (~200) and Incorrect (~55 training) responses

References

- Reference: `Mood Manipulation and PST, Experiment 1`

109. openneuro_ds004317 - Mood Manipulation and PST, Experiment 2

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004317 | Year: 2022 | URL: https://openneuro.org/datasets/ds004317
Category: High-level Mental | Subjects: 50 | Files (Completed): 50
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task (Probabilistic Selection Task) performed by participants following sad or happy mood manipulations. The paradigm mirrors Experiment 1 but compares sad and happy mood conditions rather than sad versus neutral. EEG was recorded to examine mood-dependent differences in reinforcement learning neural signatures.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004317

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`STIM`	50x~240	short	raw	Stimulus onset (stimulus pair presented)
`RESP`	50x~230	short	raw	Participant button-press response
`FB`	50x~240	short	raw	Feedback onset (reward/punishment)
`Correct`	50x~198	short	raw	Correct response outcome
`Incorrect`	50x~100	short	raw	Incorrect response outcome
`TESTING`	50x~1664	short	raw	Testing phase trial marker
`STATUS`	50	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STIM, STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Feedback Outcome Classification (Correct vs Incorrect)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -2.0 s, tmax = 6.0 s, target_events = [Correct, Incorrect]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
Correct	0	Reward feedback (correct response)
Incorrect	1	Punishment feedback (incorrect response)

- Evidence:
- DOI: 10.1111/psyp.14276 - Used OpenNeuro dataset ds004317 (Experiment 2, N=50) to examine how mood induction affects Reward Positivity (RewP) and reward learning during a probabilistic selection task, epoching EEG data from -2000 to 6000 ms around feedback onset and classifying correct vs incorrect feedback trials.
- Kernel Notes: The feedback events (FB) in the raw stream mark feedback onset. The outcome type (Correct/Incorrect) is embedded in the trial_type column of events.tsv. Kernel must join FB events with trial_type to extract labels. This task is the same analysis as the Normal Task below, so it replaces that entry.

Data Quality Notes

- STATUS is an admin/system marker and must not be included in target events
- TESTING events contain rich sub-labels in events.tsv; the trial_type column provides full label strings
- Better class balance than Experiment 1

References

- Reference: Mood Manipulation and PST, Experiment 2

110. openneuro_ds004324 - ToonFaces

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004324 | Year: 2023 | URL: https://openneuro.org/datasets/ds004324
Category: High-level Mental | Subjects: 26 | Files (Completed): 26
Channels: 38 | Sampling Rate: 500 Hz

Paradigm

Participants viewed rapid serial visual presentation (RSVP) streams of three types of face images - real photographs (img), stylized toon faces (toon), and schematic stick-figure faces (sf) - each shown at different emotional intensity levels (0-4). Subjects evaluated emotion detection and affective states, allowing ERP-based comparison across face types and intensity levels.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004324

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`img/0`	26x45	short	raw	Real face image, intensity 0 (neutral)
`img/1`	26x195	short	raw	Real face image, intensity 1
`img/2`	26x15	short	raw	Real face image, intensity 2
`img/3`	26x15	short	raw	Real face image, intensity 3
`img/4`	26x30	short	raw	Real face image, intensity 4 (most intense)
`toon/0`	26x45	short	raw	Toon face, intensity 0 (neutral)
`toon/1`	26x195	short	raw	Toon face, intensity 1
`toon/2`	26x15	short	raw	Toon face, intensity 2
`toon/3`	26x15	short	raw	Toon face, intensity 3
`toon/4`	26x30	short	raw	Toon face, intensity 4
`sf/0`	26x45	short	raw	Schematic face, intensity 0 (neutral)
`sf/1`	26x195	short	raw	Schematic face, intensity 1

Event ID	Count (total)	Type	Meta Level	Meaning
sf/2	26x15	short	raw	Schematic face, intensity 2
sf/3	26x15	short	raw	Schematic face, intensity 3
sf/4	26x30	short	raw	Schematic face, intensity 4
BlockEnd	26x3	admin	raw	End of block marker
block1	26	admin	raw	Block 1 start marker
block2	26	admin	raw	Block 2 start marker
block3	26	admin	raw	Block 3 start marker
Intro	22	admin	raw	Introduction screen marker
Instructions1	25	admin	raw	Instructions screen 1
Instructions2	25	admin	raw	Instructions screen 2
Instructions3	25	admin	raw	Instructions screen 3
PracticeInstructions	25	admin	raw	Practice instructions marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Three-Class Face Type Classification (IMG vs TOON vs SF)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 0.8 s, target_events = [img/0, img/1, img/2, img/3, img/4, toon/0, toon/1, toon/2, toon/3, toon/4, sf/0, sf/1, sf/2, sf/3, sf/4]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
img/0, img/1, img/2, img/3, img/4	0	Real photograph faces (IMG)
toon/0, toon/1, toon/2, toon/3, toon/4	1	Stylized toon faces (TOON)
sf/0, sf/1, sf/2, sf/3, sf/4	2	Schematic stick-figure faces (SF)

- Evidence:
- DOI: 10.3389/fnhum.2023.1160800 - RSVP paradigm with three face categories analyzed using ERP features (N170, LCP, differential entropy) for emotion detection. Epoch parameters: baseline correction -200ms, epoch -200ms to 800ms.
- Kernel Notes: Apply baseline correction from -200ms to stimulus onset. Exclude block markers (block1, block2, block3, BlockEnd) and instruction markers (Intro, Instructions1, Instructions2, Instructions3, PracticeInstructions) from epoching. (replaces Normal Task 1 from original

analysis.md)

Data Quality Notes

- Block/instruction markers must be excluded from epoching
- Unequal counts across intensity levels (intensity 1: 195 vs intensity 2/3: 15 per face type)
- 26 subjects; consistent event structure across subjects

References

- Reference: `Images of stylized faces improve ERP features used for emotion detection`

111. openneuro_ds004350 - Executive Functioning Study for Assessing the Effect of Neurofeedback

Source: analysis.md

Dataset: `openneuro_ds004350` | Year: 2025 | URL: https://openneuro.org/datasets/ds004350
Category: High-level Mental | Subjects: 24 | Files (Completed): 480
Channels: 64 | Sampling Rate: 256 Hz

Paradigm

Participants performed a battery of executive function tasks including a visual N-back working memory task (stimuli A/B/C/D), a Sustained Attention to Response Task (SART, target/nontarget), and a Local-Global hierarchical stimulus task (largeH/largeT x smallH/smallT). The study assessed the effect of neurofeedback training on executive function performance.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: reaction time
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004350

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`target`	44xx	short	raw	Target stimulus (requires response)
`nontarget`	44xx	short	raw	Non-target stimulus (no-go)
`A`	144xx	short	raw	N-back stimulus A
`B`	144xx	short	raw	N-back stimulus B
`C`	144xx	short	raw	N-back stimulus C
`D`	144xx	short	raw	N-back stimulus D
`largeHSmallH`	48xx	short	raw	Large H / Small H (local-global congruent)
`largeHSmallT`	48xx	short	raw	Large H / Small T (local-global incongruent)
`largeTSmallH`	48xx	short	raw	Large T / Small H (local-global incongruent)

Event ID	Count (total)	Type	Meta Level	Meaning
largeTSmallT	48xx	short	raw	Large T / Small T (local-global congruent)
responseLargeH	48xx	short	raw	Response: Large H selected
responseLargeT	48xx	short	raw	Response: Large T selected
responseSmallH	48xx	short	raw	Response: Small H selected
responseSmallT	48xx	short	raw	Response: Small T selected
rt	188xx	short	raw	Reaction time marker
255	2	admin	raw	System trigger code 255

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: reaction_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: NB1

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004350 DOI: https://doi.org/10.1038/nn1233

Primary Task 2: Publisher-defined BIDS task: NB3

- Task Type: classification/segmentation

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004350 DOI: https://doi.org/10.1038/nn1233

Primary Task 3: Publisher-defined BIDS task: LG

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004350 DOI: https://doi.org/10.1038/nn1233

Normal Task 1: SART Target Detection (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numerical codes (1, 2, 4, 5, 6, 8, 9, 10, 11, 12, 128, 255) are likely system/artifact markers - exclude from target_events. Multiple tasks in one dataset; files are per-task. Separate HDF5 files per task recommended.

Event	y	Meaning
`nontarget`	0	Non-target (go) stimulus
`target`	1	Target (no-go) stimulus
## Data Quality Notes		

- Multiple distinct tasks (SART, N-back, Local-Global) in same dataset; files are per-task
- Numerical trigger codes (1-255) are system markers, not task events
- 480 files across 24 subjects (multiple sessions/tasks per subject)

References

- Reference: `Executive Functionning Study for Assessing the Effect of Neurofeedback`

112. openneuro_ds004367 - Meta-rdk: Raw EEG data

Source: analysis.md

Dataset: `openneuro_ds004367` | Year: 2022 | URL: https://openneuro.org/datasets/ds004367
Category: High-level Mental | Subjects: 42 | Files (Completed): 40
Channels: 68 | Sampling Rate: 1200 Hz

Paradigm

A visual motion discrimination task involving left/right direction decisions on random dot kinematogram (RDK) stimuli followed by a metacognitive confidence assessment using a visual analog scale. The numerical trigger codes encode stimulus direction, coherence level, response direction, confidence level, and feedback. The dataset supports both motion perception classification and confidence/metacognition regression.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, confidence, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004367

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`3072`	40x300	short	raw	Motion stimulus onset (specific direction/coherence)
`3456`	40x300	short	raw	Motion stimulus onset (specific direction/coherence)
`3584`	40x300	short	raw	Motion stimulus onset (specific direction/coherence)
`3968`	40x300	short	raw	Motion stimulus onset (specific direction/coherence)
`4096`	40x300	short	raw	Motion stimulus onset (specific direction/coherence)
`4480`	40x300	short	raw	Motion stimulus onset (specific direction/coherence)
`31744...25984`	40x10 each	short	raw	Response/confidence marker (numerical code)

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `confidence`, `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Rdk

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004367 DOI: https://doi.org/10.1101/2022.10.19.22281249

Normal Task 1: Motion Direction Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numerical trigger codes require specific kernel decoding. Regression target: response_time and confidence from misc channels.

Event	y	Meaning
`motion_stimulus_onset`	0	Motion stimulus onset
`left_response`	1	Left direction response
`right_response`	2	Right direction response
`confidence_low`	3	Low confidence rating
`confidence_medium`	4	Medium confidence rating
`confidence_high`	5	High confidence rating

Event	y	Meaning
`feedback_onset`	6	Feedback onset
## Data Quality Notes		

- All event IDs are numerical codes requiring kernel-level decoding
- 68 channels includes EOG/ECG channels
- 42 subjects but only 40 files completed

References

- Reference: `Meta-rdk: Raw EEG data`

113. openneuro_ds004369 - Blink-Pause-Relation (Competing Speaker Paradigm)

Source: analysis.md

Dataset: openneuro_ds004369 | Year: 2023 | URL: https://openneuro.org/datasets/ds004369
Category: High-level Mental | Subjects: 41 | Files (Completed): 41
Channels: 7 | Sampling Rate: 500 Hz

Paradigm

A two-competing speaker paradigm where participants selectively attended to one of two concurrently presented audiobooks while ignoring the other. EEG was recorded to study the relationship between eye blinks and speech pauses during continuous auditory attention. No trial-level event markers are present in the EEG files.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004369

Event Structure

No event markers detected (`no_events`). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
`no_events`	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Publisher-defined BIDS task: Attended Speaker Paradigm

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004369 DOI: https://doi.org/10.18112/openneuro.ds004369.v1.0.1

Normal Task 1: Continuous Segmentation (resting/segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (unsupervised segmentation)
- Notes: No trial markers in EEG files; use sliding window segmentation. Only 7 channels (very low-density). Subject-level labels (attended speaker, blink/pause occurrence) may be available from behavioral logs.

Data Quality Notes

- Very sparse channel setup (7 channels only)
- No trial-level event markers in the EEG files
- Continuous naturalistic listening paradigm (audiobook)

References

- Reference: `Jaeger et al. (2020); Holtze et al. (2021)`

114. openneuro_ds004395 - Penn Electrophysiology of Encoding and Retrieval Study (PEERS)

Source: analysis.md

Dataset: `openneuro_ds004395` | Year: 2023 | URL: https://openneuro.org/datasets/ds004395
Category: High-level Mental | Subjects: 300 | Files (Completed): 6480
Channels: 129 | Sampling Rate: 500 Hz

Paradigm

A large-scale study investigating the electrophysiological correlates of memory encoding and retrieval using various free recall tasks. Participants encoded word lists and subsequently recalled them (free recall), also performing recognition memory tasks. The study provides a rich dataset for ERP analysis of encoding success, retrieval effort, and memory-related oscillations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: confidence
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004395

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`WORD`	6480x~256	short	raw	Word presentation onset (encoding)
`REC_WORD`	6480x~100	short	raw	Recalled word onset (retrieval)
`RECOG_TARGET`	3581x~160	short	raw	Recognition target word onset
`RECOG_LURE`	3581x~80	short	raw	Recognition lure word onset
`RECOG_RESP`	3581x~320	short	raw	Recognition response
`RECOG_CONF`	3563x~320	short	raw	Confidence rating
`DISTRACTOR`	2439xx	short	raw	Distractor task stimulus
`REC_WORD_VV`	5001xx	short	raw	Recalled word (voice-verified)
`REST_REWET`	5955x1	rest	raw	Rest/reward-with-eye-tracking marker
`SESS_START`	6478	admin	raw	Session start marker
`SESS_END`	6474	admin	raw	Session end marker

Event ID	Count (total)	Type	Meta Level	Meaning
REC_START	6480	admin	raw	Recall period start
START	3668	admin	raw	Start marker
STOP	3668	admin	raw	Stop marker
BREAK_START	459	admin	raw	Break start marker
BREAK_STOP	459	admin	raw	Break stop marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `confidence`
- Kernel Note: Keep derivation source, unit, and version metadata.

Tasks

Primary Task 1: Publisher-defined BIDS task: Ltp FR

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004395 DOI: https://doi.org/10.18112/openneuro.ds004395.v2.0.0

Primary Task 2: Publisher-defined BIDS task: Ltp FR2

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004395 DOI: https://doi.org/10.18112/openneuro.ds004395.v2.0.0

Primary Task 3: Publisher-defined BIDS task: VFFR

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)

- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004395 DOI: https://doi.org/10.18112/openneuro.ds004395.v2.0.0

Normal Task 1: Memory Encoding Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: SESS_START/END, REC_START, START/STOP, BREAK_START/STOP are admin markers - exclude. REST_REWET is a rest marker. Very large dataset (300 subjects, 6480 files); suitable for large-scale ERP studies.

Event	y	Meaning
`RECOG_TARGET`	0	Recognition target (previously seen)
`RECOG_LURE`	1	Recognition lure (not previously seen)
## Data Quality Notes		

- Very large dataset (300 subjects, 6480 files); processing time is substantial
- Session-level admin markers (SESS_START/END) must be excluded from epoching
- Multiple sub-tasks with different event structures; separate HDF5 per sub-task recommended

References

- Reference: `Penn Electrophysiology of Encoding and Retrieval Study (PEERS)`

115. openneuro_ds004408 - EEG responses to continuous naturalistic speech

Source: analysis.md

Dataset: openneuro_ds004408 | Year: 2023 | URL: https://openneuro.org/datasets/ds004408
Category: High-level Mental | Subjects: 380 | Files (Completed): 380
Channels: 128 | Sampling Rate: 512 Hz

Paradigm

Participants listened to continuous segments of an audiobook version of 'The Old Man and the Sea' to record EEG responses to naturalistic narrative speech and language processing. The study investigates low-frequency cortical entrainment to speech and phoneme-level processing. No trial-level event markers are present; recordings are continuous.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004408

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings during naturalistic speech listening without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Publisher-defined BIDS task: Listening

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004408 DOI: https://doi.org/10.18112/openneuro.ds004408.v1.0.8

Normal Task 1: Continuous Segmentation (segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (unsupervised segmentation)
- Notes: No trial markers; use sliding window segmentation. Large dataset (380 files). Suitable for EEG-speech alignment analysis using stimulus audio envelope.

Data Quality Notes

- No trial-level event markers; continuous listening paradigm
- Very large dataset (380 files); batch processing required
- 128-channel EEG at 512 Hz

References

- Reference: `Low-frequency cortical entrainment to speech reflects phoneme-level processing`

116. openneuro_ds004511 - Deception_data

Source: analysis.md

Dataset: openneuro_ds004511 | Year: 2023 | URL: https://openneuro.org/datasets/ds004511
Category: High-level Mental | Subjects: 29 | Files (Completed): 29 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed a deception task while EEG was recorded. The dataset contains continuous EEG recordings; only a single hardware synchronization marker (Sync) is present per recording rather than trial-level event markers. No meaningful trial annotations are available for epoch-based analysis.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004511

Event Structure

No task event markers detected (no_events). Only hardware sync marker Sync(1) present - this is a system/hardware event, not a task event.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Publisher-defined BIDS task: CC

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)

- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004511 DOI: https://doi.org/10.3758/s13428-020-01516-y

Primary Task 2: Publisher-defined BIDS task: GG

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004511 DOI: https://doi.org/10.3758/s13428-020-01516-y

Primary Task 3: Publisher-defined BIDS task: Rest

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004511 DOI: https://doi.org/10.3758/s13428-020-01516-y

Normal Task 1: Continuous Segmentation (resting/segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (unsupervised segmentation)
- Notes: Sync(1) is a hardware synchronization marker, not a task event. Use sliding window segmentation for feature extraction.

Data Quality Notes

- No trial-level event markers
- Sync(1) is a hardware-only event and should not be used for epoching

References

- Reference: `Deception_data`

117. openneuro_ds004514 - Simultaneous EEG and fNIRS recordings for semantic decoding of imagined animals and tools

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004514` | Year: 2023 | URL: https://openneuro.org/datasets/ds004514
Category: High-level Mental | Subjects: 22 | Files (Completed): 22 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Participants imagined animals and tools in response to visual cues while simultaneous EEG and fNIRS were recorded. Events encode the specific animal or tool category being imagined across four modalities: visual imagery, silent naming, auditory imagery, and drawing imagery. The study investigates cross-modal semantic representations of living vs non-living concepts.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004514

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`animal_visual_imagery_task`	~22xN	long	raw	Visual imagery of animals
`tool_visual_imagery_task`	~22xN	long	raw	Visual imagery of tools
`animal_silent_naming_task`	~22xN	long	raw	Silent naming imagery of animals
`tool_silent_naming_task`	~22xN	long	raw	Silent naming imagery of tools
`animal_auditory_imagery_task`	~22xN	long	raw	Auditory imagery of animals
`tool_auditory_imagery_task`	~22xN	long	raw	Auditory imagery of tools
`animal_drawing_imagery_task`	~22xN	long	raw	Drawing imagery of animals
`tool_drawing_imagery_task`	~22xN	long	raw	Drawing imagery of tools
`rest`	~22xN	rest	raw	Rest period between imagery blocks

Event ID	Count (total)	Type	Meta Level	Meaning
animal_visual_image	~22xN	short	raw	Visual image of animal (cue)
tool_visual_image	~22xN	short	raw	Visual image of tool (cue)
audio_animal	~22xN	short	raw	Audio cue for animal
audio_tool	~22xN	short	raw	Audio cue for tool
written_animal	~22xN	short	raw	Written word cue for animal
written_tool	~22xN	short	raw	Written word cue for tool
animal_drawing	~22xN	short	raw	Drawing cue for animal
tool_drawing	~22xN	short	raw	Drawing cue for tool
animal_text_cue	~22xN	short	raw	Text cue for animal
tool_text_cue	~22xN	short	raw	Text cue for tool

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Animal vs Tool Semantic Decoding from Mental Imagery

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 3.0 s, overlap = 0.0 *, target_events = [animal_visual_imagery_task, tool_visual_imagery_task, animal_silent_naming_task, tool_silent_naming_task, animal_auditory_imagery_task, tool_auditory_imagery_task, animal_drawing_imagery_task, tool_drawing_imagery_task]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
animal_visual_imagery_task	0	Animal mental imagery
animal_silent_naming_task	0	Animal mental imagery
animal_auditory_imagery_task	0	Animal mental imagery
animal_drawing_imagery_task	0	Animal mental imagery
tool_visual_imagery_task	1	Tool mental imagery
tool_silent_naming_task	1	Tool mental imagery
tool_auditory_imagery_task	1	Tool mental imagery
tool_drawing_imagery_task	1	Tool mental imagery

- Evidence:
- DOI: 10.1038/s41597-025-04967-0 - Data descriptor confirms 3-second mental task epochs for Dataset 1 (5s for Dataset 2), binary animal vs tool classification across silent naming, visual, auditory, and drawing

imagery modalities.
- Kernel Notes: Use long_event epoching since sustained imagery tasks are >=2s. The tmax=3.0s aligns with the paper's Dataset 1 mental task duration. Overlap=0.0 as each trial is distinct. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Modality-Specific Multi-class Semantic Decoding

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 3.0 s, overlap = 0.0 *, target_events = [`animal_visual_imagery_task`, `tool_visual_imagery_task`, `animal_silent_naming_task`, `tool_silent_naming_task`, `animal_auditory_imagery_task`, `tool_auditory_imagery_task`, `animal_drawing_imagery_task`, `tool_drawing_imagery_task`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`animal_silent_naming_task`	0	Animal silent naming
`tool_silent_naming_task`	1	Tool silent naming
`animal_visual_imagery_task`	2	Animal visual imagery
`tool_visual_imagery_task`	3	Tool visual imagery
`animal_auditory_imagery_task`	4	Animal auditory imagery
`tool_auditory_imagery_task`	5	Tool auditory imagery
`animal_drawing_imagery_task`	6	Animal drawing imagery
`tool_drawing_imagery_task`	7	Tool drawing imagery

- Evidence:
- DOI: 10.1038/s41597-025-04967-0 - Paper describes 8-class mental task classification (4 modalities x 2 categories) with modality encoded in event names.
- Kernel Notes: Multi-class classification distinguishing both semantic category (animal/tool) and imagery modality (silent naming, visual, auditory, drawing). Same epoch parameters as Primary Task 1.

Data Quality Notes

- Simultaneous EEG and fNIRS; only EEG channels used here
- 32 EEG channels (lower density than typical 64-ch systems)
- 4 imagery modalities x 2 categories = 8 sustained imagery conditions

References

- Reference: `Simultaneous EEG and fNIRS recordings for semantic decoding of imagined animals and tools`

118. openneuro_ds004515 - EEG: Alcohol imagery reinforcement learning task with light and heavy drinker participants

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004515` | Year: 2023 | URL: https://openneuro.org/datasets/ds004515
Category: High-level Mental | Subjects: 42 | Files (Completed): 42 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants (light and heavy drinkers) performed a reinforcement learning task involving alcohol-related and neutral imagery cues. On each trial, participants chose between left and right buttons in response to cues, with feedback (FB) indicating correct/incorrect responses. The study investigates how reward learning interacts with alcohol cue reactivity across drinking groups.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004515

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`CUE`	~42xN	short	raw	Stimulus cue onset
`Left Button`	~42xN	short	raw	Left button press response
`Right Button`	~42xN	short	raw	Right button press response
`FB`	~42xN	short	raw	Feedback onset
`STATUS`	~42xN	admin	raw	BrainVision hardware status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`

- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Reward vs Punishment Feedback Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 1.0 s, target_events = [FB]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
FB (reward/green screen)	1	Reward feedback (correct/target)
FB (punishment/red screen)	0	Punishment feedback (incorrect/non-target)

- Evidence:
- DOI: `10.1016/j.pscychresns.2023.111685` - Paper analyzes EEG from ds004515 examining RewP and delta band power in response to reward vs punishment feedback, using feedback-locked epochs cut to -500 to +1000 ms
- Kernel Notes: The FB event marker must be parsed to distinguish reward (green screen) vs punishment (red screen) conditions. Verify event annotations or stim channel contains reward/punishment sub-codes. (replaces vague Primary Task 1 from original analysis.md)

Primary Task 2: Alcohol vs Neutral Imagery Classification (within feedback context)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 1.0 s, target_events = [CUE]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
CUE (alcohol imagery)	1	Alcohol-related cue
CUE (neutral/puppy imagery)	0	Neutral control cue

- Evidence:
- DOI: `10.1016/j.pscychresns.2023.111685` - Paper examines affective imagery type (puppy vs alcohol) effects on RewP and delta power in the reinforcement learning task
- Kernel Notes: CUE events must be classified by accompanying image type (alcohol vs neutral). Verify cue annotation includes image category or link to stimulus file.

Normal Task 1: Response Direction Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: STATUS is an admin event - exclude from target_events. CUE and FB are trial markers but not the primary classification target. Left vs Right button press provides a binary motor/decision classification.

Event	y	Meaning
`Left Button`	0	Left button press
`Right Button`	1	Right button press

Data Quality Notes

- 42 subjects split between light and heavy drinkers
- STATUS markers are BrainVision hardware events, not task-related

References

- Reference: `EEG: Alcohol imagery reinforcement learning task with light and heavy drinker participants`

119. openneuro_ds004517 - EEG recordings for semantic decoding of imagined animals and tools during auditory imagery task

Source: analysis.md

Dataset: `openneuro_ds004517` | Year: 2023 | URL: https://openneuro.org/datasets/ds004517
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Participants imagined animals and tools in response to auditory cues while EEG was recorded. Each trial consisted of an audio cue followed by a sustained auditory imagery period (~5 s). The study investigates EEG correlates of semantic imagery for living (animals) vs non-living (tools) concepts in the auditory domain.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004517

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`animal_auditory_imagery_task`	~20xN	long	raw	Sustained auditory imagery of animals
`tool_auditory_imagery_task`	~20xN	long	raw	Sustained auditory imagery of tools
`audio_animal`	~20xN	short	raw	Auditory cue onset for animal
`audio_tool`	~20xN	short	raw	Auditory cue onset for tool
`rest`	~20xN	rest	raw	Rest period between imagery blocks

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Publisher-defined BIDS task: Eeg

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004517 DOI: https://doi.org/10.18112/openneuro.ds004517.v1.0.2

Normal Task 1: Animal vs Tool Auditory Imagery (binary, long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Imagery tasks are sustained (~5 s) - use long_event epoching. audio_animal/audio_tool are short cue onsets; rest events are inter-block resting periods.

Event	y	Meaning
`animal_auditory_imagery_task`	0	Auditory imagery of animals
`tool_auditory_imagery_task`	1	Auditory imagery of tools
## Data Quality Notes		

- 32 EEG channels
- Sustained imagery periods (~5 s) are suitable for long_event epoching
- 20 subjects with balanced animal/tool imagery conditions

References

- Reference: `EEG recordings for semantic decoding of imagined animals and tools during auditory imagery task`

120. openneuro_ds004532 - EEG: Probabilistic Selection Task (PST) + PST with Cabergoline Challenge

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004532` | Year: 2023 | URL: https://openneuro.org/datasets/ds004532
Category: High-level Mental | Subjects: 40 | Files (Completed): 80 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants performed the Probabilistic Selection Task (PST) in two sessions: a placebo session and a session with cabergoline (dopamine D2/D3 agonist). The PST involves learning to choose symbols with higher reward probability through feedback. Events encode training stimuli (Trn Stim), test stimuli (Tst Stim), feedback (FB), and outcome (correct/incorrect). The study investigates dopaminergic modulation of reinforcement learning.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004532

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Trn Stim`	~80xN	short	raw	Training stimulus onset
`Tst Stim`	~80xN	short	raw	Test stimulus onset
`FB`	~80xN	short	raw	Feedback onset
`correct`	~80xN	short	raw	Correct response outcome
`incorrect`	~80xN	short	raw	Incorrect response outcome
`STATUS`	~80xN	admin	raw	BrainVision hardware status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`

- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Trn Stim`, `Tst Stim`, `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Binary Outcome Classification (correct vs incorrect responses)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.5 s , tmax = 5.0 s , target_events = [`correct`, `incorrect`]
- Label Route: `Annotation`
- Target Definition:

Event	y	Meaning
`correct`	0	Correct response (reward)
`incorrect`	1	Incorrect response (punishment)

- Evidence:
- DOI: 10.1038/ncomms6394 - Shows EEG classification of correct vs incorrect outcomes in PST with 64-channel 500 Hz recordings; epochs time-locked to feedback/response events with binary reward/punishment labels.
- Kernel Notes: Epoch around `FB` (feedback) or `correct`/`incorrect` events. Baseline: -300 to -200 ms. The paper also examines conflict cost bias (B>C vs C>B choice), which could be added as a super-meta derivation. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Stimulus-Evoked Response Classification (training vs test stimuli)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s , tmax = 1.5 s , target_events = [`Trn Stim`, `Tst Stim`]
- Label Route: `Annotation`
- Target Definition:

Event	y	Meaning
`Trn Stim`	0	Training stimulus onset
`Tst Stim`	1	Test stimulus onset

- Evidence:
- DOI: 10.1038/ncomms6394 - Original PST paradigm uses training (Trn Stim) and test (Tst Stim) phases; EEG analysis epochs around these stimulus onsets to examine conflict processing and reinforcement learning.
- Kernel Notes: The PST has distinct training and test phases. Training stimuli involve congruent/incongruent trials; test stimuli assess learned selection biases.

Primary Task 3: Feedback-Related Potentials

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s , tmax = 1.0 s , target_events = [`FB`]
- Label Route: `stim`
- Meta Level: super
- Target Definition:

Event	y	Meaning

| FB | 0 | Positive feedback (reward) |
| FB | 1 | Negative feedback (punishment) |
- Super Meta Notes: Derived from raw event FB combined with subsequent `correct/incorrect` outcome. Kernel step: align FB event with following correct/incorrect marker to determine feedback valence (reward vs punishment).
- Evidence:
- DOI: 10.1038/ncomms6394 - Feedback-related potentials (FRP) analyzed time-locked to FB onset; examines dopaminergic modulation of reward/punishment processing under cabergoline.
- Kernel Notes: Must correlate each FB event with the subsequent outcome (correct/incorrect) to assign reward/punishment label. The dataset has separate correct/incorrect events that follow FB onset.

Data Quality Notes

- 40 subjects x 2 sessions (placebo + cabergoline) = 80 files
- STATUS markers are BrainVision hardware events, not task-related
- Two sessions per subject enable within-subject drug effect analysis

References

- Reference: `EEG: Probabilistic Selection Task (PST) + PST with Cabergoline Challenge`

121. openneuro_ds004561 - Illusion of Agency over Electrically-Actuated Movements

Source: analysis.md

Dataset: `openneuro_ds004561` | Year: 2023 | URL: https://openneuro.org/datasets/ds004561
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants experienced electrical muscle stimulation that actuated their hand movements in three conditions: stimulation (involuntary movement), pretest (voluntary movement before stimulation), and posttest (voluntary movement after stimulation). EEG was recorded to investigate illusory sense of agency over externally-driven movements. The study explores how prior voluntary movement influences agency attribution over stimulation-induced movements.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004561

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stimulation`	~20xN	short	raw	Electrical stimulation onset (involuntary movement)
`pretest`	~20xN	short	raw	Pre-stimulation voluntary movement
`posttest`	~20xN	short	raw	Post-stimulation voluntary movement

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulation`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Agency RT

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004561 DOI: https://doi.org/10.18112/openneuro.ds004561.v1.0.0

Normal Task 1: Agency Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 3-class classification of agency conditions. stimulation = externally actuated movement, pretest/posttest = voluntary movements before/after stimulation session.

Event	y	Meaning
`stimulation`	0	Electrical stimulation (involuntary movement)
`pretest`	1	Pre-stimulation voluntary movement
`posttest`	2	Post-stimulation voluntary movement
## Data Quality Notes		

- 20 subjects with balanced condition structure
- 64 channels at 512 Hz provides good temporal resolution for movement-related potentials

References

- Reference: `Illusion of Agency over Electrically-Actuated Movements`

122. openneuro_ds004582 - FakeFaceEmo_data

Source: analysis.md

Dataset: openneuro_ds004582 | Year: 2023 | URL: https://openneuro.org/datasets/ds004582
Category: High-level Mental | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants viewed fake face images with different emotional expressions while EEG was recorded. The dataset contains continuous EEG recordings; only hardware/metadata markers (New Segment, Comment) are present in the event files rather than trial-level stimulus markers. No trial-level event markers are available for epoch-based analysis.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004582

Event Structure

No task event markers detected (no_events). Only hardware/metadata markers present.

Event ID	Count (total)	Type	Meta Level	Meaning
New Segment/	~30x1	admin	raw	Recording segment boundary (hardware marker)

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Publisher-defined BIDS task: FF

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004582 DOI: https://doi.org/10.18112/openneuro.ds004582.v1.0.0

Normal Task 1: Continuous Segmentation (resting/segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (unsupervised segmentation)
- Notes: New Segment/ is a BrainVision hardware recording marker, not a task event. Use sliding window segmentation for feature extraction.

Data Quality Notes

- No trial-level event markers
- New Segment/ is a hardware-only event and should not be used for epoching
- 30 subjects with continuous EEG recordings

References

- Reference: `FakeFaceEmo_data`

123. openneuro_ds004595 - EEG: RL Task (3-Armed Bandit) with Alcohol Cues in Hazardous Drinkers

Source: analysis.md

Dataset: `openneuro_ds004595` | Year: 2023 | URL: https://openneuro.org/datasets/ds004595
Category: High-level Mental | Subjects: 53 | Files (Completed): 53 | Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Participants completed a reinforcement learning task using a 3-armed bandit paradigm incorporating alcohol-related and neutral beverage cues to study decision-making in hazardous drinkers versus controls. On each trial, a cue was presented followed by a three-arm choice and probabilistic win/loss feedback. The study examines electrophysiological markers of cue-driven exploration and aberrant reinforcement learning.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004595

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Fixation`	18550	short	raw	Fixation cross onset (trial start)
`FB: Win`	9713	short	raw	Win feedback delivered
`FB: Loss`	7254	short	raw	Loss feedback delivered
`Left Button`	6276	short	raw	Left arm choice response
`Up Button`	6188	short	raw	Up arm choice response
`Right Button`	5553	short	raw	Right arm choice response
`STATUS`	4271	admin	raw	System status marker (exclude)
`CUE: 123`	2955	short	raw	Cue onset - arm order permutation 1-2-3
`CUE: 231`	2934	short	raw	Cue onset - arm order permutation 2-3-1

Event ID	Count (total)	Type	Meta Level	Meaning
CUE: 132	2931	short	raw	Cue onset - arm order permutation 1-3-2
CUE: 321	2919	short	raw	Cue onset - arm order permutation 3-2-1
CUE: 312	2908	short	raw	Cue onset - arm order permutation 3-1-2
CUE: 213	2853	short	raw	Cue onset - arm order permutation 2-1-3
No Response	533	admin	raw	No-response trial (exclude)
FB: No Resp	533	admin	raw	No-response feedback (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Three Armed Bandit

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004595 DOI: https://doi.org/10.18112/openneuro.ds004595.v1.0.0

Normal Task 1: Feedback Classification (binary)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: STATUS, No Response, and FB: No Resp are admin - exclude. CUE events encode arm-order permutations; aggregate to generic CUE if order-agnostic analysis desired.

Event	y	Meaning
FB: Win	0	Win feedback
FB: Loss	1	Loss feedback

Event	y	Meaning
## Data Quality Notes		

- STATUS markers must be excluded from training.
- CUE events encode arm order permutations - group to single CUE class if order irrelevant.

References

- Reference: `Electrophysiological markers of aberrant cue-specific exploration in hazardous drinkers`

124. openneuro_ds004626 - Can We Dissociate Hypervigilance to Social Threats from Altered Perceptual Decision-Making in Lonely Individualsx

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004626` | Year: 2023 | URL: https://openneuro.org/datasets/ds004626
Category: High-level Mental | Subjects: 52 | Files (Completed): 52
Channels: 68 | Sampling Rate: 1000 Hz

Paradigm

A perceptual decision-making task investigating whether lonely individuals exhibit hypervigilance toward social threats, combining Drift Diffusion Modeling (DDM) with ERP analysis. Participants viewed social and non-social threat stimuli and made rapid perceptual judgements. ERP components (P1, N2pc, LPP) and computational decision parameters were compared across loneliness severity groups.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004626

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`208`	36977	short	raw	High-frequency stimulus/response trigger
`998`	35001	short	raw	Secondary trigger (trial boundary/ITI)
`236`	6240	short	raw	Stimulus trigger code
`146`	6239	short	raw	Stimulus trigger code
`206`	6238	short	raw	Stimulus trigger code
`186`	6238	short	raw	Stimulus trigger code
`142`	6237	short	raw	Stimulus trigger code
`210`	6233	short	raw	Stimulus trigger code
`999`	2363	admin	raw	Trial-end/admin marker (exclude)
`248`	1560	short	raw	Trigger code
`255`	1560	admin	raw	S255 admin sync trigger (exclude)
`300`	1528	short	raw	Trigger code

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Dot-Probe Stimulus Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 0.5 s, target_events = [208, 998, 236, 146, 206, 186, 142, 210]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
208	0	High-frequency stimulus/response trigger
998	1	Secondary trigger (trial boundary/ITI)
236	2	Stimulus trigger code
146	3	Stimulus trigger code
206	4	Stimulus trigger code
186	5	Stimulus trigger code
142	6	Stimulus trigger code
210	7	Stimulus trigger code

- Evidence:
- DOI: 10.1111/psyp.14406 - The paper analyzes EEG data from 52 participants performing a dot-probe task with angry/neutral face stimuli, extracting N2pc ERP components in 700ms epochs (tmin=-0.2s, tmax=0.5s relative to probe onset).
- Kernel Notes: Exclude admin markers 255 and 999 from epoching. Baseline correction: 200ms pre-stimulus. Event IDs map directly to raw triggers in the EEG stream.

Normal Task 1: Stimulus classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Trigger 255 is a known admin sync code - exclude. Trigger 999 is trial-end/admin - exclude. Full decode requires events.tsv inspection.

Event	y	Meaning
208	0	Primary stimulus trigger
998	1	Secondary trigger
236	2	Trigger code
146	3	Trigger code
206	4	Trigger code
186	5	Trigger code
142	6	Trigger code
210	7	Trigger code

Data Quality Notes

- 30+ distinct numeric trigger codes - check events.tsv for full decode table.
- Codes 255 and 999 are administrative and must be excluded.

References

- Reference: Can we dissociate hypervigilance to social threats from altered perceptual decision-making processes in lonely individualsx DOI: 10.1111/psyp.14406

125. openneuro_ds004706 - Spatial Memory and Non-Invasive Closed-Loop Stimulus Timing

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004706` | Year: 2023 | URL: https://openneuro.org/datasets/ds004706
Category: High-level Mental | Subjects: 149 | Files (Completed): 298
Channels: 137 | Sampling Rate: 2048 Hz

Paradigm

A hybrid spatial-navigation and free recall task where participants delivered items to locations in a virtual town environment, with EEG recorded during both encoding and retrieval phases. The paradigm incorporates closed-loop stimulus timing to investigate the relationship between memory encoding, theta oscillations, and stimulation-driven memory enhancement. Sessions included free recall and cued recall phases with video stimuli.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004706

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`FFR_REC_WORD`	55013	short	raw	Free-form recall word event
`CUED_REC_STOP`	44064	short	raw	Cued recall stop marker
`CUED_REC_CUE`	44064	short	raw	Cued recall cue onset
`WORD`	43260	short	raw	Word stimulus onset (encoding)
`REC_WORD`	39061	short	raw	Recalled word event
`CUED_REC_WORD`	34381	short	raw	Cued recall word event
`CLASSIFIER_WAIT`	22890	short	raw	Closed-loop classifier waiting
`POINTER_ON`	20872	short	raw	Spatial navigation pointer onset
`CLASSIFIER`	19005	short	raw	Closed-loop classifier event
`EFR_MARK`	11498	short	raw	EFR (encoding-phase) marker

Event ID	Count (total)	Type	Meta Level	Meaning
CUED_REC_WORD_VV	11328	short	raw	Cued recall word (video verified)
TRIAL_END	2888	short	raw	Trial end marker
TRIAL_START	2884	short	raw	Trial start marker
REC_STOP	2952	short	raw	Recording stop marker
REC_START	2952	short	raw	Recording start marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Memory Encoding Classification (Remembered vs Forgotten)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = 0.3 s, tmax = 1.3 s, target_events = [WORD]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
WORD	0	Word presented but later forgotten
WORD	1	Word presented and subsequently remembered
- Evidence:
- DOI: 10.1037/xlm0001341 - Uses dataset ds004706 to train classifiers on spectral features to decode successful versus failed memory encoding based on subsequent recall
- Kernel Notes: Requires label fusion: merge WORD events with recall outcome labels (REC_WORD/FFR_REC_WORD) to determine remembered (1) vs forgotten (0) labels. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Memory Retrieval Classification (Correct Recall vs Deliberation)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -1.0 s, tmax = -0.05 s, target_events = [REC_WORD]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
REC_WORD	0	Deliberation/incorrect recall
REC_WORD	1	Correct recall
- Evidence:

- DOI: 10.1037/xlm0001341 - Uses dataset ds004706 to train classifiers to decode successful retrieval versus deliberation periods
- Kernel Notes: Requires label fusion: REC_WORD events need to be paired with timing/behavioral data to distinguish correct recall from deliberation periods.

Primary Task 3: Publisher-defined BIDS task: Nicls Courier Closed Loop

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004706 DOI: https://doi.org/10.18112/openneuro.ds004706.v1.0.0

Primary Task 4: Publisher-defined BIDS task: Nicls Courier Read Only

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004706 DOI: https://doi.org/10.18112/openneuro.ds004706.v1.0.0

Data Quality Notes

- High sampling rate (2048 Hz) - downsample for efficiency.
- 30+ event types; focus on WORD/REC_WORD for memory encoding analysis.

References

- Reference: `Spatial memory and non-invasive closed-loop stimulus timing`

126. openneuro_ds004752 - Dataset of Intracranial EEG, Scalp EEG and Beamforming Sources from Epilepsy Patients - Verbal Working Memory

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004752 | Year: 2023 | URL: https://openneuro.org/datasets/ds004752
Category: High-level Mental | Subjects: 15 | Files (Completed): 136
Channels: 64 | Sampling Rate: 2000 Hz

Paradigm

Epilepsy patients performed a modified Sternberg verbal working memory task consisting of encoding, maintenance, and recall phases while simultaneous scalp and intracranial EEG were recorded. The dataset includes iEEG from depth electrodes, scalp EEG, single-unit neuron spike times, and electrode localizations, enabling multi-level analysis of memory processes. No trial-level event annotations are available in the BIDS events.tsv.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004752

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(no events)	-	-	raw	No event markers detected in events.tsv

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Verbal Working Memory Phase Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -5.0 s , tmax = 0.0 s , target_events = [`encoding`, `maintenance`]
- Label Route: `Annotation`
- Meta Level: super

- Target Definition:
Event	y	Meaning
`encoding`	0	Encoding phase (stimulus presentation, cortex->hippocampus information flow)
`maintenance`	1	Maintenance phase (memory retention, hippocampus->cortex information flow)
- Super Meta Notes: Derived from experimental design timing structure (fixed encoding 2s, maintenance 3s periods). The original data contains behavioral trial metadata (set size, match/mismatch, correct/incorrect) in the data records. Kernel must reconstruct event timing from trial onsets and apply fixed epoch windows relative to encoding/maintenance periods.
- Evidence:
- DOI: 10.7554/eLife.78677 - Shows directed functional connectivity analysis classifying encoding vs maintenance periods using Granger causality between hippocampus and auditory cortex
- Kernel Notes: Use trial onset markers to reconstruct epoch timing. Encoding: [trial_onset, trial_onset+2s], Maintenance: [trial_onset+2s, trial_onset+5s]. Focus analysis on last 2s of maintenance [-2, 0]s relative to probe.

Primary Task 2: Memory Workload Classification (Set Size)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s , tmax = 5.0 s , target_events = [`set_size_4`, `set_size_6`, `set_size_8`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`set_size_4`	0	Low memory workload (4 items)
`set_size_6`	1	Medium memory workload (6 items)
`set_size_8`	2	High memory workload (8 items)
- Super Meta Notes: Derived from trial-level metadata (Table 2 in original paper) recording set size (4/6/8). Not encoded as raw BIDS event markers. Kernel must extract set size from behavioral log and map to epoch labels.
- Evidence:
- DOI: 10.1038/s41597.020.0364-3 - Dataset paper describes Sternberg task with set sizes 4, 6, 8 as memory workload manipulation
- Kernel Notes: Extract set size information from per-trial data structure. Epoch spans full trial from encoding onset (0s) to end of maintenance period (5s total: 2s encoding + 3s maintenance).

Primary Task 3: Probe Type Classification (Match vs Mismatch)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s , tmax = 0.0 s , target_events = [`match_IN`, `mismatch_OUT`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`match_IN`	0	Probe matches encoded set (IN trial)
`mismatch_OUT`	1	Probe does not match encoded set (OUT trial)
- Super Meta Notes: Derived from behavioral response metadata (IN=match, OUT=mismatch). Probe presented after maintenance period. Kernel must extract probe type from trial records.

- Evidence:
- DOI: 10.1038/s41597.020.0364-3 - Trial metadata includes match/mismatch (IN/OUT) probe classification
- Kernel Notes: Epoch window relative to probe onset. Extract probe type from behavioral log. Use response period [-1, 0]s relative to probe for classification.

Primary Task 4: Trial Accuracy Classification (Correct vs Incorrect)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -2.0 s , tmax = 0.0 s , target_events = [`correct`, `incorrect`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`correct`	0	Subject correctly identified match/mismatch
`incorrect`	1	Subject incorrectly identified match/mismatch

- Super Meta Notes: Derived from behavioral response accuracy. Trial-level metadata includes correct/incorrect labels and response times. Kernel must compute accuracy from response vs ground truth.
- Evidence:
- DOI: 10.7554/eLife.78677 - Analysis compares correct vs incorrect trials for information flow direction classification
- DOI: 10.1038/s41597.020.0364-3 - Dataset includes accuracy labels for each trial
- Kernel Notes: Extract response accuracy from behavioral log. Epoch window covers probe and response period [-2, 0]s relative to response execution.

Normal Task 1: Continuous Segmentation

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: None (no event tag)
- Notes: No trial-level events available in events.tsv. EEG is continuous during task performance. Use sliding-window segmentation. Trial timing may be available in accompanying behavioral logs not captured in BIDS events.tsv.

Data Quality Notes

- No event markers - trial-epoch structure not available from events.tsv.
- Very high sampling rate (2000 Hz) - downsample before analysis.
- Epilepsy patients; ictal episodes should be identified and excluded.

References

- Reference: `Dataset of human medial temporal lobe neurons, scalp and intracranial EEG during a verbal working memory task. DOI: 10.1038/s41597-020-0364-3`

127. openneuro_ds004796 - PEARL-Neuro Database (Alzheimer's Risk Genes, Lifestyle, Neuroimaging)

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004796 | Year: 2025 | URL: https://openneuro.org/datasets/ds004796
Category: High-level Mental | Subjects: 79 | Files (Completed): 233
Channels: 127 | Sampling Rate: 1000 Hz

Paradigm

A multimodal neuroimaging database of healthy middle-aged adults (50-63 years) carrying Alzheimer's risk genes (APOE, PICALM). The EEG protocol includes resting-state recordings (eyes open and closed) and two active cognitive tasks: the Multi-Source Interference Task (MSIT) and Sternberg's memory task. Stimulus and button-press response events are recorded across task sessions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004796

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stimulus`	27862	short	raw	Stimulus onset (MSIT or Sternberg item)
`response`	24051	short	raw	Participant button-press response

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: APOE4 genetic status classification from theta oscillation propagation

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 0.84 s *, target_events = [`stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
derived_APOE4_minus	0	No APOE4 allele (APOE4-)
derived_APOE4_plus	1	At least one APOE4 allele (APOE4+)

- Super Meta Notes: Derived from participant genetic testing data (APOE genotype). Kernel step: Extract MSIT epochs (tmin=0, tmax=0.84s around stimulus onset) or Sternberg epochs (tmin=0, tmax=12.36s), compute theta oscillation propagation features (e.g., median LPG length), then classify APOE4+ vs APOE4-.
- Evidence:
- DOI: 10.3389/fnins.2025.1579329 - Used PEARL-Neuro dataset to classify APOE4+ vs APOE4- participants based on theta oscillation propagation differences; MSIT epoch duration ~0.84s, Sternberg ~12.36s.
- Kernel Notes: Requires participant genotype metadata. Use MSIT epochs by default (shorter, more consistent). Extract theta band (4-8 Hz) oscillatory propagation from frontal midline electrodes.

Primary Task 2: Dementia risk stratification from resting-state spectral power ratios

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
rest_segment	continuous	Clinical dementia risk scores (attention deficit, hyperactivity)

- Super Meta Notes: Derived from resting-state EEG segments (eyes open or closed). Kernel step: Apply Hann windowing with 50% overlap to create 4s epochs, compute delta/beta and theta/beta power ratios via Welch's method, correlate with clinical scores.
- Evidence:
- DOI: 10.1109/icdmw69685.2025.00218 - Applied 4s Hann-windowed epochs with 50% overlap to resting-state EEG to compute spectral power ratios and correlate with dementia risk clinical scores.
- Kernel Notes: Filter for resting-state files (task-rest in filename). Compute absolute/relative power in delta/theta/alpha/beta bands. Target variables: delta/beta ratio, theta/beta ratio correlated with clinical scores.

Normal Task 1: Stimulus vs response classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 233 files = 79 subjects x ~3 sessions. Resting-state sessions have no stimulus events; only task sessions contribute stimulus/response epochs.

Event	y	Meaning
`stimulus`	0	Stimulus onset
`response`	1	Button-press response

Data Quality Notes

- Mixed resting-state and task sessions; filter by file task label.
- High channel count (127); may include non-EEG channels.

References

- Reference: `PEARL-Neuro Database: EEG, fMRI, health and lifestyle data of middle-aged people at risk of dementia. DOI: 10.1038/s41597-024-03106-5`

128. openneuro_ds004802 - Loneliness in the Brain: Distinguishing Between Hypersensitivity and Hyperalertness (Pilot)

Source: analysis.md

Dataset: `openneuro_ds004802` | Year: 2023 | URL: https://openneuro.org/datasets/ds004802
Category: High-level Mental | Subjects: 79 | Files (Completed): 79
Channels: 73 | Sampling Rate: 512 Hz

Paradigm

A pilot EEG study investigating the neural underpinnings of loneliness by contrasting social hypersensitivity (amplified neural responses to social stimuli) and hyperalertness (heightened attention to potential social threats). Participants viewed social and non-social stimuli while EEG was recorded. Trigger codes encode stimulus type, valence, and response conditions. Code 20 dominates the event structure (49,726 total) and may represent a high-frequency refresh or fixation signal.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004802

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
20	49726	short	raw	Dominant trigger (likely fixation/refresh signal)
111	4144	short	raw	Trigger code 111
25	2823	short	raw	Trigger code 25
24	2817	short	raw	Trigger code 24
21	2747	short	raw	Trigger code 21
23	2667	short	raw	Trigger code 23
22	2666	short	raw	Trigger code 22
42	2319	short	raw	Trigger code 42
44	2316	short	raw	Trigger code 44
43	2246	short	raw	Trigger code 43
61	2221	short	raw	Trigger code 61
62	2185	short	raw	Trigger code 62
65	2182	short	raw	Trigger code 65

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Target vs non-target stimulus events
- Evidence: P300/oddball paradigm appears as core study objective. Source: https://openneuro.org/datasets/ds004802 DOI: https://doi.org/10.18112/openneuro.ds004802.v1.0.0

Normal Task 1: Stimulus classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 30+ distinct trigger codes; full decode from events.tsv required. Code 20 dominates (49,726 total) - likely a high-frequency fixation or visual onset marker.

Event	y	Meaning
`20`	0	Primary stimulus trigger
`111`	1	Trigger code 111
`25`	2	Trigger code 25
`24`	3	Trigger code 24
`21`	4	Trigger code 21
`23`	5	Trigger code 23
`22`	6	Trigger code 22
## Data Quality Notes		

- 30+ distinct trigger codes - check events.tsv decode table.
- Code 20 (49,726 total) is anomalously high - may be a fixation/refresh signal rather than stimulus.

References

- Reference: Loneliness in the Brain: Distinguishing Between Hypersensitivity and Hyperalertness

129. openneuro_ds005114 - EEG: DPX Cognitive Control Task in Acute Mild TBI

Source: analysis.md

Dataset: `openneuro_ds005114` | Year: 2024 | URL: https://openneuro.org/datasets/ds005114
Category: High-level Mental | Subjects: 62 | Files (Completed): 217 | Channels: 65 | Sampling Rate: 500 Hz

Paradigm

The Dot Pattern Expectancy (DPX) task, a version of the Continuous Performance Task, was used to assess proactive versus reactive cognitive control in sub-acute mild traumatic brain injury (mTBI) patients and healthy controls. Participants responded to cue-probe pairs (CueA/CueB x Probe aX/aY/bY), enabling decomposition of context-maintenance and response-inhibition processes via P3a/P3b ERP components.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005114

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Response: CueCorr`	50287	short	raw	Correct response to cue
`Response: Probe_Corr`	46594	short	raw	Correct response to probe
`CueAOnset`	43580	short	raw	Cue A stimulus onset
`Probe_aX_onset`	35272	short	raw	Probe aX onset (congruent with CueA)
`STATUS`	19463	admin	raw	System status marker (exclude)
`Practice: filler trigger`	15787	admin	raw	Practice trial filler (exclude)
`Probe_aY_onset`	6443	short	raw	Probe aY onset
`Probe_bY_onset`	5428	short	raw	Probe bY onset
`Response: Probe_Incorr`	3626	short	raw	Incorrect probe response
`Feedback: ProbeErr`	3626	short	raw	Error feedback
`CueBOnset`	2661	short	raw	Cue B stimulus onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`, `Practice: filler trigger`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: DPX

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://doi.org/10.1016/j.neuropsychologia.2019.107125

Normal Task 1: Cue and Probe Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: STATUS (19,463) and Practice: filler trigger (15,787) are admin - exclude. mTBI patients and healthy controls included; participant group labels in participants.tsv.

Event	y	Meaning
`CueAOnset`	0	Cue A onset
`CueBOnset`	1	Cue B onset
`Probe_aX_onset`	2	Probe aX (AX trial - expected)
`Probe_aY_onset`	3	Probe aY (AY trial - unexpected)
`Probe_bY_onset`	4	Probe bY (BY trial)
`Response: CueCorr`	5	Correct cue response
`Response: Probe_Corr`	6	Correct probe response
## Data Quality Notes		

- STATUS and Practice filler triggers must be excluded from training.
- Dataset includes mTBI patients and healthy controls - group label in participants.tsv.

References

- Reference: `ERPs predict symptomatic distress and recovery in sub-acute mild traumatic brain injury. DOI: 10.1016/j.neuropsychologia.2019.107125`

130. openneuro_ds005262 - ArEEG: Arabic Inner Speech EEG Dataset

Source: analysis.md

Dataset: `openneuro_ds005262` | Year: 2024 | URL: https://openneuro.org/datasets/ds005262
Category: High-level Mental | Subjects: 12 | Files (Completed): 186 | Channels: 8 | Sampling Rate: 250 Hz

Paradigm

EEG recordings of 12 subjects performing imagined (inner) speech tasks based on visual cues presented in Modern Standard Arabic. Subjects silently imagined saying directional words (Up, Down, Left, Right), a Select command, or remained at Rest. The study aims to develop Arabic-language BCI systems using inner speech decoding without overt vocalization.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005262

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Comment/Wait`	8949	short	raw	Wait period (inter-trial interval)
`Comment/Rest`	4590	rest	raw	Rest condition
`Comment/Up`	918	short	raw	Inner speech - "Up" command
`Comment/Select`	918	short	raw	Inner speech - "Select" command
`Comment/Right`	918	short	raw	Inner speech - "Right" command
`Comment/Left`	917	short	raw	Inner speech - "Left" command
`Comment/Down`	916	short	raw	Inner speech - "Down" command
`Comment/WarmUp`	174	admin	raw	Warm-up period (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Arabic inner speech command classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Six-class labels: Rest, Up, Down, Left, Right, Select from cue/comment events.
- Evidence: ArEEG dataset is explicitly constructed for Arabic inner-speech BCI command decoding. Source: https://openneuro.org/datasets/ds005262 DOI: https://doi.org/10.18112/openneuro.ds005262.v1.0.1

Normal Task 1: Inner Speech Command Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Comment/Wait is a high-frequency inter-trial marker - exclude from classification. Comment/WarmUp is admin - exclude. Only 8 EEG channels.

Event	y	Meaning
`Comment/Rest`	0	Rest condition
`Comment/Up`	1	Inner speech "Up"
`Comment/Down`	2	Inner speech "Down"
`Comment/Left`	3	Inner speech "Left"
`Comment/Right`	4	Inner speech "Right"
`Comment/Select`	5	Inner speech "Select"
## Data Quality Notes		

- Only 8 EEG channels - limited spatial resolution.
- Comment/Wait (8,949 total) is inter-trial interval - exclude from classification.
- Comment/WarmUp is admin - exclude.

References

- Reference: `ArEEG: Arabic Inner Speech EEG dataset`

131. openneuro_ds005305 - EEG Resting-State Microstates and Executive Functions

Source: analysis.md

Dataset: openneuro_ds005305 | Year: 2024 | URL: https://openneuro.org/datasets/ds005305
Category: High-level Mental | Subjects: 165 | Files (Completed): 165 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Resting-state EEG microstates were recorded to correlate spontaneous brain dynamics with performance on a nine-task battery of executive functions (dual N-back, Stroop, task-switching, etc.). Each file contains alternating eyes-closed (RS_EC, 825 total) and eyes-open (RS_EO, 660 total) resting blocks. The study investigates whether EEG microstate metrics predict individual differences in executive function. boundary (165) is a EEGLAB admin marker; key press events are incidental.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005305

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
RS_EC	825	rest	raw	Resting-state eyes-closed block onset
RS_EO	660	rest	raw	Resting-state eyes-open block onset
boundary	165	admin	raw	EEGLAB boundary marker (exclude)
RIGHT pressed/RIGHT released	26	admin	raw	Incidental key presses (exclude)
DOWN pressed/DOWN released	22	admin	raw	Incidental key presses (exclude)
LEFT pressed/LEFT released	16	admin	raw	Incidental key presses (exclude)
Other key events	rare	admin	raw	Incidental key presses (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Resting State

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds005305 DOI: https://doi.org/10.1016/j.cortex.2024.05.019

Normal Task 1: Eyes Open vs Eyes Closed (binary)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `boundary` (165) and all key press events are admin - exclude. Each resting block is ~30-60 s; sliding windows capture microstate dynamics. RS_EC (825) > RS_EO (660) - slight imbalance.

Event	y	Meaning
RS_EC	0	Eyes-closed resting state
RS_EO	1	Eyes-open resting state
## Data Quality Notes		

- `boundary` is EEGLAB admin marker - exclude.
- Key press events are incidental - exclude.
- Executive function scores in participants.tsv for regression analyses.

References

- Reference: `Investigating the relationship between Resting-State EEG Microstates and Executive Functions: A Null Finding`

132. openneuro_ds005383 - TMNRED: A Chinese Language EEG Dataset for Fuzzy Semantic Target Identification in Natural Reading Environments

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005383` | Year: 2024 | URL: https://openneuro.org/datasets/ds005383
Category: High-level Mental | Subjects: 30 | Files (Completed): 720
Channels: 30 | Sampling Rate: 200 Hz

Paradigm

Participants engaged in self-paced natural reading of Chinese sentences and news headlines to identify fuzzy semantic targets embedded in text. The study investigates neural correlates of semantic processing during naturalistic reading, recording EEG with 30 channels at 200 Hz. Numerical BioSemi trigger codes mark word/sentence onset events across three sessions per subject (3 x 240 files = 720 total). The trigger code structure encodes semantic conditions, target presence, and reading context.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005383

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
55	240/file	short	raw	Stimulus condition code 55 (decode from events.tsv)
31	240/file	short	raw	Stimulus condition code 31
56	240/file	short	raw	Stimulus condition code 56
36	240/file	short	raw	Stimulus condition code 36
41	240/file	short	raw	Stimulus condition code 41
51	240/file	short	raw	Stimulus condition code 51
16	240/file	short	raw	Stimulus condition code 16
61	240/file	short	raw	Stimulus condition code 61

Event ID	Count (total)	Type	Meta Level	Meaning
26	240/file	short	raw	Stimulus condition code 26
46	240/file	short	raw	Stimulus condition code 46
21	240/file	short	raw	Stimulus condition code 21
22	240/file	short	raw	Stimulus condition code 22
32	240/file	short	raw	Stimulus condition code 32
42	240/file	short	raw	Stimulus condition code 42
52	240/file	short	raw	Stimulus condition code 52

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Binary Semantic Target Identification (Target vs Non-target Sentence)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 2.0 s, target_events = [16, 21, 22, 26, 31, 32, 36, 41, 42, 46, 51, 52, 55, 56, 61]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
55, 56	0	Non-target sentence (no fuzzy semantic target word)
51, 52	1	Target sentence (contains fuzzy semantic target word: names, means of transportation, animals, fruits)
- Evidence:
- DOI: `10.1038/s41597-025-05036-2` - TMNRED dataset paper confirms binary semantic target identification (target word present vs absent in Chinese sentences) with 2-second stimulus presentation windows
- Kernel Notes: Verify event ID mapping from events.tsv to confirm which trigger codes correspond to target vs non-target conditions. Default mapping assumes codes ending in 5/6 = non-target, codes ending

in 1/2 = target based on BioSemi convention. Consider grouping: odd-ending codes (21,31,41,51) = target conditions, even-ending codes (22,32,42,52,56) = non-target conditions.

Normal Task 1: Semantic Target Identification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 30 distinct numerical codes across all files; full decode from events.tsv or paper required. 720 files from 30 subjects x 24 sessions/files. Low sampling rate (200 Hz) limits high-frequency analysis.

Event	y	Meaning
16	0	Condition code 16
21	1	Condition code 21
22	2	Condition code 22
26	3	Condition code 26
31	4	Condition code 31
32	5	Condition code 32
36	6	Condition code 36
41	7	Condition code 41
42	8	Condition code 42
46	9	Condition code 46
51	10	Condition code 51
52	11	Condition code 52
55	12	Condition code 55
56	13	Condition code 56
61	14	Condition code 61

Data Quality Notes

- Numerical trigger codes require decode from events.tsv or dataset paper for semantic interpretation.
- 30 distinct codes; group by semantic condition (target/non-target) when possible.
- Low sampling rate (200 Hz) - suitable for N400/P600 ERP components but not high-gamma analysis.

References

- Reference: `TMNRED, A Chinese Language EEG Dataset for Fuzzy Semantic Target Identification in Natural Reading Environments`

133. openneuro_ds005403 - Delayed Auditory Feedback EEG/EGG

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005403 | Year: 2024 | URL: https://openneuro.org/datasets/ds005403
Category: High-level Mental | Subjects: 66 | Files (Completed): 66
Channels: 66 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a speech production task while receiving delayed auditory feedback (DAF) to study neural processing of self-generated speech and auditory-motor monitoring. Three DAF conditions were used: an adaptation condition (fixed delay) and two random delay conditions (random1, random2), compared against a baseline (no delay). EEG was recorded alongside electroglottography (EGG) to capture vocal fold activity simultaneously. The event markers label continuous speech production blocks.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005403

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`adaption`	~64/file	long	raw	Adaptation DAF condition block
`random1`	~64/file	long	raw	Random delay condition 1 block
`random2`	~64/file	long	raw	Random delay condition 2 block
`baseline`	~64/file	rest	raw	Baseline (no delay) speech block

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: EEG Encoding of Glottal and Audio Features During Speech Production

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`

- Epoch Parameters: segment_length = 4.0 s, overlap = 0.5 s
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`baseline`	continuous	Baseline (no delay) speech production block
- Super Meta Notes: Derived from raw audio and EGG signals through signal processing (gammatone filter bank, envelope computation, onset detection). Kernel step: Extract continuous features (audio envelope, audio onsets, EGG envelope, EGG onsets) and train time-lagged regression (TRF) to predict EEG from these features.
- Evidence:
- DOI: 10.1523/jneurosci.2294.24.2025 - This paper uses the OpenNeuro ds005403 dataset to train encoding models that predict EEG from glottal and audio features and decoding models that predict glottal event onsets from EEG, demonstrating cortical tracking of glottal events during speech production and its recalibration following exposure to delayed auditory feedback.
- Kernel Notes: Use epoch_by_segmentation_hdf5 since this is a continuous encoding task (TRF-based regression) rather than event-triggered epoching. Time-lagged features span -0.5s to +0.5s around each EEG time point. Note: The paper references 'pretest', 'exposure', 'posttest' events that do not exist in this dataset's raw event IDs - only `baseline` is available. Verify event ID mapping from events.tsv if attempting to use other conditions.

Primary Task 2: DAF Condition Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = [`baseline`, `adaption`, `random1`, `random2`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`baseline`	0	Baseline (no delay)
`adaption`	1	Adaptation DAF condition
`random1`	2	Random delay condition 1
`random2`	3	Random delay condition 2
- Evidence:
- DOI: 10.1523/jneurosci.2294.24.2025 - The study investigates neural processing of self-generated speech and auditory-motor monitoring under delayed auditory feedback conditions.
- Kernel Notes: All four conditions are sustained speech production blocks (>=2 s). Use epoch_by_long_event_hdf5 to slide windows through each long event annotation. Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Simultaneous EGG recording alongside EEG - verify channel type labels.
- Speech production task - EMG artifacts from jaw/throat movement expected.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Delayed Auditory Feedback EEG/EGG`

134. openneuro_ds005815 - YOTO (You Only Think Once): An EEG Dataset for Tracking Spontaneous Cognitive Transitions and Multimodal Stimuli Responses in Resting-State and Task Phases

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005815` | Year: 2025 | URL: https://openneuro.org/datasets/ds005815
Category: High-level Mental | Subjects: 20 | Files (Completed): 205 | Channels: 30 | Sampling Rate: 250 Hz

Paradigm

Participants engaged in resting-state (eyes open/closed) and task phases involving unimodal and multimodal stimuli (visual and auditory) and mental imagery, providing self-reported vividness ratings. The YOTO paradigm investigates spontaneous cognitive transitions and mental imagery across stimulus modalities with numeric condition codes and named state markers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005815

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`6-9`	6720 each	short	raw	Stimulus condition codes (decode from events.tsv)
`TrialProc`	6528	long	raw	Trial procedure marker
`23`	840	short	raw	Stimulus/response code 23
`22`	420	short	raw	Stimulus/response code 22
`21`	420	short	raw	Stimulus/response code 21
`StartInstruction`	66	long	raw	Instruction block onset
`Eyeclosed`	66	long	raw	Eyes-closed resting onset
`FixationBreak`	66	short	raw	Fixation break event
`EyeOpen`	66	long	raw	Eyes-open resting onset

Event ID	Count (total)	Type	Meta Level	Meaning
`EndInstruction`	66	long	raw	Instruction block offset
`88`	64	short	raw	Response/trigger code 88
`30-35`	280 each	short	raw	Imagery/rating condition codes
`99`	101	admin	raw	Admin/session marker
`1`	81	short	raw	Condition code 1
`2-5`	68 each	short	raw	Condition codes 2-5

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Multi-class Mental Imagery Classification (Visual vs Auditory vs Multimodal)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -2.0 s, tmax = 4.0 s, target_events = [`6`, `7`, `8`, `9`, `21`, `22`, `23`, `30`, `31`, `32`, `33`, `34`, `35`]
- Label Route: `Annotation` | `stim`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`6`	0	Visual stimulus condition code *
`7`	0	Visual stimulus condition code *
`8`	0	Auditory stimulus condition code *
`9`	0	Auditory stimulus condition code *
`21`	0	Visual stimulus *
`22`	0	Auditory stimulus *
`23`	0	Multimodal stimulus *
`30`	1	Visual imagery (code 30)
`31`	1	Visual imagery (code 31) *
`32`	1	Auditory imagery (code 32) *
`33`	1	Auditory imagery (code 33) *
`34`	1	Multimodal imagery (code 34) *
`35`	1	Multimodal imagery (code 35) *

- Evidence:
- DOI: `10.1038/s41597.025.05881.1` - Multi-class mental imagery classification (visual=1, auditory=2, multimodal=3) using stimulus presentation (2s) followed by 4s imagery phases
- Kernel Notes:

- Epoch window spans from -2s (fixation/stimulus onset) to +4s (end of imagery period)
- Map numeric condition codes to visual/auditory/multimodal labels using events.tsv decoding table
- Verify event ID mapping from events.tsv: confirm which codes (6-9, 21-23, 30-35) correspond to visual/auditory/multimodal categories
- Filter: bandpass 1-40 Hz, artifact rejection recommended

Primary Task 2: Eyes-Open vs Eyes-Closed Resting-State Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 60 s *, overlap = 0.0, target_events = [`EyeOpen`, `Eyeclosed`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`EyeOpen`	0	Eyes-open resting-state
`Eyeclosed`	1	Eyes-closed resting-state

- Evidence:
- DOI: `10.1038/s41597.025.05881.1` - YOTO paradigm includes resting-state phases with eyes open/closed for baseline EEG comparison
- Kernel Notes:
  - Duration from Event Structure: each EyeOpen/Eyeclosed marker is ~66 occurrences, likely ~60s each
  - Apply frequency band analysis (alpha power for eyes-closed vs eyes-open)

Normal Task 1: Multi-task Cognitive (long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numerical codes require decoding from events.tsv. Eyes-open/closed markers (`EyeOpen`, `Eyeclosed`) can be used for resting-state analysis.

Event	y	Meaning
`99`	0	Admin/session marker
`1`	1	Condition code 1
`2`	2	Condition code 2
`3`	3	Condition code 3
`5`	4	Condition code 5
`StartInstruction`	5	Instruction onset
`Eyeclosed`	6	Eyes-closed onset
`FixationBreak`	7	Fixation break
`EyeOpen`	8	Eyes-open onset
`EndInstruction`	9	Instruction offset
`88`	10	Response code 88
`6`	11	Condition code 6

Event	y	Meaning
7	12	Condition code 7
8	13	Condition code 8
9	14	Condition code 9
23	15	Code 23
22	16	Code 22
21	17	Code 21
30	18	Imagery code 30
35	19	Imagery code 35
25	20	Imagery code 25
32	21	Imagery code 32
31	22	Imagery code 31
27	23	Imagery code 27
28	24	Imagery code 28
24	25	Imagery code 24
33	26	Imagery code 33
26	27	Imagery code 26
29	28	Imagery code 29
34	29	Imagery code 34

Data Quality Notes

- Numerical event codes require mapping from dataset events.tsv or README.
- Small dataset (20 subjects); low statistical power for rare conditions.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: YOTO (You Only Think Once): An EEG Dataset for Tracking Spontaneous Cognitive Transitions and Multimodal Stimuli Responses in Resting-State and Task Phases

135. openneuro_ds006018 - Cognitive Electrophysiology in Socioeconomic Context in Adulthood: An EEG dataset

Source: analysis_minimax_v2.md

Dataset: openneuro_ds006018 | Year: 2025 | URL: https://openneuro.org/datasets/ds006018
Category: High-level Mental | Subjects: 127 | Files (Completed): 357 | Channels: 31 | Sampling Rate: 500 Hz

Paradigm

A suite of standardized ERP tasks from the ERP CORE compendium (Kappenman et al. 2021) measuring neural correlates of perception, cognition, and action in adults in relation to socioeconomic status and ADHD symptoms. Tasks include N170 (face perception), MMN (mismatch negativity), N2pc (visual attention), P3 (cognitive updating), N400 (semantic processing), ERN (error monitoring), and LRP (lateralized readiness potential).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006018

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
S201	40081	short	raw	ERP CORE stimulus code 201
S 80	30199	short	raw	ERP CORE stimulus code 80
S212	10400	short	raw	ERP CORE stimulus code 212
S111	9971	short	raw	ERP CORE stimulus code 111
S222	9017	short	raw	ERP CORE stimulus code 222
S121	8449	short	raw	ERP CORE stimulus code 121
S 22	8268	short	raw	ERP CORE stimulus code 22
S 11	8263	short	raw	ERP CORE stimulus code 11
S 12	8252	short	raw	ERP CORE stimulus code 12
S 21	8244	short	raw	ERP CORE stimulus code 21

Event ID	Count (total)	Type	Meta Level	Meaning
S 70	7977	short	raw	ERP CORE stimulus code 70
S202	6292	short	raw	ERP CORE stimulus code 202
S221	5197	short	raw	ERP CORE stimulus code 221
S122	4665	short	raw	ERP CORE stimulus code 122
S211	3707	short	raw	ERP CORE stimulus code 211

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: S201, S 80, S212, S111, S222, S121, S 22, S 11
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Active Visual Oddball P300 classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s, target_events = [S201, S202, S221, S122, S212, S222, S111, S121, S211]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
S201	0	standard stimulus (non-target)
S202	0	standard stimulus (non-target)
S221	0	standard stimulus (non-target)
S122	0	standard stimulus (non-target)
S212	0	standard stimulus (non-target)
S222	0	standard stimulus (non-target)
S111	1	oddball/target stimulus (P300)
S121	1	oddball/target stimulus (P300)
S211	1	oddball/target stimulus (P300)

- Evidence:
- DOI: 10.1109/bibm66473.2025.11356119 - Binary single-trial classification of oddball versus standard stimuli under P300 paradigm with epochs -100ms to +1000ms
- Kernel Notes: Verify event ID mapping from events.tsv. Uses oddball (target=1) vs standard (non-target=0) binary labels. Apply baseline correction [-0.1, 0] s. Filter 0.1-40 Hz. (replaces Primary Task 1 from original analysis.md)

Primary Task 2: ERP CORE Multi-paradigm classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [`S201`, `S 80`, `S212`, `S111`, `S222`, `S121`, `S 22`, `S 11`, `S 12`, `S 21`, `S 70`, `S202`, `S221`, `S122`, `S211`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`S201`	0	Passive Auditory Oddball: standard tone
`S 80`	1	Passive Auditory Oddball: deviant/mismatch (MMN)
`S212`	2	Active Visual Oddball: standard (non-target)
`S111`	3	Active Visual Oddball: target (oddball)
`S222`	4	Active Visual Oddball: non-target variant
`S121`	5	Active Visual Oddball: target variant
`S 22`	6	Flanker: congruent (incongruent=22)
`S 11`	7	Flanker: incongruent (congruent=11)
`S 12`	8	N170: faces
`S 21`	9	N170: cars
`S 70`	10	Visual Search: non-target
`S202`	11	Visual Search: target
`S221`	12	N400: unrelated prime-target
`S122`	13	N400: related prime-target
`S211`	14	LRP/ERN: response-related

- Evidence:
- DOI: `10.1038/s41597.025.05209.z` - Dataset containing EEG from 127 adults during ERP CORE paradigms (passive auditory oddball, active visual oddball, visual search, flanker) with stimulus event codes for target vs non-target conditions
- DOI: `10.1016/j.neuroimage.2020.117465` - ERP CORE resource with six optimized paradigms (N170, MMN, N2pc, N400, P3, LRP/ERN)
- Kernel Notes: Use ERP CORE standard epoch parameters (-0.2 to 1.0s). Apply 0.1-40 Hz bandpass filter and artifact rejection. Baseline correct [-0.2, 0] s. `boundary` markers are administrative and excluded. (replaces Primary Task 2 and Primary Task 3 from original analysis.md)

Normal Task 1: ERP CORE Multi-paradigm (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Multi-paradigm ERP CORE dataset. `boundary` is admin and excluded. Codes encode task x condition combinations per ERP CORE documentation.

Event	y	Meaning
`S 12`	0	ERP code 12
`S 22`	1	ERP code 22
`S 21`	2	ERP code 21
`S 11`	3	ERP code 11
`S201`	4	ERP code 201

Event	y	Meaning
S202	5	ERP code 202
S221	6	ERP code 221
S122	7	ERP code 122
S212	8	ERP code 212
S222	9	ERP code 222
S111	10	ERP code 111
S121	11	ERP code 121
S211	12	ERP code 211
S112	13	ERP code 112
S 80	14	ERP code 80
S 70	15	ERP code 70
S180	16	ERP code 180
S 1	17	ERP code 1
S 25	18	ERP code 25
S 23	19	ERP code 23
S 24	20	ERP code 24
S 53	21	ERP code 53
S 43	22	ERP code 43
S 45	23	ERP code 45
S 44	24	ERP code 44
S 41	25	ERP code 41
S 13	26	ERP code 13
S 14	27	ERP code 14
S 42	28	ERP code 42

Data Quality Notes

- Multi-paradigm dataset - participants completed multiple ERP CORE tasks; segment by task if needed.
- `boundary` is an administrative marker, excluded from target_events.
- Apply bandpass filter (0.1-40 Hz) and artifact rejection; baseline correct [-0.2, 0] s.

References

- Reference: `Isbell, E., Peters, A. N., Richardson, D. M., & Rodas De Len, N. E. (2025). Cognitive electrophysiology in socioeconomic context in adulthood. Scientific Data, 12(1), 19.`

136. openneuro_ds006040 - Sustained Attention Task (gradCPT) Dataset using simultaneous EEG-fMRI and DTI

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds006040` | Year: 2025 | URL: https://openneuro.org/datasets/ds006040
Category: High-level Mental | Subjects: 28 | Files (Completed): 1064 | Channels: 64 | Sampling Rate: 5000 Hz

Paradigm

The dataset uses the Gradual Continuous Performance Task (gradCPT) to measure sustained attention via simultaneous EEG and fMRI recording. Participants detect rare city scenes (targets, S245) within a stream of common mountain scenes (standards, S155). Additional tasks include resting state, visual checkerboard stimulation, and mental imagery. MRI synchronization pulses (TR, Sync On) and cardiac (qrs) signals are also recorded.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006040

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`S155`	201922	short	raw	Standard stimulus (mountain scene, frequent)
`S245`	201110	short	raw	Target stimulus (city scene, rare)
`Sync On`	195242	admin	raw	MRI sync pulse onset
`qrs`	170358	admin	raw	Cardiac QRS complex
`TR`	106494	admin	raw	MRI TR repetition marker
`T 1`	52960	admin	raw	Timing marker
`S254`	813	short	raw	Response / button press
`[]`	15	admin	raw	Empty/null event

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.

- Candidate Trigger Labels: S155, S245, S254
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: gradCPT Correct vs Incorrect Trial Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.5 s *, tmax = 2.0 s, target_events = [S155, S245, S254]
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
S155 + no S254 within trial	0	Correct omission (standard, withhold)
S155 + S254 within trial	1	Commission error (incorrect)
S245 + S254 within trial	2	Correct commission (target, press)
S245 + no S254 within trial	3	Omission error (incorrect)

- Super Meta Notes: Derived from raw events [S155, S245, S254] by combining stimulus events with response events to determine trial correctness. Kernel step: For each S155 or S245 epoch, check if an S254 (button press) occurs within the trial window. If S245 has S254 -> correct commission (y=2); if S245 no S254 -> omission error (y=3); if S155 no S254 -> correct omission (y=0); if S155 has S254 -> commission error (y=1). Classification is correct (y=0,2) vs incorrect (y=1,3).
- Evidence:
- DOI: 10.1038/s41597.026.06616.6 - Paper publishes the gradCPT EEG-fMRI-DTI dataset and analyzes sustained attention by classifying correct vs incorrect trials (commission/omission errors) based on pre-target brain activity.
- Kernel Notes: Requires epoching around both S155 and S245 stimuli, then applying response detection logic to assign super labels. Verify event ID mapping from events.tsv.

Primary Task 2: In-the-Zone vs Out-of-the-Zone Attention State Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.5 s *, tmax = 2.0 s, target_events = [S155, S245, S254]
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
Correct trial with low RT variability	0	In-the-zone (sustained attention)
Correct trial with high RT variability	1	Out-of-the-zone (attentional lapses)

- Super Meta Notes: Derived from raw events [S155, S245, S254] by computing reaction time (RT) variability across correct trials within a block/session. Kernel step: Extract RT from S245 stimulus onset to S254 response for correct trials. Compute rolling RT variability (e.g., standard deviation or coefficient of variation). Trials with low variability -> in-the-zone (y=0); trials with high variability -> out-of-the-zone (y=1).
- Evidence:
- DOI: 10.1038/s41597.026.06616.6 - Paper analyzes sustained attention by classifying in-the-zone vs out-of-the-zone attentional states based on RT variability during correct trials.

- Kernel Notes: Requires computing RT for each S245->S254 correct trial, then computing RT variability metric per block. Verify event ID mapping from events.tsv.

Primary Task 3: gradCPT Stimulus Type Classification (Standard vs Target)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s *, tmax = 2.0 s, target_events = [S155, S245]
- Label Route: `stim`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
S155	0	Standard stimulus (mountain scene)
S245	1	Target stimulus (city scene)

- Evidence:
- DOI: 10.1038/s41597.026.06616.6 - Paper uses S155 (mountain/standard) and S245 (city/target) as the core stimulus events for gradCPT sustained attention task.
- Kernel Notes: Simple stimulus-locked epoching for binary classification of mountain vs city scenes. (replaces Primary Task 1 from original analysis.md)

Normal Task 1: gradCPT Sustained Attention (long_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s *, tmax = 2.0 s, target_events = [S155, S245, S254]
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Primary scientific contrast is S155 (standard) vs S245 (target) for gradCPT attention decoding. `Sync On`, `TR`, `qrs`, `T 1` are MRI/cardiac markers. Very high sampling rate (5000 Hz); downsample before processing.

Event	y	Meaning
`Sync On`	0	MRI sync pulse
`S155`	1	Standard (mountain)
`TR`	2	MRI TR marker
`S245`	3	Target (city)
`qrs`	4	Cardiac QRS
`T 1`	5	Timing marker
`S254`	6	Button press/response
`[]`	7	Null event

Data Quality Notes

- Very high sampling rate (5000 Hz); downsample to <=1000 Hz for most analyses.
- `Sync On`, `TR`, `qrs` are MRI/cardiac markers and should be excluded from stimulus-locked analysis.
- Simultaneous EEG-fMRI data requires MRI artifact correction (template subtraction or similar).

References

- Reference: Sustained Attention Task (gradCPT) Dataset using simultaneous EEG-fMRI and DTI

137. openneuro_ds006104 - EEG dataset for speech decoding

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds006104` | Year: 2025 | URL: https://openneuro.org/datasets/ds006104
Category: High-level Mental | Subjects: 24 | Files (Completed): 80 | Channels: 62 | Sampling Rate: 2000 Hz

Paradigm

A phoneme discrimination task where participants listened to speech sounds - including single phonemes, phoneme pairs, and words - while undergoing transcranial magnetic stimulation (TMS) to test the role of speech motor cortex in auditory speech perception. Subjects provided button-press responses after each sound. The dataset supports decoding of speech perception from TMS-EEG.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006104

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`TMS`	29260	short	raw	TMS pulse delivery
`stimulus`	29260	short	raw	Auditory speech stimulus onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Phoneme Category Classification (Consonant vs Vowel)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`stimulus`]
- Label Route: `Annotation`
- Meta Level: super

- Target Definition:
Event	y	Meaning
`stimulus` (consonant phoneme)	0	Consonant phoneme (/b/,/p/,/d/,/t/,/s/,/z/)
`stimulus` (vowel phoneme)	1	Vowel phoneme (/i/,/□/,/ɑ/,/u/,/o□/)
- Super Meta Notes: Derived from raw event `stimulus` by matching stimulus onset timestamps with stimulus presentation log to extract phoneme category labels (consonant vs vowel). Kernel step: Load stimulus metadata file (stimuli.csv or similar), join with stimulus events by timing, extract phoneme_category column.
- Evidence:
- DOI: 10.1038/s41597.025.05187.2 - Paper describes multiclass classification of phoneme identity with consonant and vowel categories from TMS-EEG during speech perception
- Kernel Notes: Requires access to stimulus presentation logs; verify phoneme-to-category mapping from dataset documentation. (replaces Placeholder Primary Tasks 1-3 from original analysis.md)

Primary Task 2: Word Type Classification (Real Word vs Pseudoword)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`stimulus` (real word)	0	Real word stimulus
`stimulus` (pseudoword)	1	Pseudoword stimulus
- Super Meta Notes: Derived from raw event `stimulus` by matching stimulus onset timestamps with stimulus presentation log to extract word type labels (real_word vs pseudoword). Kernel step: Load stimulus metadata file, join with stimulus events by timing, extract word_type column.
- Evidence:
- DOI: 10.1038/s41597.025.05187.2 - Paper demonstrates classification of real words vs pseudowords from TMS-EEG recordings
- Kernel Notes: Requires access to stimulus presentation logs; verify word-type mapping from dataset documentation.

Normal Task 1: TMS vs Stimulus Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Binary classification (TMS vs stimulus). TMS pulses produce large artifacts; apply TMS artifact rejection before epoching.

Event	y	Meaning
`TMS`	0	TMS pulse
`stimulus`	1	Speech stimulus onset

Data Quality Notes

- Very high sampling rate (2000 Hz); downsample to 500 Hz for most analyses.

- TMS pulses produce large EEG artifacts requiring specialized artifact removal.
- Apply bandpass filter (1-80 Hz) after TMS artifact correction.

References

- Reference: `Moreira et al. "An open-access EEG dataset for speech decoding: Exploring the role of articulation and coarticulation"`

138. openneuro_ds006142 - Essex EEG Movie Memory dataset

Source: analysis_minimax_v2.md

Dataset: openneuro_ds006142 | Year: 2025 | URL: https://openneuro.org/datasets/ds006142
Category: High-level Mental | Subjects: 27 | Files (Completed): 27 | Channels: 64 | Sampling Rate: 2048 Hz

Paradigm

Memory recognition task where participants watched 10-second movie clips and identified whether they remembered them from previous viewings, including time-stamping recognition instants. Clips were shown twice; the first watch (`startOfRememberedClipFirstWatch`, `startOfRecognisedClipFirstWatch`) established encoding, while the second watch (`startOfSecondWatch`) tested recognition. Participants clicked when they recognized clips (`firstRecognitionClick`, `recognitionClick`).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006142

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`recognitionClick`	6675	short	raw	Participant clicks upon recognition
`endOfClip`	6094	short	raw	Movie clip ends
`trialStart`	4153	long	raw	Trial onset
`trialEnd`	4153	long	raw	Trial offset
`startOfNotRecognisedClip`	2212	long	raw	Clip onset - not recognized
`startOfSecondWatch`	1941	long	raw	Second viewing onset
`firstRecognitionClick`	1881	short	raw	First recognition click in second watch
`startOfRememberedClipFirstWatch`	1426	long	raw	First viewing onset - later remembered
`startOfRecognisedClipFirstWatch`	515	long	raw	First viewing onset - later recognized

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Movie Memory Recognition Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`startOfRememberedClipFirstWatch`, `startOfRecognisedClipFirstWatch`, `startOfNotRecognisedClip`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`startOfNotRecognisedClip`	0	Clip not recognized during first watch
`startOfRecognisedClipFirstWatch`	1	Clip recognized during first watch
`startOfRememberedClipFirstWatch`	2	Clip remembered from first watch

- Evidence:
- DOI: 10.1038/s41597.025.06203.1 - Three-class memory recognition classification (not-recognised vs recognised vs remembered) using EEG epochs from first-watch clip onsets
- Kernel Notes: Epoch around first-watch clip onset events; extract ERP features (tmin=-0.2s, tmax=1.0s) or PSD features (tmin=-1.0s, tmax=10.0s) for classification into three memory states. This is the core memory retrieval paradigm. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- High sampling rate (2048 Hz); downsample to 512 Hz for standard ERP analysis.
- Small dataset (27 subjects); limited statistical power.
- Apply bandpass filter (0.1-40 Hz) and artifact rejection prior to epoching.

References

- Paper: `Annotating, Understanding, and Predicting Long-term Video Memorability`

139. openneuro_ds006159 - Implicit Learning EEG (BioSemi)

Source: analysis.md

Dataset: openneuro_ds006159 | Year: 2025 | URL: https://openneuro.org/datasets/ds006159
Category: High-level Mental | Subjects: 35 | Files (Completed): 35 | Channels: 69 | Sampling Rate: 1024 Hz

Paradigm

An experimental study investigating the neural correlates of implicit learning using a BioSemi EEG system. Participants underwent repeated exposure to structured sequences without explicit instructions to learn the underlying patterns. No trial-level event annotations are present in the dataset; the EEG is recorded as a continuous stream during the implicit learning protocol.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006159

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Publisher-defined BIDS task: Implicitlearn

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006159 DOI: https://doi.org/10.18112/openneuro.ds006159.v1.0.0

Normal Task 1: Continuous Segmentation (segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5 s
- Label Route: `Annotation`
- Target Definition: N/A (no event markers; segmentation only)
- Notes: No trial-level events available; purely continuous EEG. Labels must be derived from external metadata (e.g., block condition labels from participants.tsv if available).

Data Quality Notes

- No event markers present; continuous-only processing.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `MNE-BIDS: Organizing electrophysiological data into the BIDS format and facilitating their analysis`

140. openneuro_ds006465 - 3M-CPSEED: EEG Dataset for Chinese Pinyin Production in Overt, Silent-Intended, and Imagined Speech

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds006465` | Year: 2025 | URL: https://openneuro.org/datasets/ds006465
Category: High-level Mental | Subjects: 20 | Files (Completed): 84 | Channels: 33 | Sampling Rate: 500 Hz

Paradigm

Participants performed three modes of speech production - overt (spoken aloud), silent-intended (mouthed without sound), and imagined (mental speech without movement) - for specific Chinese Pinyin sounds including initials and finals. The dataset enables comparison of neural signatures across overt, silent, and imagined speech conditions. Each trial consisted of a visual cue (stimulus_presentation) followed by a production phase and a response confirmation. Numeric codes 1-10 encode specific phoneme or condition types.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006465

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stimulus_presentation`	42	short	raw	Visual cue onset prompting speech production
`1`	302	short	raw	Phoneme/condition code 1
`2`	200	short	raw	Phoneme/condition code 2
`3`	200	short	raw	Phoneme/condition code 3
`4`	200	short	raw	Phoneme/condition code 4
`5`	200	short	raw	Phoneme/condition code 5
`6`	200	short	raw	Phoneme/condition code 6
`7`	200	short	raw	Phoneme/condition code 7

Event ID	Count (total)	Type	Meta Level	Meaning
8	200	short	raw	Phoneme/condition code 8
9	200	short	raw	Phoneme/condition code 9
10	200	short	raw	Phoneme/condition code 10
response	84	short	raw	Participant response / confirmation

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stimulus_presentation
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Speech Mode Classification (Overt vs Silent vs Imagined)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, target_events = [stimulus_presentation]
- Note: Epochs extracted at multiple phase windows per trial: Speak [3-5s, 5-7s, 7-9s], Silently articulated [10-12s, 12-14s, 14-16s], Imagine [17-19s, 19-21s, 21-23s] - total 9 epochs per trial
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
stimulus_presentation (position 3-5s, 5-7s, 7-9s in trial)	0	Overt (spoken) speech production
stimulus_presentation (position 10-12s, 12-14s, 14-16s in trial)	1	Silent-intended (mouthed) speech
stimulus_presentation (position 17-19s, 19-21s, 21-23s in trial)	2	Imagined (mental) speech
- Evidence:
- DOI: 10.1038/s41597-025-06346-1 - Paper validates EEGNet-based binary classification for speech mode discrimination (overt/silent/imagined) achieving 60-73% accuracy on 3M-CPSEED dataset
- Kernel Notes: Extract epochs from stimulus_presentation events and derive speech mode from temporal position within trial structure (overt phase: 3-9s, silent phase: 10-16s, imagine phase: 17-23s). Requires trial segmentation to identify phase boundaries. (replaces Primary Task 1 from original analysis.md)

Primary Task 2: Pinyin Syllable Pairwise Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, target_events = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
- Label Route: stim
- Meta Level: raw

- Target Definition:
Event	y	Meaning
1	0	Pinyin syllable code 1
2	1	Pinyin syllable code 2
3	2	Pinyin syllable code 3
4	3	Pinyin syllable code 4
5	4	Pinyin syllable code 5
6	5	Pinyin syllable code 6
7	6	Pinyin syllable code 7
8	7	Pinyin syllable code 8
9	8	Pinyin syllable code 9
10	9	Pinyin syllable code 10
- Evidence:
- DOI: 10.1038/s41597-025-06346-1 - Paper validates pairwise pinyin syllable classification achieving >55% accuracy on 3M-CPSEED dataset
- Kernel Notes: Use numeric trigger codes (1-10) directly from events.tsv as labels. These correspond to specific Pinyin syllables (a, i, u, ü, m, f, j, l, k, ch with their pairings).

Normal Task 1: Speech Production Mode Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numeric codes 1-10 correspond to specific Pinyin sounds or speech modes; decode from events.tsv. `response` is a motor response event. Some files show only numeric triggers; mixed annotation format across subjects.

Event	y	Meaning
`stimulus_presentation`	0	Visual cue for speech production
`1`	1	Phoneme/condition code 1
`5`	2	Phoneme/condition code 5
`6`	3	Phoneme/condition code 6
`7`	4	Phoneme/condition code 7
`4`	5	Phoneme/condition code 4
`3`	6	Phoneme/condition code 3
`2`	7	Phoneme/condition code 2
`10`	8	Phoneme/condition code 10
`8`	9	Phoneme/condition code 8
`9`	10	Phoneme/condition code 9

Data Quality Notes

- Some files show only numeric triggers while others include `stimulus_presentation`; mixed annotation format across sessions.
- Derivatives (preprocessed) files available alongside raw data.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection before epoching.

References

- Reference: 3M-CPSEED: An EEG-based Dataset for Chinese Pinyin Production in Overt, Silent-intended, and Imagined Speech

141. openneuro_ds006647 - Poetry EEG Study

Source: analysis.md

Dataset: `openneuro_ds006647` | Year: 2025 | URL: https://openneuro.org/datasets/ds006647
Category: High-level Mental | Subjects: 51 | Files (Completed): 4 | Channels: 70 | Sampling Rate: 512 Hz

Paradigm

Participants read and evaluated 210 short poetic (Haiku, Senryu) and non-poetic control texts on five subjective dimensions: Aesthetic Appeal, Vivid Imagery, Being Moved, Originality, and Creativity. EEG was recorded during reading using a 64-channel BioSemi system. The study investigates neural correlates of aesthetic and creative cognition during poetry processing. BioSemi EDF trigger codes (65280-65288) encode stimulus onset and condition types.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006647

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`65284`	852	short	raw	BioSemi trigger - stimulus/condition onset
`65281`	848	short	raw	BioSemi trigger - stimulus/condition onset
`65282`	848	short	raw	BioSemi trigger - stimulus/condition onset
`65283`	848	short	raw	BioSemi trigger - stimulus/condition onset
`65280`	848	short	raw	BioSemi trigger - stimulus/condition onset
`65285`	848	short	raw	BioSemi trigger - stimulus/condition onset
`65286`	848	short	raw	BioSemi trigger - stimulus/condition onset
`65287`	848	short	raw	BioSemi trigger - stimulus/condition onset

Event ID	Count (total)	Type	Meta Level	Meaning
65288	848	short	raw	BioSemi trigger - stimulus/condition onset
boundary	5	admin	raw	Segment boundary marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Readpoetry

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006647 DOI: https://doi.org/10.18112/openneuro.ds006647.v1.0.1

Normal Task 1: Poetry Condition Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: BioSemi trigger codes 65280-65288; decode from events.tsv to determine poetry vs. control mapping. boundary is an admin segment marker (exclude). Only 4 files present despite 51 subjects reported - dataset may be a subset or pilot.

Event	y	Meaning
65284	0	BioSemi condition code 65284
65281	1	BioSemi condition code 65281
65282	2	BioSemi condition code 65282
65283	3	BioSemi condition code 65283
65280	4	BioSemi condition code 65280
65285	5	BioSemi condition code 65285
65286	6	BioSemi condition code 65286
65287	7	BioSemi condition code 65287
65288	8	BioSemi condition code 65288

Event	y	Meaning
## Data Quality Notes		

- Only 4 completed files despite 51 subjects reported; this appears to be a partial dataset.
- `boundary` is a segment boundary marker and must be excluded from analysis.
- BioSemi EDF trigger codes 65280-65288 need decoding from events.tsv.

References

- Reference: `Poetry EEG Study`

142. openneuro_ds006648 - Poetry Assessment EEG Dataset 1

Source: analysis.md

Dataset: `openneuro_ds006648` | Year: 2025 | URL: https://openneuro.org/datasets/ds006648
Category: High-level Mental | Subjects: 47 | Files (Completed): 47 | Channels: 70 | Sampling Rate: 512 Hz

Paradigm

Participants read and evaluated 210 short English-language texts including Haikus and Senryus on five subjective dimensions related to aesthetic and creative cognition: Aesthetic Appeal, Vivid Imagery, Being Moved, Originality, and Creativity. EEG was recorded during reading using a 64-channel BioSemi system (70 channels total including references). This is the first of two poetry datasets (companion to ds006647) with full N=47 subject coverage.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006648

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`65284`	9993	short	raw	BioSemi trigger - stimulus/condition D onset
`65281`	9964	short	raw	BioSemi trigger - stimulus/condition A onset
`65282`	9964	short	raw	BioSemi trigger - stimulus/condition B onset
`65283`	9964	short	raw	BioSemi trigger - stimulus/condition C onset
`65286`	47	short	raw	BioSemi trigger - stimulus/condition F onset
`65287`	47	short	raw	BioSemi trigger - stimulus/condition G onset
`65288`	47	short	raw	BioSemi trigger - stimulus/condition H onset

Event ID	Count (total)	Type	Meta Level	Meaning
65285	46	short	raw	BioSemi trigger - stimulus/condition E onset
65280	18	short	raw	BioSemi trigger - rare/practice condition

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Readpoetry

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006648 DOI: https://doi.org/10.18112/openneuro.ds006648.v1.0.0

Normal Task 1: Poetry Aesthetic Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: BioSemi EDF trigger codes 65280-65288; decode from events.tsv for exact poetry/non-poetry mapping. Codes 65281-65284 are highly frequent (~9964-9993 each); codes 65285-65288 appear rarely (~46-47 total). Code 65280 appears in only 18 files.

Event	y	Meaning
65284	0	BioSemi condition code 65284
65281	1	BioSemi condition code 65281
65282	2	BioSemi condition code 65282
65283	3	BioSemi condition code 65283
65286	4	BioSemi condition code 65286
65287	5	BioSemi condition code 65287
65288	6	BioSemi condition code 65288
65285	7	BioSemi condition code 65285

Event	y	Meaning
65280	8	BioSemi condition code 65280
## Data Quality Notes		

- Codes 65285-65288 and 65280 are rare events; consider excluding from classification due to imbalance.
- BioSemi trigger codes need decoding from events.tsv to determine poetry/non-poetry mapping.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: Poetry Assessment EEG Dataset 1

143. openneuro_ds006768 - Multiple Object Monitoring EEG

Source: analysis.md

Dataset: `openneuro_ds006768` | Year: 2025 | URL: https://openneuro.org/datasets/ds006768
Category: High-level Mental | Subjects: 30 | Files (Completed): 210 | Channels: 65 | Sampling Rate: 1000 Hz

Paradigm

Subjects completed a Multiple Object Monitoring (MOM) task requiring sustained visual attention to track multiple moving objects simultaneously. In each trial, a subset of objects is designated as targets at the start, then all objects move during a tracking period, and subjects report which objects were targets. EEG captures neural correlates of visual attention load and tracking accuracy. Events mark trial onset, target designation, and distractor appearances.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006768

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`event`	20384	short	raw	Generic trial event marker
`target`	11004	short	raw	Target object onset (object to be tracked)
`distractor`	11004	short	raw	Distractor object onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: MOM

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006768 DOI: https://doi.org/10.1101/2025.07.10.663816

Normal Task 1: Target vs Distractor Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Well-balanced binary contrast: target vs. distractor (11004 each). `event` is a generic trial marker. 210 files from 30 subjects (7 runs/subject on average).

Event	y	Meaning
`event`	0	Generic trial event marker
`target`	1	Target object to be tracked
`distractor`	2	Distractor object
## Data Quality Notes		

- 210 files from 30 subjects suggests ~7 runs per subject.
- `event` is a generic trial marker; prefer specific `target`/`distractor` events for classification.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Multiple Object Monitoring (EEG)`

144. openneuro_ds006840 - IACKD: Intention Action Conflict EEG-Hand Kinematics Dataset

Source: analysis.md

Dataset: `openneuro_ds006840` | Year: 2025 | URL: https://openneuro.org/datasets/ds006840
Category: High-level Mental | Subjects: 128 | Files (Completed): 128 | Channels: 29 | Sampling Rate: 1024 Hz

Paradigm

This dataset investigates the neural correlates and hand kinematics associated with conflicts between intended and executed actions. In an intention-action conflict paradigm, participants are cued to perform one action but must sometimes execute a competing action, creating conflict between motor intention and execution. EEG is recorded alongside hand kinematic data during pre-movement planning (`premove`), execution (`execution`), and successful hit trials (`hit`).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006840

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`execution`	7032	short	raw	Motor execution onset
`hit`	7032	short	raw	Successful action completion marker
`premove`	7029	short	raw	Pre-movement preparation period onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Intention action congruency

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006840 DOI: https://doi.org/10.18112/openneuro.ds006840.v1.0.0

Normal Task 1: Motor Phase Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Well-balanced 3-class design (~7032 each). Hand kinematic data available alongside EEG for multimodal analysis. BrainVision format.

Event	y	Meaning
`premove`	0	Pre-movement intention/preparation
`execution`	1	Motor execution onset
`hit`	2	Successful trial completion
## Data Quality Notes		

- Consistent event structure (~7029-7032 events per type); well-controlled paradigm.
- Kinematic data channels (VEOG, HEOG, Trigger) should be excluded from EEG analysis.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `IACKD: Intention Action Conflict EEG-Hand Kinematics Dataset`

145. openneuro_ds006848 - AlphaDirection1: EEG, ECG, PPG in the resting state and working memory for sequentially and simultaneously presented digits

Source: analysis_minimax_v2.md

Dataset: openneuro_ds006848 | Year: 2025 | URL: https://openneuro.org/datasets/ds006848/versions/1.0.0
Category: High-level Mental | Subjects: 30 | Files (Completed): 52
Channels: 65 | Sampling Rate: 1000 Hz

Paradigm

This dataset studies the directional characteristics of brain alpha waves during resting state and working memory tasks. Participants completed a verbal working memory digit span task in two encoding modes: serial presentation (one digit is displayed at a time) and simultaneous presentation (an entire set of digits are displayed at once), followed by serial recall; before the experiment, a resting state (eyes open/eyes closed alternating) recording was included, and cartoon clips were used to separate different experimental segments. Data is recorded simultaneously with EEG, ECG and PPG.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006848/versions/1.0.0

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
Encoding_DigitValue_0	3163	short	raw	Sequence encoding: number 0 presents onset
Encoding_DigitValue_1	3134	short	raw	Sequence encoding: number 1 presents onset
Encoding_DigitValue_2	3077	short	raw	Sequence encoding: number 2 presents onset
Encoding_DigitValue_3	3124	short	raw	Sequence encoding: number 3 presents onset
Encoding_DigitValue_4	3164	short	raw	Sequence encoding: number 4 presents onset
Encoding_DigitValue_5	3104	short	raw	Sequence encoding: number 5 presents onset

Event ID	Count (total)	Type	Meta Level	Meaning
Encoding_DigitValue_6	3103	short	raw	Sequence encoding: number 6 presents onset
Encoding_DigitValue_7	3131	short	raw	Sequence encoding: number 7 presents onset
Encoding_DigitValue_8	3117	short	raw	Sequence encoding: number 8 rendering onset
Encoding_DigitValue_9	3131	short	raw	Sequence encoding: number 9 presents onset
Encoding_Set_Simultaneous	1495	short	raw	Simultaneous encoding: the entire set of numbers is presented at once onset
Digits_Retrieval	5960	short	raw	The recall phase begins (the subject begins to report verbally)
Baseline_2s	5956	short	raw	2 s before each trial, the beginning of the baseline segment
Retention_Fast	1499	long	raw	Fast delay hold period (sequence encoding, short delay)
Retention_FastDelay	1499	long	raw	Fast delay additional waiting period (sequence encoding)
Retention_Slow	1466	long	raw	Slow delay hold period (sequence encoding, long delay)
Retention_Simultaneous	1495	long	raw	Retention period after simultaneous encoding
Rest	209	rest	raw	Resting state segment start mark
Eyes_Closed	87	rest	raw	Close your eyes and start to rest
Eyes_Opened	88	rest	raw	Open your eyes and start to rest
Boundary	157	admin	raw	BIDS file boundary mark
Experiment_End	29	admin	raw	Experiment end mark
Start_Cartoon	22	admin	raw	Cartoon short film starts (intersegment interference)
End_Cartoon	22	admin	raw	End of cartoon short

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Verbal Working Memory - Digit Recall Prediction (Regression)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -10.0 s, tmax = 8.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Retention_Fast`	0-7	Fast encoding (400ms/digit) - predict digits recalled
`Retention_FastDelay`	0-7	Fast encoding + delay (400ms+600ms gap) - predict digits recalled
`Retention_Slow`	0-7	Slow encoding (1000ms/digit) - predict digits recalled
`Retention_Simultaneous`	0-7	Simultaneous encoding (all digits 2800ms) - predict digits recalled

- Evidence:
- DOI: 10.1101/2025.05.04.652111 - This paper analyzes EEG data from OpenNeuro dataset ds006848 to examine how alpha, theta, and beta oscillations during encoding and retention periods predict working memory performance (number of digits recalled) across four digit presentation modes.
- Kernel Notes: Extract epochs from -10s to +8s relative to retention onset. The paper analyzes EEG spectral power in windows: Encoding, Retention 1 (0.6-3s), Retention 2 (3-6s), excluding first 0.6s of retention. Labels are the actual number of digits recalled (0-7), obtained from behavioral data. Verify event ID mapping from events.tsv: `Retention_Fast` maps to "Fast" condition, `Retention_FastDelay` maps to "Fast+delay" condition, `Retention_Slow` maps to "Slow" condition, `Retention_Simultaneous` maps to "Simultaneous" condition.

Primary Task 2: Publisher-defined BIDS task: Rest

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006848/versions/1.0.0 DOI: https://doi.org/10.18112/openneuro.ds006848.v1.0.0

Primary Task 3: Publisher-defined BIDS task: Verbalwm

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)

- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006848/versions/1.0.0 DOI: https://doi.org/10.18112/openneuro.ds006848.v1.0.0

Normal Task 1: Task 1 (main task): coding mode two classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: The categories are seriously unbalanced (sequence ~31000 times vs. simultaneous ~1495 times), downstream needs to downsample to 1:1

Event	y	Meaning
`Encoding_DigitValue_0`	0	Sequence code (number 0)
`Encoding_DigitValue_1`	0	Sequence code (number 1)
`Encoding_DigitValue_2`	0	Sequence code (number 2)
`Encoding_DigitValue_3`	0	Sequence code (number 3)
`Encoding_DigitValue_4`	0	Sequence code (number 4)
`Encoding_DigitValue_5`	0	Sequence code (number 5)
`Encoding_DigitValue_6`	0	Sequence code (number 6)
`Encoding_DigitValue_7`	0	Sequence code (number 7)
`Encoding_DigitValue_8`	0	Sequence code (number 8)
`Encoding_DigitValue_9`	0	Sequence code (number 9)
`Encoding_Set_Simultaneous`	1	Simultaneous encoding (whole group)

Normal Task 2: Digital Identification (11-class multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Each number category ~3100 times, Encoding_Set_Simultaneous ~1495 times, the overall balance is relatively balanced; This is the actual configuration in task_config.json

Event	y	Meaning
`Encoding_DigitValue_0`	0	Number 0 (sequence condition)
`Encoding_DigitValue_1`	1	number 1
...	...	...
`Encoding_DigitValue_9`	9	Number 9
`Encoding_Set_Simultaneous`	10	Present the entire group simultaneously

Normal Task 3: Retention period type classification (4-class, long event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: This task needs to be configured separately (different from the epoch method of task 2). The current main entry of task_config.json does not contain this task.

Event	y	Meaning
`Retention_Fast`	0	Sequence encoding, short delay hold period
`Retention_FastDelay`	1	Sequence encoding, short delay and extra wait
`Retention_Slow`	2	Sequence encoding, long delay hold period
`Retention_Simultaneous`	3	Retention period after simultaneous encoding

Normal Task 4: Resting state: open/closed eyes classification (binary, long event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Each file has about 1-2 switches (87/88 times in total), the amount of data is limited; `Rest` is the segment start mark and is not directly used for epoch

Event	y	Meaning
`Eyes_Closed`	0	Close eyes and rest
`Eyes_Opened`	1	Rest with eyes open

Data Quality Notes

- The ratio of sequence encoding vs. simultaneous encoding categories is about 20:1, and downsampling is required to do a fair binary classification evaluation.
- `Boundary` is the BIDS file boundary, which will appear at the beginning of each file and must be excluded from the epoch target
- Some subjects' files lack the `Eyes_Closed`/`Eyes_Opened` mark (only 22/30 subjects have resting eye state records)
- Sampling rate 1000 Hz, stage5 will downsample to 200 Hz

References

- Reference: `AlphaDirection1: EEG, ECG, PPG in the resting state and working memory for sequentially and simultaneously presented digits`

146. openneuro_ds007172 - EEG-Asymmetries Dataset

Source: analysis.md

Dataset: `openneuro_ds007172` | Year: 2026 | URL: https://openneuro.org/datasets/ds007172
Category: High-level Mental | Subjects: 100 | Files (Completed): 501 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Multi-task EEG battery investigating hemispheric asymmetries across resting state and four cognitive categorization tasks: emotion recognition (face stimuli with emotional valence), language comprehension, face perception (identity discrimination), and visual discrimination (house images). The dataset examines relationships between N1 lateralization and alpha asymmetry across task domains. BrainVision trigger codes (Stimulus/S1xx and S2xx families) encode stimulus conditions per task.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds007172

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S111`	12800	long	raw	Emotion/language task condition S111
`Stimulus/S223`	12800	long	raw	Emotion/language task condition S223
`Stimulus/S212`	12800	long	raw	Emotion/language task condition S212
`Stimulus/S214`	12800	long	raw	Emotion/language task condition S214
`Stimulus/S211`	12800	long	raw	Emotion/language task condition S211
`Stimulus/S123`	12800	long	raw	Emotion/language task condition S123
`Stimulus/S114`	12800	long	raw	Emotion/language task condition S114
`Stimulus/S112`	12800	long	raw	Emotion/language task condition S112
`Stimulus/S222`	12000	long	raw	Emotion/language task condition S222
`Stimulus/S113`	12000	long	raw	Emotion/language task condition S113

Event ID	Count (total)	Type	Meta Level	Meaning
Stimulus/S121	12000	long	raw	Emotion/language task condition S121
Stimulus/S122	12000	long	raw	Emotion/language task condition S122
Stimulus/S124	12000	long	raw	Emotion/language task condition S124
Stimulus/S221	12000	long	raw	Emotion/language task condition S221
Stimulus/S213	12000	long	raw	Emotion/language task condition S213
Stimulus/S224	12000	long	raw	Emotion/language task condition S224
Stimulus/S217	11998	long	raw	Face task condition S217
Stimulus/S218	11998	long	raw	Face task condition S218
Stimulus/S118	11998	long	raw	Face task condition S118
Stimulus/S117	11998	long	raw	Face task condition S117
Stimulus/S227	11998	long	raw	Face task condition S227
Stimulus/S228	11998	long	raw	Face task condition S228
Stimulus/S127	11998	long	raw	Face task condition S127
Stimulus/S128	11998	long	raw	Face task condition S128
Stimulus/S 21	11998	long	raw	Houses/visual task condition S21
Stimulus/S 11	11998	long	raw	Houses/visual task condition S11
Stimulus/S 22	11998	long	raw	Houses/visual task condition S22
New Segment/	501	admin	raw	Segment boundary marker
Comment/actiCAP Data On	383	admin	raw	Hardware status marker
Comment/actiCAP USB Power On	134	admin	raw	Hardware status marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.

- Candidate Trigger Labels: `Stimulus/S111`, `Stimulus/S223`, `Stimulus/S212`, `Stimulus/S214`, `Stimulus/S211`, `Stimulus/S123`, `Stimulus/S114`, `Stimulus/S112`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Houses

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds007172 DOI: https://doi.org/10.1080/1357650x.2025.2591660

Primary Task 2: Publisher-defined BIDS task: Words

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds007172 DOI: https://doi.org/10.1080/1357650x.2025.2591660

Primary Task 3: Publisher-defined BIDS task: Emotions

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds007172 DOI: https://doi.org/10.1080/1357650x.2025.2591660

Normal Task 1: Multi-Condition EEG Asymmetry (long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 5 task files per subject (emotions, faces, houses, language, rest). `New Segment/`, `Comment/actiCAP Data On`, `Comment/actiCAP USB Power On` are hardware/admin markers; exclude from stimulus analysis.

Event	y	Meaning
`Comment/actiCAP Data On`	0	Hardware marker
`Comment/actiCAP USB Power On`	1	Hardware power marker
`Stimulus/S111`	2	Condition S111

Event	y	Meaning
Stimulus/S223	3	Condition S223
Stimulus/S212	4	Condition S212
Stimulus/S214	5	Condition S214
Stimulus/S211	6	Condition S211
Stimulus/S123	7	Condition S123
Stimulus/S114	8	Condition S114
Stimulus/S112	9	Condition S112
Stimulus/S222	10	Condition S222
Stimulus/S113	11	Condition S113
Stimulus/S121	12	Condition S121
Stimulus/S122	13	Condition S122
Stimulus/S124	14	Condition S124
Stimulus/S221	15	Condition S221
Stimulus/S213	16	Condition S213
Stimulus/S224	17	Condition S224
Stimulus/S217	18	Condition S217
Stimulus/S218	19	Condition S218
Stimulus/S118	20	Condition S118
Stimulus/S117	21	Condition S117
Stimulus/S227	22	Condition S227
Stimulus/S228	23	Condition S228
Stimulus/S127	24	Condition S127
Stimulus/S128	25	Condition S128
Stimulus/S 21	26	Condition S21
Stimulus/S 11	27	Condition S11
Stimulus/S 22	28	Condition S22
## Data Quality Notes		

- 501 files from 100 subjects includes 5 task types per subject.
- `New Segment/` and all `Comment/` markers are administrative; never use as classification targets.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Hemispheric asymmetries in the EEG: Is there an association between N1 lateralization and alpha asymmetryx`

147. other_CockPartDataset - CockPart Multi-Paradigm EEG Dataset

Source: analysis.md

Dataset: `unknown_CockPartDataset` | Year: N/A | URL: N/A
Category: High-level Mental | Subjects: varies by paradigm | Files (Completed): 1590 (Case B kernel) | Channels: 128 | Sampling Rate: 128 Hz (N400/Natural Speech) / 512 Hz (Speech in Noise)

Paradigm

A multi-paradigm EEG corpus comprising four experimental subdatasets, all recorded from a 128-channel EEG cap. The four paradigms are:

1. N400: Subjects read or heard sentence contexts ending in high-cloze (predictable) or low-cloze (unpredictable) words; pre-epoched EEG trials at 128 Hz stored as `eegData_hiCloze` and `eegData_loCloze` arrays (128 channels x 103 timepoints x 150 trials) for 9 subjects.
2. Natural Speech: Subjects listened to natural continuous speech; raw continuous EEG stored at 128 Hz for 19 subjects (multiple runs per subject, up to ~420 run files total).
3. Natural Speech - Reverse: Subjects listened to time-reversed (unnatural) speech as a control condition; raw continuous EEG at 128 Hz for 10 subjects (200 run files).
4. Speech in Noise (SiN): Subjects listened to speech presented with background noise in audio-only (_a_) and video-only (_v_) modality conditions; raw continuous EEG at 512 Hz for 21 subjects (315 audio + 315 video files = 630 total SiN files).

The Case B kernel dispatches loading to the appropriate paradigm-specific method based on the file path.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: N/A (local/private dataset description)

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`N400_hiCloze`	150/subject x 9 = 1350	long	raw	N400 paradigm: high-cloze (predictable) word epoch
`N400_loCloze`	150/subject x 9 = 1350	long	raw	N400 paradigm: low-cloze (unpredictable) word epoch
`NaturalSpeech`	1/file x ~420	long	raw	Natural speech paradigm: whole-run annotation

Event ID	Count (total)	Type	Meta Level	Meaning
NaturalSpeechReverse	1/file x ~200	long	raw	Reversed speech paradigm: whole-run annotation
SiN_audio	1/file x 315	long	raw	Speech-in-Noise audio-only condition
SiN_video	1/file x 315	long	raw	Speech-in-Noise video-only condition

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Case B kernel: N400 epoch arrays are concatenated into a single continuous raw signal; each epoch onset is annotated individually. Natural Speech and SiN paradigms provide single whole-recording annotations per file.

Tasks

Normal Task 1: N400 Language ERP - High-Cloze vs. Low-Cloze (epoch_by_event)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 0.8 s (= 103/128 Hz; matches pre-epoched trial length), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Filter to N400 paradigm files (N400/EEG/Subject{N}.mat). Case B kernel concatenates hiCloze (150) then loCloze (150) epochs; epoch window (0.8 s) matches the original pre-epoched length. sfreq = 128 Hz.

Event	y	Meaning
N400_hiCloze	0	High-cloze (predictable) word - N400 attenuated
N400_loCloze	1	Low-cloze (unpredictable) word - N400 enhanced

Normal Task 2: Natural Speech Auditory Processing - Speech vs. Reversed Speech (epoch_by_long_event)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, overlap = 0.0 s, use full recording duration from annotation, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Filter to Natural Speech and Natural Speech - Reverse paradigm files. sfreq = 128 Hz. Multiple runs per subject; run files are independent.

Event	y	Meaning
NaturalSpeech	0	Listening to natural (forward) speech
NaturalSpeechReverse	1	Listening to reversed (backward) speech

Normal Task 3: Speech-in-Noise Modality - Audio vs. Video (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, overlap = 0.0 s, use full recording duration from annotation, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Filter to Speech in Noise paradigm files. sfreq = 512 Hz. Condition (audio/video) inferred from filename suffix (_a_ or _v_).

Event	y	Meaning
`SiN_audio`	0	Speech-in-Noise audio-only condition
`SiN_video`	1	Speech-in-Noise video-only condition

Normal Task 4: N400 cloze-condition decoding (high vs low cloze)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: `N400_hiCloze` vs `N400_loCloze` trial labels from the N400 subset.
- Evidence: Primary task inferred from internal dataset documentation only; external publication/reference is not yet available.
- Notes: Demoted from Primary Task because no external DOI/URL source is available.

Data Quality Notes

- Case B kernel: `.mat` files contain pre-epoched arrays or raw continuous EEG depending on the paradigm; paradigm is inferred from the directory structure in the file path.
- Channel names in reconstructed raw are generic (`EEG001-EEG128`); no electrode layout metadata is provided.
- Variable sampling rates across paradigms: N400 and Natural Speech = 128 Hz; Speech in Noise = 512 Hz. Do not mix paradigms without resampling.
- Dataset origin and publication unknown; provenance requires verification before publication.
- No age or sex information available.

References

- Reference: Unknown provenance. Multi-paradigm EEG corpus (CockPart Dataset). No associated publication identified.

148. other_seed - SJTU Emotion EEG Dataset (SEED)

Source: analysis_minimax_v2.md

Dataset: `other_seed` | Year: 2015 | URL: https://bcmi.sjtu.edu.cn/home/seed/
Category: High-level Mental | Subjects: 15 | Files (Completed): 81 | Channels: 62 | Sampling Rate: 1000 Hz

Paradigm

The SEED (SJTU Emotion EEG Dataset) was collected from 15 subjects who watched 15 emotional film clips (positive, neutral, negative) across three sessions (Zheng & Lu, 2015). EEG was recorded from 62 channels using an ESI NeuroScan system (Neuroscan cap) at 1000 Hz. Fifteen emotional video clips (~4 minutes each) were shown in each session, with sessions repeated on different days (up to 3 sessions per subject). Valence labels (+1 positive, 0 neutral, -1 negative) for each clip are stored in separate `*_label.mat` files; they are not embedded in the `.cnt` EEG recordings. The kernel does not extract emotion labels from EEG annotations; labels must be loaded externally.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://bcmi.sjtu.edu.cn/home/seed/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`0`	~81	admin	raw	File-level marker from `.cnt` format (no experimental meaning)
`255`	rare	admin	raw	End-of-file or reset marker in `.cnt` format

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Critical: Emotion valence labels (+1, 0, -1) are stored in companion `*_label.mat` files, NOT in the EEG event annotations. Events `0` and `255` are format-level markers with no experimental significance.

Tasks

Primary Task 1: Three-class emotion valence classification (positive/neutral/negative)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 2.0 s
- Label Route: `Annotation`

- Meta Level: super
- Target Definition:
Label	y	Meaning
positive	0	Positive valence (+1 from SEED labels)
neutral	1	Neutral valence (0 from SEED labels)
negative	2	Negative valence (-1 from SEED labels)
- Super Meta Notes: Derived from experimental stimulus protocol (15 film clips per session designed to elicit specific emotions). Labels (+1, 0, -1) obtained from SEED companion *_label.mat files; not from EEG event markers. Kernel must load external label files and align with EEG epochs by trial order (clip index). EEG events 0 and 255 are format-level markers and excluded.
- Evidence:
- DOI: 10.1007_s12559-025-10516-z - Fuzzy linguistic summarization applied to SEED for 3-class valence classification using frontal alpha/beta bands from AF3/AF4 channels.
- DOI: 10.3389/fnins.2018.00162 - Binary cross-subject emotion classification using 4-s sliding windows with 2-s overlap from middle 1-minute of each 4-min trial; labels from external SEED label files.
- Kernel Notes: Load valence labels from *_label.mat files (provided with SEED dataset). Segment EEG into sliding windows during each 4-minute video clip. Map each epoch to the clip's valence label. The dataset contains 15 subjects x 3 sessions x 15 clips = 675 total trials. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Binary emotion classification (positive vs negative)

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 4.0 s, overlap = 2.0 s
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Label	y	Meaning
positive	0	Positive valence (+1 from SEED labels)
negative	1	Negative valence (-1 from SEED labels)
- Super Meta Notes: Derived from experimental stimulus protocol. Labels (+1, -1) obtained from SEED companion *_label.mat files; neutral trials (label=0) are discarded. Kernel loads external label files, filters to positive/negative classes only, and aligns with EEG epochs. Middle 1-minute portion of each 4-minute trial used per paper methodology.
- Evidence:
- DOI: 10.3389/fnins.2018.00162 - Binary cross-subject emotion classification (positive vs negative) using EEG features from 4-s sliding windows with 2-s overlap, using middle 1-minute segment of each 4-min trial.
- Kernel Notes: Load valence labels from *_label.mat files. Segment EEG into 4-s windows with 2-s overlap. Filter out neutral (label=0) epochs. Use only middle 60 seconds of each 4-minute clip as per paper methodology. (replaces Normal Task 2 from original analysis.md)

Data Quality Notes

- Emotion labels are not embedded in EEG files; they must be loaded from companion *_label.mat files and aligned with EEG epochs by trial order.
- EEG events 0 and 255 are file-format markers; exclude from all analyses.
- 3 sessions per subject on different days; within-session and cross-session analyses are common.
- 15 video clips per session x 3 sessions x 15 subjects = 675 trials total.
- NeuroScan .cnt format; 62-channel standard 10-20 EEG montage.

References

- Reference: Zheng W-L, Lu B-L (2015). Investigating Critical Frequency Bands and Channels for EEG-based Emotion Recognition with Deep Neural Networks. IEEE Trans. Autonomous Mental Development, 7(3), 162-175. https://bcmi.sjtu.edu.cn/home/seed/

149. physionet_auditoryeeg - Auditory EEG Dataset (Biometric Identification)

Source: analysis_minimax_v2.md

Dataset: `physionet_auditoryeeg` | Year: 2021 | URL: https://physionet.org/content/auditory-eeg/1.0.0/
Category: High-level Mental | Subjects: 19 | Files (Completed): 205 | Channels: 13 | Sampling Rate: 200 Hz

Paradigm

Nineteen subjects (s01-s20, one subject absent) aged 21-31 participated in a biometric EEG identification study using resting-state and auditory stimulus paradigms (Trujillo et al., 2021). EEG was recorded using an OpenBCI Cyton board (13 channels, gold-cup electrodes at T7, F8, Cz, P4 and surrounding positions). Each subject completed multiple recording sessions across different experimental conditions: two resting-state conditions (eyes open, eyes closed), eyes-open with noise isolation, and six auditory stimulus conditions (native song, non-native song, neutral music - via in-ear headphones or bone-conduction headphones). Each recording is labeled by experiment code (ex01-ex12) encoded in the filename. Files from subject 5 (ex05) lack structured event codes and retain raw experiment codes as annotations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: questionnaire
> Web Reference Used for Decision: https://physionet.org/content/auditory-eeg/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`resting_eyes_open`	~50	long	raw	Resting state with eyes open (ex01)
`resting_eyes_closed`	~25	long	raw	Resting state with eyes closed (ex02)
`resting_eyes_open_noise_isolation`	~50	long	raw	Eyes open with acoustic isolation (ex04, ex06)
`auditory_native_song_in_ear`	~25	long	raw	Native-language song via in-ear headphones (ex07)
`auditory_nonnative_song_in_ear`	~25	long	raw	Non-native language song via in-ear headphones (ex08)
`auditory_neutral_music_in_ear`	~25	long	raw	Neutral instrumental music via in-ear headphones (ex09)

Event ID	Count (total)	Type	Meta Level	Meaning
auditory_native_song_bone	~25	long	raw	Native song via bone-conduction headphones (ex10)
auditory_nonnative_song_bone	~25	long	raw	Non-native song via bone-conduction headphones (ex11)
ex05	~19	long	raw	Uncoded experiment (raw experiment code, exclude from classification)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Each annotation spans the entire recording file. Counts are approximate.

MISC Label Summary

- Route: misc
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Biometric Subject Identification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 5.0 s, overlap = 2.5 s
- Label Route: misc
- Meta Level: super
- Target Definition:

Event	y	Meaning
biometric_identity	0-19	Subject identity (20 subjects)

- Super Meta Notes: Derived from file metadata (subject IDs s01-s20 embedded in filenames). Kernel step: Extract subject ID from filename pattern (e.g., s01_ex01.csv -> subject s01), map to integer y=0-19.
- Evidence:
- DOI: 10.1523/eneuro.0265.25.2025 - Uses Auditory-Evoked Potential EEG Biometric Dataset for TriNet-MTL biometric identification across 20 subjects with sliding window segmentation (1,000-sample windows, 500-sample stride)
- Kernel Notes: Parse filename for subject identifier. Map s01->0, s02->1, ... s20->19. Exclude files with ex05 (uncoded). (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Stimulus Language Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 5.0 s, overlap = 2.5 s
- Label Route: misc
- Meta Level: super
- Target Definition:

| Event | y | Meaning |

|-------|---|---------|
| stimulus_language | 0 | Native language stimulus |
| stimulus_language | 1 | Non-native language stimulus |
- Super Meta Notes: Derived from raw event(s) [auditory_native_song_in_ear, auditory_nonnative_song_in_ear, auditory_native_song_bone, auditory_nonnative_song_bone] by mapping native songs to y=0, non-native songs to y=1. Kernel step: Map events containing 'native' -> 0, 'nonnative' -> 1. Neutral music excluded from this task.
- Evidence:
- DOI: 10.1523/eneuro.0265.25.2025 - Trains simultaneous stimulus language classification (native vs non-native) using sliding window segmentation
- Kernel Notes: Filter epochs to only auditory_native_song_ and auditory_nonnative_song_ events. Map native->0, non-native->1. Exclude neutral music events.

Primary Task 3: Device Modality Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 5.0 s, overlap = 2.5 s
- Label Route: misc
- Meta Level: super
- Target Definition:
Event	y	Meaning
device_modality	0	In-ear headphones
device_modality	1	Bone-conduction headphones
- Super Meta Notes: Derived from raw event(s) [auditory_native_song_in_ear, auditory_nonnative_song_in_ear, auditory_neutral_music_in_ear, auditory_native_song_bone, auditory_nonnative_song_bone, auditory_neutral_music_bone] by mapping _in_ear -> 0, _bone -> 1. Kernel step: Detect keyword in event ID.
- Evidence:
- DOI: 10.1523/eneuro.0265.25.2025 - Trains simultaneous device modality recognition (in-ear vs bone conduction) using sliding window segmentation
- Kernel Notes: Parse all auditory events. If event contains '_in_ear' -> y=0, if contains '_bone' -> y=1.

Normal Task 2: Resting-State vs. Auditory Stimulus (epoch_by_long_event)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 5.0 s, overlap = 2.5 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
resting_eyes_open	0	Resting state
resting_eyes_closed	0	Resting state
resting_eyes_open_noise_isolation	0	Resting state
auditory_native_song_in_ear	1	Any auditory condition
auditory_nonnative_song_in_ear	1	Any auditory condition
auditory_neutral_music_in_ear	1	Any auditory condition

Event	y	Meaning
auditory_native_song_bone	1	Any auditory condition
auditory_nonnative_song_bone	1	Any auditory condition

Data Quality Notes

- Only 13 EEG channels (non-standard OpenBCI placement); spatial coverage is limited.
- ex05 files (subject 5 recordings in some conditions) retain raw experiment codes and should be excluded from condition-labeled classification.
- Primary use case is biometric identification (same person across sessions), not condition classification; adapt label source accordingly.
- Some subjects may be missing recordings for certain experiment codes.

References

- Reference: Trujillo LT et al. (2021). EEG recordings for biometric identification. Physionet. https://physionet.org/content/auditory-eeg/1.0.0/

150. physionet_eegmat - EEG During Mental Arithmetic Task

Source: analysis_minimax_v2.md

Dataset: `physionet_eegmat` | Year: 2020 | URL: https://physionet.org/content/eegmat/1.0.0/
Category: High-level Mental | Subjects: 36 | Files (Completed): 72 | Channels: 21 | Sampling Rate: 500 Hz

Paradigm

Thirty-six healthy subjects (Subject00-Subject35, ages 16-26, predominantly female) completed a mental arithmetic task while EEG was recorded from 21 channels using a Neurocom 23-channel system with Ag/AgCl electrodes (Zyma et al., 2019). Each subject contributed two files: a background resting-state recording (suffix `_1.edf`) and a mental arithmetic task recording (suffix `_2.edf`). During the mental arithmetic task, subjects performed serial subtraction (e.g., counting backwards by 7s) while EEG was recorded. A separate `Count quality` field in `subject-info.csv` rates the subject's arithmetic performance as Good or Bad, providing a behavioral quality label.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/eegmat/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Background EEG;Count Quality: Good`	27	long	raw	Resting-state recording, subject's arithmetic quality was Good
`Background EEG;Count Quality: Bad`	9	long	raw	Resting-state recording, subject's arithmetic quality was Bad
`Mental Arithmetic Task EEG;Count Quality: Good`	27	long	raw	Mental arithmetic recording, quality Good
`Mental Arithmetic Task EEG;Count Quality: Bad`	9	long	raw	Mental arithmetic recording, quality Bad

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Each annotation spans the entire recording. Counts are approximate (36 subjects x 2 files = 72 total; Good/Bad split estimated from subject distribution).

Tasks

Primary Task 1: Mental Arithmetic vs Resting State Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 60 s, overlap = 2.5 s *, target_events = [`Background EEG;Count Quality: Good`, `Background EEG;Count Quality: Bad`, `Mental Arithmetic Task EEG;Count Quality: Good`, `Mental Arithmetic Task EEG;Count Quality: Bad`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Background EEG;Count Quality: Good`	0	Resting-state/baseline EEG
`Background EEG;Count Quality: Bad`	0	Resting-state/baseline EEG
`Mental Arithmetic Task EEG;Count Quality: Good`	1	Mental arithmetic task (serial subtraction)
`Mental Arithmetic Task EEG;Count Quality: Bad`	1	Mental arithmetic task (serial subtraction)

- Evidence:
- DOI: 10.1109/indicon49873_2020.9342140 - VMD-based mental task classification achieves 93% average accuracy on EEGMAT for BST vs SRT
- DOI: 10.1109/jsen_2023.3337519 - XGBoost with GFT features achieves 99.75% accuracy on EEGMAT for baseline vs serial subtraction
- Kernel Notes: Parse event prefix before semicolon to extract task label (Background=0, Mental Arithmetic=1). Use sliding window (tmax=5s, overlap=2.5s) on the 60s recording. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Mental Arithmetic Performance Quality Prediction

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 60 s, overlap = 2.5 s *, target_events = [`Mental Arithmetic Task EEG;Count Quality: Good`, `Mental Arithmetic Task EEG;Count Quality: Bad`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Mental Arithmetic Task EEG;Count Quality: Bad`	0	Poor arithmetic performance
`Mental Arithmetic Task EEG;Count Quality: Good`	1	Good arithmetic performance

- Evidence:
- DOI: 10.1038/s41598.026.36434.0 - PC-entropy analysis shows good counters (n=26) vs poor counters (n=10) have significantly different beta band synchronization
- Kernel Notes: Filter to Mental Arithmetic events only, parse suffix after semicolon for quality label. Class imbalance expected (~73% Good, ~27% Bad). (replaces Normal Task 2 from original analysis.md)

Primary Task 3: Mental Stress Classification (Super Meta)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 5 s, overlap = 0 s, target_events = N/A (sliding window on raw signal)
- Label Route: `Annotation`
- Meta Level: super

- Target Definition:
Event	y	Meaning
`low_stress`	0	Low mental stress (clusters with Good quality performers)
`high_stress`	1	High mental stress (clusters with Poor quality performers)
- Super Meta Notes: Derived from raw events [`Mental Arithmetic Task EEG;Count Quality: Good`, `Mental Arithmetic Task EEG;Count Quality: Bad`] by k-means clustering (k=2) on log-band power features (delta, theta, alpha, beta). Kernel step: Extract 5s EEG segments, compute band power in delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), apply k-means clustering, map clusters to stress labels based on overlap with performance quality metrics.
- Evidence:
- DOI: 10.1109/wccest66994_2025.11389948 - CNN/LSTM/hybrid models achieve 98% accuracy on stress classification using 5-second epochs and k-means derived labels

Data Quality Notes

- Two files per subject (`_1.edf` = background, `_2.edf` = task); pair them by Subject ID.
- Count quality (Good/Bad) is embedded in the annotation description after the semicolon; always parse carefully.
- ECG channel present (channel `A2-A1`) in some recordings; exclude from EEG feature extraction.
- Small dataset (36 subjects x 2 = 72 files); limited generalization.

References

- Reference: Zyma I, Tukaev S, Seleznov I, Kiyono K, Popov A, Chernykh M, Shpenkov O (2019). Electroencephalograms during mental arithmetic task performance. Data, 4(1), 14. https://physionet.org/content/eegmat/1.0.0/

151. zenodo_3618205 - Selective Auditory Attention EEG Dataset (Normal-Hearing and Hearing-Impaired)

Source: analysis.md

Dataset: zenodo_3618205 | Year: 2020 | URL: https://zenodo.org/records/3618205
Category: High-level Mental | Subjects: 44 | Files (Completed): 74 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

Forty-four participants (22 normal-hearing, 22 hearing-impaired) underwent EEG recording in three tasks (Fuglsang et al., 2020): (1) selective auditory attention - two-talker competing speech; attend to one of two simultaneously presented speakers; (2) MMN - mismatch negativity; passive listening to a standard tone sequence with occasional deviant tones; (3) resting state - eyes open and closed. EEG was recorded with BioSemi ActiveTwo (64 channels + reference + auxiliary = ~73 total channels, 512 Hz). Nineteen of the 44 subjects also had in-ear EEG recorded. Event annotations are stored in BIDS-format _events.tsv sidecar files.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: questionnaire
> Web Reference Used for Decision: https://zenodo.org/records/3618205

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
attend_left	varies	long	raw	Subject attends to the left-ear speaker (selective attention trial)
attend_right	varies	long	raw	Subject attends to the right-ear speaker (selective attention trial)
(MMN deviant events)	varies	short	raw	Deviant tone stimulus onset (MMN paradigm)
(resting state marker)	varies	rest	raw	Resting state block onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Exact event codes from BIDS _events.tsv sidecars. Selective attention trial labels and MMN deviant markers are both present; parse task type from filename.

MISC Label Summary

- Route: `misc`
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Resting-state analysis baseline

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Condition/group labels from metadata when available
- Evidence: Dataset is primarily resting-state or spontaneous-state EEG. Source: https://zenodo.org/records/3618205 DOI: https://doi.org/10.5281/zenodo.3618205

Normal Task 1: Auditory Attention Decoding - Left vs. Right (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 60.0 s (or full trial duration), overlap = 30.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Use only files from the selective attention task. MMN and resting state files have different event structures.

Event	y	Meaning
`attend_left`	0	Attend to left-ear speaker
`attend_right`	1	Attend to right-ear speaker

Normal Task 2: Normal-Hearing vs. Hearing-Impaired Group Comparison (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: As in Task 1, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Needs verification.
- Notes: Group labels (normal-hearing / hearing-impaired) must be derived from subject ID and companion demographics file. Primary comparison is neural tracking of attended speech across the two groups.

Data Quality Notes

- Three distinct task types (selective attention, MMN, rest); always filter files by task type before analysis.
- 19 subjects also have in-ear EEG recordings; these appear as additional channels or separate files.
- Hearing-impaired subjects may have different audiological profiles; audiogram data is available in the dataset.

References

- Reference: Fuglsang SA et al. (2020). Midfrontal slow waves causally mediate the effect of pre-target attention on visual task performance. bioRxiv. Zenodo. https://zenodo.org/records/3618205

152. zenodo_4518754 - Ultra High-Density 256-Channel EEG AAD Dataset (Subset)

Source: analysis_minimax_v2.md

Dataset: `zenodo_4518754` | Year: 2021 | URL: https://zenodo.org/records/4518754
Category: High-level Mental | Subjects: 13 | Files (Completed): 156 | Channels: 256 | Sampling Rate: 1000 Hz

Paradigm

Thirteen subjects listened to two simultaneously presented Dutch stories (left vs. right ear) and were asked to attend to one speaker while ignoring the other (auditory attention decoding, AAD). EEG was recorded from 256 channels (BioSemi ActiveTwo ultra-high-density, Ag/AgCl) at 1000 Hz. Stories were presented via insert earphones at 60 dBA, with 180 deg spatial separation (left vs. right). Each subject completed multiple trials per session; each recording file corresponds to one trial labeled by attended direction (L/R) encoded in the filename (`S{N}_AAD_{trial}{L/R}.dap`). The kernel creates a single whole-trial `attend_left` or `attend_right` annotation per file.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://zenodo.org/records/4518754

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`attend_left`	~78	long	raw	Subject attends to the left-ear story for the entire recording
`attend_right`	~78	long	raw	Subject attends to the right-ear story for the entire recording

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Each file has exactly one annotation spanning the entire recording (~6 minutes per trial).

Tasks

Primary Task 1: Auditory attention decoding (attend left vs right)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 10.0 s, overlap = 5.0 s *, target_events = [`attend_left`, `attend_right`]
- Label Route: `Annotation`
- Meta Level: raw

- Target Definition:
Event	y	Meaning
`attend_left`	0	Attend to left-ear speaker
`attend_right`	1	Attend to right-ear speaker
- Evidence:
- DOI: `10.1088/1741-2552.ac2629` - Uses the ultra-high density 255-channel EEG-AAD dataset to classify whether subjects attend to the left or right speaker, analyzing with 10s decision windows and testing epoch durations from 1-60s.
- Kernel Notes: Paper used leave-one-epoch-out cross-validation with sliding windows. Verify overlap parameter matches paper's sliding window approach. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Ultra-high-density 256-channel EEG; spatial filtering and source analysis are computationally intensive.
- High sampling rate (1000 Hz); consider downsampling to 128 or 64 Hz for typical AAD decoding.
- Stimulus audio files are required for stimulus-reconstruction AAD methods; check availability.
- Only `.dap` files are EEG; auxiliary `.dat / .rs3` files are skipped by the kernel.

References

- Reference: Ultra high-density 255-channel EEG-AAD dataset. Zenodo. https://zenodo.org/records/4518754

153. zenodo_dtu - DTU Auditory Attention Decoding Dataset

Source: analysis_minimax_v2.md

Dataset: `zenodo_dtu` | Year: 2018 | URL: https://zenodo.org/records/1199011
Category: High-level Mental | Subjects: 18 | Files (Completed): 18 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

Eighteen subjects (S1-S18) participated in an auditory attention decoding (AAD) task in which two competing speakers were presented simultaneously and subjects were asked to attend to one speaker (Fuglsang et al., 2018). EEG was recorded from 64 channels (BioSemi ActiveTwo, 512 Hz) plus additional auxiliary channels (total 73). Each subject's recording is stored as a single MATLAB `.mat` file containing EEG data, stimulus onsets (`data.stimu`), and attended speaker labels (`data.attended`). The kernel extracts per-trial onset times and attended speaker indices to create trial-level annotations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://zenodo.org/records/1199011

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`attend_speaker_1`	varies	long	raw	Subject attends to speaker 1 for the trial duration
`attend_speaker_2`	varies	long	raw	Subject attends to speaker 2 for the trial duration

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Note: The status_summary indicates that `stimu` and `attended` fields may be missing in some versions of the dataset (Needs verification:Needs verification); verify field availability before processing.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimu`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Auditory attention decoding to competing speakers

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.0 s *, target_events = [`attend_speaker_1`, `attend_speaker_2`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`attend_speaker_1`	0	Attend to speaker 1 (left)
`attend_speaker_2`	1	Attend to speaker 2 (right)

- Evidence:
- DOI: 10.1088/2057.1976.ae09b2 - The paper uses the DTU dataset (18 subjects, Danish language, binary left/right attention classification) to validate TFDISNet, achieving 88.2% accuracy with a 2-second window for decoding which speaker the subject is attending to.
- Kernel Notes: Use 2-second non-overlapping windows within each trial annotation. Trial boundaries are derived from `data.stimu` onset times. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- MATLAB `.mat` field names (`data.stimu`, `data.attended`) must be present; verify before processing.
- One file per subject (18 files total); each file contains all trials for that subject.
- Stimulus audio files (speech signals) are required for full stimulus-reconstruction AAD decoding.

References

- Reference: Fuglsang SA et al. (2018). Noise-robust cortical tracking of attended speech in real-world acoustic scenes. NeuroImage, 156, 435-444. https://zenodo.org/records/1199011

154. zenodo_kul - KU Leuven Auditory Attention Decoding Dataset

Source: analysis.md

Dataset: `zenodo_kul` | Year: 2020 | URL: https://zenodo.org/records/4004271
Category: High-level Mental | Subjects: 15 | Files (Completed): 15 (Case B kernel) | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

Fifteen subjects performed a dichotic listening auditory attention decoding (AAD) task (Biesmans et al., 2017). Two competing story narratives were presented simultaneously, one to each ear, and subjects were instructed to attend to one. EEG was recorded from 64 channels (BioSemi ActiveTwo, standard 10-20 layout) at 128 Hz. Each MATLAB `.mat` file contains 20 trials (10 attended-left + 10 attended-right), stored as a Python list of trial dictionaries. The Case B kernel concatenates all 20 trials from each file into a single continuous raw signal and creates per-trial annotations indicating the attended ear.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://zenodo.org/records/4004271

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`attended_left`	150 (10/subject x 15 subjects)	long	raw	Subject attends to the left-ear speaker
`attended_right`	150 (10/subject x 15 subjects)	long	raw	Subject attends to the right-ear speaker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Case B kernel: 20 trials (each ~62.5 s at 128 Hz) concatenated into one continuous raw per file.

Tasks

Primary Task 1: Dichotic auditory attention decoding (left vs right ear)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Attended ear/speaker label in dichotic listening trials.

- Evidence: KUL AAD dataset is designed for supervised decoding of attended speech stream. Source: https://zenodo.org/records/4004271 DOI: https://doi.org/10.5281/zenodo.4004271

Normal Task 1: Auditory Attention Decoding - Left vs. Right Ear (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, use trial duration from annotation, overlap = 0.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Case B kernel uses `scipy.io.loadmat(simplify_cells=True)` and a custom `load()` method. Annotation durations match individual trial lengths. Balanced design: 10 trials per class per subject.

Event	y	Meaning
`attended_left`	0	Attended ear: left
`attended_right`	1	Attended ear: right
## Data Quality Notes		

- Low sampling rate (128 Hz); limits high-frequency analysis but is standard for EEG-AAD.
- 15 subjects only; relatively small cohort for robust model evaluation.
- No age or sex information available in the public release.
- Each file contains 20 trials; the Case B kernel creates discontinuities at trial boundaries (artificial concatenation).

References

- Reference: Biesmans W et al. (2017). Auditory-Inspired Speech Envelope Extraction Methods for Auditory Attention Detection in N-of-1 Trials. IEEE Trans. Neural Syst. Rehabil. Eng., 25(5), 428-437. https://zenodo.org/records/4004271

155. zenodo_saa - Spatial Auditory Attention EEG Dataset

Source: analysis.md

Dataset: `zenodo_saa` | Year: 2019 | URL: https://zenodo.org/records/3357164
Category: High-level Mental | Subjects: 18 | Files (Completed): 18 | Channels: 41 | Sampling Rate: 1024-4096 Hz (subject-variable)

Paradigm

Eighteen subjects (S12, S13, S15, S21, S27, S28, S31, S33, S36, S37, S38, S39, S40, S42, S46, S47, S50, S51) participated in a spatial auditory attention decoding (SAA) task in which a target speaker was positioned at a fixed azimuth (30 deg) among distractors. EEG was recorded from 32 BioSemi ActiveTwo channels (A1-A32, Ag/AgCl cap) plus 8 external channels (total 41 channels) at variable sampling rates (1024, 2048, or 4096 Hz depending on subject). Stimulus events are encoded as BioSemi trigger codes in the Status channel; the kernel extracts these using `mne.find_events()` and converts them to text annotations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://zenodo.org/records/3357164

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(numeric BioSemi trigger codes)	varies	short/admin	raw	Stimulus onset, response, and trial boundary markers encoded in Status channel

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Trigger codes include stimulus onset markers (~3856-3966), control codes (65530-65532), and other administrative codes. Exact mapping requires dataset-specific documentation. Primary task: detect spatial attention direction from EEG.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Spatial auditory attention decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification] | tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Spatial attention condition labels from cue/stimulus protocol.
- Evidence: SAA dataset targets decoding of attended spatial auditory stream/location. Source: https://zenodo.org/records/3357164 DOI: https://doi.org/10.5281/zenodo.3357164

Normal Task 1: Spatial Attention Decoding - Present Stimulus (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s (per stimulus onset event), target_events = see Target Definition
- Label Route: `stim`
- Target Definition: Requires mapping trigger codes to stimulus/attention labels from the dataset documentation. Stimulus-onset codes (~3856-3966) mark individual sound presentations.
- Notes: Trigger code semantics must be verified against the dataset README or companion paper. Variable sampling rates across subjects require per-subject resampling to a common rate.

Data Quality Notes

- Variable sampling rates (1024-4096 Hz) across subjects; resample to a common rate before cross-subject analysis.
- Trigger code interpretation requires dataset-specific documentation (Status channel values need decoding).
- 41 channels (32 EEG + 8 external); identify and exclude non-EEG channels before analysis.

References

- Reference: Spatial Auditory Attention Dataset. Zenodo. https://zenodo.org/records/3357164

156. zenodo_sin - Speech-in-Noise (SiN) Auditory Attention Decoding Dataset (Zenodo Mirror)

Source: analysis.md

Dataset: `zenodo_sin` | Year: 2021 | URL: N/A (Zenodo mirror of Mendeley SiN dataset)
Category: High-level Mental | Subjects: 26 | Files (Completed): 26 (Case B kernel) | Channels: 66 | Sampling Rate: 256 Hz

Paradigm

Twenty-six normal-hearing subjects (NH_subject01-NH_subject26) participated in a cocktail-party auditory attention decoding (AAD) task in which competing speech was presented with background noise at two signal-to-noise conditions (Zenodo mirror of Mendeley SiN dataset). EEG was recorded from 64 channels plus 2 mastoid references (total 66 channels) at 256 Hz. Each subject's `.mat` file contains pre-epoched EEG data as a 3D array `(896, 66, 100)` - 896 time points x 66 channels x 100 trials - for each of two Speech-in-Noise conditions (condition 7 and condition 13). The Case B kernel concatenates the 100 trials for each condition into a single continuous raw signal and annotates each trial with its condition label.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://zenodo.org/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`SiN_condition_7`	~1300 (~50/subject x 26)	long	raw	Speech-in-Noise condition 7 trial (one cocktail-party condition)
`SiN_condition_13`	~1300 (~50/subject x 26)	long	raw	Speech-in-Noise condition 13 trial (second cocktail-party condition)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Case B kernel: pre-epoched `.mat` data; each file reconstructed into continuous raw by concatenating 100 trials per condition (~3.5 s each at 256 Hz = 896 samples per trial).

Tasks

Primary Task 1: Speech-in-noise auditory attention decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification] | tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Condition or attended-stream labels (released split commonly uses condition 7 vs 13).
- Evidence: Speech-in-noise AAD dataset is explicitly provided for supervised attention decoding tasks. Source: https://zenodo.org/

Normal Task 1: Speech-in-Noise Condition Classification - Condition 7 vs. 13 (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 3.5 s (= 896/256 Hz; matches pre-epoched trial length), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Case B kernel reconstructs continuous raw from pre-epoched data; trial boundaries are annotated. Epoch window (3.5 s) matches the original pre-epoched length; use exactly 896/256 = 3.5 s to avoid boundary effects.

Event	y	Meaning
`SiN_condition_7`	0	Speech-in-Noise condition 7
`SiN_condition_13`	1	Speech-in-Noise condition 13
## Data Quality Notes		

- Case B kernel: original `.mat` data is pre-epoched (not continuous); reconstruction introduces trial boundary discontinuities.
- Zenodo mirror of the Mendeley SiN dataset (`mendeley_sin`); the two datasets are identical in content and structure.
- No age or sex information available.

References

- Reference: Speech-in-Noise Auditory Attention Decoding Dataset. Zenodo mirror of Mendeley SiN dataset. See also `mendeley_sin`.

157. zenodo_uhd - Ultra High-Density 256-Channel EEG AAD Dataset

Source: analysis_minimax_v2.md

Dataset: zenodo_uhd | Year: 2021 | URL: N/A
Category: High-level Mental | Subjects: 13 | Files (Completed): 156 | Channels: 256 | Sampling Rate: 1000 Hz

Paradigm

Thirteen subjects listened to two simultaneously presented stories (one in each ear) and attended to one speaker while ignoring the other (auditory attention decoding, AAD). Ultra-high-density EEG was recorded from 256 channels (BioSemi ActiveTwo, Ag/AgCl) at 1000 Hz. Subjects completed multiple trials; each recording file corresponds to one trial, with the attended direction (left/right) encoded in the filename (`S{N}_AAD_{trial}{L/R}.dap`). The kernel creates a whole-trial `attend_left` or `attend_right` annotation per file. This dataset has the same structure as `zenodo_4518754` and uses an identical kernel implementation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://zenodo.org/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`attend_left`	~78	long	raw	Subject attends to the left-ear story for the entire recording
`attend_right`	~78	long	raw	Subject attends to the right-ear story for the entire recording

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Each file has exactly one annotation spanning the entire recording.

Tasks

Primary Task 1: Ultra-high-density auditory attention decoding (left vs right)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 10.0 s, overlap = 5.0 s, target_events = [`attend_left`, `attend_right`]
- Label Route: `Annotation`

- Target Definition:
Event	y	Meaning
--------|
`attend_left`	0	Attend to left-ear speaker
`attend_right`	1	Attend to right-ear speaker
- Evidence:
- DOI: `10.1088/1741-2552.ac2629` - Uses a 255-channel ultra-high-density EEG dataset for auditory attention decoding where subjects attend to one of two simultaneously presented stories, and a linear decoder reconstructs speech envelopes from EEG to identify the attended speaker via correlation comparison.
- Kernel Notes: The attended direction is encoded in the filename (`S{N}_AAD_{trial}{L/R}.dap`). The kernel extracts this from file naming convention and creates whole-trial annotations. The dataset uses long annotations spanning entire recordings (approximately 10 seconds each). Multiple epoch lengths (10s, 5s, 2s, 1s) can be used for cross-validation as reported in the literature. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Ultra-high-density 256-channel EEG; computationally intensive.
- High sampling rate (1000 Hz); consider downsampling to 64 or 128 Hz for typical AAD decoding.
- Same structure and kernel as `zenodo_4518754`; may be processed identically.
- No age or sex information available.

References

- Reference: Ultra High-Density 256-Channel EEG Auditory Attention Decoding Dataset. Zenodo.

Chapter 3. Action/Output

Datasets in this chapter: 28. Motor intention/execution and BCI-output focused tasks.

158. bcic_iii_1 - BCI Competition III Dataset 1 (Directional Motor Imagery EEG)

Source: analysis.md

Dataset: `bcic_iii_1` | Year: 2005 | URL: https://www.bbci.de/competition/iii/download/
Category: Action/Output | Subjects: 1 | Files (Completed): 1 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

A single subject performed directional motor imagery (BCI Competition III, 2005). Visual arrow cues prompted 4-second motor imagery periods (two classes: left and right/foot). The dataset is stored in a single GDF file (`BCI2005_I.gdf`). The GDF format embeds event annotations natively; MNE reads them automatically. No age or sex information is available in the public release.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://www.bbci.de/competition/iii/download/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`768`	377	admin	raw	Start of a trial (pre-cue marker)
`769`	138	short	raw	Cue onset - class 1 (left hand / motor imagery direction 1)
`772`	139	short	raw	Cue onset - class 2 (right hand or foot, motor imagery direction 2)
`783`	100	admin	raw	Cue unknown (evaluation-set placeholder)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)
- Evidence: The official BCI Competition III dataset description identifies directional motor-imagery decoding as the benchmark objective for this dataset. Source: https://www.bbci.de/competition/iii/ Official description file: `references/website_des.txt`

Normal Task 1: Binary Motor Imagery Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Event codes follow the GDF/BCIC standard but are not renamed by the kernel; they appear as numeric strings. Event `768` is a trial-onset marker preceding the cue and should be excluded from classification targets. Event `783` (cue unknown) appears in unlabeled evaluation data and should also be excluded unless the test labels are available.

Event	y	Meaning
`769`	0	Motor imagery class 1 (left-hand direction)
`772`	1	Motor imagery class 2 (right-hand or foot direction)
## Data Quality Notes		

- Single-subject dataset; results cannot be generalized across subjects.
- Very small dataset (1 file, 377 trials); statistical power is limited.
- High sampling rate (1000 Hz); consider downsampling to <=250 Hz.
- No demographic information available.
- GDF event codes are kept as numeric strings in MNE annotations; ensure correct integer parsing.

References

- Reference: BCI Competition III (2005), Dataset 1 - Directionally Modulated EEG Signals. https://www.bbci.de/competition/iii/

159. bcic_iii_2 - BCI Competition III Dataset II (P300 Speller)

Source: analysis_minimax_v2.md

Dataset: `bcic_iii_2` | Year: 2005 | URL: https://www.bbci.de/competition/iii/download/
Category: Action/Output | Subjects: 2 | Files (Completed): 4 (Case B kernel; 2 Train + 2 Test) | Channels: 64 | Sampling Rate: 240 Hz

Paradigm

Two subjects (A and B) attended to individual letters in a 6x6 character matrix while rows and columns flashed sequentially (BCI Competition III, 2005). Each character trial consisted of 15 sequences, each sequence comprising 12 flashes (6 rows + 6 columns). The P300 event-related potential, peaking ~300 ms after the target row or column flash, encodes the attended character. The training set contains 85 labeled characters per subject; the test set contains 100 characters without disclosed labels. The original `.mat` files are in a pre-epoched 3D format and are reconstructed into continuous raw data by the Case B kernel, which concatenates the 85 training sequences end-to-end.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://www.bbci.de/competition/iii/download/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`seq_onset`	85	admin	raw	Onset of each of the 85 character sequences (boundary marker)
`target_flash`	2550	short	raw	Flash of the row or column containing the target character
`nontarget_flash`	~12750	short	raw	Flash of a row or column not containing the target character

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Counts per Train file (1 file per subject). Test files have no class labels -> all annotations set to `unknown`.

Tasks

Primary Task 1: P300 Target Detection (ERP Classification)

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 0.667 s, target_events = [`target_flash`, `nontarget_flash`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`target_flash`	1	Flash of row/column containing attended character (produces P300)
`nontarget_flash`	0	Flash of row/column not containing attended character

- Evidence:
- DOI: 10.1109/TBME.2008.915728 - Uses BCI Competition III Dataset II for binary classification of post-stimulus EEG signals as target (P300 present) vs non-target (no P300), extracting epochs from 0 to 667ms after stimulus onset.
- DOI: 10.1016/j.asoc.2021.108176 - Extracts epochs in [0, 800] ms from stimulus onsets and classifies target versus non-target flashes using LDA with channel selection optimization.
- DOI: 10.1016/j.heliyon.2023.e15380 - Evaluates P300 detection on Dataset II, using 667 ms epochs downsampled to 40 Hz with band-pass filter 0.1-30 Hz.
- DOI: 10.1088/1741.2560/12.2.026009 - Uses 667 ms epoch length for standard target vs non-target classification on Dataset II.
- Kernel Notes: Extract epochs aligned to flash onsets. Apply baseline correction (e.g., pre-stimulus -0.1s to 0s). Handle class imbalance (~1:5 target:nontarget ratio). Use train files only; test files have unknown labels. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Class imbalance: ~1 target flash per 5-6 non-target flashes; balance before classification.
- Test labels are not publicly available; test files yield `unknown` annotations after kernel processing.
- Case B kernel: original `.mat` data is pre-epoched (3D array) and is reconstructed into a continuous raw signal; treat sequence boundaries marked by `seq_onset` as artificial discontinuities.
- Only 2 subjects; model generalization cannot be assessed from this dataset alone.

References

- Reference: BCI Competition III (2005), Dataset II - P300 Speller. Wadsworth Center, NYS Department of Health. Wolpaw JR, Schalk G, Krusienski DJ. https://www.bbci.de/competition/iii/

160. bcic_iv_1 - BCI Competition IV Dataset 1 (Motor Imagery, Uncued Classifier Application)

Source: analysis_minimax_v2.md

Dataset: `bcic_iv_1` | Year: 2008 | URL: https://www.bbci.de/competition/iv/desc_1.html
Category: Action/Output | Subjects: 7 | Files (Completed): 14 | Channels: 59 | Sampling Rate: 1000 Hz

Paradigm

Seven subjects performed two-class motor imagery without feedback in an asynchronous (self-paced) BCI paradigm (BCI Competition IV, 2008). The two movement classes differed per subject, selected from: left hand, right hand, or foot imagery. A calibration phase used visual arrow cues with 4-second imagery intervals. The evaluation phase used acoustic cues (spoken words) with variable-duration imagery (1.5-8 s) and rest periods (1.5-8 s), simulating a realistic asynchronous BCI scenario. Both calibration and evaluation `.mat` files are provided per subject; the kernel reads class names and event positions directly from the `.mat` structure.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://www.bbci.de/competition/iv/desc_1.html

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`left`	~100/subject	short	raw	Cue onset for left-hand motor imagery
`right`	~100/subject	short	raw	Cue onset for right-hand motor imagery (subjects assigned right class)
`foot`	~100/subject	short	raw	Cue onset for foot motor imagery (subjects assigned foot class)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Class labels vary per subject: some subjects have `left + right`, others have `left + foot`. Event names are loaded dynamically from `nfo.className` in the `.mat` file. Counts shown are approximate per calibration file.

Tasks

Primary Task 1: Two-Class Motor Imagery Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 2.0 s, tmax = 6.0 s, target_events = [`left`, `right`, `foot`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`left`	0	Left-hand motor imagery (class 1, subject-specific)
`right`	1	Right-hand motor imagery (class 2, for subjects with left+right)
`foot`	1	Foot motor imagery (class 2, for subjects with left+foot)

- Evidence:
- DOI: 10.1109/tnsre.2020.3020975 - Uses epochs extracted from the 2-6s time window relative to cue onset for two-class motor imagery classification
- DOI: 10.1109/tnsre.2021.3071140 - Applies CTFSP algorithm with sliding windows [2-5]s, [2.5-5.5]s, [3-6]s relative to visual cue onset
- DOI: 10.1016/j.neuroimage.2007.01.051 - Dataset description defining two-class motor imagery with 4s cue intervals during calibration
- Kernel Notes: Subject-specific class assignment (some subjects have left+right, others have left+foot); kernel must read class names from `.mat` file `nfo.className` field. Evaluation phase contains variable 1.5-8s imagery periods; this Primary Task uses the standard 4s MI window (2-6s post-cue) from calibration data as supported by papers.

Normal Task 1: Two-Class Motor Imagery Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Class names are subject-specific; the kernel reads them from the `.mat` file and creates annotations accordingly. Evaluation files have continuous recordings with variable-length trials and idle periods; only annotated cue positions should be used as epoch anchors. High sampling rate (1000 Hz); consider downsampling to <=250 Hz.

Event	y	Meaning
`left`	0	Left-hand motor imagery
`right` or `foot`	1	Second motor imagery class (subject-specific)

Data Quality Notes

- Two-class labels vary per subject (left+right vs. left+foot); verify class names per file from the `eegunity_description` field or the `.mat` `nfo.className` field.
- No age or sex information is available in the public release.
- Evaluation phase has variable trial durations and idle periods; the idle state is unlabeled and should be identified by absence of annotations.
- High sampling rate (1000 Hz); memory-intensive; downsampling recommended.

References

- Reference: BCI Competition IV (2008), Dataset 1 - Motor Imagery, Uncued Classifier Application. https://www.bbci.de/competition/iv/desc_1.html

161. bcic_iv_2a - BCI Competition IV Dataset 2a (4-Class Motor Imagery)

Source: analysis_minimax_v2.md

Dataset: `bcic_iv_2a` | Year: 2008 | URL: https://www.bbci.de/competition/iv/desc_2a.html
Category: Action/Output | Subjects: 9 | Files (Completed): 18 | Channels: 25 | Sampling Rate: 250 Hz

Paradigm

Nine subjects (A01-A09) performed four-class cued motor imagery (BCI Competition IV, 2008): left hand, right hand, feet, and tongue. Each session contained 6 runs of 48 trials each (288 trials per session), balanced across 4 classes (72 trials per class). A fixation cross preceded each trial, followed by a visual arrow cue indicating the imagery class, and a 4-second imagery period. Short resting intervals separated trials. Each subject was recorded in two sessions: a training session (suffix `T.gdf`) with event labels, and an evaluation session (suffix `E.gdf`) with initially unknown labels replaced by kernel-loaded `.mat` companion files. EEG was recorded from 22 channels (standard 10-20, sensorimotor focus) plus 3 bipolar EOG channels (filtered 0.5-100 Hz, notch at 50 Hz).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://www.bbci.de/competition/iv/desc_2a.html

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Start of a trial`	~576/subject	admin	raw	Trial-onset fixation cross (pre-cue marker)
`Cue onset left (class 1)`	~144/subject	short	raw	Left-hand motor imagery cue
`Cue onset right (class 2)`	~144/subject	short	raw	Right-hand motor imagery cue
`Cue onset foot (class 3)`	~144/subject	short	raw	Feet motor imagery cue
`Cue onset tongue (class 4)`	~144/subject	short	raw	Tongue motor imagery cue
`Idling EEG (eyes open)`	~9/subject	admin	raw	Resting state with eyes open
`Idling EEG (eyes closed)`	~9/subject	admin	raw	Resting state with eyes closed
`Rejected trial`	variable	admin	raw	Trial rejected by experimenter

Event ID	Count (total)	Type	Meta Level	Meaning
`Start of a new run`	~12/subject	admin	raw	Run boundary marker
`Eye movements`	rare	admin	raw	Artifactual eye movement event

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Counts are approximate per subject (Training + Evaluation combined). The kernel remaps all GDF numeric codes to human-readable strings.

Tasks

Primary Task 1: Four-Class Motor Imagery Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = [`Cue onset left (class 1)`, `Cue onset right (class 2)`, `Cue onset foot (class 3)`, `Cue onset tongue (class 4)`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Cue onset left (class 1)`	0	Left-hand motor imagery
`Cue onset right (class 2)`	1	Right-hand motor imagery
`Cue onset foot (class 3)`	2	Feet motor imagery
`Cue onset tongue (class 4)`	3	Tongue motor imagery

- Evidence:
- DOI: 10.1088/1741-2552.ab3471 - CNN-LSTM hybrid for four-class MI classification extracting 4-second epochs from cue onset
- DOI: 10.1109/access.2021.3056088 - TSGL-EEGNet achieving 81.34% average 4-class accuracy on BCI Competition IV 2a
- DOI: 10.1109/smc42975.2020.9283028 - EEG-TCNET achieving 77.35% accuracy with epochs from -0.5s to 4.0s
- DOI: 10.1109/tnnls.2018.2789927 - CNN trained on 0-4s MI period after cue onset
- DOI: 10.1109/tnsre.2022.3156076 - SincNet-based hybrid achieving 74.26% accuracy
- DOI: 10.3389/fnins.2012.00039 - FBCSP algorithm using 0.5-2.5s after cue for training
- DOI: 10.3389/fnins.2012.00151 - Four-class MI classification using epochs 0.5-2.5s after cue
- DOI: 10.3390/diagnostics12040995 - MBEEGSE achieving 82.87% accuracy with -0.5s to 4.0s epochs
- Kernel Notes: Use 22 EEG channels (exclude EOG). Epoch window starts 0.5s before cue (baseline) and spans full 4s imagery period. Reject trials marked `Rejected trial`. Training session (T) has labeled cues; evaluation session (E) requires loading labels from companion `.mat` files. (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Binary Motor Imagery - Left vs. Right Hand (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

- Notes: Subset of Task 1. Discard feet and tongue trials.

Event	y	Meaning
`Cue onset left (class 1)`	0	Left-hand motor imagery
`Cue onset right (class 2)`	1	Right-hand motor imagery

Data Quality Notes

- Rejected trials (variable per subject/session) should be excluded from analysis.
- EOG channels present; apply artifact rejection or regression before motor imagery classification.
- Evaluation files initially contain `Cue unknown (class 783)` entries that the kernel replaces with proper class labels from `.mat` companion files; verify companion file presence.
- Class balance is enforced by design (72 trials per class per session); minor imbalance may arise from rejected trials.
- Subject age range: 17-26 years; 7 female, 2 male subjects.

References

- Reference: Brunner C, Leeb R, Muller-Putz G, Schlogl A, Pfurtscheller G (2008). BCI Competition 2008 - Graz Data Set A. Institute for Knowledge Discovery, Graz University of Technology. https://www.bbci.de/competition/iv/desc_2a.html

162. bcic_iv_2b - BCI Competition IV Dataset 2b (Left/Right Hand Motor Imagery)

Source: analysis_minimax_v2.md

Dataset: `bcic_iv_2b` | Year: 2008 | URL: https://www.bbci.de/competition/iv/desc_2b.html
Category: Action/Output | Subjects: 9 | Files (Completed): 46 | Channels: 6 | Sampling Rate: 250 Hz

Paradigm

Nine subjects (B01-B09) performed two-class cued motor imagery of the left and right hand (BCI Competition IV, 2008). Only 3 bipolar EEG channels over the sensorimotor cortex (C3, Cz, C4) and 3 EOG channels were recorded (0.5-100 Hz, notch at 50 Hz). Each subject underwent 5 sessions over two days: sessions 1-2 without feedback, sessions 3-5 with BCI feedback. Each session contained multiple runs of ~60-80 trials per class. A fixation cross preceded each trial, followed by a visual left/right arrow cue for 1.25 s, then a 4-second imagery period with optional BCI cursor feedback. Evaluation sessions (suffix `E.gdf`) have unknown cue labels that the kernel resolves from companion `.mat` files.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://www.bbci.de/competition/iv/desc_2b.html

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Start of a trial`	~1520/subject	admin	raw	Trial-onset fixation cross (pre-cue)
`Cue onset left (class 1)`	~380/subject	short	raw	Left-hand motor imagery cue
`Cue onset right (class 2)`	~380/subject	short	raw	Right-hand motor imagery cue
`BCI feedback (continuous)`	~760/subject	admin	raw	BCI feedback period (sessions 3-5 only)
`Idling EEG (eyes open)`	rare	admin	raw	Resting state with eyes open
`Idling EEG (eyes closed)`	rare	admin	raw	Resting state with eyes closed
`Rejected trial`	variable	admin	raw	Experimenter-rejected trial
`Start of a new run`	~40/subject	admin	raw	Run boundary marker
`Horizontal eye movement`	rare	admin	raw	Eye artifact marker

Event ID	Count (total)	Type	Meta Level	Meaning
`Vertical eye movement`	rare	admin	raw	Eye artifact marker
`Eye rotation`	rare	admin	raw	Eye artifact marker
`Eye blinks`	rare	admin	raw	Eye artifact marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Counts are approximate per subject (all sessions combined). GDF numeric codes remapped to human-readable strings by kernel.

Tasks

Primary Task 1: Binary Motor Imagery Classification - Left vs. Right Hand

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = [`Cue onset left (class 1)`, `Cue onset right (class 2)`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Cue onset left (class 1)`	0	Left-hand motor imagery
`Cue onset right (class 2)`	1	Right-hand motor imagery

- Evidence:
- DOI: 10.1038/s41598.025.96611-5 - Uses BCI IV-2b for binary motor imagery (left vs right), epoching 3-7s relative to cue onset (0-4s post-cue imagery period).
- DOI: 10.1109/tnsre.2022.3156076 - Uses BCI Competition IV dataset 2b with 720 trials from nine participants, segmented into overlapping 2-second windows for classification.
- DOI: 10.1109/tnsre.2023.3241846 - Uses Dataset 2b with 3 channels (C3, Cz, C4), epoching from 3rd to 7th second relative to cue onset (4s imagery period).
- DOI: 10.3389/fnins.2020.00918 - Trains CNN classifiers on BCI Competition IV-2b, using STFT windows within 3-7s post-trial onset (0-4s post-cue).
- DOI: 10.3390/s19010210 - Validates deep neural networks on Graz dataset 2b, using 4-second windows for motor imagery classification.
- DOI: 10.3390/s20164485 - Uses BCI competition IV dataset 2b with 4s imagery period filtered at 8-30 Hz for left vs right classification.
- Kernel Notes: Standard MI epoch captures baseline (-0.5 to 0s) and imagery period (0-4s post-cue). Sessions 1-2 (no feedback) vs 3-5 (with feedback) may exhibit learning effects; some papers train on sessions 1-3 and test on 4-5. EOG channels should be excluded; CSP spatial filtering recommended given only 3 EEG channels (C3, Cz, C4). (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Only 3 EEG channels (C3, Cz, C4): very limited spatial resolution; spatial filtering (CSP) is critical.
- EOG channels present; apply artifact regression before MI classification.
- Feedback sessions (3-5) may show learning effects; sessions 1-2 provide cleaner baseline.
- Evaluation files use `Cue unknown (class 783)` initially; companion `.mat` files are required by the kernel to resolve labels.

- Subject demographics: B01-B09, ages 20-32, mixed sex.
- 46 files total across 9 subjects x 5 sessions; some sessions may have extra evaluation runs.

References

- Reference: Leeb R, Lee F, Keinrath C, Scherer R, Bischoff H, Pfurtscheller G (2008). BCI Competition 2008 - Graz Data Set B. Institute for Knowledge Discovery, Graz University of Technology. https://www.bbci.de/competition/iv/desc_2b.html

163. bcic_iv_4 - BCI Competition IV Dataset 4 (ECoG Finger Movement Decoding)

Source: analysis_minimax_v2.md

Dataset: `bcic_iv_4` | Year: 2008 | URL: https://www.bbci.de/competition/iv/desc_4.html
Category: Action/Output | Subjects: 3 | Files (Completed): 3 (Case B kernel) | Channels: 48-64 (variable per subject; ECoG) | Sampling Rate: 1000 Hz

Paradigm

Three subjects (sub1-sub3) performed individual finger flexions under instruction while ECoG (electrocorticography) was recorded from 48-64 electrodes implanted over sensorimotor cortex (BCI Competition IV, 2008). Subjects flexed each of the five fingers individually on cue. Continuous finger flexion amplitude was measured via a data glove and stored as 5 labels per time sample. The dataset contains a training segment (~400 s per subject) with labeled finger flexion signals and a test segment (~200 s) with labels withheld. The Case B kernel concatenates train and test segments into a single continuous raw signal per subject; finger flexion signals (5 channels) are stored as `misc`-type stim channels (`finger1-finger5`).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://www.bbci.de/competition/iv/desc_4.html

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`train_segment`	1/subject	admin	raw	Annotates the entire training segment (with labeled finger flexions)
`test_segment_onset`	1/subject	admin	raw	Marks the onset of the unlabeled test segment

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Finger flexion labels are not stored as event annotations but as 5 continuous `misc` stim channels (`finger1-finger5`). This is a regression/continuous decoding task.

MISC Label Summary

- Route: `misc`
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification

- Kernel Note: Document derivation logic, source fields, units, and version tags.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Continuous Finger Flexion Trajectory Decoding

- Task Type: regression
- Epoch Method: `epoch_by_long_event_hdf5` (sliding window on continuous segment)
- Epoch Parameters: tmin = 0.0 s, tmax = duration of segment, overlap = 0.5 s *, target_events = [`train_segment`]
- Label Route: `misc` (finger1-finger5 channels)
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`finger1` misc channel	continuous [0-1]	Thumb flexion amplitude
`finger2` misc channel	continuous [0-1]	Index flexion amplitude
`finger3` misc channel	continuous [0-1]	Middle flexion amplitude
`finger4` misc channel	continuous [0-1]	Ring flexion amplitude
`finger5` misc channel	continuous [0-1]	Little flexion amplitude

- Super Meta Notes: Labels are continuous finger flexion amplitudes recorded via data glove (25Hz, upsampled to 1000Hz), stored as misc stim channels in the original dataset. Kernel must extract these 5 channels as regression targets.
- Evidence:
- DOI: 10.1109/ieeeconf58974.2023.10405112 - FingerFlex convolutional encoder-decoder achieves correlation up to 0.74 on continuous finger trajectory prediction
- DOI: 10.1109/tim.2025.3644562 - DeepFingerNet achieves average correlation 0.69 using 2.56s windows with 0.01s overlap
- DOI: 10.1109/i2ct61223.2024.10543868 - Random Forest classifier achieves 80% accuracy on binary left/right hand classification
- Kernel Notes:
- Extract EEG features from ECoG channels only (exclude finger1-finger5 misc channels)
- Use sliding windows of 2-4s duration over train_segment; predict all 5 finger trajectories per window
- Apply temporal alignment offset (~0.2s) to account for neural-to-movement delay
- Evaluate using Pearson correlation coefficient between predicted and actual finger trajectories
- Test segment has withheld labels; use train_segment only for supervised learning

Primary Task 2: Binary Hand Classification (Left vs Right)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5` with threshold-derived event labels
- Epoch Parameters: tmin = 0.0 s, tmax = duration of segment, overlap = 0.5 s *, target_events = [`train_segment`]
- Label Route: `Annotation` (derived) | `misc` (source)
- Meta Level: super

- Target Definition:
Event	y	Meaning
Derived from finger1-finger5	0	Left hand movement (fingers on left hand)
Derived from finger1-finger5	1	Right hand movement (fingers on right hand)
- Super Meta Notes:
- Derived from raw misc channels (`finger1-finger5`) by thresholding amplitude to detect active movement periods, then classifying as left-hand (thumb+index+middle) vs right-hand (ring+little) based on finger groupings
- Kernel step: Apply amplitude threshold (>0.1 *) to identify movement epochs; assign label based on which finger group shows higher activity
- Evidence:
- DOI: 10.1109/i2ct61223.2024.10543868 - Binary left-hand vs right-hand classification achieves 80% accuracy with Random Forest
- Kernel Notes:
- Threshold finger channel amplitudes to detect movement onset
- Map fingers 1-3 to left hand, fingers 4-5 to right hand (or verify mapping from original data glove configuration)
- Epoch detection should identify discrete movement trials within the continuous segment

Normal Task 1: Finger Flexion Decoding - Training Segment (epoch_by_long_event)

- Task Type: regression
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = duration of `train_segment`, overlap = 0.5 s (sliding window), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: N/A (target labels are continuous; see notes below)
- Notes: This is a regression task, not a standard classification task. The `train_segment` annotation spans the entire training portion (~400 s); use `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5` to extract fixed-length windows. Within each epoch, extract the corresponding `finger1-finger5` misc-channel values as regression targets. The `test_segment_onset` annotation marks where labels become unavailable. Finger channels must be excluded from the EEG feature matrix but retained as label sources.

Normal Task 2: Active vs. Rest Finger Movement Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: Requires custom thresholding of finger channel amplitudes to define discrete movement epochs; not directly supported by standard `epoch_by_event_hdf5`.
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `misc`
- Target Definition: Needs verification.
- Notes: Threshold the `finger1-finger5` misc channels to identify movement onset events, then apply `epoch_by_event_hdf5`. Active finger movement (amplitude > threshold) vs. rest (amplitude <= threshold) yields a binary classification problem per finger.

Data Quality Notes

- ECoG data, not scalp EEG: electrode locations are patient-specific implants over sensorimotor cortex; no standard montage applies.
- Channel count varies per subject: sub1 = 62, sub2 = 64, sub3 = 48 ECoG channels.

- Each file additionally contains 5 `misc` channels (`finger1-finger5`) storing continuous flexion amplitudes; exclude these from EEG analysis but use them as regression labels.
- No age or sex information available.
- High sampling rate (1000 Hz); consider downsampling to <=250 Hz for classification tasks.
- Test-segment labels are withheld; only the `train_segment` portion is usable for supervised learning.

References

- Reference: BCI Competition IV (2008), Dataset 4 - Finger Movements in ECoG. https://www.bbci.de/competition/iv/desc_4.html

164. figshare_largemi - A Large EEG Motor Imagery Dataset for EEG-BCI

Source: analysis_minimax_v2.md

Dataset: `figshare_largemi` | Year: 2018 | URL: https://figshare.com/collections/A_large_electroencephalographic_motor_imagery_dataset_for_electroencephalographic_brain_computer_interfaces/3917698
Category: Action/Output | Subjects: 13 | Files (Completed): 73 | Channels: 22 | Sampling Rate: 200 or 1000 Hz (protocol-dependent)

Paradigm

Thirteen subjects (A-M) participated in multiple EEG motor imagery sessions using five different paradigms (Kaya et al., 2018). EEG was recorded from 22 scalp channels. The paradigms are:
- CLA: 3-class MI - left hand, right hand, passive state.
- HaLT: 6-class MI - left hand, right hand, passive state, left leg, right leg, tongue.
- FREEFORM / FREEDOM: 2-class MI - left hand, right hand (free imagery style).
- 5F: 5-class finger MI - thumb, index, middle, ring, and pinkie.
- NoMT: no-motor-task control sessions (resting state, no imagery).

Trials were cue-based with a blank screen rest period, followed by an arrow/cue presentation indicating the imagery target. Each trial lasted approximately 3 seconds of imagery. NoMT sessions serve as control recordings without motor task events.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://figshare.com/collections/A_large_electroencephalographic_motor_imagery_dataset_for_electroencephalographic_brain_computer_interfaces/3917698

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Left hand MI`	varies by protocol	short	raw	Left-hand motor imagery cue
`Right hand MI`	varies by protocol	short	raw	Right-hand motor imagery cue
`Passive state`	varies by protocol	short	raw	Passive / idle state (CLA/HaLT)
`Left leg MI`	varies by protocol	short	raw	Left-leg motor imagery cue (HaLT)
`Right leg MI`	varies by protocol	short	raw	Right-leg motor imagery cue (HaLT)
`Tongue MI`	varies by protocol	short	raw	Tongue motor imagery cue (HaLT)

Event ID	Count (total)	Type	Meta Level	Meaning
`Thumb MI`	varies by protocol	short	raw	Thumb finger imagery cue (5F)
`Index finger MI`	varies by protocol	short	raw	Index finger imagery cue (5F)
`Middle finger MI`	varies by protocol	short	raw	Middle finger imagery cue (5F)
`Ring finger MI`	varies by protocol	short	raw	Ring finger imagery cue (5F)
`Pinkie MI`	varies by protocol	short	raw	Little finger imagery cue (5F)
`none`	all NoMT files	admin	raw	No motor task (NoMT protocol control)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Multi-class Motor Imagery (CLA paradigm)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s *, tmax = 0.85 s, target_events = [`Left hand MI`, `Right hand MI`, `Passive state`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Left hand MI`	0	Left-hand motor imagery cue
`Right hand MI`	1	Right-hand motor imagery cue
`Passive state`	2	Passive/idle state (no imagery)

- Evidence:
- DOI: 10.1038/sdata.2018.211 - Original dataset paper validates SVM classifier on 0.85s epoch windows from stimulus onset for 3-class motor imagery discrimination (left hand, right hand, passive).
- Kernel Notes: Filter files by protocol type `CLA` (check file path). Parameters tmin/tmax from original validation; tmin marked * as not explicitly stated but standard is stimulus onset (0.0s). (replaces Normal Task 1 from original analysis.md for CLA subset)

Primary Task 2: 6-Class Motor Imagery (HaLT paradigm)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`

- Epoch Parameters: tmin = 0.0 s *, tmax = 0.85 s, target_events = [`Left hand MI`, `Right hand MI`, `Passive state`, `Left leg MI`, `Right leg MI`, `Tongue MI`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Left hand MI`	0	Left-hand motor imagery
`Right hand MI`	1	Right-hand motor imagery
`Passive state`	2	Passive/idle state
`Left leg MI`	3	Left-leg motor imagery
`Right leg MI`	4	Right-leg motor imagery
`Tongue MI`	5	Tongue motor imagery

- Evidence:
- DOI: 10.1038/sdata.2018.211 - Dataset descriptor validates 6-class MI discrimination (left/right hand, legs, tongue, passive) using 0.85s stimulus-locked epochs.
- Kernel Notes: Filter files by protocol type `HaLT` (check file path). Marker codes: 1=left hand, 2=right hand, 3=passive, 4=left leg, 5=tongue, 6=right leg. (replaces Normal Task 2 from original analysis.md)

Primary Task 3: 5-Finger Motor Imagery (5F paradigm)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s *, tmax = 0.85 s, target_events = [`Thumb MI`, `Index finger MI`, `Middle finger MI`, `Ring finger MI`, `Pinkie MI`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Thumb MI`	0	Thumb finger motor imagery
`Index finger MI`	1	Index finger motor imagery
`Middle finger MI`	2	Middle finger motor imagery
`Ring finger MI`	3	Ring finger motor imagery
`Pinkie MI`	4	Little finger motor imagery

- Evidence:
- DOI: 10.1038/sdata.2018.211 - Validates 5-finger MI classification using marker codes 1-5 (thumb, index, middle, ring, pinkie) on 0.85s epochs.
- Kernel Notes: Filter files by protocol type `5F` (check file path). Sampling rate for 5F protocol is 200Hz (170 samples = 0.85s). (replaces Normal Task 3 from original analysis.md)

Primary Task 4: Binary Motor Imagery (FREEFORM paradigm)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 3.0 s, target_events = [`Left hand MI`, `Right hand MI`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Left hand MI`	0	Left-hand motor imagery

| `Right hand MI` | 1 | Right-hand motor imagery |
- Evidence:
- DOI: 10.1109/tnsre.2022.3186442 - Uses figshare_largemi dataset for inter-subject binary MI classification (left vs right hand) with 3-second epoch window extracted after stimulus onset.
- Kernel Notes: Filter files by protocol type `FREEFORM` or `FREEDOM` (check file path). Alternative: can also use CLA protocol's left/right classes for binary classification. (replaces Normal Task 1 from original analysis.md for FREEFORM subset)

Normal Task 4: Resting-State Segmentation - NoMT Control (epoch_by_segmentation)

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: length = 3.0 s, overlap = 0.5 s
- Label Route: `Annotation`
- Target Definition: N/A (no events; use as unlabeled or negative class)
- Notes: NoMT files have event annotation `none` set by kernel; use `epoch_by_segmentation_hdf5` for these files.

Data Quality Notes

- Multiple paradigms in one dataset; always filter by protocol type before analysis (use file path to identify protocol: `CLA`, `HaLT`, `FREEFORM`, `5F`, `NoMT`).
- Sampling rates differ: 200 Hz (5F protocol), 1000 Hz (CLA/HaLT/FREEFORM); process separately or resample to a common rate.
- Subject demographics: 13 subjects (A-M), 9 female / 4 male, age ranges 20-35.
- NoMT sessions have `none` annotations; exclude from MI analysis or use as rest class.

References

- Reference: Kaya M, Binli MK, Ozbay E, Yanar H, Mishchenko Y (2018). A large electroencephalographic motor imagery dataset for electroencephalographic brain computer interfaces. Scientific Data, 5, 180211. https://doi.org/10.1038/sdata.2018.211

165. figshare_meng2019 - EEG Dataset for Cursor BCI (Meng 2019)

Source: analysis_minimax_v2.md

Dataset: `figshare_meng2019` | Year: 2019 | URL: https://dx.doi.org/10.6084/m9.figshare.7959572
Category: Action/Output | Subjects: N/A (see notes) | Files (Completed): 126 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

Participants performed a one-dimensional cursor BCI task using mental imagery in three experiments (Meng et al., 2019). EEG was recorded from 64 channels. Subjects imagined motor or other mental activities to move a cursor toward a target positioned above ("+") or below ("-") center, or at rest (center target). The dataset covers three experiments (Exp1, Exp2, Exp3); Exp1 and Exp2 used a Neuroscan SynAmps RT amplifier, while Exp3 used a Biosemi Active Two system. The paradigm recorded both target intent events (what the subject was trying to do) and result events (what the BCI decoded), enabling analysis of mental imagery decoding accuracy.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://dx.doi.org/10.6084/m9.figshare.7959572

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`target cursor plus`	varies	short	raw	Target positioned above center; subject imagines upward/+ movement
`target cursor minus`	varies	short	raw	Target positioned below center; subject imagines downward/- movement
`target rest`	varies	short	raw	Target at center; subject imagines neutral/rest state
`result cursor plus`	varies	admin	raw	BCI decoded + direction (feedback signal)
`result cursor minus`	varies	admin	raw	BCI decoded - direction (feedback signal)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> `result cursor *` events encode BCI feedback and are not ground-truth labels; use only `target cursor *` events for supervised classification.

Tasks

Primary Task 1: Motor imagery cursor direction decoding (left-hand vs right-hand)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5"
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, target_events = [`target cursor plus`, `target cursor minus`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`target cursor plus`	0	Left-hand motor imagery (cursor moves up/+ toward left target)
`target cursor minus`	1	Right-hand motor imagery (cursor moves down/- toward right target)

- Evidence:
- DOI: 10.3389/fnhum.2019.00128 - This study analyzes a pooled dataset of 42 subjects across three BCI training sessions using motor imagery (left-hand vs right-hand) to control a cursor toward left or right targets, with behavioral performance and ERD/ERS metrics evaluated across sessions.
- Kernel Notes: Use only `target cursor *` events as ground-truth labels. Exclude `result cursor *` events as these encode BCI feedback/output, not the true mental imagery label. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Three-class cursor BCI classification (including rest)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, target_events = [`target cursor plus`, `target cursor minus`, `target rest`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`target cursor plus`	0	Left-hand motor imagery (cursor moves toward + target)
`target cursor minus`	1	Right-hand motor imagery (cursor moves toward - target)
`target rest`	2	Rest/neutral mental state (cursor at center)

- Evidence:
- DOI: 10.3389/fnhum.2019.00128 - This paper analyzes the same figshare dataset from 42 subjects performing left/right hand motor imagery to control a cursor BCI across three training sessions, evaluating behavioral performance, R2 values, and ERD/ERS patterns.
- Kernel Notes: (replaces Normal Task 2 from original analysis.md)

Data Quality Notes

- Sex information is not available in the dataset metadata.
- Three experiments (Exp1, Exp2, Exp3) use different amplifier systems; process per-experiment or normalize across systems.

- `result cursor *` events are BCI feedback signals, not labels; do not use for classification targets.
- Subject count is not explicitly stated; estimate from unique subject folders in file paths.

References

- Reference: Meng J et al. (2019). EEG dataset for cursor BCI (Figshare). https://dx.doi.org/10.6084/m9.figshare.7959572

166. figshare_shudb - SHU Motor Imagery EEG Dataset

Source: analysis_minimax_v2.md

Dataset: `figshare_shudb` | Year: 2022 | URL: https://figshare.com/articles/software/shu_dataset/19228725/1
Category: Action/Output | Subjects: 21 | Files (Completed): 232 | Channels: 32 | Sampling Rate: 250 Hz

Paradigm

Twenty-one subjects (sub-001 to sub-021) performed cued left/right hand motor imagery across 4-5 sessions each (SHU dataset). EEG was recorded from 32 channels using a BrainAmp MR plus amplifier with Ag/AgCl cap (standard 10-20 system). Each session contained approximately 44-50 trials per class (left hand MI and right hand MI). Events are stored in TSV files following a BIDS-like structure and loaded by the kernel from adjacent `events/` directories. The dataset is well-suited for studying longitudinal motor imagery performance and cross-session transfer.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://figshare.com/articles/software/shu_dataset/19228725/1

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`left`	~10900 (total)	short	raw	Left-hand motor imagery cue onset
`right`	~10900 (total)	short	raw	Right-hand motor imagery cue onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Total counts estimated as 232 files x ~47 trials per class. Exact per-session counts vary slightly (44-50 per class).

Tasks

Primary Task 1: Binary Motor Imagery Classification - Left vs Right Hand

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 4.0 s, target_events = [`left`, `right`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning

| `left` | 0 | Left-hand motor imagery |
| `right` | 1 | Right-hand motor imagery |
- Evidence:
- DOI: 10.1038/s41597-022-01647-1 - Presents SHU Motor Imagery Dataset with 25 subjects performing 5 sessions of left/right hand MI, using 4-second EEG epochs for binary classification between left (label 1) and right (label 2).
- DOI: 10.1016/j.heliyon.2024.e37343 - Validates RST method for multi-session left vs right hand MI classification using CNN-based feature extraction.
- DOI: 10.1109/jbhi.2024.3467090 - Evaluates MFRC-Net on SHU dataset for binary motor imagery classification of left vs right hand.
- DOI: 10.1016/j.jneumeth.2025.110483 - Validates models on SHU dataset achieving 87.2% within-session accuracy for left/right hand MI classification.
- Kernel Notes: Extract 4s epochs from cue onset (tmin=0s to tmax=4.0s). Apply bandpass filtering (0.5-40 Hz). Use sliding window augmentation (3s window, 0.5s shift) as in paper_10_1016_j_heliyon_2024_e37343. Map `left` -> 0 and `right` -> 1 for binary classification. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Age and sex information are not available in this dataset.
- 4-5 sessions per subject; slight trial count variation per session (44-50 per class) may introduce minor class imbalance within a session.
- BrainAmp MR plus amplifier with Ag/AgCl cap; standard 32-channel 10-20 montage.
- BIDS-like event structure; kernel reads TSV files from the `events/` subdirectory.

References

- Reference: SHU Motor Imagery Dataset (Figshare, 2022).
https://figshare.com/articles/software/shu_dataset/19228725/1

167. figshare_stroke - EEG Datasets of Stroke Patients (Motor Imagery)

Source: analysis_minimax_v2.md

Dataset: `figshare_stroke` | Year: 2022 | URL: https://figshare.com/articles/dataset/EEG_datasets_of_stroke_patients/21679035/5
Category: Action/Output | Subjects: 50 | Files (Completed): 50 | Channels: 33 | Sampling Rate: 500 Hz

Paradigm

Fifty stroke patients (sub-01 to sub-50) performed a cued left/right hand motor imagery task (ZhenTec NT1 amplifier, Ag/AgCl cap). EEG was recorded from 31 scalp channels plus 2 horizontal EOG channels (HEOL, HEOR). Each recording session contained 20 trials per hand class (20 left + 20 right = 40 trials total). Each trial comprised three phases: an instruction period (visual cue indicating the target hand), a motor imagery period (subject imagines the indicated hand movement), and a break period (rest). The dataset is designed to study MI-based BCI performance in stroke rehabilitation. Subject ages range from 31 to 77 years (median ~55), with predominantly right-handed subjects (49/50) and a male-majority cohort (~36M/14F).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://figshare.com/articles/dataset/EEG_datasets_of_stroke_patients/21679035/5

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Instruction beginning - Left hand`	1000	admin	raw	Instruction cue onset for left-hand trial
`Instruction beginning - Right hand`	1000	admin	raw	Instruction cue onset for right-hand trial
`Motor imagery beginning - Left hand`	1000	short	raw	Left-hand motor imagery period onset
`Motor imagery beginning - Right hand`	1000	short	raw	Right-hand motor imagery period onset
`Break beginning - Left hand`	1000	admin	raw	Rest/break period following a left-hand trial
`Break beginning - Right hand`	1000	admin	raw	Rest/break period following a right-hand trial

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Counts are totals across all 50 subjects (20 per class per recording x 50 subjects).

Tasks

Primary Task 1: Left vs. Right Hand Motor Imagery Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 1.0 s, tmax = 4.0 s, target_events = [`Motor imagery beginning - Left hand`, `Motor imagery beginning - Right hand`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Motor imagery beginning - Left hand`	0	Left-hand motor imagery
`Motor imagery beginning - Right hand`	1	Right-hand motor imagery

- Evidence:
- DOI: 10.1016/j.compbiomed.2025.111285 - Uses figshare stroke dataset (50 subjects) for left/right hand MI classification, segmenting EEG trials from 1 to 4 seconds post-cue with multi-domain features (EMD, HHT, CSP, Riemannian) and SCSO-optimized BLS classifier achieving 96.41% accuracy.
- Kernel Notes: Extract epochs from MI onset events (1.0-4.0 s window). Exclude EOG channels (HEOL, HEOR) before classification. Apply artifact rejection for stroke patient data quality. (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Stroke Patient Subgroup Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 4.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Same as Task 1 (MI events); group labels (age, lesion side, etc.) from `participants.tsv`.
- Notes: Clinical metadata (age, sex, stroke side, handedness) is embedded in the kernel's `_SUBJECT_DICT`. Cross-subject analysis with clinical covariates is the primary use case.

Data Quality Notes

- Stroke patient population: highly variable EEG quality due to neurological deficits and movement impairments.
- Very small per-subject trial count (20 per class); cross-subject aggregation is essential.
- EOG channels present (HEOL, HEOR); apply artifact rejection or EOG regression before MI classification.
- One left-handed subject (sub-06); consider excluding or annotating for lateralization analyses.
- Age range 31-77 years (mean ~55); age-related EEG changes may confound MI classification.

References

- Reference: EEG datasets of stroke patients for motor imagery BCI (Figshare, 2022). https://figshare.com/articles/dataset/EEG_datasets_of_stroke_patients/21679035/5

168. kaggle_inria - Kaggle BCI Challenge (INRIA) Dataset

Source: analysis.md

Dataset: `kaggle_inria` | Year: 2015 | URL: https://www.kaggle.com/c/inria-bci-challenge
Category: Action/Output | Subjects: 26 | Files (Completed): 130 | Channels: 58 | Sampling Rate: 200 Hz

Paradigm

Twenty-six subjects (ages 20-37) performed a P300-based BCI feedback task using a 56-channel Ag/AgCl EEG cap at 200 Hz (plus 1 EOG channel and 1 FeedBackEvent channel, total 58). The task was a P300 matrix speller in which subjects attended to a target character while row/column flashes elicited P300 responses. Feedback was provided after each trial: subjects received either a correct ("good") or incorrect ("wrong") response to the P300 classifier output. Each session file (CSV format) contains one session for one subject; each subject has 5 sessions. Feedback event times are encoded in the dedicated `FeedBackEvent` channel; labels ("good" or "wrong") are sourced from a companion `TrainLabels.csv` file. Test sessions have unknown labels.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://www.kaggle.com/c/inria-bci-challenge

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`good feedback to P300`	varies (~1300)	short	raw	Feedback event: BCI output was correct
`wrong feedback to P300`	varies (~650)	short	raw	Feedback event: BCI output was incorrect
`unknown feedback to P300`	varies (~1560)	short	raw	Feedback event: label unknown (test sessions)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Feedback events are 1 s in duration (as set in kernel annotations). Test sessions (Sess01-Sess04 for most subjects) have `unknown` labels; labeled sessions are typically Sess05. Approximately 60 feedback events per session.

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Target vs non-target stimulus events
- Evidence: The reference paper evaluates online error correction in a P300-based spelling task using this challenge setting. Source: https://www.kaggle.com/c/inria-bci-challenge DOI: https://doi.org/10.1155/2012/578295

Normal Task 1: P300 Feedback Correctness - Good vs. Wrong (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.9 s (1 s feedback window), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Exclude files with `unknown feedback to P300` events only (test sessions without labels). Kernel requires `TrainLabels.csv` to be present at the dataset root directory. FeedBackEvent channel (column index 57) encodes event onset times as a binary channel; labels are matched by feedback ID (e.g., `S01_Sess01_FB001`).

Event	y	Meaning
`good feedback to P300`	0	Correct BCI feedback
`wrong feedback to P300`	1	Incorrect BCI feedback
## Data Quality Notes		

- Data stored as CSV files (not standard EEG format); MNE loader produces unsupported-file-type warnings that are non-fatal.
- 58 channels include 56 EEG + 1 EOG + 1 FeedBackEvent auxiliary channel; remove FeedBackEvent and EOG before training.
- Many sessions have only unknown labels (test set); only sessions with known labels in TrainLabels.csv can be used for supervised learning.
- Age range 20-37 available from dataset metadata; no sex information.

References

- Reference: Inria BCI Challenge. Kaggle Competition 2015. https://www.kaggle.com/c/inria-bci-challenge
- Reference: Margaux P, Maby E, Daligault S, Bertrand O, Mattout J. (2012). Objective and Subjective Evaluation of Online Error Correction during P300-Based Spelling. Advances in Human-Computer Interaction. DOI: https://doi.org/10.1155/2012/578295

169. openneuro_ds002034 - Real-time EEG feedback on alpha power lateralization leads to behavioral improvements in a covert attention task

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002034` | Year: 2022 | URL: https://openneuro.org/datasets/ds002034
Category: Action/Output | Subjects: 14 | Files (Completed): 167 | Channels: 82.0 | Sampling Rate: 512.0 Hz

Paradigm

Participants performed a covert spatial attention task where they were trained to lateralize their alpha-band EEG power using real-time neurofeedback. The study evaluated whether closed-loop neurofeedback of alpha lateralization improves behavioral performance in a spatial attention paradigm. No trial-level event markers are stored in the BIDS EEG files; the task condition is reflected continuously in the EEG signal itself.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002034

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(none)	-	-	raw	No event markers detected in EEG files; continuous neurofeedback paradigm

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Covert Visuospatial Attention Classification (Left vs Right Hemifield)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 1.0 s, overlap = 0.9375 (62.5ms step)
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

| Event | y | Meaning |

|-------|---|---------|
| `left_attention` | 0 | Subject instructed to attend to left hemifield (cue: left half-circle) |
| `right_attention` | 1 | Subject instructed to attend to right hemifield (cue: right half-circle) |
- Super Meta Notes: Derived from experimental paradigm metadata (cue timing and direction). No raw event markers present in BIDS files. Kernel must derive trial epochs from available task metadata or use sliding-window approach on continuous data, then compute alpha-band lateralization index (alpha power right - left) / (alpha power right + left) as feature for binary classification of attended hemifield.
- Evidence:
- DOI: 10.1007_s10548-019-00725-9 - Real-time EEG neurofeedback study on alpha-band lateralization for covert visuospatial attention; post-hoc single-trial classification decodes attended hemifield (left vs right) using alpha-LI features.
- Kernel Notes:
  1. Verify presence of trial timing metadata in dataset sidecars or task description.
  2. If trial timing available: extract epochs around cue onset using `epoch_by_event_hdf5` (tmin=0s, tmax=variable per trial).
  3. If no trial markers: use sliding segmentation (segment_length=1.0s, overlap=0.9375).
  4. Compute alpha-band (8-13 Hz) power in left and right hemisphere channels; compute Lateralization Index = (right - left) / (right + left).
  5. Train binary classifier (LDA as per paper) to decode left vs right attention.
  (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Publisher-defined BIDS task: Task offline

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds002034 DOI: https://doi.org/10.1007/s10548-019-00725-9

Primary Task 3: Publisher-defined BIDS task: Task offlinecatch

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds002034 DOI: https://doi.org/10.1007_s10548-019-00725-9

Primary Task 4: Publisher-defined BIDS task: Task onlinesham

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)

- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds002034 DOI: https://doi.org/10.1007/s10548-019-00725-9

Data Quality Notes

- No trial-level event markers detected in any EEG files; only continuous recordings available.
- Real-time neurofeedback paradigm - EEG signal itself contains the task-relevant information.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Real-time EEG Feedback on Alpha Power Lateralization Leads to Behavioral Improvements in a Covert Attention Task`

170. openneuro_ds002336 - A multi-modal human neuroimaging dataset for data integration: simultaneous EEG and fMRI acquisition during a motor imagery neurofeedback task: XP1

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002336` | Year: 2021 | URL: https://openneuro.org/datasets/ds002336
Category: Action/Output | Subjects: 10 | Files (Completed): 107
Channels: 64 | Sampling Rate: 5000 Hz

Paradigm

Simultaneous EEG-fMRI acquisition during a kinesthetic motor imagery neurofeedback task. Subjects performed right-hand motor imagery or motor execution during sustained task blocks alternating with rest blocks, while receiving unimodal (EEG or fMRI) or bimodal neurofeedback. Each task block (Task-NF: neurofeedback, Task-ME: motor execution, Task-MI: motor imagery) is a sustained mental state lasting several seconds. BrainVision format with MRI artifact markers (TPULSE, R128) that must be excluded.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002336

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Rest`	93 (files)	long	raw	Rest / baseline block
`Task-NF`	60 (files)	long	raw	Neurofeedback task block
`Task-ME`	19 (files)	long	raw	Motor execution task block
`Task-MI`	14 (files)	long	raw	Motor imagery task block
`Response/R128`	14	admin	raw	BrainVision response trigger / MRI sync - exclude
`Stimulus/S 99`	14	admin	raw	BrainVision stimulus code 99 - likely admin/sync
`Stimulus/S 2`	14	admin	raw	BrainVision stimulus code 2 - likely admin/sync

Event ID	Count (total)	Type	Meta Level	Meaning
`Pulse Artifact/R`	7	admin	raw	MRI gradient pulse artifact marker - exclude
`Time 0/`	7	admin	raw	MRI timing reference marker - exclude
`TPULSE/TSTART`	7	admin	raw	MRI T-pulse start - exclude
`TPULSE/TPEAK`	7	admin	raw	MRI T-pulse peak - exclude
`TPULSE/TEND`	7	admin	raw	MRI T-pulse end - exclude

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/S 99`, `Stimulus/S 2`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Four-class motor imagery/neurofeedback state classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 20.0 s, overlap = 0.0, target_events = [`Rest`, `Task-NF`, `Task-ME`, `Task-MI`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Rest`	0	Rest / baseline block (20s)
`Task-NF`	1	Neurofeedback task block
`Task-ME`	2	Motor execution block
`Task-MI`	3	Motor imagery block

- Evidence:
- DOI: 10.1038/s41597-020-0498-3 - Dataset paper confirming 20s rest/20s task block design with markers S 99 (rest) and S 2 (task), label mapping Rest/Task-ME/Task-MI/Task-NF
- DOI: 10.1101/862375 - Describes dataset XP1 with block design and S 99/S 2 markers
- Kernel Notes: High sampling rate (5000 Hz) requires downsampling to ~200 Hz. MRI artifact markers (TPULSE/*, Pulse Artifact/R) must be excluded from epochs. Each annotation represents a 20s sustained block; use full 20s window with no overlap to capture block-level features. (replaces existing Primary Tasks 1-3)

Primary Task 2: Binary rest vs task classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 20.0 s, overlap = 0.0, target_events = [`Rest`, `Task-MI`, `Task-ME`, `Task-NF`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Rest`	0	Rest / baseline block
`Task-MI`	1	Any motor imagery task
`Task-ME`	1	Any motor execution task
`Task-NF`	1	Any neurofeedback task

- Super Meta Notes: Derived from raw events [`Rest`, `Task-MI`, `Task-ME`, `Task-NF`] by collapsing Task-MI, Task-ME, and Task-NF into a single positive class (1) vs Rest (0). Kernel step: Map Rest->0, and all task types (Task-MI, Task-ME, Task-NF)->1.
- Evidence:
- DOI: 10.3389/fnhum.2017.00193 - Describes binary classification of rest vs motor imagery (20s blocks alternating), confirming right-hand kinesthetic motor imagery paradigm
- Kernel Notes: Same preprocessing as Primary Task 1; binary collapse can be done in kernel or as super meta derivation.

Normal Task 1: Motor Imagery / Neurofeedback State Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 20.0 s, overlap = 0.0, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: MRI artifact markers (TPULSE/*, Pulse Artifact/R, Time 0/, R128) are all admin markers - excluded. High sampling rate (5000 Hz); downsample before epoching. Task-ME and Task-MI are sparse (14-19 files).

Event	y	Meaning
`Rest`	0	Rest / baseline block
`Task-NF`	1	Neurofeedback task block
`Task-ME`	2	Motor execution block
`Task-MI`	3	Motor imagery block

Data Quality Notes

- MRI gradient artifact contamination present; TPULSE markers indicate artifact regions.
- High sampling rate (5000 Hz); downsampling to 250-1000 Hz recommended.
- Task-ME and Task-MI blocks are sparsely present (subset of sessions only).
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Lioi, G., et al. (2019). Simultaneous MRI-EEG during a motor imagery neurofeedback task: an open access brain imaging dataset for multi-modal data integration. BioRxiv.`

171. openneuro_ds002338 - A multi-modal human neuroimaging dataset for data integration: simultaneous EEG and fMRI acquisition during a motor imagery neurofeedback task: XP2

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002338` | Year: 2022 | URL: https://openneuro.org/datasets/ds002338
Category: Action/Output | Subjects: 16 | Files (Completed): 169 | Channels: 64.0 | Sampling Rate: 5000.0 Hz

Paradigm

Participants performed right-hand motor imagery and received neurofeedback (1D or 2D display) during simultaneous EEG-fMRI acquisition to learn to regulate sensorimotor rhythms. The task alternated between sustained Rest blocks and Task-NF (neurofeedback) blocks. MRI gradient artifacts are marked with TPULSE markers which must be excluded. This is XP2 of a two-experiment series; XP1 is ds002336.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002338

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Rest`	101 (files)	long	raw	Rest / baseline block
`Task-NF`	101 (files)	long	raw	Neurofeedback motor imagery task block
`Response/R128`	68	admin	raw	BrainVision response trigger / MRI sync - exclude
`Stimulus/S 99`	68	admin	raw	BrainVision admin code - exclude
`Stimulus/S 2`	68	admin	raw	BrainVision admin code - exclude
`Stimulus/S101`	39	admin	raw	BrainVision admin code - exclude
`Pulse Artifact/R`	34	admin	raw	MRI gradient pulse artifact - exclude
`Time 0/`	34	admin	raw	MRI timing reference - exclude

Event ID	Count (total)	Type	Meta Level	Meaning
TPULSE/TSTART	34	admin	raw	MRI T-pulse start - exclude
TPULSE/TPEAK	34	admin	raw	MRI T-pulse peak - exclude
TPULSE/TEND	34	admin	raw	MRI T-pulse end - exclude
Stimulus/S 35	2	admin	raw	Sparse BrainVision code - exclude

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Stimulus/S 99, Stimulus/S 2, Stimulus/S101, Stimulus/S 35
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Motor Imagery Neurofeedback State Classification (binary)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 20.0 s, overlap = 0.0, target_events = [Rest, Task-NF]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
Rest	0	Rest / baseline block
Task-NF	1	Motor imagery neurofeedback block (right-hand kinaesthetic MI)

- Evidence:
- DOI: 10.1038/s41597.020.0498-3 - Dataset paper confirms XP2 (ds002338) uses 20s alternating Rest and Task-NF blocks with binary classification of motor imagery vs rest
- DOI: 10.18112/openneuro.ds002338.v1.0.0 - Confirms 20s block design with EEG NF scores updated every 250ms
- Kernel Notes: Epoch window matches full 20s block duration (no overlap needed for block-level classification). Exclude all MRI-related markers (TPULSE/*, Pulse Artifact/R, Time 0/) during preprocessing. Downsample from 5000 Hz to 200 Hz before epoching per paper recommendation. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- MRI gradient artifacts contaminate EEG; TPULSE markers indicate artifact windows.
- High sampling rate (5000 Hz); downsampling recommended.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Simultaneous MRI-EEG during a motor imagery neurofeedback task: an open access brain imaging dataset for multi-modal data integration

172. openneuro_ds002720 - A dataset recorded during development of a tempo-based brain-computer music interface

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002720` | Year: 2020 | URL: https://openneuro.org/datasets/ds002720
Category: Action/Output | Subjects: 19 | Files (Completed): 165 | Channels: 19.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants used kinaesthetic motor imagery (imagining squeezing a ball) or relaxation to intentionally modulate the tempo of music in a continuous brain-computer music interface (BCMI) task. Event code 1 = motor_imagery (ball squeezing imagination) and code 2 = relaxation (rest), each lasting several seconds. The BCI decoded the EEG to adjust music tempo in real-time, creating a closed-loop feedback system.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002720

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`motor_imagery` (code 1)	143 (files)	long	raw	Sustained motor imagery - imagining squeezing a ball
`relaxation` (code 2)	143 (files)	long	raw	Sustained relaxation period

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Motor imagery vs relaxation binary classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 16 s, overlap = 0.0, target_events = [`motor_imagery`, `relaxation`]
- Label Route: `Annotation`
- Meta Level: raw

- Target Definition:
Event	y	Meaning
`motor_imagery`	0	Kinaesthetic motor imagery - imagining squeezing a ball
`relaxation`	1	Relaxation / rest period
- Evidence:
- DOI: 10.1038/s41597.020.0507-6 - The paper describes the 'BCMI tempo' dataset (openneuro_ds002720) recorded during development of a tempo-based BCMI where participants used kinaesthetic motor imagery to control music tempo, with EEG recorded from 19 channels during trials consisting of 4s fixation followed by 12s motor imagery tasks.
- Kernel Notes: Kernel extracts epochs centered on each motor_imagery and relaxation annotation. Epochs are 16 seconds long (matching 4s fixation + 12s task structure). For binary classification, labels are derived directly from event type: motor_imagery=0, relaxation=1. Verify that overlapping epochs are handled correctly if both event types co-occur in the same recording. (replaces incomplete Primary Task 1 from original analysis.md)

Normal Task 1: Motor Imagery BCI Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s (motor imagery window), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Motor imagery BCI task with multi-second imagination periods. Long 4-second epoch window captures the full motor imagery event. Specific kernel decodes raw codes (1->motor_imagery, 2->relaxation).

Event	y	Meaning
`motor_imagery`	0	Motor imagery - imagining squeezing a ball
`relaxation`	1	Relaxation / rest period

Data Quality Notes

- 19-channel EEG is low density but suitable for BCI motor imagery decoding.
- Count column (143 files) indicates files-with-event, not trial count per file.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `A dataset recorded during development of a tempo-based brain-computer music interface`

173. openneuro_ds003810 - Motor Imagery vs Rest - Low-Cost EEG System

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003810` | Year: 2021 | URL: https://openneuro.org/datasets/ds003810
Category: Action/Output | Subjects: 10 | Files (Completed): 50
Channels: 15 | Sampling Rate: 125 Hz

Paradigm

Participants performed kinesthetic motor imagery of grasping movements with the dominant hand during EEG recording using a low-cost consumer-grade EEG system. Sessions included a preliminary run of actual hand movements followed by motor imagery blocks alternating with resting-state baselines. OpenViBE (OVTK) event triggers mark trial onset, cue type, feedback, and session boundaries. The study evaluates BCI performance using motor imagery and demonstrates applicability of affordable EEG hardware.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003810

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`OVTK_GDF_Right`	50	long	raw	Right-hand motor imagery cue onset
`OVTK_GDF_Tongue`	50	long	raw	Tongue motor imagery cue onset (second class)
`OVTK_GDF_Start_Of_Trial`	50	short	raw	Trial start marker
`OVTK_GDF_Cross_On_Screen`	50	short	raw	Fixation cross display (inter-trial)
`OVTK_GDF_Feedback_Continuous`	50	short	raw	Continuous BCI feedback onset
`OVTK_GDF_End_Of_Trial`	50	short	raw	Trial end marker
`OVTK_GDF_Correct`	50	short	raw	Correct classification feedback
`OVTK_GDF_Incorrect`	50	short	raw	Incorrect classification feedback
`OVTK_StimulationId_BaselineStart`	50	rest	raw	Resting baseline period start

Event ID	Count (total)	Type	Meta Level	Meaning
OVTK_StimulationId_BaselineStop	50	rest	raw	Resting baseline period end
OVTK_StimulationId_Beep	50	short	raw	Auditory beep cue for trial start
OVTK_StimulationId_Train	48	short	raw	Training phase marker
OVTK_StimulationId_ExperimentStart	46	admin	raw	Experiment start marker
OVTK_StimulationId_ExperimentStop	48	admin	raw	Experiment stop marker
OVTK_GDF_End_Of_Session	48	admin	raw	Session end marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: OVTK_StimulationId_BaselineStart, OVTK_StimulationId_BaselineStop, OVTK_StimulationId_Beep, OVTK_StimulationId_Train, OVTK_StimulationId_ExperimentStart, OVTK_StimulationId_ExperimentStop
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Motor Imagery vs Rest Binary Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = 0.5 s, tmax = 2.5 s, target_events = [OVTK_GDF_Right, OVTK_GDF_Tongue]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
OVTK_GDF_Right	1	Right-hand motor imagery (hand grasp kinesthetic imagination)
OVTK_GDF_Tongue	0	Tongue motor imagery (used as rest/control condition per original study design)

- Evidence:
- DOI: https://doi.org/10.1016/j.heliyon.2020.e03425 - Consumer-grade low-cost MI-BCI system classifying hand motor imagery vs rest with epochs extracted from 0.5-2.5s after stimulus onset using CSP and filter-bank methods
- DOI: 10.1016/j.dib.2022.108225 - Data in Brief paper describing motor imagery vs rest EEG with event markers OVTK_GDF_Right for motor imagery and OVTK_GDF_Tongue for rest conditions acquired during 4-second imagery periods

- Kernel Notes: Extract epochs from 0.5-2.5s after cue onset (2s window within 4s MI period). Label: OVTK_GDF_Right -> 1 (motor imagery), OVTK_GDF_Tongue -> 0 (rest/control). (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Low-cost EEG system; lower signal quality and spatial resolution than research-grade systems.
- Low sampling rate (125 Hz); limits temporal resolution and applicable frequency range.
- `OVTK_StimulationId_ExperimentStart/Stop` and `OVTK_GDF_End_Of_Session` are admin markers; exclude from epoching.

References

- Reference: `Motor Imagery vs Rest using a low-cost EEG system. OpenNeuro ds003810.`

174. openneuro_ds004022 - Multimodal EEG and fNIRS Biosignal Acquisition during Motor Imagery Tasks in Patients with Orthopedic Impairment

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004022` | Year: 2022 | URL: https://openneuro.org/datasets/ds004022
Category: Action/Output | Subjects: 7 | Files (Completed): 21
Channels: 18 | Sampling Rate: 500 Hz

Paradigm

Participants with orthopedic impairment performed four distinct motor imagery tasks (Reach, Twist, Lift, and Grasp) guided by visual cues across three experimental sessions. EEG and fNIRS were recorded simultaneously to study motor imagery neural patterns in a clinical population for BCI development. Each trial began with a rest cue (S 1) followed by a task cue (S 2-S 10), with S 7-S 10 encoding the four MI conditions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004022

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`S 2`	21 files	short	raw	Cue onset (task start)
`S 7`	21 files	short	raw	Motor imagery cue - task 1 (Reach)
`S 8`	21 files	short	raw	Motor imagery cue - task 2 (Twist)
`S 9`	21 files	short	raw	Motor imagery cue - task 3 (Lift)
`S 10`	21 files	short	raw	Motor imagery cue - task 4 (Grasp)
`S 3`	21 files	short	raw	Rest period cue
`S 4`	21 files	short	raw	Rest period cue (variant)
`S 5`	21 files	short	raw	Feedback/response marker
`S 6`	21 files	short	raw	Feedback/response marker

Event ID	Count (total)	Type	Meta Level	Meaning
S 1	21 files	short	raw	Trial start / baseline cue
condition 12	21 files	admin	raw	Condition block marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: S 2, S 7, S 8, S 9, S 10, S 3, S 4, S 5
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Four-Class Motor Imagery Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [S 7, S 8, S 9, S 10]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
S 7	0	Motor imagery - Reach
S 8	1	Motor imagery - Twist
S 9	2	Motor imagery - Lift
S 10	3	Motor imagery - Grasp
- Evidence:
- DOI: https://doi.org/10.18112/openneuro.ds004022.v1.0.0 - Dataset description confirms four MI tasks (Reach, Twist, Lift, Grasp) with event markers S 7-S 10
- Kernel Notes: Extract epochs centered on MI cue onsets (S 7-S 10); exclude rest markers (S 1, S 3, S 4) and feedback markers (S 5, S 6). (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Binary Anomaly Detection (Healthy vs Anomalous EEG)

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 0.512 s (256 samples at 500 Hz), overlap = 0.2, target_events = N/A (sliding window)
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning

| healthy segment | 0 | Normal EEG segment |
| anomalous segment | 1 | Abnormal EEG segment (detected by BiLSTM) |
- Super Meta Notes: Derived from raw EEG by sliding 256-point windows with 20% overlap. Kernel step: Apply pre-trained BiLSTM classifier (88% accuracy) to each window; labels generated automatically, validated with manual spot checks. Note: This is the "Unhealthy Dataset" from the paper - all subjects have orthopedic impairment.
- Evidence:
- DOI: 10.1016/j.bspc.2025.108597 - Explicitly uses openneuro_ds004022 as the "Unhealthy Dataset" (7 participants with orthopedic impairment), performs anomaly detection with 256-point windows and BiLSTM classifier
- Kernel Notes: Requires pre-trained BiLSTM model for inference; labels are model-derived, not human-annotated.

Primary Task 3: Binary Health Classification (Healthy vs Orthopedic Impairment)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 1.0 s, overlap = 0.0, target_events = [`S 7`, `S 8`, `S 9`, `S 10`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
healthy	0	Healthy subject EEG (from other dataset)
orthopedic impairment	1	Orthopedic impairment subject EEG (ds004022)
- Super Meta Notes: Derived from raw event(s) [`S 7`, `S 8`, `S 9`, `S 10`] by extracting 1-s windows from MI intervals only. Binary health condition label is derived from subject group membership (this dataset = orthopedic, other dataset = healthy). Kernel step: Combine this dataset with healthy MI dataset; extract non-overlapping 1-s windows from MI intervals; train classifier to distinguish health status.
- Evidence:
- DOI: 10.1371/journal.pone.0335364 - Uses OpenNeuro ds004022 combined with healthy MI dataset for binary health classification using Conditional VAE; extracts 1-s EEG windows from motor imagery intervals (S 7-S 10)
- Kernel Notes: Requires combining this dataset with external healthy MI dataset; MI task identity (S 7-S 10) used only for segment extraction, not as classification target.

Data Quality Notes

- Only 7 subjects with orthopedic impairment - very small clinical sample
- 18-channel EEG with motor cortex coverage; suitable for BCI but limited spatial resolution
- `condition 12` is a block marker to exclude from epoching

References

- Reference: `Multimodal EEG and fNIRS Biosignal Acquisition during Motor Imagery Tasks in Patients with Orthopedic Impairment`

175. openneuro_ds004444 - The BMI-HDEEG dataset 1

Source: analysis.md

Dataset: openneuro_ds004444 | Year: 2023 | URL: https://openneuro.org/datasets/ds004444
Category: Action/Output | Subjects: 465 | Files (Completed): 465
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

Brain-computer interfacing task based on sensorimotor rhythms (SMR) using high-density scalp EEG. Subjects performed motor imagery of four limb movements (left hand, right hand, feet, tongue) following visual cues. This is the first of four BMI-HDEEG datasets, enabling classification of motor imagery across all four classes.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004444

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
1	465xx	long	raw	Motor imagery cue - class 1 (left hand MI)
2	465xx	long	raw	Motor imagery cue - class 2 (right hand MI)
3	465xx	long	raw	Motor imagery cue - class 3 (feet MI)
4	465xx	long	raw	Motor imagery cue - class 4 (tongue MI)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)
- Evidence: Dataset references indicate MI BCI paradigm as core benchmark. Source: https://openneuro.org/datasets/ds004444 DOI: https://doi.org/10.18112/openneuro.ds004444.v1.0.1

Normal Task 1: Motor Imagery Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numerical codes 1-4 decoded by specific kernel to semantic MI labels. Extended epoch window (-0.5 to 4.0 s) captures full motor imagery period. High-density 130-channel EEG.

Event	y	Meaning
`left_hand_MI`	0	Left hand motor imagery
`right_hand_MI`	1	Right hand motor imagery
`feet_MI`	2	Feet motor imagery
`tongue_MI`	3	Tongue motor imagery
## Data Quality Notes		

- Numerical event codes (1-4) require kernel-level decoding to semantic MI labels
- 130-channel high-density EEG with extended epochs
- Large dataset (465 files); batch processing required

References

- Reference: `High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023)`

176. openneuro_ds004446 - The BMI-HDEEG dataset 2

Source: analysis.md

Dataset: openneuro_ds004446 | Year: 2023 | URL: https://openneuro.org/datasets/ds004446
Category: Action/Output | Subjects: 237 | Files (Completed): 237
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

A high-density EEG dataset focused on sensorimotor rhythm-based brain-computer interfacing (BCI) with three motor imagery classes (left hand, right hand, feet). This is the second of four BMI-HDEEG datasets, using the same high-density setup as dataset 1 but with a three-class motor imagery design.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004446

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
1	237xx	long	raw	Motor imagery cue - class 1 (left hand MI)
2	237xx	long	raw	Motor imagery cue - class 2 (right hand MI)
3	237xx	long	raw	Motor imagery cue - class 3 (feet MI)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)

- Evidence: Dataset references indicate MI BCI paradigm as core benchmark. Source: https://openneuro.org/datasets/ds004446 DOI: https://doi.org/10.18112/openneuro.ds004446.v1.0.1

Normal Task 1: Motor Imagery Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Numerical codes 1-3 decoded by specific kernel to semantic MI labels. Same paradigm as dataset 1 (BMI-HDEEG) but 3-class design and different subjects.

Event	y	Meaning
`left_hand_MI`	0	Left hand motor imagery
`right_hand_MI`	1	Right hand motor imagery
`feet_MI`	2	Feet motor imagery
## Data Quality Notes		

- Numerical event codes (1-3) require kernel-level decoding to semantic MI labels
- 130-channel high-density EEG with extended epochs
- 237 files; smaller than dataset 1

References

- Reference: `High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023)`

177. openneuro_ds004447 - The BMI-HDEEG dataset 3

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004447 | Year: 2023 | URL: https://openneuro.org/datasets/ds004447
Category: Action/Output | Subjects: 418 | Files (Completed): 418
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

A brain-computer interface (BCI) task based on sensorimotor rhythms (SMR) recorded with high-density scalp EEG. Subjects performed motor imagery of four limb movements (left hand, right hand, feet, tongue) following visual cues. This is the third of four BMI-HDEEG datasets with a four-class design, though tongue and feet classes appear in fewer files.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004447

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
1	418xx	long	raw	Motor imagery cue - class 1 (left hand MI)
2	418xx	long	raw	Motor imagery cue - class 2 (right hand MI)
3	91xx	long	raw	Motor imagery cue - class 3 (feet MI)
4	91xx	long	raw	Motor imagery cue - class 4 (tongue MI)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Rest vs Right-Hand Motor Imagery Binary Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s *, tmax = 4.0 s, target_events = [2]

- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
(implicit rest)	0	Rest/fixation baseline
2	1	Right-hand motor imagery
- Super Meta Notes: Derived from raw event(s) [2]. Binary classification: rest (implicit baseline) vs right-hand MI. Kernel step: Treat all epochs from class 2 as positive (y=1), create synthetic rest epochs or treat inter-trial intervals as baseline (y=0) for binary classification. Verify: Rest condition may require deriving from inter-trial intervals if no explicit rest event exists.
- Evidence:
- DOI: 10.1038/s41597.023.02260.6 - Validates BMI-HDEEG dataset 3 using binary classification of rest vs right-hand motor imagery periods with spectral power-based features and random forest
- Kernel Notes: Validate existence of rest events in events.tsv; if absent, derive rest epochs from inter-trial intervals (typically 2-4s between trials) or exclude this task. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Four-Class Motor Imagery Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s *, tmax = 4.0 s, target_events = [1, 2, 3, 4]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
1	0	Left hand motor imagery
2	1	Right hand motor imagery
3	2	Feet motor imagery
4	3	Tongue motor imagery
- Evidence:
- DOI: 10.1038/s41597.023.02260.6 - Dataset paper describing four-class motor imagery paradigm with left hand, right hand, feet, tongue classes
- Kernel Notes: Classes 3 and 4 (feet, tongue) appear in only 91/418 files - kernel should handle class imbalance. Consider binary left/right classification as balanced alternative subset.

Data Quality Notes

- Numerical event codes (1-4) require kernel-level decoding
- Classes 3 and 4 (feet/tongue) only appear in 91 of 418 files
- 130-channel high-density EEG with extended epochs

References

- Reference: `High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023).`

178. openneuro_ds004448 - The BMI-HDEEG dataset 4

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004448 | Year: 2023 | URL: https://openneuro.org/datasets/ds004448
Category: Action/Output | Subjects: 280 | Files (Completed): 280
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a sensorimotor rhythm-based brain-computer interface task using high-density scalp EEG. This is the fourth BMI-HDEEG dataset with a two-class (binary) motor imagery design (left hand vs. right hand). Only one event code is present in the data, suggesting the class label is encoded differently.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004448

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
1	280xx	long	raw	Motor imagery cue - class 1

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Motor imagery vs Relaxation binary classification

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 9.0 s *, overlap = 0.0, target_events = [1]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
1	1	Right hand motor imagery
1	0	Relaxation (rest) period

- Evidence:

- DOI: 10.1038/s41597.023.02260.6 - Describes the BMI-HDEEG dataset 4 (OpenNeuro ds004448) with binary motor imagery (right hand vs relaxation), using long-event epochs with imagery periods of 6-9s and relaxation periods of 5-9s, validated with binary classification of spectral power features to decode resting vs imagery periods.
- Kernel Notes: The dataset contains only event ID `1` representing motor imagery cues. Class labels (motor imagery vs relaxation) are derived from experimental timing segments. Verify event ID mapping from events.tsv to confirm imagery vs relaxation epoch boundaries. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Only one event code in metadata; class label may be in events.tsv columns rather than trial_type
- 130-channel high-density EEG with extended epochs
- 280 files; binary MI design

References

- Reference: `High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023)`

179. openneuro_ds004505 - Real World Table Tennis

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004505 | Year: 2023 | URL: https://openneuro.org/datasets/ds004505
Category: Action/Output | Subjects: 19 | Files (Completed): 19 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants played real table tennis matches in two conditions: cooperative (rallying) and competitive (point-scoring). EEG was recorded during actual gameplay using a mobile EEG system. Events mark the condition type and specific ball-hit moments (moving_serve vs stationary_serve shots). The study investigates motor cortex dynamics during ecological real-world sport.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004505

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
moving_hit	~19xN	short	raw	Ball hit during moving serve condition
stationary_hit	~19xN	short	raw	Ball hit during stationary serve condition
cooperative	~19xN	long	raw	Cooperative play condition block
competitive	~19xN	long	raw	Competitive play condition block
moving_serve	~19xN	long	raw	Moving serve condition block
stationary_serve	~19xN	long	raw	Stationary serve condition block
STATUS	~19xN	admin	raw	BrainVision hardware status marker
M 1	~19xN	admin	raw	Miscellaneous hardware marker

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5

- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Hit Type Classification (moving vs stationary serve)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [moving_hit, stationary_hit]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
moving_hit	0	Ball hit during moving serve condition
stationary_hit	1	Ball hit during stationary serve condition
- Evidence:
- DOI: 10.1016/j.dib.2023.110024 - This Data in Brief paper describes a dual-layer EEG dataset from 25 participants playing real-world table tennis with synchronized IMUs and video, where hit events were marked when IMU acceleration exceeded 0.75g and verified against video.
- Kernel Notes: Use event-based epoching around hit markers. The paper confirms this two-class classification task (moving_hit vs stationary_hit) with epoch window [-0.2s, 1.0s] relative to hit onset.

Normal Task 1: Hit Type Classification (binary)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: STATUS and M 1 are admin events - exclude from target_events. cooperative/competitive/moving_serve/stationary_serve are long condition blocks. Focus on discrete hit events for epoch-based analysis.

Event	y	Meaning
moving_hit	0	Ball hit in moving serve condition
stationary_hit	1	Ball hit in stationary serve condition

Data Quality Notes

- Real-world mobile EEG with significant motion artifacts expected
- STATUS and M 1 are hardware markers, not task events

References

- Reference: `Real World Table Tennis`

180. openneuro_ds006519 - Dataset of Intracranial EEG during Cortical Stimulations Evoking Negative Motor Responses

Source: analysis.md

Dataset: `openneuro_ds006519` | Year: 2025 | URL: https://openneuro.org/datasets/ds006519
Category: Action/Output | Subjects: 23 | Files (Completed): 35 | Channels: 33 | Sampling Rate: 4096 Hz

Paradigm

Patients undergoing presurgical evaluation for drug-resistant epilepsy received intracranial electrical stimulation to map cortical function. The stimulation protocol targeted regions that, when stimulated, evoked negative motor responses (involuntary movement inhibition or arrest). EEG was recorded from intracranial electrodes during pre-stimulation (prestim) and post-stimulation (poststim) periods to characterize neural activity changes associated with negative motor mapping.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006519

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Comment/prestim`	35	short	raw	Pre-stimulation baseline period onset
`Comment/poststim`	35	short	raw	Post-stimulation period onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Comment/prestim`, `Comment/poststim`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Dcs

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006519 DOI: https://doi.org/10.18112/openneuro.ds006519.v1.0.0

Normal Task 1: Pre- vs Post-Stimulation Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Binary classification of pre- vs post-stimulation neural states. Intracranial EEG from epilepsy patients; electrode locations vary across subjects. Very high sampling rate (4096 Hz); consider downsampling.

Event	y	Meaning
`Comment/prestim`	0	Pre-stimulation baseline period
`Comment/poststim`	1	Post-stimulation response period
## Data Quality Notes		

- Intracranial EEG dataset; electrode montage is patient-specific and non-standard.
- Very high sampling rate (4096 Hz); downsampling to <=1000 Hz recommended.
- Small dataset (35 files, 23 subjects); limited sample size.

References

- Reference: `Dataset of intracranial EEG during cortical stimulations evoking negative motor responses`

181. openneuro_ds007327 - EEG During Sport-Related Motor Imagery and Observation Tasks

Source: analysis.md

Dataset: openneuro_ds007327 | Year: 2026 | URL: https://openneuro.org/datasets/ds007327
Category: Action/Output | Subjects: 35 | Files (Completed): 105 | Channels: 88 | Sampling Rate: 1000 Hz

Paradigm

EEG was recorded during two sports-related tasks (ball passing and dribbling) to study motor intention and motor observation. The paradigm uses a 2x2 design crossing motor intention (MI = motor intention, MO = motor observation) with movement execution modality, yielding four compound conditions: MIMI (imagine intention), MIMO (imagine observation), MOMI (observe intention), MOMO (observe observation). Additional single-factor conditions (MI, MO, MO_MI) and a static/rest baseline are included.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds007327

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
MOMI	8396	short	raw	Motor observation of intention trials
MIMO	8394	short	raw	Motor imagination of observation trials
MOMO	8394	short	raw	Motor observation of observation trials
MIMI	8335	short	raw	Motor imagination of intention trials
static	4313	short	raw	Static baseline condition
MO	4200	short	raw	Motor observation condition
MO_MI	4198	short	raw	Motor observation with motor intention
MI	4195	short	raw	Motor intention condition
rest	4073	short	raw	Resting/baseline period

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Imagery class labels (e.g., left/right hand, feet, tongue; dataset-specific)
- Evidence: The official dataset page describes sport-related motor imagery and motor observation conditions that support a primary motor-intention decoding benchmark. Source: https://openneuro.org/datasets/ds007327

Normal Task 1: Motor Intention/Observation Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 9-class design; 2x2 factorial core (MIMI/MIMO/MOMI/MOMO ~8300 each) plus single-factor and baseline conditions (~4000 each). 105 files = 35 subjects x 3 task types (ballpass, dribble, passing).

Event	y	Meaning
`MIMI`	0	Motor imagination x intention
`MIMO`	1	Motor imagination x observation
`MOMI`	2	Motor observation x intention
`MOMO`	3	Motor observation x observation
`MI`	4	Motor intention
`MO_MI`	5	Motor observation with intention
`static`	6	Static baseline
`MO`	7	Motor observation
`rest`	8	Rest baseline
## Data Quality Notes		

- High channel count (88) includes 24 BIP channels for EMG/kinematic data; exclude BIP channels from EEG analysis.
- 35 subjects x 3 task files = 105 files; verify task structure.
- Apply bandpass filter (0.5-40 Hz), remove BIP channels, and artifact reject prior to epoching.

References

- No paper reference available

182. other_highgammadataset - High-Gamma EEG Motor Imagery Dataset

Source: analysis_minimax_v2.md

Dataset: `other_highgammadataset` | Year: 2017 | URL: https://gin.g-node.org/robintibor/high-gamma-dataset/src/master/data
Category: Action/Output | Subjects: 14 | Files (Completed): 28 | Channels: 133 | Sampling Rate: 500 Hz

Paradigm

Fourteen subjects performed four-class cued motor imagery (left hand, right hand, feet, tongue) while high-density EEG was recorded from 128+ channels using a NeurOne amplifier (Mega Electronics Ltd, Kuopio, FI) with a WaveGuard Original cap (ANT, Enschede, NL) at 500 Hz (Schirrmeister et al., 2017). Each subject contributed two sessions: a training set and a test set. Per session, 40-225 trials per class were recorded depending on subject. This dataset is notably used to evaluate deep learning models (EEGNet, ShallowConvNet, DeepConvNet) for motor imagery BCI and is one of the highest-density publicly available MI datasets.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://gin.g-node.org/robintibor/high-gamma-dataset/src/master/data

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`left_hand`	~4200 total	short	raw	Left-hand motor imagery cue onset
`right_hand`	~4200 total	short	raw	Right-hand motor imagery cue onset
`feet`	~1400 total	short	raw	Feet motor imagery cue onset
`tongue`	~1400 total	short	raw	Tongue motor imagery cue onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Trial counts per session are variable (40-225 per class). Approximate totals across all 28 files.

Tasks

Primary Task 1: Four-class Motor Imagery Classification

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = [`left_hand`, `right_hand`, `feet`, `tongue`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`left_hand`	0	Left-hand motor imagery
`right_hand`	1	Right-hand motor imagery
`feet`	2	Feet motor imagery
`tongue`	3	Tongue motor imagery
- Evidence:
- DOI: `10.1002/hbm.23730` - Uses the High-Gamma Dataset (HGD) with 14 subjects performing four-class motor imagery (left hand, right hand, feet, tongue) across two sessions per subject, evaluating deep ConvNets against FBCSP baseline for EEG decoding with 4.5-second epochs from -0.5s to 4.0s relative to trial cue onset.
- Kernel Notes: Use tmin=-0.5s to capture 500ms baseline before cue. Trials are 4s duration per paper. Preprocessing may include highpass filtering and resampling to 250Hz as described in the original paper. Verify event ID mapping from events.tsv. (replaces incomplete Primary Task 1 from original analysis.md)

Primary Task 2: Left vs. Right Hand Motor Imagery Binary Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = [`left_hand`, `right_hand`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`left_hand`	0	Left-hand motor imagery
`right_hand`	1	Right-hand motor imagery
- Evidence:
- DOI: `10.1002/hbm.23730` - The four-class paradigm includes left and right hand motor imagery as distinct classes that can be isolated for binary classification.
- Kernel Notes: Filter epochs to only include left_hand and right_hand trials; discard feet and tongue trials.

Normal Task 1: 4-Class Motor Imagery Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
`left_hand`	0	Left-hand motor imagery
`right_hand`	1	Right-hand motor imagery
`feet`	2	Feet motor imagery
`tongue`	3	Tongue motor imagery

Normal Task 2: Left vs. Right Hand Motor Imagery (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 4.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: `left_hand` -> 0, `right_hand` -> 1 (discard feet and tongue trials).

Data Quality Notes

- No age or sex information available in the public release.
- High channel count (128+ EEG); high-gamma frequency analysis (60-300 Hz) is a primary use case.
- Trial counts are imbalanced across classes (more left/right hand than feet/tongue); balance before 4-class training.
- Two sessions per subject (train/test): separate by filename suffix for proper evaluation.

References

- Reference: Schirrmeister RT et al. (2017). Deep learning with convolutional neural networks for EEG decoding and visualization. Human Brain Mapping, 38(11), 5391-5420. https://doi.org/10.1002/hbm.23730

183. other_openbmi - OpenBMI Motor Imagery EEG Dataset

Source: analysis_minimax_v2.md

Dataset: `other_openbmi` | Year: 2019 | URL: https://gigadb.org/dataset/view/id/100542/File_page/5
Category: Action/Output | Subjects: 54 | Files (Completed): 108 | Channels: 62 | Sampling Rate: 1000 Hz

Paradigm

Fifty-four subjects (ages 19-32, mixed sex) performed two-class cued motor imagery of the left and right hand across two sessions each (Lee et al., 2019). EEG was recorded from 62 channels (BrainAmp, Brain Products; Ag/AgCl cap) at 1000 Hz. Each session contained 100 trials per class (50 left + 50 right hand MI), yielding 200 trials per session per subject. The motor imagery epoch lasts 3 seconds. Trial onset times and class labels are stored in MATLAB struct fields (`EEG_MI_train.t` for sample indices and `EEG_MI_train.y_class` for labels); the kernel extracts these and creates 3-second MNE annotations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://gigadb.org/dataset/view/id/100542/File_page/5

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`left hand MI`	5400	short	raw	Left-hand motor imagery trial onset (50/session x 2 sessions x 54 subjects)
`right hand MI`	5400	short	raw	Right-hand motor imagery trial onset (50/session x 2 sessions x 54 subjects)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Motor imagery decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 3.0 s, target_events = [`left hand MI`, `right hand MI`]
- Label Route: `Annotation`
- Meta Level: raw

- Target Definition:
Event	y	Meaning
`left hand MI`	0	Left-hand motor imagery
`right hand MI`	1	Right-hand motor imagery
- Evidence:
- DOI: 10.1016/j.neunet.2020.12.013 - Uses OpenBMI dataset with 54 subjects for binary motor imagery (left/right hand), proposes CNN-based transfer learning achieving 84.19% accuracy.
- DOI: 10.1093/gigascience/giz002 - Introduces OpenBMI dataset with 54 subjects for binary MI classification, validates CSP and LDA methods.
- Kernel Notes: The 3-second epoch aligns with the dataset paradigm specification. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Subject demographics: 54 subjects, ages 19-32, mixed sex.
- Two sessions per subject enable cross-session analysis and session-to-session learning stability.
- High sampling rate (1000 Hz); consider downsampling to 250 Hz for computational efficiency.
- MATLAB struct-based event extraction; requires `EEG_MI_train` field to be present in `.mat` file.
- 62-channel standard 10-20 montage with Ag/AgCl electrodes.

References

- Reference: Lee MH et al. (2019). EEG Dataset and OpenBMI Toolbox for Three BCI Paradigms: An Investigation into BCI Illiteracy. GigaScience, 8(5), giz002. https://doi.org/10.1093/gigascience/giz002

184. physionet_eegmmidb - EEG Motor Movement/Imagery Database

Source: analysis_minimax_v2.md

Dataset: `physionet_eegmmidb` | Year: 2009 | URL: https://physionet.org/content/eegmmidb/1.0.0/
Category: Action/Output | Subjects: 109 | Files (Completed): 1526 | Channels: 64 | Sampling Rate: 160 Hz

Paradigm

One hundred and nine volunteers performed a series of motor movement and motor imagery tasks, recorded with 64 EEG electrodes (international 10-10 system) at 160 Hz using the BCI2000 system (Schalk et al., 2004). Each subject completed 14 runs (R01-R14): two baseline runs (eyes open and eyes closed), and twelve task runs organized in three repeating cycles of four run types. Task runs alternate between real movements and mental imagery of the same movements. Two movement classes are used: left/right fist, and both fists/both feet. Events T0 (rest), T1, and T2 are annotated within each run; T1/T2 meanings differ by run number.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/eegmmidb/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`eyes open`	109	rest	raw	Baseline recording with eyes open (R01)
`eyes closed`	109	rest	raw	Baseline recording with eyes closed (R02)
`Baseline`	~1200	rest	raw	Rest/inter-trial interval in task runs (T0)
`Open Left Fist Movement`	~327	short	raw	Real left-fist movement onset (R03/07/11 T1)
`Open Right Fist Movement`	~327	short	raw	Real right-fist movement onset (R03/07/11 T2)
`Imagine Opening Left Fist Movement`	~327	short	raw	Left-fist motor imagery onset (R04/08/12 T1)
`Imagine Opening Right Fist Movement`	~327	short	raw	Right-fist motor imagery onset (R04/08/12 T2)

Event ID	Count (total)	Type	Meta Level	Meaning
`Open Both Fists Movement`	~327	short	raw	Real both-fists movement onset (R05/09/13 T1)
`Open Both Feet Movement`	~327	short	raw	Real both-feet movement onset (R05/09/13 T2)
`Imagine Opening Both Fists Movement`	~327	short	raw	Both-fists motor imagery onset (R06/10/14 T1)
`Imagine Opening Both Feet Movement`	~327	short	raw	Both-feet motor imagery onset (R06/10/14 T2)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Total counts are approximate across all 109 subjects x 3 repetition cycles. Per-subject per-run counts: ~20 T1 + 20 T2 per task run.

Tasks

Primary Task 1: 5-Class Motor Imagery (including baseline)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 4.0 s, target_events = [`eyes closed`, `Imagine Opening Both Feet Movement`, `Imagine Opening Both Fists Movement`, `Imagine Opening Left Fist Movement`, `Imagine Opening Right Fist Movement`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`eyes closed`	0	Eye closed baseline
`Imagine Opening Both Feet Movement`	1	Both-feet motor imagery
`Imagine Opening Both Fists Movement`	2	Both-fists motor imagery
`Imagine Opening Left Fist Movement`	3	Left-fist motor imagery
`Imagine Opening Right Fist Movement`	4	Right-fist motor imagery

- Evidence:
- DOI: 10.1109/embc.2018.8512590 - Evaluates 5-class motor imagery classification (eye closed, both feet, both fists, left fist, right fist) with sliding window length=10s, 50% overlap, achieving validation on first 12 subjects
- Kernel Notes: Use only imagery runs (R04, R06, R08, R10, R12, R14). Exclude real movement runs (R03, R05, R07, R09, R11, R13) and baseline runs (R01, R02). Epoch window from cue onset (t=0) to 4s post-cue.

Primary Task 2: 4-Class Motor Imagery Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 4.0 s, target_events = [`Imagine Opening Left Fist Movement`, `Imagine Opening Right Fist Movement`, `Imagine Opening Both Fists Movement`, `Imagine Opening Both Feet Movement`]
- Label Route: `Annotation`

- Meta Level: raw
- Target Definition:
Event	y	Meaning
`Imagine Opening Left Fist Movement`	0	Left-fist motor imagery
`Imagine Opening Right Fist Movement`	1	Right-fist motor imagery
`Imagine Opening Both Fists Movement`	2	Both-fists motor imagery
`Imagine Opening Both Feet Movement`	3	Both-feet motor imagery
- Evidence:
- DOI: 10.1109/tla_2018_8291481 - Uses 4-class motor imagery (left fist, right fist, both fists, both feet) with window=1.0s, stride=0.5s for continuous sliding window classification
- Kernel Notes: Use only imagery runs (R04, R06, R08, R10, R12, R14). Each run type alternates between fist (R04/08/12) and feet/fists (R06/10/14) tasks. (replaces Normal Task 3 from original analysis.md)

Primary Task 3: Binary Left vs. Right Fist Motor Imagery

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 4.0 s, target_events = [`Imagine Opening Left Fist Movement`, `Imagine Opening Right Fist Movement`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`Imagine Opening Left Fist Movement`	0	Left-fist motor imagery (R04/08/12 T1)
`Imagine Opening Right Fist Movement`	1	Right-fist motor imagery (R04/08/12 T2)
- Evidence:
- DOI: 10.15368/theses_2017_62 - Uses 4.1-second EEG epochs (tmin=0.0s, tmax=4.1s) of left vs. right hand motor imagery achieving 79.4% average accuracy across 40 subjects
- Kernel Notes: Use only runs R04, R08, R12 (imagery sessions). Filter by run number in filename (positions 5-7). (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Both Fists vs. Both Feet Motor Imagery (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 4.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Use only runs R06, R10, R14 (imagery sessions). Run numbers are encoded in the filename (positions 5-7, e.g., `S001R06.edf`). Exclude baseline runs (R01, R02), movement runs (R05, R09, R13), and fist-only imagery runs (R04, R08, R12) for this analysis.

Event	y	Meaning
`Imagine Opening Both Fists Movement`	0	Both-fists motor imagery (R06/10/14)
`Imagine Opening Both Feet Movement`	1	Both-feet motor imagery (R06/10/14)

Data Quality Notes

- Age and sex information are not available in the public metadata.

- Known data quality issues: several subjects have noisy recordings; EEG.physionet.org community notes suggest subjects S088, S089, S092, S100, S104 may have unusual recordings.
- T1/T2 event meaning changes by run number; always filter by run before assigning class labels.
- `Baseline` (T0) events in task runs are rest periods and should be excluded from MI classification targets.
- Low sampling rate (160 Hz); sufficient for motor imagery but limits high-frequency analysis.

References

- Reference: Schalk G, McFarland DJ, Hinterberger T, Birbaumer N, Wolpaw JR (2004). BCI2000: A general-purpose brain-computer interface (BCI) system. IEEE Transactions on Biomedical Engineering, 51(6), 1034-1043. https://physionet.org/content/eegmmidb/1.0.0/

185. physionet_erpbci - ERP-Based Brain-Computer Interface Recordings

Source: analysis_minimax_v2.md

Dataset: `physionet_erpbci` | Year: 2013 | URL: https://physionet.org/content/erpbci/1.0.0/
Category: Action/Output | Subjects: 21 | Files (Completed): 245 | Channels: 70 | Sampling Rate: 2048 Hz

Paradigm

Participants spelled characters using a standard 6x6 matrix speller while EEG was recorded with a BioSemi ActiveTwo system (64 EEG + 2 earlobe reference + 4 EOG channels) at 2048 Hz (Riccio et al., 2013). Each row and column of the 6x6 character matrix was randomly intensified for 100 ms with a 50 ms gap (150 ms SOA). Each row/column flashed approximately 20 times per trial, and participants mentally counted target character flashes to sustain attention. The kernel processes raw row/column flash events and classifies each as `seen` (target - the row or column containing the attended character flashed) or `unseen` (non-target) for both rows and columns separately.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/erpbci/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`seen (row)`	varies	short	raw	Target row flash (attended character's row lit up)
`seen (column)`	varies	short	raw	Target column flash (attended character's column lit up)
`unseen (row)`	varies	short	raw	Non-target row flash
`unseen (column)`	varies	short	raw	Non-target column flash

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Each 6x6 speller has 6 rows + 6 columns = 12 flashes per sequence; 2 are target (1 row + 1 column), 10 are non-target.

Tasks

Primary Task 1: P300 Target Detection (Binary Classification)

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [`seen (row)`, `seen (column)`, `unseen (row)`, `unseen (column)`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`seen (row)`	1	Target (attended row flash)
`seen (column)`	1	Target (attended column flash)
`unseen (row)`	0	Non-target row flash
`unseen (column)`	0	Non-target column flash

- Evidence:
- DOI: 10.13026/C2101S - Two-class P300 oddball classification: target (attended character flash) vs non-target (non-attended flash) in 6x6 matrix speller with epoch parameters tmin=-0.1s, tmax=0.8s around each row/column intensification event
- Kernel Notes: Extract epochs around row/column flash events. Combine `seen (row)` and `seen (column)` as positive class (y=1); `unseen (row)` and `unseen (column)` as negative class (y=0). Class imbalance is ~1:5 (2 target vs 10 non-target per sequence). Exclude EOG channels (VEOGL, VEOGR, HEOGL, HEOGR) from features. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Very high sampling rate (2048 Hz); consider downsampling to 256 or 512 Hz for computational efficiency.
- Class imbalance: 2 target flashes vs. 10 non-target flashes per sequence (1:5 ratio); balance with oversampling or cost-sensitive learning.
- EOG channels present (4 channels); apply artifact rejection before P300 classification.
- Age and sex information are not available.

References

- Reference: Riccio A et al. (2013). Interfacing Brain with Computer to Improve Communication and Rehabilitation after Spinal Cord Injury. Frontiers in Neuroengineering. Goldberger AL et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. https://physionet.org/content/erpbci/1.0.0/

Chapter 4. Sensory/Response

Datasets in this chapter: 52. Basic stimulus-response, ERP/SSVEP/RSVP, or simple attention tasks.

186. openneuro_ds001810 - EEG study of the attentional blink; before, during, and after transcranial Direct Current Stimulation (tDCS)

Source: analysis.md

Dataset: `openneuro_ds001810` | Year: 2019 | URL: https://openneuro.org/datasets/ds001810
Category: Sensory/Response | Subjects: 40 | Files (Completed): 527 | Channels: 73.0 | Sampling Rate: 512.0 Hz

Paradigm

Participants performed an attentional blink (AB) task using a Rapid Serial Visual Presentation (RSVP) stream to identify two targets (T1 and T2) embedded in a stream of distractors. EEG was recorded before, during, and after receiving anodal or cathodal transcranial Direct Current Stimulation (tDCS) to study how cortical excitability modulates the attentional blink - the impaired detection of T2 when it follows T1 within ~500 ms. BioSemi STATUS channel encodes trial conditions as numeric codes.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds001810

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`T1_present_T2_present_lag3`	large	short	raw	T1 and T2 both present, short lag (within AB window)
`T1_present_T2_absent_lag3`	large	short	raw	T1 present, T2 absent, short lag
`T1_absent_T2_present`	large	short	raw	T1 absent, T2 present
`T1_T2_lag8`	large	short	raw	T1 and T2 at long lag (outside AB window)
`T1_onset`	large	short	raw	T1 target onset
`T2_onset`	large	short	raw	T2 target onset
`CM_baseline (61440)`	261 files	admin	raw	BioSemi CM_in_range baseline - exclude

Event ID	Count (total)	Type	Meta Level	Meaning
CM_out_of_range (61694)	263 files	admin	raw	BioSemi CM status - exclude
stimulus_onset (61468)	large	short	raw	General stimulus onset

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: response_time
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stimulus_onset
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Attentional blink

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds001810 DOI: https://doi.org/10.18112/openneuro.ds001810.v1.1.0

Normal Task 1: Attentional Blink Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: BioSemi STATUS codes decoded via specific kernel (TODO: verify exact code mapping from original paper doi.org/10.1111/j.1469-8986.2010.01081.x). CM_baseline (61440) and CM_out_of_range (61694) are admin markers excluded. Regression target: response_time.

Event	y	Meaning
T1_present_T2_present_lag3	0	T2 detected within AB window
T1_present_T2_absent_lag3	1	T2 missed within AB window

Event	y	Meaning
T1_absent_T2_present	2	T2 present without T1
T1_T2_lag8	3	T2 detected outside AB window (control)
T1_onset	4	T1 target onset
T2_onset	5	T2 target onset
## Data Quality Notes		

- BioSemi STATUS channel encodes all triggers numerically; specific kernel maps codes to human-readable labels.
- 61440 (CM_in_range) and 61694 (CM_out_of_range) are BioSemi hardware status codes - exclude from analysis.
- Large dataset (527 files, 40 subjects x pre/during/post x anodal/cathodal conditions).
- Verify that event code mapping is correct against the original paper before downstream use.

References

- Reference: EEG study of the attentional blink; before, during, and after transcranial Direct Current Stimulation (tDCS)

187. openneuro_ds002218 - Auditory and Visual Rhythm Omission EEG

Source: analysis_minimax_v2.md

Dataset: openneuro_ds002218 | Year: 2019 | URL: https://openneuro.org/datasets/ds002218
Category: Sensory/Response | Subjects: 18 | Files (Completed): 18
Channels: 32 | Sampling Rate: 256 Hz

Paradigm

Participants passively perceived rhythmic auditory tone sequences and visual flashing stimuli that occasionally omitted an expected beat, to study predictive neural activity (mismatch responses) versus sensory evoked responses. The paradigm investigates how the brain generates predictions based on temporal regularities and detects violations of expected rhythmic patterns. Only BioSemi STATUS admin markers are detectable in the locator metadata; no decoded trial events are available from the locator pipeline.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002218

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
STATUS	18 (files)	admin	raw	BioSemi system status marker - exclude from epoch

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Auditory/Visual Rhythm Omission Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -1.67 s, tmax = +1.67 s, target_events = [tone_onset, flash_onset] *
- Label Route: Annotation | stim
- Meta Level: super
- Target Definition:

Event	y	Meaning
auditory_non-omission	0	Tone/auditory stimulus presented at expected time
auditory_omission	1	Expected auditory stimulus omitted
visual_non-omission	2	Flash/visual stimulus presented at expected time
visual_omission	3	Expected visual stimulus omitted

- Super Meta Notes: Derived from raw event(s) [tone_onset, flash_onset] by mapping experimental design: each stimulus train had 100 events with 7 randomly placed omissions per train. Kernel step: Researchers controlled omission timing; labels must be reconstructed from paradigm structure (100 trials per run, 7 omissions randomly positioned). Verify raw event IDs from events.tsv.
- Evidence:
- DOI: 10.1101/2020.11.29.402701 - Uses same ds002218 dataset (18 subjects) with omission paradigm, epochs -1.67s to +1.67s, four-class classification (auditory/visual x omission/non-omission)
- DOI: 10.1111/ejn.15314 - Analyzes beta band activity comparing auditory vs visual rhythms and omission vs non-omission conditions with 3.34s epochs
- Kernel Notes: Only BioSemi STATUS admin markers detected in locator metadata. Actual trial events (tone/flash onset markers) must be verified in events.tsv files. The experimental design (100 events per train, 7 random omissions) allows reconstruction of labels if trial timing is available. Verify raw event ID names from events.tsv (likely tone_onset, flash_onset, or similar). (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Only STATUS events found in locator metadata - BioSemi admin markers only.
- The actual auditory/visual rhythm and omission events may be encoded in the events.tsv files but were not decoded by the locator pipeline; check events.tsv directly.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Comstock DC, Ross JM, Balasubramaniam R. Modality-specific frequency band activity during neural entrainment to auditory and visual rhythms. European Journal of Neuroscience. DOI: https://doi.org/10.1111/ejn.15314

188. openneuro_ds002578 - Visual Oddball Task (256 channels)

Source: analysis_minimax_v2.md

Dataset: openneuro_ds002578 | Year: 2021 | URL: https://openneuro.org/datasets/ds002578
Category: Sensory/Response | Subjects: N/A | Files (Completed): 2 | Channels: 256.0 | Sampling Rate: 256.0 Hz

Paradigm

A visual oddball task designed to study selective attention using event-related potentials with high-density 256-channel electrode coverage. Despite the task description implying trial-level stimuli (standard vs. oddball), no event markers were detected in the BIDS EEG files. The dataset contains only 2 completed files and is a very small subset.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002578

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(none)	-	-	raw	No event markers detected in EEG files

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Visual Oddball Target Response Classification (Fast vs Slow RT)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 2.0 s, target_events = [`target_visual_stimulus`, `button_press_response`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
fast_response	1	Reaction time below median (fast motor response)

| slow_response | 0 | Reaction time above median (slow motor response) |
- Super Meta Notes: Derived from raw events [`target_visual_stimulus`, `button_press_response`] by computing reaction time (RT = button_press_timestamp - target_visual_stimulus_timestamp) for each trial, then classifying by median split. Kernel step: extract target onset and button press events per trial, compute RT per trial, apply median split to create binary labels (fast=1, slow=0).
- Evidence:
- DOI: 10.1523/jneurosci.3477.07.2007 - This 2007 J Neurosci paper analyzes visual oddball EEG data to show that larger medial prefrontal theta (4.5 Hz) complexes precede faster motor responses (button presses), using ICA to decompose 31-channel (and 253-channel) target response epochs and correlate single-trial low-theta power with reaction time quantiles.
- Kernel Notes: Verify event ID mapping from events.tsv. Compute RT per trial: (button_press_time - target_stimulus_time). Apply median split across all trials to define fast/slow labels. Epoch window is 3s total (1s pre-stimulus to 2s post-stimulus). (replaces Primary Task 1 from original analysis.md)

Normal Task 1: Resting / Continuous Segmentation (resting)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no trial-level labels available))
- Notes: Only 2 completed files available - very small dataset. High-density 256-channel EEG provides spatial richness. Check events.tsv directly for potential trial markers not captured in locator pipeline.

Data Quality Notes

- Only 2 completed EEG files - extremely limited dataset size.
- No event markers detected despite visual oddball paradigm description.
- High-density 256-channel array; verify electrode positions from channel info files.

References

- Reference: `Visual Oddball Task (256 channels)`

189. openneuro_ds002680 - Go-nogo categorization and detection task

Source: analysis_minimax_v2.md

Dataset: openneuro_ds002680 | Year: 2021 | URL: https://openneuro.org/datasets/ds002680
Category: Sensory/Response | Subjects: N/A | Files (Completed): 350 | Channels: 31.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants performed a visual go/no-go paradigm involving rapid animal categorization and image recognition tasks with brief 20 ms stimulus presentations. Go trials required a button-press response when an animal was detected (or a target category matched); no-go trials required withholding response. Each run contains approximately 100 stimulus events and ~50 response events, suggesting roughly 50% go rate. The paradigm studies rapid visual processing and response inhibition.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, reaction time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002680

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stimulus`	~35,000 (350x100)	short	raw	Visual stimulus onset (20 ms image presentation)
`response`	~18,000 (350x52)	short	raw	Button-press go response

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `reaction_time`, `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Go/No-Go Visual Categorization Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.3 s, tmax = 0.7 s, target_events = [`stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`go`	1	Animal target stimulus followed by button-press response (go trial)
`no-go`	0	Non-animal distractor stimulus without response (no-go trial)

- Super Meta Notes: Derived from raw events [`stimulus`, `response`] by matching each stimulus to subsequent response within 1.5s window: if response follows stimulus, label = go (1); if no response, label = no-go (0). Kernel step: For each stimulus event, check for presence of response event within 1.5s window; create go/no-go label accordingly.
- Evidence:
- DOI: 10.1038/s41598.023.27528.0 - Shows classification of correct animal targets (go responses) vs correct non-animal distractors (no-go responses) using epoch extraction from -0.3 to 0.7s
- DOI: 10.1016/S0925-2312(02)00415-0 - Uses rapid go-nogo visual categorization with 20ms stimuli, classifying target (animal) vs non-target (no animal) based on response presence
- Kernel Notes: Must implement response-matching logic to derive go/no-go labels from raw stimulus and response events. Exclude incorrect trials (response on non-target or no response on target) if classification targets only correct responses.

Primary Task 2: Stimulus-Locked ERP Classification (Extended Window)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -1.0 s, tmax = 2.0 s, target_events = [`stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`target`	1	Animal target stimulus (go trial with response)
`distractor`	0	Non-animal distractor (no-go trial without response)

- Super Meta Notes: Derived from raw events [`stimulus`, `response`] by same response-matching logic. Kernel step: For each stimulus epoch, assign target=1 if response event follows within 1.5s, else distractor=0.
- Evidence:
- DOI: 10.1038/s41598.023.27528.0 - Uses epoch extraction from -1 to 2 seconds relative to stimulus onset for initial analysis of go vs no-go brain potentials
- Kernel Notes: Same response-matching derivation as Primary Task 1 but with extended epoch window for capturing later ERP components.

Normal Task 1: Stimulus-Locked ERP / Reaction Time Regression (regression)

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to stimulus onset), target_events = see Target Definition
- Label Route: `misc`
- Target Definition: N/A (null (regression only; stimulus serves as epoch anchor))
- Notes: Go/no-go structure: ~100 stimuli per run, ~52 responses (go trials). Regression target is response_time on go trials. For classification, go vs no-go labels would need to be inferred from presence/absence of a response after each stimulus.
- Regression target: `response_time` (reaction time from stimulus to response, from misc channel)

Data Quality Notes

- Large dataset: 350 runs; verify subject/session structure from BIDS metadata.
- ~50% go rate per run based on stimulus (100) vs response (52) counts.
- 1000 Hz sampling rate; consider downsampling to 250 Hz for efficiency.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Go-nogo categorization and detection task`

190. openneuro_ds002814 - A Multimodal Neuroimaging Dataset to Study Spatiotemporal Dynamics of Visual Processing in Humans

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002814` | Year: 2021 | URL: https://openneuro.org/datasets/ds002814
Category: Sensory/Response | Subjects: N/A | Files (Completed): 168 | Channels: 68.0 | Sampling Rate: 1200.0 Hz

Paradigm

A multimodal neuroimaging study (EEG + MEG/MRI) investigating the spatiotemporal dynamics of visual processing in humans. Despite the dataset title implying an active visual processing task, no event markers are detectable in the BIDS EEG files. The visual stimulation protocol may be encoded in other modality files or events are not available in the EEG files processed by the locator pipeline.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002814

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(none)	-	-	raw	No event markers detected in EEG files

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Visual Category Classification (5-class)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.6 s, target_events = `[stimulus_onset]`
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning

`stimulus_onset` (animals trial)	0	Visual stimulus presentation of animal category
`stimulus_onset` (chairs trial)	1	Visual stimulus presentation of chair category
`stimulus_onset` (faces trial)	2	Visual stimulus presentation of face category
`stimulus_onset` (fruits trial)	3	Visual stimulus presentation of fruit category
`stimulus_onset` (vehicles trial)	4	Visual stimulus presentation of vehicle category

- Super Meta Notes: (N/A - meta level is raw)
- Evidence:
- DOI: 10.1101/2022.05.12.491595 - Paper describes this dataset and performs EEG category classification decoding among 5 visual object categories (animals, chairs, faces, fruits, vehicles) using SVM classifiers on epochs extracted from -100ms to +600ms relative to stimulus onset, achieving significant pairwise decoding accuracy.
- Kernel Notes: Extract epochs time-locked to stimulus onset events. The 5 visual categories (animals, chairs, faces, fruits, vehicles) serve as class labels. Verify event ID mapping from events.tsv - paper uses 'stimulus_onset' but actual raw event ID may differ (e.g., 'stimulus', 'stim', or numeric trigger code). Consider downsampling from 1200 Hz before classification.

Normal Task 1: Resting / Continuous Segmentation (resting)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no trial-level labels available))
- Notes: No event markers found in EEG locator metadata. High sampling rate (1200 Hz); downsampling recommended. Part of a multimodal dataset - check MEG/MRI files for corresponding event timing.

Data Quality Notes

- No event markers detected in any of the 168 EEG files.
- High sampling rate (1200 Hz); downsampling required before most analyses.
- Part of multimodal dataset; MEG or MRI-derived event timing may be applicable if aligned.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `A Multimodal Neuroimaging Dataset to Study Spatiotemporal Dynamics of Visual Processing in Humans`

191. openneuro_ds002893 - Auditory-Visual Shift Study

Source: analysis.md

Dataset: openneuro_ds002893 | Year: 2022 | URL: https://openneuro.org/datasets/ds002893
Category: Sensory/Response | Subjects: 49 | Files (Completed): 52 | Channels: 36.0 | Sampling Rate: 250.0 Hz

Paradigm

An interleaved dual-oddball task in which subjects selectively attended to either auditory or visual stimuli while switching between modalities based on spoken and printed attention cue instructions. The study investigated modality-specificity of sensory aging in vision and audition using event-related potentials. Attention states (attend_auditory, attend_visual) are sustained conditions lasting multiple trials; shift_attention marks the transition cue between modalities.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002893

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
attend_auditory	51 (files)	long	raw	Sustained auditory attention block
attend_visual	51 (files)	long	raw	Sustained visual attention block
shift_attention	49 (files)	admin	raw	Attention shift cue - transition marker between blocks

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
> - rest: resting-state segment marker, for epoch_by_segmentation_hdf5
> - admin: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Target vs non-target stimulus events

- Evidence: P300/oddball paradigm appears as core study objective. Source: https://openneuro.org/datasets/ds002893 DOI: https://doi.org/10.18112/openneuro.ds002893.v2.0.0

Normal Task 1: Attention Modality Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `shift_attention` is a transition cue - it marks the moment of modality switching, not a sustained state; treat as admin. The `attend_*` events mark the onset of sustained attention blocks of multiple seconds.

Event	y	Meaning
`attend_auditory`	0	Sustained auditory attention block
`attend_visual`	1	Sustained visual attention block
## Data Quality Notes		

- `shift_attention` marks attention shift cues - not a sustained state; exclude from classification target.
- Count column reflects files-with-event (51 subjects out of 52 files), not per-session trial count.
- Older adult participants (aging study); verify age distribution from participants.tsv.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Modality-specificity of sensory aging in vision and audition: evidence from event-related potentials. Brain research, 1215, pp.53-68.`

192. openneuro_ds003061 - EEG data from an auditory oddball task

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003061 | Year: 2022 | URL: https://openneuro.org/datasets/ds003061
Category: Sensory/Response | Subjects: 39 | Files (Completed): 39
Channels: 79 | Sampling Rate: 256 Hz

Paradigm

An auditory three-stimulus oddball task in which participants pressed a key in response to infrequent target tones among frequent standard tones and rare distractor tones. The paradigm is designed to elicit P300 event-related potentials. Event labels `stimulus` (tone onset) and `response` (keypress) are present in all 39 files; `STATUS` appears as a BioSemi admin marker in 2 files.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, reaction time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003061

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stimulus`	39 (files)	short	raw	Auditory tone stimulus onset
`response`	39 (files)	short	raw	Participant keypress response
`STATUS`	2 (files)	admin	raw	BioSemi system status marker - exclude

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `reaction_time`, `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`, `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Auditory Oddball ERP Classification (Target vs Standard)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.3 s , tmax = 0.7 s , target_events = [`stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`stimulus` (with response)	1	Oddball target tone (1000Hz) with correct keypress
`stimulus` (no response)	0	Standard non-target tone (500Hz) with no response

- Super Meta Notes: Derived from raw events [`stimulus`, `response`] by: label=1 if stimulus event is followed by a response event within 2s (oddball target correct response); label=0 if stimulus event has no subsequent response within 2s (standard non-target correct rejection). Kernel step: For each stimulus epoch, check if a response event occurs within the trial window; assign label based on response presence. Excludes distractor tones (15%) and incorrect responses.
- Evidence:
- DOI: 10.1038/s41598.023.27528.0 - Uses openneuro_ds003061 auditory oddball dataset comparing brain-evoked potentials between correct responses to infrequent oddball sounds (1000Hz, 15%) and frequent standard sounds (500Hz, 70%) with epochs extracted from -1 to 2s around stimulus onset. Later analyses used -0.3 to 0.7s window.
- Kernel Notes: Verify events.tsv contains trial_type column for tone type verification. Filter to only correct responses as per paper methodology. Consider excluding distractor tones from classification. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- `STATUS` is a BioSemi hardware marker - not a task event; exclude from target_events.
- `response` events mark keypresses; sub-classification by tone type (standard/target/distractor) requires events.tsv lookup.
- 79-channel BioSemi EEG; check consistent montage across subjects.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG data from an auditory oddball task`

193. openneuro_ds003190 - Assessment of the visual stimuli properties in P300 paradigm

Source: analysis.md

Dataset: `openneuro_ds003190` | Year: 2020 | URL: https://openneuro.org/datasets/ds003190
Category: Sensory/Response | Subjects: 19 | Files (Completed): 384
Channels: 9 | Sampling Rate: 256 Hz

Paradigm

A P300 ERP experiment evaluating the effects of stimulus count and visual presentation styles (standard flash vs. cartoon face) on EEG brain activity. The experiment used a P300-based BCI paradigm with target and non-target stimulus flashes. Codes 1-9 represent different stimulus conditions or counts; codes 100-203 range encode specific stimulus configurations. The dataset has 19 subjects with many runs (384 files total).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003190

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1`	382 (files)	short	raw	P300 stimulus condition 1 (non-target flashx)
`2`	382 (files)	short	raw	P300 stimulus condition 2 (non-target flashx)
`3`	382 (files)	short	raw	P300 stimulus condition 3
`4`	382 (files)	short	raw	P300 stimulus condition 4
`200`	382 (files)	short	raw	P300 stimulus configuration 200
`201`	382 (files)	short	raw	P300 stimulus configuration 201
`202`	382 (files)	short	raw	P300 stimulus configuration 202
`203`	382 (files)	short	raw	P300 stimulus configuration 203

Event ID	Count (total)	Type	Meta Level	Meaning
101	363 (files)	short	raw	P300 target flash condition 101
102	354 (files)	short	raw	P300 target flash condition 102
103	368 (files)	short	raw	P300 target flash condition 103
104	370 (files)	short	raw	P300 target flash condition 104
5	326 (files)	short	raw	P300 stimulus condition 5
105	305 (files)	short	raw	P300 target flash condition 105
6	224 (files)	short	raw	P300 stimulus condition 6
106	212 (files)	short	raw	P300 target flash condition 106
7	169 (files)	short	raw	P300 stimulus condition 7
107	161 (files)	short	raw	P300 target flash condition 107
8	112 (files)	short	raw	P300 stimulus condition 8
108	110 (files)	short	raw	P300 target flash condition 108
9	56 (files)	short	raw	P300 stimulus condition 9
109	47 (files)	short	raw	P300 target flash condition 109

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Target vs non-target stimulus events
- Evidence: P300/oddball paradigm appears as core study objective. Source: https://openneuro.org/datasets/ds003190 DOI: https://doi.org/10.18112/openneuro.ds003190.v1.0.1

Normal Task 1: P300 Target vs Non-Target Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Specific kernel decodes numeric codes into semantic labels (target_flash, nontarget_flash). 9-channel EEG is very sparse; focus on occipital and parietal channels for P300. Regression target `response_time` available.

Event	y	Meaning
`target_flash`	0	Target stimulus flash (P300 target)
`nontarget_flash`	1	Non-target stimulus flash
## Data Quality Notes		

- Very sparse 9-channel EEG; limited spatial coverage.
- 384 files across 19 subjects = ~20 runs/subject; verify session structure.
- Numeric trigger codes require specific kernel to decode semantic P300 target/non-target labels.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Assessment of the visual stimuli properties in P300 paradigm`

194. openneuro_ds003490 - EEG: 3-Stim Auditory Oddball and Rest in Parkinson's

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003490 | Year: 2021 | URL: https://openneuro.org/datasets/ds003490
Category: Sensory/Response | Subjects: 50 | Files (Completed): 75
Channels: 67 | Sampling Rate: 500 Hz

Paradigm

A three-stimulus auditory oddball task (standard tones, target tones requiring button-press response, and novel distractor tones) combined with resting-state (eyes open and eyes closed) EEG recordings in Parkinson's disease patients and age-matched healthy controls. The study aims to identify auditory P300 and mismatch negativity differences between PD patients and controls. STATUS is a BioSemi admin marker.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003490

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Standard Tone`	75 (files)	short	raw	Frequent standard auditory tone (non-target)
`Target Tone`	75 (files)	short	raw	Infrequent target tone (requires button press)
`Novel Tone`	75 (files)	short	raw	Rare novel distractor tone
`Eyes Closed`	75 (files)	long	raw	Eyes-closed resting state block
`Eyes Open`	75 (files)	long	raw	Eyes-open resting state block
`STATUS`	75 (files)	admin	raw	BioSemi system status marker - exclude

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Group-level classification (PD patients vs Controls)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -2.0 s, tmax = 2.0 s, target_events = [`Standard Tone`, `Target Tone`, `Novel Tone`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Standard Tone`	0	Frequent standard tone (non-target)
`Target Tone`	1	Infrequent target tone (requires button press)
`Novel Tone`	2	Rare novel distractor tone

- Super Meta Notes: Derived from participant clinical status in participants.tsv. Labels: Control (healthy age-matched) -> 0, Parkinson's Disease (PD) -> 1. Also contains medication condition (ON/OFF) as secondary variable. Kernel step: Merge participant diagnosis from participants.tsv with epoch metadata to produce group labels.
- Evidence:
- DOI: 10.1016/j.clinph.2017.11.023 - SVM pattern classifiers on EEG responses to 3-stimulus auditory oddball paradigm to classify PD patients from age-matched controls, achieving 82% accuracy using novelty condition responses.
- Kernel Notes: Verify event ID mapping from events.tsv. Use any of the three tone types (Standard/Target/Novel) as stimulus events; paper uses all three tone conditions. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Medication state classification (PD ON vs OFF vs Controls)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -2.0 s, tmax = 2.0 s, target_events = [`Standard Tone`, `Target Tone`, `Novel Tone`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Standard Tone`	0	Frequent standard tone (non-target)
`Target Tone`	1	Infrequent target tone (requires button press)
`Novel Tone`	2	Rare novel distractor tone

- Super Meta Notes: Derived from participant metadata in participants.tsv. Labels: ON (Parkinson's on medication) -> 0, OFF (Parkinson's off medication) -> 1, CTRL (healthy control) -> 2. Kernel step: Merge medication condition and diagnosis from participants.tsv with epoch metadata to produce three-class labels.

- Evidence:
- DOI: 10.21528/cbic2023.027 - Uses Haralick and Fractal texture features from EEG epochs (±2s around auditory stimuli) to classify PD patients ON vs OFF medication vs healthy controls.
- Kernel Notes: Verify event ID mapping from events.tsv. Uses all three tone types as stimulus events.

Data Quality Notes

- STATUS is BioSemi hardware marker - exclude.
- Dataset includes both Parkinson's patients and age-matched controls; group label from participants.tsv.
- Three-stimulus oddball design provides Standard/Target/Novel tone categories for ERP analysis.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG: 3-Stim Auditory Oddball and Rest in Parkinson's`

195. openneuro_ds003522 - EEG: Three-Stimulus Auditory Oddball and Rest in Acute and Chronic TBI

Source: analysis.md

Dataset: openneuro_ds003522 | Year: 2021 | URL: https://openneuro.org/datasets/ds003522
Category: Sensory/Response | Subjects: 62 | Files (Completed): 196
Channels: 65 | Sampling Rate: 500 Hz

Paradigm

This study investigates EEG markers in mild traumatic brain injury (mTBI) using a three-stimulus auditory oddball task and resting-state recordings. Participants (sub-acute mTBI patients, chronic mTBI patients, and healthy controls) passively listened to standard, novel, and target tones while performing a target detection task. Resting-state blocks with eyes open and eyes closed were also recorded. The P3a and P3b ERP components - reflecting bottom-up and top-down attentional processes - are the primary measures examined as TBI biomarkers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003522

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Standard Tone`	194	short	raw	Frequent standard auditory tone (non-target)
`Novel Tone`	194	short	raw	Rare novel/distractor tone eliciting P3a
`Target Tone`	194	short	raw	Rare target tone requiring button press (P3b)
`Eyes Open`	196	rest	raw	Eyes-open resting-state block marker
`Eyes Closed`	196	rest	raw	Eyes-closed resting-state block marker
`STATUS`	196	admin	raw	System status marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`

- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Target vs non-target stimulus events
- Evidence: P300/oddball paradigm appears as core study objective. Source: https://doi.org/10.1016/j.neuropsychologia.2019.107125

Normal Task 1: Oddball three-category classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Classic 3-stimulus oddball. Standard tones are ~80% frequent (~184/file), novel and target each ~10% (~38/file). Class imbalance: undersample standards or weight loss. Eyes-open/closed events are resting-state block markers lasting several minutes each and are not trial-level stimuli.

Event	y	Meaning
`Standard Tone`	0	Standard frequent tone
`Novel Tone`	1	Novel interference tone (P3a)
`Target Tone`	2	Target pitch (P3b)
## Data Quality Notes		

- Eyes Open/Closed markers are resting block delimiters, not instantaneous trial events; do not epoch them as stimuli.
- Some files contain "Eyes Open: Every 1000 ms" style sub-event annotations; these are timing markers within rest blocks.
- Clinical groups (acute mTBI, chronic mTBI, controls) should be used as stratification variables.

References

- Reference: `ERPs predict symptomatic distress and recovery in sub-acute mild traumatic brain injury. Neuropsychologia, 2019. DOI: 10.1016/j.neuropsychologia.2019.107125`

196. openneuro_ds003570 - EEG: Improvisation and Musical Structures (Auditory Oddball Chords)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003570` | Year: 2021 | URL: https://openneuro.org/datasets/ds003570
Category: Sensory/Response | Subjects: 34 | Files (Completed): 34
Channels: 64 | Sampling Rate: 2048 Hz

Paradigm

Participants listened to sequences of three-chord musical progressions in an oddball paradigm designed to detect functional and exemplar deviants occurring within a stream of standard chord progressions. The study investigates neural responses (mismatch negativity, P600) to harmonic violations and musical structure deviants, probing the auditory system's predictive coding of musical sequences. Only STATUS system markers were detected in the EEG data; the chord-level event annotations may be stored externally.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003570

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
STATUS	34	admin	raw	System status marker (only detected event)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: STATUS
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Auditory Oddball Regression (Brain Connectivity Predicting Musical Training)

- Task Type: regression

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.4 s, tmax = 1.2 s, target_events = [STATUS]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`standard` (derived)	N/A	85% of trials: no harmonic change (predictor baseline)
`exemplar_deviant` (derived)	N/A	7.5% of trials: same functional class but different chord inversion
`function_deviant` (derived)	N/A	7.5% of trials: different functional class (harmonic violation)
- Super Meta Notes: Derived from experimental paradigm annotations. Kernel must: (1) Identify chord onsets from STATUS markers using external stimulus timing log; (2) Classify each trial as standard/exemplar_deviant/function_deviant based on chord sequence; (3) Compute connectivity (e.g., coherence, PLV) and/or bandpower for each condition; (4) Compute difference features: Psi(Ex) - Psi(Fu) and log.BP(Ex) - log.BP(Fu); (5) Train regression model to predict: imp.hours (weekly improvisation hours), cl.hours (weekly classical hours), dIES (difference in inverse efficiency score).
- Evidence:
- DOI: `10.1016/j.neuroimage.2021.118458` - Uses OpenNeuro ds003570 EEG data to predict self-reported improvisation and classical training hours from brain connectivity differences between exemplar and function deviants in an auditory oddball paradigm with epochs extracted -400 to 1200ms relative to second chord onset.
- Kernel Notes: Requires external events.tsv or stimulus timing log to map STATUS markers to chord types. The raw EEG contains only STATUS admin markers; chord-level event annotations must be reconstructed. Three analysis windows: 0-400ms between chords, -400 to 0ms pre-response, 0-400ms post-response. (replaces Primary Task 1 from original analysis.md)

Normal Task 1: Continuous EEG segmentation

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `stim`
- Target Definition: N/A (null (no labels, for unsupervised analysis))
- Notes: Only STATUS admin markers detected; no chord-level event annotations in EEG triggers. High sampling rate (2048 Hz). If events.tsv contains chord onset information, epoching by event may be feasible with custom preprocessing.

Data Quality Notes

- Very high sampling rate (2048 Hz); high memory requirements.
- No usable task events detected in EEG triggers beyond STATUS admin marker.
- Chord-level event timing may require reconstruction from audio or behavioral logs.

References

- Reference: `EEG dataset on improvisation and musical structures. OpenNeuro ds003570.`

197. openneuro_ds003645 - Face Processing MEEG Dataset with HED Annotation

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003645 | Year: 2023 | URL: https://openneuro.org/datasets/ds003645
Category: Sensory/Response | Subjects: 18 | Files (Completed): 114
Channels: 395 | Sampling Rate: 1100 Hz

Paradigm

Participants performed a simple perceptual task judging the spatial symmetry of famous, unfamiliar, and scrambled faces during simultaneous MEG and EEG (MEEG) recording. This is a replication of the Wakeman and Henson (2015) dataset with Hierarchical Event Descriptor (HED) annotations added. The study serves as a reference dataset for multimodal neuroimaging analysis methods, with face symmetry judgment as the overt task while examining neural responses to face identity and familiarity.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003645

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
-	-	-	raw	No EEG trigger markers detected; events encoded in HED annotations or MEEG-specific channels

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Three-Class Face Type Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 1.2 s, target_events = [`famous`, `unfamiliar`, `scrambled`]
- Label Route: `Annotation`
- Meta Level: raw

- Target Definition:
Event	y	Meaning
`famous`	0	Famous/familiar face stimulus
`unfamiliar`	1	Unfamiliar (non-famous) face stimulus
`scrambled`	2	Scrambled face stimulus (control)
- Evidence:
- DOI: `10.1038/sdata.2015.1` - Wakeman and Henson (2015) multi-modal dataset; participants performed symmetry judgment on familiar, unfamiliar, and scrambled faces; epochs extracted from -500 to +1200 ms around stimulus onset for ERP analysis comparing face types.
- Kernel Notes: Event IDs derived from HED-annotated trial_type in events.tsv. Verify face condition labels from dataset's events.tsv (likely values: 'famous', 'unfamiliar', 'scrambled'). Extract EEG channels only (exclude MEG channels). High sampling rate (1100 Hz) may require decimation for standard ERP analysis. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- MEEG dataset; 395 channels include both MEG and EEG sensors. Isolate EEG channels before analysis.
- No EEG trigger events detected in the EEG .set files; events may be in MEEG-format files or HED sidecar.
- Very high sampling rate (1100 Hz) and channel count; high computational requirements.

References

- Reference: `Wakeman and Henson (2015) / Robbins et al. (2021). Face processing MEEG dataset with HED annotation.`

198. openneuro_ds003690 - EEG, ECG and Pupil Data from Young and Older Adults: Rest and Auditory Cued Reaction Time Tasks

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003690` | Year: 2021 | URL: https://openneuro.org/datasets/ds003690
Category: Sensory/Response | Subjects: 75 | Files (Completed): 375
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Young and older adults performed auditory-cued simple reaction time and go/no-go tasks to investigate age-related differences in neural and physiological activity. An auditory cue preceded each trial; participants pressed a button to go stimuli and withheld responses to no-go stimuli. Error feedback tones were provided after commission errors. Resting-state EEG was also recorded. The study examines how aging affects preparatory neural activity, inhibitory control, and pupil dilation responses.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: reaction time
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003690

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`cue`	375	short	raw	Auditory warning cue preceding stimulus
`go`	300	short	raw	Go stimulus (respond with button press)
`no-go`	150	short	raw	No-go stimulus (withhold response)
`button press`	300	short	raw	Participant button-press response
`error of commission`	64	short	raw	Commission error (response to no-go)
`error feedback tone`	179	short	raw	Auditory error feedback tone
`task start`	320	short	raw	Task block start marker
`task end`	375	short	raw	Task block end marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: reaction_time
- Kernel Note: Keep derivation source, unit, and version metadata.

Tasks

Primary Task 1: Go/No-go Response Inhibition Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 6.0 s (relative to cue onset), target_events = [cue]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:
Event	y	Meaning
go	1	Go stimulus (response required)
no-go	0	No-go stimulus (withhold response)
- Evidence:
- DOI: 10.1016/j.neuroimage.2019.06.004 - EEG/ECG/pupil data from young and older adults performing cued go/no-go auditory tasks; cue-locked epochs from -0.2s to 6s used for ERP analysis
- Kernel Notes: Epochs are time-locked to cue events; labels derived from subsequent go vs no-go events within the trial. (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Correct/Error Suppression Classification (binary)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Commission errors (64 total) are rare compared to correct inhibitions; severe class imbalance requires oversampling or weighted loss.

Event	y	Meaning
no-go (correct inhibition)	0	Correct inhibition no-go
error of commission	1	Error response (key pressed during no-go)

Data Quality Notes

- 66-channel EEG; includes ECG and pupil data (non-EEG channels should be excluded).
- Longitudinal design with 5 sessions per subject; verify session consistency.

- Age group comparison (young vs older) is the primary experimental variable.

References

- Reference: `Age-related differences in EEG, ECG and pupilography during auditory cued reaction time tasks. OpenNeuro ds003690.`

199. openneuro_ds003800 - Auditory Gamma Entrainment

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003800 | Year: 2021 | URL: https://openneuro.org/datasets/ds003800
Category: Sensory/Response | Subjects: 8 | Files (Completed): 24
Channels: 19 | Sampling Rate: 250 Hz

Paradigm

Elderly participants with dementia received 40 Hz amplitude-modulated (AM) auditory stimulation to induce gamma-band entrainment of brain oscillations. The paradigm alternated between active 40 Hz AM stimulation (Stimulus) periods and quiet rest periods. The study investigates whether auditory gamma entrainment can modulate neural oscillations in a dementia population, potentially as a therapeutic intervention. EEG was recorded using a standard 10-20 system.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003800

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus`	13	long	raw	40 Hz AM auditory stimulation period onset
`Rest`	13	long	raw	Resting/quiet period

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Auditory Gamma Entrainment - Stimulus vs Rest Classification

- Task Type: classification

- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 40.0 s (Stimulus) / 20.0 s (Rest), overlap = 0.0, target_events = [`Stimulus`, `Rest`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Stimulus`	1	40 Hz AM auditory stimulation period
`Rest`	0	Resting/quiet period

- Evidence:
- DOI: 10.1016/j.dib.2022.107839 - Describes binary classification of stimulus vs rest during 40 Hz auditory gamma entrainment with stimulus trials of 40s duration and rest trials of 20s duration, interleaved in a 340s session (6x40s stimulus + 5x20s rest)
- Kernel Notes: Extract epochs spanning each Stimulus period (40 s) and Rest period (20 s). Stimulus and Rest are long events; use tmax equal to actual annotation duration for each event. (replaces Primary Task 1 from original analysis.md)

Primary Task 2: Entrained vs Non-Entrained Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 40.0 s, overlap = 0.0, target_events = [`Stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Entrained`	1	Participants showing gamma response >=3 SD above mean at 40 Hz
`Non-Entrained`	0	Participants showing gamma response <3 SD above mean at 40 Hz

- Super Meta Notes: Derived from raw event `Stimulus` by computing power spectral density in 38-42 Hz band for each Stimulus epoch and comparing peak amplitude to 3 standard deviations above the mean of adjacent frequencies. Kernel step: Compute FFT for each Stimulus epoch, extract 40 Hz amplitude, derive threshold from adjacent frequencies (38-42 Hz), classify each subject as Entrained (1) or Non-Entrained (0). (replaces Primary Task 2 from original analysis.md)
- Evidence:
- DOI: 10.18178/ijpmbs.13.1.14-18 - Uses Auditory Gamma Entrainment dataset to classify patients as entrained or non-entrained based on gamma power response to 40 Hz auditory stimulation, where entrained = peak 40Hz response >=3 SD above mean of adjacent frequencies

Normal Task 1: Stimulation period vs resting period (binary, long event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Stimulation and rest periods are sustained states (likely 1-5 min each); use long-event epoching. Only 8 subjects and 24 files - very small dataset. 19-channel EEG (10-20 system).

Event	y	Meaning
`Stimulus`	1	40 Hz AM auditory stimulation activation period

Event	y	Meaning
Rest	0	Rest/quiet period

Data Quality Notes

- Very small dataset (8 subjects, 24 files); results may not generalize.
- Elderly dementia population; data quality may be variable.
- 19-channel 10-20 EEG; limited spatial resolution.
- 250 Hz sampling rate; adequate for gamma entrainment analysis up to 125 Hz.

References

- Reference: Auditory Gamma Entrainment in Elderly with Dementia. OpenNeuro ds003800.

200. openneuro_ds003825 - Human electroencephalography recordings from 50 subjects for 22,248 images from 1,854 object concepts

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003825 | Year: 2022 | URL: https://openneuro.org/datasets/ds003825
Category: Sensory/Response | Subjects: 50 | Files (Completed): 100
Channels: 64 | Sampling Rate: 250 Hz

Paradigm

Participants viewed a large-scale set of 22,248 images spanning 1,854 object concepts presented in rapid serial visual presentation (RSVP) streams at 10 Hz while performing a fixation color change detection task. The study aimed to record high-resolution neural representations of everyday objects across a large stimulus set. Each image was presented for 100 ms in continuous streams, and participants responded to a concurrent fixation task to maintain attention.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003825

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`E 1`	~1,237,200	short	raw	Image onset (RSVP stimulus)
`E 2`	~1,237,200	short	raw	Second trial-type marker (redundant or condition flag)
`E 3`	~4,200	short	raw	Block/sequence boundary or target event

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: RSVP Image Onset Classification (Raw Events)

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s, target_events = [`E 1`, `E 2`, `E 3`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`E 1`	0	RSVP image onset (primary stimulus stream)
`E 2`	1	Secondary condition marker (stimulus offset)
`E 3`	2	Sequence boundary/target event

- Evidence:
- DOI: 10.1038/s41597.021.01102.7 - Paper describes creating epochs from -100 to 1000 ms relative to stimulus onset (E1 triggers) and using linear discriminant classifiers to decode between 1,854 object concepts and 200 validation images presented in RSVP streams at 10 Hz.
- Kernel Notes: Verify event ID mapping from events.tsv. E 1 is the primary stimulus onset marker for RSVP image presentation. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: RSVP Object Concept Classification (Super Meta)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s, target_events = [`E 1`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`E 1` + stimulus design file	0-1853	One of 1,854 object concepts (image identity)

- Super Meta Notes: Derived from raw event `E 1` combined with experimental design/stimulus order files (stimuli definitions). Kernel step: Load experimental design file to map each trial's timestamp to its corresponding image concept ID (0-1853), then use this as the class label for each epoch.
- Evidence:
- DOI: 10.1038/s41597.021.01102.7 - Paper presents THINGS-EEG dataset of EEG responses to 22,248 images from 1,854 object concepts presented in RSVP streams at 10Hz, with epochs extracted from -100 to 1000ms relative to stimulus onset, and performs pairwise classification decoding between images and concepts.
- Kernel Notes: Requires reading external stimulus design file (provided with dataset) to map each E 1 timestamp to its image concept class label (1,854 classes). This is a super-meta task because image identity is not stored in the EEG events.

Data Quality Notes

- Data from derivatives/eeglab directory; preprocessed with EEGLAB
- Very high epoch count (~24,648 E 1 events per subject); memory management required during HDF5 generation
- Sub-06 has fewer events (~12,360), suggesting a partial recording

References

- Reference: `Grootswagers, T., Zhou, I., Robinson, A.K. et al. Human EEG recordings for 1,854 concepts presented in rapid serial visual presentation streams. Sci Data 9, 3 (2022).`

201. openneuro_ds004010 - MAVIS

Source: analysis.md

Dataset: openneuro_ds004010 | Year: 2022 | URL: https://openneuro.org/datasets/ds004010
Category: Sensory/Response | Subjects: 24 | Files (Completed): 24
Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

A multisensory detection task (MAVIS) in which participants identified auditory and visual targets embedded within simultaneous audio-visual noise streams to investigate selective attention and crossmodal sensory processing. The dataset contains continuous EEG recordings without discrete event markers, suggesting the stimuli were presented as ongoing streams rather than discrete trials.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004010

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Publisher-defined BIDS task: MAVIS

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation

- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004010 DOI: https://doi.org/10.18112/openneuro.ds004010.v1.0.0

Normal Task 1: Continuous EEG Segmentation (resting/continuous)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: None (no trial-level events available)
- Notes: No event markers present. Use continuous segmentation for downstream tasks such as subject identification or spectral analysis. Task condition labels (if any) must come from external behavioral logs or stimulus delivery system.

Data Quality Notes

- No trial-level event markers; only continuous EEG available
- Stimulus delivery information may be in auxiliary files outside the BIDS structure
- 64 channels at 1000 Hz provides good temporal resolution for auditory processing analysis

References

- Reference: `MAVIS`

202. openneuro_ds004018 - EEG recordings for 200 object images presented in RSVP sequences at 5Hz or 20Hz

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004018` | Year: 2022 | URL: https://openneuro.org/datasets/ds004018
Category: Sensory/Response | Subjects: 2 | Files (Completed): 2
Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

Rapid Serial Visual Presentation (RSVP) of 200 object images at two presentation frequencies (5 Hz and 20 Hz) to investigate the temporal dynamics of visual object representations. Images were categorized as "stars" or "boats" (two superordinate categories), allowing investigation of how presentation rate affects category-level neural decoding. This is a very small dataset (2 subjects).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004018

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stars`	2 files	short	raw	RSVP onset - star/celestial category image
`boats`	2 files	short	raw	RSVP onset - boat/vehicle category image

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: RSVP Category Decoding (Stars vs Boats)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s, target_events = [`stars`, `boats`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`stars`	0	RSVP onset - star/celestial category image
`boats`	1	RSVP onset - boat/vehicle category image

- Evidence:
- DOI: 10.1016/j.neuroimage.2018.12.046 - Category-level decoding of RSVP stimuli including "stars" and "boats" superordinate categories using LDA classifiers on epochs from -100 to 1000ms
- Kernel Notes: Extract epochs aligned to stimulus onset markers for each category. Labels 0/1 correspond to celestial vs vehicle superordinate categories. (replaces Normal Task 1 and Normal Task 2 from original analysis.md)

Normal Task 1: Raw-meta task

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Needs verification
- Kernel Notes: Exclude admin markers from target labels.

Normal Task 2: Super-meta task

- Task Type: regression/classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target / Label: Needs verification
- Kernel Notes: Derive and version super-meta in kernel.

Data Quality Notes

- Only 2 subjects - extremely limited statistical power for group-level analysis
- RSVP at 5 Hz and 20 Hz: at 20 Hz, images appear for only 50 ms (near threshold for conscious processing)
- Verify that `stars` and `boats` events correspond to onset markers and not sustained states

References

- Reference: `The representational dynamics of visual objects in rapid serial visual processing streams. NeuroImage, 188, 668-679`

203. openneuro_ds004024 - TMS-EEG-MRI-fMRI-DWI data on paired associative stimulation and connectivity (Shirley Ryan AbilityLab, Chicago, IL)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004024` | Year: 2022 | URL: https://openneuro.org/datasets/ds004024
Category: Sensory/Response | Subjects: 497 files | Files (Completed): 497
Channels: 69 | Sampling Rate: 20000 Hz

Paradigm

A multi-modal neuroimaging study using Paired Associative Stimulation (PAS) combined with TMS-EEG to investigate cortical connectivity and neuroplasticity. Participants received precisely timed TMS pulses paired with peripheral nerve stimulation, and EEG was recorded at very high sampling rate (20 kHz) to capture TMS-evoked potentials. The study also collected MRI, fMRI, and DWI data. This is a clinical/neurophysiology dataset focusing on brain connectivity assessment.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004024

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/A`	345 files	short	raw	TMS or PAS stimulus onset
`Out/A`	378 files	short	raw	Stimulus output marker
`SB/A`	33 files	short	raw	Sub-threshold baseline stimulus
`Video/B`	32 files	short	raw	Video instruction marker
`New Segment/`	497 files	admin	raw	Recording segment boundary - exclude

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/A`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: ccPAS Asynchrony Classification (LTP vs Control vs LTD)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`Stimulus/A`] *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
(derived)	0	LTD - 4ms asynchrony (decreased connectivity)
(derived)	1	No-change - 9ms asynchrony (control)
(derived)	2	LTP - 14ms asynchrony (increased connectivity)

- Super Meta Notes: Derived from raw event `Stimulus/A` by extracting asynchrony condition markers from the trial metadata. Kernel step: Identify which trials correspond to 4ms, 9ms, and 14ms asynchrony conditions based on experimental design files; assign labels 0/1/2 accordingly.
- Evidence:
- DOI: 10.1016/j.neurom.2022.10.045 - Three-class ccPAS asynchrony detection (14ms LTP vs 9ms no-change vs 4ms LTD) based on TMS-evoked EEG connectivity changes in the 5-20ms post-stimulus window
- Kernel Notes: Verify event ID mapping from events.tsv. The asynchrony condition labels may need to be derived from separate sidecar JSON/BIDS files or extracted from the event stream metadata. Consider the Before vs After[1] vs After[2] comparison epochs as described in the paper.

Primary Task 2: Single-Pulse TMS Evoked Potential Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`Stimulus/A`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Stimulus/A`	0	TMS/PAS stimulus pulse - main evoked potential event

- Evidence:
- DOI: 10.1016/j.neurom.2022.10.045 - Uses single-pulse TMS (spTMS) before and after ccPAS to test cortico-cortical evoked potentials (ccEPs)
- Kernel Notes: The paper uses spTMS epochs to assess connectivity changes. Very high sampling rate (20 kHz) - downsampling recommended. TMS artifact cleaning required for first ~10ms post-stimulus.

Normal Task 1: TMS-Evoked Potential Epoching (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`

- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: New Segment/ is an admin boundary marker and must be excluded. Very high sampling rate (20 kHz) - downsampling to 1-5 kHz strongly recommended before HDF5 generation. The TMS artifact (first ~10 ms post-pulse) requires specialized cleaning before analysis.

Event	y	Meaning
`Stimulus/A`	0	TMS/PAS stimulus pulse - main evoked potential event
`SB/A`	1	Sub-threshold baseline TMS pulse

Data Quality Notes

- Very high sampling rate (20,000 Hz) - downsampling required
- TMS artifact contaminates first ~10 ms post-stimulus; specialized TMS-EEG artifact removal (e.g., SSP, ICA) is required
- New Segment/ is an admin boundary marker to exclude

References

- Paper: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. Scientific Data, 6, 103`

204. openneuro_ds004043 - The time-course of feature-based attention effects dissociated from temporal expectation and target-related processes

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004043` | Year: 2022 | URL: https://openneuro.org/datasets/ds004043
Category: Sensory/Response | Subjects: 20 | Files (Completed): 40
Channels: 63 | Sampling Rate: 250 Hz

Paradigm

Participants viewed rapid sequences of overlaid oriented grating pairs (presented at ~10 Hz) and detected a target orientation while attention was cued to different feature dimensions (orientation or color). The study dissociates the time-course of feature-based attention effects from temporal expectation and target-detection processes using EEG frequency-tagging and ERP methods.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004043

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`E 1`	20 files	short	raw	Grating stimulus onset - condition 1 (e.g., attended feature)
`E 2`	20 files	short	raw	Grating stimulus onset - condition 2
`E 3`	20 files	short	raw	Grating stimulus onset - condition 3 (e.g., target/distractor)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Feature-Based Attention - Cued vs Uncued Orientation Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [`E 1`, `E 2`, `E 3`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`E 1`	0	Cued/attended orientation stimulus (non-target trial)
`E 2`	1	Uncued/ignored orientation stimulus (non-target trial)
`E 3`	2	Target or distractor stimulus (excluded from primary analysis)

- Super Meta Notes: (None - raw meta)
- Evidence:
- DOI: 10.1038/s41598.022.10687.x - MVPA decoding of cued vs uncued orientation from EEG epochs (-100ms to 800ms) in non-target trials, using LDA classifiers
- DOI: 10.1101/2021.05.24.445376 - Multivariate pattern analysis decoding cued vs uncured orientation information over time from -100ms to 800ms post-stimulus
- Kernel Notes: Extract epochs from -100ms to 800ms relative to stimulus onset. Filter to non-target trials only (exclude target-related trials). Binary classification: cued (E 1) vs uncued (E 2) orientation. E 3 events represent targets/foils and should be excluded per the paper's analysis focus on non-target trials to control for target-related confounds. Verify event ID mapping from events.tsv - E 1 represents cued/attended, E 2 represents uncued/ignored based on paper methodology.

Normal Task 1: Attention Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 20 subjects x 2 sessions = 40 files. E 1/E 2/E 3 are EEGLAB-style markers; full decoding requires consulting the original study events.tsv. Balanced across conditions per session.

Event	y	Meaning
`E 1`	0	Attention condition 1
`E 2`	1	Attention condition 2
`E 3`	2	Attention condition 3 (target or distractor)

Data Quality Notes

- Events encoded as generic E markers; semantic meaning requires events.tsv inspection
- 20 subjects x 2 files suggests two sessions or two task conditions per subject
- 250 Hz sampling rate; adequate for ERP/steady-state visual evoked potential analysis

References

- Reference: `Undivided attention: The temporal effects of attention dissociated from decision, memory, and expectation`

205. openneuro_ds004151 - Effect of obesity on inhibitory control in preadolescents during stop-signal task. An event-related potentials study

Source: analysis.md

Dataset: `openneuro_ds004151` | Year: 2022 | URL: https://openneuro.org/datasets/ds004151
Category: Sensory/Response | Subjects: 57 | Files (Completed): 57
Channels: 128 | Sampling Rate: 500 Hz

Paradigm

Preadolescents (obese and normal-weight, ages 9-10) performed a stop-signal task to investigate the impact of obesity on reactive inhibitory control using ERPs. Participants respond to a go stimulus (trigger code 1) but must suppress the response when a stop signal follows (trigger code 2). The study tests whether obese children show electrophysiological differences in inhibition-related brain activity (N2, P3 components) compared to normal-weight peers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004151

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1`	~14,364	short	raw	Go trial stimulus onset
`2`	~6,156	short	raw	Stop-signal trial (inhibition required)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Stop signal task

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://doi.org/10.1016/j.ijpsycho.2021.04.003

Normal Task 1: Raw-meta task

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Needs verification
- Kernel Notes: Exclude admin markers from target labels.

Normal Task 2: Super-meta task

- Task Type: regression/classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target / Label: Needs verification
- Kernel Notes: Derive and version super-meta in kernel.

Data Quality Notes

- Event IDs are numeric (1, 2); decoded from stop-signal paradigm description
- 128-channel EGI cap; verify consistent montage across subjects
- Class imbalance: ~70% go trials vs ~30% stop-signal trials

References

- Reference: `Effect of obesity on inhibitory control in preadolescents during stop-signal task (DOI: 10.1016/j.ijpsycho.2021.04.003)`

206. openneuro_ds004252 - Rotation-tolerant representations elucidate the time course of high-level object processing

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004252 | Year: 2022 | URL: https://openneuro.org/datasets/ds004252
Category: Sensory/Response | Subjects: 1 | Files (Completed): 1
Channels: 127 | Sampling Rate: 1000 Hz

Paradigm

A single participant viewed visual stimuli of objects at various rotations to investigate the neural time course of rotation-tolerant, high-level object representations. A rapid serial visual presentation (RSVP) design was used with three event types encoding distinct stimulus categories or conditions. The study is a single-subject pilot or proof-of-concept recording.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004252

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Event/E 1`	14400	short	raw	Stimulus condition 1 (object/rotation category A)
`Event/E 2`	14400	short	raw	Stimulus condition 2 (object/rotation category B)
`Event/E 3`	120	short	raw	Stimulus condition 3 (object/rotation category C)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Rotation-tolerant object representation decoding

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [`Event/E 1`, `Event/E 2`, `Event/E 3`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Event/E 1`	0	Stimulus condition 1 (object/rotation category A)
`Event/E 2`	1	Stimulus condition 2 (object/rotation category B)
`Event/E 3`	2	Stimulus condition 3 (object/rotation category C)

- Evidence:
- DOI: 10.31234/osf.io.wp73u - This study used EEG data from OpenNeuro ds004252 to decode 60-way object classification using multivariate pattern analysis, comparing fixed-rotation decoding with rotation-tolerant decoding, with epochs from -100ms to 800ms relative to stimulus onset.
- Kernel Notes: Epoch parameters (tmin=-0.1s, tmax=0.8s) are derived from the paper's stated epoch window (-100ms to 800ms). The paper performs 60-way object classification (60 objects from 12 categories x 5 exemplars), but the raw data contains only 3 event markers. For full 60-way classification, a derived label approach would be needed. Verify event ID mapping from events.tsv and clarify whether additional stimulus identity information is available in companion files. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Single subject; results not generalizable across subjects
- 127 channels (1 reference electrode excluded)
- Severe class imbalance: E 1 and E 2 have 14400 trials each vs. E 3 with only 120

References

- Reference: `Rotation-tolerant representations elucidate the time course of high-level object processing`

207. openneuro_ds004256 - Encoding of Sound Source Elevation in Human Cortex

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004256` | Year: 2022 | URL: https://openneuro.org/datasets/ds004256
Category: Sensory/Response | Subjects: 53 | Files (Completed): 53
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants performed an auditory deviant detection task and a one-back sound localization task to investigate the cortical encoding of sound source elevation. Stimuli consisted of sounds presented from different vertical elevations, with subjects indicating perceived location. The study uses a multiclass design with conditions likely corresponding to different elevation angles.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004256

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S 1`	53x320	short	raw	Auditory stimulus condition 1 (elevation angle)
`Stimulus/S 2`	53x320	short	raw	Auditory stimulus condition 2 (elevation angle)
`Stimulus/S 3`	53x320	short	raw	Auditory stimulus condition 3 (elevation angle)
`Stimulus/S 4`	53x320	short	raw	Auditory stimulus condition 4 (elevation angle)
`Stimulus/S 5`	23x320	short	raw	Auditory stimulus condition 5 (elevation angle)
`Stimulus/S 6`	23x320	short	raw	Auditory stimulus condition 6 (elevation angle)
`Stimulus/S 7`	23x96	short	raw	Auditory stimulus condition 7 (deviant/rare)

Event ID	Count (total)	Type	Meta Level	Meaning
`New Segment/`	53	admin	raw	Recording segment boundary
`BAD boundary`	53	admin	raw	Bad epoch boundary marker
`EDGE boundary`	53	admin	raw	Edge boundary artifact marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/S 1`, `Stimulus/S 2`, `Stimulus/S 3`, `Stimulus/S 4`, `Stimulus/S 5`, `Stimulus/S 6`, `Stimulus/S 7`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sound Elevation Classification (4-class)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s (probe onset), tmax = 0.9 s (covers 150-900ms analysis window), target_events = [`Stimulus/S 1`, `Stimulus/S 2`, `Stimulus/S 3`, `Stimulus/S 4`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Stimulus/S 1`	0	Elevation angle 37.5°
`Stimulus/S 2`	1	Elevation angle 12.5°
`Stimulus/S 3`	2	Elevation angle -12.5°
`Stimulus/S 4`	3	Elevation angle -37.5°

- Evidence:
- DOI: 10.1101/2023.05.03.539222 - Uses OpenNeuro dataset ds004256 to decode sound elevation from EEG responses using multivariate logistic regression, analyzing ERP differences between adapter-probe pairs at different vertical elevations (37.5°, 12.5°, -12.5°, -37.5°).
- Kernel Notes: Epochs extracted relative to probe onset (tmin=0s). Paper analyzes 100-300ms and 150-900ms post-probe windows; tmax=0.9s covers the later window. Exclude S5/S6/S7 from this task as they are not part of the 4-class elevation decoding. (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Full 7-class Stimulus Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition

- Label Route: `Annotation`
- Target Definition: Complete event-to-label mapping including all stimulus conditions.
- Notes: S5/S6/S7 present in only 23/53 subjects; requires subject-level filtering. BAD/EDGE boundaries excluded.

Event	y	Meaning
`Stimulus/S 1`	0	Auditory stimulus condition 1
`Stimulus/S 2`	1	Auditory stimulus condition 2
`Stimulus/S 3`	2	Auditory stimulus condition 3
`Stimulus/S 4`	3	Auditory stimulus condition 4
`Stimulus/S 5`	4	Auditory stimulus condition 5
`Stimulus/S 6`	5	Auditory stimulus condition 6
`Stimulus/S 7`	6	Auditory stimulus condition 7 (deviant)

Data Quality Notes

- S 5, S 6, S 7 appear in only 23 subjects (possibly a different task condition block)
- BAD boundary and EDGE boundary markers must not be used as target events
- Balance classes by undersampling; verify event counts per subject before pooling

References

- Reference: `Encoding of Sound Source Elevation in Human Cortex`

208. openneuro_ds004347 - Symmetry perception and affective responses: a combined EEG/EMG study

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004347 | Year: 2022 | URL: https://openneuro.org/datasets/ds004347
Category: Sensory/Response | Subjects: 24 | Files (Completed): 24
Channels: 73 | Sampling Rate: 512 Hz

Paradigm

Participants performed a visual perception task identifying whether stimuli were regular (symmetric) or random (asymmetric) patterns to investigate symmetry perception and affective responses. EEG and EMG were simultaneously recorded to capture both neural and facial muscle activity. No trial-level event markers are present in the EEG files; the dataset is processed as continuous recordings.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004347

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Visual Symmetry Perception Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5

- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`symmetry_block`	0	Symmetry condition (reflection/rotational patterns)
`asymmetry_block`	1	Asymmetry condition (random patterns)
- Evidence:
- DOI: `10.1016/j.neuropsychologia.2012.10.003` - This 2012 paper describes the experimental paradigm (symmetry vs random pattern classification) that underlies the openneuro_ds004347 dataset, using epochs time-locked to stimulus onset (1-3s duration) with 80 trials per condition per participant.
- Kernel Notes: No discrete event markers present in EEG files (`no_events`). Use sliding window segmentation on continuous recordings. Subject-level condition labels must be derived from BIDS sidecar files or experimental design metadata. Verify block timing from dataset annotations.

Normal Task 1: Continuous Segmentation (resting/segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (unsupervised segmentation)
- Notes: No trial markers in EEG files; use sliding window segmentation. Subject-level labels (symmetric vs. random condition blocks) may be available from experimental design metadata.

Data Quality Notes

- No trial-level event markers in the EEG files
- 73 channels includes EMG/EOG channels that should be excluded from EEG analysis
- Continuous recordings suitable for spectral or connectivity analysis

References

- Reference: `Symmetry perception and affective responses: a combined EEG/EMG study.`

209. openneuro_ds004356 - Subcortical responses to music and speech are alike while cortical responses diverge

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004356 | Year: 2024 | URL: https://openneuro.org/datasets/ds004356
Category: Sensory/Response | Subjects: 22 | Files (Completed): 24
Channels: 34 | Sampling Rate: 10000 Hz

Paradigm

The study measures subcortical Auditory Brainstem Responses (ABR) to continuous clicks, music genres, and speech stimuli using a deconvolution method. Participants passively listened to 12 different audio categories (Chinese speech, English speech, 8 music genres, acoustic guitar, and click trains) to compare subcortical and cortical neural encoding across stimulus types.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004356

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`click`	22x10	long	raw	Continuous click train stimulus
`acoustic`	22x40	long	raw	Acoustic guitar stimulus
`classical`	22x40	long	raw	Classical music stimulus
`jazz`	22x40	long	raw	Jazz music stimulus
`pop`	22x40	long	raw	Pop music stimulus
`metal`	22x40	long	raw	Metal music stimulus
`hiphop`	22x40	long	raw	Hip-hop music stimulus
`chn_aud`	22x40	long	raw	Chinese speech stimulus
`eng_aud`	22x40	long	raw	English speech stimulus
`news`	22x40	long	raw	News speech stimulus
`talk`	22x40	long	raw	Talk speech stimulus

Event ID	Count (total)	Type	Meta Level	Meaning
`interview`	22x40	long	raw	Interview speech stimulus
`lecture`	22x40	long	raw	Lecture speech stimulus

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Music vs Speech Classification from Subcortical and Cortical EEG Responses

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 12 s *, overlap = 0.0, target_events = [`acoustic`, `classical`, `jazz`, `pop`, `metal`, `hiphop`, `chn_aud`, `eng_aud`, `news`, `talk`, `interview`, `lecture`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`acoustic`	0	Music stimulus
`classical`	0	Music stimulus
`jazz`	0	Music stimulus
`pop`	0	Music stimulus
`metal`	0	Music stimulus
`hiphop`	0	Music stimulus
`chn_aud`	1	Speech stimulus
`eng_aud`	1	Speech stimulus
`news`	1	Speech stimulus
`talk`	1	Speech stimulus
`interview`	1	Speech stimulus
`lecture`	1	Speech stimulus

- Kernel Notes: Continuous 12s audio stimulus blocks. For subcortical ABR analysis: apply bandpass filter (100-2000 Hz) and extract [0, 15]ms window post-stimulus. For cortical analysis: apply bandpass filter (0.1-40 Hz) and extract [0, 200]ms window. Use deconvolution methods as described in the paper to model neural responses to continuous stimuli. (replaces Normal Task 1 from original analysis.md)
- Evidence:
- DOI: 10.1038/s41598.023.50438.0 - Binary classification of music vs speech stimuli from subcortical (ABR) and cortical EEG responses using deconvolution on continuous 12s stimulus blocks; finds highly correlated subcortical responses between music and speech but divergent cortical responses.

Data Quality Notes

- Very high sampling rate (10 kHz) designed for subcortical ABR; bandpass filtering required for cortical analysis
- 2 subjects have 2 files each (24 files, 22 subjects)
- Balanced stimulus presentation (40 trials per condition for most categories)

References

- Reference: Subcortical responses to music and speech are alike while cortical responses diverge (Shan et al., 2024)

210. openneuro_ds004357 - Features-EEG

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004357 | Year: 2024 | URL: https://openneuro.org/datasets/ds004357
Category: Sensory/Response | Subjects: 16 | Files (Completed): 16
Channels: 127 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed rapid serial visual presentation (RSVP) streams of Gabor-like stimuli at 6.67 Hz and 20 Hz while performing an orthogonal fixation color change detection task. Three event types encode different feature or condition categories, enabling investigation of how visual features are dynamically encoded in EEG responses.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004357

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Event/E 1`	16x~16000	short	raw	Stimulus feature condition 1
`Event/E 2`	16x~16000	short	raw	Stimulus feature condition 2
`Event/E 3`	16x~60	short	raw	Stimulus feature condition 3 (rare)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Visual Feature Decoding from RSVP Stimuli

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.6 s, target_events = [`Event/E 1`, `Event/E 2`, `Event/E 3`]
- Label Route: `Annotation`
- Meta Level: raw

- Target Definition:
Event	y	Meaning
`Event/E 1`	0	Stimulus feature condition 1
`Event/E 2`	1	Stimulus feature condition 2
`Event/E 3`	2	Stimulus feature condition 3 (rare)
- Evidence:
- DOI: 10.1038/s41598.025.27703.5 - MVPA decoding of visual grating stimuli (orientation, contrast, spatial frequency, color) from RSVP epochs -0.1s to 0.6s post-stimulus using OpenNeuro ds004357
- Kernel Notes: Extract epochs aligned to stimulus onset markers. The severe class imbalance (E3 has ~60 trials vs ~16000 for E1/E2) should be addressed in model training (class weights, oversampling, or focal loss).

Normal Task 1: Visual Feature Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Event/E 3 count (~60 per file) is far lower than E 1 and E 2 (~16000 each) - severe class imbalance. Decode exact condition mapping from paper. Similar structure to ds004252.

Event	y	Meaning
`Event/E 1`	0	Stimulus feature condition 1
`Event/E 2`	1	Stimulus feature condition 2
`Event/E 3`	2	Stimulus feature condition 3

Data Quality Notes

- Severe class imbalance: E 1 and E 2 have ~16000 trials each vs. E 3 with ~60 per subject
- 127-channel high-density EEG
- RSVP paradigm at 6.67 Hz and 20 Hz; stimuli are very brief

References

- Reference: `Mapping the Dynamics of Visual Feature Coding: Insights into Perception and Integration`

211. openneuro_ds004381 - Intraoperative EEG dataset during medianus-tibialis stimulation with 8 different rates

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004381 | Year: 2024 | URL: https://openneuro.org/datasets/ds004381
Category: Sensory/Response | Subjects: 18 | Files (Completed): 770
Channels: 4 | Sampling Rate: 20000 Hz

Paradigm

Intraoperative recording of Somatosensory Evoked Potentials (SEP) elicited by medianus and tibial nerve stimulation at eight different repetition rates to optimize signal-to-noise ratio for intraoperative monitoring. Recordings were made during neurosurgery using a minimal-channel invasive EEG setup with very high sampling rate for subcortical potential resolution.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004381

Event Structure

No event markers detected (no_events). This dataset contains continuous SEP recordings without trial-level BIDS annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: SEP SNR Regression by Stimulation Rate

- Task Type: regression
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 5.0 s *, overlap = 0.0 *

- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
continuous	0-1	Normalized SNR value (signal_amplitude / (0.5 * noise_rms))
- Super Meta Notes: Derived from stimulation rate metadata encoded in filenames (8 rates: 2.7, 4.7, 8.7, 12.7, 16.7, 20.7, 24.7, 28.7 Hz) by computing SNR from averaged post-stimulus sweeps (50 ms for medianus, 100 ms for tibial). Kernel step: (1) Parse stimulation rate from filename metadata, (2) High-pass filter (200 Hz) for artifact detection, (3) Sweep classification with amplitude threshold (>10 µV rejection), (4) Compute SNR = signal_amplitude / (0.5 * noise_rms), (5) Fit amplitude decay model Amplitude = a * Rate^k.
- Evidence:
- DOI: 10.1016/j.clinph.2023.03.008 - Uses openneuro_ds004381 dataset to optimize SEP stimulation rate by analyzing SNR as a function of stimulation rate (2.7-28.7 Hz) and recording duration (5-20 s), finding optimal rates of 12.7 Hz for short medianus SEP and 4.7 Hz for tibial SEP recordings
- Kernel Notes: No raw event markers exist in data (`no_events`). Stimulus onsets must be inferred from continuous data via high-pass filtering (200 Hz) and local peak detection for sweep extraction. Segment length should match analysis durations (5s, 10s, 20s). Verify stimulation rate mapping from filename metadata.

Normal Task 1: Continuous Segmentation (segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (unsupervised segmentation)
- Notes: No BIDS trial event markers; stimulation rate (8 levels) encoded in filename/metadata. SEP analysis requires specialized averaging pipeline rather than standard epoch classification.

Normal Task 2: Raw-meta extraction for metadata-driven epoching

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 0.05 s (medianus) / 0.1 s (tibial) *, target_events = [`no_events`] *
- Label Route: `Annotation`
- Target Definition:
Event	y	Meaning
`no_events`	N/A	Placeholder - actual epoching requires external trigger reconstruction
- Super Meta Notes: Raw data contains `no_events`. Stimulus timing must be reconstructed from metadata (stimulation rate) and high-pass filtered artifact detection. Kernel step: (1) Read stimulation rate from BIDS sidecar or filename, (2) Generate trigger timing from rate, (3) Apply to epoch_by_event_hdf5.
- Evidence:
- DOI: 10.1016/j.clinph.2023.03.008 - Confirms sweep length parameters (50 ms medianus, 100 ms tibial) and artifact rejection criteria
- Kernel Notes: This is a placeholder task. Actual implementation requires reconstructing stimulus triggers from metadata. (replaces Primary Task 1 from original analysis.md)

Data Quality Notes

- Very high sampling rate (20 kHz) for subcortical SEP; downsampling required for standard EEG analysis

- Only 4 channels (intraoperative minimal setup)
- 770 files across 18 subjects (many runs per subject, 8 stimulation rates)

References

- Reference: `Optimization of signal-to-noise ratio in short-duration SEP recordings by variation of stimulation rate`

212. openneuro_ds004388 - Somatosensory evoked potentials in the human spinal cord to mixed nerve stimulation

Source: analysis.md

Dataset: `openneuro_ds004388` | Year: 2023 | URL: https://openneuro.org/datasets/ds004388
Category: Sensory/Response | Subjects: 40 | Files (Completed): 399
Channels: 115 | Sampling Rate: 10000 Hz

Paradigm

The dataset records somatosensory evoked potentials (SEPs) in response to mixed median and tibial nerve stimulation, alongside a resting state condition. Simultaneous EEG and epidural spinal cord recordings (ESG) were acquired to characterize the spinal and cortical processing of somatosensory signals.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004388

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stimulus`	399x~500	long	raw	Electrical nerve stimulation onset

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Alternating

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004388 DOI: https://doi.org/10.18112/openneuro.ds004388.v1.0.0

Primary Task 2: Publisher-defined BIDS task: Median

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004388 DOI: https://doi.org/10.18112/openneuro.ds004388.v1.0.0

Primary Task 3: Publisher-defined BIDS task: Tibial

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004388 DOI: https://doi.org/10.18112/openneuro.ds004388.v1.0.0

Normal Task 1: SEP Stimulus Epoching (long event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s (relative to event onset), overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Only one event class; suitable for evoked potential analysis or regression on stimulation parameters. Multiple nerve sites and stimulation rates encoded in file naming conventions.

Event	y	Meaning
`stimulus`	0	Nerve stimulation onset
## Data Quality Notes		

- Very high sampling rate (10 kHz) for SEP/ESG; downsampling required for standard EEG analysis
- 115 channels includes spinal cord (ESG) and peripheral nerve (ENG/EMG) channels
- 399 files across 40 subjects (multiple runs/conditions per subject)

References

- Reference: `Somatosensory evoked potentials in the human spinal cord to mixed nerve stimulation`

213. openneuro_ds004389 - Somatosensory evoked potentials in the human spinal cord to mixed and sensory nerve stimulation

Source: analysis.md

Dataset: `openneuro_ds004389` | Year: 2023 | URL: https://openneuro.org/datasets/ds004389
Category: Sensory/Response | Subjects: 26 | Files (Completed): 260
Channels: 90 | Sampling Rate: 10000 Hz

Paradigm

Simultaneous EEG, epidural spinal cord (ESG), nerve conduction (ENG), and EMG recordings during resting state and various conditions of mixed and sensory nerve stimulation of the arms and legs to evoke somatosensory potentials. The study compares spinal cord vs. cortical potentials for different stimulation types and sites.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004389

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`stimulus`	260x~1500	long	raw	Electrical nerve stimulation onset

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Mediansensory

- Task Type: classification/segmentation

- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004389 DOI: https://doi.org/10.18112/openneuro.ds004389.v1.0.0

Primary Task 2: Publisher-defined BIDS task: Medianmixed

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004389 DOI: https://doi.org/10.18112/openneuro.ds004389.v1.0.0

Primary Task 3: Publisher-defined BIDS task: Tibialmixed

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004389 DOI: https://doi.org/10.18112/openneuro.ds004389.v1.0.0

Normal Task 1: SEP Stimulus Epoching (long event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s (relative to event onset), overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Only one event class; suitable for evoked potential analysis. Multiple nerve sites (median mixed, median sensory, tibial) and stimulation conditions encoded in file/run naming.

Event	y	Meaning
`stimulus`	0	Nerve stimulation onset
## Data Quality Notes		

- Very high sampling rate (10 kHz) for SEP/ESG; downsampling required for standard EEG analysis
- 90 channels includes ESG, ENG, EMG, EOG, ECG channels - filter to EEG channels only
- 260 files across 26 subjects (multiple runs/conditions per subject)

References

- Reference: `Somatosensory evoked potentials in the human spinal cord to mixed and sensory nerve stimulation.`

214. openneuro_ds004502 - Anticipatory differences between Attention and Expectation

Source: analysis.md

Dataset: `openneuro_ds004502` | Year: 2023 | URL: https://openneuro.org/datasets/ds004502
Category: Sensory/Response | Subjects: 38 | Files (Completed): 38
Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

An experimental task investigating neural mechanisms of anticipation by comparing stimulus-driven attention and predictive expectation. Participants attended to auditory or visual stimuli with varying validity cues to disentangle attention (stimulus-driven) from expectation (predictive) effects on neural anticipatory activity.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004502

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`S 17`	38xx	short	raw	BrainVision trigger - decode from events.tsv
`S 18`	38xx	short	raw	BrainVision trigger - decode from events.tsv
`S 19`	38xx	short	raw	BrainVision trigger - decode from events.tsv
`S 20`	38xx	short	raw	BrainVision trigger - decode from events.tsv
`S 21`	38xx	short	raw	BrainVision trigger - decode from events.tsv
`S 22`	38xx	short	raw	BrainVision trigger - decode from events.tsv
`S 23`	38xx	short	raw	BrainVision trigger - decode from events.tsv

Event ID	Count (total)	Type	Meta Level	Meaning
S 24	38xx	short	raw	BrainVision trigger - decode from events.tsv
S 33...S 57	38xx	short	raw	BrainVision trigger codes (22 codes) - decode from events.tsv

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: S 17, S 18, S 19, S 20, S 21, S 22, S 23, S 24
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Attexp

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004502 DOI: https://doi.org/10.18112/openneuro.ds004502.v1.0.1

Normal Task 1: Attention vs Expectation Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: 30 distinct BrainVision trigger codes (S 17-S 57 range) require semantic decoding from events.tsv. Grouping into attention vs. expectation conditions requires understanding the trigger code mapping.

Event	y	Meaning
S 17...S 24	0	Attention condition trials (S1x range)
S 33...S 57	1	Expectation condition trials (S3x-S5x range)
## Data Quality Notes		

- 30 distinct trigger codes require semantic decoding from events.tsv metadata

- All 30 codes appear in all 38 files with consistent counts
- 63 channels at 1000 Hz (BrainVision system)

References

- Reference: `Anticipatory differences between Attention and Expectation`

215. openneuro_ds004563 - Vicarious touch: overlapping neural patterns between seeing and feeling touch

Source: analysis.md

Dataset: openneuro_ds004563 | Year: 2023 | URL: https://openneuro.org/datasets/ds004563
Category: Sensory/Response | Subjects: 29 | Files (Completed): 29 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants either saw someone else being touched (vicarious touch, event 4) or felt touch themselves (direct touch, event 1). EEG was recorded to investigate overlapping neural patterns between observed and experienced tactile stimulation. The study tests whether vicarious touch engages somatosensory cortex similarly to directly felt touch.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004563

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
4	~29xN	short	raw	Seeing touch (vicarious touch condition)
1	~29xN	short	raw	Feeling touch (direct touch condition)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Publisher-defined BIDS task: Touch decoding

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004563 DOI: https://doi.org/10.1101/2022.06.21.497107

Normal Task 1: Raw-meta task

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Needs verification
- Kernel Notes: Exclude admin markers from target labels.

Normal Task 2: Super-meta task

- Task Type: regression/classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target / Label: Needs verification
- Kernel Notes: Derive and version super-meta in kernel.

Data Quality Notes

- 29 subjects with binary condition structure
- 64 channels at 512 Hz

References

- Reference: `Vicarious touch: overlapping neural patterns between seeing and feeling touch`

216. openneuro_ds004574 - Cross-modal Oddball Task

Source: analysis.md

Dataset: openneuro_ds004574 | Year: 2023 | URL: https://openneuro.org/datasets/ds004574
Category: Sensory/Response | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed an oddball task with cross-modal stimuli (visual and auditory). Standard stimuli (S 1), deviant stimuli (S 2), and rare stimuli (S 3) were presented in mixed modality streams. The study investigates the mismatch negativity (MMN) and P300 components elicited by deviant stimuli across sensory modalities.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004574

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
S 1	~30xN	short	raw	Standard stimulus
S 2	~30xN	short	raw	Deviant stimulus
S 3	~30xN	short	raw	Rare/target stimulus
boundary	~30xN	admin	raw	Recording boundary marker
S255	~30xN	admin	raw	BrainVision overflow/system marker

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: S 1, S 2, S 3, S255
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: ERP target detection (P300/Oddball)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Target vs non-target stimulus events
- Evidence: P300/oddball paradigm appears as core study objective. Source: https://openneuro.org/datasets/ds004574 DOI: https://doi.org/10.1101/2022.07.26.22278079

Normal Task 1: Oddball Stimulus Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: boundary and S255 are admin events - exclude from target_events. S255 is a BrainVision system marker (overflow/sync), not a task event.

Event	y	Meaning
`S 1`	0	Standard stimulus
`S 2`	1	Deviant stimulus
`S 3`	2	Rare/target stimulus
## Data Quality Notes		

- boundary and S255 must be excluded from epoching
- S255 is a BrainVision system marker, not a stimulus event
- 30 subjects with standard oddball structure

References

- Reference: `Cross-modal Oddball Task`

217. openneuro_ds004579 - Interval Timing Task

Source: analysis.md

Dataset: openneuro_ds004579 | Year: 2023 | URL: https://openneuro.org/datasets/ds004579
Category: Sensory/Response | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed an interval timing task where they estimated durations of different intervals. BrainVision S-codes (S 1-S 7) encode different timing conditions or interval lengths. Participants produced or reproduced time intervals by button press (R 3). The study investigates neural oscillations and temporal processing mechanisms.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004579

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
S 1	~30xN	short	raw	Timing condition 1
S 2	~30xN	short	raw	Timing condition 2
S 3	~30xN	short	raw	Timing condition 3
S 4	~30xN	short	raw	Timing condition 4
S 5	~30xN	short	raw	Timing condition 5
S 6	~30xN	short	raw	Timing condition 6
S 7	~30xN	short	raw	Timing condition 7
R 3	~30xN	short	raw	Button press response
boundary	~30xN	admin	raw	Recording boundary marker
S255	~30xN	admin	raw	BrainVision overflow/system marker

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `S 1`, `S 2`, `S 3`, `S 4`, `S 5`, `S 6`, `S 7`, `S255`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Interval Timing

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004579 DOI: https://doi.org/10.1101/2022.07.26.22278079

Normal Task 1: Timing Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: boundary and S255 are admin events - exclude from target_events. R 3 is a response event - can be used as an additional classification target. S255 is a BrainVision system marker (overflow/sync), not a task event.

Event	y	Meaning
`S 1`	0	Timing condition 1
`S 2`	1	Timing condition 2
`S 3`	2	Timing condition 3
`S 4`	3	Timing condition 4
`S 5`	4	Timing condition 5
`S 6`	5	Timing condition 6
`S 7`	6	Timing condition 7
## Data Quality Notes		

- boundary and S255 must be excluded from epoching
- S255 is a BrainVision system marker, not a task event
- 30 subjects with 7 timing conditions

References

- Reference: `Interval Timing Task`

218. openneuro_ds004580 - Simon-conflict Task

Source: analysis.md

Dataset: openneuro_ds004580 | Year: 2023 | URL: https://openneuro.org/datasets/ds004580
Category: Sensory/Response | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed the Simon conflict task in which stimulus location (left/right) could be congruent or incongruent with the required response direction. S 1 and S 2 encode congruent and incongruent trial types respectively; S 3 marks rare/catch trials. The study investigates conflict monitoring and executive control neural signatures (N2, P3 ERP components).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004580

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`S 1`	30x384	short	raw	Congruent trial (stimulus side matches response side)
`S 2`	30x383	short	raw	Incongruent trial (stimulus side opposite to response side)
`S 3`	~30xN	short	raw	Rare/catch trial
`boundary`	~30xN	admin	raw	Recording boundary marker
`S255`	~30xN	admin	raw	BrainVision overflow/system marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`

- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `S 1`, `S 2`, `S 3`, `S255`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Simon

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004580 DOI: https://doi.org/10.1101/2022.07.26.22278079

Normal Task 1: Congruent vs Incongruent Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s (relative to event onset), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: boundary and S255 are admin events - exclude from target_events. S255 is a BrainVision system marker (overflow/sync), not a task event. S 3 (rare trials) can be optionally included as a third class.

Event	y	Meaning
`S 1`	0	Congruent trial
`S 2`	1	Incongruent trial
## Data Quality Notes		

- boundary and S255 must be excluded from epoching
- Balanced S 1 and S 2 counts (~384 vs ~383 per subject)
- 30 subjects with consistent congruent/incongruent structure

References

- Reference: `Simon-conflict Task`

219. openneuro_ds004587 - IllusionGameEEG_data

Source: analysis.md

Dataset: `openneuro_ds004587` | Year: 2024 | URL: https://openneuro.org/datasets/ds004587
Category: Sensory/Response | Subjects: 103 | Files (Completed): 114 | Channels: 64 | Sampling Rate: 10000 Hz

Paradigm

Participants performed perceptual judgements on visual targets embedded within three classic visual illusions - Ebbinghaus, Muller-Lyer, and Vertical-Horizontal - to measure individual differences in illusion sensitivity and perceptual bias. The study investigates how prior experience and personality shape perceptual decision-making. EEG was recorded at 10 kHz; actual stimulus-event markers appear not to have been stored in events.tsv.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004587

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`New Segment/LostSamples: 2`	370	admin	raw	Lost-samples boundary marker (exclude)
`New Segment/`	115	admin	raw	BrainVision segment boundary (exclude)
`New Segment/LostSamples: 3`	111	admin	raw	Lost-samples boundary marker (exclude)
`New Segment/LostSamples: 4`	6	admin	raw	Lost-samples boundary marker (exclude)
`Comment/motion`	9	admin	raw	Motion artifact annotation (exclude)
`Comment/ro post`	6	admin	raw	Post-recording comment (exclude)
`Comment/error`	3	admin	raw	Error annotation (exclude)
`Comment/rc pre`	3	admin	raw	Pre-recording comment (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: IG

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004587 DOI: https://doi.org/10.18112/openneuro.ds004587.v1.0.0

Normal Task 1: Continuous Task EEG (segmentation)

- Task Type: regression
- Epoch Method: `segmentation`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: None
- Notes: All events in events.tsv are administrative markers or artifact annotations. No stimulus/response trial events were captured. Trial structure may be recoverable from separate behavioral logs. Use sliding-window segmentation.

Data Quality Notes

- All events are admin; no stimulus-locked epochs possible from events.tsv alone.
- Very high sampling rate (10 kHz) - downsample before analysis.
- Paper reference (ARSQ 2.0) may be a metadata mismatch; verify README.

References

- Reference: `The ARSQ 2.0 reveals age and personality effects on mind-wandering experiences`

220. openneuro_ds004602 - Registered Replication Report of ERN/Pe Psychometrics

Source: analysis.md

Dataset: openneuro_ds004602 | Year: 2023 | URL: https://openneuro.org/datasets/ds004602
Category: Sensory/Response | Subjects: 182 | Files (Completed): 546 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

Participants completed Flanker, Stroop, and Go/no-go tasks to evaluate the psychometric reliability and convergent/divergent validity of error-related ERP components (ERN and Pe). The study tests whether error-related components are stable within individuals across tasks and sessions. Three task files per subject yield the 546-file count. cor/err mark correct/error responses; con/inc mark congruent/incongruent Flanker/Stroop trials; go/ng/neu mark Go/no-go trial types.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004602

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
cor	169236	short	raw	Correct response (all tasks)
go	58251	short	raw	Go trial (Go/no-go task)
con	54498	short	raw	Congruent trial (Flanker/Stroop)
inc	47242	short	raw	Incongruent trial (Flanker/Stroop)
err	24198	short	raw	Error response (all tasks)
neu	20890	short	raw	Neutral trial condition
ng	8636	short	raw	No-go trial (Go/no-go task)
boundary	545	admin	raw	Segment boundary marker (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Mixed

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004602 DOI: https://doi.org/10.18112/openneuro.ds004602.v1.0.1

Normal Task 1: Error vs Correct Classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: boundary (545) is admin - exclude. cor (169,236) >> err (24,198) - significant class imbalance; undersample cor or use weighted loss. con/inc/go/ng/neu usable for separate task-level analyses.

Event	y	Meaning
`cor`	0	Correct response (ERN/Pe absent)
`err`	1	Error response (ERN/Pe present)

Normal Task 2: Congruent vs Incongruent (binary, Flanker/Stroop)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 54,498 vs 47,242 - reasonably balanced. Filter by Flanker/Stroop task files only.

Event	y	Meaning
`con`	0	Congruent trial
`inc`	1	Incongruent trial
## Data Quality Notes		

- boundary (545) is admin - exclude.
- 3 task files per subject (Flanker, Stroop, Go/no-go); filter by task for task-specific analyses.
- High channel count (129) may include non-EEG channels - verify channel types.

References

- Reference: `Registered Replication Report of ERN/Pe Psychometrics`

221. openneuro_ds004603 - Visual Attribute-Specific Contextual Trajectory Paradigm

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004603` | Year: 2023 | URL: https://openneuro.org/datasets/ds004603
Category: Sensory/Response | Subjects: 37 | Files (Completed): 37 | Channels: 65 | Sampling Rate: 1024 Hz

Paradigm

Participants observed sequences of visual stimuli that either followed or violated an established contextual trajectory in one of three visual attributes - brightness (color), size, or orientation. The paradigm tests predictive coding by contrasting EEG responses to expected (step) stimuli against unexpected violations and a no-trajectory control condition. Each sequence comprised four ordered steps followed by a target or violation stimulus.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004603

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`step_1`	31368	short	raw	Sequence step 1 - first ordered stimulus in trajectory
`step_2`	31367	short	raw	Sequence step 2
`step_3`	31367	short	raw	Sequence step 3
`step_4`	31367	short	raw	Sequence step 4 - final step before target/violation
`control`	14203	short	raw	Control condition - no established trajectory
`siz_vio`	4736	short	raw	Size trajectory violation
`ori_vio`	4735	short	raw	Orientation trajectory violation
`col_vio`	4735	short	raw	Color/brightness trajectory violation
`catch`	2958	short	raw	Catch trial requiring button-press response

Event ID	Count (total)	Type	Meta Level	Meaning
initalising_triggers	1	admin	raw	System initialisation trigger (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: initalising_triggers
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Visual Violation Detection (Multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 0.7 s, target_events = [control, siz_vio, ori_vio, col_vio]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
control	0	No-trajectory control condition
siz_vio	1	Size trajectory violation
ori_vio	2	Orientation trajectory violation
col_vio	3	Color/brightness trajectory violation

- Evidence:
- DOI: 10.1016/j.cortex.2023.08.004 - Linear SVM classifiers decoded expectation violation signals by comparing each violation type (brightness/size/orientation) to physically identical control trials using epoched data time-locked to final stimulus onset (segmented -200 to 700 ms, baselined -100 to 0 ms).
- Kernel Notes: Exclude initalising_triggers (admin event). Baseline correction should be applied from -100 ms to 0 ms per the paper. Catch trials (catch) should be handled separately as they require motor response. Sequence steps (step_1 to step_4) can be used for prediction build-up ERP analysis but are not part of this violation classification task. (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Catch Trial Motor Response Detection

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.5 s, target_events = [catch]
- Label Route: Annotation
- Target Definition: Detection of motor response requirement in catch trials.

Event	y	Meaning
catch	0	Catch trial requiring button-press response

- Notes: Catch trials require motor response - analyze separately from violation detection.

Data Quality Notes

- Exclude `initalising_triggers` from all analysis.
- Catch trials have a motor response component - treat separately or exclude from violation analysis.

References

- Reference: `Visual Attribute-Specific Contextual Trajectory Paradigm`

222. openneuro_ds004745 - 8-Channel SSVEP EEG Dataset with Artifact Trials

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004745 | Year: 2023 | URL: https://openneuro.org/datasets/ds004745
Category: Sensory/Response | Subjects: 6 | Files (Completed): 6
Channels: 8 | Sampling Rate: 1000 Hz

Paradigm

Participants performed Steady-State Visual Evoked Potential (SSVEP) tasks at stimulation frequencies of 2 Hz, 4 Hz, and 8 Hz, under both static artifact-free and voluntary movement artifact conditions. The dataset was created to test artifact removal algorithms for SSVEP-based BCIs under realistic conditions with deliberate motion contamination. Multiple trigger codes mark different frequency and artifact conditions.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004745

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
10	44	short	raw	Trigger code 10 - likely artifact or rest condition
4	38	short	raw	4 Hz SSVEP frequency onset
2	36	short	raw	2 Hz SSVEP frequency onset
1	36	short	raw	Trigger code 1
3	36	short	raw	Trigger code 3
5	36	short	raw	Trigger code 5
6	36	short	raw	Trigger code 6
7	36	short	raw	Trigger code 7
8	38	short	raw	8 Hz SSVEP frequency onset

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: SSVEP frequency condition classification with artifact detection

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 10.0 s, target_events = [2, 4, 8, 10]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
2	0	2 Hz SSVEP condition
4	1	4 Hz SSVEP condition
8	2	8 Hz SSVEP condition
10	3	Artifact/rest condition

- Evidence:
- DOI: 10.1109/embc46164.2021.9629771 - Evaluates ASR artifact removal on 8-channel SSVEP EEG data from 6 subjects performing tasks at 2Hz, 4Hz, and 8Hz stimulation frequencies under both artifact-free and artifact conditions, computing Frequency-Tagged Response (FTR) as the SSVEP measure.
- Kernel Notes: Kernel should apply bandpass filter 0.15-40Hz before epoching. The 10-second window captures multiple cycles of the 2Hz stimulus for robust SSVEP detection. Overlap factor 0.5-0.75 recommended to increase sample count from limited 6 subjects. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Only 6 subjects and 8 channels - limited for generalization.
- Deliberate motion artifacts present in some trials; note artifact condition labels.

References

- Reference: Kumaravel et al., (IEEE EMBC 2021)

223. openneuro_ds004816 - EEG Attention RSVP Experiment 1

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004816 | Year: 2023 | URL: https://openneuro.org/datasets/ds004816
Category: Sensory/Response | Subjects: N/A | Files (Completed): 40 | Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

A Rapid Serial Visual Presentation (RSVP) task investigating the neural dynamics of prioritizing task-relevant information using stimuli composed of small letters on large objects. Event codes `E 1` and `E 2` are the dominant condition markers (69,120 each total); codes `1` and `0` appear 31,104 times each; `E 3` appears 1,920 times. The `E 1`/`E 2` pair likely marks the two main stimulus streams (letter-prioritized vs object-prioritized). Codes `1`/`0` mark target-present vs target-absent trials. `Event/E x` variants are the same conditions from different recordings.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004816

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`E 1`	69120	short	raw	RSVP condition 1 (primary stream)
`E 2`	69120	short	raw	RSVP condition 2 (secondary stream)
`1`	31104	short	raw	Target-present marker
`0`	31104	short	raw	Target-absent marker
`Event/E 1`	6912	short	raw	RSVP condition 1 (alternate format)
`Event/E 2`	6912	short	raw	RSVP condition 2 (alternate format)
`E 3`	1920	short	raw	RSVP condition 3 (rare)
`Event/E 3`	192	short	raw	RSVP condition 3 (alternate format, rare)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: RSVP Condition Classification (Object vs Letter Task)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s, target_events = [`E 1`, `E 2`, `Event/E 1`, `Event/E 2`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`E 1`	0	Object-prioritized RSVP condition
`E 2`	1	Letter-prioritized RSVP condition
`Event/E 1`	0	Object-prioritized condition (alternate format)
`Event/E 2`	1	Letter-prioritized condition (alternate format)

- Evidence:
- DOI: 10.51628/001c.21174 - MVPA decoding comparing object-task vs letter-task conditions from epochs time-locked to stimulus onset (tmin=-0.1s, tmax=1.0s)
- Kernel Notes: Merge `E 1` with `Event/E 1` (label 0), and `E 2` with `Event/E 2` (label 1) before epoching. Epochs time-locked to stimulus onset. Exclude rare `E 3`/`Event/E 3` codes (1920/192 events) as they represent catch trials insufficient for reliable decoding.

Normal Task 1: RSVP Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `Event/E 1` and `Event/E 2` are alternate-format versions of the same codes; merge with `E 1`/`E 2` if needed. `E 3` (1,920) and `Event/E 3` (192) are rare - exclude or treat as separate class. 5 Hz RSVP; verify stimulus duration from events.tsv.

Event	y	Meaning
`E 1`	0	RSVP condition 1
`E 2`	1	RSVP condition 2
`1`	2	Target-present
`0`	3	Target-absent

Data Quality Notes

- `Event/E x` variants likely from different recording format; merge with `E x` before epoching.
- `E 3` and `Event/E 3` are rare codes - may represent catch trials or errors.
- High-count RSVP paradigm; very large epoch counts per subject.

References

- Reference: `Grootswagers T., Robinson A.K., Shatek S.M., Carlson T.A. (2021). The neural dynamics underlying prioritisation of task-relevant information. Neurons, Behaviour, Data Analysis, and Theory, 5(1)`

224. openneuro_ds004817 - EEG-attention-rsvp-exp2

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004817 | Year: 2023 | URL: https://openneuro.org/datasets/ds004817
Category: Sensory/Response | Subjects: 40 | Files (Completed): 40
Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

A Rapid Serial Visual Presentation (RSVP) task (Experiment 2) investigating the neural dynamics of prioritizing task-relevant information, using stimuli consisting of small objects superimposed on large letters. Participants performed target detection in rapid streams. Experiment 2 is a companion to Experiment 1 (ds004816) with different stimulus configurations (objects on letters rather than letters on objects).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004817

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`E 1`	69137	short	raw	RSVP event type 1 (task-relevant stimulus)
`E 2`	69137	short	raw	RSVP event type 2 (task-irrelevant stimulus)
`1`	29376	short	raw	Trigger code 1 (response or condition marker)
`0`	29376	short	raw	Trigger code 0
`Event/E 1`	10385	short	raw	RSVP event type 1 (alternate format)
`Event/E 2`	10385	short	raw	RSVP event type 2 (alternate format)
`E 3`	1921	short	raw	RSVP event type 3
`Event/E 3`	289	short	raw	RSVP event type 3 (alternate format)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`

- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: RSVP Stimulus Identity Classification (MVPA)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s, target_events = [`E 1`, `E 2`, `E 3`, `Event/E 1`, `Event/E 2`, `Event/E 3`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`E 1`	0	RSVP event type 1 (attended/task-relevant stimulus)
`E 2`	1	RSVP event type 2 (unattended/task-irrelevant stimulus)
`E 3`	2	RSVP event type 3
`Event/E 1`	0	RSVP event type 1 (alternate format)
`Event/E 2`	1	RSVP event type 2 (alternate format)
`Event/E 3`	2	RSVP event type 3 (alternate format)

- Evidence:
- DOI: 10.51628/001c.21174 - EEG epochs extracted from -100ms to 1000ms relative to stimulus onset with MVPA decoding to classify which of 16 objects or 16 letters was presented, as well as object category and animacy.
- Kernel Notes: Aggregate E 1/Event/E 1 and E 2/Event/E 2 as attended vs unattended classes. The 16 object and 16 letter identities can be derived from stimulus file metadata if super-meta labels are needed. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: RSVP Category and Animacy Classification (Super Meta)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 1.0 s, target_events = [`E 1`, `E 2`, `Event/E 1`, `Event/E 2`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`E 1` / `Event/E 1`	0	Attended - category: bird
`E 1` / `Event/E 1`	1	Attended - category: fish
`E 1` / `Event/E 1`	2	Attended - category: boat
`E 1` / `Event/E 1`	3	Attended - category: plane
`E 1` / `Event/E 1`	4	Attended - animacy: animal

E 1 / Event/E 1	5	Attended - animacy: vehicle
E 2 / Event/E 2	0	Unattended - category: bird
E 2 / Event/E 2	1	Unattended - category: fish
E 2 / Event/E 2	2	Unattended - category: boat
E 2 / Event/E 2	3	Unattended - category: plane
E 2 / Event/E 2	4	Unattended - animacy: animal
E 2 / Event/E 2	5	Unattended - animacy: vehicle

- Super Meta Notes: Derived from raw event(s) [E 1, E 2, Event/E 1, Event/E 2] by mapping stimulus file names to category (bird/fish/boat/plane) and animacy (animal/vehicle) labels. Kernel step: Parse stimulus metadata files to extract object identity, then compute category and animacy labels as super-meta features.
- Evidence:
- DOI: 10.51628/001c.21174 - MVPA decoding performed to classify object category (bird/fish/boat/plane) and animacy (animal/vehicle) in addition to stimulus identity.
- Kernel Notes: Requires access to stimulus presentation files to map trial indices to category/animacy labels. Verify stimulus file naming convention from dataset documentation.

Data Quality Notes

- E 1/Event/E 1 and E 2/Event/E 2 are duplicate event formats - aggregate for analysis.
- Companion dataset: ds004816 is Experiment 1.

References

- Reference: Grootswagers T., Robinson A.K., Shatek S.M., Carlson T.A. (2021). The neural dynamics underlying prioritisation of task-relevant information. Neurons, Behaviour, Data Analysis, and Theory, 5(1)

225. openneuro_ds004951 - Braille Letters EEG

Source: analysis.md

Dataset: `openneuro_ds004951` | Year: 2024 | URL: https://openneuro.org/datasets/ds004951
Category: Sensory/Response | Subjects: 23 | Files (Completed): 23 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

Participants identified Braille letters presented via tactile stimulation to left or right index fingers while EEG was recorded. The study investigates cortical plasticity by comparing sighted controls and congenitally blind individuals in processing Braille letter information. Stimulus codes S1-S16 mark individual Braille letter presentations (3,316-3,319 each); S101 and S222 (63,609/63,607) dominate and likely mark response-correct/response-incorrect feedback or block boundaries; S200 (10,535) is a block marker; S253 (9,211) and S254/S255 are admin/system markers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004951

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S101`	63609	short	raw	Response feedback / block marker A
`Stimulus/S222`	63607	short	raw	Response feedback / block marker B
`Stimulus/S200`	10535	short	raw	Block boundary / condition marker
`Stimulus/S253`	9211	admin	raw	Admin / response marker (exclude)
`Stimulus/S 16`	3319	short	raw	Braille letter 16 onset
`Stimulus/S 6`	3319	short	raw	Braille letter 6 onset
`Stimulus/S 15`	3319	short	raw	Braille letter 15 onset
`Stimulus/S 11`	3318	short	raw	Braille letter 11 onset
`Stimulus/S 3`	3318	short	raw	Braille letter 3 onset
`Stimulus/S 10`	3317	short	raw	Braille letter 10 onset
`Stimulus/S 1-S 16`	~3316-3319 each	short	raw	Individual Braille letter onsets (16 letters)
`Stimulus/S255`	~437	admin	raw	BrainVision admin sync marker (exclude)

Event ID	Count (total)	Type	Meta Level	Meaning
`New Segment/`	rare	admin	raw	BrainVision segment boundary (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/S101`, `Stimulus/S222`, `Stimulus/S200`, `Stimulus/S253`, `Stimulus/S 16`, `Stimulus/S 6`, `Stimulus/S 15`, `Stimulus/S 11`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Letters

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004951 DOI: https://doi.org/10.18112/openneuro.ds004951.v1.0.0

Normal Task 1: Braille Letter Identification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `New Segment/`, `Stimulus/S253`, `Stimulus/S254`, `Stimulus/S255` are admin - exclude. S101/S222 (63k each) are very high-frequency feedback/block markers - exclude from classification. S200 is a block boundary - exclude.

Event	y	Meaning
`Stimulus/S 1`	0	Braille letter 1
`Stimulus/S 2`	1	Braille letter 2
`Stimulus/S 3`	2	Braille letter 3
`Stimulus/S 4`	3	Braille letter 4
`Stimulus/S 5`	4	Braille letter 5
`Stimulus/S 6`	5	Braille letter 6
`Stimulus/S 7`	6	Braille letter 7
`Stimulus/S 8`	7	Braille letter 8
`Stimulus/S 9`	8	Braille letter 9
`Stimulus/S 10`	9	Braille letter 10

Event	y	Meaning
Stimulus/S 11	10	Braille letter 11
Stimulus/S 12	11	Braille letter 12
Stimulus/S 13	12	Braille letter 13
Stimulus/S 14	13	Braille letter 14
Stimulus/S 15	14	Braille letter 15
Stimulus/S 16	15	Braille letter 16
## Data Quality Notes		

- Stimulus/S101 and Stimulus/S222 are very high-frequency (63k each) - likely feedback or block markers, exclude from classification.
- New Segment/, Stimulus/S253, Stimulus/S254, Stimulus/S255 are admin/system - exclude.
- Dataset includes sighted controls and congenitally blind participants; group label in participants.tsv.

References

- Reference: The transformation of sensory to perceptual braille letter representations in the visually deprived brain

226. openneuro_ds005087 - Rapid Hemifield Object EEG

Source: analysis.md

Dataset: openneuro_ds005087 | Year: 2024 | URL: https://openneuro.org/datasets/ds005087
Category: Sensory/Response | Subjects: 80 | Files (Completed): 80 | Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed object and word stimuli presented at 5 Hz in rapid visual streams to the left hemifield, right hemifield, or central visual field while performing a red target detection task. Event codes E 1/E 2 (184,320 each) and E 3 (960) mark the three presentation positions (left/right/center hemifield). Numeric codes 2/3/1 (92,160/46,080/46,080) likely encode stimulus categories or object identities within each presentation stream.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005087

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
E 1	184320	short	raw	Hemifield/position condition 1
E 2	184320	short	raw	Hemifield/position condition 2
2	92160	short	raw	Stimulus category/identity code 2
3	46080	short	raw	Stimulus category/identity code 3
1	46080	short	raw	Stimulus category/identity code 1
E 3	960	short	raw	Hemifield/position condition 3 (rare)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Publisher-defined BIDS task: Rsvp

- Task Type: classification/segmentation

- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds005087 DOI: https://doi.org/10.31234/osf.a3bpd

Normal Task 1: Visual Hemifield Presentation Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: E 3 (960) is rare - exclude or treat as separate class. 5 Hz RSVP streams; very short inter-stimulus interval. Verify code meanings from events.tsv (Robinson et al., 2025).

Event	y	Meaning
E 1	0	Hemifield/position condition 1
E 2	1	Hemifield/position condition 2
1	2	Stimulus category code 1
2	3	Stimulus category code 2
3	4	Stimulus category code 3
## Data Quality Notes		

- Verify full event decode from events.tsv or Robinson et al., 2025.
- 5 Hz RSVP - confirm exact stimulus duration from events.tsv.
- E 3 (960) is rare relative to E 1/E 2 (184k each); exclude or treat separately.

References

- Reference: Robinson A.K., Grootswagers T., Shatek S., Behrmann M., Carlson T.A. (2025). Dynamics of visual object coding within and across the hemispheres: Objects in the periphery. Science Advances, 11, eadq0889

227. openneuro_ds005106 - Human Infant EEG for 200 Object Images in Rapid Visual Streams

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005106` | Year: 2024 | URL: https://openneuro.org/datasets/ds005106
Category: Sensory/Response | Subjects: 42 | Files (Completed): 42 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Infants viewed 200 distinct object images presented in rapid visual streams (RSVP) while EEG was recorded to study the development of visual object representations. Each numeric trigger code corresponds to one of 200 object IDs; code -1 is a baseline/inter-stimulus marker. The dataset enables decoding of object-specific neural representations in the infant brain.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005106

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`-1`	43105	short	raw	Baseline / inter-stimulus marker
`155`	220	short	raw	Object image 155 onset
`123`	220	short	raw	Object image 123 onset
`72`	219	short	raw	Object image 72 onset
`147`	219	short	raw	Object image 147 onset
`(~196 more object codes)`	~219 each	short	raw	Individual object image onsets (IDs 1-200)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Object Image Classification (200-way) from Infant EEG during RSVP

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [1, 2, 3, 4, 5, ... up to 200]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
1	0	Object image 1 onset
2	1	Object image 2 onset
...	...	...
200	199	Object image 200 onset

- Evidence:
- DOI: 10.1038/s41597-025-04744-z - This paper describes the dataset of 42 infant EEG recordings during rapid serial visual presentation of 200 object images, using epochs from -100 to 800 ms relative to stimulus onset for representational similarity analysis and multivariate pattern classification to decode object-specific neural representations.
- DOI: 10.31234/osf.io.dzrkq - This paper presents the OpenNeuro ds005106 dataset of 42 infant EEG recordings during RSVP of 200 object images and performs time-resolved multivariate pattern analysis to decode object identity from neural responses.
- Kernel Notes: Code `-1` is baseline/inter-trial interval marker - exclude from classification. Use stimulus onset codes 1-200 as event triggers for epoching. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Object Category Classification (Super Meta Level)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [1, 2, 3, 4, 5, ... up to 200]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
(objects in category 1-50)	0	Category 1
(objects in category 51-100)	1	Category 2
...	...	...
(objects in category 151-200)	49	Category 50

- Super Meta Notes: Derived from raw event(s) [1-200] by mapping each object ID to one of 50 semantic categories (e.g., 'dog', 'car', 'apple'). Kernel step: Load category metadata from stimulus dataset and create super-level label column by mapping object trigger code to category label.
- Evidence:
- DOI: 10.31234/osf.io.dzrkq - Supplementary labels include category (50 categories) and superordinate (animate/inanimate) levels, derived from stimulus metadata.
- Kernel Notes: Requires external stimulus metadata file mapping object IDs 1-200 to category labels. Verify category mapping schema from dataset documentation.

Primary Task 3: Animate/Inanimate Classification (Super Meta Level)

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [1, 2, 3, 4, 5, ... up to 200]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
(objects 1-200)	0	Inanimate object
(objects 1-200)	1	Animate object

- Super Meta Notes: Derived from raw event(s) [1-200] by mapping each object ID to superordinate level (animate/inanimate). Kernel step: Load superordinate category metadata and create binary label column.
- Evidence:
- DOI: 10.31234/osf.io.dzrkq - Supplementary labels include category (50 categories) and superordinate (animate/inanimate) levels, derived from stimulus metadata.
- Kernel Notes: Requires external stimulus metadata file. Binary classification of object animacy from neural responses.

Data Quality Notes

- ~200 unique object trigger codes - each code represents one object image.
- Code -1 is a baseline/ITI marker and should be excluded.
- Infant EEG - expect high noise and movement artifacts.

References

- Reference: `Human Infant EEG Recordings for 200 Object Images Presented in Rapid Visual Streams`

228. openneuro_ds005169 - Intracranial EEG During Cortical Stimulation Evoking Visual Effects

Source: analysis.md

Dataset: openneuro_ds005169 | Year: 2024 | URL: https://openneuro.org/datasets/ds005169
Category: Sensory/Response | Subjects: 22 | Files (Completed): 112 | Channels: 82 | Sampling Rate: 4096 Hz

Paradigm

Intracranial EEG was recorded from drug-resistant epilepsy patients undergoing presurgical evaluation during direct electrical stimulation (DES) of cortical sites that evoke visual hallucinations and perceptual effects. Comment/prestim (112 total) marks pre-stimulation baseline periods and Comment/poststim (111) marks post-stimulation response windows. UUID-format Comment events (2-10 per file) are patient-specific or trial-specific annotations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005169

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
Comment/prestim	112	long	raw	Pre-stimulation baseline period
Comment/poststim	111	long	raw	Post-stimulation response window
Comment/[UUID]	2-10/file	admin	raw	Patient/trial-specific annotation (UUID format, exclude)
Comment/[electrode] [mA] SYM	~2/file	admin	raw	Stimulation parameter annotation (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Comment/prestim, Comment/poststim
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Dcs

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://doi.org/10.1002/hbm.25749

Normal Task 1: Pre vs Post Stimulation (binary)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: UUID-format and electrode/mA Comment events are admin - exclude. Very high sampling rate (4096 Hz) - downsample before processing. iEEG data; stimulation artifact present in poststim windows.

Event	y	Meaning
`Comment/prestim`	0	Pre-stimulation baseline
`Comment/poststim`	1	Post-stimulation response
## Data Quality Notes		

- Very high sampling rate (4096 Hz) - downsample to <=1000 Hz before analysis.
- Intracranial EEG - different noise profile from scalp EEG; stimulation artifacts expected.
- UUID-format Comment events are patient-specific annotations - exclude from epoching.

References

- Reference: `Imaging the effective networks associated with cortical function through intracranial high-frequency stimulation. DOI: 10.1002/hbm.25749`

229. openneuro_ds005363 - Object Recognition in Healthy Aging (ORHA) - EEG

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005363 | Year: 2024 | URL: https://openneuro.org/datasets/ds005363
Category: Sensory/Response | Subjects: 43 | Files (Completed): 43 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed images across five visual categories (faces, animals, places, objects, and likely one additional) to investigate visual object representations and how they change with healthy aging. S65-S69 encode the five stimulus categories; S99 is a high-frequency event likely marking fixations or trial separators. The study uses representational similarity analysis to track age-related changes in neural object coding.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005363

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
Stimulus/S 99	21460	short	raw	Fixation/trial separator (high-frequency onset)
Stimulus/S 66	4508	short	raw	Visual category 2
Stimulus/S 67	4463	short	raw	Visual category 3
Stimulus/S 65	4421	short	raw	Visual category 1
Stimulus/S 69	4351	short	raw	Visual category 5
Stimulus/S 68	4285	short	raw	Visual category 4
Stimulus/S 4-Stimulus/S 57	~1377 each	short	raw	Individual exemplar image onsets (60+ items)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.

- Candidate Trigger Labels: `Stimulus/S 99`, `Stimulus/S 66`, `Stimulus/S 67`, `Stimulus/S 65`, `Stimulus/S 69`, `Stimulus/S 68`, `Stimulus/S 4`, `Stimulus/S 57`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Animate vs Inanimate Visual Object Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 0.75 s *, target_events = [`Stimulus/S 65`, `Stimulus/S 66`, `Stimulus/S 67`, `Stimulus/S 68`, `Stimulus/S 69`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Stimulus/S 65`	0	Animate (faces)
`Stimulus/S 66`	0	Animate (animals) *
`Stimulus/S 67`	0	Animate (animals) *
`Stimulus/S 68`	1	Inanimate (places/objects) *
`Stimulus/S 69`	1	Inanimate (places/objects) *

- Super Meta Notes: Derived from raw events S65-S69 by grouping visual categories into animate (faces+animals) vs inanimate (places+objects). Kernel step: Map category codes to binary animate/inanimate labels. Verify category-to-code mapping from events.tsv or dataset documentation.
- Evidence:
- DOI: `10.1162/imag.a.1033` - Uses openneuro_ds005363 as control dataset for 2-class decoding of EEG responses to animate vs inanimate images with 43 participants (~88,112 trials)
- Kernel Notes: Use epoch window 0-0.75s based on visual object recognition paradigm. Verify animate/inanimate grouping of S65-S69 from dataset documentation.

Primary Task 2: Publisher-defined BIDS task: Objrecog

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 0.75 s *, target_events = [`Stimulus/S 65`, `Stimulus/S 66`, `Stimulus/S 67`, `Stimulus/S 68`, `Stimulus/S 69`]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task
- Evidence:
- DOI: `https://doi.org/10.18112/openneuro.ds005363.v1.0.0` - OpenNeuro metadata indicates task name "Objrecog"
- Kernel Notes: Default epoch parameters inferred from visual object recognition paradigm (tmax ~0.75s covers visual evoked response). Verify exact parameters from dataset readme if available.

Normal Task 1: Visual Category Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

- Notes: S99 (21,460 total) is a high-frequency fixation/separator marker - exclude. S4-S57 encode individual exemplar images (60+ codes, ~1377 each); use for RSA analysis.

Event	y	Meaning
`Stimulus/S 65`	0	Visual category 1
`Stimulus/S 66`	1	Visual category 2
`Stimulus/S 67`	2	Visual category 3
`Stimulus/S 68`	3	Visual category 4
`Stimulus/S 69`	4	Visual category 5

Data Quality Notes

- S99 is a very high-frequency marker (21,460 total) - likely fixation onset, exclude from category decoding.
- 60+ exemplar codes (~1377 each) suitable for RSA.

References

- Reference: `Healthy aging delays and dedifferentiates high-level visual representations`

230. openneuro_ds005406 - EEG Frequency Tagging Reveals the Integration of Dissimilar Observed Actions

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005406` | Year: 2024 | URL: https://openneuro.org/datasets/ds005406
Category: Sensory/Response | Subjects: 29 | Files (Completed): 29 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

A frequency-tagging EEG study measuring steady-state visual evoked potentials (SSVEP) as participants observed dissimilar human actions presented in rapid visual sequences. The study uses neural entrainment at different tagging frequencies to measure the integration of distinct action components (e.g., arm and leg movements). Four action-pair conditions are encoded as S-code pairs (S11/12, S21/22, S31/32, S41/42).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005406

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S 41`	231	short	raw	Action condition 4, sequence start
`Stimulus/S 21`	231	short	raw	Action condition 2, sequence start
`Stimulus/S 22`	231	short	raw	Action condition 2, sequence end
`Stimulus/S 42`	230	short	raw	Action condition 4, sequence end
`Stimulus/S 31`	230	short	raw	Action condition 3, sequence start
`Stimulus/S 32`	230	short	raw	Action condition 3, sequence end
`Stimulus/S 11`	230	short	raw	Action condition 1, sequence start
`Stimulus/S 12`	230	short	raw	Action condition 1, sequence end
`New Segment/`	29	admin	raw	BrainVision segment boundary (exclude)
`Stimulus/S255`	24	admin	raw	S255 admin sync trigger (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/S 41`, `Stimulus/S 21`, `Stimulus/S 22`, `Stimulus/S 42`, `Stimulus/S 31`, `Stimulus/S 32`, `Stimulus/S 11`, `Stimulus/S 12`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: SSVEP Action Condition Classification (4-class)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -2.0 s , tmax = 48.6 s , target_events = [`Stimulus/S 11`, `Stimulus/S 12`, `Stimulus/S 21`, `Stimulus/S 22`, `Stimulus/S 31`, `Stimulus/S 32`, `Stimulus/S 41`, `Stimulus/S 42`]
- Label Route: `stim`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Stimulus/S 11`	0	Condition 1: same movement, 1 participant
`Stimulus/S 12`	0	Condition 1: same movement, 1 participant
`Stimulus/S 21`	1	Condition 2: different movement, 1 participant
`Stimulus/S 22`	1	Condition 2: different movement, 1 participant
`Stimulus/S 31`	2	Condition 3: same movement, 2 participants
`Stimulus/S 32`	2	Condition 3: same movement, 2 participants
`Stimulus/S 41`	3	Condition 4: different movement, 2 participants
`Stimulus/S 42`	3	Condition 4: different movement, 2 participants

- Super Meta Notes: Derived from raw event pairs by grouping start/end markers of each action condition (S11/S12->condition1, S21/S22->condition2, S31/S32->condition3, S41/S42->condition4). Kernel step: Collapse 8 raw event IDs to 4 condition labels using mapping table.
- Evidence:
- DOI: 10.1101/2024.05.10.593473 - EEG frequency-tagging study with 29 participants analyzing SSVEP responses across four action conditions varying in movement type (same/different) and agent number (1P/2P). Epochs extracted from -2s to 48.6s post-stimulus onset.
- Kernel Notes: Raw events S11/S12, S21/S22, S31/S32, S41/S42 each mark condition start/end. Use either start or end marker for epoching; both map to same condition label. (replaces Primary Task 1 from original analysis.md)

Normal Task 1: Action Integration Condition Classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: New Segment/ (29) and Stimulus/S255 (24) are admin - exclude. Four action conditions with start/end markers each.

Event	y	Meaning
Stimulus/S 11	0	Action condition 1 start
Stimulus/S 12	1	Action condition 1 end
Stimulus/S 21	2	Action condition 2 start
Stimulus/S 22	3	Action condition 2 end
Stimulus/S 31	4	Action condition 3 start
Stimulus/S 32	5	Action condition 3 end
Stimulus/S 41	6	Action condition 4 start
Stimulus/S 42	7	Action condition 4 end

Data Quality Notes

- New Segment/ and Stimulus/S255 are administrative - exclude.
- Frequency-tagging paradigm; SSVEP analysis may be more appropriate than traditional ERP epoching.

References

- Reference: EEG frequency tagging reveals the integration of dissimilar observed actions

231. openneuro_ds005586 - EEG responses to the number of objects in partially occluded and uncovered scenes

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005586` | Year: 2024 | URL: https://openneuro.org/datasets/ds005586
Category: Sensory/Response | Subjects: 23 | Files (Completed): 23 | Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

Participants passively viewed game boards depicting varying numbers of game pieces under partial occlusion and subsequent uncovering conditions. The study investigates EEG responses to numerosity perception under occlusion, measuring visual evoked potentials to quantify object tracking. Stimulus codes encode number of objects (1 or 2 digits) and condition (occluded vs uncovered).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005586

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S 95`	14399	short	raw	Catch / fixation stimulus
`Stimulus/S 21`	6884	short	raw	Stimulus condition (2-digit code: object count x condition)
`Stimulus/S 11`	6875	short	raw	Stimulus condition (2-digit code: object count x condition)
`Stimulus/S 15`	2241	short	raw	Stimulus condition
`Stimulus/S127`	1413	short	raw	Stimulus condition (3-digit code)
`Stimulus/S223`	1299	short	raw	Stimulus condition (3-digit code)
`Stimulus/S111`	903	short	raw	Stimulus condition (3-digit code)
`Stimulus/S124`	866	short	raw	Stimulus condition (3-digit code)
`Stimulus/S224`	866	short	raw	Stimulus condition (3-digit code)
`Stimulus/S226`	864	short	raw	Stimulus condition (3-digit code)

Event ID	Count (total)	Type	Meta Level	Meaning
Stimulus/S126	864	short	raw	Stimulus condition (3-digit code)
Stimulus/S215	864	short	raw	Stimulus condition (3-digit code)
Stimulus/S115	864	short	raw	Stimulus condition (3-digit code)
Stimulus/S118	863	short	raw	Stimulus condition (3-digit code)
Stimulus/S218	863	short	raw	Stimulus condition (3-digit code)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Stimulus/S 95, Stimulus/S 21, Stimulus/S 11, Stimulus/S 15, Stimulus/S127, Stimulus/S223, Stimulus/S111, Stimulus/S124
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: ERP Classification - Numerosity Perception under Occlusion

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [Stimulus/S 95, Stimulus/S 21, Stimulus/S 11, Stimulus/S 15, Stimulus/S127, Stimulus/S223, Stimulus/S111, Stimulus/S124, Stimulus/S224, Stimulus/S226, Stimulus/S126, Stimulus/S215, Stimulus/S115, Stimulus/S118, Stimulus/S218]
- Label Route: stim
- Meta Level: raw
- Target Definition:

Event	y	Meaning
Stimulus/S 95	0	Catch/fixation baseline
Stimulus/S 21	1	2-digit condition code (numerosity x occlusion)
Stimulus/S 11	2	2-digit condition code (numerosity x occlusion)
Stimulus/S 15	3	2-digit condition code (numerosity x occlusion)
Stimulus/S127	4	Visible=4 pieces, uncovered=28
Stimulus/S223	5	Visible=32 pieces, uncovered=2
Stimulus/S111	6	Visible=4 pieces, uncovered=1
Stimulus/S124	7	Visible=4 pieces, uncovered=4
Stimulus/S224	8	Visible=32 pieces, uncovered=4
Stimulus/S226	9	Visible=32 pieces, uncovered=6
Stimulus/S126	10	Visible=4 pieces, uncovered=6
Stimulus/S215	11	Visible=32 pieces, uncovered=15
Stimulus/S115	12	Visible=4 pieces, uncovered=15
Stimulus/S118	13	Visible=4 pieces, uncovered=18

| Stimulus/S218 | 14 | Visible=32 pieces, uncovered=18 |
- Evidence:
- DOI: 10.1162/jocn_a_02264 - Uses openneuro_ds005586 to measure EEG responses during passive viewing of partially occluded game boards, segmenting data from -0.2 to 1.0s relative to stimulus onset for ERP analysis of P1, N1, P2 (occluded phase) and P1, N1, P3 (uncovered phase) components.
- Kernel Notes: Baseline correction from -0.2 to 0s. Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching. The 3-digit codes encode: 1st digit=visible pieces (1=4, 2=32), 2nd-3rd digits=uncovered piece count (00-32). Exclude New Segment/ admin events.

Primary Task 2: Binary Occlusion Classification (High vs Low Visibility)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [Stimulus/S127, Stimulus/S111, Stimulus/S124, Stimulus/S126, Stimulus/S115, Stimulus/S118, Stimulus/S223, Stimulus/S224, Stimulus/S226, Stimulus/S215, Stimulus/S218]
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
Stimulus/S127, Stimulus/S111, Stimulus/S124, Stimulus/S126, Stimulus/S115, Stimulus/S118	0	Low visibility (4 visible pieces, codes 1xx)
Stimulus/S223, Stimulus/S224, Stimulus/S226, Stimulus/S215, Stimulus/S218	1	High visibility (32 visible pieces, codes 2xx)

- Super Meta Notes: Derived from raw event(s) by extracting first digit of 3-digit stimulus code: codes starting with 1 = low visibility (4 pieces), codes starting with 2 = high visibility (32 pieces). Kernel step: decode stimulus codes to extract visibility level.
- Evidence:
- DOI: 10.1162/jocn_a_02264 - Paper analyzes ERP components by condition, distinguishing between occluded scenes with few vs many visible pieces.
- Kernel Notes: Baseline correction from -0.2 to 0s. Filter 1-40 Hz. Task investigates whether numerosity processing differs based on visual access to objects.

Normal Task 1: Object Numerosity / Occlusion (epoch_by_event)

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Stimulus codes encode object count x occlusion condition; decode from events.tsv. New Segment/ is admin and excluded.

Event	y	Meaning
Stimulus/S 95	0	Catch/fixation
Stimulus/S127	1	Condition code 127
Stimulus/S223	2	Condition code 223
Stimulus/S111	3	Condition code 111
Stimulus/S 21	4	Condition code 21

Event	y	Meaning
Stimulus/S 11	5	Condition code 11
Stimulus/S117	6	Condition code 117
Stimulus/S124	7	Condition code 124
Stimulus/S224	8	Condition code 224
Stimulus/S227	9	Condition code 227
Stimulus/S118	10	Condition code 118
Stimulus/S218	11	Condition code 218
Stimulus/S122	12	Condition code 122
Stimulus/S217	13	Condition code 217
Stimulus/S222	14	Condition code 222
Stimulus/S212	15	Condition code 212
Stimulus/S123	16	Condition code 123
Stimulus/S113	17	Condition code 113
Stimulus/S213	18	Condition code 213
Stimulus/S112	19	Condition code 112
Stimulus/S226	20	Condition code 226
Stimulus/S211	21	Condition code 211
Stimulus/S126	22	Condition code 126
Stimulus/S221	23	Condition code 221
Stimulus/S121	24	Condition code 121
Stimulus/S216	25	Condition code 216
Stimulus/S116	26	Condition code 116
Stimulus/S214	27	Condition code 214
Stimulus/S114	28	Condition code 114

Data Quality Notes

- Small dataset (23 subjects, 23 files); verify that all subjects completed the same number of trials.
- `New Segment/` is an admin event and should be excluded.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG responses to the number of objects in partially occluded and uncovered scenes`

232. openneuro_ds005589 - A multi-subject and multi-session EEG dataset for modelling human visual object recognition

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005589` | Year: 2024 | URL: https://openneuro.org/datasets/ds005589
Category: Sensory/Response | Subjects: 32 | Files (Completed): 1592 | Channels: 133 | Sampling Rate: 1000 Hz

Paradigm

Participants underwent Rapid Serial Visual Presentation (RSVP) at 5 Hz and a low-speed visual presentation task to record brain responses to visual object recognition across multiple image categories. Codes 1-20 correspond to object categories in the low-speed task; codes 21-27 to RSVP categories; code 99 is catch trials; codes 252/254 are feedback. Multiple sessions were recorded per subject.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005589

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`252`	141374	short	raw	Correct response feedback
`1-20`	~80332 each	short	raw	Object category codes 1-20 (low-speed presentation)
`99`	~157200	short	raw	Catch trial
`254`	~40000	short	raw	Incorrect response feedback
`21-27`	~798 each	short	raw	Object category codes 21-27 (RSVP task)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Visual object recognition category decoding

- Task Type: classification

- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 0.5 s, target_events = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 99, 252, 254]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
1	0	Object category 1 (low-speed)
2	1	Object category 2 (low-speed)
3	2	Object category 3 (low-speed)
4	3	Object category 4 (low-speed)
5	4	Object category 5 (low-speed)
6	5	Object category 6 (low-speed)
7	6	Object category 7 (low-speed)
8	7	Object category 8 (low-speed)
9	8	Object category 9 (low-speed)
10	9	Object category 10 (low-speed)
11	10	Object category 11 (low-speed)
12	11	Object category 12 (low-speed)
13	12	Object category 13 (low-speed)
14	13	Object category 14 (low-speed)
15	14	Object category 15 (low-speed)
16	15	Object category 16 (low-speed)
17	16	Object category 17 (low-speed)
18	17	Object category 18 (low-speed)
19	18	Object category 19 (low-speed)
20	19	Object category 20 (low-speed)
99	20	Catch trial
252	21	Correct feedback
254	22	Incorrect feedback
21	23	RSVP category 21
22	24	RSVP category 22
23	25	RSVP category 23
24	26	RSVP category 24
25	27	RSVP category 25
26	28	RSVP category 26
27	29	RSVP category 27

- Evidence:
- DOI: 10.1038/s41597.025.04843.x - Multi-subject EEG dataset for visual object recognition using RSVP (5Hz) and low-speed (1s) paradigms with 20 semantic categories; epochs of 0-500ms relative to stimulus onset used for classification.
- Kernel Notes: Filter bandpass 1-40 Hz and apply baseline correction. For RSVP task at 5Hz, consider excluding epochs overlapping with adjacent stimuli. Verify event ID mapping from events.tsv (replaces Primary Task 1 from original analysis.md)

Normal Task 1: Visual Object Recognition (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`

- Target Definition: Event-to-label mapping table below.
- Notes: 30 event classes. Primary task: decode object category from EEG (codes 1-20 or 21-27). Codes 252/254 and 99 may be used for filtering.

Event	y	Meaning
1	0	Object category 1
2	1	Object category 2
3	2	Object category 3
4	3	Object category 4
5	4	Object category 5
6	5	Object category 6
7	6	Object category 7
8	7	Object category 8
9	8	Object category 9
10	9	Object category 10
11	10	Object category 11
12	11	Object category 12
13	12	Object category 13
14	13	Object category 14
15	14	Object category 15
16	15	Object category 16
17	16	Object category 17
18	17	Object category 18
19	18	Object category 19
20	19	Object category 20
99	20	Catch trial
252	21	Correct feedback
254	22	Incorrect feedback
21	23	RSVP category 21
22	24	RSVP category 22
23	25	RSVP category 23
24	26	RSVP category 24
25	27	RSVP category 25
26	28	RSVP category 26
27	29	RSVP category 27

Data Quality Notes

- Large dataset (32 subjects x 1592 files); multi-session design - account for session effects.
- Apply bandpass filter (1-40 Hz) and baseline correction prior to analysis.

References

- Reference: A multi-subject and multi-session EEG dataset for modelling human visual object recognition

233. openneuro_ds005662 - A high-quality EEG dataset for studying visual touch perception

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005662 | Year: 2025 | URL: https://openneuro.org/datasets/ds005662
Category: Sensory/Response | Subjects: N/A | Files (Completed): 80 | Channels: 65 | Sampling Rate: 2048 Hz

Paradigm

Participants viewed rapid streams of videos depicting touch to a hand or an object to investigate visual touch perception. The experiment used target and non-target trials to measure brain responses to observed touch. No event markers are present in the BIDS metadata; the EEG is stored as continuous recordings requiring segmentation for downstream analysis.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005662

Event Structure

No event markers detected (`no_events`). This dataset contains continuous EEG recordings without trial-level annotations in events.tsv.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Event ID	Count (total)	Type	Meta Level	Meaning
`no_events`	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Visual Touch Perception - Perspective Decoding (Self vs Other)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [`stimulus_onset` *]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning

| stimulus_onset | 0 | Self-perspective video (viewer sees own hand being touched) |
| stimulus_onset | 1 | Other-perspective video (viewer sees another person's hand being touched) |
- Super Meta Notes: Derived from video metadata (perspective field). Kernel must extract perspective label from video metadata files and map to binary classification: self=0, other=1.
- Evidence:
- DOI: 10.1038/s41597.026.06714.5 - MVPA decoding of perspective (self vs other) from EEG epochs -100 to 800 ms relative to stimulus onset
- DOI: 10.1101/2025.07.10.664069 - MVPA decoding of self vs other perspective from EEG epochs time-locked to stimulus onset
- Kernel Notes: Verify event ID mapping from events.tsv. Since dataset shows no_events, kernel must locate stimulus timing from video metadata or derive from presentation logs. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Visual Touch Perception - Material Classification

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [stimulus_onset *]
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning
stimulus_onset	0-7	Material category (8 types: wood, metal, plastic, fabric, rubber, glass, skin, ceramic)
- Super Meta Notes: Derived from video metadata (material field). Kernel must extract material label from video metadata files (8 categories).
- Evidence:
- DOI: 10.1038/s41597.026.06714.5 - Classification of 8 material categories from EEG epochs -100 to 800 ms
- DOI: 10.1101/2025.07.10.664069 - Classification of material (8 categories) from EEG epochs
- Kernel Notes: Verify event ID mapping from events.tsv.

Primary Task 3: Visual Touch Perception - Hand Orientation Decoding

- Task Type: classification
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [stimulus_onset *]
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning
stimulus_onset	0	Left hand being touched
stimulus_onset	1	Right hand being touched
- Super Meta Notes: Derived from video metadata (hand_orientation field: left/right hand, self/other perspective). Kernel must extract hand laterality from video metadata.
- Evidence:
- DOI: 10.1038/s41597.026.06714.5 - Decoding hand orientation (4 categories: left/right hand x self/other perspective) from EEG
- DOI: 10.1101/2025.07.10.664069 - Classification of hand orientation from EEG

- Kernel Notes: Verify event ID mapping from events.tsv.

Primary Task 4: Visual Touch Perception - Valence Regression

- Task Type: regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.1 s, tmax = 0.8 s, target_events = [`stimulus_onset` *]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`stimulus_onset`	continuous	Valence rating (continuous, from validated database)

- Super Meta Notes: Derived from validated database ratings. Kernel must extract valence scores from stimulus metadata files.
- Evidence:
- DOI: 10.1038/s41597.026.06714.5 - Regression of valence (continuous) from EEG epochs -100 to 800 ms
- DOI: 10.1101/2025.07.10.664069 - Valence regression from EEG epochs
- Kernel Notes: Verify event ID mapping from events.tsv.

Data Quality Notes

- No event markers available - cannot perform event-locked epoching without additional annotation files.
- High sampling rate (2048 Hz); downsample to 500 Hz before processing.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `Validated Touch-Video Database (Smit & Rich, 2025)`

234. openneuro_ds006171 - EEG during Near-Threshold Visual Detection Tasks (No-Cue, 50%-Cue, 100%-Cue)

Source: analysis.md

Dataset: `openneuro_ds006171` | Year: 2025 | URL: https://openneuro.org/datasets/ds006171
Category: Sensory/Response | Subjects: 26 | Files (Completed): 104 | Channels: 144 | Sampling Rate: 1024 Hz

Paradigm

Participants performed near-threshold visual detection tasks across three cueing conditions: no-cue (0% validity), noninformative cue (50% validity), and informative cue (100% validity). The study examines how spatial attention modulates near-threshold detection performance and associated EEG signatures. Stimuli were brief visual Gabor patches presented at or near detection threshold, and participants reported detection via button press. Behavioral columns (gabor_contrast, subjective_r, objective_r, response_time) stored in events.tsv enable regression analyses.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006171

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`128`	69653	admin	raw	BioSemi status boundary marker
`125`	23250	short	raw	100% validity cue - correct rejection (100pct_CR)
`124`	18239	short	raw	100% validity cue - false alarm (100pct_FA)
`2`	13793	short	raw	Response (detection report)
`1`	13790	short	raw	Trial start
`22`	6299	short	raw	50%-validity cue condition, stimulus B
`21`	6299	short	raw	50%-validity cue condition, stimulus A
`11`	6297	short	raw	No-cue condition, stimulus A

Event ID	Count (total)	Type	Meta Level	Meaning
12	6295	short	raw	No-cue condition, stimulus B
122	2753	short	raw	50% validity cue - miss (50pct_miss)
121	2650	short	raw	50% validity cue - hit (50pct_hit)
103	1530	short	raw	No-cue - false alarm (nocue_FA)
101	1530	short	raw	No-cue - hit (nocue_hit)
111	1529	short	raw	50% validity cue - hit (50pct_hit)
104	1529	short	raw	No-cue - correct rejection (nocue_CR)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Near threshold visual detection task

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006171 DOI: https://doi.org/10.18112/openneuro.ds006171.v1.0.0

Normal Task 1: Near-Threshold Detection Outcome (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

- Notes: Specific kernel maps numeric BioSemi codes to semantic labels. `regression_target`: response_time. `continuous_cols`: response_time, gabor_contrast, subjective_r, objective_r. Code `128` is a BioSemi boundary marker (admin; exclude). Key contrast: hit vs miss across cueing conditions.

Event	y	Meaning
`catch_trial`	0	Catch trial (no stimulus)
`catch_response`	1	Response during catch trial
`nocue_hit`	2	No-cue condition - hit
`nocue_miss`	3	No-cue condition - miss
`nocue_FA`	4	No-cue condition - false alarm
`nocue_CR`	5	No-cue condition - correct rejection
`50pct_hit`	6	50%-validity cue - hit
`50pct_miss`	7	50%-validity cue - miss
`50pct_FA`	8	50%-validity cue - false alarm
`50pct_CR`	9	50%-validity cue - correct rejection
`100pct_hit`	10	100%-validity cue - hit
`100pct_miss`	11	100%-validity cue - miss
`100pct_FA`	12	100%-validity cue - false alarm
`100pct_CR`	13	100%-validity cue - correct rejection
## Data Quality Notes		

- BioSemi 128/256-channel system; code `128` is a hardware boundary marker, not a stimulus.
- Continuous behavioral columns (gabor_contrast, subjective_r, objective_r, response_time) available for regression.
- Apply bandpass filter (0.1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG data during three near-threshold visual detection tasks: a no-cue task, a noninformative cue task (50% validity), and an informative cue task (100% validity)`

235. openneuro_ds006547 - Visual EEG Study (BrainVision -> BIDS)

Source: analysis.md

Dataset: `openneuro_ds006547` | Year: 2025 | URL: https://openneuro.org/datasets/ds006547
Category: Sensory/Response | Subjects: 31 | Files (Completed): 31 | Channels: 67 | Sampling Rate: 500 Hz

Paradigm

High-density EEG recordings were collected during a visual stimulation task using a BrainVision system, subsequently converted to BIDS format. Participants viewed visual stimuli under multiple conditions encoded by numeric BrainVision trigger codes (S2 through S56). The 30 distinct stimulus codes suggest a factorial or multi-condition visual paradigm covering perceptual and attentional manipulations.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006547

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S 5`	31	short	raw	Visual stimulus condition S5
`Stimulus/S 2`	31	short	raw	Visual stimulus condition S2
`Stimulus/S 42`	31	short	raw	Visual stimulus condition S42
`Stimulus/S 11`	31	short	raw	Visual stimulus condition S11
`Stimulus/S 51`	31	short	raw	Visual stimulus condition S51
`Stimulus/S 8`	31	short	raw	Visual stimulus condition S8
`Stimulus/S 45`	31	short	raw	Visual stimulus condition S45
`Stimulus/S 56`	31	short	raw	Visual stimulus condition S56
`Stimulus/S 3`	31	short	raw	Visual stimulus condition S3
`Stimulus/S 46`	31	short	raw	Visual stimulus condition S46

Event ID	Count (total)	Type	Meta Level	Meaning
Stimulus/S 21	31	short	raw	Visual stimulus condition S21
Stimulus/S 7	31	short	raw	Visual stimulus condition S7
Stimulus/S 12	31	short	raw	Visual stimulus condition S12
Stimulus/S 6	31	short	raw	Visual stimulus condition S6
Stimulus/S 16	31	short	raw	Visual stimulus condition S16
Stimulus/S 19	31	short	raw	Visual stimulus condition S19
Stimulus/S 44	31	short	raw	Visual stimulus condition S44
Stimulus/S 10	31	short	raw	Visual stimulus condition S10
Stimulus/S 55	31	short	raw	Visual stimulus condition S55
Stimulus/S 17	31	short	raw	Visual stimulus condition S17
Stimulus/S 41	31	short	raw	Visual stimulus condition S41
Stimulus/S 53	31	short	raw	Visual stimulus condition S53
Stimulus/S 50	31	short	raw	Visual stimulus condition S50
Stimulus/S 22	31	short	raw	Visual stimulus condition S22
Stimulus/S 20	31	short	raw	Visual stimulus condition S20
Stimulus/S 52	31	short	raw	Visual stimulus condition S52
Stimulus/S 14	31	short	raw	Visual stimulus condition S14
Stimulus/S 13	31	short	raw	Visual stimulus condition S13
Stimulus/S 54	31	short	raw	Visual stimulus condition S54
Stimulus/S 4	31	short	raw	Visual stimulus condition S4

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.

- Candidate Trigger Labels: `Stimulus/S 5`, `Stimulus/S 2`, `Stimulus/S 42`, `Stimulus/S 11`, `Stimulus/S 51`, `Stimulus/S 8`, `Stimulus/S 45`, `Stimulus/S 56`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Visual

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds006547 DOI: https://doi.org/10.18112/openneuro.ds006547.v1.0.0

Normal Task 1: Visual Condition Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: 30 distinct stimulus codes; decode from events.tsv to determine condition grouping. All codes appear in all 31 files (perfectly balanced). BrainVision format; verify event alignment with EEG data.

Event	y	Meaning
`Stimulus/S 5`	0	Visual condition S5
`Stimulus/S 2`	1	Visual condition S2
`Stimulus/S 42`	2	Visual condition S42
`Stimulus/S 11`	3	Visual condition S11
`Stimulus/S 51`	4	Visual condition S51
`Stimulus/S 8`	5	Visual condition S8
`Stimulus/S 45`	6	Visual condition S45
`Stimulus/S 56`	7	Visual condition S56
`Stimulus/S 3`	8	Visual condition S3
`Stimulus/S 46`	9	Visual condition S46
`Stimulus/S 21`	10	Visual condition S21
`Stimulus/S 7`	11	Visual condition S7
`Stimulus/S 12`	12	Visual condition S12
`Stimulus/S 6`	13	Visual condition S6
`Stimulus/S 16`	14	Visual condition S16
`Stimulus/S 19`	15	Visual condition S19
`Stimulus/S 44`	16	Visual condition S44
`Stimulus/S 10`	17	Visual condition S10
`Stimulus/S 55`	18	Visual condition S55

Event	y	Meaning
`Stimulus/S 17`	19	Visual condition S17
`Stimulus/S 41`	20	Visual condition S41
`Stimulus/S 53`	21	Visual condition S53
`Stimulus/S 50`	22	Visual condition S50
`Stimulus/S 22`	23	Visual condition S22
`Stimulus/S 20`	24	Visual condition S20
`Stimulus/S 52`	25	Visual condition S52
`Stimulus/S 14`	26	Visual condition S14
`Stimulus/S 13`	27	Visual condition S13
`Stimulus/S 54`	28	Visual condition S54
`Stimulus/S 4`	29	Visual condition S4
## Data Quality Notes		

- BrainVision format converted to BIDS; verify event alignment with EEG data.
- Large number of distinct event types (30); group by experimental condition before classification.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

236. openneuro_ds006777 - SFARI AVSRT EEG

Source: analysis.md

Dataset: openneuro_ds006777 | Year: 2025 | URL: https://openneuro.org/datasets/ds006777
Category: Sensory/Response | Subjects: 132 | Files (Completed): 132 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

An audiovisual simple reaction-time task (AVSRT) in which participants press a button as quickly as possible upon detecting auditory, visual, or simultaneous audiovisual stimuli. The paradigm measures unimodal vs. multimodal stimulus processing and associated reaction times. The dataset is from the SFARI (Simons Foundation Autism Research Initiative) collection, likely including participants with autism spectrum disorder and neurotypical controls.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: reaction time
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006777

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Response_button`	46353	short	raw	Participant button-press response
`Visual_stim`	16250	short	raw	Visual stimulus onset
`Audiovisual_stim`	16130	short	raw	Simultaneous audiovisual stimulus onset
`Auditory_stim`	15995	short	raw	Auditory stimulus onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: `reaction_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`

- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Visual_stim`, `Audiovisual_stim`, `Auditory_stim`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Audiovisual reaction-time decoding and RT modeling

- Task Type: classification/regression
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation` and `misc`
- Target Definition: Stimulus modality labels (auditory/visual/audiovisual) and reaction-time target derived from stimulus-response pairing.
- Evidence: AVSRT paradigm is designed to compare modality effects and reaction time in simple response trials. Source: https://openneuro.org/datasets/ds006777 DOI: https://doi.org/10.18112/openneuro.ds006777.v1.0.0

Normal Task 1: Stimulus Modality Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Three balanced modality conditions (Auditory/Visual/Audiovisual ~16000 each) well-suited for multiclass decoding. `Response_button` is a motor response event; exclude from stimulus classification if needed.

Event	y	Meaning
`Response_button`	0	Button-press response
`Audiovisual_stim`	1	Simultaneous audiovisual stimulus
`Auditory_stim`	2	Auditory-only stimulus
`Visual_stim`	3	Visual-only stimulus
## Data Quality Notes		

- SFARI autism cohort; likely includes participants with ASD and neurotypical controls.
- `Response_button` is a motor event marker, not a stimulus; consider excluding from stimulus decoding.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `SFARI AVSRT EEG`

237. physionet_mssvepdb - Multi-Stimulus SSVEP Database

Source: analysis.md

Dataset: `physionet_mssvepdb` | Year: 2021 | URL: N/A
Category: Sensory/Response | Subjects: 11 | Files (Completed): 235 | Channels: 14 (Emotiv) or 256 (BrainProducts) | Sampling Rate: 128 Hz (dataset3) or 250 Hz (datasets 1-2)

Paradigm

Eleven subjects performed steady-state visual evoked potential (SSVEP) tasks across three recording datasets with different EEG hardware (Nakanishi et al., 2018). Subjects fixated on flickering visual stimuli at different frequencies while EEG was recorded. Datasets 1 and 2 used a BrainProducts system (256 channels, 250 Hz); dataset 3 used an Emotiv EPOC (14 channels, 128 Hz). Each recording corresponds to one SSVEP stimulus condition (frequency). The kernel creates a single whole-recording `SSVEP` annotation for each file; the stimulus frequency is encoded in the filename/subdirectory.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://www.physionet.org/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
SSVEP	235	long	raw	SSVEP stimulus period (spans entire recording); frequency encoded in filename

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Stimulus frequency information must be parsed from the filename or subdirectory (e.g., frequency in Hz). Each file corresponds to one stimulus frequency condition.

Tasks

Primary Task 1: SSVEP target/frequency decoding

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification] | tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification]
- Label Route: `stim` or `Annotation`
- Target Definition: Stimulus frequency or target index

- Evidence: SSVEP paradigm is explicitly stated in dataset metadata/reference. Source: https://www.physionet.org/ DOI: https://doi.org/10.1109/tbme.2017.2694818

Normal Task 1: SSVEP Frequency Classification (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 4.0 s, overlap = 2.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Frequency label derived from filename; e.g., map each stimulus frequency to a class index.
- Notes: Stimulus frequencies and their mapping to class indices must be parsed from file paths or metadata. Process datasets 1/2 (BrainProducts, 256ch, 250Hz) and dataset 3 (Emotiv, 14ch, 128Hz) separately due to different channel counts and sampling rates.

Data Quality Notes

- Hardware differences across datasets: datasets 1-2 (BrainProducts, 256 channels, 250 Hz) vs. dataset 3 (Emotiv EPOC, 14 channels, 128 Hz); do not mix datasets in cross-subject analysis without normalization.
- Frequency labels are encoded in filenames, not event annotations; parse filenames to obtain classification targets.
- Age range: 24-39 years; sex: ~171 male / 64 female recording sessions.
- For SSVEP analysis, focus on occipital channels (Oz, O1, O2, POz) from the BrainProducts recordings.

References

- Reference: Nakanishi M et al. (2018). Enhancing Detection of SSVEPs for a High-Speed Brain Speller Using Task-Related Component Analysis. IEEE Trans. Biomed. Eng. https://www.physionet.org/

Chapter 5. Biological State

Datasets in this chapter: 43. Task-free or state-monitoring data (rest/sleep/clinical monitoring).

238. openneuro_ds001787 - EEG meditation study

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds001787` | Year: 2024 | URL: https://openneuro.org/datasets/ds001787
Category: Biological State | Subjects: 24 | Files (Completed): 40 | Channels: 80.0 | Sampling Rate: 256.0 Hz

Paradigm

Subjects engaged in focused meditation sessions that were interrupted every two minutes by a brief probe assessing concentration level and mind-wandering. EEG was recorded continuously with stimulus markers at probe onsets and response markers at button-press responses. The study examined neural correlates of sustained attention and mind-wandering during meditation using ERP and time-frequency analysis.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: response_time, regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds001787

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`response`	40 (files)	short	raw	Participant button-press response to attention probe
`stimulus`	40 (files)	short	raw	Probe stimulus onset (attention check)
`STATUS`	5	admin	raw	BioSemi system status marker - exclude

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

MISC Label Summary

- Route: `misc`
- Role: Stores derived/continuous labels that are not native raw event IDs.

- Candidate Labels: `response_time`
- Kernel Note: Keep derivation source, unit, and version metadata.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `stimulus`, `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: EEG-GSR Phase-Amplitude Coupling Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = [`stimulus`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`stimulus`	0	Low phase-amplitude coupling (d-prime below median across subjects)
`stimulus`	1	High phase-amplitude coupling (d-prime above median across subjects)

- Super Meta Notes: Derived from coupling analysis between EEG infraslow activity (0.01-0.1 Hz) phase and GSR amplitude during first 30s after meditation probe onset. Kernel step: (1) Extract 30-s epochs aligned to each `stimulus` event, (2) compute EEG ISA phase via Hilbert transform (0.01-0.1 Hz bandpass), (3) compute GSR amplitude envelope, (4) calculate d-prime of phase-amplitude coupling per channel, (5) average across channels, (6) threshold at median to produce binary high/low coupling labels. Also compute expert vs novice labels based on self-reported meditation experience.
- Evidence:
- DOI: 10.3389/fnins.2022.765585 - The paper analyzes coupling between EEG infraslow activity (0.01-0.1 Hz) phase and GSR amplitude during the first 30 seconds of each meditation trial from openneuro dataset ds001787, computing d-prime values and dividing subjects into high/low coupling groups.
- Kernel Notes: Verify event ID mapping: paper references 'meditation_onset' but dataset uses `stimulus` (probe onset) as proxy for meditation trial start. Kernel must compute d-prime from EEG-GSR coupling; raw event `stimulus` provides epoch timing only. The label (high/low coupling) is derived, not directly present in raw events.

Primary Task 2: Publisher-defined BIDS task: Meditation

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s , tmax = 1.0 s , target_events = [`stimulus`, `response`]
- Label Route: `Annotation`
- Target Definition:

Event	y	Meaning
`stimulus`	0	Probe stimulus onset (attention check)
`response`	1	Button-press response to probe

- Evidence:
- DOI: https://doi.org/10.18112/openneuro.ds001787.v1.1.1 - OpenNeuro metadata confirms tasksCompleted and dataset BIDS structure.

- Kernel Notes: Parameters require verification from dataset readme or publication. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- STATUS events are BioSemi system markers, not task events - do not epoch on these.
- Multiple `response` events may occur per `stimulus` probe; consider windowing to first response only.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Arnaud Delorme (October 17, 2018; updated June 2024)`

239. openneuro_ds001849 - RS_TMSEEG_Data

Source: analysis_minimax_v2.md

Dataset: openneuro_ds001849 | Year: 2020 | URL: https://openneuro.org/datasets/ds001849
Category: Biological State | Subjects: N/A | Files (Completed): 120 | Channels: 30.0 | Sampling Rate: 5000.0 Hz

Paradigm

Resting-state EEG data recorded in conjunction with Transcranial Magnetic Stimulation (TMS). Subjects were at rest (no active task) while TMS pulses were delivered; the R128 marker in BrainVision format marks a system/response trigger present in all files but does not indicate a meaningful task event. The primary scientific interest is resting-state oscillatory dynamics and TMS-evoked potentials.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds001849

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
R128	120 (files)	admin	raw	BrainVision system/response trigger present in every file - not a task event

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
> - rest: resting-state segment marker, for epoch_by_segmentation_hdf5
> - admin: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Publisher-defined BIDS task: Tmseegrest

- Task Type: classification/segmentation
- Epoch Method: epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: Annotation
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds001849 DOI: https://doi.org/10.18112/openneuro.ds001849.v1.0.2

- Kernel Notes: The dataset contains only one raw event ID (`R128`), which is a BrainVision system marker present in all 120 files and classified as admin (not a task event). The paper (DOI: 10.1371/journal.pone.0216185) analyzes TMS-evoked potentials requiring differentiation between stimulation sites (M1, PPC, DLPFC) and active vs sham conditions. Since these condition labels are not encoded in the raw event stream, meaningful epoch-level classification cannot be performed without external metadata.

Normal Task 1: Resting / Continuous Segmentation (resting)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no task events; R128 is admin))
- Notes: R128 appears in all 120 files and is a BrainVision recording artifact marker. High sampling rate (5000 Hz); consider downsampling. TMS artifact rejection may be needed if TMS pulses are captured in recording.

Data Quality Notes

- R128 is a BrainVision system marker present in every file - it is not a stimulus or response event.
- High sampling rate (5000 Hz); significant downsampling required before most analyses.
- TMS artifacts may contaminate EEG within the first ~50 ms post-pulse; apply TMS artifact rejection if analyzing TMS-evoked potentials.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `RS_TMSEEG_Data`

240. openneuro_ds002778 - UC San Diego Resting State EEG Data from Patients with Parkinson's Disease

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds002778` | Year: 2021 | URL: https://openneuro.org/datasets/ds002778
Category: Biological State | Subjects: N/A | Files (Completed): 46 | Channels: 41.0 | Sampling Rate: 512.0 Hz

Paradigm

Spontaneous scalp EEG recorded from patients with Parkinson's disease (PD) and healthy controls during resting-state conditions. The study investigated abnormal beta-band synchrony and beta waveform shape features (phase amplitude coupling, non-sinusoidal characteristics) as biomarkers of PD. All events in the BioSemi recording are system status markers (65536 = CM_in_range baseline, 130817 = CM status); no task events exist.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds002778

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`1`	44 (files)	admin	raw	BioSemi recording marker (onset of continuous record) - admin
`65536`	43 (files)	admin	raw	BioSemi CM_in_range baseline status code - exclude
`2`	1 (file)	admin	raw	BioSemi marker - sparse, likely admin
`130817`	1 (file)	admin	raw	BioSemi CM status code - exclude

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

Tasks

Primary Task 1: Parkinson's Disease vs Healthy Control Classification

- Task Type: classification

- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`HC`	0	Healthy control
`PD`	1	Parkinson's disease patient
- Super Meta Notes: Derived from raw events (none in raw data) by reading participant metadata from participants.tsv. The dataset contains only BioSemi system status markers (`1`, `65536`, `2`, `130817`) as raw events - all are admin codes. Clinical diagnosis labels (PD vs HC) are obtained from subject demographic metadata. Kernel step: Read participants.tsv, map subject ID to group label (PD/HC), assign label to each epoch.
- Evidence:
- DOI: 10.1016/j.bspc.2022.104116 - Uses openneuro_ds002778 for automated Parkinson's disease diagnosis, segmenting 3-minute resting-state EEG into 2-second non-overlapping windows and classifying PD vs HC using FAWT entropy with KNN
- DOI: 10.1016/j.compbiomed.2023.107031 - Uses openneuro_ds002778 for two-class PD vs HC classification with 1-s epochs and CSP features achieving 99.31% accuracy
- Kernel Notes: Epoch entire continuous recording using sliding window segmentation. Map each epoch's subject ID to group label from participants.tsv. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Medication State Classification (PD On vs Off Medication)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`off`	0	Parkinson's disease patient off medication
`on`	1	Parkinson's disease patient on medication
- Super Meta Notes: Derived from raw events (none in raw data) by reading participant metadata from participants.tsv. The dataset contains only BioSemi system status markers as raw events. Medication status (on/off) is stored in participant demographic data. Kernel step: Read participants.tsv, extract medication_state column, map subject ID to on/off label, assign to epochs.
- Evidence:
- DOI: 10.1038/s41598-022-26644-7 - Uses openneuro_ds002778 to classify off-PD vs on-PD vs HC using 10-second segments and DWT-based entropy features
- DOI: 10.1523/ENEURO.0151-19.2019 - Analyzes resting-state EEG from PD patients on and off medication to determine if beta waveform shape features can differentiate medication states
- Kernel Notes: Only applicable to epochs from PD subjects (exclude HC subjects). Map subject IDs to medication status from participants.tsv.

Data Quality Notes

- Pure resting-state dataset; all events are BioSemi hardware status codes.
- 65536 (CM_in_range) and 130817 are BioSemi system codes - do not epoch on these.
- PD vs HC group labels available in participants.tsv for disease classification.

- Verify channel count consistency across subjects before pooling.

References

- Paper: Characteristics of Waveform Shape in Parkinson's Disease Detected with Scalp Electroencephalography (DOI/ref: `10.1523/ENEURO.0151-19.2019`)

241. openneuro_ds003194 - Neuroepo multisession

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds003194` | Year: 2022 | URL: https://openneuro.org/datasets/ds003194
Category: Biological State | Subjects: 25 | Files (Completed): 29
Channels: 19 | Sampling Rate: 200 Hz

Paradigm

Resting-state EEG recorded before and after a six-month Neuroepo drug administration period to evaluate neuroprotective effects in Parkinson's disease patients. The protocol includes resting-state (eyes closed / eyes open), hyperventilation periods, and photic stimulation sections. This is a clinical neurological assessment battery recorded in Spanish (event labels in Spanish).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003194

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`ojos cerrados`	29 (files)	long	raw	Eyes-closed resting state
`ojos abiertos`	28 (files)	long	raw	Eyes-open resting state
`recuperacion`	28 (files)	long	raw	Recovery period after hyperventilation
`hiperventilacion 1`	27 (files)	long	raw	Hyperventilation period 1
`hiperventilacion 2`	28 (files)	long	raw	Hyperventilation period 2
`hiperventilacion 3`	28 (files)	long	raw	Hyperventilation period 3
`hiperventilacion 4`	15 (files)	long	raw	Hyperventilation period 4
`photic stimulation`	11 (files)	long	raw	Photic stimulation block
`fotoestimulacion`	11 (files)	long	raw	Photic stimulation (Spanish label)
`discontinuity`	9 (files)	admin	raw	Recording discontinuity marker - exclude

Event ID	Count (total)	Type	Meta Level	Meaning
HPVOJOS ABIERTOS	6 (files)	long	raw	Eyes-open during hyperventilation
HEPERVENTILACION OJOS ABIERTOS	1 (file)	long	raw	Hyperventilation eyes-open variant

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: photic stimulation, fotoestimulacion
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: qEEG-Mediated Neuroepo Effect on Cognition

- Task Type: regression
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 2.56 s, overlap = 0.0 *, target_events = [ojos cerrados]
- Label Route: super
- Meta Level: super
- Target Definition:
Event	y	Meaning
continuous	0-1	Cognitive test scores (MMSE, DRS, FAB, Sequencing, Memory, Recognition)
continuous	0-1	Neuroepo dose (0 at baseline, 5mg at 6 months for treatment group; 0 for placebo)
- Super Meta Notes: Derived from raw event ojos cerrados by extracting 2.56-s artifact-free epochs (512 samples at 200 Hz), computing power spectral density (PSD) in 49 frequency bins (0.78-19.14 Hz) via Welch's method, applying VARETA inverse solution for qEEG source estimation (3,244 source spectra), then performing probabilistic CCA (pCCA) with 6 cognitive test scores. Kernel step: Extract epochs -> compute PSD -> source localization -> pCCA latent variables -> predict cognitive scores from Neuroepo dose.
- Evidence:
- DOI: 10.3389/fnins.2022.841428 - Uses openneuro_ds003194 EEG data to test whether qEEG source activity mediates the effect of Neuroepo dosage on cognitive performance in PD patients, using pCCA to extract latent variables from 2.56-s resting-state epochs and linear mixed-effects mediation models.
- Kernel Notes: 1) Extract 2.56-s epochs from ojos_closed annotation events. 2) Compute PSD using Welch's method (0.78-19.14 Hz, 49 bins). 3) Apply VARETA or equivalent qEEG source imaging. 4) Perform pCCA to derive latent variables from source spectra and cognitive scores. 5) Use mediation analysis (linear mixed-effects) to test qEEG mediation of Neuroepo-cognition relationship. Note: overlap=0.0 as epochs are artifact-free selections (24 epochs averaged per subject). Verify epoch count matches paper methodology.

Normal Task 1: Brain State Classification (multiclass)

- Task Type: classification

- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `discontinuity` is a recording discontinuity marker - excluded. All other events are sustained clinical states (>=2 s). `photic stimulation` and `fotoestimulacion` are equivalent labels (Spanish/English) for the same condition.

Event	y	Meaning
`ojos cerrados`	0	Eyes-closed resting state
`ojos abiertos`	1	Eyes-open resting state
`recuperacion`	2	Recovery period
`hiperventilacion 2`	3	Hyperventilation period 2
`hiperventilacion 3`	4	Hyperventilation period 3
`hiperventilacion 1`	5	Hyperventilation period 1
`hiperventilacion 4`	6	Hyperventilation period 4
`photic stimulation`	7	Photic stimulation
`fotoestimulacion`	8	Photic stimulation (Spanish)
`HPVOJOS ABIERTOS`	9	Eyes-open during hyperventilation
`HEPERVENTILACION OJOS ABIERTOS`	10	Hyperventilation eyes-open variant

Data Quality Notes

- Clinical recording from Parkinson's patients; expect higher artifact rate than healthy subjects.
- `photic stimulation` / `fotoestimulacion` are duplicate labels for the same condition.
- `discontinuity` events mark recording gaps - exclude from epoch analysis.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. Sci. data 6, 103 (2019)`

242. openneuro_ds003195 - Placebo Neuroepo multisession

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003195 | Year: 2022 | URL: https://openneuro.org/datasets/ds003195
Category: Biological State | Subjects: 20 | Files (Completed): 20
Channels: 19 | Sampling Rate: 200 Hz

Paradigm

Resting-state EEG recorded from Parkinson's disease patients who received a placebo (not active Neuroepo) during a six-month drug evaluation trial. Recordings were taken before and after the administration period. This is the placebo control arm companion to ds003194 (Neuroepo treatment). Protocol includes resting-state (eyes open/closed), hyperventilation, and recovery periods.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003195

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
ojos cerrados	20 (files)	long	raw	Eyes-closed resting state
ojos abiertos	19 (files)	long	raw	Eyes-open resting state
hiperventilacion 1	19 (files)	long	raw	Hyperventilation period 1
hiperventilacion 2	19 (files)	long	raw	Hyperventilation period 2
hiperventilacion 3	19 (files)	long	raw	Hyperventilation period 3
recuperacion	19 (files)	long	raw	Recovery period after hyperventilation
hiperventilacion 4	12 (files)	long	raw	Hyperventilation period 4
discontinuity	8 (files)	admin	raw	Recording discontinuity marker - exclude
photic stimulation	3 (files)	long	raw	Photic stimulation block
fotoestimulacion	3 (files)	long	raw	Photic stimulation (Spanish label)

Event ID	Count (total)	Type	Meta Level	Meaning
HPVOJOS ABIERTOS	2 (files)	long	raw	Eyes-open during hyperventilation
hipervantilacion 4	1 (file)	long	raw	Hyperventilation period 4 (variant spelling)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: photic stimulation, fotoestimulacion
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Cognitive Performance Prediction from Resting-state EEG

- Task Type: regression
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 2.56 s, overlap = 0.0 *, target_events = [ojos cerrados] (implicit, see Kernel Notes)
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning
(continuous)	0-1 (normalized)	Cognitive performance latent score from pCCA
- Super Meta Notes: Derived from raw EEG segments at ojos cerrados (eyes-closed) periods. Kernel computes: (1) extract 2.56 s epochs (512 samples at 200 Hz) from eyes-closed resting EEG, (2) compute spectral power across frequency bands, (3) derive latent qEEG variable via probabilistic CCA from spectra x frequency points, (4) predict cognitive latent variable (Cog) from qEEG latent using linear mixed-effects model.
- Evidence:
- DOI: 10.3389/fnins.2022.841428 - Uses EEG from ds003195 (placebo) + ds003194 (treatment) with pCCA-derived latent variables from qEEG spectra and cognitive test scores to test mediation model
- Kernel Notes: Paper states eyes-closed resting was "implicit, not labeled in events" - kernel must identify contiguous eyes-closed segments from ojoscerrados annotations and extract artifact-free 2.56 s epochs. 24 epochs averaged per subject for PSD calculation per paper. Verify event ID mapping: ojoscerrados in events.tsv maps to Spanish "ojos cerrados" in annotation.

Normal Task 1: Brain State Classification (multiclass)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: Annotation

- Target Definition: Event-to-label mapping table below.
- Notes: `discontinuity` is a recording gap marker - excluded. Sparse counts for photic stimulation (3 files) and HPVOJOS (2 files); consider merging equivalent Spanish/English labels. Placebo arm of Neuroepo trial (NCT04110678).

Event	y	Meaning
`ojos cerrados`	0	Eyes-closed resting state
`ojos abiertos`	1	Eyes-open resting state
`hiperventilacion 1`	2	Hyperventilation period 1
`hiperventilacion 2`	3	Hyperventilation period 2
`hiperventilacion 3`	4	Hyperventilation period 3
`recuperacion`	5	Recovery period
`hiperventilacion 4`	6	Hyperventilation period 4
`photic stimulation`	7	Photic stimulation
`fotoestimulacion`	8	Photic stimulation (Spanish)
`HPVOJOS ABIERTOS`	9	Eyes-open during hyperventilation
`hipervantilacion 4`	10	Hyperventilation period 4 (variant spelling)

Data Quality Notes

- Clinical recording from Parkinson's patients; expect higher artifact rate.
- `photic stimulation` and `fotoestimulacion` are equivalent labels - merge if needed.
- `hipervantilacion 4` is a spelling variant of `hiperventilacion 4` - treat as same condition.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `NCT04110678`

243. openneuro_ds003478 - EEG: Depression rest

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003478 | Year: 2021 | URL: https://openneuro.org/datasets/ds003478
Category: Biological State | Subjects: 122 | Files (Completed): 243
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recording featuring alternating one-minute intervals of eyes-open and eyes-closed conditions in participants with varying scores on the Beck Depression Inventory (BDI). The study aims to identify EEG biomarkers of depression severity. Each session includes multiple alternating eyes-open and eyes-closed blocks, with `Start and Finish` markers indicating session boundaries. STATUS is a BioSemi admin marker.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, stim, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003478

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Eyes Open`	368 (files)	long	raw	Eyes-open resting state block (~1 min)
`Eyes Closed`	368 (files)	long	raw	Eyes-closed resting state block (~1 min)
`STATUS`	243 (files)	admin	raw	BioSemi system status marker - exclude
`Start and Finish`	184 (files)	admin	raw	Session start/end boundary marker - exclude

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

MISC Label Summary

- Route: `misc`
- Role: Stores derived labels (super meta) not directly present in raw event IDs.

- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `STATUS`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Depression Status Classification (cMDD vs pMDD vs None vs Control)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 73.68 s (after removing 10s artifact from ~93.68s avg), segment_length = 71.82 s (after removing 10s artifact from ~91.82s avg), overlap = 0.0
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Control`	0	BDI<=7, healthy control (N=75)
`Depressive`	1	BDI>=13, with further subclassification
`cMDD`	2	Current Major Depressive Disorder (within depressive group)
`pMDD`	3	Past Major Depressive Disorder (within depressive group)
`None`	4	Depressive symptoms but no MDD history (within depressive group)

- Super Meta Notes: Derived from raw events [`Eyes Open`, `Eyes Closed`] by using entire segments of each condition (excluding first/last 10s artifact). Labels derived from BDI scores in participants.tsv: BDI<10=Control, BDI>=13=Depressive. Clinical interview further subdivided depressive group into cMDD, pMDD, None. Kernel step: extract participant BDI from metadata, map to depression status label, then segment EEG by condition and assign label.
- Evidence:
- DOI: 10.3389/fpsyt.2023.1132996 - Multi-class classification of depression status (cMDD vs pMDD vs None vs Control) using spatial correlations of EEG amplitude fluctuations (0.05-0.1Hz) during eyes-open and eyes-closed rest segments
- Kernel Notes: Extract participant-level BDI scores from participants.tsv. Map BDI<=7->Control, BDI>=13->Depressive subclass (cMDD/pMDD/None). Segment EEG by condition (`Eyes Open`, `Eyes Closed`), remove initial and final 10s of each segment. Compute spatial correlation features from theta/alpha/beta bands. (replaces Normal Task 1 from original analysis.md: Eyes Open vs Eyes Closed binary is superseded by this more clinically meaningful depression classification task using the same event markers)

Data Quality Notes

- `STATUS` is BioSemi hardware marker - exclude.
- `Start and Finish` marks session boundaries - admin marker, exclude.
- Large dataset: 122 subjects, 243 files; subject-level BDI scores available for depression prediction.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG: Depression rest`

244. openneuro_ds003555 - Dataset of EEG Recordings of Pediatric Patients with Epilepsy Based on the 10-20 System

Source: analysis.md

Dataset: openneuro_ds003555 | Year: 2021 | URL: https://openneuro.org/datasets/ds003555
Category: Biological State | Subjects: 30 | Files (Completed): 257
Channels: 52 | Sampling Rate: 2000 Hz

Paradigm

Continuous sleep EEG recordings from pediatric patients with epilepsy, focused on identifying High-Frequency Oscillations (HFOs) as clinical biomarkers during N3 slow-wave sleep stages. The dataset was recorded using the standard 10-20 electrode system and is intended for automated HFO detection algorithm development and sleep stage analysis. No task-based paradigm was employed; recordings capture spontaneous brain activity during sleep.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003555

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
-	-	-	raw	No event markers detected; continuous sleep EEG recording

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Sleep stage classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Sleep stage labels (W/N1/N2/N3/REM; mapping dataset-specific)

- Evidence: Dataset and references explicitly center on PSG/EEG sleep scoring. Source: https://openneuro.org/datasets/ds003555 DOI: https://doi.org/10.18112/openneuro.ds003555.v1.0.1

Normal Task 1: Continuous sleep EEG segmentation

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no label, for unsupervised or HFO detection))
- Notes: High sampling rate (2000 Hz) is appropriate for HFO detection (80-500 Hz). Sleep stage labels (if available from hypnogram) can serve as y labels for sleep classification tasks. No trial-level events present.

Data Quality Notes

- Very high sampling rate (2000 Hz); high memory requirements for processing.
- Pediatric epilepsy population; inter-individual variability in seizure history and medications.
- HFO detection requires high-frequency filtering (80-500 Hz); standard 0.5-40 Hz bandpass not appropriate for primary analysis goal.

References

- Reference: `Dataset of EEG recordings of pediatric patients with epilepsy based on the 10-20 system`

245. openneuro_ds003710 - APPLESEED Example Dataset

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003710 | Year: 2022 | URL: https://openneuro.org/datasets/ds003710
Category: Biological State | Subjects: 13 | Files (Completed): 48
Channels: 32 | Sampling Rate: 5000 Hz

Paradigm

Longitudinal spontaneous EEG recordings from infants at 4, 8, and 12 months of age during passive viewing of social and non-social stimuli (faces, speech, objects, water). The dataset validates the APPLESEED (Automated Preprocessing Pipe-Line for the Estimation of Scale-wise Entropy from EEG Data) pipeline for pediatric EEG preprocessing and entropy estimation. Stimuli include social (face, speech) and non-social (object, water) categories presented as short video clips, with attention-getter events to orient the infant.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003710

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
face	48	short	raw	Face stimulus onset (social)
speech	48	short	raw	Speech stimulus onset (social)
object	48	short	raw	Object stimulus onset (non-social)
water	48	short	raw	Water stimulus onset (non-social)
attention-getter	48	short	raw	Attention-orienting stimulus before trial
experimenter-pause	45	admin	raw	Experimenter pause marker
New Segment	48	admin	raw	Recording segment boundary marker
Comment	48	admin	raw	Experimenter annotation

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5

- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Social vs non-social stimulus classification (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [`face`, `speech`, `object`, `water`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`face`	1	Face stimulus (social)
`speech`	1	Speech stimulus (social)
`object`	0	Object stimulus (non-social)
`water`	0	Water stimulus (non-social)

- Evidence:
- DOI: `10.1016/j.dcn.2022.101163` - This paper develops and validates the APPLESEED preprocessing pipeline using infant EEG data from OpenNeuro ds003710, applying it to analyze scale-wise entropy differences between face viewing (social) and object viewing (non-social) conditions across developmental stages.
- Kernel Notes: Downsample from 5000 Hz to 250-500 Hz before epoching. Use epoch_by_event_hdf5 with tmin=-0.2s and tmax=1.0s relative to stimulus onset. Exclude admin markers (`New Segment`, `Comment`, `experimenter-pause`). (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Extremely high sampling rate (5000 Hz); downsampling required before standard EEG analysis.
- Infant population; high motion artifacts and data loss expected.
- Small sample (13 subjects); longitudinal design with 3-4 files per subject across age points.
- `New Segment` and `Comment` markers should be excluded from epoching.

References

- Reference: `The Automated Preprocessing Pipe-Line for the Estimation of Scale-wise Entropy from EEG Data (APPLESEED): Development and validation for use in pediatric populations. OpenNeuro ds003710.`

246. openneuro_ds003768 - Simultaneous EEG and fMRI Signals During Sleep from Humans

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003768 | Year: 2026 | URL: https://openneuro.org/datasets/ds003768
Category: Biological State | Subjects: 33 | Files (Completed): 252
Channels: 32 | Sampling Rate: 5000 Hz

Paradigm

Simultaneous EEG-fMRI recordings were acquired during resting-state wakefulness and multiple sleep sessions (NREM, REM) to analyze spontaneous brain activity and sleep stage dynamics. Participants underwent extended sleep sessions in an MRI scanner while 32-channel EEG was recorded concurrently with fMRI. BrainVision numeric stimulus codes (S1, S2, S3) and response codes (R128) mark sleep stage transitions or scoring events. The dataset supports multimodal sleep neuroscience research.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003768

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Stimulus/S 1`	183	long	raw	BrainVision S1 trigger (sleep stage marker)
`Stimulus/S 2`	193	long	raw	BrainVision S2 trigger (sleep stage marker)
`Stimulus/S 3`	183	long	raw	BrainVision S3 trigger (sleep stage marker)
`Response/R128`	252	admin	raw	BrainVision R128 response trigger

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `Stimulus/S 1`, `Stimulus/S 2`, `Stimulus/S 3`

- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sleep stage classification (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = [`sleep_stage_label`] *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`sleep_stage_label`	0	Wake (W)
`sleep_stage_label`	1	NREM1 (N1)
`sleep_stage_label`	2	NREM2 (N2)
`sleep_stage_label`	3	NREM3 (N3)

- Super Meta Notes: Derived from raw EEG by polysomnographic scoring. Sleep stage labels are stored in the sourcedata folder (annotated by Registered Polysomnographic Technologist per AASM guidelines). Raw trigger codes `Stimulus/S 1`, `Stimulus/S 2`, `Stimulus/S 3` are button-press synchronization markers, not sleep stage labels themselves. Kernel step: Load sleep stage annotations from sourcedata, align with EEG epochs using timestamps, produce Annotation channel with sleep stage labels.
- Evidence:
- DOI: `10.1016/j.dib.2023.109059` - Data in Brief paper describing openneuro_ds003768 dataset with manually annotated sleep stage labels (W, N1, N2, N3, REM) scored by RPT on 30-sec epochs
- DOI: `10.1016/j.neuroimage.2025.121384` - Sleep stage classification study using 30-sec AASM-standard epochs on this dataset
- Kernel Notes: Load sleep stage annotations from dataset's sourcedata folder. Verify exact annotation file format and timestamp alignment. Preprocess: MRI gradient artifact removal required (EEG-fMRI simultaneous recording). Downsample from 5000 Hz to 250 Hz. (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Binary arousal state classification (awake vs sleep)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 20 s, overlap = 10 s, target_events = [`arousal_state`] *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`arousal_state`	0	Awake
`arousal_state`	1	Asleep (any NREM stage)

- Super Meta Notes: Derived from polysomnographic sleep stage labels (W vs N1/N2/N3). Kernel step: Collapse multiclass sleep stage labels (W=0, N1/N2/N3=1) into binary arousal labels.
- Evidence:
- DOI: `10.1016/j.neuroimage.2025.121384` - Binary classification of arousal states (awake vs anesthesia; awake vs sleep) using 20-second epochs with 10-second overlaps on this dataset
- Kernel Notes: Requires completing Primary Task 1 first to obtain sleep stage labels. Then collapse to binary: W->0 (awake), N1/N2/N3->1 (asleep). Downsample to 250 Hz. MRI artifact removal required.

Data Quality Notes

- Very high sampling rate (5000 Hz); downsample to 250-500 Hz before analysis.
- EEG-fMRI simultaneous recording; gradient artifacts present; requires MRI artifact removal.
- R128 is a BrainVision response/sync marker; treat as admin.
- 252 files from 33 subjects across multiple sleep sessions.

References

- Reference: `Albouy et al., Journal of Sleep Research, 2013. Simultaneous EEG and fMRI during sleep. OpenNeuro ds003768.`

247. openneuro_ds003775 - SRM Resting-State EEG

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003775 | Year: 2022 | URL: https://openneuro.org/datasets/ds003775
Category: Biological State | Subjects: 111 | Files (Completed): 153
Channels: 64 | Sampling Rate: 1024 Hz

Paradigm

Four minutes of uninterrupted eyes-closed resting-state EEG recorded from healthy adult participants as part of a larger experimental paradigm. This dataset was collected to provide a normative baseline for comparing resting EEG characteristics across studies. The resting-state recording provides a standardized, task-free EEG sample suitable for spectral power analysis, functional connectivity, and individual difference studies.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003775

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
-	-	-	raw	No event markers detected; pure resting-state EEG

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Age Regression from Resting-State EEG Spectral Features

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 5.0 s, overlap = 0.0 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning

`age`	continuous	participant age in years
`sex`	categorical	male/female
`site`	categorical	research center (Oslo)
- Super Meta Notes: Derived from dataset metadata (not EEG events). Labels include biological covariates (age, sex) and batch variable (site) extracted from participant demographics. Kernel step: Extract spectral power features (delta, theta, alpha, beta, gamma bands) from each 5-second epoch, apply ComBat harmonization to remove site effects, train regression model to predict age.
- Evidence:
- DOI: 10.1016/j.clinph.2024.09.019 - Shows Oslo dataset (n=111) from openneuro_ds003775 was used for resting-state EEG spectral feature harmonization and age-correlation analysis using ComBat methods; used 5-second epochs from preprocessed signals.
- Kernel Notes: Preprocess: bandpass filter 0.5-45 Hz, re-reference to average. Feature extraction: compute spectral power in standard frequency bands per epoch. Apply ComBat harmonization across sites (if combining with other datasets). Train regression model (e.g., ridge, SVR) to predict age. Report correlation between predicted and actual age.

Normal Task 1: Resting state EEG segmentation

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (null (no labels, used for unsupervised analysis or individual differences research))
- Notes: Pure 4-minute eyes-closed resting-state; 111 subjects, 153 files (some subjects may have multiple sessions). Suitable for individual alpha frequency, spectral power, and connectivity analyses. Large normative dataset.

Primary Task 2: Publisher-defined BIDS task: Resteyesc

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.5 *
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds003775 DOI: https://doi.org/10.1111/ejn.14964

Data Quality Notes

- Purely resting-state; no task events or stimuli.
- 153 files from 111 subjects; some subjects have multiple files (repeat sessions).
- 64-channel EEG at 1024 Hz; standard configuration for resting-state analysis.

References

- Reference: `Hatlestad-Hall, C., Rygvold, T. W., & Andersson, S. (2022). BIDS-structured resting-state electroencephalography (EEG) data extracted from an experimental paradigm. Data in Brief, 45, 108647.`

248. openneuro_ds003944 - EEG: First Episode Psychosis vs. Control Resting Task 1

Source: analysis_minimax_v2.md

Dataset: openneuro_ds003944 | Year: 2022 | URL: https://openneuro.org/datasets/ds003944
Category: Biological State | Subjects: 82 | Files (Completed): 82
Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

A 5-minute eyes-open resting-state EEG recording comparing spontaneous brain activity between individuals with First Episode Psychosis (FEP) and age-matched healthy controls. The dataset is designed to identify EEG biomarkers of psychosis onset. No trial-level stimuli are presented; the label of interest is the clinical group membership (FEP vs. control) stored in participants.tsv.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003944

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
unknown	82	admin	raw	Unidentified system marker - not suitable for epoching

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: FEP vs Control Classification from Resting-State EEG

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 30 s, overlap = 0.0 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning

| `Control` | 0 | Healthy control subject |
| `FEP` | 1 | First Episode Psychosis patient |
- Super Meta Notes: Derived from clinical group membership in participants.tsv (FEP diagnosis vs healthy control). Kernel step: Extract group label from participants metadata and assign to all segments from each subject. Both papers use subject-level diagnosis labels applied uniformly to all EEG segments from that participant.
- Evidence:
- DOI: 10.1038/s41598.024.66110.0 - Uses OpenNeuro ds003944 dataset to classify FEP vs healthy controls using PSD features from 30-second segments of 5-minute resting-state EEG, achieving 95.51% accuracy with Gaussian Process Classifier.
- DOI: 10.3390/biomedicines11123223 - Uses ds003944 to classify FEP vs healthy controls from resting-state EEG using ciSSA decomposition and entropy/statistical/frequency features, achieving up to 96.23% accuracy.
- Kernel Notes: Segment the continuous 5-minute resting-state recording into non-overlapping 30-second windows. Extract the group label (FEP=1, Control=0) from participants.tsv and assign to all segments from that subject. Features: PSD, entropy, statistical, or frequency-domain features from each segment. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- No meaningful event markers; `unknown` events should be excluded
- Subject count not specified in header; 82 files suggests ~82 subjects
- Verify FEP/control group balance in participants.tsv before training

References

- No paper reference available

249. openneuro_ds003947 - EEG: First Episode Psychosis vs. Control Resting Task 2

Source: analysis.md

Dataset: openneuro_ds003947 | Year: 2022 | URL: https://openneuro.org/datasets/ds003947
Category: Biological State | Subjects: 61 | Files (Completed): 61
Channels: 64 | Sampling Rate: 3000 Hz

Paradigm

Five minutes of resting-state EEG recording used to compare spontaneous brain activity between individuals with First Episode Psychosis (FEP) and healthy controls. This is a second independent resting-state dataset from the same clinical study series as ds003944, recorded at a higher sampling rate (3000 Hz). Group membership (FEP vs. control) serves as the classification label.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds003947

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
unknown	61	admin	raw	Unidentified system marker - not suitable for epoching

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Resting-state analysis baseline

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Condition/group labels from metadata when available
- Evidence: Dataset is primarily resting-state or spontaneous-state EEG. Source: https://openneuro.org/datasets/ds003947 DOI: https://doi.org/10.18112/openneuro.ds003947.v1.0.1

Normal Task 1: FEP vs. Control Classification (binary, resting-state segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: None (label from participants.tsv group column)
- Notes: Purely resting-state dataset; segment continuously and assign subject-level diagnosis labels (FEP=1, control=0). High sampling rate (3000 Hz) - consider downsampling to 500 Hz before processing to reduce computational load.

Data Quality Notes

- No meaningful event markers; `unknown` events should be excluded
- Very high sampling rate (~3000 Hz) may require downsampling
- Companion dataset to ds003944 (Task 1); can be combined for larger resting FEP cohort

References

- No paper reference available

250. openneuro_ds004348 - Ear-EEG Sleep Monitoring 2017 (EESM17)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004348` | Year: 2024 | URL: https://openneuro.org/datasets/ds004348
Category: Biological State | Subjects: 9 | Files (Completed): 36
Channels: 35 | Sampling Rate: 200 Hz

Paradigm

Nightly sleep monitoring and pre-sleep resting recordings using ear-EEG and partial polysomnography in a naturalistic home environment. Subjects wore ear-mounted EEG devices during sleep, allowing investigation of sleep stage classification and comparison between ear-EEG and conventional PSG signals.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004348

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Lights Off`	9	long	raw	Lights-off marker (sleep onset reference)
`Lights On`	9	long	raw	Lights-on marker (sleep end reference)
`Awake`	1	long	raw	Wakefulness annotation
`Artifact`	9	admin	raw	Artifact annotation
`Arousal`	9	admin	raw	Arousal annotation
`Beginning of Recording`	9	admin	raw	Recording start marker
`End of Study`	9	admin	raw	Recording end marker
`Montage`	9	admin	raw	Montage configuration event
`Started Analyzer - Sleep Events`	9	admin	raw	Analyzer start marker
`Started Analyzer - ECG`	9	admin	raw	ECG analyzer start
`Stopped Analyzer - ECG`	9	admin	raw	ECG analyzer stop

Event ID	Count (total)	Type	Meta Level	Meaning
`Started Analyzer - Auto-Staging`	9	admin	raw	Auto-staging start
`Stopped Analyzer - Auto-Staging`	9	admin	raw	Auto-staging stop
`512`	9	admin	raw	Numerical trigger code 512

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sleep stage classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = [`Awake`, `Lights Off`] *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
W (Wake)	0	Awake state during sleep period
NREM1	1	Light sleep stage 1
NREM2	2	Light sleep stage 2
NREM3	3	Deep sleep (slow wave)
REM	4	Rapid eye movement sleep

- Super Meta Notes: Derived from raw event(s) [`Awake`, `Lights Off`, `Lights On`, `Started Analyzer - Auto-Staging`, `Stopped Analyzer - Auto-Staging`] by reading PSG-derived sleep stage annotation files (stored separately from raw events). Kernel step: Load annotation file containing sleep stage labels (W/NREM1/NREM2/NREM3/REM) and merge with epoched EEG data. The annotation files contain manual expert scoring following AASM guidelines.
- Evidence:
- DOI: 10.1186/s12938-017-0400-5 - Uses a dataset of nine healthy subjects recorded simultaneously with PSG and ear-EEG for one night each, training a random forest classifier to automatically score sleep stages (W/REM/NREM1/NREM2/NREM3) from ear-EEG data and comparing to manual expert scoring.
- Kernel Notes: Use 30-s epochs per AASM guidelines. Exclude epochs containing `Artifact` or `Arousal` markers. Verify sleep stage annotation file locations in dataset directory structure. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Sleep stage annotations are in PSG annotation files, not directly as trial events
- Artifact markers must be excluded from analysis windows
- 35 channels includes non-EEG channels (ECG, respiratory belt, etc.)
- Only 9 subjects; limited generalization

References

- Reference: `Ear-EEG Sleep Monitoring 2017 (EESM17)`

251. openneuro_ds004504 - A dataset of EEG recordings from: Alzheimer's disease, Frontotemporal dementia and Healthy subjects

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004504 | Year: 2024 | URL: https://openneuro.org/datasets/ds004504
Category: Biological State | Subjects: 88 | Files (Completed): 176
Channels: 19 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings with eyes closed collected from healthy controls and patients with Alzheimer's disease (AD) and Frontotemporal Dementia (FTD). The study provides a clinical EEG dataset for developing diagnostic biomarkers and machine learning classifiers to differentiate neurodegenerative conditions from healthy aging.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004504

Event Structure

No event markers detected (`no_events`). This dataset contains resting-state EEG recordings without trial-level annotations.

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
`no_events`	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Three-class Dementia Classification (AD vs FTD vs HC)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`

- Epoch Parameters: segment_length = 4 s, overlap = 0.5
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Label	y	Meaning
HC	0	Healthy Control
AD	1	Alzheimer's Disease
FTD	2	Frontotemporal Dementia
- Super Meta Notes: Derived from subject-level diagnosis in `participants.tsv`. Each 4-second EEG epoch inherits the diagnostic group label (AD/FTD/CN) of the participant it belongs to. Kernel step: For each epoch, look up the subject_id from the epoch's parent recording, then assign the diagnosis label from participants.tsv. (replaces Normal Task 1 from original analysis.md)
- Evidence:
- DOI: 10.3390/data8060095 - Benchmarks classification using 4-second windows with 50% overlap, labeling epochs as AD, FTD, or CN based on subject diagnosis for binary classification (AD-CN and FTD-CN) with Leave-One-Subject-Out validation
- DOI: 10.1016/j.clinph.2024.12.008 - Analyzes resting-state EEG from ds004504 (36 AD, 23 FTD, 29 HC) using SVM with time-frequency functional connectivity, achieving 95% accuracy for AD vs HC and 86% for FTD vs HC
- DOI: 10.1109/access.2023.3294618 - Uses 30-second EEG windows with 15-second overlap for binary classification (AD-CN and FTD-CN) with DICE-net on the same dataset
- Kernel Notes: Since this dataset contains no raw event markers (`no_events`), use `epoch_by_segmentation_hdf5` with sliding windows. Each epoch must be labeled with the subject's diagnosis from participants.tsv. Verify subject-to-epoch mapping from the HDF5 file structure.

Data Quality Notes

- No trial-level event markers; pure resting-state dataset
- Subject diagnosis labels (HC/AD/FTD) available in participants.tsv
- Standard 10-20 montage (19 channels)
- 2 recordings per subject (176 files / 88 subjects)

References

- Reference: `A Dataset of Scalp EEG Recordings of Alzheimers Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG`

252. openneuro_ds004577 - Dataset containing resting EEG for a sample of 103 normal infants in the first year of life

Source: analysis.md

Dataset: `openneuro_ds004577` | Year: 2023 | URL: https://openneuro.org/datasets/ds004577
Category: Biological State | Subjects: 103 | Files (Completed): 103 | Channels: 128 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings from 103 healthy infants (aged 1-12 months) during the first year of life. Recordings were made during natural infant rest/sleep states. Eyes-open and eyes-closed state markers are present but the primary data is continuous infant resting EEG. The dataset supports developmental neuroscience research on early brain maturation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004577

Event Structure

No task event markers. State markers (eyes_closed, eyes_open) and artifact markers (discontinuity) are present but the dataset is primarily resting-state continuous EEG.

Event ID	Count (total)	Type	Meta Level	Meaning
`eyes_closed`	~103xN	rest	raw	Eyes-closed resting state
`eyes_open`	~103xN	rest	raw	Eyes-open resting state
`discontinuity`	~103xN	admin	raw	Recording discontinuity/artifact marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Publisher-defined BIDS task: EEG

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`

- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004577 DOI: https://doi.org/10.1016/j.earlhumdev.2011.05.010

Normal Task 1: Continuous Segmentation (resting/segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: N/A (unsupervised segmentation)
- Notes: Infant resting EEG without trial structure. discontinuity markers indicate artifact segments. Use sliding window segmentation; consider excluding segments surrounding discontinuity markers.

Data Quality Notes

- 103 infants aged 1-12 months; high variability in developmental stage
- 128 channels; high-density pediatric EEG cap
- discontinuity markers indicate artifact/movement artifacts in infant recordings

References

- Reference: `Dataset containing resting EEG for a sample of 103 normal infants in the first year of life`

253. openneuro_ds004584 - Rest Eyes Open (Parkinson's Disease)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds004584` | Year: 2023 | URL: https://openneuro.org/datasets/ds004584
Category: Biological State | Subjects: 149 | Files (Completed): 149 | Channels: 63 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG was collected from 149 participants (100 with Parkinson's disease and 49 healthy controls) who sat quietly with eyes open for approximately two minutes. The study characterizes spontaneous neural oscillatory patterns to differentiate PD patients from controls. No active task was required; participants fixated on a cross and remained still.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004584

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`boundary`	149	admin	raw	Segment boundary marker inserted by recording software (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Cognitive Impairment Detection from Resting-State EEG

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 150 s , overlap = 0.0 , target_events = [`resting_state_eog`] *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`cognitively_normal`	0	MoCA >= 26
`cognitive_impairment`	1	MoCA < 26

- Super Meta Notes: Derived from raw event(s) none (resting-state recording) by extracting MoCA scores from participants.tsv. Kernel step: Load participants.tsv, filter rows where MoCA < 26

(cognitive_impairment=1) vs MoCA >= 26 (cognitively_normal=0), merge with corresponding EEG recordings by participant_id.
- Evidence:
- DOI: 10.1038/s41531.023.00602.0 - Uses resting-state EEG from 100 PD and 49 control participants to develop LEAPD, a linear predictive coding-based biomarker that correlates with MoCA scores (rho=0.68) and detects cognitive impairment with 80% accuracy.
- DOI: 10.21203/rs.3.rs-2666578/v1 - Uses LEAPD (Linear Predictive Coding EEG Algorithm for Parkinson's Disease) to classify cognitive impairment (MoCA<26) versus normal cognition (MoCA>=26) from resting-state EEG in 100 PD and 49 control participants.
- Kernel Notes: Load continuous resting-state EEG recording (~2-3 min), apply bandpass filter 2-29 Hz, extract features (LPC encoding or equivalent), classify using MoCA-derived labels. Verify tmax matches actual recording duration per subject.

Normal Task 1: Resting State Segmentation (segmentation)

- Task Type: classification
- Epoch Method: `segmentation`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: None (no task events)
- Notes: Only `boundary` admin markers present - pure resting-state dataset. Use sliding-window segmentation. PD vs. control group labels available from participants.tsv for downstream classification.

Data Quality Notes

- Only boundary markers present; no trial events for epoch-locked classification.
- PD vs. control labels available in participants.tsv.

References

- Reference: `Rest eyes open EEG dataset - Parkinson's disease and healthy controls`

254. openneuro_ds004635 - Gaffrey Lab Infant EEG Microstates Reliability

Source: analysis.md

Dataset: openneuro_ds004635 | Year: 2024 | URL: https://openneuro.org/datasets/ds004635
Category: Biological State | Subjects: 48 | Files (Completed): 1
Channels: 129 | Sampling Rate: 1000 Hz

Paradigm

Infants watched a series of engaging and relaxing videos to facilitate recording of spontaneous EEG activity. The dataset was collected to assess the test-retest reliability of EEG microstate metrics in infants. The single aggregated file contains data from 48 infants with recording-boundary markers; no active task was required.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004635

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`bgin`	10	long	raw	Beginning of a recording segment
`TRSP`	10	long	raw	Transition / segment boundary marker

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

Tasks

Primary Task 1: Publisher-defined BIDS task: Resting

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)

- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004635 DOI: https://doi.org/10.1038/35081509

Normal Task 1: Long-term classification of paragraph tags (binary)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Only 10 events of each type - very limited for supervised classification. Primary scientific use is resting-state microstate analysis. Single aggregated file covers all 48 infants.

Event	y	Meaning
`bgin`	0	Recording segment start
`TRSP`	1	Transition segment marker
## Data Quality Notes		

- Single file aggregating 48 infants - individual subject effects not separable.
- Only 10 events of each type; insufficient for robust supervised classification.

References

- Reference: `Gaffrey Lab Infant EEG Microstates Reliability`

255. openneuro_ds004902 - Resting-State EEG Dataset for Sleep Deprivation

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004902 | Year: 2023 | URL: https://openneuro.org/datasets/ds004902
Category: Biological State | Subjects: 71 | Files (Completed): 218 | Channels: 61 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings with eyes open and eyes closed were conducted under two conditions: normal sleep and 24-hour sleep deprivation. The study investigates how sleep loss modulates spontaneous brain activity. Subjects participated in two sessions (normal sleep vs sleep-deprived). Files are split by condition (eyesopen/eyesclosed) with 5-minute continuous blocks. Event marker S 33 (60 total) appears as a resting-state session marker in a subset of files; all other S-codes are rare system markers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004902

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
S 33	60	rest	raw	Resting-state session marker (BrainVision)
S 45	7	admin	raw	Rare system marker (exclude)
S 44	7	admin	raw	Rare system marker (exclude)
S 55	7	admin	raw	Rare system marker (exclude)
S119	7	admin	raw	Rare system marker (exclude)
S 66	7	admin	raw	Rare system marker (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: S 33, S 45, S 44, S 55, S119, S 66
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sleep Deprivation Classification from Resting-State EEG

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.0
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`normal_sleep`	0	Resting-state EEG recorded after normal sleep (session type: ses-normal)
`sleep_deprived`	1	Resting-state EEG recorded after 24-hour sleep deprivation (session type: ses-sleepdeprived)

- Super Meta Notes: Labels derived from session metadata (ses-normal vs ses-sleepdeprived folder structure and participants.tsv condition field), not from raw event markers. Raw marker `S 33` appears in subset of files but is NOT used for labeling. Kernel step: Extract condition label from folder structure (ses-normal -> 0, ses-sleepdeprived -> 1) and assign to all 4-second epochs from the corresponding 5-minute recording.
- Evidence:
- DOI: 10.1109/tnsre.2025.3547616 - Binary classification between normal and sleep-deprived states using 4-second non-overlapping windows from 5-minute recordings
- DOI: 10.1038/s41597-024-03268-2 - Dataset descriptor validating 4-second epoch segmentation for NS vs SD classification
- Kernel Notes: Segment 5-minute continuous EEG into 4-second non-overlapping windows (75 epochs per file). Assign label based on session metadata: if filename contains "ses-normal" -> y=0; if "ses-sleepdeprived" -> y=1. (replaces Primary Task 1, 2, 3 from original analysis.md)

Normal Task 1: Sleep Deprivation Resting-State Segmentation (segmentation)

- Task Type: regression
- Epoch Method: `segmentation`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `Annotation`
- Target Definition: None
- Notes: Majority of files have no event markers - purely continuous resting-state. Use filename-derived or participants.tsv condition labels (eyesopen/eyesclosed x normal/deprived) for downstream classification. S 44/45/55/66/119 are rare admin system markers - exclude.

Data Quality Notes

- Most files have no event markers; condition derived from filename (eyesopen/eyesclosed).
- Session condition (normal sleep / sleep deprivation) in participants.tsv.
- S 44/45/55/66/119 are rare admin BrainVision system markers - exclude.

References

- Reference: `Xiang, C., Fan, X., Bai, D. et al. A resting-state EEG dataset for sleep deprivation. Sci Data 11, 427 (2024).`

256. openneuro_ds005178 - Ear-EEG Sleep Monitoring 2023 (EESM23)

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005178 | Year: 2024 | URL: https://openneuro.org/datasets/ds005178
Category: Biological State | Subjects: 10 | Files (Completed): 159 | Channels: 5 | Sampling Rate: 250 Hz

Paradigm

Naturalistic nightly sleep monitoring using ear-EEG (in-ear electrodes) and polysomnography (PSG) in an ecological home environment where participants self-applied the recording equipment. The study investigates the feasibility of home-based sleep staging using miniaturized ear-EEG compared to gold-standard PSG. The `trigger` event marks synchronization pulses between the ear-EEG and PSG systems.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005178

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`trigger`	175	long	raw	Synchronization trigger between ear-EEG and PSG

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `trigger`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sleep stage classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = [`sleep_stage_W`, `sleep_stage_N1`, `sleep_stage_N2`, `sleep_stage_N3`, `sleep_stage_REM`]
- Label Route: `Annotation`

- Meta Level: super
- Target Definition:
Event	y	Meaning
`sleep_stage_W`	0	Wake stage
`sleep_stage_N1`	1	Light sleep stage 1
`sleep_stage_N2`	2	Light sleep stage 2
`sleep_stage_N3`	3	Deep sleep (slow wave)
`sleep_stage_REM`	4	Rapid eye movement sleep
- Super Meta Notes: Derived from manual PSG sleep scoring by trained sleep technicians according to AASM standard. Labels stored in BIDS-format .tsv event files. Epochs marked as 'unscored' (approximately 3%) should be excluded.
- Evidence:
- DOI: 10.1038/s41597-025-04579-8 - Data descriptor confirms EESM23 dataset contains manual PSG sleep stage annotations (W/N1/N2/N3/REM) scored by trained technicians in .tsv event files
- DOI: 10.3389/fnins.2023.987578 - Dataset paper presents automatic sleep scoring using 30-second epochs and random forest classifier trained on ear-EEG features
- Kernel Notes: Load sleep stage labels from BIDS events .tsv files. Filter out unscored epochs (approximately 3%). Map labels to integer classes 0-4. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Only 5 channels (ear-EEG) - very limited spatial resolution.
- No sleep stage labels in events.tsv; PSG annotations needed for staging.
- Home-recorded data - subject-specific motion and environmental artifacts expected.

References

- Reference: `Ear-EEG Sleep Monitoring 2023 (EESM23)`

257. openneuro_ds005185 - Ear-EEG Sleep Monitoring 2019 (EESM19)

Source: analysis_minimax_v2.md

Dataset: `openneuro_ds005185` | Year: 2024 | URL: https://openneuro.org/datasets/ds005185
Category: Biological State | Subjects: 20 | Files (Completed): 516 | Channels: 25 | Sampling Rate: 500 Hz

Paradigm

Naturalistic multi-night sleep monitoring using ear-EEG and polysomnography (PSG) in a home environment collected in 2019. Participants self-applied in-ear EEG electrodes over multiple nights for sleep staging and feasibility assessment. Event markers are Danish-language protocol annotations (lights out, going to bed, waking up), synchronization pulses, and calibration periods. The dominant event is `trigger` (30,726 total) - a PSG synchronization pulse. Sleep stage labels are derived from PSG annotations rather than EEG event markers.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005185

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`trigger`	30726	admin	raw	Synchronization trigger between ear-EEG and PSG (exclude)
`Lights Out`	94	rest	raw	Lights out - sleep onset marker
`Gik i seng`	80	rest	raw	"Went to bed" (Danish)
`Biocalibrations`	79	admin	raw	Biocalibration period (exclude)
`Sync`	77	admin	raw	Synchronization marker (exclude)
`Start af maling`	75	admin	raw	"Start of recording" (Danish, exclude)
`Stod op`	70	rest	raw	"Got up" (Danish) - wake marker
`Biocalibrations start`	11	admin	raw	Biocalibration start (exclude)
`Vagen / vagen`	13	rest	raw	"Awake" (Danish)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `trigger`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Sleep Stage Classification (W/N1/N2/N3/REM)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = [`sleep_stage_W`, `sleep_stage_N1`, `sleep_stage_N2`, `sleep_stage_N3`, `sleep_stage_REM`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`sleep_stage_W`	0	Wake stage (AASM)
`sleep_stage_N1`	1	N1 sleep stage
`sleep_stage_N2`	2	N2 sleep stage
`sleep_stage_N3`	3	N3 (deep) sleep stage
`sleep_stage_REM`	4	REM sleep stage

- Super Meta Notes: Labels derived from manual PSG sleep scoring by trained sleep technicians following AASM v2.5 guidelines. Two independent scorers per recording for EESM19; 5% unscored epochs excluded. Epochs marked 'unscored' by manual scorers were disregarded in analysis.
- Kernel step: Load PSG-derived sleep stage annotations from dataset-sidecar files, map to 30-second epoch boundaries aligned with EEG recording timestamps, produce label array matching epoch count.
- Evidence:
- DOI: 10.1038/s41598-019-53115-3 - Shows 80 full-night ear-EEG recordings from 20 subjects with automatic 5-class sleep staging (W/N1/N2/N3/REM) using 30-second non-overlapping epochs, achieving Cohen's kappa of 0.73
- DOI: 10.1038/s41597-025-04579-8 - Data descriptor confirming EESM19 (openneuro_ds005185) with 320 sleep recordings, 30-second epochs, manual PSG scoring as ground truth
- Kernel Notes: Verify epoch alignment between PSG sleep stages and EEG recording timestamps. Exclude epochs labeled 'unscored' (~5% of data). Use AASM v2.5 scoring guidelines. (replaces Primary Tasks 1, 2, 3 from original analysis.md - no evidence found for ASSR or Comfort BIDS tasks)

Normal Task 1: Sleep Monitoring Segmentation (segmentation)

- Task Type: classification
- Epoch Method: `segmentation`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5
- Label Route: `stim`
- Target Definition: None
- Notes: All event markers are Danish-language protocol annotations or sync triggers - not suitable as EEG classification targets. Sleep stage labels (N1/N2/N3/REM/Wake) should be obtained from PSG annotation files. `trigger` (30,726) is a high-frequency sync pulse - exclude.

Data Quality Notes

- `trigger` (30,726) is a sync pulse - exclude from classification.
- All event markers are protocol annotations, not neural response labels.
- 25-channel ear-EEG with limited spatial resolution.
- PSG reference annotations required for sleep staging labels.

References

- Reference: `Ear-EEG Sleep Monitoring 2019 (EESM19)`

258. openneuro_ds005385 - Resting-State EEG Before and After Cognitive Activity Across the Adult Lifespan (5-Year Follow-Up)

Source: analysis.md

Dataset: `openneuro_ds005385` | Year: 2024 | URL: https://openneuro.org/datasets/ds005385
Category: Biological State | Subjects: 608 | Files (Completed): 3264 | Channels: 65 | Sampling Rate: 1000 Hz

Paradigm

Resting-state EEG was recorded with eyes open and eyes closed before and after a two-hour block of cognitive tasks (working memory, attention, processing speed) across a lifespan cohort of adults. A 5-year longitudinal follow-up is included. BrainVision S-codes (S1-S207+) encode the specific resting-state condition (pre/post x eyes open/closed) and session number.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005385

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`boundary`	3375	admin	raw	BrainVision segment boundary (exclude)
`S 4-S 7`	~207 each	long	raw	Resting-state session markers (BrainVision S-codes)
`S 11-S 19`	~204-205 each	long	raw	Resting-state condition codes
`S 14-S 16`	68 each	long	raw	Resting-state condition (higher count subset)
`S196-S207`	~205-207 each	long	raw	Session/condition index codes

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `S 4`, `S 7`, `S 11`, `S 19`, `S 14`, `S 16`, `S196`, `S207`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Resting-state analysis baseline

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Condition/group labels from metadata when available
- Evidence: Dataset is primarily resting-state or spontaneous-state EEG. Source: https://doi.org/10.2196/32352

Normal Task 1: Resting Condition Segmentation (long event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `boundary` (3375) is admin - exclude. S-codes encode resting condition and session; full decode from README/events.tsv required. 3264 files = 608 subjects x multiple sessions.

Event	y	Meaning
`S 4`	0	Resting condition S4
`S 5`	1	Resting condition S5
`S 6`	2	Resting condition S6
`S 7`	3	Resting condition S7
## Data Quality Notes		

- boundary markers (3375) are admin - exclude.
- Full S-code decode needed from README to identify pre/post and eyes-open/closed conditions.
- Large dataset (3264 files, 608 subjects); high variability across sessions and age groups.

References

- Reference: `Gajewski et al., 2022, doi: 10.2196/32352`

259. openneuro_ds005420 - Resting State EEG with Closed Eyes and Open Eyes in Females from 60 to 80 Years Old

Source: analysis_minimax_v2.md

Dataset: openneuro_ds005420 | Year: 2024 | URL: https://openneuro.org/datasets/ds005420
Category: Biological State | Subjects: 72 | Files (Completed): 72
Channels: 20 | Sampling Rate: 500 Hz

Paradigm

Spontaneous resting-state EEG was recorded in a cohort of females aged 60-80 years during alternating eyes-open and eyes-closed conditions. The dataset investigates age-related changes in spontaneous brain activity in older women. No event markers were included in events.tsv; the EEG is stored as continuous recordings. Eyes-open vs. eyes-closed condition can be inferred from the file naming or participants.tsv.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005420

Event Structure

No event markers detected (no_events). This dataset contains continuous resting-state EEG recordings without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Eyes-Open vs Eyes-Closed Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 10.0 s, overlap = 1.0 s

- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`eyes-open (EO)`	0	Eyes-open resting condition
`eyes-closed (EC)`	1	Eyes-closed resting condition
- Super Meta Notes: Derived from file metadata (filename or participants.tsv) by extracting BIDS task label (task-Oa for eyes-closed, task-Oc for eyes-open). Kernel step: Parse filename pattern `*_task-0a_*` for EC and `*_task-0c_*` for EO, or read task label from participants.tsv column 'task'.
- Evidence:
- DOI: 10.1109/ubmk67458.2025.11207051 - The paper used OpenNeuro ds005420 for cross-dataset validation, segmenting each raw file into 10-s epochs with 1.0-s overlap to generate sufficient samples for binary eyes-open vs eyes-closed classification using VGG-19 and custom CNN models.
- Kernel Notes: Since the dataset contains no event markers (`no_events`), labels must be derived from BIDS task identifiers in filenames. Files with `task-0a` correspond to eyes-closed (class 1), files with `task-0c` correspond to eyes-open (class 0). Apply bandpass filter (0.5-40 Hz) and artifact rejection before segmentation.

Normal Task 1: Resting-State Segmentation (segmentation)

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5 s
- Label Route: `Annotation`
- Target Definition: N/A (null(no events; use file-level condition label for eyes-open/closed from filename or participants.tsv))
- Notes: No event markers. Only 20 EEG channels. Female-only sample aged 60-80. Eyes-open/closed condition may be in filename.

Data Quality Notes

- No event markers - purely continuous resting-state recordings.
- 20-channel montage; limited spatial coverage.
- Female-only cohort aged 60-80; demographic info in participants.tsv.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography`

260. openneuro_ds006269 - Tethered EEG Recordings in Syngap1 Rats

Source: analysis.md

Dataset: `openneuro_ds006269` | Year: 2025 | URL: https://openneuro.org/datasets/ds006269
Category: Biological State | Subjects: 40 | Files (Completed): 40 | Channels: 33 | Sampling Rate: 1000 Hz

Paradigm

Continuous 6-hour tethered EEG recordings were collected from wildtype and Syngap1 mutant rats to monitor spontaneous brain activity and examine circadian-specific neural states. The Syngap1 gene encodes a synaptic GTPase-activating protein and its haploinsufficiency is a model of intellectual disability. The dataset examines spontaneous EEG patterns (sleep/wake states, oscillations) across genotypes.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006269

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`unknown`	40	admin	raw	Unknown system marker - not usable for classification

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Resting-state analysis baseline

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Condition/group labels from metadata when available
- Evidence: Dataset is primarily resting-state or spontaneous-state EEG. Source: https://doi.org/10.1016/j.celrep.2024.114733

Normal Task 1: Continuous Segmentation (long_event)

- Task Type: regression
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition

- Label Route: Annotation
- Target Definition: N/A (no usable event markers; target_events: null)
- Notes: unknown is an uninformative system marker. Use subject-level metadata (wildtype vs. Syngap1 mutant genotype) from participants.tsv as classification labels.

Data Quality Notes

- Only unknown events present - no meaningful trial-level annotation.
- Labels must be derived from external genotype metadata (participants.tsv).
- Animal (rat) EEG data; electrode placement and artifact characteristics differ from human EEG.

References

- Paper: Key roles of C2/GAP domains in SYNGAP1-related pathophysiology (DOI: 10.1016/j.celrep.2024.114733)

261. openneuro_ds006525 - Resting EEG with Structural MRI

Source: analysis_minimax_v2.md

Dataset: openneuro_ds006525 | Year: 2025 | URL: https://openneuro.org/datasets/ds006525
Category: Biological State | Subjects: 34 | Files (Completed): 34 | Channels: 128 | Sampling Rate: 250 Hz

Paradigm

Spontaneous brain activity was recorded during resting-state EEG, supplemented by structural MRI scans. The dataset captures baseline neural dynamics in the absence of any task. Event markers include resting-state block onset (rsts), offset (rstf), and a session-begin marker (bgin), enabling segmentation of clean resting periods.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006525

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
bgin	20400	long	raw	Session begin marker
rsts	20400	rest	raw	Resting-state block start
rstf	34	rest	raw	Resting-state block finish/end
NRST	34	admin	raw	Non-resting state marker
DIN	2	admin	raw	Digital input trigger

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Resting-state EEG cCAP clustering (human)

- Task Type: segmentation
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 1.0 s, overlap = 0.0 s
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning

cCAP_1	0	Cluster 1 co-activation pattern
cCAP_2	1	Cluster 2 co-activation pattern
cCAP_3	2	Cluster 3 co-activation pattern
cCAP_4	3	Cluster 4 co-activation pattern
cCAP_5	4	Cluster 5 co-activation pattern
cCAP_6	5	Cluster 6 co-activation pattern
cCAP_7	6	Cluster 7 co-activation pattern
cCAP_8	7	Cluster 8 co-activation pattern

- Super Meta Notes: Derived from raw EEG signals by ICA decomposition followed by K-means clustering (k=8) on time-frequency features of independent components. Kernel step: Extract 1-s epochs, apply ICA, compute time-frequency features (power spectral density across frequency bands), run K-means clustering to assign each epoch to one of 8 cCAP clusters.
- Evidence:
- DOI: 10.1016/j.neuroimage.2022.119461 - Analyzes resting-state EEG from 34 participants using ICA and K-means clustering to identify 8 component-based co-activation patterns (cCAPs) representing recurring brain states and their dynamic transitions
- Kernel Notes: Filter raw EEG (0.5-40 Hz), segment into 1-s non-overlapping epochs, apply ICA to decompose epochs into independent components, extract time-frequency features from components, perform K-means clustering (k=8), assign cluster labels to epochs. Session boundaries can be identified using `bgin` events.

Primary Task 2: Resting-state EEG CAP clustering (alpha band)

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 1.0 s *, overlap = 0.0 s *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
CAP_0	0	Cluster 0 alpha co-activation pattern
CAP_1	1	Cluster 1 alpha co-activation pattern
CAP_2	2	Cluster 2 alpha co-activation pattern
CAP_3	3	Cluster 3 alpha co-activation pattern
CAP_4	4	Cluster 4 alpha co-activation pattern
CAP_5	5	Cluster 5 alpha co-activation pattern
CAP_6	6	Cluster 6 alpha co-activation pattern
CAP_7	7	Cluster 7 alpha co-activation pattern

- Super Meta Notes: Derived from raw EEG signals by computing instantaneous amplitude in the alpha band (8-12 Hz) via Hilbert transform, then applying K-means clustering (k=8) on z-scored amplitude patterns. Kernel step: Bandpass filter EEG to alpha band (8-12 Hz), compute analytic amplitude via Hilbert transform, z-score amplitude across channels/time, run K-means clustering to assign each time frame to one of 8 CAP clusters.
- Evidence:
- DOI: 10.1016/j.neuroimage.2025.121408 - Uses resting-state EEG data from openneuro_ds006525 (34 participants) to identify transient co-activation patterns (CAPs) via K-means clustering on alpha-band (8-12 Hz) instantaneous amplitude data, comparing human EEG CAPs to monkey ECoG CAPs
- Kernel Notes: Filter raw EEG to alpha band (8-12 Hz), compute Hilbert transform to get instantaneous amplitude, z-score amplitude values across time, segment into 1-s epochs, perform K-means clustering (k=8), assign CAP labels to each epoch.

Normal Task 1: Resting-State Segmentation (long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `rsts` and `rstf` mark the boundaries of resting blocks; segment within these boundaries. `NRST` may indicate periods outside resting state that should be excluded.

Event	y	Meaning
`bgin`	0	Session begin
`rsts`	1	Resting-state block start
`rstf`	2	Resting-state block end
`NRST`	3	Non-resting state period
`DIN`	4	Digital input trigger

Data Quality Notes

- High channel count (128) with 250 Hz sampling rate; EGI/Geodesic system likely.
- `NRST` may indicate periods outside resting state that should be excluded from analysis.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `Resting EEG`

262. openneuro_ds006695 - Validation of Sleep Staging with Forehead EEG Patch

Source: analysis_minimax_v2.md

Dataset: openneuro_ds006695 | Year: 2025 | URL: https://openneuro.org/datasets/ds006695
Category: Biological State | Subjects: 19 | Files (Completed): 19 | Channels: 3 | Sampling Rate: 500 Hz

Paradigm

This dataset validates a spectral sleep staging approach using a three-electrode forehead EEG patch against manual scoring from standard polysomnography (PSG). Overnight sleep recordings were collected using a minimally intrusive forehead device, and sleep stages (Wake, N1, N2, N3, REM) were scored both manually and algorithmically. The study examines whether simplified forehead EEG can reliably reproduce standard PSG hypnogram classifications.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006695

Event Structure

No event markers detected (no_events). This dataset contains continuous overnight EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Sleep stage classification from PSG hypnogram

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0 *, target_events = [Needs verification - sleep stage annotations from external PSG hypnogram]
- Label Route: Annotation
- Meta Level: super

- Target Definition:
Event	y	Meaning
Wake	0	Awake state
N1	1	Light sleep stage 1
N2	2	Light sleep stage 2
N3	3	Deep sleep (slow wave)
REM	4	Rapid eye movement sleep
- Super Meta Notes: Derived from external PSG hypnogram scoring files aligned with forehead EEG recordings. Kernel step: Load PSG hypnogram scoring file (visual labels), extract timestamp-aligned 30-sec epoch labels, map to EEG channel Annotation data. No raw EEG event markers exist; labels come from manual visual scoring per AASM rules.
- Evidence:
- DOI: 10.3389/frsle.2024.1349537 - Validates spectral sleep scoring from 3-channel forehead EEG patch against manual visual scoring of full PSG using 30-sec epoch alignment for 5-stage classification (Wake/N1/N2/N3/REM).
- Kernel Notes: Load external PSG hypnogram scoring file containing sleep stage labels with timestamps. Align each 30-sec epoch with corresponding EEG data segment. Map visual sleep stage labels (Wake/N1/N2/N3/REM) to numeric class labels. Verify event ID mapping from hypnogram files.

Normal Task 1: Continuous Segmentation (segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5 s
- Label Route: `Annotation`
- Target Definition: N/A (no event markers; sleep stage labels from external PSG scoring)
- Notes: No EEG event markers available. Sleep stage labels (Wake/N1/N2/N3/REM) must be derived from PSG hypnogram annotations stored as separate scoring files. Very few channels (3); limited spatial resolution for advanced decoding.

Data Quality Notes

- Only 3 channels (forehead placement); very limited spatial information.
- Overnight recordings are long (hours); memory-efficient processing needed.
- Sleep stage labels require alignment with PSG hypnogram files.

References

- Paper: `Validation of spectral sleep scoring with polysomnography using forehead EEG device` (DOI: 10.3389/frsle.2024.1349537)

263. openneuro_ds006801 - Resting-State EEG Before and After Different Study Methods

Source: analysis.md

Dataset: openneuro_ds006801 | Year: 2025 | URL: https://openneuro.org/datasets/ds006801
Category: Biological State | Subjects: 21 | Files (Completed): 42 | Channels: 31 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG was recorded before and after an educational intervention in which participants engaged in active retrieval practice, passive review, or a control condition. The design examines how different study strategies affect resting-state neural oscillations. Each participant completed eyes-open and eyes-closed resting blocks. The 42 files correspond to pre- and post-intervention recordings for 21 subjects.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006801

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`New Segment/LostSamples`	1	admin	raw	Segment discontinuity marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Resting-state analysis baseline

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Condition/group labels from metadata when available
- Evidence: Dataset is primarily resting-state or spontaneous-state EEG. Source: https://openneuro.org/datasets/ds006801 DOI: https://doi.org/10.18112/openneuro.ds006801.v1.0.0

Normal Task 1: Resting-State Segmentation (long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition
- Label Route: `Annotation`

- Target Definition: N/A (no trial events; target_events: null)
- Notes: `New Segment/LostSamples` is an admin marker indicating signal discontinuity; never use as classification target. Labels for pre/post intervention and study condition must be derived from participants.tsv metadata.

Data Quality Notes

- `New Segment/LostSamples` is a signal discontinuity marker - exclude from all analyses.
- 42 files = 21 subjects x 2 sessions (pre/post intervention).
- Eyes-open vs eyes-closed blocks may be distinguished by duration or metadata.

References

- Reference: `The RECAP-EEG: Retrieval with Feedback and Cognitive Adaptation EEG Dataset`

264. openneuro_ds006921 - High-Density Resting-State EEG of Phantom Limb Pain and Controls

Source: analysis_minimax_v2.md

Dataset: openneuro_ds006921 | Year: 2025 | URL: https://openneuro.org/datasets/ds006921
Category: Biological State | Subjects: 152 | Files (Completed): 152 | Channels: 128 | Sampling Rate: 2400 Hz

Paradigm

High-density resting-state EEG recordings were collected from three groups: amputees with phantom limb pain (PLP), amputees without phantom limb pain, and healthy controls. Recordings were made during eyes-open and eyes-closed conditions. The dataset enables investigation of neural correlates associated with phantom limb pain, cortical reorganization following amputation, and resting-state oscillatory differences across groups.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds006921

Event Structure

No event markers detected (no_events). This dataset contains continuous resting-state EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Event ID	Count (total)	Type	Meta Level	Meaning
no_events	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Phantom Limb Pain Group Classification (Resting-State)

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 5 s, overlap = 0.0 *
- Label Route: Annotation
- Meta Level: super
- Target Definition:

| Event | y | Meaning |

|-------|---|---------|
| eyes_open | 0 | Eyes-open resting condition |
| eyes_closed | 1 | Eyes-closed resting condition |
(within each condition: AmpPLP=PLP amputee, AmpNoPLP=non-PLP amputee, IC=healthy control)
- Super Meta Notes: Derived from raw event no_events by segmenting continuous EEG into non-overlapping 5-second windows. Group labels (AmpPLP/AmpNoPLP/IC) must be obtained from participants.tsv metadata. The paper uses first/last 30s of each 7-min recording, epoch length 5s.
- Evidence:
- DOI: 10.21203/rs.3.rs-8645281.v1 - "This pre-registered study analyzes high-density resting-state EEG from openneuro_ds006921 to compare theta power and peak alpha frequency between amputees with phantom limb pain, amputees without pain, and healthy controls using mixed-effects models."
- Kernel Notes: Load participants.tsv to map subject IDs to group labels (AmpPLP, AmpNoPLP, IC). Segment continuous recordings into 5-s windows. Use first/last 30s of each 7-minute recording per the paper. Train classifier to distinguish three groups (PLP vs non-PLP vs controls). Downsample from 2400 Hz to 500-1000 Hz before feature extraction.

Primary Task 2: Eyes-Open vs Eyes-Closed Condition Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 5 s, overlap = 0.0 *
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning
eyes_open	0	Eyes-open resting condition
eyes_closed	1	Eyes-closed resting condition
- Super Meta Notes: Derived from raw event no_events by segmenting continuous EEG into non-overlapping 5-second windows. Condition labels (eyes-open/eyes-closed) must be obtained from the dataset's task-*-events.tsv or scan parameters.
- Evidence:
- DOI: 10.21203/rs.3.rs-8645281.v1 - Paper analyzes both eyes-open and eyes-closed conditions in the same dataset.
- Kernel Notes: Identify condition labels from BIDS-sidecar JSON or events.tsv files. Segment continuous recordings into 5-s windows. Binary classification: eyes-open vs eyes-closed. This task can be combined with Primary Task 1 for multi-class groupxcondition classification.

Normal Task 1: Resting-State Segmentation (segmentation)

- Task Type: segmentation
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5 s
- Label Route: Annotation
- Target Definition: N/A (no event markers; group labels from participants.tsv)
- Notes: No trial-level events. Group labels (PLP amputee / non-PLP amputee / healthy control) must be derived from participants.tsv metadata. Very high sampling rate (2400 Hz); downsample before analysis.

Data Quality Notes

- Very high sampling rate (2400 Hz); downsample to 500-1000 Hz for most analyses.
- 152 files may include multiple sessions per subject; verify session structure.

- High-density 128-channel system (likely EGI/Geodesic); non-standard montage.

References

- Reference: `High Density Resting State EEG of Phantom Limb Pain and Controls`

265. openneuro_ds007020 - EEG Mortality Dataset in Parkinson's Disease

Source: analysis.md

Dataset: openneuro_ds007020 | Year: 2025 | URL: https://openneuro.org/datasets/ds007020
Category: Biological State | Subjects: 94 | Files (Completed): 94 | Channels: 63 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings (eyes open) were collected from Parkinson's disease (PD) patients and healthy controls to evaluate EEG-based biomarkers for mortality classification. The dataset enables investigation of spectral and connectivity biomarkers that predict mortality risk in PD. The sole event marker (`New Segment/`) indicates a recording segment boundary and carries no experimental meaning.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds007020

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`New Segment/`	94	admin	raw	Recording segment boundary marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Resting-state analysis baseline

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Condition/group labels from metadata when available
- Evidence: Dataset is primarily resting-state or spontaneous-state EEG. Source: https://openneuro.org/datasets/ds007020 DOI: https://doi.org/10.1016/j.prdoa.2025.100409

Normal Task 1: Resting-State Segmentation (long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition
- Label Route: `Annotation`

- Target Definition: N/A (no trial events; target_events: null)
- Notes: `New Segment/` is a BrainVision segment boundary marker and must never be used as a classification target. Group labels (PD patient / healthy control, mortality outcome) must be derived from clinical metadata in participants.tsv.

Data Quality Notes

- `New Segment/` is a BrainVision format boundary marker; exclude from all analyses.
- Labels for mortality prediction must be derived from clinical metadata in participants.tsv.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `EEG Mortality Dataset in Parkinson's Disease`

266. openneuro_ds007176 - Longitudinal EEG Test-Retest Reliability in Healthy Individuals

Source: analysis.md

Dataset: `openneuro_ds007176` | Year: 2026 | URL: https://openneuro.org/datasets/ds007176
Category: Biological State | Subjects: 43 | Files (Completed): 300 | Channels: 60 | Sampling Rate: 1000 Hz

Paradigm

Longitudinal resting-state EEG recordings collected over four sessions spanning two years to evaluate test-retest reliability of quantitative EEG measures. Recordings are continuous resting-state without structured task events. The dataset enables assessment of EEG biomarker stability across sessions and time points.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds007176

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`0`	2	admin	raw	Null trigger code - no experimental meaning

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: CE

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)

- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://doi.org/10.1016/j.bspc.2026.109484

Primary Task 2: Publisher-defined BIDS task: OE

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://doi.org/10.1016/j.bspc.2026.109484

Normal Task 1: Resting-State Segmentation (long_event)

- Task Type: segmentation
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s, target_events = see Target Definition
- Label Route: `stim`
- Target Definition: N/A (no trial events; target_events: null)
- Notes: Only event code `0` present (2 total); this is a null trigger with no experimental meaning. Resting-state data; use segmentation across full recording. Session/time-point labels available from participants.tsv for longitudinal analysis.

Data Quality Notes

- No meaningful event markers; purely continuous resting-state EEG.
- 43 subjects x ~7 sessions/subject = 300 files; multi-session longitudinal design.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Paper: `Longitudinal test-retest reliability of quantitative EEG in healthy individuals using an automated preprocessing approach` (DOI: 10.1016/j.bspc.2026.109484)

267. openneuro_ds007181 - Structural MRI, Resting-State fMRI, and PSG/EEG Dataset of Zoster-Associated Neuralgia

Source: analysis.md

Dataset: `openneuro_ds007181` | Year: 2026 | URL: https://openneuro.org/datasets/ds007181
Category: Biological State | Subjects: 59 | Files (Completed): 59 | Channels: 24 | Sampling Rate: 1024 Hz

Paradigm

Resting-state EEG (part of a PSG recording) was collected from patients with zoster-associated neuralgia and healthy controls, alongside structural MRI and resting-state fMRI. Manual sleep staging is available from PSG annotations. The dataset enables investigation of brain activity and sleep architecture differences between healthy controls and patients with herpes zoster-associated chronic pain.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds007181

Event Structure

No event markers detected (`no_events`). This dataset contains continuous PSG/EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Event ID	Count (total)	Type	Meta Level	Meaning
`no_events`	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.

Tasks

Primary Task 1: Sleep stage classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Sleep stage labels (W/N1/N2/N3/REM; mapping dataset-specific)
- Evidence: Dataset and references explicitly center on PSG/EEG sleep scoring. Source: https://openneuro.org/datasets/ds007181 DOI: https://doi.org/10.18112/openneuro.ds007181.v1.0.1

Primary Task 2: Publisher-defined BIDS task: Rest

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds007181 DOI: https://doi.org/10.18112/openneuro.ds007181.v1.0.1

Normal Task 1: Continuous Segmentation (segmentation)

- Task Type: regression
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 2.0 s, overlap = 0.5 s
- Label Route: `Annotation`
- Target Definition: N/A (no event markers; sleep stage labels from PSG scoring files)
- Notes: No EEG event markers available. Sleep stage labels (Wake/N1/N2/N3/REM) must be derived from PSG hypnogram annotations if present. Group labels (zoster neuralgia / healthy control) from participants.tsv.

Data Quality Notes

- No event markers present; continuous-only processing.
- PSG-derived sleep stage labels require alignment with EEG data if available.
- 24-channel PSG EEG system; non-standard montage compared to research-grade EEG.

References

- Reference: `Structural MRI, Resting-state fMRI, and PSG/EEG Dataset of Zoster-associated Neuralgia`

268. other_migrainedb - Ultra High-Density EEG - Interictal Migraine and Controls

Source: analysis_minimax_v2.md

Dataset: `other_migrainedb` | Year: 2020 | URL: https://kilthub.cmu.edu/articles/dataset/Ultra_high-density_EEG_recording_of_interictal_migraine_and_controls_sensory_and_rest/12636731
Category: Biological State | Subjects: 39 | Files (Completed): 117 | Channels: 144 | Sampling Rate: 512 Hz

Paradigm

Thirty-nine subjects (18 interictal migraine patients: M1-M18; 21 healthy controls: C1-C21) underwent high-density EEG recording (BioSemi ActiveTwo, 144 channels, 512 Hz) in three task conditions (Chiang et al., 2020): (1) SSVEP - steady-state visual evoked potential, flickering stimuli at 4 Hz and 6 Hz; (2) SSAEP - steady-state auditory evoked potential, tone sequences at 4 Hz and 6 Hz; (3) resting state. The dataset enables comparison of sensory processing in migraine patients vs. healthy controls in the interictal (between-attack) period.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://kilthub.cmu.edu/articles/dataset/Ultra_high-density_EEG_recording_of_interictal_migraine_and_controls_sensory_and_rest/12636731

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`4Hz vision stimulation start`	varies	short	raw	SSVEP 4 Hz flicker onset
`6Hz vision stimulation start`	varies	short	raw	SSVEP 6 Hz flicker onset
`4Hz vision feedback end`	varies	admin	raw	SSVEP 4 Hz block end marker
`6Hz vision feedback end`	varies	admin	raw	SSVEP 6 Hz block end marker
`4Hz auditory stimulation start`	varies	short	raw	SSAEP 4 Hz tone onset
`6Hz auditory stimulation start`	varies	short	raw	SSAEP 6 Hz tone onset
`4Hz auditory feedback end`	varies	admin	raw	SSAEP 4 Hz block end marker
`6Hz auditory feedback end`	varies	admin	raw	SSAEP 6 Hz block end marker

Event ID	Count (total)	Type	Meta Level	Meaning
`resting state block start`	varies	rest	raw	Resting state block onset
`resting state block end`	varies	admin	raw	Resting state block end marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: `4Hz vision stimulation start`, `6Hz vision stimulation start`, `4Hz auditory stimulation start`, `6Hz auditory stimulation start`
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Migraine vs Control Classification (Stimulus Epochs)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, target_events = [`4Hz vision stimulation start`, `6Hz vision stimulation start`, `4Hz auditory stimulation start`, `6Hz auditory stimulation start`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`4Hz vision stimulation start`	0	Healthy control (C-series subject)
`6Hz vision stimulation start`	0	Healthy control (C-series subject)
`4Hz auditory stimulation start`	0	Healthy control (C-series subject)
`6Hz auditory stimulation start`	0	Healthy control (C-series subject)
`4Hz vision stimulation start`	1	Migraine patient (M-series subject)
`6Hz vision stimulation start`	1	Migraine patient (M-series subject)
`4Hz auditory stimulation start`	1	Migraine patient (M-series subject)
`6Hz auditory stimulation start`	1	Migraine patient (M-series subject)

- Super Meta Notes: Derived from subject ID prefix: 'M' prefix in filename indicates migraine patient, 'C' prefix indicates healthy control. Kernel step: Extract epoch around each stimulus event, then apply group label based on subject ID parsed from filename.
- Evidence:
- DOI: 10.1093/braincomms.fcab061 - Uses 2-s stimulus epochs (4Hz/6Hz visual and auditory) to classify migraine patients vs healthy controls based on EEG coherence patterns
- Kernel Notes: 1) Epoch using tmin=0.0s, tmax=2.0s around each of the four stimulus event types. 2) Parse subject ID from filename to determine group (M=migraine, C=control). 3) Apply appropriate label (0 for control, 1 for migraine) to each epoch. 4) Compute EEG coherence features (e.g., coherence in alpha band) or use raw epochs for classification.

Primary Task 2: Migraine vs Control Classification (Resting State)

- Task Type: classification

- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.0, target_events = [`resting state block start`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`resting state block start`	0	Healthy control (C-series subject)
`resting state block start`	1	Migraine patient (M-series subject)

- Super Meta Notes: Derived from subject ID prefix: 'M' prefix in filename indicates migraine patient, 'C' prefix indicates healthy control. Kernel step: Segment resting state blocks into 2-s non-overlapping windows, then apply group label based on subject ID.
- Evidence:
- DOI: 10.1093/braincomms.fcab061 - Uses 2-s non-overlapping resting state segments to classify migraine patients vs healthy controls based on EEG coherence patterns
- Kernel Notes: 1) Use resting state block start to identify resting periods. 2) Create 2-s non-overlapping segments (tmin=0, tmax=2, overlap=0). 3) Parse subject ID from filename to determine group. 4) Apply appropriate label and compute features (e.g., alpha-band coherence).

Normal Task 1: SSVEP Frequency Classification - 4 Hz vs. 6 Hz (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 2.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
`4Hz vision stimulation start`	0	4 Hz flicker stimulus
`6Hz vision stimulation start`	1	6 Hz flicker stimulus

Normal Task 2: Migraine Patient vs. Control - Resting State (epoch_by_segmentation)

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: length = 5.0 s, overlap = 2.5 s (within `resting state block start` / end boundaries)
- Label Route: `Annotation`
- Target Definition: N/A (group label from subject ID: M-series = migraine, C-series = healthy)
- Notes: Group labels (migraine vs. control) are encoded in the subject ID (M/C prefix in filename). Apply the group label post-hoc after segmentation.

Data Quality Notes

- Group labels must be derived from the subject ID prefix (M for migraine, C for control).
- Age range: 19-54 years; sex: mixed male/female.
- High channel count (144); spatial filtering / source localization may be useful.
- Sensory task and resting state recordings provide complementary views of migraine-related cortical changes.

References

- Reference: Chiang AKI et al. (2020). Ultra high-density EEG recording of interictal migraine and controls sensory and rest. KiltHub CMU. https://kilthub.cmu.edu/articles/dataset/12636731

269. physionet_capslpdb - CAP Sleep Database

Source: analysis_minimax_v2.md

Dataset: `physionet_capslpdb` | Year: 2014 | URL: https://physionet.org/content/capslpdb/1.0.0/
Category: Biological State | Subjects: 108 | Files (Completed): 108 | Channels: 5-36 (variable) | Sampling Rate: 100-512 Hz (variable)

Paradigm

One hundred and eight patients from the Sleep Disorders Center, Ospedale Civile di Parma, Italy, underwent overnight polysomnographic (PSG) recording with detailed expert annotations (Terzano et al., 2002). The cohort includes 16 healthy controls and 92 patients with various sleep disorders: nocturnal frontal lobe epilepsy (NFLE, n=40), REM sleep behavior disorder (RBD, n=22), periodic limb movements (PLM, n=10), insomnia (n=9), narcolepsy (n=5), sleep-disordered breathing (SDB, n=4), and bruxism (n=2). Beyond standard sleep staging (Rechtschaffen & Kales), annotations include Cyclic Alternating Pattern (CAP) subtypes: phase A1 (high-amplitude synchronized EEG activity), A2 (mixed), and A3 (desynchronized EEG). Channel count and sampling rate vary across subjects.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/capslpdb/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`SLEEP-S0`	19495	long	raw	Wake stage (S0 = W)
`SLEEP-S1`	4642	long	raw	NREM Stage 1 - light sleep
`SLEEP-S2`	39192	long	raw	NREM Stage 2 - sleep spindles/K-complexes
`SLEEP-S3`	11953	long	raw	NREM Stage 3 - SWS
`SLEEP-S4`	13780	long	raw	NREM Stage 4 - deep SWS
`SLEEP-REM`	18935	long	raw	REM sleep
`SLEEP-MT`	11	admin	raw	Movement time (artifact/transition)
`MCAP-A1`	26788	short	raw	CAP phase A1 onset (synchronized EEG)
`MCAP-A2`	10519	short	raw	CAP phase A2 onset (mixed pattern)

Event ID	Count (total)	Type	Meta Level	Meaning
MCAP-A3	14324	short	raw	CAP phase A3 onset (desynchronized EEG)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Sleep Stage Classification

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 30.0 s, overlap = 0.0 s, target_events = [SLEEP-S0, SLEEP-S1, SLEEP-S2, SLEEP-S3, SLEEP-S4, SLEEP-REM]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
SLEEP-S0	0	Wake
SLEEP-S1	1	NREM Stage 1
SLEEP-S2	2	NREM Stage 2
SLEEP-S3	3	NREM Stage 3
SLEEP-S4	4	NREM Stage 4
SLEEP-REM	5	REM

- Evidence:
- DOI: 10.1109/tnsre.2019.2934828 - LSTM classifiers for CAP A-phase detection using CAP Sleep Database with 30s R&K epoch segmentation
- Kernel Notes: Exclude SLEEP-MT epochs (movement time artifacts). Use 30-second epochs aligned to Rechtschaffen & Kales standard scoring.

Primary Task 2: Insomnia vs Healthy Control Classification

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0.0 s, tmax = 30.0 s, overlap = 0.0 s *, target_events = [SLEEP-S0, SLEEP-S1, SLEEP-S2, SLEEP-S3, SLEEP-S4, SLEEP-REM] *
- Label Route: Annotation
- Meta Level: super
- Target Definition:

Event	y	Meaning
healthy	0	Healthy control subject
insomnia	1	Insomnia patient

- Super Meta Notes: Derived from clinical subject diagnosis metadata in CAP database. Subjects selected from 9 insomniacs and 16 healthy controls. Kernel must join epoch labels with subject-level diagnosis metadata to assign per-epoch labels.
- Evidence:
- DOI: 10.1088/1361-6579/adfda8 - Uses CAP sleep database to extract C4-A1 EEG from 6 healthy and 6 insomniac subjects, performing 30s epoch segmentation and wavelet augmentation to classify insomnia vs healthy controls

- Kernel Notes: Requires subject metadata lookup. Select only subjects with clinical diagnosis of insomnia or healthy controls. Verify that target_events covers all sleep stages in the selected epochs. (replaces Normal Task 1 from original analysis.md - same 30s R&K epoch parameters now properly evidenced)

Normal Task 2: CAP Phase A Subtype Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 3.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
`MCAP-A1`	0	CAP Phase A1 (synchronized)
`MCAP-A2`	1	CAP Phase A2 (mixed)
`MCAP-A3`	2	CAP Phase A3 (desynchronized)

Data Quality Notes

- Channel count and sampling rate vary substantially across subjects (5-36 channels, 100-512 Hz); normalize before cross-subject analysis.
- Mixed clinical population: NFLE, RBD, PLM, insomnia, narcolepsy, SDB, bruxism, and healthy controls; group labels are encoded in the subject metadata.
- CAP annotations are specialized biomarkers of sleep instability; fewer publications use them compared to standard staging.

References

- Reference: Terzano MG et al. (2002). Atlas, rules, and recording techniques for the scoring of cyclic alternating pattern (CAP) in human sleep. Sleep Medicine, 3(2), 187-199. Goldberger AL et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. https://physionet.org/content/capslpdb/1.0.0/

270. physionet_chbmit - CHB-MIT Scalp EEG Database

Source: analysis_minimax_v2.md

Dataset: `physionet_chbmit` | Year: 2010 | URL: https://physionet.org/content/chbmit/1.0.0/
Category: Biological State | Subjects: 22 | Files (Completed): 686 | Channels: 23 | Sampling Rate: 256 Hz

Paradigm

Twenty-two pediatric patients (5 males aged 3-22; 17 females aged 1.5-19) with intractable seizures were monitored at Children's Hospital Boston following withdrawal of anti-seizure medication to evaluate surgical candidacy (Shoeb & Guttag, 2010). Long-term continuous EEG was recorded from 23 scalp channels (bipolar montage following the 10-20 system) at 256 Hz with 16-bit resolution. The dataset contains 686 EEG files and 182 seizure events in total. Seizure onset and offset times are documented in per-subject summary text files (`chbNN-summary.txt`); the kernel parses these to create `seizure` annotations spanning each ictal period.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://physionet.org/content/chbmit/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`seizure`	182	long	raw	Seizure period (onset to offset, variable duration)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Files without seizures have no annotations; these ictal-free recordings serve as the negative (interictal) class.

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Binary seizure detection (ictal vs interictal)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4 s *, overlap = 0.5 *

- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`seizure`	1	Ictal (seizure period)
(no annotation)	0	Interictal (seizure-free)
- Evidence:
- DOI: 10.1016/j.bspc.2024.107379 - Classifies 2-second seizure vs non-seizure EEG segments from CHB-MIT, achieving 96.72% accuracy
- DOI: 10.1038/s41598.025.98389.y - Divides CHB-MIT EEG into 5-second fragments classified as Seizure or No Seizure
- DOI: 10.1038/s41598.025.18479_9 - Binary classification of seizure vs non-seizure EEG segments from CHB-MIT, 96.94% accuracy
- Kernel Notes: Use `epoch_by_segmentation_hdf5` to extract sliding windows from all recordings. Label windows overlapping any `seizure` annotation as ictal (y=1); all other windows as interictal (y=0). (replaces Normal Task 1 from original analysis.md)

Primary Task 2: Three-class seizure state classification (pre-ictal vs ictal vs interictal)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4 s, overlap = 0.75 *
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
pre-ictal	0	20-30 min before seizure onset
`seizure`	1	Ictal (during seizure)
interictal	2	Seizure-free periods (>=4 hours before/after)
- Super Meta Notes: Derived from raw `seizure` annotation by computing time windows: pre-ictal = 20-30 min before seizure onset, ictal = during seizure annotation, interictal = all other times >=4 hours from any seizure. Kernel step: for each 4s window, compute time to nearest seizure onset; assign pre-ictal label if within 30 min before, ictal if overlapping seizure annotation, interictal otherwise.
- Evidence:
- DOI: 10.1088/1741-2552.ada0e5 - Uses CHB-MIT for seizure detection (ictal vs preictal) and prediction (preictal vs interictal), with preictal defined as 30 min before seizure
- DOI: 10.54525_bbmd.1526069 - Three-class classification (pre-seizure/ictal/non-seizure) on CHB-MIT using 4-second windows with 3s overlap
- Kernel Notes: Compute temporal relationship of each 4s window to all seizure onsets in the same subject. Pre-ictal: window ending 0-30 min before seizure onset. Ictal: window overlapping seizure annotation. Interictal: window >4 hours from any seizure.

Primary Task 3: Seizure prediction (preictal vs interictal)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 30 s *, overlap = 0.0 *
- Label Route: `Annotation`
- Meta Level: super

- Target Definition:
Event	y	Meaning
preictal	1	30-90 minutes before seizure onset
interictal	0	>=4 hours before seizure onset or after seizure offset
- Super Meta Notes: Derived from raw `seizure` annotation by defining preictal window as 30-90 min before seizure onset. Kernel step: for each 30s window, compute time to nearest seizure onset; assign preictal if in [30,90) min before, interictal otherwise.
- Evidence:
- DOI: 10.1109_access_2020.2976866 - Deep learning for seizure prediction on CHB-MIT, preictal state defined as 30-90 minutes before seizure onset, 92.7% sensitivity
- DOI: 10.1088/1741-2552.ada0e5 - Seizure prediction (preictal vs interictal) with 30 min pre-ictal matched to ictal duration
- Kernel Notes: Extract 30-second non-overlapping windows from all recordings. Label windows in the 30-90 min pre-seizure window as preictal (y=1), all other seizure-free windows as interictal (y=0).

Normal Task 2: Patient-Specific Seizure Detection (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event`
- Epoch Parameters: Needs verification
- Label Route: `Annotation`
- Target Definition: Needs verification.
- Notes: Train per patient; subjects chb17 appears in three sub-cases (chb17a, chb17b, chb17c) and should be treated as a single subject.

Data Quality Notes

- Severe class imbalance: the majority of recordings are seizure-free; balance by undersampling interictal or using cost-sensitive learning.
- Channel count varies: most files have 23 channels; some cases include 24 or 26 channels (extra ECG or vagal nerve stimulus channels); exclude non-EEG channels.
- Patient-specific model training is recommended; cross-patient generalization is difficult.
- Subject chb24 has unknown age and sex.
- Age range: 1.5-22 years (pediatric); EEG morphology differs significantly from adult norms.

References

- Reference: Shoeb AH, Guttag JV (2010). Application of machine learning to epileptic seizure detection. In Proceedings of the 27th ICML. Goldberger AL et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. Circulation, 101(23), e215-e220. https://physionet.org/content/chbmit/1.0.0/

271. physionet_hmcsleepstaging - HMC Sleep Staging Database

Source: analysis_minimax_v2.md

Dataset: `physionet_hmcsleepstaging` | Year: 2022 | URL: https://physionet.org/content/hmc-sleep-staging/1.1/
Category: Biological State | Subjects: 77 | Files (Completed): 77 | Channels: 8 | Sampling Rate: 256 Hz

Paradigm

Seventy-seven overnight polysomnographic recordings from the Haaglanden Medisch Centrum (HMC) Sleep Center, Netherlands, were collected and annotated using the AASM (American Academy of Sleep Medicine) scoring rules. EEG was recorded from 4 bipolar derivations (F4-M1, C4-M1, O2-M1, C3-M2), supplemented by chin EMG, two EOG channels (E1-M2, E2-M2), and ECG, all at 256 Hz. Expert sleep technicians scored each 30-second epoch as Wake, N1, N2, N3, or REM. Light-on and light-off events mark the intended sleep period boundaries.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/hmc-sleep-staging/1.1/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Sleep stage W`	12217	long	raw	Wake (30-second epoch)
`Sleep stage N1`	8103	long	raw	NREM Stage 1 (AASM)
`Sleep stage N2`	24969	long	raw	NREM Stage 2 (AASM)
`Sleep stage N3`	13243	long	raw	NREM Stage 3 - slow-wave sleep (AASM)
`Sleep stage R (REM)`	10974	long	raw	REM sleep (AASM)
`Lights off`	76	admin	raw	Start of intended sleep period
`Lights on`	76	admin	raw	End of intended sleep period

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Counts are totals across all 77 recordings. Each stage annotation spans exactly 30 seconds.

Tasks

Primary Task 1: 5-Class Sleep Stage Classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 30.0 s, overlap = 0.0 *, target_events = [`Sleep stage W`, `Sleep stage N1`, `Sleep stage N2`, `Sleep stage N3`, `Sleep stage R (REM)`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Sleep stage W`	0	Wake
`Sleep stage N1`	1	NREM Stage 1
`Sleep stage N2`	2	NREM Stage 2
`Sleep stage N3`	3	NREM Stage 3 / Slow-wave sleep
`Sleep stage R (REM)`	4	REM sleep

- Evidence:
- DOI: 10.3390/s23052458 - Uses HMC sleep staging database, trains 1D-CNN for 5-class sleep staging (Wake/N1/N2/N3/REM) on 30-s epochs with 22.5-s overlap.
- DOI: 10.1109/embc58623_2025.11252818 - Uses HMC sleep dataset, extracts C4/M1 EEG data segmented into non-overlapping 30-second epochs for 5-class classification (W/N1/N2/N3/R).
- Kernel Notes: Extract epochs aligned with AASM sleep stage annotations. Each 30-s epoch corresponds to one label. Consider excluding epochs outside the lights-off to lights-on interval. Overlap parameter set to 0.0 based on paper 2; paper 1 used 22.5-s overlap (0.75) but this is not the recommended default. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- AASM scoring (N1/N2/N3/REM/W) as opposed to R&K (S1/S2/S3/S4/REM/W) used in older datasets; not directly comparable.
- ECG channel present; exclude from EEG features.
- Age and sex information are not available in the public metadata.
- N1 is relatively rare; class imbalance between stages is expected.

References

- Reference: Alvarez-Estevez D, Rijsman RM (2022). HMC Sleep Staging Database. PhysioNet. https://physionet.org/content/hmc-sleep-staging/1.1/

272. physionet_sleepedfx - Sleep-EDF Database Expanded

Source: analysis_minimax_v2.md

Dataset: `physionet_sleepedfx` | Year: 1992/2013 | URL: https://physionet.org/content/sleep-edfx/1.0.0/
Category: Biological State | Subjects: 58 | Files (Completed): 197 | Channels: 7 | Sampling Rate: 100 Hz

Paradigm

The Sleep-EDF Database Expanded contains whole-night polysomnographic (PSG) sleep recordings with expert-annotated sleep stage hypnograms (Kemp et al., 2000). Two study types are included: Sleep Cassette (SC, 39 subjects, 2 nights each = 78 recordings) and Sleep Treatment (ST, 19 subjects). EEG was recorded from 2 channels (Fpz-Cz and Pz-Oz) along with a horizontal EOG, submental EMG, and event/temperature markers. Sleep stages (W, 1, 2, 3, 4, REM) were scored in 30-second epochs by sleep technicians using Rechtschaffen & Kales criteria. Hypnogram annotations are loaded by the kernel from companion `*Hypnogram.edf` files adjacent to each PSG file.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/sleep-edfx/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Sleep stage W`	~24000	long	raw	Wake stage (30-second epoch)
`Sleep stage 1`	~3000	long	raw	NREM Stage 1 - light sleep
`Sleep stage 2`	~40000	long	raw	NREM Stage 2 - sleep spindles/K-complexes
`Sleep stage 3`	~5000	long	raw	NREM Stage 3 - slow-wave sleep (SWS)
`Sleep stage 4`	~3000	long	raw	NREM Stage 4 - deep slow-wave sleep
`Sleep stage R`	~9000	long	raw	REM sleep
`Movement time`	rare	admin	raw	Movement artifact period (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Counts are approximate totals across all 197 recordings. Each stage annotation spans exactly 30 seconds.

Tasks

Primary Task 1: Sleep Stage Classification (5-Class AASM)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 30.0 s, overlap = 0.0 s, target_events = [`Sleep stage W`, `Sleep stage 1`, `Sleep stage 2`, `Sleep stage 3`, `Sleep stage R`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`Sleep stage W`	0	Wake
`Sleep stage 1`	1	NREM Stage 1 (N1)
`Sleep stage 2`	2	NREM Stage 2 (N2)
`Sleep stage 3`	3	NREM Stage 3 (N3) - merged S3 and S4
`Sleep stage R`	4	REM sleep

- Evidence:
- DOI: 10.7717_peerjcs.1561 - 5-class sleep staging (W, N1, N2, N3, REM) with 30-second epochs, using AASM-standard merged N3 class
- Kernel Notes: Exclude `Sleep stage 4` annotations (merged into N3 per AASM standard). Exclude `Movement time` epochs. Verify that each 30-second epoch window aligns exactly with annotation boundaries from Hypnogram.edf files.

Normal Task 1: Sleep Stage Classification - 5-Class (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 30.0 s (one full 30-second epoch), overlap = 0.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Exclude `Movement time` epochs. Each annotation spans exactly 30 seconds - match epoch window exactly to annotation duration. Only PSG files (filenames containing `PSG` or `SC`) are processed by the kernel; Hypnogram files return unchanged raw objects.

Event	y	Meaning
`Sleep stage W`	0	Wake
`Sleep stage 1`	1	NREM Stage 1
`Sleep stage 2`	2	NREM Stage 2
`Sleep stage 3`	3	NREM Stage 3
`Sleep stage 4`	4	NREM Stage 4
`Sleep stage R`	5	REM

Normal Task 2: Wake vs. Sleep Binary Classification (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 30.0 s, overlap = 0.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
`Sleep stage W`	0	Wake
`Sleep stage 1`	1	Any sleep (stages 1-4 and R)
`Sleep stage 2`	1	Any sleep
`Sleep stage 3`	1	Any sleep
`Sleep stage 4`	1	Any sleep
`Sleep stage R`	1	Any sleep

Data Quality Notes

- Only PSG files should be processed; Hypnogram files (matched companion files) are used for annotation only and are returned unchanged by the kernel.
- Class imbalance: Stage 2 is the most frequent stage; Wake also contributes large counts from lights-on periods at start/end of recording.
- Two EEG channels only (Fpz-Cz, Pz-Oz); limited spatial resolution.
- Age range is wide (18-101 years); SC study subjects are healthy adults, ST subjects had sleep disorders.
- SC study: 2 nights per subject; subject IDs encode night number in filename.

References

- Reference: Kemp B, Varri A, Rosa AC, Nielsen KD, Gade J (1992). A simple format for exchange of digitized polygraphic recordings. Electroencephalography and Clinical Neurophysiology, 82(5), 391-393. Goldberger AL et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. Circulation, 101(23), e215-e220. https://physionet.org/content/sleep-edfx/1.0.0/

273. physionet_ucddb - UCD Sleep Apnea Database

Source: analysis_minimax_v2.md

Dataset: `physionet_ucddb` | Year: 2011 | URL: https://physionet.org/content/ucddb/1.0.0/
Category: Biological State | Subjects: 25 | Files (Completed): 50 (Case B kernel; 25 PSG .rec + 25 ECG _lifecard.edf) | Channels: variable (PSG) | Sampling Rate: variable

Paradigm

Twenty-five patients suspected of sleep-disordered breathing underwent overnight polysomnographic (PSG) monitoring at University College Dublin (UCD). PSG files are stored in `.rec` format (EDF-compatible) and contain multiple physiological signals. Sleep stages were scored by expert technicians and stored in companion `_stage.txt` files (30-second epochs: Wake, Stage 1-4, REM). Companion `_lifecard.edf` ECG files are co-located per subject but are not EEG recordings and are skipped by the kernel (`-> unknown` annotations). The Case B kernel reconstructs continuous raw signals from the non-standard `.rec` format.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/ucddb/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`Sleep stage W`	4707	long	raw	Wake (30-second epoch)
`Sleep stage 1`	3016	long	raw	NREM Stage 1 (R&K)
`Sleep stage 2`	3403	long	raw	NREM Stage 2 (R&K)
`Sleep stage 3`	6985	long	raw	NREM Stage 3 - slow-wave sleep (R&K)
`Sleep stage 4`	673	long	raw	NREM Stage 4 - deep SWS (R&K)
`Sleep stage REM`	1990	long	raw	REM sleep
`Sleep stage 8`	15	admin	raw	Undefined/artifact epoch (exclude from classification)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Counts are totals across all 25 PSG files. `Sleep stage 8` indicates undefined or movement-time epochs. ECG `_lifecard.edf` files yield `unknown` annotations and should be excluded.

Tasks

Primary Task 1: Sleep stage classification (5-class)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0 *, target_events = [`Sleep stage W`, `Sleep stage 1`, `Sleep stage 2`, `Sleep stage 3`, `Sleep stage 4`, `Sleep stage REM`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`Sleep stage W`	0	Wake
`Sleep stage 1`	1	NREM Stage 1
`Sleep stage 2`	2	NREM Stage 2
`Sleep stage 3`	3	NREM Stage 3 (merged from R&K stages 3 and 4)
`Sleep stage REM`	4	REM

- Super Meta Notes: Derived from raw events [`Sleep stage 3`, `Sleep stage 4`] by merging both into a single N3 class per AASM standard. Kernel step: map labels y=3 to combined Sleep stage 3/4 output. Exclude `Sleep stage 8` (undefined epochs).
- Evidence:
- DOI: 10.1371/journal.pone.0256111 - Uses the St. Vincent's Hospital/University College Dublin Sleep Apnea Database (25 overnight PSGs) for sleep stage classification, with R&K scored stages mapped to AASM 5-class format (W/N1/N2/N3/R).
- Kernel Notes: Use 30-second epoch windows matching the scoring granularity. Resample to 100Hz (input size 4x3000) as per paper. (replaces Normal Task 1 from original analysis.md)

Normal Task 2: Sleep Apnea Context - Wake vs. Sleep (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event`
- Epoch Parameters: Needs verification
- Label Route: `Annotation`
- Target Definition: Needs verification.
- Notes: Sleep apnea-related breathing events are not included in the current EEG annotations; the dataset can be used for stage classification as a secondary target. Respiratory signals in the PSG recordings can supplement EEG for apnea detection but require custom parsing.

Data Quality Notes

- Case B kernel; `.rec` files require EDF-via-tempfile workaround; `_lifecard.edf` ECG files are excluded (-> `unknown` annotations in locator).
- Age and sex information may be in `SubjectDetails.xls` within the dataset; the kernel loads demographics from this file if available.
- R&K scoring system (stages 1-4 + REM); not directly comparable to AASM (N1/N2/N3) datasets.
- Small dataset (25 subjects); combine with other sleep databases for robust model training.

References

- Reference: Heneghan C et al. (2011). UCD Sleep Apnea Database. PhysioNet. DOI: https://doi.org/10.13026/C26C7D

274. sleepdata_shhs - sleepdata_shhs

Source: analysis_minimax_v2.md

Dataset: `sleepdata_shhs` | Year: 1995-2003 (SHHS-1: 1995-1998; SHHS-2: 2001-2003) | URL: https://sleepdata.org/datasets/shhs
Category: Biological State | Subjects: 6,441 (SHHS-1); 3,295 follow-up (SHHS-2) | Files (Completed): 5,786 EDF (SHHS-1) + 532 EDF (SHHS-2)
Channels: 14 (per EDF) | Sampling Rate: 125 Hz (EEG/EMG/ECG primary), 50 Hz (EOG), 10 Hz (respiratory), 1 Hz (SpO2/HR); exported EDF channels are resampled to 125 Hz

Paradigm

The Sleep Heart Health Study (SHHS) is a multi-center NIH/NHLBI cohort investigating cardiovascular and other outcomes of sleep-disordered breathing. Participants aged 40+ underwent unattended overnight home polysomnography (PSG), collected by certified technicians. SHHS-1 enrolled 6,441 participants from major epidemiological cohorts, and 3,295 participants completed a second PSG approximately 5 years later (SHHS-2). The dataset includes full-night EEG, EOG, EMG, ECG, respiratory, and oximetry signals with expert sleep-stage annotation in 30-second epochs.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://sleepdata.org/datasets/shhs

Event Structure

Annotations are available in both original Profusion XML and NSRR-processed XML (`*-nsrr.xml`). The NSRR XML files contain three high-level `EventType` groups:

Event ID	Count (total)	Type	Meta Level	Meaning
`Stages|Stages`	~100-400 annotations/file	long	raw	Sleep-stage labels in 30 s scoring epochs
`Arousals|Arousals`	variable	short/long	raw	Cortical arousal events
`Respiratory|Respiratory`	variable	short/long	raw	Respiratory events (apnea/hypopnea/desaturation-related)

Sleep Stage Labels (inside `EventConcept`)

Event ID	Count (total)	Type	Meta Level	Meaning
`Wake|0`	frequent	long	raw	Wake
`Stage 1|1`	lower frequency	long	raw	NREM Stage 1 (N1)
`Stage 2|2`	high frequency	long	raw	NREM Stage 2 (N2)
`Stage 3|3`	moderate	long	raw	Slow-wave sleep

Event ID	Count (total)	Type	Meta Level	Meaning
Stage 4|4	rare	long	raw	Slow-wave sleep (legacy R&K stage)
REM|5	moderate	long	raw	REM sleep

Event Type Description:
- short: brief event marker (for event-locked analysis)
- long: sustained-state marker (for long-event epoching, including sleep stages)
- rest: resting-state segment marker when explicitly present
- admin: boundary/system marker, exclude from target labels

Tasks

Primary Task 1: Sleep Spindle Characterization (Regression)

- Task Type: regression
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = [Stage 2|2]
- Label Route: Annotation
- Meta Level: super
- Target Definition:
Event	y	Meaning
N2 epoch	continuous (spindle density)	Number of spindles per 30s N2 epoch
N2 epoch	continuous (spindle amplitude)	Mean spindle amplitude (μV)
N2 epoch	continuous (spindle duration)	Mean spindle duration (s)
N2 epoch	continuous (spindle frequency)	Mean spindle frequency (Hz)
- Super Meta Notes: Derived from raw event Stage 2|2 (N2 sleep) by extracting spindle characteristics using wavelet bandpass detection (0.3-3.0s duration windows, FC=8-18Hz). Kernel step: Apply wavelet decomposition or bandpass filter to EEG, detect spindle events using amplitude/duration thresholds, compute aggregate statistics (density, amplitude, duration, frequency) per 30s N2 epoch.
- Evidence:
- DOI: 10.1038/ncomms15930 - Characterizes sleep spindles in 11,630 individuals from NSRR including SHHS (N=5,793), analyzing spindle density, amplitude, duration and frequency during N2 sleep as heritable quantitative traits
- Kernel Notes: Input: raw EEG from N2-annotated 30s epochs. Step 1: Bandpass filter (e.g., 8-18 Hz or wavelet decomposition). Step 2: Detect spindle events using amplitude threshold and duration criteria (0.3-3.0 s). Step 3: Compute per-epoch statistics: spindle count (density), mean amplitude, mean duration, dominant frequency. Output: 4 continuous regression targets. (replaces incomplete Primary Task 1 from original analysis.md)

Normal Task 1: Sleep Staging (5-class)

- Task Type: classification
- Epoch Method: epoch_by_long_event_hdf5
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0 (standard staging), target_events = see Target Definition
- Label Route: Annotation
- Target Definition: Event-to-label mapping table below.
- Notes: Stage 4 is rare in many files; merging Stage 3 and Stage 4 is recommended for robust modeling.

Event	y	Meaning
Wake|0	0	Wake

Event	y	Meaning
`Stage 1|1`	1	N1
`Stage 2|2`	2	N2
`Stage 3|3` or `Stage 4|4`	3	N3/SWS
`REM|5`	4	REM

Normal Task 2: Sleep vs Wake Detection (binary)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = 30 s, overlap = 0.0, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Useful baseline task when class granularity is reduced.

Event	y	Meaning
`Wake|0`	0	Awake
Any sleep stage (`1/2/3/4/5`)	1	Asleep

Normal Task 3: Arousal Detection (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` (or long-event segmentation if arousal duration labels are available)
- Epoch Parameters: tmin = -5.0 s, tmax = 5.0 s (event-centered), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Negative sampling strategy should be explicitly controlled to avoid label leakage.

Event	y	Meaning
`Arousals|Arousals`	1	Cortical arousal
Background non-arousal segments	0	Non-arousal

Normal Task 4: Respiratory Event Detection (binary)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5` (depends on event duration granularity)
- Epoch Parameters: task-dependent (for example, 10-30 s windows), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Respiratory subtype definitions should be verified per XML file and NSRR conversion version.

Event	y	Meaning
`Respiratory|Respiratory` (event windows)	1	Respiratory event
Non-event windows	0	Normal breathing/no event

Data Quality Notes

- Mixed-rate channels are exported at 125 Hz in EDF; originally low-rate channels (EOG/respiratory/SpO2/HR) are upsampled during export.
- Channel naming may vary across files (`NEW AIR` vs `AIRFLOW`), so channel selection should be robust to aliases.
- Annotation counts can differ between XML-derived event totals and summary CSV fields due to historical software conversion effects.
- Desaturation labels may be missing or partially inconsistent in some files; consider deriving desaturation from raw `Sa02` traces when needed.
- Class imbalance is expected (N2 dominant; Wake/N1 relatively sparse; Stage 4 rare).
- Recordings are unattended home PSG and therefore may have higher motion/environmental artifact than in-lab PSG.
- SHHS-2 ECG can differ in effective native sampling relative to SHHS-1; handle ECG separately if exact spectral comparability is required.

References

- Quan, S.F. et al. (1997). The Sleep Heart Health Study: design, rationale, and methods. Sleep, 20(12), 1077-1085.
- Redline, S. et al. (1998). Methods for obtaining and analyzing unattended polysomnography data for a multicenter study. Sleep, 21(7), 759-767.
- Dean, D.A. et al. (2016). Scaling Up Scientific Discovery in Sleep Medicine: The National Sleep Research Resource. Sleep, 39(5), 1151-1164. DOI: https://doi.org/10.5665/sleep.5774
- Zhang, G.Q. et al. (2018). The National Sleep Research Resource: making data findable, accessible, interoperable, reusable and promoting new research. Sleep Medicine, 10(5), 367-376.
- NSRR SHHS dataset page: https://sleepdata.org/datasets/shhs

275. tuh_eeg - TUH EEG Corpus (Base)

Source: analysis_minimax_v2.md

Dataset: `tuh_eeg` | Year: 2020 | URL: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg/v2.0.1/
Category: Biological State | Subjects: ~14987 | Files (Completed): 69652 | Channels: 17-128 | Sampling Rate: 250-1000 Hz

Paradigm

The Temple University Hospital EEG Corpus (v2.0.1) is the largest publicly available clinical EEG archive, containing approximately 69,652 EDF recordings from ~14,987 de-identified patients (Obeid & Picone, 2016). Recordings span routine clinical EEG sessions performed for neurological evaluation (epilepsy monitoring, encephalopathy, post-operative assessment, etc.). Three standard TCP (Temporal Central Parasagittal) referential montages are used: `01_tcp_ar` (average reference), `02_tcp_le` (linked ears), and `03_tcp_ar_a` (alternating average reference). Channel counts vary (17-128) and multiple sampling rates are present (250, 256, 400, 512, 1000 Hz). No structured experiment events exist; EDF annotations are clinical technician notes in free text.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg/v2.0.1/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`unknown`	69652	admin	raw	Clinical EEG with unstructured technician annotations; no ML-ready event labels

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> No structured classification annotations. The base corpus is not labeled for any specific ML task; use specialized TUH subsets (tuh_eeg_abnormal, tuh_eeg_seizure, etc.) for supervised learning.

Tasks

Primary Task 1: Normal vs Abnormal EEG Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 60 s, overlap = 0.0 s, target_events = [`unknown`] *

- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`unknown`	0	Normal EEG (no abnormality detected)
`unknown`	1	Abnormal EEG (epileptic or pathological)
- Super Meta Notes: Derived from clinical reports attached to each EEG recording in the TUH database. The dataset includes reports with patient medical history and medication details that provide the diagnostic classification (normal vs abnormal). Kernel step: Parse clinical report text to extract diagnostic label and apply to all epochs from that recording.
- Evidence:
- DOI: `10.1016/j.patrec.2020.03.009` - Uses the TUH EEG dataset to classify normal and abnormal EEG signals, training on 2717 EEG signals from 2130 unique patients and evaluating on 276 signals from 253 patients, achieving 79.34% accuracy using SVM with wavelet-based features.
- Kernel Notes: Extract 60-second non-overlapping epochs from raw EEG signals at 250 Hz. Map clinical diagnosis from report metadata to each epoch. Ensure montage consistency (process 01_tcp_ar, 02_tcp_le, 03_tcp_ar_a separately or re-reference uniformly).

Normal Task 1: Resting-State Segmentation (epoch_by_segmentation)

- Task Type: segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: length = 5.0 s, overlap = 2.5 s
- Label Route: `Annotation`
- Target Definition: N/A (no labels; use for pretraining / unsupervised learning)
- Notes: The base corpus is best used for self-supervised pretraining or large-scale representation learning. Labels (patient age, sex, montage type) can be derived from the EDF header or `eegunity_description` field parsed by the kernel. Ensure montage consistency before cross-recording analysis: process each montage type (01_tcp_ar, 02_tcp_le, 03_tcp_ar_a) separately or re-reference uniformly.

Data Quality Notes

- No structured ML annotations; this corpus is used as a pretraining source, not for supervised classification.
- Three TCP montages with differing channel layouts; normalize before cross-recording analysis.
- Clinical recordings: variable length, variable channel count, variable sampling rate.
- Sex available from EDF header (M/F/unknown).
- Age is not reliably available in the public release.
- Very large dataset (70k files): batch-processing and memory management are critical.

References

- Reference: Obeid I, Picone J (2016). The Temple University Hospital EEG Data Corpus. Frontiers in Neuroscience, 10, 196. https://isip.piconepress.com/projects/tuh_eeg/

276. tuh_eeg_abnormal - TUH EEG Abnormal Corpus

Source: analysis.md

Dataset: `tuh_eeg_abnormal` | Year: 2022 | URL: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_abnormal/v3.0.1/
Category: Biological State | Subjects: ~2983 sessions | Files (Completed): 2983 | Channels: 27-36 | Sampling Rate: 250-512 Hz

Paradigm

The TUH EEG Abnormal Corpus (v3.0.1) is a clinically annotated subset of the TUH EEG Corpus where each recording has been classified as normal or abnormal by board-certified neurologists (Lopez et al., 2015). Recordings are routine clinical EEG sessions acquired in an outpatient setting at Temple University Hospital. The normal/abnormal label reflects the overall clinical impression of the EEG recording, not individual event annotations. The label is encoded in the directory path (`normal/` or `abnormal/`) and extracted by the kernel to create a single whole-recording annotation.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_abnormal/v3.0.1/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`normal:1`	1517	long	raw	Normal EEG recording (full-recording label)
`abnormal:1`	1466	long	raw	Abnormal EEG recording (full-recording label)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Each recording has exactly one annotation spanning its full duration. `normal:1` = one "normal" annotation. Label derived from parent directory (`normal/` or `abnormal/`).

Tasks

Primary Task 1: Normal vs abnormal EEG recording classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification

- Label Route: `Annotation`
- Target Definition: Recording-level normal/abnormal diagnostic label.
- Evidence: TUH Abnormal Corpus is explicitly curated for normal-versus-abnormal EEG classification.
Source: https://isip.piconepress.com/projects/tuh_eeg/ DOI: https://doi.org/10.3389/fnins.2016.00196

Normal Task 1: Normal vs. Abnormal EEG Classification (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 5.0 s, overlap = 2.5 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: The annotation spans the entire recording; use `epoch_by_long_event_hdf5` with sliding windows. Parse the label prefix before the colon (`:`) to extract the class name. Near-balanced classes: 1517 normal vs. 1466 abnormal.

Event	y	Meaning
`normal:1`	0	Normal EEG
`abnormal:1`	1	Abnormal EEG
## Data Quality Notes		

- Sex is not reliably available in this subset.
- Variable channel count (27-36) and sampling rate (250-512 Hz) across recordings; normalize before cross-recording analysis.
- Whole-recording global labels; local event patterns within recordings are not annotated.
- Train/dev split is encoded in directory path; respect the split for proper evaluation.

References

- Reference: Lopez S et al. (2015). Analysis of Factors that Affect EEG Signal Quality. EMBC 2015. Obeid I, Picone J (2016). The Temple University Hospital EEG Data Corpus.
https://isip.piconepress.com/projects/tuh_eeg/

277. tuh_eeg_epilepsy - TUH EEG Epilepsy Corpus

Source: analysis.md

Dataset: `tuh_eeg_epilepsy` | Year: 2022 | URL: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_epilepsy/v2.0.1/
Category: Biological State | Subjects: ~2298 sessions | Files (Completed): 2298 | Channels: 17-41 | Sampling Rate: 250-1000 Hz

Paradigm

The TUH EEG Epilepsy Corpus (v2.0.1) is a clinically labeled subset of the TUH EEG Corpus where each recording is classified as epilepsy or no epilepsy based on patients' established clinical diagnoses (not real-time EEG pattern) recorded in hospital notes. Unlike tuh_eeg_seizure, the epilepsy label reflects a patient-level diagnosis, not the presence of ictal activity in the specific recording. The label is extracted from the directory path (`00_epilepsy/` or `01_no_epilepsy/`) and assigned as a single whole-recording annotation by the kernel.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_epilepsy/v2.0.1/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`epilepsy:1`	1785	long	raw	Patient has a diagnosis of epilepsy (whole-recording label)
`no epilepsy:1`	513	long	raw	Patient does not have epilepsy (whole-recording label)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Each recording has exactly one annotation spanning its full duration.

Tasks

Primary Task 1: Epilepsy vs non-epilepsy classification

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification

- Label Route: `Annotation`
- Target Definition: Patient/record-level epilepsy label from TUH Epilepsy corpus metadata.
- Evidence: TUH Epilepsy Corpus is specifically assembled for epilepsy diagnosis classification research. Source: https://isip.piconepress.com/projects/tuh_eeg/ DOI: https://doi.org/10.3389/fnins.2016.00196

Normal Task 1: Epilepsy Patient Classification (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 5.0 s, overlap = 2.5 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Parse label prefix before the colon. Class imbalance: 1785 epilepsy vs. 513 no-epilepsy (~3.5:1); apply class-balancing. Label is a patient-level diagnosis, not an EEG event; the EEG recording may or may not contain ictal activity.

Event	y	Meaning
`epilepsy:1`	1	Epilepsy diagnosis
`no epilepsy:1`	0	No epilepsy diagnosis
## Data Quality Notes		

- Class imbalance: epilepsy (77.6%) vs. no epilepsy (22.4%); balance before training.
- Label caveat: The epilepsy label is a clinical diagnosis from hospital records, not annotation of seizure events within the recording; some epilepsy-labeled recordings may be interictal.
- Variable channels (17-41) and sampling rates (250-1000 Hz); normalize before cross-recording analysis.
- Sex is not reliably available.

References

- Reference: TUH EEG Epilepsy Corpus v2.0.1. Obeid I, Picone J (2016). The Temple University Hospital EEG Data Corpus. https://isip.piconepress.com/projects/tuh_eeg/

278. tuh_eeg_events - TUH EEG Events Corpus

Source: analysis_minimax_v2.md

Dataset: `tuh_eeg_events` | Year: 2022 | URL: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_events/v2.0.1/
Category: Biological State | Subjects: ~370 | Files (Completed): 518 | Channels: 24-33 | Sampling Rate: 250 Hz

Paradigm

The TUH EEG Events Corpus (v2.0.1) provides time-stamped annotations of neurological events in 518 clinical EEG recordings from ~370 unique patients (Shah et al., 2018). Expert neurologists annotated six event types at temporal and spatial resolution using companion `.rec` sidecar files (4-column format: channel, start_time, stop_time, label_code). Events include epileptiform discharges, background EEG, and artifact. All recordings are sampled at 250 Hz with 24-33 channels.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_events/v2.0.1/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`background`	varies	long	raw	Normal background EEG activity
`spike and sharp wave`	varies	short	raw	Spike or sharp wave discharge
`generalized periodic epileptiform discharges`	varies	short	raw	GPEDs - generalized periodic epileptiform pattern
`periodic lateralized epileptiform discharges`	varies	short	raw	PLEDs - lateralized epileptiform pattern
`eye movement`	varies	short	raw	Eye movement artifact
`artifact`	varies	short	raw	Non-specific artifact

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Label codes from `.rec` files: 1=spike and sharp wave, 2=generalized periodic epileptiform discharges, 3=periodic lateralized epileptiform discharges, 4=eye movement, 5=artifact,

6=background.

Tasks

Primary Task 1: Seizure vs. Non-Seizure EEG Classification

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4 s, overlap = 0.0, target_events = [`seizure`, `non-seizure`]
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:

Event	y	Meaning
`non-seizure`	0	Normal/non-epileptiform EEG segments
`seizure`	1	Segments containing epileptiform activity (spikes, GPEDs, PLEDs)

- Super Meta Notes: Derived from raw event(s) [`background`, `spike and sharp wave`, `generalized periodic epileptiform discharges`, `periodic lateralized epileptiform discharges`]. Kernel step: Segment raw EEG into 4-second windows. Assign y=1 if window overlaps with any epileptiform annotation (spike/sharp wave, GPEDs, PLEDs); otherwise assign y=0 (background).
- Evidence:
- DOI: 10.1109/tii.2023.3274913 - Uses TUH EEG dataset and classifies 4-second EEG segments into seizure and non-seizure classes using SS-CSV features with CNN, SVM, and kNN classifiers, achieving 98.48% accuracy with CNN.
- Kernel Notes: Requires `.rec` sidecar files for annotation extraction. Verify event ID mapping from events.tsv. Windowing approach: 4-second fixed segments, no overlap. All seizure-related annotations (spike/sharp wave, GPEDs, PLEDs) map to class 1; background maps to class 0. This is a binary simplification of the full multi-class event detection task.

Primary Task 2: Multi-Class EEG Event Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s , tmax = 1.0 s , target_events = [`background`, `spike and sharp wave`, `generalized periodic epileptiform discharges`, `periodic lateralized epileptiform discharges`]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`background`	0	Normal background EEG
`spike and sharp wave`	1	Epileptiform spike/sharp wave
`generalized periodic epileptiform discharges`	2	GPEDs
`periodic lateralized epileptiform discharges`	3	PLEDs

- Evidence:
- DOI: 10.1109/tii.2023.3274913 - Confirms TUH EEG Events Corpus provides time-stamped multi-class event annotations as benchmark target (seizure vs non-seizure uses underlying event classes).
- Kernel Notes: Default parameters (tmin=-0.5s, tmax=1.0s) are assumed based on typical EEG epoch windows. Verify against original paper implementation. Excludes eye movement and artifact events per dataset design. (replaces Normal Task 1 and Normal Task 2 from original analysis.md)

Normal Task 1: EEG Event Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Exclude `eye movement` and `artifact` events from classification targets. For binary detection: background (y=0) vs. any epileptiform event (y=1). `.rec` sidecar files must be present alongside `.edf` files.

Event	y	Meaning
`background`	0	Normal background EEG
`spike and sharp wave`	1	Epileptiform spike/sharp wave
`generalized periodic epileptiform discharges`	2	GPEDs
`periodic lateralized epileptiform discharges`	3	PLEDs

Normal Task 2: Epileptiform vs. Normal EEG Detection (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.5 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
`background`	0	Normal EEG
`spike and sharp wave`	1	Epileptiform
`generalized periodic epileptiform discharges`	1	Epileptiform
`periodic lateralized epileptiform discharges`	1	Epileptiform

Data Quality Notes

- `.rec` sidecar files required; if missing, no annotations are set.
- Background annotations are long events; epileptiform events are short - class length mismatch may cause imbalance.
- Uniform sampling rate (250 Hz) and consistent channel count (24-33); relatively homogeneous compared to other TUH subsets.
- Sex is not reliably available.

References

- Reference: Shah V et al. (2018). The Temple University Hospital EEG Event Corpus. Frontiers in Neuroscience, 12, 862. https://isip.piconepress.com/projects/tuh_eeg/

279. tuh_eeg_seizure - TUH EEG Seizure Corpus

Source: analysis_minimax_v2.md

Dataset: `tuh_eeg_seizure` | Year: 2022 | URL: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_seizure/v2.0.3/
Category: Biological State | Subjects: ~675 | Files (Completed): 7361 | Channels: 17-128 | Sampling Rate: 250-1000 Hz

Paradigm

The TUH EEG Seizure Corpus (v2.0.3) is the largest publicly available seizure detection dataset, containing 7,361 clinical EEG recordings from ~675 unique patients with time-stamped seizure annotations (Shah et al., 2020). Expert neurologists labeled each segment as background (bckg) or one of five seizure types using companion CSV sidecar files (channel, start_time, stop_time, label). The dataset is pre-split into train and dev sets (directory path contains `/train/` or `/dev/`). Recording lengths range from a few minutes to several hours.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_seizure/v2.0.3/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`bckg`	~75000	long	raw	Background - no seizure activity
`gnsz`	varies	long	raw	Generalized non-specific seizure
`fnsz`	varies	long	raw	Focal non-specific seizure
`cpsz`	varies	long	raw	Complex partial seizure
`absz`	varies	long	raw	Absence seizure
`tnsz`	varies	long	raw	Tonic seizure

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> `bckg` (background) is overwhelmingly the most frequent label. Seizure events are annotated with onset, offset, and type from CSV sidecars.

Tasks

Primary Task 1: Binary Seizure Detection via Fixed 4-Second Segmentation

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 4.0 s, overlap = 0.0 s
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`bckg`	0	Background / non-seizure
`gnsz`	1	Generalized non-specific seizure
`fnsz`	1	Focal non-specific seizure
`cpsz`	1	Complex partial seizure
`absz`	1	Absence seizure
`tnsz`	1	Tonic seizure

- Evidence:
- DOI: 10.1109/tii.2023.3274913 - Uses TUH EEG Corpus, segments EEG signals into 4-second epochs without overlap and classifies each segment as seizure or non-seizure using SS-CSV-derived features with CNN/SVM/kNN classifiers.
- Kernel Notes: Apply segmentation to full recording (no annotations needed). For labels, cross-reference each segment's timestamp with seizure annotation CSV sidecars; if segment overlaps any seizure annotation, label=1, else label=0 (bckg).

Primary Task 2: Binary Seizure Detection via 1-Second Sliding Windows

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = 1.0 s, overlap = 0.5 s
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
`bckg`	0	Background / non-seizure
`gnsz`	1	Generalized non-specific seizure
`fnsz`	1	Focal non-specific seizure
`cpsz`	1	Complex partial seizure
`absz`	1	Absence seizure
`tnsz`	1	Tonic seizure

- Evidence:
- DOI: 10.1088/1741-2552/ac7d0d - Uses TUH EEG dataset with 12 selected cases for cross-subject seizure detection, segmenting signals into 1-second windows with 50% overlap and using binary classification between seizure and non-seizure EEG.
- Kernel Notes: Apply sliding window segmentation (1s window, 0.5s overlap) to raw EEG. Label each window by checking overlap with seizure annotations from CSV sidecars.

Primary Task 3: Binary Seizure Detection via Annotation-Based Epoching

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = annotation duration, overlap = 0.0 s, target_events = [`bckg`, `gnsz`, `fnsz`, `cpsz`, `absz`, `tnsz`]

- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:
Event	y	Meaning
`bckg`	0	Background / non-seizure
`gnsz`	1	Generalized non-specific seizure
`fnsz`	1	Focal non-specific seizure
`cpsz`	1	Complex partial seizure
`absz`	1	Absence seizure
`tnsz`	1	Tonic seizure
- Evidence:
- DOI: 10.1117/12.2670106 - Uses TUH EEG Seizure Corpus (TUSZ) version 1.5.2 with train/dev split to train XGBoost, Random Forest, KNN, and SVC classifiers for binary seizure vs background detection using channel selection based on cross-correlation analysis.
- Kernel Notes: Extract epochs directly from seizure and background annotation segments. Apply 0.5-40 Hz bandpass filter as mentioned in paper. Label bckg=0, all seizure types=1.

Normal Task 1: Seizure Detection - bckg vs. Seizure (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s (annotation onset), duration = annotation duration, overlap = 0.0 s for initial segmentation; or sliding window 5.0 s, overlap = 2.5 s on annotated segments, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Extreme class imbalance: `bckg` >> all seizure types combined. Undersample background or use cost-sensitive learning. CSV sidecars required alongside EDF files.

Event	y	Meaning
`bckg`	0	Background / no seizure
`gnsz`	1	Generalized seizure
`fnsz`	1	Focal seizure
`cpsz`	1	Complex partial seizure
`absz`	1	Absence seizure
`tnsz`	1	Tonic seizure

Normal Task 2: Seizure Type Classification (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: Use annotation duration (seizure-type segments only), target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Use only seizure-type annotations (gnsz=0, fnsz=1, cpsz=2, absz=3, tnsz=4); exclude `bckg`.
- Notes: Severe class imbalance across seizure types; some types (absz, tnsz) are rare.

Data Quality Notes

- Extreme class imbalance: background constitutes >80% of annotations; severely balance before training.
- Variable channel count (17-128) and sampling rate (250-1000 Hz); normalize per recording.
- Train/dev split is pre-defined; respect it for evaluation. Split encoded in file path (`/train/`, `/dev/`).
- Sex is not reliably available.
- Long recordings may span multiple seizure episodes; temporal context is important.

References

- Reference: Shah V et al. (2020). The Temple University Hospital EEG Seizure Corpus. Frontiers in Neural Engineering. https://isip.piconepress.com/projects/tuh_eeg/

280. tuh_eeg_slowing - TUH EEG Slowing Corpus

Source: analysis.md

Dataset: `tuh_eeg_slowing` | Year: 2022 | URL: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_slowing/v2.0.1/
Category: Biological State | Subjects: ~47 | Files (Completed): 112 | Channels: 27-41 | Sampling Rate: 250-512 Hz

Paradigm

The TUH EEG Slowing Corpus (v2.0.1) contains 112 clinical EEG recordings with time-stamped annotations for EEG slowing, seizure activity, and background segments. Expert annotators labeled each segment using companion CSV sidecar files (channel, start_time, stop_time, label). EEG slowing (diffuse or focal slow-wave activity) is a neurological biomarker of conditions such as encephalopathy, stroke, or post-ictal states. The corpus enables the development of algorithms to automatically detect and localize EEG slowing.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision:
> https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_slowing/v2.0.1/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`seiz`	varies	long	raw	Seizure activity
`slow`	varies	long	raw	EEG slowing (pathological slow-wave activity)
`bckg`	varies	long	raw	Background - normal EEG

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Multiple annotations may appear in a single recording (e.g., `bckg + slow`, `bckg + seiz`). Per the CSV analysis, top events include `seiz:22`, `seiz:43`, `slow:64`, and `bckg:43`, `slow:63`.

Tasks

Primary Task 1: EEG slowing detection (focal/generalized vs background)

- Task Type: classification/detection
- Epoch Method: `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, overlap = Needs verification, target_events = [Needs verification] | segment_length = Needs verification, overlap = Needs verification
- Label Route: `Annotation`

- Target Definition: Slowing-related event labels against normal background segments.
- Evidence: TUH Slowing Corpus provides curated slowing annotations for automated slowing detection tasks. Source: https://isip.piconepress.com/projects/tuh_eeg/ DOI: https://doi.org/10.3389/fnins.2016.00196

Normal Task 1: EEG Slowing Detection - Normal vs. Slowing (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, use annotation duration, overlap = 0.0 s; or sliding window 5.0 s, overlap = 2.5 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Exclude `seiz` events if slowing-specific detection is the goal, or include as a third class. CSV sidecars required.

Event	y	Meaning
`bckg`	0	Normal background EEG
`slow`	1	EEG slowing

Normal Task 2: 3-Class EEG State Classification (epoch_by_long_event)

- Task Type: classification
- Epoch Method: `epoch_by_long_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, use annotation duration, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
`bckg`	0	Background
`slow`	1	Slowing
`seiz`	2	Seizure
## Data Quality Notes		

- Small dataset (112 files, ~47 subjects); limited sample size.
- CSV sidecars required; if missing, the kernel raises a ValueError.
- Variable sampling rate (250-512 Hz); normalize before cross-recording analysis.
- Annotations may overlap (e.g., slowing + seizure in the same segment); handle compound annotations carefully.
- Sex is not reliably available.

References

- Reference: TUH EEG Slowing Corpus v2.0.1. Obeid I, Picone J (2016). The Temple University Hospital EEG Data Corpus. https://isip.piconepress.com/projects/tuh_eeg/

Chapter 6. Others

Datasets in this chapter: 6. Does not clearly fit the above categories.

281. openneuro_ds004075 - what_are_we_talking_about

Source: analysis.md

Dataset: `openneuro_ds004075` | Year: 2022 | URL: https://openneuro.org/datasets/ds004075
Category: Others | Subjects: N/A | Files (Completed): 115
Channels: 64.0 | Sampling Rate: 1000.0 Hz

Paradigm

The provided title 'what_are_we_talking_about' without an accompanying README is too vague to determine a specific experimental paradigm.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: regression target
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004075

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`no_events`	N/A	admin	raw	No discrete event markers detected; use continuous segmentation or external metadata labels.
## MISC Label Summary				

- Route: `misc`
- Role: Stores derived labels (super meta) not directly present in raw event IDs.
- Candidate Labels: Needs verification
- Kernel Note: Document derivation logic, source fields, units, and version tags.

Tasks

Primary Task 1: Publisher-defined BIDS task: Context03

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]

- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004075 DOI: https://doi.org/10.18112/openneuro.ds004075.v1.0.0

Primary Task 2: Publisher-defined BIDS task: Context09

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004075 DOI: https://doi.org/10.18112/openneuro.ds004075.v1.0.0

Primary Task 3: Publisher-defined BIDS task: Context10

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004075 DOI: https://doi.org/10.18112/openneuro.ds004075.v1.0.0

Normal Task 1: Raw-meta task

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Needs verification
- Kernel Notes: Exclude admin markers from target labels.

Normal Task 2: Super-meta task

- Task Type: regression/classification
- Epoch Method: `epoch_by_event_hdf5` or `epoch_by_long_event_hdf5`
- Epoch Parameters: Needs verification, target_events = see Target Definition
- Label Route: `misc`
- Target Definition: Continuous target value as defined by kernel and label route.
- Regression Target / Label: Needs verification
- Kernel Notes: Derive and version super-meta in kernel.

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

282. openneuro_ds004784 - Phantom EEG Dataset with Motion, Muscle, and Eye Artifacts

Source: analysis_minimax_v2.md

Dataset: openneuro_ds004784 | Year: 2023 | URL: https://openneuro.org/datasets/ds004784
Category: Others | Subjects: N/A | Files (Completed): 10 | Channels: 264 | Sampling Rate: 512 Hz

Paradigm

Technical validation study using an electrically conductive head phantom to simulate brain activity under various artifact conditions (motion, muscle, eye movements) for testing EEG cleaning algorithms. Code 65471 appears in all 10 files (20 total events); code 65343 appears in all 10 files (10 total events). These likely mark artifact-on vs artifact-off conditions or different artifact types.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004784

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
65471	20	short	raw	Artifact condition marker A (e.g., artifact-on)
65343	10	short	raw	Artifact condition marker B (e.g., clean condition)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Artifact Condition Classification

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = [65471, 65343]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

Event	y	Meaning
65471	0	Artifact condition A (artifact-on)
65343	1	Artifact condition B (artifact-off/clean)

- Evidence:
- DOI: https://doi.org/10.3390/s23198214 - Validates iCanClean artifact removal algorithm on OpenNeuro ds004784 phantom EEG dataset containing ground-truth brain signals and various artifact conditions (motion, muscle, eye, line-noise), comparing it to ASR, Auto-CCA, and Adaptive Filtering methods.
- Kernel Notes: Binary classification between artifact-present (65471) and artifact-absent (65343) conditions. Verify epoch timing from dataset README; paper also mentions testing window sizes of 2-4s for real-time applications. (replaces Normal Task 1 from original analysis.md)

Data Quality Notes

- Very small dataset (10 files, 30 events total) - insufficient for standard classification.
- 264-channel phantom EEG; not standard scalp EEG configuration.
- Designed for artifact algorithm validation, not neural decoding.

References

- Reference: `Phantom EEG Dataset with Motion, Muscle, and Eye Artifacts and Example Scripts`

283. openneuro_ds004843 - Visual Situational Awareness EEG (T16)

Source: analysis.md

Dataset: openneuro_ds004843 | Year: 2023 | URL: https://openneuro.org/datasets/ds004843
Category: Others | Subjects: 7 | Files (Completed): 7 | Channels: 70 | Sampling Rate: 256 Hz

Paradigm

The VisualSituationalAwareness task investigates neural correlates of situational awareness during visual monitoring. Subjects viewed structured visual scenes with EEG recorded across multiple runs. The 5-digit hierarchical trigger codes encode stimulus category, load level, and response type. Codes 22301/22302 (3,384 each) and 22501/22502 (2,331/2,325) dominate; code 32000 (1,099) likely marks scene/trial boundaries. Codes ending in 01/02 mark paired conditions (onset/offset or left/right).

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds004843

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
22301	3384	short	raw	Condition 2-2-3, variant 01
22302	3384	short	raw	Condition 2-2-3, variant 02
22501	2331	short	raw	Condition 2-2-5, variant 01
22502	2325	short	raw	Condition 2-2-5, variant 02
22102	1327	short	raw	Condition 2-2-1, variant 02
22101	1325	short	raw	Condition 2-2-1, variant 01
32000	1099	admin	raw	Scene/trial boundary marker (exclude)
22601	436	short	raw	Condition 2-2-6, variant 01
22602	436	short	raw	Condition 2-2-6, variant 02
42001	382	short	raw	Condition 4-2-0, variant 01

Event ID	Count (total)	Type	Meta Level	Meaning
42002	382	short	raw	Condition 4-2-0, variant 02
22001	372	short	raw	Condition 2-2-0, variant 01
22002	372	short	raw	Condition 2-2-0, variant 02
12201	306	short	raw	Condition 1-2-2, variant 01
12202	306	short	raw	Condition 1-2-2, variant 02

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: `stim`
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: Needs verification
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Visual Situational Awareness

- Task Type: classification/segmentation
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: `Annotation`
- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds004843 DOI: https://doi.org/10.18112/openneuro.ds004843.v1.0.0

Normal Task 1: Visual Situational Awareness (multiclass)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: `32000` (1,099) is a scene/trial boundary - exclude. Very small dataset (7 files); balance classes by undersampling. 5-digit codes require README or paper decode.

Event	y	Meaning
22301	0	Condition 22301
22302	1	Condition 22302
22501	2	Condition 22501
22502	3	Condition 22502
22101	4	Condition 22101

Event	y	Meaning
22102	5	Condition 22102
22601	6	Condition 22601
22602	7	Condition 22602
## Data Quality Notes		

- Only 7 files - very small dataset for training.
- 5-digit hierarchical codes require decode from README or paper.
- 32000 is a boundary/transition marker - exclude from epoching.

References

- Reference: T16 - Visual Situational Awareness EEG

284. openneuro_ds005787 - NOD-EEG

Source: analysis.md

Dataset: openneuro_ds005787 | Year: 2025 | URL: https://openneuro.org/datasets/ds005787
Category: Others | Subjects: N/A | Files (Completed): 896 | Channels: 64 | Sampling Rate: 250 Hz

Paradigm

The NOD-EEG dataset contains 64-channel EEG recordings from a stimulus-response paradigm. Each trial includes a stimulus onset (stim_on), participant button-press response (resp), and block boundary markers (begin, end). The precise experimental task and stimuli are described in the dataset README.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (Annotation, stim, misc) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation, stim
> Web Reference Used for Decision: https://openneuro.org/datasets/ds005787

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
resp	2273060	short	raw	Participant button-press response
stim_on	112000	short	raw	Stimulus onset
begin	896	long	raw	Block begin marker
end	896	long	raw	Block end marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

STIM Label Summary

- Route: stim
- Role: Trigger-channel or stimulus-marker based labeling.
- Candidate Trigger Labels: stim_on
- Kernel Note: Keep trigger decoding table versioned and auditable.

Tasks

Primary Task 1: Publisher-defined BIDS task: Image Net

- Task Type: classification/segmentation
- Epoch Method: epoch_by_event_hdf5
- Epoch Parameters: tmin = Needs verification, tmax = Needs verification, target_events = [Needs verification]
- Label Route: Annotation

- Target Definition: Event/condition labels for this BIDS task (dataset-specific mapping)
- Evidence: OpenNeuro metadata.tasksCompleted and dataset readme. Source: https://openneuro.org/datasets/ds005787 DOI: https://doi.org/10.18112/openneuro.ds005787.v1.0.0

Normal Task 1: Stimulus-Response (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = -0.2 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Very high count for `resp` (2.27M total); verify that events are correctly timestamped. `begin/end` mark block boundaries.

Event	y	Meaning
`resp`	0	Button-press response
`stim_on`	1	Stimulus onset
`begin`	2	Block begin
`end`	3	Block end
## Data Quality Notes		

- Very high total event count for `resp`; verify no duplicate events.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. Scientific Data, 6, 103`

285. physionet_motionartifact - EEG/fNIRS Motion Artifact Dataset

Source: analysis_minimax_v2.md

Dataset: `physionet_motionartifact` | Year: 2018 | URL: https://physionet.org/content/motion-artifact/1.0.0/
Category: Others | Subjects: 32 | Files (Completed): 32 | Channels: 10 (EEG) or 12 (fNIRS) | Sampling Rate: 2048 Hz (EEG), 200 Hz (fNIRS)

Paradigm

Thirty-two recordings (23 EEG + 9 fNIRS) were collected from subjects performing various physical activities: cycling, running, walking, push-ups, pull-ups, stretching, and rest (Schultz & Kluge, 2018). The purpose is to provide a benchmark dataset for motion artifact detection and removal algorithms. EEG was recorded from 10 channels (gel electrodes) at 2048 Hz; fNIRS from 12 channels at 200 Hz. Each recording corresponds to one subject performing one activity. Activity labels are encoded in the filename (recording ID: `eeg_1` through `eeg_23`; `fnirs_1` through `fnirs_9`) and documented in a companion README.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 1 + Type 3
> Why: Raw markers are sufficient for primary tasks.
> Primary Label Routes: Annotation
> Web Reference Used for Decision: https://physionet.org/content/motion-artifact/1.0.0/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
(none)	-	-	raw	No event annotations present; activity label is implicit in recording ID

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

Tasks

Primary Task 1: Binary Motion Artifact Segmentation

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0 s, tmax = * (full artifact/clean interval duration, typically ~2 min), target_events = [R, F]
- Label Route: `Annotation`
- Meta Level: raw
- Target Definition:

| Event | y | Meaning |

|-------|---|---------|
| R | 0 | Clean interval (rise annotation) |
| F | 1 | Motion artifact-contaminated interval (fall annotation) |
- Evidence:
- DOI: https://doi.org/10.13026/C2988P - PhysioNet dataset contains trigger annotations marking transitions between clean (R) and artifact-contaminated (F) intervals for evaluating artifact removal methods.
- Kernel Notes: Verify exact event ID strings ('R' and 'F') from events.tsv; these correspond to "rise" (clean) and "fall" (artifact) markers in the trigger stream. (replaces Primary Task 1 from original analysis.md)

Primary Task 2: Motion Artifact Correction via Wavelet Decomposition

- Task Type: regression/segmentation
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: segment_length = * (full recording length ~9 min), overlap = 0.0
- Label Route: `Annotation`
- Meta Level: super
- Target Definition:
Event	y	Meaning
`motion_corrupted_signal`	1	Motion-artifact contaminated sub-band component (WPD decomposition)
`reference_ground_truth`	0	Clean reference signal from synchronized undisturbed channel
- Super Meta Notes: Derived from simultaneous dual-channel recording setup: one motion-impacted channel paired with one reference channel placed very close (20mm EEG, 30mm fNIRS). Kernel must use wavelet packet decomposition (WPD level j=4) to decompose signals into 16 sub-band components and identify artifact-contaminated components using the reference channel as ground truth.
- Evidence:
- DOI: 10.3390/s22093169 - Uses PhysioNet motion artifact dataset with paired motion-corrupted and reference ground truth signals to evaluate WPD methods for single-channel motion artifact correction.
- Kernel Notes: Requires access to both disturbed and reference channels within each recording; WPD decomposition parameters (level j=4) must be applied to generate sub-band components for classification.

Normal Task 1: Physical Activity Classification - Resting-State Segmentation (epoch_by_segmentation)

- Task Type: classification
- Epoch Method: `epoch_by_segmentation_hdf5`
- Epoch Parameters: length = 5.0 s, overlap = 2.5 s
- Label Route: `Annotation`
- Target Definition: N/A - activity labels must be assigned post-hoc from filename metadata (see notes)
- Notes: No event annotations are present; the kernel only records the recording ID in the description field. To assign activity labels for supervised learning, map recording IDs to physical activities using the dataset README. For artifact detection tasks, all recordings can be treated as motion-contaminated (except rest recordings).

Data Quality Notes

- No structured event annotations; classification labels must be derived from filename-based recording IDs.
- Mixed signal types: EEG (eeg_1-eeg_23, 10 channels, 2048 Hz) and fNIRS (fnirs_1-fnirs_9, 12 channels, 200 Hz) in the same locator; process these file types separately.
- Age and sex information are not available.

- Primary use case is motion artifact removal/detection benchmarking, not cognitive task decoding.

References

- Reference: Schultz T, Kluge T. Motion artifact dataset for EEG and fNIRS. PhysioNet. DOI: https://doi.org/10.13026/C2988P

286. tuh_eeg_artifact - TUH EEG Artifact Corpus

Source: analysis_minimax_v2.md

Dataset: `tuh_eeg_artifact` | Year: 2022 | URL: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_artifact/v3.0.1/
Category: Others | Subjects: ~310 sessions | Files (Completed): 310 | Channels: 27-128 | Sampling Rate: 250-1000 Hz

Paradigm

The TUH EEG Artifact Corpus (v3.0.1) provides time-stamped, channel-level artifact annotations on 310 clinical EEG recordings selected from the TUH EEG Corpus. Expert annotators labeled artifact segments with type and temporal boundaries, stored in companion CSV sidecar files. Each annotation includes: channel name, start time, stop time, label, and confidence score. Multiple artifact types may co-occur within a recording.

Kernel Responsibilities

- Type 1 (Raw-meta extraction only): Extract and normalize raw labels from events/trigger streams.
- Type 2 (Raw + Super meta integration): Derive and merge high-level labels that are not directly represented in raw events.
- Type 3 (Label-route orchestration): Route final labels to EEGUnity-compatible channels (`Annotation`, `stim`, `misc`) with reproducible schema.

> Selected Mode: Type 2 + Type 3
> Why: Derived targets are needed beyond raw markers.
> Primary Label Routes: Annotation, misc
> Evidence Keywords: confidence
> Web Reference Used for Decision: https://isip.piconepress.com/projects/tuh_eeg/downloads/tuh_eeg_artifact/v3.0.1/

Event Structure

Event ID	Count (total)	Type	Meta Level	Meaning
`eyem`	varies	short	raw	Eye movement artifact
`musc`	varies	short	raw	Muscle (EMG) artifact
`elec`	varies	short	raw	Electrode artifact (pop, movement)
`chew`	varies	short	raw	Chewing artifact
`shiv`	varies	short	raw	Shivering artifact
`elpp`	varies	short	raw	Electrode pop
`null`	varies	admin	raw	No artifact (clean segment)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude
>
> Artifact annotations are channel-level and time-stamped. Multiple artifact types may appear in a single recording; combined labels (e.g., `musc_elec`, `eyem_musc`) are also present.

MISC Label Summary

- Route: misc
- Role: Stores derived/continuous labels that are not native raw event IDs.
- Candidate Labels: confidence
- Kernel Note: Keep derivation source, unit, and version metadata.

Tasks

Primary Task 1: Artifact Binary Detection (artifact vs clean)

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 1.0 s, overlap = 0.0, target_events = [eyem, musc, elec, chew, shiv, elpp, null, eyem_musc, combined_artifacts, other_combined]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
null	0	Clean EEG segment
eyem	1	Eye movement artifact
musc	1	Muscle (EMG) artifact
elec	1	Electrode artifact
chew	1	Chewing artifact
shiv	1	Shivering artifact
elpp	1	Electrode pop
eyem_musc	1	Combined artifact
combined_artifacts	1	Combined artifact
other_combined	1	Combined artifact

- Evidence:
- DOI: 10.1109/embc48229.2022.9871413 - Binary classification (artifact vs clean) on 1-second windows achieves 93.95% accuracy
- DOI: 10.1038/s41598_024_52551_0 - Binary classification with artifact->1, null->0 mapping on 1s windows
- Kernel Notes: Sliding window segmentation extracts 1s epochs from raw signal; each window labeled based on whether any artifact annotation overlaps that segment. Combined labels (eyem_musc, combined_artifacts, etc.) mapped to artifact class (y=1). (replaces Primary Task 1 from original analysis.md)

Primary Task 2: Multi-class Artifact Type Classification

- Task Type: classification
- Epoch Method: epoch_by_segmentation_hdf5
- Epoch Parameters: segment_length = 1.0 s, overlap = 0.0, target_events = [eyem, musc, elec, chew, shiv, elpp, null]
- Label Route: Annotation
- Meta Level: raw
- Target Definition:

Event	y	Meaning
null	0	Clean EEG segment
eyem	1	Eye movement artifact
musc	2	Muscle (EMG) artifact
elec	3	Electrode artifact
chew	4	Chewing artifact

| shiv | 5 | Shivering artifact |
| elpp | 6 | Electrode pop |
- Evidence:
- DOI: 10.1109/embc48229.2022.9871413 - Multi-class multi-output classification assigns 12 artifact types per channel
- DOI: 10.1038/s41598_024_52551_0 - Multiclass-multioutput classification with artifact types 1-12 per channel
- Kernel Notes: Sliding window segmentation with 1s epochs; each window labeled with the dominant artifact type if multiple overlap. Exclude windows with combined/mixed labels or assign to a separate class. Standard TUAR mapping: eyem->0, musc->1, elec->2, chew->3, shiv->4, elpp->5.

Normal Task 1: Artifact Type Classification (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.
- Notes: Exclude `null` segments. Combined labels (e.g., `musc_elec`) require a decision: either assign to the primary artifact type or treat as a separate class. CSV sidecars must be present alongside EDF files.

Event	y	Meaning
`eyem`	0	Eye movement
`musc`	1	Muscle (EMG)
`elec`	2	Electrode artifact
`chew`	3	Chewing
`shiv`	4	Shivering
`elpp`	5	Electrode pop

Normal Task 2: Artifact vs. Clean EEG Detection (epoch_by_event)

- Task Type: classification
- Epoch Method: `epoch_by_event_hdf5`
- Epoch Parameters: tmin = 0.0 s, tmax = 1.0 s, target_events = see Target Definition
- Label Route: `Annotation`
- Target Definition: Event-to-label mapping table below.

Event	y	Meaning
`null`	0	Clean EEG segment
`eyem`	1	Any artifact
`musc`	1	Any artifact
`elec`	1	Any artifact
`chew`	1	Any artifact

Data Quality Notes

- Annotations are at the channel level; aggregate to recording level or use channel-specific labels depending on the task.
- Combined label types (e.g., `eyem_musc`) create label ambiguity; standardize before training.

- Variable channel count and sampling rate; normalize before cross-recording analysis.
- Sex is not reliably available.

References

- Reference: Buckwalter G, Chhin S, Rahman S, Obeid I, Picone J. (2021). Recent Advances in the TUH EEG Corpus: Improving the Interrater Agreement for Artifacts and Epileptiform Events. IEEE Signal Processing in Medicine and Biology Symposium (SPMB). DOI: https://doi.org/10.1109/SPMB52430.2021.9672302